Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			31	ARPEN	R/W	0x0	ARP Offload Enable When this bit is set, the MAC can recognize an incoming ARP request packet and schedules the ARP packet for transmission. It forwards the ARP packet to the application and also indicate the events in the RxStatus. When this bit is reset, the MAC receiver does not recognize any ARP packet and indicates them as Type frame in the RxStatus. This bit is available only when the Enable IPv4 ARP Offload is selected.Values: 0x0 (DISABLE): ARP Offload is disabled 0x1 (ENABLE): ARP Offload is enabled Value After Reset: 0x0
			30:28	SARC	R/W	0x0	Source Address Insertion or Replacement Control This field controls the source address insertion or replacement for all transmitted packets. Bit 30 specifies which MAC Address register (0 or 1) is used for source address insertion or replacement based on the values of Bits[29:28]: 2'b0x: The mti_sa_ctrl_i and ati_sa_ctrl_i input signals control the SA field generation. 2'b10: If Bit 30 is set to 0, the MAC inserts the content of the MAC Address 0 registers in the SA field of all transmitted packets. If Bit 30 is set to 1 and the Enable MAC Address Register 1 option is selected while configuring the IP, the MAC inserts the content of the MAC Address 1 registers in the SA field of all transmitted packets. 2'b11: If Bit 30 is set to 0, the MAC replaces the content of the MAC Address 0 registers in the SA field of all transmitted packets. If Bit 30 is set to 1 and the MAC Address Register 1 is enabled, the MAC replaces the content of the MAC Address 1 registers in the SA field of all transmitted packets. Note: Changes to this field take effect only on the start of a packet. If you write to this register field when a packet is being transmitted, only the subsequent packet can use the updated value, that is, the current packet does not use the updated value.Values: 0x0 (SA_CTRL_IN): mti_sa_ctrl_i and ati_sa_ctrl_i input signals control the SA field generation 0x2 (MAC0_INS_SA): Contents of MAC Addr-0 inserted in SA field 0x3 (MAC0_REP_SA): Contents of MAC Addr-0 replaces SA field 0x6 (MAC1_INS_SA): Contents of MAC Addr-1 inserted in SA field 0x7 (MAC1_REP_SA): Contents of MAC Addr-1 replaces SA field Value After Reset: 0x0
			27	IPC	R/W	0x0	Checksum Offload When set, this bit enables the IPv4 header checksum checking and IPv4 or IPv6 TCP, UDP, or ICMP payload checksum checking. When this bit is reset, the COE function in the receiver is disabled. The Layer 3 and Layer 4 Packet Filter and Enable Split Header features automatically selects the IPC Full Checksum Offload Engine on the Receive side. When any of these features are enabled, you must set the IPC bit.Values: 0x0 (DISABLE): IP header/payload checksum checking is disabled 0x1 (ENABLE): IP header/payload checksum checking is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			26:24	IPG	R/W	0x0	Inter-Packet Gap These bits control the minimum IPG between packets during transmission. This range of minimum IPG is valid in full-duplex mode. In the half-duplex mode, the minimum IPG can be configured only for 64-bit times (IPG = 100). Lower values are not considered. When a JAM pattern is being transmitted because of backpressure activation, the MAC does not consider the minimum IPG. This function (IPG less than 96 bit times) is valid only when EIPGEN bit in MAC_Ext_Configuration register is reset. When EIPGEN is set, then the minimum IPG (greater than 96 bit times) is controlled as per the description given in EIPG field in MAC_Ext_Configuration register.Values: 0x0 (IPG96): 96 bit times IPG 0x1 (IPG88): 88 bit times IPG 0x2 (IPG80): 80 bit times IPG 0x3 (IPG72): 72 bit times IPG 0x4 (IPG64): 64 bit times IPG 0x5 (IPG56): 56 bit times IPG 0x6 (IPG48): 48 bit times IPG 0x7 (IPG40): 40 bit times IPG Value After Reset: 0x0
			23	GPSLCE	R/W	0x0	Giant Packet Size Limit Control Enable When this bit is set, the MAC considers the value in GPSL field in MAC_Ext_Configuration register to declare a received packet as Giant packet. This field must be programmed to more than 1,518 bytes. Otherwise, the MAC considers 1,518 bytes as giant packet limit. When this bit is reset, the MAC considers a received packet as Giant packet when its size is greater than 1,518 bytes (1522 bytes for tagged packet). The watchdog timeout limit, Jumbo Packet Enable and 2K Packet Enable have higher precedence over this bit, that is the MAC considers a received packet as Giant packet when its size is greater than 9,018 bytes (9,022 bytes for tagged packet) with Jumbo Packet Enabled and greater than 2,000 bytes with 2K Packet Enabled. The watchdog timeout, if enabled, terminates the received packet when watchdog limit is reached. Therefore, the programmed giant packet limit must be less than the watchdog limit to get the giant packet status.Values: 0x0 (DISABLE): Giant Packet Size Limit Control is disabled 0x1 (ENABLE): Giant Packet Size Limit Control is enabled Value After Reset: 0x0
			22	S2KP	R/W	0x0	IEEE 802.3as Support for 2K Packets When this bit is set, the MAC considers all packets with up to 2,000 bytes length as normal packets. When the JE bit is not set, the MAC considers all received packets of size more than 2K bytes as Giant packets. When this bit is reset and the JE bit is not set, the MAC considers all received packets of size more than 1,518 bytes (1,522 bytes for tagged) as giant packets. For more information about how the setting of this bit and the JE bit impact the Giant packet status, see the Table, Giant Packet Status based on S2KP and JE Bits. Note: When the JE bit is set, setting this bit has no effect on the giant packet status.Values: 0x0 (DISABLE): Support upto 2K packet is disabled 0x1 (ENABLE): Support upto 2K packet is Enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x0	MAC_Configuration	The MAC Configuration Register establishes the operating mode of the MAC.	21	CST	R/W	0x0	CRC stripping for Type packets 1: MAC strips and drops the last four bytes (FCS) of all Ether type packets, as indicated by the Length/Type field of the packet, before forwarding the packet to the application. Note: For more details about impact of CST and ACS bit settings on the packet length, see "Packet Length based on the CST and ACS Bits" table in the Architecture chapter.Values: 0x0 (DISABLE): CRC stripping for Type packets is disabled 0x1 (ENABLE): CRC stripping for Type packets is enabled Value After Reset: 0x0
			20	ACS	R/W	0x0	Automatic Pad or CRC Stripping 1: MAC strips the pad or the FCS field of the incoming Ethernet Length packets. For the Ethernet Type packets, MAC transfers the packets to the application without stripping the Pad or FCS field. 0: MAC passes all incoming packets to the application, without any modification. Note: For more details about impact of CST and ACS bit settings on the packet length, see "Packet Length based on the CST and ACS Bits" table in the Architecture chapter.Values: 0x0 (DISABLE): Automatic Pad or CRC Stripping is disabled 0x1 (ENABLE): Automatic Pad or CRC Stripping is enabled Value After Reset: 0x0
			19	WD	R/W	0x0	Watchdog Disable When this bit is set, the MAC disables the watchdog timer on the receiver. The MAC can receive packets of up to 16,383 bytes. When this bit is reset, the MAC does not allow more than 2,048 bytes (10,240 if JE is set high) of the packet being received. The MAC cuts off any bytes received after 2,048 bytes.Values: 0x0 (ENABLE): Watchdog is enabled 0x1 (DISABLE): Watchdog is disabled Value After Reset: 0x0
			18			#VALUE!	Reserved Field: Yes
			17	JD	R/W	0x0	Jabber Disable When this bit is set, the MAC disables the jabber timer on the transmitter. The MAC can transfer packets of up to 16,383 bytes. When this bit is reset, if the application sends more than 2,048 bytes of data (10,240 if JE is set high) during transmission, the MAC does not send rest of the bytes in that packet.Values: 0x0 (ENABLE): Jabber is enabled 0x1 (DISABLE): Jabber is disabled Value After Reset: 0x0
			16	JE	R/W	0x0	Jumbo Packet Enable When this bit is set, the MAC allows jumbo packets of 9,018 bytes (9,022 bytes for VLAN tagged packets) without reporting a giant packet error in the Rx packet status.Values: 0x0 (DISABLE): Jumbo packet is disabled 0x1 (ENABLE): Jumbo packet is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			15	PS	R/W	0x0	Port Select This bit selects the Ethernet line speed. This bit, along with Bit 14, selects the exact line speed. In the 10/100 Mbps-only (always 1) or 1000 Mbps-only (always 0) configurations, this bit is read-only (RO) with appropriate value. In default 10/100/1000 Mbps configurations, this bit is read-write (R/W). The mac_speed_o[1] signal reflects the value of this bit.Values: 0x0 (M_1000_2500M): For 1000 or 2500 Mbps operations 0x1 (M_10_100M): For 10 or 100 Mbps operations Value After Reset: 0x0
			14	FES	R/W	0x0	Speed This bit selects the speed mode. The mac_speed_o[0] signal reflects the value of this bit.Values: 0x0 (M_10_1000M): 10 Mbps when PS bit is 1 and 1 Gbps when PS bit is 0 0x1 (M_100_2500M): 100 Mbps when PS bit is 1 and 2.5 Gbps when PS bit is 0 Value After Reset: 0x0
			13	DM	R	0x1	Duplex Mode When this bit is set, the MAC operates in the full-duplex mode in which it can transmit and receive simultaneously. This bit is RO with default value of 1'b1 in the full-duplex-only configurations.Values: 0x0 (HDUPLX): Half-duplex mode 0x1 (FDUPLX): Full-duplex mode Value After Reset: 0x1
			12	LM	R/W	0x0	Loopback Mode When this bit is set, the MAC operates in the loopback mode at GMII or MII. The (G)MII Rx clock input (clk_rx_i) is required for the loopback to work properly. This is because the Tx clock is not internally looped back.Values: 0x0 (DISABLE): Loopback is disabled 0x1 (ENABLE): Loopback is enabled Value After Reset: 0x0
			11	ECRSFD	R/W	0x0	Enable Carrier Sense Before Transmission in Full-Duplex Mode When this bit is set, the MAC transmitter checks the CRS signal before packet transmission in the full-duplex mode. The MAC starts the transmission only when the CRS signal is low. When this bit is reset, the MAC transmitter ignores the status of the CRS signal.Values: 0x0 (DISABLE): ECRSFD is disabled 0x1 (ENABLE): ECRSFD is enabled Value After Reset: 0x0
			10:8			#VALUE!	Reserved Field: Yes
			7	Reserved_7	R	0x0	Reserved. Value After Reset: 0x0
			6:4			#VALUE!	Reserved Field: Yes
			3:2	PRELEN	R/W	0x0	Preamble Length for Transmit packets These bits control the number of preamble bytes that are added to the beginning of every Tx packet. The preamble reduction occurs only when the MAC is operating in the full-duplex mode.Values: 0x0 (M_7BYTES): 7 bytes of preamble 0x1 (M_5BYTES): 5 bytes of preamble 0x2 (M_3BYTES): 3 bytes of preamble 0x3 (RESERVED): Reserved Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			1	TE	R/W	0x0	Transmitter Enable When this bit is set, the Tx state machine of the MAC is enabled for transmission on the GMII or MII interface. When this bit is reset, the MAC Tx state machine is disabled after it completes the transmission of the current packet. The Tx state machine does not transmit any more packets.Values: 0x0 (DISABLE): Transmitter is disabled 0x1 (ENABLE): Transmitter is enabled Value After Reset: 0x0
			0	RE	R/W	0x0	Receiver Enable When this bit is set, the Rx state machine of the MAC is enabled for receiving packets from the GMII or MII interface. When this bit is reset, the MAC Rx state machine is disabled after it completes the reception of the current packet. The Rx state machine does not receive any more packets from the GMII or MII interface.Values: 0x0 (DISABLE): Receiver is disabled 0x1 (ENABLE): Receiver is enabled Value After Reset: 0x0
			31			#VALUE!	Reserved Field: Yes
			30	APDIM	R/W	0x0	ARP Packet Drop if IP Address Mismatch When this bit is set, Packet for which Target Protocol Address does not match IPv4 address is dropped in the MTL layer. When this bit is reset, when target Protocol Address does not match, packet is forwarded to MTL maintaining backward compatibilityValues: 0x0 (DISABLE): mux select to drop the arp packet if target protocol address mismatches IPv4 address disabled 0x1 (ENABLE): mux select to drop the arp packet if target protocol address mismatches IPv4 address enabled Value After Reset: 0x0
			29:25	EIPG	R/W	0x0	Extended Inter-Packet Gap The value in this field is applicable when the EIPGEN bit of this register is 1. The 5 bits of this field are the most significant bits; IPG field of the MAC_Configuration register are the least significant 3 bits. Together, these 8 bits represent the value of the minimum IPG greater than 96-bit times, in steps of 8-bit times. {EIPG, IPG} 8'h00 - 104 bit times 8'h01 - 112 bit times 8'h02 - 120 bit times ... 8'hFF - 2144 bit times Value After Reset: 0x0
			24	EIPGEN	R/W	0x0	Extended Inter-Packet Gap Enable 1: MAC uses the EIPG field for the extended inter-packet gap. For more details of how the MAC uses the EIPG field, see the description of the EIPG field. 0: MAC ignores EIPG field. MAC uses the IPG field of the MAC_Configuration register as the minimum IPG less than or equal to 96-bit times, in steps of 8-bit times. Note: Use this feature only when operating in full-duplex mode. If you enable this feature in half-duplex mode, you might see undesirable effects on back-pressure function and frame transmission.Values: 0x0 (DISABLE): Extended Inter-Packet Gap is disabled 0x1 (ENABLE): Extended Inter-Packet Gap is enabled Value After Reset: 0x0
			23	Reserved_23	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x4	MAC_Ext_Configuration	The MAC Extended Configuration Register establishes the operating mode of the MAC.	22:20	HDSMS	R/W	0x0	Maximum Size for Splitting the Header Data These bits indicate the maximum header size allowed for splitting the header data in the received packet.Values: 0x0 (M_64BYTES): Maximum Size for Splitting the Header Data is 64 bytes 0x1 (M_128BYTES): Maximum Size for Splitting the Header Data is 128 bytes 0x2 (M_256BYTES): Maximum Size for Splitting the Header Data is 256 bytes 0x3 (M_512BYTES): Maximum Size for Splitting the Header Data is 512 bytes 0x4 (M_1024BYTES): Maximum Size for Splitting the Header Data is 1024 bytes 0x5 (RSVD): Reserved Value After Reset: 0x0
			19	PDC	R/W	0x0	Packet Duplication Control 1: MAC interprets the DCS field of the MAC_Address(#i)_High register as a one-hot value; each bit of the DCS field corresponds to a Rx DMA channel . 0: MAC interprets the DSC field of the MAC_Address(#i)_High_register as a binary value. Values: 0x0 (DISABLE): Packet Duplication Control is disabled 0x1 (ENABLE): Packet Duplication Control is enabled Value After Reset: 0x0
			18	USP	R/W	0x0	Unicast Slow Protocol Packet Detect When this bit is set, the MAC detects the Slow Protocol packets with unicast address of the station specified in the MAC_Address0_High and MAC_Address0_Low registers. The MAC also detects the Slow Protocol packets with the Slow Protocols multicast address (01-80-C2-00-00-02). When this bit is reset, the MAC detects only Slow Protocol packets with the Slow Protocol multicast address specified in the IEEE 802.3-2015, Section 5.Values: 0x0 (DISABLE): Unicast Slow Protocol Packet Detection is disabled 0x1 (ENABLE): Unicast Slow Protocol Packet Detection is enabled Value After Reset: 0x0
			17	SPEN	R/W	0x0	Slow Protocol Detection Enable 1: MAC processes the Slow Protocol packets (Ether Type = 0x8809) and provides the Slow Protocol Sub-Type and Code fields in the Rx status. 0: MAC forwards all the error-free Slow Protocol packets to the application. The MAC considers such packets as normal Type packets. Values: 0x0 (DISABLE): Slow Protocol Detection is disabled 0x1 (ENABLE): Slow Protocol Detection is enabled Value After Reset: 0x0
			16	DCRCC	R/W	0x0	Disable CRC Checking for Received Packets When this bit is set, the MAC receiver does not check the CRC field in the received packets. When this bit is reset, the MAC receiver always checks the CRC field in the received packets.Values: 0x0 (ENABLE): CRC Checking is enabled 0x1 (DISABLE): CRC Checking is disabled Value After Reset: 0x0
			15:14	Reserved_15_14	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			13:0	GPSL	R/W	0x0	Giant Packet Size Limit If the received packet size is greater than the value programmed in this field in units of bytes, the MAC declares the received packet as Giant packet. The value programmed in this field must be greater than or equal to 1,518 bytes. Any other programmed value is considered as 1,518 bytes. For VLAN tagged packets, the MAC adds 4 bytes to the programmed value. The MAC adds 8 bytes to the programmed value for double VLAN tagged packets. The value in this field is applicable when the GPSLCE bit is set in MAC_Configuration register. Value After Reset: 0x0
			31	RA	R/W	0x0	Receive All 1: The MAC receiver transfers all the received packets to the application regardless of the address filter status. MAC updates the SA or DA filtering status in the corresponding bit of the MAC filter status word. For more details, see the "MAC Filter Status Word Format" table in the Architecture chapter. 0: The MAC receiver transfers only the packets that pass the SA or DA address filter, to the application. Values: 0x0 (DISABLE): Receive All is disabled 0x1 (ENABLE): Receive All is enabled Value After Reset: 0x0
			30:22	Reserved_30_22	R	0x0	Reserved. Value After Reset: 0x0
			21	DNTU	R/W	0x0	Drop Non-TCP/UDP over IP Packets When this bit is set, the MAC drops the non-TCP or UDP over IP packets. The MAC forward only those packets that are processed by the Layer 4 filter. When this bit is reset, the MAC forwards all non-TCP or UDP over IP packets.Values: 0x0 (FWD): Forward Non-TCP/UDP over IP Packets 0x1 (DROP): Drop Non-TCP/UDP over IP Packets Value After Reset: 0x0
			20	IPFE	R/W	0x0	Layer 3 and Layer 4 Filter Enable When this bit is set, the MAC drops packets that do not match the enabled Layer 3 and Layer 4 filters. If Layer 3 or Layer 4 filters are not enabled for matching, this bit does not have any effect. When this bit is reset, the MAC forwards all packets irrespective of the match status of the Layer 3 and Layer 4 fields.Values: 0x0 (DISABLE): Layer 3 and Layer 4 Filters are disabled 0x1 (ENABLE): Layer 3 and Layer 4 Filters are enabled Value After Reset: 0x0
			19:17	Reserved_19_17	R	0x0	Reserved. Value After Reset: 0x0
			16	VTFE	R/W	0x0	VLAN Tag Filter Enable When this bit is set, the MAC drops the VLAN tagged packets that do not match the VLAN Tag. When this bit is reset, the MAC forwards all packets irrespective of the match status of the VLAN Tag.Values: 0x0 (DISABLE): VLAN Tag Filter is disabled 0x1 (ENABLE): VLAN Tag Filter is enabled Value After Reset: 0x0
			15:11	Reserved_15_11	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x8	MAC_Packet_Filter	The MAC Packet Filter register contains the filter controls for receiving packets. Some of the controls from this register go to the address check block of the MAC which performs the first level of address filtering. The second level of filtering is performed on the incoming packet based on other controls such as Pass Bad Packets and Pass Control Packets.	10	HPF	R/W	0x0	Hash or Perfect Filter When this bit is set, the address filter passes a packet if it matches either the perfect filtering or hash filtering as set by the HMC or HUC bit. When this bit is reset and the HUC or HMC bit is set, the packet is passed only if it matches the Hash filter.Values: 0x0 (DISABLE): Hash or Perfect Filter is disabled 0x1 (ENABLE): Hash or Perfect Filter is enabled Value After Reset: 0x0
			9	SAF	R/W	0x0	Source Address Filter Enable 1: MAC compares the SA field of the received packets with the values you programmed in the MAC_Address(#i)_High and MAC_Address(#i)_Low registers. If the comparison fails, the MAC drops the packet. 0: MAC forwards the received packet to the application. Also, MAC updates the result of SA address comparison in the SA Filter Fail, bit[48] of the MAC Rx status word. For more details about MAC Rx status word, see "Receive Status Word at the MAC Interface" table in the Architecture chapter. Note: According to the IEEE specification, Bit 47 of the SA is reserved. However, in DWC_ether_qos, the MAC compares all 48 bits. The application must consider this while programming the MAC address registers for SA.Values: 0x0 (DISABLE): SA Filtering is disabled 0x1 (ENABLE): SA Filtering is enabled Value After Reset: 0x0
			8	SAIF	R/W	0x0	SA Inverse Filtering When this bit is set, the Address Check block operates in the inverse filtering mode for SA address comparison. If the SA of a packet matches the values programmed in the SA registers, it is marked as failing the SA Address filter. When this bit is reset, if the SA of a packet does not match the values programmed in the SA registers, it is marked as failing the SA Address filter.Values: 0x0 (DISABLE): SA Inverse Filtering is disabled 0x1 (ENABLE): SA Inverse Filtering is enabled Value After Reset: 0x0
			7:6	PCF	R/W	0x0	Pass Control Packets These bits control the forwarding of all control packets (including unicast and multicast Pause packets).Values: 0x0 (FLTR_ALL): MAC filters all control packets from reaching the application 0x1 (FW_XCPT_PAU): MAC forwards all control packets except Pause packets to the application even if they fail the Address filter 0x2 (FW_ALL): MAC forwards all control packets to the application even if they fail the Address filter 0x3 (FW_PASS): MAC forwards the control packets that pass the Address filter Value After Reset: 0x0
			5	DBF	R/W	0x0	Disable Broadcast Packets When this bit is set, the AFM module blocks all the incoming broadcast packets. In addition, it overrides all other filter settings. When this bit is reset, the AFM module passes all received broadcast packets.Values: 0x0 (ENABLE): Enable Broadcast Packets 0x1 (DISABLE): Disable Broadcast Packets Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			4	PM	R/W	0x0	Pass All Multicast When this bit is set, it indicates that all the received packets with a multicast destination address (first bit in the destination address field is '1') are passed. When this bit is reset, filtering of multicast packet depends on HMC bit.Values: 0x0 (DISABLE): Pass All Multicast is disabled 0x1 (ENABLE): Pass All Multicast is enabled Value After Reset: 0x0
			3	DAIF	R/W	0x0	DA Inverse Filtering When this bit is set, the Address Check block operates in inverse filtering mode for the DA address comparison for both unicast and multicast packets. When this bit is reset, normal filtering of packets is performed.Values: 0x0 (DISABLE): DA Inverse Filtering is disabled 0x1 (ENABLE): DA Inverse Filtering is enabled Value After Reset: 0x0
			2	HMC	R/W	0x0	Hash Multicast 1: MAC performs the destination address filtering of the received multicast packets according to the hash table. 0: MAC performs perfect filtering of destination address for multicast packets, that is, it compares the DA field of the packet with the values programmed in the MAC_Address(#i)_High and MAC_Address(#i)_Low registers. Values: 0x0 (DISABLE): Hash Multicast is disabled 0x1 (ENABLE): Hash Multicast is enabled Value After Reset: 0x0
			1	HUC	R/W	0x0	Hash Unicast 1: MAC performs the destination address filtering of unicast packets according to the hash table. 0: MAC performs a perfect filtering of destination address for unicast packets, that is, it compares the DA field with the values you programmed in the MAC_Address(#i)_High and MAC_Address(#i)_Low registers. Values: 0x0 (DISABLE): Hash Unicast is disabled 0x1 (ENABLE): Hash Unicast is enabled Value After Reset: 0x0
			0	PR	R/W	0x0	Promiscuous Mode 1: MAC transfers all incoming packets to the application regardless of the destination or source address filtering status. MAC clears the SA or DA Filter Fail status bits of the MAC Rx status word. For more details about MAC Rx status word, see "Receive Status Word at the MAC Interface" table in the Architecture chapter. 0: MAC transfers to the application only the packets that pass destination or source address filtering. Values: 0x0 (DISABLE): Promiscuous Mode is disabled 0x1 (ENABLE): Promiscuous Mode is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xc	MAC_WD_JB_Timeout	The Watchdog and Jabber Timeout register controls the watchdog timeout limit for received packets and jabber timeout limit for transmitted packets.	31:25	Reserved_31_25	R	0x0	Reserved. Value After Reset: 0x0
			24	PJE	R/W	0x0	Programmable Jabber Enable When this bit is set and the JD bit of the MAC_Configuration register is reset, the JTO field is used as jabber timeout limit for a transmitted packet. When this bit is cleared, the jabber timeout for a received packet is controlled by setting of JD and JE bits in MAC_Configuration register.Values: 0x0 (DISABLE): Programmable Jabber is disabled 0x1 (ENABLE): Programmable Jabber is enabled Value After Reset: 0x0
			23:20	Reserved_23_20	R	0x0	Reserved. Value After Reset: 0x0
			19:16	JTO	R/W	0x0	Jabber Timeout When the PJE bit is set and the JD bit of the MAC_Configuration register is reset, this field is used as jabber timeout limit for a transmitted packet. If the length of a transmitted packet exceeds the value of this field, such packet is terminated and declared as an error packet. Note: When the PJE bit is set, the value in this field must be more than 1,522 (0x05F2). Otherwise, the IEEE 802.3-specified valid tagged packets are declared as error packets and terminated.Values: 0x0 (M_2KBYTES): 2 KB 0x1 (M_3KBYTES): 3 KB 0x2 (M_4KBYTES): 4 KB 0x3 (M_5KBYTES): 5 KB 0x4 (M_6KBYTES): 6 KB 0x5 (M_7KBYTES): 7 KB 0x6 (M_8KBYTES): 8 KB 0x7 (M_9KBYTES): 9 KB 0x8 (M_10KBYTES): 10 KB 0x9 (M_11KBYTES): 11 KB 0xa (M_12KBYTES): 12 KB 0xb (M_13KBYTES): 13 KB 0xc (M_14KBYTES): 14 KB 0xd (M_15KBYTES): 15 KB 0xe (M_16383BYTES): 16383 Bytes 0xf (RESERVED): Reserved Value After Reset: 0x0
			15:9	Reserved_15_9	R	0x0	Reserved. Value After Reset: 0x0
			8	PWE	R/W	0x0	Programmable Watchdog Enable When this bit is set and the WD bit of the MAC_Configuration register is reset, the WTO field is used as watchdog timeout limit for a received packet. When this bit is cleared, the watchdog timeout for a received packet is controlled by setting of WD and JE bits in MAC_Configuration register.Values: 0x0 (DISABLE): Programmable Watchdog is disabled 0x1 (ENABLE): Programmable Watchdog is enabled Value After Reset: 0x0
			7:4	Reserved_7_4	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			3:0	WTO	R/W	0x0	Watchdog Timeout When the PWE bit is set and the WD bit of the MAC_Configuration register is reset, this field is used as watchdog timeout limit for a received packet. If the length of a received packet exceeds the value of this field, such packet is terminated and declared as an error packet. Note: When the PWE bit is set, the value in this field must be more than 1,522 (0x05F2). Otherwise, the IEEE 802.3-specified valid tagged packets are declared as error packets and then dropped.Values: 0x0 (M_2KBYTES): 2 KB 0x1 (M_3KBYTES): 3 KB 0x2 (M_4KBYTES): 4 KB 0x3 (M_5KBYTES): 5 KB 0x4 (M_6KBYTES): 6 KB 0x5 (M_7KBYTES): 7 KB 0x6 (M_8KBYTES): 8 KB 0x7 (M_9KBYTES): 9 KB 0x8 (M_10KBYTES): 10 KB 0x9 (M_11KBYTES): 11 KB 0xa (M_12KBYTES): 12 KB 0xb (M_13KBYTES): 13 KB 0xc (M_14KBYTES): 14 KB 0xd (M_15KBYTES): 15 KB 0xe (M_16383BYTES): 16383 Bytes 0xf (RESERVED): Reserved Value After Reset: 0x0
0x10	MAC_Hash_Table_Reg0		31:0	HT31T0	R/W	0x0	The Hash Table Register 0 contains the first 32 bits of the hash table, when the width of the hash table is 128 or 256 bits. You can specify the width of the hash table by using the Hash Table Size option in coreConsultant.The Hash table is used for group address filtering. For hash filtering, the content of the destination address in the incoming packet is passed through the CRC logic and the upper six (seven in 128-bit Hash or eight in 256-bit Hash) bits of the CRC register are used to index the content of the Hash table. The most significant bits determines the register to be used (Hash Table Register X), and the least significant five bits determine the bit within the register. For example, a hash value of 6'b100000 (in 64-bit Hash) selects Bit 0 of the Hash Table Register 1, a value of 7b'1110000 (in 128-bit Hash) selects Bit 16 of the Hash Table Register 3 and a value of 8b'10111111 (in 256-bit Hash) selects Bit 31 of the Hash Table Register 5.The hash value of the destination address is calculated in the following way:Calculate the 32-bit CRC for the DA (See IEEE 802.3, Section 3.2.8 for the steps to calculate CRC32).Perform bitwise reversal for the value obtained in Step 1.Take the upper 6 (or 7 or 8) bits from the value obtained in Step 2.If the corresponding bit value of the register is 1'b1, the packet is accepted. Otherwise, it is rejected. If the PM bit is set in MAC_Packet_Filter, all multicast packets are accepted regardless of the multicast hash values.If the Hash Table register is configured to be double-synchronized to the (G)MII clock domain, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the Hash Table Register X registers are written. If double-synchronization is enabled, consecutive writes to this register must be performed after at least four clock cycles in the destination clock domain. MAC Hash Table First 32 Bits This field contains the first 32 Bits [31:0] of the Hash table. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x14	MAC_Hash_Table_Reg1		31:0	HT63T32	R/W	0x0	The Hash Table Register 1 contains the second 32 bits of the hash table. You can specify the width of the hash table by using the Hash Table Size option in coreConsultant.The Hash table is used for group address filtering. For hash filtering, the content of the destination address in the incoming packet is passed through the CRC logic and the upper six (seven in 128-bit Hash or eight in 256-bit Hash) bits of the CRC register are used to index the content of the Hash table. The most significant bits determines the register to be used (Hash Table Register X), and the least significant five bits determine the bit within the register. For example, a hash value of 6'b100000 (in 64-bit Hash) selects Bit 0 of the Hash Table Register 1, a value of 7b'1110000 (in 128-bit Hash) selects Bit 16 of the Hash Table Register 3 and a value of 8b'10111111 (in 256-bit Hash) selects Bit 31 of the Hash Table Register 5.The hash value of the destination address is calculated in the following way:Calculate the 32-bit CRC for the DA (See IEEE 802.3, Section 3.2.8 for the steps to calculate CRC32).Perform bitwise reversal for the value obtained in Step 1.Take the upper 6 (or 7 or 8) bits from the value obtained in Step 2.If the corresponding bit value of the register is 1'b1, the packet is accepted. Otherwise, it is rejected. If the PM bit is set in MAC_Packet_Filter, all multicast packets are accepted regardless of the multicast hash values.If the Hash Table register is configured to be double-synchronized to the (G)MII clock domain, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the Hash Table Register X registers are written. If double-synchronization is enabled, consecutive writes to this register must be performed after at least four clock cycles in the destination clock domain. MAC Hash Table Second 32 Bits This field contains the second 32 Bits [63:32] of the Hash table. Value After Reset: 0x0
			31	EIVLRXS	R/W	0x0	Enable Inner VLAN Tag in Rx Status 1: MAC provides the inner VLAN Tag in the Rx status. 0: MAC does not provide the inner VLAN Tag in Rx status. Values: 0x0 (DISABLE): Inner VLAN Tag in Rx status is disabled 0x1 (ENABLE): Inner VLAN Tag in Rx status is enabled Value After Reset: 0x0
			30	Reserved_30	R	0x0	Reserved. Value After Reset: 0x0
			29:28	EIVLS	R/W	0x0	Enable Inner VLAN Tag Stripping on Receive This field indicates the stripping operation on inner VLAN Tag in received packet.Values: 0x0 (DONOT): Do not strip 0x1 (IFPASS): Strip if VLAN filter passes 0x2 (IFFAIL): Strip if VLAN filter fails 0x3 (ALWAYS): Always strip Value After Reset: 0x0
			27	ERIVLT	R/W	0x0	Enable Inner VLAN Tag Comparison When this bit, VTHM bit and the EDVLP field are set, the MAC receiver enables VLAN Hash filtering operation on the inner VLAN Tag (if present). When this bit is reset and VTHM bit is set, the MAC receiver enables VLAN Hash filtering operation on the outer VLAN Tag (if present). The ERSVLM bit and DOVLTC bit determines which VLAN type is enabled for filtering.Values: 0x0 (DISABLE): Inner VLAN tag is disabled 0x1 (ENABLE): Inner VLAN tag is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x50	MAC_VLAN_Tag_Ctrl	This register is the redefined format of the MAC VLAN Tag Register. It is used for indirect addressing. It contains the address offset, command type and Busy Bit for CSR access of the Per VLAN Tag registers.	26	EDVLP	R/W	0x0	Enable Double VLAN Processing 1: MAC enables processing of up to two VLAN Tags on Tx and Rx (if present). 0: MAC enables processing of up to one VLAN Tag on Tx and Rx (if present). Values: 0x0 (DISABLE): Double VLAN Processing is disabled 0x1 (ENABLE): Double VLAN Processing is enabled Value After Reset: 0x0
			25	VTHM	R/W	0x0	VLAN Tag Hash Table Match Enable 1: MAC uses the most significant four bits of CRC of VLAN Tag to index the content of the MAC_VLAN_Hash_Table register. A value of 1 in the MAC_VLAN_Hash_Table register, corresponding to the index, indicates that the packet matched the VLAN hash table. When ETV = 1, MAC uses the CRC of the 12-bit VLAN Identifier (VID) for comparison. When ETV = 0, MAC uses the CRC of the 16-bit VLAN Identifier (VID) for comparison. 0: MAC does not perform the VLAN Hash Match operation. For more details, see "VLAN Tag Hash Filtering" section in the Packet Filtering chapter.Values: 0x0 (DISABLE): VLAN Tag Hash Table Match is disabled 0x1 (ENABLE): VLAN Tag Hash Table Match is enabled Value After Reset: 0x0
			24	EVLRXS	R/W	0x0	Enable VLAN Tag in Rx status 1: MAC provides the outer VLAN Tag in the Rx status. 0: MAC does not provide the outer VLAN Tag in Rx status. Values: 0x0 (DISABLE): VLAN Tag in Rx status is disabled 0x1 (ENABLE): VLAN Tag in Rx status is enabled Value After Reset: 0x0
			23	Reserved_23	R	0x0	Reserved. Value After Reset: 0x0
			22:21	EVLS	R/W	0x0	Enable VLAN Tag Stripping on Receive This field indicates the stripping operation on the outer VLAN Tag in received packet.Values: 0x0 (DONOT): Do not strip 0x1 (IFPASS): Strip if VLAN filter passes 0x2 (IFFAIL): Strip if VLAN filter fails 0x3 (ALWAYS): Always strip Value After Reset: 0x0
			20	DOVLTC	R/W	0x0	Disable VLAN Type Check for VLAN Hash Filtering 1: MAC VLAN Hash Filter does not check whether the VLAN Tag specified by the ERIVLT bit is of type S-VLAN or C-VLAN. 0: MAC VLAN Hash Filter filters or matches the VLAN Tag specified by the ERIVLT bit only when VLAN Tag type is similar to the one specified by the ERSVLM bit. Values: 0x0 (ENABLE): VLAN Type Check is enabled 0x1 (DISABLE): VLAN Type Check is disabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			19	ERSVLM	R/W	0x0	Enable Receive S-VLAN Match for VLAN Hash Filtering 1: MAC receiver enables VLAN Hash filtering or matching for S-VLAN (Type = 0x88A8) packets. 0: MAC receiver enables VLAN Hash filtering or matching for C-VLAN (Type = 0x8100) packets. The ERIVLT bit determines the VLAN tag position considered for VLAN Hash filtering or matching.Values: 0x0 (DISABLE): Receive S-VLAN Match is disabled 0x1 (ENABLE): Receive S-VLAN Match is enabled Value After Reset: 0x0
			18	ESVL	R/W	0x0	Enable S-VLAN 1: MAC transmitter and receiver consider the S-VLAN packets (Type = 0x88A8) as valid VLAN tagged packets. Values: 0x0 (DISABLE): S-VLAN is disabled 0x1 (ENABLE): S-VLAN is enabled Value After Reset: 0x0
			17	VTIM	R/W	0x0	VLAN Tag Inverse Match Enable 1: Enables the VLAN Tag inverse matching. The packets without matching VLAN Tag are marked as matched. 0: Enables the VLAN Tag perfect matching. The packets with matched VLAN Tag are marked as matched. Values: 0x0 (DISABLE): VLAN Tag Inverse Match is disabled 0x1 (ENABLE): VLAN Tag Inverse Match is enabled Value After Reset: 0x0
			16	ETV	R/W	0x0	Enable 12-Bit VLAN Tag Comparison for VLAN Hash Filtering 1: A 12-bit VLAN identifier is used for VLAN Hash filtering instead of the complete 16-bit VLAN tag. Bits[11:0] of VLAN tag in the received VLAN-tagged packet are used for hash-based VLAN filtering. 0: All 16 bits of the 15th and 16th bytes of the received VLAN packet are used for VLAN hash filtering. Values: 0x0 (DISABLE): 12-Bit VLAN Tag Comparison is disabled 0x1 (ENABLE): 12-Bit VLAN Tag Comparison is enabled Value After Reset: 0x0
			15:5	Reserved_15_y	R	0x0	Reserved. Value After Reset: 0x0
			4:2	OFS	R/W	0x0	Offset This field holds the address offset of the MAC VLAN Tag Filter Register which the application is trying to access. The width of the field depends on the number of MAC VLAN Tag Registers enabled. Value After Reset: 0x0
			1	CT	R/W	0x0	Command Type This bit indicates if the current register access is a read or a write. 1: Indicates a read operation. 0: Indicates a write operation. Values: 0x0 (WRITE): Write operation 0x1 (READ): Read operation Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	OB	R/W	0x0	Operation Busy Software writes 1 in this bit to initiate an indirect read or write access to the MAC_VLAN_Tag register. MAC writes 0 in this bit when it completes the indirect read or write access to the MAC_VLAN_Tag register. MAC can initiate the next indirect register access when this bit has a value 0. During a write operation, MAC writes 0 in this bit after it completes writing the data into the MAC_VLAN_Tag register. During a read operation, the application must read the data from the MAC_VLAN_Tag_Data register only after MAC writes 0 in this bit. Values: 0x0 (DISABLE): Operation Busy is disabled 0x1 (ENABLE): Operation Busy is enabled Value After Reset: 0x0 Testable: untestable
		This register holds the read or write data for the indirect access of the MAC_VLAN_Tag_Filter(#i) registers. For	31:28	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			27:25	DMACHN	R/W	0x0	DMA Channel Number The DMA Channel number to which the VLAN Tagged Frame is to be routed if it passes this VLAN Tag Filter is programmed in this field. If the Routing based on VLAN Tag Filter is not necessary, this field need not be programmed. Value After Reset: 0x0
			24	DMACHEN	R/W	0x0	DMA Channel Number Enable 1: Enables the DMA channel number that you program in the DMACHN field. Values: 0x0 (DISABLE): DMA Channel Number is disabled 0x1 (ENABLE): DMA Channel Number is enabled Value After Reset: 0x0
			23:21	Reserved_23_21	R	0x0	Reserved. Value After Reset: 0x0
			20	ERIVLT	R/W	0x0	Enable Inner VLAN Tag Comparison This bit is valid only when Double VLAN Tag Enable of the Filter is set. When this bit and the EDVLP field are set, the MAC receiver enables operation on the inner VLAN Tag (if present). When this bit is reset, the MAC receiver enables operation on the outer VLAN Tag (if present).Values: 0x0 (DISABLE): Inner VLAN tag comparison is disabled 0x1 (ENABLE): Inner VLAN tag comparison is enabled Value After Reset: 0x0
			19	ERSVLM	R/W	0x0	Enable S-VLAN Match for received Frames This bit is valid only when you enable the VLAN Tag by programming the VEN field of the MAC_VLAN_Tag_Filter(#i) register. When this bit is set, the MAC receiver enables filtering or matching for S-VLAN (Type = 0x88A8) packets. When this bit is reset, the MAC receiver enables filtering or matching for C-VLAN (Type = 0x8100) packets.Values: 0x0 (DISABLE): Receive S-VLAN Match is disabled 0x1 (ENABLE): Receive S-VLAN Match is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x54	MAC_VLAN_Tag_Data	the read operation, this register contains valid read data only after the MAC writes 0 to the OB bit of the MAC_VLAN_Tag_Ctrl register.For the write operation, Software must ensure valid data in this register before writing 1 to the OB bit of the MAC_VLAN_Tag_Ctrl register.	18	DOVLTC	R/W	0x0	Disable VLAN Type Comparison This bit is valid only when you enable the VLAN Tag by programming the VEN field of the MAC_VLAN_Tag_Filter(#i) register. 1: MAC does not check the type of the VLAN Tag during the VLAN tag comparison. MAC performs either inner or outer VLAN Tag comparison based on the value you program in the ERIVLT field of the MAC_VLAN_Tag_Filter(#i) register. 0: MAC considers the VLAN Tag type during the comparison. The VLAN Tag type, S-VLAN or C-VLAN, used by the MAC for comparison is determined by the value you program in the ERSVLM field of the MAC_VLAN_Tag_Ctrl register Values: 0x0 (ENABLE): VLAN type comparison is enabled 0x1 (DISABLE): VLAN type comparison is disabled Value After Reset: 0x0
			17	ETV	R/W	0x0	12-Bit or 16-Bit VLAN Comparison This bit is valid only when you enable the VLAN Tag by programming the VEN field of the corresponding MAC_VLAN_Tag_Filter(#i) register. 1: MAC uses a 12-bit VLAN tag for the comparison and filtering instead of the 16-bit VLAN tag. MAC compares the bits [11:0] of the VLAN tag with the corresponding field in the received VLAN-tagged packet. Values: 0x0 (M_16BIT): 16 bit VLAN comparison 0x1 (M_12BIT): 12 bit VLAN comparison Value After Reset: 0x0
			16	VEN	R/W	0x0	VLAN Tag Enable This bit is used to enable or disable the VLAN Tag. 1: MAC compares the VLAN Tag of the received packet with the VLAN Tag ID. 0: MAC does not compare the VLAN Tag of received packet with the VLAN Tag ID, irrespective of the programming of the other fields. Values: 0x0 (DISABLE): VLAN Tag is disabled 0x1 (ENABLE): VLAN Tag is enabled Value After Reset: 0x0
			15:0	VID	R/W	0x0	VLAN Tag ID This field holds the VLAN Tag value which is used by the MAC for perfect comparison. It is valid when VLAN Tag Enable is set. Value After Reset: 0x0
			31:28	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			27:25	DMACHN	R/W	0x0	DMA Channel Number The DMA Channel number to which the VLAN Tagged Frame is to be routed if it passes this VLAN Tag Filter is programmed in this field. If the Routing based on VLAN Tag Filter is not necessary, this field need not be programmed. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x54	MAC_VLAN_Tag_Filter(#i)(fori=0;i<=7)	This register contains VLAN Tag filter 0 control information.	24	DMACHEN	R/W	0x0	DMA Channel Number Enable This bit is the Enable for the DMA Channel Number value programmed in the field DMACH. When this bit is reset, the Routing does not occur based on VLAN Filter result. The frame is routed based on DA Based DMA Channel Routing.Values: 0x0 (DISABLE): DMA Channel Number is disabled 0x1 (ENABLE): DMA Channel Number is enabled Value After Reset: 0x0
			23:21	Reserved_23_21	R	0x0	Reserved. Value After Reset: 0x0
			20	ERIVLT	R/W	0x0	Enable Inner VLAN Tag Comparison This bit is valid only when Double VLAN Tag Enable of the Filter is set. When this bit and the EDVLP field are set, the MAC receiver enables operation on the inner VLAN Tag (if present). When this bit is reset, the MAC receiver enables operation on the outer VLAN Tag (if present).Values: 0x0 (DISABLE): Inner VLAN tag comparison is disabled 0x1 (ENABLE): Inner VLAN tag comparison is enabled Value After Reset: 0x0
			19	ERSVLM	R/W	0x0	Enable S-VLAN Match for received Frames This bit is valid only when VLAN Tag Enable of the Filter is set. When this bit is set, the MAC receiver enables filtering or matching for S-VLAN (Type = 0x88A8) packets. When this bit is reset, the MAC receiver enables filtering or matching for C-VLAN (Type = 0x8100) packets.Values: 0x0 (DISABLE): Receive S-VLAN Match is disabled 0x1 (ENABLE): Receive S-VLAN Match is enabled Value After Reset: 0x0
			18	DOVLTC	R/W	0x0	Disable VLAN Type Comparison This bit is valid only when VLAN Tag Enable of the Filter is set. When this bit is set, the MAC does not check whether the VLAN Tag specified by the Enable Inner VLAN Tag Comparison bit is of type S-VLAN or C-VLAN. When this bit is reset, the MAC filters or matches the VLAN Tag specified by the Enable Inner VLAN Tag Comparison bit only when VLAN Tag type is similar to the one specified by the Enable S-VLAN Match for received Frames bit.Values: 0x0 (ENABLE): VLAN type comparison is enabled 0x1 (DISABLE): VLAN type comparison is disabled Value After Reset: 0x0
			17	ETV	R/W	0x0	12-bit or 16-bit VLAN Comparison This bit is valid only when VEN of the Filter is set. When this bit is set, a 12-bit VLAN identifier is used for comparing and filtering instead of the complete 16-bit VLAN tag. Bits [11:0] of VLAN tag are compared with the corresponding field in the received VLAN-tagged packet.Values: 0x0 (M_16BIT): 16 bit VLAN comparison 0x1 (M_12BIT): 12 bit VLAN comparison Value After Reset: 0x0
			16	VEN	R/W	0x0	VLAN Tag Enable 1: MAC compares the VLAN Tag of the received packet with the VLAN Tag ID. 0: MAC does not perform VLAN Tag comparison. Values: 0x0 (DISABLE): VLAN Tag is disabled 0x1 (ENABLE): VLAN Tag is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			15:0	VID	R/W	0x0	VLAN Tag ID This field holds the VLAN Tag value which is used by the MAC for perfect comparison. It is valid when VLAN Tag Enable is set. Value After Reset: 0x0
0x58	MAC_VLAN_Hash_Table	When VTHM bit of the MAC_VLAN_Tag register is set, the 16-bit VLAN Hash Table register is used for group address filtering based on the VLAN tag. For hash filtering, the content of the 16-bit VLAN tag or 12-bit VLAN ID (based on the ETV bit of MAC_VLAN_Tag Register) in the incoming packet is passed through the CRC logic. The upper four bits of the calculated hash value are used to index the contents of the VLAN Hash table. For example, hash value of 4b'1000 selects Bit 8 of the VLAN Hash table.The hash value of the destination address is calculated in the following way:Calculate the 32-bit CRC for the VLAN tag or ID (For steps to calculate CRC32, see Section 3.2.8 of IEEE 802.3).Perform bitwise reversal for the value obtained in step 1.Take the upper four bits from the value obtained in step 2.If the VLAN hash Table register is configured to be double-synchronized to the (G)MII clock domain, the synchronization is triggered only when Bits[15:8] (in little-endian mode) or Bits[7:0] (in big-endian mode) of this register are written.If double-synchronization is enabled, consecutive writes to this register must be performed after at least four clock cycles in the destination clock domain.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	VLHT	R/W	0x0	VLAN Hash Table This field contains the 16-bit VLAN Hash Table. Value After Reset: 0x0
			31	BUSY	R	0x0	Busy Indicates the status of the read operation from, and write operation to the indirect access of the queue or channel specific VLAN inclusion register. Write Operation: When MAC completes writing to the indirect access register, MAC writes 0 to this bit. Read Operation: When the data is valid or ready for reading from the indirect access register, MAC writes 0 to this bit. Application must access the indirect access registers only when this bit is 0.Values: 0x0 (INACTIVE): Busy status not detected 0x1 (ACTIVE): Busy status detected Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x60	MAC_VLAN_Incl	The VLAN Tag Inclusion or Replacement register contains the VLAN tag for insertion or replacement in the Transmit packets. It also contains the VLAN tag insertion controls.	30	RDWR	R/W	0x0	Read write control This bit controls the read or write operation for indirectly accessing the queue/channel specific VLAN Inclusion register. When set indicates write operation and when reset indicates read operation. This does not have any effect when CBTI is reset.Values: 0x0 (READ): Read operation of indirect access 0x1 (WRITE): Write operation of indirect access Value After Reset: 0x0
			29:27	Reserved_29_y	R	0x0	Reserved. Value After Reset: 0x0
			26:24	ADDR	R/W	0x0	Address This field selects one of the queue/channel specific VLAN Inclusion register for read/write access. This does not have any effect when CBTI is reset. Value After Reset: 0x0
			23:22	Reserved_23_22	R	0x0	Reserved. Value After Reset: 0x0
			21	CBTI	R/W	0x0	Channel based tag insertion When this bit is set, outer VLAN tag is inserted for every packets transmitted by the MAC. The tag value is taken from the queue/channel specific VLAN tag register. The VLTI, VLP, VLC, and VLT fields of this register are ignored when this bit is set. When this bit is set, a write operation to byte 3 of this register initiates the read/write access to the indirect register. When reset, outer VLAN operation is based on the setting of VLTI, VLP, VLC and VLT fields of this register.Values: 0x0 (DISABLE): Channel based tag insertion is disabled 0x1 (ENABLE): Channel based tag insertion is enabled Value After Reset: 0x0
			20	VLTI	R/W	0x0	VLAN Tag Input When this bit is set, it indicates that the VLAN tag to be inserted or replaced in Tx packet must be taken from: The Tx descriptor Values: 0x0 (DISABLE): VLAN Tag Input is disabled 0x1 (ENABLE): VLAN Tag Input is enabled Value After Reset: 0x0
			19	CSVL	R/W	0x0	C-VLAN or S-VLAN When this bit is set, S-VLAN type (0x88A8) is inserted or replaced in the 13th and 14th bytes of transmitted packets. When this bit is reset, C-VLAN type (0x8100) is inserted or replaced in the 13th and 14th bytes of transmitted packets.Values: 0x0 (C-VLAN): C-VLAN type (0x8100) is inserted or replaced 0x1 (S-VLAN): S-VLAN type (0x88A8) is inserted or replaced Value After Reset: 0x0
			18	VLP	R/W	0x0	VLAN Priority Control When this bit is set, the control bits[17:16] are used for VLAN deletion, insertion, or replacement. When this bit is reset, the mti_vlan_ctrl_i control input is used and bits[17:16] are ignored.Values: 0x0 (DISABLE): VLAN Priority Control is disabled 0x1 (ENABLE): VLAN Priority Control is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			17:16	VLC	R/W	0x0	VLAN Tag Control in Transmit Packets 2'b00: No VLAN tag deletion, insertion, or replacement 2'b01: VLAN tag deletion The MAC removes the VLAN type (bytes 13 and 14) and VLAN tag (bytes 15 and 16) of all transmitted packets with VLAN tags. 2'b10: VLAN tag insertion The MAC inserts VLT in bytes 15 and 16 of the packet after inserting the Type value (0x8100 or 0x88a8) in bytes 13 and 14. This operation is performed on all transmitted packets, irrespective of whether they already have a VLAN tag. 2'b11: VLAN tag replacement The MAC replaces VLT in bytes 15 and 16 of all VLAN-type transmitted packets (Bytes 13 and 14 are 0x8100 or 0x88a8). Note: Changes to this field take effect only on the start of a packet. If you write this register field when a packet is being transmitted, only the subsequent packet can use the updated value, that is, the current packet does not use the updated value.Values: 0x0 (NONE): No VLAN tag deletion, insertion, or replacement 0x1 (DELETE): VLAN tag deletion 0x2 (INSERT): VLAN tag insertion 0x3 (REPLACE): VLAN tag replacement Value After Reset: 0x0
			15:0	VLT	R/W	0x0	VLAN Tag for Transmit Packets This field contains the value of the VLAN tag to be inserted or replaced. The value must only be changed when the transmit lines are inactive or during the initialization phase. The following list describes the bits of this field: Bits[15:13]: User Priority Bit 12: Canonical Format Indicator (CFI) or Drop Eligible Indicator (DEI) Bits[11:0]: VLAN Identifier (VID) field of VLAN tag Value After Reset: 0x0
0x60	MAC_VLAN_Incl(#i)(fori=0;i<=7)	The Tx Queue 0 VLAN Tag Inclusion register contains the VLAN tag for insertion in the Transmit packets from Tx Queue 0. It also contains the VLAN tag insertion controls.	31:20	Reserved_31_20	R	0x0	Reserved. Value After Reset: 0x0
			19	CSVL	R/W	0x0	C-VLAN or S-VLAN 1: S-VLAN type (0x88A8) is inserted in the 13th and 14th bytes of transmitted packets. 0: C-VLAN type (0x8100) is inserted in the 13th and 14th bytes of transmitted packets. Values: 0x0 (C-VLAN): C-VLAN type (0x8100) is inserted 0x1 (S-VLAN): S-VLAN type (0x88A8) is inserted Value After Reset: 0x0
			18:16	Reserved_18_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	VLT	R/W	0x0	VLAN Tag for Transmit Packets This field contains the value of the VLAN tag to be inserted. The value must only be changed when the transmit lines are inactive or during the initialization phase. The following list describes the bits of this field: Bits[15:13]: User Priority Bit 12: Canonical Format Indicator (CFI) or Drop Eligible Indicator (DEI) Bits[11:0]: VLAN Identifier (VID) field of VLAN tag Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x64	MAC_Inner_VLAN_Incl	The Inner VLAN Tag Inclusion or Replacement register contains the inner VLAN tag to be inserted or replaced in the Transmit packet. It also contains the inner VLAN tag insertion controls.	31:21	Reserved_31_21	R	0x0	Reserved. Value After Reset: 0x0
			20	VLTI	R/W	0x0	VLAN Tag Input When this bit is set, it indicates that the VLAN tag to be inserted or replaced in Tx packet must be taken from: The Tx descriptor Values: 0x0 (DISABLE): VLAN Tag Input is disabled 0x1 (ENABLE): VLAN Tag Input is enabled Value After Reset: 0x0
			19	CSVL	R/W	0x0	C-VLAN or S-VLAN 1: S-VLAN type (0x88A8) is inserted or replaced in the 17th and 18th bytes of transmitted packets. 0: C-VLAN type (0x8100) is inserted or replaced in the 17th and 18th bytes of transmitted packets. Values: 0x0 (C-VLAN): C-VLAN type (0x8100) is inserted 0x1 (S-VLAN): S-VLAN type (0x88A8) is inserted Value After Reset: 0x0
			18	VLP	R/W	0x0	VLAN Priority Control 1: MAC uses the VLC field for VLAN deletion, insertion, or replacement. 0: MAC uses the mti_vlan_ctrl_i signal for VLAN deletion, insertion, or replacement. MAC ignores the VLC field. Values: 0x0 (DISABLE): VLAN Priority Control is disabled 0x1 (ENABLE): VLAN Priority Control is enabled Value After Reset: 0x0
			17:16	VLC	R/W	0x0	VLAN Tag Control in Transmit Packets 2'b00: No VLAN tag deletion, insertion, or replacement 2'b01: VLAN tag deletion The MAC removes the VLAN type (bytes 17 and 18) and VLAN tag (bytes 19 and 20) of all transmitted packets with VLAN tags. 2'b10: VLAN tag insertion The MAC inserts VLT in bytes 19 and 20 of the packet after inserting the Type value (0x8100 or 0x88a8) in bytes 17 and 18. This operation is performed on all transmitted packets, irrespective of whether they already have a VLAN tag. 2'b11: VLAN tag replacement The MAC replaces VLT in bytes 19 and 20 of all VLAN-type transmitted packets (Bytes 17 and 18 are 0x8100 or 0x88a8). Note: Changes to this field take effect only on the start of a packet. If you write this register field when a packet is being transmitted, only the subsequent packet can use the updated value, that is, the current packet does not use the updated value.Values: 0x0 (NONE): No VLAN tag deletion, insertion, or replacement 0x1 (DELETE): VLAN tag deletion 0x2 (INSERT): VLAN tag insertion 0x3 (REPLACE): VLAN tag replacement Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			15:0	VLT	R/W	0x0	VLAN Tag for Transmit Packets This field contains the value of the VLAN tag to be inserted or replaced. The value must only be changed when the transmit lines are inactive or during the initialization phase. The following list describes the bits of this field: Bits[15:13]: User Priority Bit 12: Canonical Format Indicator (CFI) or Drop Eligible Indicator (DEI) Bits[11:0]: VLAN Identifier (VID) field of VLAN tag Value After Reset: 0x0
0x70	MAC_Q0_Tx_Flow_Ctrl	The Flow Control register controls the generation and reception of the Control (Pause Command) packets by the Flow control module of the MAC. A Write to a register with the Busy bit set to 1 triggers the Flow Control block to generate a Pause packet. The fields of the control packet are selected as specified in the 802.3x specification, and the Pause Time value from this register is used in the Pause Time field of the control packet. The Busy bit remains set until the control packet is transferred onto the cable. The application must make sure that the Busy bit is cleared before writing to the register.When the PFCE bit in the MAC_Rx_Flow_Ctrl register is enabled, this register controls the generation of Priority Flow Control (PFC) frames with priorities mapped according to PSRQ0 in the MAC_RxQ_Ctrl2 register.	31:16	PT	R/W	0x0	Pause Time This field holds the value to be used in the Pause Time field in the Tx control packet. If the Pause Time bits are configured to be double-synchronized to the (G)MII clock domain, consecutive writes to this register should be performed only after at least four clock cycles in the destination clock domain. Value After Reset: 0x0
			15:8	Reserved_15_8	R	0x0	Reserved. Value After Reset: 0x0
			7	DZPQ	R/W	0x0	Disable Zero-Quanta Pause When this bit is set, it disables the automatic generation of the zero-quanta Pause packets on de-assertion of the flow-control signal from the FIFO layer (MTL or external sideband flow control signal sbd_flowctrl_i or mti_flowctrl_i). When this bit is reset, normal operation with automatic zero-quanta Pause packet generation is enabled.Values: 0x0 (ENABLE): Zero-Quanta Pause packet generation is enabled 0x1 (DISABLE): Zero-Quanta Pause packet generation is disabled Value After Reset: 0x0
			6:4	PLT	R/W	0x0	Pause Low Threshold This field configures the threshold of the Pause timer at which the input flow control signal mti_flowctrl_i (or sbd_flowctrl_i) is checked for automatic retransmission of the Pause packet. The threshold values should be always less than the Pause Time configured in Bits[31:16]. For example, if PT = 100H (256 slot times), and PLT = 001, a second Pause packet is automatically transmitted if the mti_flowctrl_i signal is asserted at 228 (256-28) slot times after the first Pause packet is transmitted. The following list provides the threshold values for different values. The slot time is defined as the time taken to transmit 512 bits (64 bytes) on the GMII or MII interface. This (approximate) computation is based on the packet size (64, 1518, 2000, 9018, 16384, or 32768) + 2 Pause Packet Size + IPG in Slot Times.Values: 0x0 (PT4): Pause Time minus 4 Slot Times (PT -4 slot times) 0x1 (PT28): Pause Time minus 28 Slot Times (PT -28 slot times) 0x2 (PT36): Pause Time minus 36 Slot Times (PT -36 slot times) 0x3 (PT144): Pause Time minus 144 Slot Times (PT -144 slot times) 0x4 (PT256): Pause Time minus 256 Slot Times (PT -256 slot times) 0x5 (PT512): Pause Time minus 512 Slot Times (PT -512 slot times) 0x6 (RSVD): Reserved Value After Reset: 0x0
			3:2	Reserved_3_2	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			1	TFE	R/W	0x0	Transmit Flow Control Enable In the full-duplex mode, when this bit is set, the MAC enables the flow control operation to Tx Pause packets. When this bit is reset, the flow control operation in the MAC is disabled, and the MAC does not transmit any Pause packets.Values: 0x0 (DISABLE): Transmit Flow Control is disabled 0x1 (ENABLE): Transmit Flow Control is enabled Value After Reset: 0x0
			0	FCB_BPA	R/W	0x0	Flow Control Busy This bit initiates a Pause packet in the full-duplex mode if the TFE bit is set. In the full-duplex mode, this bit should be read as 1'b0 before writing to this register. To initiate a Pause packet, the application must set this bit to 1'b1. During Control packet transfer, this bit continues to be set to indicate that a packet transmission is in progress. When Pause packet transmission is complete, the MAC resets this bit to 1'b0. You should not write to this register until this bit is cleared. Access restrictions apply. MAC initiates the pause packet or activates the backpressure, when application writes 1. MAC writes 0 after the operation is complete. MAC ignores when application writes 0. Values: 0x0 (DISABLE): Flow Control Busy or Backpressure Activate is disabled 0x1 (ENABLE): Flow Control Busy or Backpressure Activate is enabled Value After Reset: 0x0 Testable: untestable
			31:16	PT	R/W	0x0	Pause Time This field holds the value to be used in the Pause Time field in the Tx control packet. If the Pause Time bits are configured to be double-synchronized to the (G)MII clock domain, consecutive writes to this register should be performed only after at least four clock cycles in the destination clock domain. Value After Reset: 0x0
			15:8	Reserved_15_8	R	0x0	Reserved. Value After Reset: 0x0
			7	DZPQ	R/W	0x0	Disable Zero-Quanta Pause When this bit is set, it disables the automatic generation of the zero-quanta Pause packets on de-assertion of the flow-control signal from the FIFO layer (MTL or external sideband flow control signal sbd_flowctrl_i or mti_flowctrl_i). When this bit is reset, normal operation with automatic zero-quanta Pause packet generation is enabled.Values: 0x0 (ENABLE): Zero-Quanta Pause packet generation is enabled 0x1 (DISABLE): Zero-Quanta Pause packet generation is disabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x74	MAC_Q1_Tx_Flow_Ctrl	The Flow Control register controls the generation and reception of the Control (Pause Command) packets by the Flow control module of the MAC. A Write to a register with the Busy bit set to 1 triggers the Flow Control block to generate a Pause packet. The fields of the control packet are selected as specified in the 802.3x specification, and the Pause Time value from this register is used in the Pause Time field of the control packet. The Busy bit remains set until the control packet is transferred onto the cable. The application must make sure that the Busy bit is cleared before writing to the register.When the PFCE bit in the MAC_Rx_Flow_Ctrl register is enabled, this register controls the generation of Priority Flow Control (PFC) frames with priorities mapped according to PSRQ0 in the MAC_RxQ_Ctrl2 register.	6:4	PLT	R/W	0x0	Pause Low Threshold This field configures the threshold of the Pause timer at which the input flow control signal mti_flowctrl_i (or sbd_flowctrl_i) is checked for automatic retransmission of the Pause packet. The threshold values should be always less than the Pause Time configured in Bits[31:16]. For example, if PT = 100H (256 slot times), and PLT = 001, a second Pause packet is automatically transmitted if the mti_flowctrl_i signal is asserted at 228 (256-28) slot times after the first Pause packet is transmitted. The following list provides the threshold values for different values. The slot time is defined as the time taken to transmit 512 bits (64 bytes) on the GMII or MII interface. This (approximate) computation is based on the packet size (64, 1518, 2000, 9018, 16384, or 32768) + 2 Pause Packet Size + IPG in Slot Times.Values: 0x0 (PT4): Pause Time minus 4 Slot Times (PT -4 slot times) 0x1 (PT28): Pause Time minus 28 Slot Times (PT -28 slot times) 0x2 (PT36): Pause Time minus 36 Slot Times (PT -36 slot times) 0x3 (PT144): Pause Time minus 144 Slot Times (PT -144 slot times) 0x4 (PT256): Pause Time minus 256 Slot Times (PT -256 slot times) 0x5 (PT512): Pause Time minus 512 Slot Times (PT -512 slot times) 0x6 (RSVD): Reserved Value After Reset: 0x0
			3:2	Reserved_3_2	R	0x0	Reserved. Value After Reset: 0x0
			1	TFE	R/W	0x0	Transmit Flow Control Enable In the full-duplex mode, when this bit is set, the MAC enables the flow control operation to Tx Pause packets. When this bit is reset, the flow control operation in the MAC is disabled, and the MAC does not transmit any Pause packets.Values: 0x0 (DISABLE): Transmit Flow Control is disabled 0x1 (ENABLE): Transmit Flow Control is enabled Value After Reset: 0x0
			0	FCB_BPA	R/W	0x0	Flow Control Busy This bit initiates a Pause packet in the full-duplex mode if the TFE bit is set. In the full-duplex mode, this bit should be read as 1'b0 before writing to this register. To initiate a Pause packet, the application must set this bit to 1'b1. During Control packet transfer, this bit continues to be set to indicate that a packet transmission is in progress. When Pause packet transmission is complete, the MAC resets this bit to 1'b0. You should not write to this register until this bit is cleared. Access restrictions apply. MAC initiates the pause packet or activates the backpressure, when application writes 1. MAC writes 0 after the operation is complete. MAC ignores when application writes 0. Values: 0x0 (DISABLE): Flow Control Busy or Backpressure Activate is disabled 0x1 (ENABLE): Flow Control Busy or Backpressure Activate is enabled Value After Reset: 0x0 Testable: untestable
			31:16	PT	R/W	0x0	Pause Time This field holds the value to be used in the Pause Time field in the Tx control packet. If the Pause Time bits are configured to be double-synchronized to the (G)MII clock domain, consecutive writes to this register should be performed only after at least four clock cycles in the destination clock domain. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x78	MAC_Q2_Tx_Flow_Ctrl	The Flow Control register controls the generation and reception of the Control (Pause Command) packets by the Flow control module of the MAC. A Write to a register with the Busy bit set to 1 triggers the Flow Control block to generate a Pause packet. The fields of the control packet are selected as specified in the 802.3x specification, and the Pause Time value from this register is used in the Pause Time field of the control packet. The Busy bit remains set until the control packet is transferred onto the cable. The application must make sure that the Busy bit is cleared before writing to the register.When the PFCE bit in the MAC_Rx_Flow_Ctrl register is enabled, this register controls the generation of Priority Flow Control (PFC) frames with priorities mapped according to PSRQ0 in the MAC_RxQ_Ctrl2 register.	15:8	Reserved_15_8	R	0x0	Reserved. Value After Reset: 0x0
			7	DZPQ	R/W	0x0	Disable Zero-Quanta Pause When this bit is set, it disables the automatic generation of the zero-quanta Pause packets on de-assertion of the flow-control signal from the FIFO layer (MTL or external sideband flow control signal sbd_flowctrl_i or mti_flowctrl_i). When this bit is reset, normal operation with automatic zero-quanta Pause packet generation is enabled.Values: 0x0 (ENABLE): Zero-Quanta Pause packet generation is enabled 0x1 (DISABLE): Zero-Quanta Pause packet generation is disabled Value After Reset: 0x0
			6:4	PLT	R/W	0x0	Pause Low Threshold This field configures the threshold of the Pause timer at which the input flow control signal mti_flowctrl_i (or sbd_flowctrl_i) is checked for automatic retransmission of the Pause packet. The threshold values should be always less than the Pause Time configured in Bits[31:16]. For example, if PT = 100H (256 slot times), and PLT = 001, a second Pause packet is automatically transmitted if the mti_flowctrl_i signal is asserted at 228 (256-28) slot times after the first Pause packet is transmitted. The following list provides the threshold values for different values. The slot time is defined as the time taken to transmit 512 bits (64 bytes) on the GMII or MII interface. This (approximate) computation is based on the packet size (64, 1518, 2000, 9018, 16384, or 32768) + 2 Pause Packet Size + IPG in Slot Times.Values: 0x0 (PT4): Pause Time minus 4 Slot Times (PT -4 slot times) 0x1 (PT28): Pause Time minus 28 Slot Times (PT -28 slot times) 0x2 (PT36): Pause Time minus 36 Slot Times (PT -36 slot times) 0x3 (PT144): Pause Time minus 144 Slot Times (PT -144 slot times) 0x4 (PT256): Pause Time minus 256 Slot Times (PT -256 slot times) 0x5 (PT512): Pause Time minus 512 Slot Times (PT -512 slot times) 0x6 (RSVD): Reserved Value After Reset: 0x0
			3:2	Reserved_3_2	R	0x0	Reserved. Value After Reset: 0x0
			1	TFE	R/W	0x0	Transmit Flow Control Enable In the full-duplex mode, when this bit is set, the MAC enables the flow control operation to Tx Pause packets. When this bit is reset, the flow control operation in the MAC is disabled, and the MAC does not transmit any Pause packets.Values: 0x0 (DISABLE): Transmit Flow Control is disabled 0x1 (ENABLE): Transmit Flow Control is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	FCB_BPA	R/W	0x0	Flow Control Busy This bit initiates a Pause packet in the full-duplex mode if the TFE bit is set. In the full-duplex mode, this bit should be read as 1'b0 before writing to this register. To initiate a Pause packet, the application must set this bit to 1'b1. During Control packet transfer, this bit continues to be set to indicate that a packet transmission is in progress. When Pause packet transmission is complete, the MAC resets this bit to 1'b0. You should not write to this register until this bit is cleared. Access restrictions apply. MAC initiates the pause packet or activates the backpressure, when application writes 1. MAC writes 0 after the operation is complete. MAC ignores when application writes 0. Values: 0x0 (DISABLE): Flow Control Busy or Backpressure Activate is disabled 0x1 (ENABLE): Flow Control Busy or Backpressure Activate is enabled Value After Reset: 0x0 Testable: untestable
0x7C	MAC_Q3_Tx_Flow_Ctrl	The Flow Control register controls the generation and reception of the Control (Pause Command) packets by the Flow control module of the MAC. A Write to a register with the Busy bit set to 1 triggers the Flow Control block to generate a Pause packet. The fields of the control packet are selected as specified in the 802.3x specification, and the Pause Time value from this register is used in the Pause Time field of the control packet. The Busy bit remains set until the control packet is transferred onto the cable. The application must make sure that the Busy bit is cleared before writing to the register.When the PFCE bit in the MAC_Rx_Flow_Ctrl register is enabled, this register controls the generation of Priority Flow Control (PFC) frames with priorities mapped according to PSRQ0 in the MAC_RxQ_Ctrl2 register.	31:16	PT	R/W	0x0	Pause Time This field holds the value to be used in the Pause Time field in the Tx control packet. If the Pause Time bits are configured to be double-synchronized to the (G)MII clock domain, consecutive writes to this register should be performed only after at least four clock cycles in the destination clock domain. Value After Reset: 0x0
			15:8	Reserved_15_8	R	0x0	Reserved. Value After Reset: 0x0
			7	DZPQ	R/W	0x0	Disable Zero-Quanta Pause When this bit is set, it disables the automatic generation of the zero-quanta Pause packets on de-assertion of the flow-control signal from the FIFO layer (MTL or external sideband flow control signal sbd_flowctrl_i or mti_flowctrl_i). When this bit is reset, normal operation with automatic zero-quanta Pause packet generation is enabled.Values: 0x0 (ENABLE): Zero-Quanta Pause packet generation is enabled 0x1 (DISABLE): Zero-Quanta Pause packet generation is disabled Value After Reset: 0x0
			6:4	PLT	R/W	0x0	Pause Low Threshold This field configures the threshold of the Pause timer at which the input flow control signal mti_flowctrl_i (or sbd_flowctrl_i) is checked for automatic retransmission of the Pause packet. The threshold values should be always less than the Pause Time configured in Bits[31:16]. For example, if PT = 100H (256 slot times), and PLT = 001, a second Pause packet is automatically transmitted if the mti_flowctrl_i signal is asserted at 228 (256-28) slot times after the first Pause packet is transmitted. The following list provides the threshold values for different values. The slot time is defined as the time taken to transmit 512 bits (64 bytes) on the GMII or MII interface. This (approximate) computation is based on the packet size (64, 1518, 2000, 9018, 16384, or 32768) + 2 Pause Packet Size + IPG in Slot Times.Values: 0x0 (PT4): Pause Time minus 4 Slot Times (PT -4 slot times) 0x1 (PT28): Pause Time minus 28 Slot Times (PT -28 slot times) 0x2 (PT36): Pause Time minus 36 Slot Times (PT -36 slot times) 0x3 (PT144): Pause Time minus 144 Slot Times (PT -144 slot times) 0x4 (PT256): Pause Time minus 256 Slot Times (PT -256 slot times) 0x5 (PT512): Pause Time minus 512 Slot Times (PT -512 slot times) 0x6 (RSVD): Reserved Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			3:2	Reserved_3_2	R	0x0	Reserved. Value After Reset: 0x0
			1	TFE	R/W	0x0	Transmit Flow Control Enable In the full-duplex mode, when this bit is set, the MAC enables the flow control operation to Tx Pause packets. When this bit is reset, the flow control operation in the MAC is disabled, and the MAC does not transmit any Pause packets.Values: 0x0 (DISABLE): Transmit Flow Control is disabled 0x1 (ENABLE): Transmit Flow Control is enabled Value After Reset: 0x0
			0	FCB_BPA	R/W	0x0	Flow Control Busy This bit initiates a Pause packet in the full-duplex mode if the TFE bit is set. In the full-duplex mode, this bit should be read as 1'b0 before writing to this register. To initiate a Pause packet, the application must set this bit to 1'b1. During Control packet transfer, this bit continues to be set to indicate that a packet transmission is in progress. When Pause packet transmission is complete, the MAC resets this bit to 1'b0. You should not write to this register until this bit is cleared. Access restrictions apply. MAC initiates the pause packet or activates the backpressure, when application writes 1. MAC writes 0 after the operation is complete. MAC ignores when application writes 0. Values: 0x0 (DISABLE): Flow Control Busy or Backpressure Activate is disabled 0x1 (ENABLE): Flow Control Busy or Backpressure Activate is enabled Value After Reset: 0x0 Testable: untestable
			31:16	PT	R/W	0x0	Pause Time This field holds the value to be used in the Pause Time field in the Tx control packet. If the Pause Time bits are configured to be double-synchronized to the (G)MII clock domain, consecutive writes to this register should be performed only after at least four clock cycles in the destination clock domain. Value After Reset: 0x0
			15:8	Reserved_15_8	R	0x0	Reserved. Value After Reset: 0x0
			7	DZPQ	R/W	0x0	Disable Zero-Quanta Pause When this bit is set, it disables the automatic generation of the zero-quanta Pause packets on de-assertion of the flow-control signal from the FIFO layer (MTL or external sideband flow control signal sbd_flowctrl_i or mti_flowctrl_i). When this bit is reset, normal operation with automatic zero-quanta Pause packet generation is enabled.Values: 0x0 (ENABLE): Zero-Quanta Pause packet generation is enabled 0x1 (DISABLE): Zero-Quanta Pause packet generation is disabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x80	MAC_Q4_Tx_Flow_Ctrl	The Flow Control register controls the generation and reception of the Control (Pause Command) packets by the Flow control module of the MAC. A Write to a register with the Busy bit set to 1 triggers the Flow Control block to generate a Pause packet. The fields of the control packet are selected as specified in the 802.3x specification, and the Pause Time value from this register is used in the Pause Time field of the control packet. The Busy bit remains set until the control packet is transferred onto the cable. The application must make sure that the Busy bit is cleared before writing to the register.When the PFCE bit in the MAC_Rx_Flow_Ctrl register is enabled, this register controls the generation of Priority Flow Control (PFC) frames with priorities mapped according to PSRQ0 in the MAC_RxQ_Ctrl2 register.	6:4	PLT	R/W	0x0	Pause Low Threshold This field configures the threshold of the Pause timer at which the input flow control signal mti_flowctrl_i (or sbd_flowctrl_i) is checked for automatic retransmission of the Pause packet. The threshold values should be always less than the Pause Time configured in Bits[31:16]. For example, if PT = 100H (256 slot times), and PLT = 001, a second Pause packet is automatically transmitted if the mti_flowctrl_i signal is asserted at 228 (256-28) slot times after the first Pause packet is transmitted. The following list provides the threshold values for different values. The slot time is defined as the time taken to transmit 512 bits (64 bytes) on the GMII or MII interface. This (approximate) computation is based on the packet size (64, 1518, 2000, 9018, 16384, or 32768) + 2 Pause Packet Size + IPG in Slot Times.Values: 0x0 (PT4): Pause Time minus 4 Slot Times (PT -4 slot times) 0x1 (PT28): Pause Time minus 28 Slot Times (PT -28 slot times) 0x2 (PT36): Pause Time minus 36 Slot Times (PT -36 slot times) 0x3 (PT144): Pause Time minus 144 Slot Times (PT -144 slot times) 0x4 (PT256): Pause Time minus 256 Slot Times (PT -256 slot times) 0x5 (PT512): Pause Time minus 512 Slot Times (PT -512 slot times) 0x6 (RSVD): Reserved Value After Reset: 0x0
			3:2	Reserved_3_2	R	0x0	Reserved. Value After Reset: 0x0
			1	TFE	R/W	0x0	Transmit Flow Control Enable In the full-duplex mode, when this bit is set, the MAC enables the flow control operation to Tx Pause packets. When this bit is reset, the flow control operation in the MAC is disabled, and the MAC does not transmit any Pause packets.Values: 0x0 (DISABLE): Transmit Flow Control is disabled 0x1 (ENABLE): Transmit Flow Control is enabled Value After Reset: 0x0
			0	FCB_BPA	R/W	0x0	Flow Control Busy This bit initiates a Pause packet in the full-duplex mode if the TFE bit is set. In the full-duplex mode, this bit should be read as 1'b0 before writing to this register. To initiate a Pause packet, the application must set this bit to 1'b1. During Control packet transfer, this bit continues to be set to indicate that a packet transmission is in progress. When Pause packet transmission is complete, the MAC resets this bit to 1'b0. You should not write to this register until this bit is cleared. Access restrictions apply. MAC initiates the pause packet or activates the backpressure, when application writes 1. MAC writes 0 after the operation is complete. MAC ignores when application writes 0. Values: 0x0 (DISABLE): Flow Control Busy or Backpressure Activate is disabled 0x1 (ENABLE): Flow Control Busy or Backpressure Activate is enabled Value After Reset: 0x0 Testable: untestable
			31:16	PT	R/W	0x0	Pause Time This field holds the value to be used in the Pause Time field in the Tx control packet. If the Pause Time bits are configured to be double-synchronized to the (G)MII clock domain, consecutive writes to this register should be performed only after at least four clock cycles in the destination clock domain. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x84	MAC_Q5_Tx_Flow_Ctrl	The Flow Control register controls the generation and reception of the Control (Pause Command) packets by the Flow control module of the MAC. A Write to a register with the Busy bit set to 1 triggers the Flow Control block to generate a Pause packet. The fields of the control packet are selected as specified in the 802.3x specification, and the Pause Time value from this register is used in the Pause Time field of the control packet. The Busy bit remains set until the control packet is transferred onto the cable. The application must make sure that the Busy bit is cleared before writing to the register.When the PFCE bit in the MAC_Rx_Flow_Ctrl register is enabled, this register controls the generation of Priority Flow Control (PFC) frames with priorities mapped according to PSRQ0 in the MAC_RxQ_Ctrl2 register.	15:8	Reserved_15_8	R	0x0	Reserved. Value After Reset: 0x0
			7	DZPQ	R/W	0x0	Disable Zero-Quanta Pause When this bit is set, it disables the automatic generation of the zero-quanta Pause packets on de-assertion of the flow-control signal from the FIFO layer (MTL or external sideband flow control signal sbd_flowctrl_i or mti_flowctrl_i). When this bit is reset, normal operation with automatic zero-quanta Pause packet generation is enabled.Values: 0x0 (ENABLE): Zero-Quanta Pause packet generation is enabled 0x1 (DISABLE): Zero-Quanta Pause packet generation is disabled Value After Reset: 0x0
			6:4	PLT	R/W	0x0	Pause Low Threshold This field configures the threshold of the Pause timer at which the input flow control signal mti_flowctrl_i (or sbd_flowctrl_i) is checked for automatic retransmission of the Pause packet. The threshold values should be always less than the Pause Time configured in Bits[31:16]. For example, if PT = 100H (256 slot times), and PLT = 001, a second Pause packet is automatically transmitted if the mti_flowctrl_i signal is asserted at 228 (256-28) slot times after the first Pause packet is transmitted. The following list provides the threshold values for different values. The slot time is defined as the time taken to transmit 512 bits (64 bytes) on the GMII or MII interface. This (approximate) computation is based on the packet size (64, 1518, 2000, 9018, 16384, or 32768) + 2 Pause Packet Size + IPG in Slot Times.Values: 0x0 (PT4): Pause Time minus 4 Slot Times (PT -4 slot times) 0x1 (PT28): Pause Time minus 28 Slot Times (PT -28 slot times) 0x2 (PT36): Pause Time minus 36 Slot Times (PT -36 slot times) 0x3 (PT144): Pause Time minus 144 Slot Times (PT -144 slot times) 0x4 (PT256): Pause Time minus 256 Slot Times (PT -256 slot times) 0x5 (PT512): Pause Time minus 512 Slot Times (PT -512 slot times) 0x6 (RSVD): Reserved Value After Reset: 0x0
			3:2	Reserved_3_2	R	0x0	Reserved. Value After Reset: 0x0
			1	TFE	R/W	0x0	Transmit Flow Control Enable In the full-duplex mode, when this bit is set, the MAC enables the flow control operation to Tx Pause packets. When this bit is reset, the flow control operation in the MAC is disabled, and the MAC does not transmit any Pause packets.Values: 0x0 (DISABLE): Transmit Flow Control is disabled 0x1 (ENABLE): Transmit Flow Control is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	FCB_BPA	R/W	0x0	Flow Control Busy This bit initiates a Pause packet in the full-duplex mode if the TFE bit is set. In the full-duplex mode, this bit should be read as 1'b0 before writing to this register. To initiate a Pause packet, the application must set this bit to 1'b1. During Control packet transfer, this bit continues to be set to indicate that a packet transmission is in progress. When Pause packet transmission is complete, the MAC resets this bit to 1'b0. You should not write to this register until this bit is cleared. Access restrictions apply. MAC initiates the pause packet or activates the backpressure, when application writes 1. MAC writes 0 after the operation is complete. MAC ignores when application writes 0. Values: 0x0 (DISABLE): Flow Control Busy or Backpressure Activate is disabled 0x1 (ENABLE): Flow Control Busy or Backpressure Activate is enabled Value After Reset: 0x0 Testable: untestable
0x88	MAC_Q6_Tx_Flow_Ctrl	The Flow Control register controls the generation and reception of the Control (Pause Command) packets by the Flow control module of the MAC. A Write to a register with the Busy bit set to 1 triggers the Flow Control block to generate a Pause packet. The fields of the control packet are selected as specified in the 802.3x specification, and the Pause Time value from this register is used in the Pause Time field of the control packet. The Busy bit remains set until the control packet is transferred onto the cable. The application must make sure that the Busy bit is cleared before writing to the register.When the PFCE bit in the MAC_Rx_Flow_Ctrl register is enabled, this register controls the generation of Priority Flow Control (PFC) frames with priorities mapped according to PSRQ0 in the MAC_RxQ_Ctrl2 register.	31:16	PT	R/W	0x0	Pause Time This field holds the value to be used in the Pause Time field in the Tx control packet. If the Pause Time bits are configured to be double-synchronized to the (G)MII clock domain, consecutive writes to this register should be performed only after at least four clock cycles in the destination clock domain. Value After Reset: 0x0
			15:8	Reserved_15_8	R	0x0	Reserved. Value After Reset: 0x0
			7	DZPQ	R/W	0x0	Disable Zero-Quanta Pause When this bit is set, it disables the automatic generation of the zero-quanta Pause packets on de-assertion of the flow-control signal from the FIFO layer (MTL or external sideband flow control signal sbd_flowctrl_i or mti_flowctrl_i). When this bit is reset, normal operation with automatic zero-quanta Pause packet generation is enabled.Values: 0x0 (ENABLE): Zero-Quanta Pause packet generation is enabled 0x1 (DISABLE): Zero-Quanta Pause packet generation is disabled Value After Reset: 0x0
			6:4	PLT	R/W	0x0	Pause Low Threshold This field configures the threshold of the Pause timer at which the input flow control signal mti_flowctrl_i (or sbd_flowctrl_i) is checked for automatic retransmission of the Pause packet. The threshold values should be always less than the Pause Time configured in Bits[31:16]. For example, if PT = 100H (256 slot times), and PLT = 001, a second Pause packet is automatically transmitted if the mti_flowctrl_i signal is asserted at 228 (256-28) slot times after the first Pause packet is transmitted. The following list provides the threshold values for different values. The slot time is defined as the time taken to transmit 512 bits (64 bytes) on the GMII or MII interface. This (approximate) computation is based on the packet size (64, 1518, 2000, 9018, 16384, or 32768) + 2 Pause Packet Size + IPG in Slot Times.Values: 0x0 (PT4): Pause Time minus 4 Slot Times (PT -4 slot times) 0x1 (PT28): Pause Time minus 28 Slot Times (PT -28 slot times) 0x2 (PT36): Pause Time minus 36 Slot Times (PT -36 slot times) 0x3 (PT144): Pause Time minus 144 Slot Times (PT -144 slot times) 0x4 (PT256): Pause Time minus 256 Slot Times (PT -256 slot times) 0x5 (PT512): Pause Time minus 512 Slot Times (PT -512 slot times) 0x6 (RSVD): Reserved Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			3:2	Reserved_3_2	R	0x0	Reserved. Value After Reset: 0x0
			1	TFE	R/W	0x0	Transmit Flow Control Enable In the full-duplex mode, when this bit is set, the MAC enables the flow control operation to Tx Pause packets. When this bit is reset, the flow control operation in the MAC is disabled, and the MAC does not transmit any Pause packets.Values: 0x0 (DISABLE): Transmit Flow Control is disabled 0x1 (ENABLE): Transmit Flow Control is enabled Value After Reset: 0x0
			0	FCB_BPA	R/W	0x0	Flow Control Busy This bit initiates a Pause packet in the full-duplex mode if the TFE bit is set. In the full-duplex mode, this bit should be read as 1'b0 before writing to this register. To initiate a Pause packet, the application must set this bit to 1'b1. During Control packet transfer, this bit continues to be set to indicate that a packet transmission is in progress. When Pause packet transmission is complete, the MAC resets this bit to 1'b0. You should not write to this register until this bit is cleared. Access restrictions apply. MAC initiates the pause packet or activates the backpressure, when application writes 1. MAC writes 0 after the operation is complete. MAC ignores when application writes 0. Values: 0x0 (DISABLE): Flow Control Busy or Backpressure Activate is disabled 0x1 (ENABLE): Flow Control Busy or Backpressure Activate is enabled Value After Reset: 0x0 Testable: untestable
			31:16	PT	R/W	0x0	Pause Time This field holds the value to be used in the Pause Time field in the Tx control packet. If the Pause Time bits are configured to be double-synchronized to the (G)MII clock domain, consecutive writes to this register should be performed only after at least four clock cycles in the destination clock domain. Value After Reset: 0x0
			15:8	Reserved_15_8	R	0x0	Reserved. Value After Reset: 0x0
			7	DZPQ	R/W	0x0	Disable Zero-Quanta Pause When this bit is set, it disables the automatic generation of the zero-quanta Pause packets on de-assertion of the flow-control signal from the FIFO layer (MTL or external sideband flow control signal sbd_flowctrl_i or mti_flowctrl_i). When this bit is reset, normal operation with automatic zero-quanta Pause packet generation is enabled.Values: 0x0 (ENABLE): Zero-Quanta Pause packet generation is enabled 0x1 (DISABLE): Zero-Quanta Pause packet generation is disabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x8C	MAC_Q7_Tx_Flow_Ctrl	The Flow Control register controls the generation and reception of the Control (Pause Command) packets by the Flow control module of the MAC. A Write to a register with the Busy bit set to 1 triggers the Flow Control block to generate a Pause packet. The fields of the control packet are selected as specified in the 802.3x specification, and the Pause Time value from this register is used in the Pause Time field of the control packet. The Busy bit remains set until the control packet is transferred onto the cable. The application must make sure that the Busy bit is cleared before writing to the register.When the PFCE bit in the MAC_Rx_Flow_Ctrl register is enabled, this register controls the generation of Priority Flow Control (PFC) frames with priorities mapped according to PSRQ0 in the MAC_RxQ_Ctrl2 register.	6:4	PLT	R/W	0x0	Pause Low Threshold This field configures the threshold of the Pause timer at which the input flow control signal mti_flowctrl_i (or sbd_flowctrl_i) is checked for automatic retransmission of the Pause packet. The threshold values should be always less than the Pause Time configured in Bits[31:16]. For example, if PT = 100H (256 slot times), and PLT = 001, a second Pause packet is automatically transmitted if the mti_flowctrl_i signal is asserted at 228 (256-28) slot times after the first Pause packet is transmitted. The following list provides the threshold values for different values. The slot time is defined as the time taken to transmit 512 bits (64 bytes) on the GMII or MII interface. This (approximate) computation is based on the packet size (64, 1518, 2000, 9018, 16384, or 32768) + 2 Pause Packet Size + IPG in Slot Times.Values: 0x0 (PT4): Pause Time minus 4 Slot Times (PT -4 slot times) 0x1 (PT28): Pause Time minus 28 Slot Times (PT -28 slot times) 0x2 (PT36): Pause Time minus 36 Slot Times (PT -36 slot times) 0x3 (PT144): Pause Time minus 144 Slot Times (PT -144 slot times) 0x4 (PT256): Pause Time minus 256 Slot Times (PT -256 slot times) 0x5 (PT512): Pause Time minus 512 Slot Times (PT -512 slot times) 0x6 (RSVD): Reserved Value After Reset: 0x0
			3:2	Reserved_3_2	R	0x0	Reserved. Value After Reset: 0x0
			1	TFE	R/W	0x0	Transmit Flow Control Enable In the full-duplex mode, when this bit is set, the MAC enables the flow control operation to Tx Pause packets. When this bit is reset, the flow control operation in the MAC is disabled, and the MAC does not transmit any Pause packets.Values: 0x0 (DISABLE): Transmit Flow Control is disabled 0x1 (ENABLE): Transmit Flow Control is enabled Value After Reset: 0x0
			0	FCB_BPA	R/W	0x0	Flow Control Busy This bit initiates a Pause packet in the full-duplex mode if the TFE bit is set. In the full-duplex mode, this bit should be read as 1'b0 before writing to this register. To initiate a Pause packet, the application must set this bit to 1'b1. During Control packet transfer, this bit continues to be set to indicate that a packet transmission is in progress. When Pause packet transmission is complete, the MAC resets this bit to 1'b0. You should not write to this register until this bit is cleared. Access restrictions apply. MAC initiates the pause packet or activates the backpressure, when application writes 1. MAC writes 0 after the operation is complete. MAC ignores when application writes 0. Values: 0x0 (DISABLE): Flow Control Busy or Backpressure Activate is disabled 0x1 (ENABLE): Flow Control Busy or Backpressure Activate is enabled Value After Reset: 0x0 Testable: untestable
			31:9	Reserved_31_9	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x90	MAC_Rx_Flow_Ctrl	The Receive Flow Control register controls the pausing of MAC Transmit based on the received Pause packet.	8	PFCE	R/W	0x0	Priority Based Flow Control Enable When this bit is set, it enables generation and reception of priority-based flow control (PFC) packets. When this bit is reset, it enables generation and reception of 802.3x Pause control packets.Values: 0x0 (DISABLE): Priority Based Flow Control is disabled 0x1 (ENABLE): Priority Based Flow Control is enabled Value After Reset: 0x0
			7:2	Reserved_7_2	R	0x0	Reserved. Value After Reset: 0x0
			1	UP	R/W	0x0	Unicast Pause Packet Detect A pause packet is processed when it has the unique multicast address specified in the IEEE 802.3. When this bit is set, the MAC can also detect Pause packets with unicast address of the station. This unicast address should be as specified in MAC_Address0_High and MAC_Address0_Low. When this bit is reset, the MAC only detects Pause packets with unique multicast address. Note: The MAC does not process a Pause packet if the multicast address is different from the unique multicast address. This is also applicable to the received PFC packet when the Priority Flow Control (PFC) is enabled. The unique multicast address (0x01_80_C2_00_00_01) is as specified in IEEE 802.1 Qbb-2011.Values: 0x0 (DISABLE): Unicast Pause Packet Detect disabled 0x1 (ENABLE): Unicast Pause Packet Detect enabled Value After Reset: 0x0
			0	RFE	R/W	0x0	Receive Flow Control Enable When this bit is set and the MAC is operating in full-duplex mode, the MAC decodes the received Pause packet and disables its transmitter for a specified (Pause) time. When this bit is reset or the MAC is operating in half-duplex mode, the decode function of the Pause packet is disabled. When PFC is enabled, flow control is enabled for PFC packets. The MAC decodes the received PFC packet and disables the Transmit queue, with matching priorities, for a duration of received Pause time.Values: 0x0 (DISABLE): Receive Flow Control is disabled 0x1 (ENABLE): Receive Flow Control is enabled Value After Reset: 0x0
			31	UDC	R/W	0x0	Unicast Packet Duplication Control When this bit is set along with PDC bit of the MAC_Ext_Configuration register, the MAC provides DMA Channel number specified as one-hot value in DCS field of MAC_Address(#i)_High register that matched the Unicast Destination Address of the received packet. Note: When the above condition is satisfied for unicast packet but the highest Receive Queue selected in configuration is not enabled or when DA/SA/VLAN filter fails and corresponding filter fail Receive Queue is enabled, it is routed to Receive Queue determined by other queuing methods. When this bit is reset, the MAC routes the received packet with Unicast Destination Address based on other queuing methods, the DMA Channel number selected is binary value from DCS field of MAC_Address(#i)_High register (if PDC bit is reset) or lowest bit set in one-hot value (if PDC is set) in DCS field of MAC_Address(#i)_High register that matched the Unicast Destination Address of the received packet.Values: 0x0 (DISABLE): Unicast packet duplication is disabled 0x1 (ENABLE): Unicast packet duplication is enabled Value After Reset: 0x0
			30:20	Reserved_30_y	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x94	MAC_RxQ_Ctrl4	The Receive Queue Control 4 register controls the routing of unicast and multicast packets that fail the Destination or Source address filter to the Rx queues.	19:17	VFFQ	R/W	0x0	VLAN Tag Filter Fail Packets Queue This field holds the Rx queue number to which the tagged packets failing the Destination or Source Address filter (and UFFQE/MFFQE not enabled) or failing the VLAN tag filter must be routed to. This field is valid only when the VFFQE bit is set. Value After Reset: 0x0
			16	VFFQE	R/W	0x0	VLAN Tag Filter Fail Packets Queuing Enable When this bit is set, the tagged packets which fail the Destination or Source address filter or fail the VLAN tag filter, are routed to the Rx Queue Number programmed in the VFFQ. When this bit is reset, the tagged packets which fail the Destination or Source address filter or fail the VLAN tag filter are routed based on other routing options. This bit is valid only when the RA bit of the MAC_Packet_Filter register is set.Values: 0x0 (DISABLE): VLAN tag Filter Fail Packets Queuing is disabled 0x1 (ENABLE): VLAN tag Filter Fail Packets Queuing is enabled Value After Reset: 0x0
			15:12	Reserved_15_y	R	0x0	Reserved. Value After Reset: 0x0
			11:9	MFFQ	R/W	0x0	Multicast Address Filter Fail Packets Queue. This field holds the Rx queue number to which the Multicast packets failing the Destination or Source Address filter are routed to. This field is valid only when the MFFQE bit is set. Value After Reset: 0x0
			8	MFFQE	R/W	0x0	Multicast Address Filter Fail Packets Queuing Enable. When this bit is set, the Multicast packets which fail the Destination or Source address filter is routed to the Rx Queue Number programmed in the MFFQ. When this bit is reset, the Multicast packets which fail the Destination or Source address filter is routed based on other routing options. This bit is valid only when the RA bit of the MAC_Packet_Filter register is set.Values: 0x0 (DISABLE): Multicast Address Filter Fail Packets Queuing is disabled 0x1 (ENABLE): Multicast Address Filter Fail Packets Queuing is enabled Value After Reset: 0x0
			7:4	Reserved_7_y	R	0x0	Reserved. Value After Reset: 0x0
			3:1	UFFQ	R/W	0x0	Unicast Address Filter Fail Packets Queue. This field holds the Rx queue number to which the Unicast packets failing the Destination or Source Address filter are routed to. This field is valid only when the UFFQE bit is set. Value After Reset: 0x0
			0	UFFQE	R/W	0x0	Unicast Address Filter Fail Packets Queuing Enable. When this bit is set, the Unicast packets which fail the Destination or Source address filter is routed to the Rx Queue Number programmed in the UFFQ. When this bit is reset, the Unicast packets which fail the Destination or Source address filter is routed based on other routing options. This bit is valid only when the RA bit of the MAC_Packet_Filter register is set.Values: 0x0 (DISABLE): Unicast Address Filter Fail Packets Queuing is disabled 0x1 (ENABLE): Unicast Address Filter Fail Packets Queuing is enabled Value After Reset: 0x0
			31:24	PSTQ3	R/W	0x0	Priorities Selected in Transmit Queue 3 This bit is similar to the PSTQ0 bit. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x98	MAC_TxQ_Prty_Map0	The Transmit Queue Priority Mapping 0 register contains the priority values assigned to Tx Queue 0 through Tx Queue 3.	23:16	PSTQ2	R/W	0x0	Priorities Selected in Transmit Queue 2 This bit is similar to the PSTQ0 bit. Value After Reset: 0x0
			15:8	PSTQ1	R/W	0x0	Priorities Selected in Transmit Queue 1 This bit is similar to the PSTQ0 bit. Value After Reset: 0x0
			7:0	PSTQ0	R/W	0x0	Priorities Selected in Transmit Queue 0 This field holds the priorities assigned to Tx Queue 0 by the software. This field determines if Tx Queue 0 should be blocked from transmitting specified pause time when a PFC packet is received with priorities matching the priorities programmed in this field. If the content of this field is not mutually exclusive to the corresponding fields of other Transmit queues, that is, same priority is mapped to multiple Tx queues, the MAC blocks all queues with matching priority, for the specified time. Value After Reset: 0x0
0x9c	MAC_TxQ_Prty_Map1	The Transmit Queue Priority Mapping 1 register contains the priority values assigned to Tx Queue 4 through Tx Queue 7.	31:24	PSTQ7	R/W	0x0	Priorities Selected in Transmit Queue 7 This bit is similar to the PSTQ4 bit. Value After Reset: 0x0
			23:16	PSTQ6	R/W	0x0	Priorities Selected in Transmit Queue 6 This bit is similar to the PSTQ4 bit. Value After Reset: 0x0
			15:8	PSTQ5	R/W	0x0	Priorities Selected in Transmit Queue 5 This bit is similar to the PSTQ4 bit. Value After Reset: 0x0
			7:0	PSTQ4	R/W	0x0	Priorities Selected in Transmit Queue 4 This field holds the priorities assigned to Tx Queue 4 by the software. This field determines if Tx Queue 4 should be blocked from transmitting specified pause time when a PFC packet is received with priorities matching the priorities programmed in this field. If the content of this field is not mutually exclusive to corresponding fields of other Transmit queues, that is, same priority is mapped to multiple Tx queues, the MAC blocks all queues with matching priority for specified time. Value After Reset: 0x0
			31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:14	RXQ7EN	R/W	0x0	Receive Queue 7 Enable This field is similar to the RXQ0EN field.Values: 0x0 (DISABLE): Queue not enabled 0x1 (EN_AV): Queue enabled for AV 0x2 (EN_DCB_GEN): Queue enabled for DCB/Generic 0x3 (RSVD): Reserved Value After Reset: 0x0
			13:12	RXQ6EN	R/W	0x0	Receive Queue 6 Enable This field is similar to the RXQ0EN field.Values: 0x0 (DISABLE): Queue not enabled 0x1 (EN_AV): Queue enabled for AV 0x2 (EN_DCB_GEN): Queue enabled for DCB/Generic 0x3 (RSVD): Reserved Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xa0	MAC_RxQ_Ctrl0	The Receive Queue Control 0 register controls the queue management in the MAC Receiver.Note: In multiple Rx queues configuration, all the queues are disabled by default. Enable the Rx queue by programming the corresponding field in this register.	11:10	RXQ5EN	R/W	0x0	Receive Queue 5 Enable This field is similar to the RXQ0EN field.Values: 0x0 (DISABLE): Queue not enabled 0x1 (EN_AV): Queue enabled for AV 0x2 (EN_DCB_GEN): Queue enabled for DCB/Generic 0x3 (RSVD): Reserved Value After Reset: 0x0
			9:8	RXQ4EN	R/W	0x0	Receive Queue 4 Enable This field is similar to the RXQ0EN field.Values: 0x0 (DISABLE): Queue not enabled 0x1 (EN_AV): Queue enabled for AV 0x2 (EN_DCB_GEN): Queue enabled for DCB/Generic 0x3 (RSVD): Reserved Value After Reset: 0x0
			7:6	RXQ3EN	R/W	0x0	Receive Queue 3 Enable This field is similar to the RXQ0EN field.Values: 0x0 (DISABLE): Queue not enabled 0x1 (EN_AV): Queue enabled for AV 0x2 (EN_DCB_GEN): Queue enabled for DCB/Generic 0x3 (RSVD): Reserved Value After Reset: 0x0
			5:4	RXQ2EN	R/W	0x0	Receive Queue 2 Enable This field is similar to the RXQ0EN field.Values: 0x0 (DISABLE): Queue not enabled 0x1 (EN_AV): Queue enabled for AV 0x2 (EN_DCB_GEN): Queue enabled for DCB/Generic 0x3 (RSVD): Reserved Value After Reset: 0x0
			3:2	RXQ1EN	R/W	0x0	Receive Queue 1 Enable This field is similar to the RXQ0EN field.Values: 0x0 (DISABLE): Queue not enabled 0x1 (EN_AV): Queue enabled for AV 0x2 (EN_DCB_GEN): Queue enabled for DCB/Generic 0x3 (RSVD): Reserved Value After Reset: 0x0
			1:0	RXQ0EN	R/W	0x0	Receive Queue 0 Enable This field indicates whether Rx Queue 0 is enabled for AV (2'b01) or DCB (2'b10) or Generic (2'b10).Values: 0x0 (DISABLE): Queue not enabled 0x1 (EN_AV): Queue enabled for AV 0x2 (EN_DCB_GEN): Queue enabled for DCB/Generic 0x3 (RSVD): Reserved Value After Reset: 0x0
			31:30	Reserved_31_30	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			29	TBRQE	R/W	0x0	Type Field Based Rx Queuing Enable When this bit is set, it enables Type Field based Rx Queuing where the Type field of received packet is compared with programmed TYP field in MAC_TMRQ_Regs(#i) and if a match occurs the packet is routed to the corresponding TMRQ field. Value After Reset: 0x0
			28	OMCBCQ	R/W	0x0	Over-riding MC-BC queue priority select 1: Priority of MCBCQ is reduced and the received packet is first routed to PTPQ, AVCPQ, DCBCPQ depending on packet type. 0: Received Multicast/Broadcast packet is routed to MCBCQ. Values: 0x0 (DISABLE): overriding MCBCQ priority disabled 0x1 (ENABLE): overriding MCBCQ priority enabled Value After Reset: 0x0
			27	Reserved_27	R	0x0	Reserved. Value After Reset: 0x0
			26:24	FPRQ	R/W	0x0	Frame Preemption Residue Queue This field holds the Rx queue number to which the residual preemption frames must be forwarded. Preemption frames that are tagged and pass the SA/DA/VLAN filtering are routed based on PSRQ and all other frames are treated as residual frames and is routed to the queue number mentioned in this field. The Queue-0 is used as a default queue for express frames, so this field cannot be programmed to a value 0. Value After Reset: 0x0
			23:22	TPQC	R/W	0x0	Tagged PTP over Ethernet Packets Queuing Control. This field controls the routing of the VLAN Tagged PTPoE packets. the following programmable options are allowed. 2'b00: VLAN Tagged PTPoE packets are routed as generic VLAN Tagged packet (based on PSRQ for only non-AV enabled Rx Queues). 2'b01: VLAN Tagged PTPoE packets are routed to Rx Queue specified by PTPQ field (That Rx Queue can be enabled for AV or non-AV traffic). 2'b10: VLAN Tagged PTPoE packets are routed to only AV enabled Rx Queues based on PSRQ. 2'b11: Reserved DCB/ Value After Reset: 0x0
			21	TACPQE	R/W	0x0	Tagged AV Control Packets Queuing Enable. When set, the MAC routes the received Tagged AV Control packets to the Rx queue specified by AVCPQ field. When reset, the MAC routes the received Tagged AV Control packets based on the tag priority matching the PSRQ fields in MAC_RxQ_Ctrl2 and MAC_RxQ_Ctrl3 register.Values: 0x0 (DISABLE): Tagged AV Control Packets Queuing is disabled 0x1 (ENABLE): Tagged AV Control Packets Queuing is enabled Value After Reset: 0x0
			20	MCBCQEN	R/W	0x0	Multicast and Broadcast Queue Enable This bit specifies that Multicast or Broadcast packets routing to the Rx Queue is enabled and the Multicast or Broadcast packets must be routed to Rx Queue specified in MCBCQ field.Values: 0x0 (DISABLE): Multicast and Broadcast Queue is disabled 0x1 (ENABLE): Multicast and Broadcast Queue is enabled Value After Reset: 0x0
			19	Reserved_19	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xa4	MAC_RxQ_Ctrl1	The Receive Queue Control 1 register controls the routing of multicast, broadcast, AV, DCB, and untagged packets to the Rx queues.	18:16	MCBCQ	R/W	0x0	Multicast and Broadcast Queue This field specifies the Rx Queue onto which Multicast or Broadcast Packets are routed. Any Rx Queue enabled for Generic/DCB/AV traffic can be used to route the Multicast or Broadcast Packets.Values: 0x0 (QUEUE0): Receive Queue 0 0x1 (QUEUE1): Receive Queue 1 0x2 (QUEUE2): Receive Queue 2 0x3 (QUEUE3): Receive Queue 3 0x4 (QUEUE4): Receive Queue 4 0x5 (QUEUE5): Receive Queue 5 0x6 (QUEUE6): Receive Queue 6 0x7 (QUEUE7): Receive Queue 7 Value After Reset: 0x0
			15	Reserved_15	R	0x0	Reserved. Value After Reset: 0x0
			14:12	UPQ	R/W	0x0	Untagged Packet Queue This field indicates the Rx Queue to which Untagged Packets are to be routed. Any Rx Queue enabled for Generic/DCB/AV traffic can be used to route the Untagged Packets.Values: 0x0 (QUEUE0): Receive Queue 0 0x1 (QUEUE1): Receive Queue 1 0x2 (QUEUE2): Receive Queue 2 0x3 (QUEUE3): Receive Queue 3 0x4 (QUEUE4): Receive Queue 4 0x5 (QUEUE5): Receive Queue 5 0x6 (QUEUE6): Receive Queue 6 0x7 (QUEUE7): Receive Queue 7 Value After Reset: 0x0
			11	Reserved_11	R	0x0	Reserved. Value After Reset: 0x0
			10:8	DCBCPQ	R/W	0x0	DCB Control Packets Queue This field specifies the Rx queue on which the received DCB control packets are routed. The DCB data packets are routed based on the PSRQ field of the Transmit Flow Control Register of corresponding queue.Values: 0x0 (QUEUE0): Receive Queue 0 0x1 (QUEUE1): Receive Queue 1 0x2 (QUEUE2): Receive Queue 2 0x3 (QUEUE3): Receive Queue 3 0x4 (QUEUE4): Receive Queue 4 0x5 (QUEUE5): Receive Queue 5 0x6 (QUEUE6): Receive Queue 6 0x7 (QUEUE7): Receive Queue 7 Value After Reset: 0x0
			7	Reserved_7	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			6:4	PTPQ	R/W	0x0	PTP Packets Queue This field specifies the Rx queue on which the PTP packets sent over the Ethernet payload (not over IPv4 or IPv6) are routed. When the AV8021ASMEN bit of MAC_Timestamp_Control register is set, only untagged PTP over Ethernet packets are routed on an Rx Queue. If the bit is not set, then based on programming of TPQC field, both tagged and untagged PTPoE packets can be routed to this Rx Queue.Values: 0x0 (QUEUE0): Receive Queue 0 0x1 (QUEUE1): Receive Queue 1 0x2 (QUEUE2): Receive Queue 2 0x3 (QUEUE3): Receive Queue 3 0x4 (QUEUE4): Receive Queue 4 0x5 (QUEUE5): Receive Queue 5 0x6 (QUEUE6): Receive Queue 6 0x7 (QUEUE7): Receive Queue 7 Value After Reset: 0x0
			3	Reserved_3	R	0x0	Reserved. Value After Reset: 0x0
			2:0	AVCPQ	R/W	0x0	AV Untagged Control Packets Queue This field specifies the Receive queue on which the received AV tagged and untagged control packets are routed. The AV tagged (when TACPQE bit is set) and untagged control packets are routed to Receive queue specified by this field.Values: 0x0 (QUEUE0): Receive Queue 0 0x1 (QUEUE1): Receive Queue 1 0x2 (QUEUE2): Receive Queue 2 0x3 (QUEUE3): Receive Queue 3 0x4 (QUEUE4): Receive Queue 4 0x5 (QUEUE5): Receive Queue 5 0x6 (QUEUE6): Receive Queue 6 0x7 (QUEUE7): Receive Queue 7 Value After Reset: 0x0
			31:24	PSRQ3	R/W	0x0	Priorities Selected in the Receive Queue 3 This field decides the priorities assigned to Rx Queue 3. Packets whose priorities match the values set in this field are routed to Rx Queue 3. For example, if PSRQ3[6, 3] have value 1, the packets whose USP field of the VLAN Tag has value 3 or 6 are routed to Rx Queue 3. Application must ensure that values in the PSRQ fields of all the Rx queues within a set that you enable generically or for AV or DCB features, are mutually exclusive. This is because multiple Rx queues cannot have the same priority within a set. this field also determines the priorities to be included in the PFC packet sent to the remote station when Rx Queue 3 crosses the flow control threshold settings. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xa8	MAC_RxQ_Ctrl2	This register controls the routing of tagged packets based on the USP (user Priority) field of the received packets to the RxQueues 0 to 3.	23:16	PSRQ2	R/W	0x0	Priorities Selected in the Receive Queue 2 This field decides the priorities assigned to Rx Queue 2. All packets with priorities that match the values set in this field are routed to Rx Queue 2. For example, if PSRQ2[1, 0] are set, packets with USP field equal to 1 or 0 are routed to Rx Queue 2. The software must ensure that the content of this field is mutually exclusive to the PSRQ fields for other queues, that is, the same priority is not mapped to multiple Rx queues. this field also determines the priorities to be included in the PFC packet sent to remote station when Rx Queue 2 crosses the flow control threshold settings. Value After Reset: 0x0
			15:8	PSRQ1	R/W	0x0	Priorities Selected in the Receive Queue 1 This field decides the priorities assigned to Rx Queue 1. All packets with priorities that match the values set in this field are routed to Rx Queue 1. For example, if PSRQ1[4] is set, packets with USP field equal to 4 are routed to Rx Queue 1. The software must ensure that the content of this field is mutually exclusive to the PSRQ fields for other queues, that is, the same priority is not mapped to multiple Rx queues. this field also determines the priorities to be included in the PFC packet sent to remote station when Rx Queue 1 crosses the flow control threshold settings. Value After Reset: 0x0
			7:0	PSRQ0	R/W	0x0	Priorities Selected in the Receive Queue 0 This field decides the priorities assigned to Rx Queue 0. All packets with priorities that match the values set in this field are routed to Rx Queue 0. For example, if PSRQ0[5] is set, packets with USP field equal to 5 are routed to Rx Queue 0. The software must ensure that the content of this field is mutually exclusive to the PSRQ fields for other queues, that is, the same priority is not mapped to multiple Rx queues. this field also determines the priorities to be included in the PFC packet sent to remote station when Rx Queue 0 crosses the flow control threshold settings. Value After Reset: 0x0
0xac	MAC_RxQ_Ctrl3	This register controls the routing of tagged packets based on the USP (user Priority) field of the received packets to the RxQueues 4 to 7.	31:24	PSRQ7	R/W	0x0	Priorities Selected in the Receive Queue 7 This field decides the priorities assigned to Rx Queue 7. All packets with priorities that match the values set in this field are routed to Rx Queue 7. For example, if PSRQ7[7, 4] are set, packets with USP field equal to 7 or 4 are routed to Rx Queue 7. The software must ensure that the content of this field is mutually exclusive to the PSRQ fields for other queues, that is, the same priority is not mapped to multiple Rx queues. this field also determines the priorities to be included in the PFC packet sent to remote station when Rx Queue 7 crosses the flow control threshold settings. Value After Reset: 0x0
			23:16	PSRQ6	R/W	0x0	Priorities Selected in the Receive Queue 6 This field decides the priorities assigned to Rx Queue 6. All packets with priorities that match the values set in this field are routed to Rx Queue 6. For example, if PSRQ6[5] are set, packets with USP field equal to 5 are routed to Rx Queue 6. The software must ensure that the content of this field is mutually exclusive to the PSRQ fields for other queues, that is, the same priority is not mapped to multiple Rx queues. this field also determines the priorities to be included in the PFC packet sent to remote station when Rx Queue 6 crosses the flow control threshold settings. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			15:8	PSRQ5	R/W	0x0	Priorities Selected in the Receive Queue 5 This field decides the priorities assigned to Rx Queue 5. All packets with priorities that match the values set in this field are routed to Rx Queue 5. For example, if PSRQ5[6] is set, packets with USP field equal to 6 are routed to Rx Queue 5. The software must ensure that the content of this field is mutually exclusive to the PSRQ fields for other queues, that is, the same priority is not mapped to multiple Rx queues. this field also determines the priorities to be included in the PFC packet sent to remote station when Rx Queue 5 crosses the flow control threshold settings. Value After Reset: 0x0
			7:0	PSRQ4	R/W	0x0	Priorities Selected in the Receive Queue 4 This field decides the priorities assigned to Rx Queue 4. All packets with priorities that match the values set in this field are routed to Rx Queue 4. For example, if PSRQ4[7:4] is set, packets with USP field equal to 7, 6, 5, or 4 are routed to Rx Queue 4. The software must ensure that the content of this field is mutually exclusive to the PSRQ fields for other queues, that is, the same priority is not mapped to multiple Rx queues. this field also determines the priorities to be included in the PFC packet sent to remote station when Rx Queue 4 crosses the flow control threshold settings. Value After Reset: 0x0
			31:24	Reserved_31_24	R	0x0	Reserved. Value After Reset: 0x0
			23			#VALUE!	Reserved Field: Yes
			22:21	Reserved_22_21	R	0x0	Reserved. Value After Reset: 0x0
			20	MFRIS	R	0x0	MMC FPE Receive Interrupt Status 1: Indicates to the application that one or more interrupt status bits are 1 in the MMC_FPE_Rx_Interrupt register. MAC writes 0 to this bit when all the interrupt status bits in the MMC_FPE_Rx_Interrupt register are 0. Values: 0x0 (INACTIVE): MMC FPE Receive Interrupt status not active 0x1 (ACTIVE): MMC FPE Receive Interrupt status active Value After Reset: 0x0
			19	MFTIS	R	0x0	MMC FPE Transmit Interrupt Status This bit is set high when an interrupt is generated in the MMC FPE Transmit Interrupt Register. This bit is cleared when all bits in this interrupt register are cleared. This bit is valid only when you select the Enable MAC Management Counters (MMC) option along with FPE support.Values: 0x0 (INACTIVE): MMC FPE Transmit Interrupt status not active 0x1 (ACTIVE): MMC FPE Transmit Interrupt status active Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			18	MDIOIS	R	0x0	MDIO Interrupt Status 1: Indicates to the application that MDIO operation is complete. MAC writes 0 to this bit when the application reads this bit. Application can write 1 to clear this bit when RCWE = 1, in the MAC_CSR_SW_Ctrl register. Following access restrictions apply. Clears on read Clears on write 1, when RCWE = 1 in the MAC_CSR_SW_Ctrl register. Self-set to 1 on internal event. Values: 0x0 (INACTIVE): MDIO Interrupt status not active 0x1 (ACTIVE): MDIO Interrupt status active Value After Reset: 0x0
			17	FPEIS	R	0x0	Frame Preemption Interrupt Status This bit indicates an interrupt event during the operation of Frame Preemption (Bits[19:16] of MAC_FPE_CTRL_STS register is set). To reset this bit, the application must clear the event in MAC_FPE_CTRL_STS that has caused the Interrupt.Values: 0x0 (INACTIVE): Frame Preemption Interrupt status not active 0x1 (ACTIVE): Frame Preemption Interrupt status active Value After Reset: 0x0
			16	Reserved_16	R	0x0	Reserved. Value After Reset: 0x0
			15			#VALUE!	Reserved Field: Yes
			14	RXSTSIS	R	0x0	Receive Status Interrupt This bit indicates the status of received packets. This bit is set when the RWT bit is set in the MAC_Rx_Tx_Status register. This bit is cleared when the corresponding interrupt source bit is read (or corresponding interrupt source bit is written to 1 when RCWE bit of MAC_CSR_SW_Ctrl register is set) in the MAC_Rx_Tx_Status register.Values: 0x0 (INACTIVE): Receive Interrupt status not active 0x1 (ACTIVE): Receive Interrupt status active Value After Reset: 0x0
			13	TXSTSIS	R	0x0	Transmit Status Interrupt This bit indicates the status of transmitted packets. This bit is set when any of the following bits is set in the MAC_Rx_Tx_Status register: Excessive Collision (EXCOL) Late Collision (LCOL) Excessive Deferral (EXDEF) Loss of Carrier (LCARR) No Carrier (NCARR) Jabber Timeout (TJT) This bit is cleared when the corresponding interrupt source bit is read (or corresponding interrupt source bit is written to 1 when RCWE bit of MAC_CSR_SW_Ctrl register is set) in the MAC_Rx_Tx_Status register.Values: 0x0 (INACTIVE): Transmit Interrupt status not active 0x1 (ACTIVE): Transmit Interrupt status active Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xb0	MAC_Interrupt_Status	The Interrupt Status register contains the status of interrupts.	12	TSIS	R	0x0	Timestamp Interrupt Status this bit is 1, when any of the following conditions is true: The system time value is equal to or exceeds the value you programmed in the MAC_PPS(#i)_Target_Time_Seconds and MAC_PPS(#i)_Target_Time_Nanoseconds. An overflow in the MAC_System_Time_Seconds register. Timestamp Target Time Error occurred. That is, the time you programmed in the MAC_PPS(#i)_Target_Time_Seconds and MAC_PPS(#i)_Target_Time_Nanoseconds has elapsed. When the auxiliary snapshot trigger is asserted. when the MAC updates the captured transmit timestamp in the MAC_Tx_Timestamp_Status_Nanoseconds and Mac_TxTimestamp_Status_Seconds registers. when the MAC updates the captured transmit timestamp in the MAC_Tx_Timestamp_Status_Nanoseconds and MAC_Tx_Timestamp_Status_Seconds registers, for the PTO generated Delay Request and Pdelay request packets. MAC writes 0 to this bit when Application reads the corresponding interrupt source bit in the MAC_Timestamp_Status register. Application writes 1 to the corresponding interrupt source bit in the MAC_Timestamp_Status register, when RCWE = 1 in the MAC_CSR_SW_Ctrl register. Values: 0x0 (INACTIVE): Timestamp Interrupt status not active 0x1 (ACTIVE): Timestamp Interrupt status active Value After Reset: 0x0
			11	MMCRXIPIS	R	0x0	MMC Receive Checksum Offload Interrupt Status This bit is set high when an interrupt is generated in the MMC Receive Checksum Offload Interrupt Register. This bit is cleared when all bits in this interrupt register are cleared.Values: 0x0 (INACTIVE): MMC Receive Checksum Offload Interrupt status not active 0x1 (ACTIVE): MMC Receive Checksum Offload Interrupt status active Value After Reset: 0x0
			10	MMCTXIS	R	0x0	MMC Transmit Interrupt Status This bit is set high when an interrupt is generated in the MMC Transmit Interrupt Register. This bit is cleared when all bits in this interrupt register are cleared.Values: 0x0 (INACTIVE): MMC Transmit Interrupt status not active 0x1 (ACTIVE): MMC Transmit Interrupt status active Value After Reset: 0x0
			9	MMCRXIS	R	0x0	MMC Receive Interrupt Status This bit is set high when an interrupt is generated in the MMC Receive Interrupt Register. This bit is cleared when all bits in this interrupt register are cleared.Values: 0x0 (INACTIVE): MMC Receive Interrupt status not active 0x1 (ACTIVE): MMC Receive Interrupt status active Value After Reset: 0x0
			8	MMCIS	R	0x0	MMC Interrupt Status This bit is set high when Bit 11, Bit 10, or Bit 9 is set high. This bit is cleared only when all these bits are low.Values: 0x0 (INACTIVE): MMC Interrupt status not active 0x1 (ACTIVE): MMC Interrupt status active Value After Reset: 0x0
			7:6	Reserved_7_6	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			5	LPIIS	R	0x0	LPI Interrupt Status When the Energy Efficient Ethernet feature is enabled, this bit is set for any LPI state entry or exit in the MAC Transmitter or Receiver. This bit is cleared when the corresponding interrupt source bit of MAC_LPI_Control_Status register is read (or corresponding interrupt source bit of MAC_LPI_Control_Status register is written to 1 when RCWE bit of MAC_CSR_SW_Ctrl register is set).Values: 0x0 (INACTIVE): LPI Interrupt status not active 0x1 (ACTIVE): LPI Interrupt status active Value After Reset: 0x0
			4	PMTIS	R	0x0	PMT Interrupt Status This bit is set when a Magic packet or Wake-on-LAN packet is received in the power-down mode (RWKPRCVD and MGKPRCVD bits in MAC_PMT_Control_Status register). This bit is cleared when corresponding interrupt source bit are cleared because of a Read operation to the MAC_PMT_Control_Status register (or corresponding interrupt source bit of MAC_PMT_Control_Status register is written to 1 when RCWE bit of MAC_CSR_SW_Ctrl register is set). This bit is valid only when you select the Enable Power Management option.Values: 0x0 (INACTIVE): PMT Interrupt status not active 0x1 (ACTIVE): PMT Interrupt status active Value After Reset: 0x0
			3	PHYIS	R	0x0	PHY Interrupt This bit is set when rising edge is detected on the phy_intr_i input. This bit is cleared when this register is read (or this bit is written to 1 when RCWE bit of MAC_CSR_SW_Ctrl register is set).Values: 0x0 (INACTIVE): PHY Interrupt not detected 0x1 (ACTIVE): PHY Interrupt detected Value After Reset: 0x0
			2:1			#VALUE!	Reserved Field: Yes
			0	RGSMIIIS	R	0x0	RGMII or SMII Interrupt Status This bit is set because of any change in value of the Link Status of RGMII or SMII interface (LNKSTS bit in MAC_PHYIF_Control_Status register). This bit is cleared when the MAC_PHYIF_Control_Status register is read (or LNKSTS bit of MAC_PHYIF_Control_Status register is written to 1 when RCWE bit of MAC_CSR_SW_Ctrl register is set). This bit is valid only when you select the optional RGMII or SMII PHY interface.Values: 0x0 (INACTIVE): RGMII or SMII Interrupt Status is not active 0x1 (ACTIVE): RGMII or SMII Interrupt Status is active Value After Reset: 0x0 Testable: untestable
			31:19	Reserved_31_19	R	0x0	Reserved. Value After Reset: 0x0
			18	MDIOIE	R/W	0x0	MDIO Interrupt Enable When this bit is set, it enables the assertion of the interrupt when MDIOIS field is set in the MAC_Interrupt_Status register.Values: 0x0 (DISABLE): MDIO Interrupt is disabled 0x1 (ENABLE): MDIO Interrupt is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xb4	MAC_Interrupt_Enable	The Interrupt Enable register contains the masks for generating the interrupts.	17	FPEIE	R/W	0x0	Frame Preemption Interrupt Enable When this bit is set, it enables the assertion of the interrupt when FPEIS field is set in the MAC_Interrupt_Status register.Values: 0x0 (DISABLE): Frame Preemption Interrupt is disabled 0x1 (ENABLE): Frame Preemption Interrupt is enabled Value After Reset: 0x0
			16	Reserved_16	R	0x0	Reserved. Value After Reset: 0x0
			15	Reserved_15	R	0x0	Reserved. Value After Reset: 0x0
			14	RXSTSIE	R/W	0x0	Receive Status Interrupt Enable When this bit is set, it enables the assertion of the interrupt signal because of the setting of RXSTSIS bit in the MAC_Interrupt_Status register.Values: 0x0 (DISABLE): Receive Status Interrupt is disabled 0x1 (ENABLE): Receive Status Interrupt is enabled Value After Reset: 0x0
			13	TXSTSIE	R/W	0x0	Transmit Status Interrupt Enable When this bit is set, it enables the assertion of the interrupt signal because of the setting of TXSTSIS bit in the MAC_Interrupt_Status register.Values: 0x0 (DISABLE): Timestamp Status Interrupt is disabled 0x1 (ENABLE): Timestamp Status Interrupt is enabled Value After Reset: 0x0
			12	TSIE	R/W	0x0	Timestamp Interrupt Enable When this bit is set, it enables the assertion of the interrupt signal because of the setting of TSIS bit in MAC_Interrupt_Status register.Values: 0x0 (DISABLE): Timestamp Interrupt is disabled 0x1 (ENABLE): Timestamp Interrupt is enabled Value After Reset: 0x0
			11:6	Reserved_11_6	R	0x0	Reserved. Value After Reset: 0x0
			5	LPIIE	R/W	0x0	LPI Interrupt Enable When this bit is set, it enables the assertion of the interrupt signal because of the setting of LPIIS bit in MAC_Interrupt_Status register.Values: 0x0 (DISABLE): LPI Interrupt is disabled 0x1 (ENABLE): LPI Interrupt is enabled Value After Reset: 0x0
			4	PMTIE	R/W	0x0	PMT Interrupt Enable When this bit is set, it enables the assertion of the interrupt signal because of the setting of PMTIS bit in MAC_Interrupt_Status register.Values: 0x0 (DISABLE): PMT Interrupt is disabled 0x1 (ENABLE): PMT Interrupt is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			3	PHYIE	R/W	0x0	PHY Interrupt Enable When this bit is set, it enables the assertion of the interrupt signal because of the setting of PHYIS bit in MAC_Interrupt_Status register.Values: 0x0 (DISABLE): PHY Interrupt is disabled 0x1 (ENABLE): PHY Interrupt is enabled Value After Reset: 0x0
			2:1			#VALUE!	Reserved Field: Yes
			0	RGSMIIIE	R/W	0x0	RGMII or SMII Interrupt Enable When this bit is set, it enables the assertion of the interrupt signal because of the setting of RGSMIIIS bit in MAC_Interrupt_Status register.Values: 0x0 (DISABLE): RGMII or SMII Interrupt is disabled 0x1 (ENABLE): RGMII or SMII Interrupt is enabled Value After Reset: 0x0
0xb8	MAC_Rx_Tx_Status	The Receive Transmit Status register contains the Receive and Transmit Error status.	31:9	Reserved_31_9	R	0x0	Reserved. Value After Reset: 0x0
			8	RWT	R	0x0	Receive Watchdog Timeout This bit is set when a packet with length greater than 2,048 bytes is received (10, 240 bytes when Jumbo Packet mode is enabled) and the WD bit is reset in the MAC_Configuration register. This bit is set when a packet with length greater than 16,383 bytes is received and the WD bit is set in the MAC_Configuration register. Access restriction applies. Clears on read (or write of 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event.Values: 0x0 (INACTIVE): No receive watchdog timeout 0x1 (ACTIVE): Receive watchdog timed out Value After Reset: 0x0
			7:6	Reserved_7_6	R	0x0	Reserved. Value After Reset: 0x0
			5:1			#VALUE!	Reserved Field: Yes
			0	TJT	R	0x0	Transmit Jabber Timeout This bit indicates that the Transmit Jabber Timer expired which happens when the packet size exceeds 2,048 bytes (10,240 bytes when the Jumbo packet is enabled) and JD bit is reset in the MAC_Configuration register. This bit is set when the packet size exceeds 16,383 bytes and the JD bit is set in the MAC_Configuration register. Access restriction applies. Clears on read (or write of 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event.Values: 0x0 (INACTIVE): No Transmit Jabber Timeout 0x1 (ACTIVE): Transmit Jabber Timeout occurred Value After Reset: 0x0
			31	RWKFILTRST	R/W	0x0	Remote Wake-Up Packet Filter Register Pointer Reset When this bit is set, the remote wake-up packet filter register pointer is reset to 3'b000. It is automatically cleared after 1 clock cycle. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect.Values: 0x0 (DISABLE): Remote Wake-Up Packet Filter Register Pointer is not Reset 0x1 (ENABLE): Remote Wake-Up Packet Filter Register Pointer is Reset Value After Reset: 0x0 Testable: untestable
			30:29	Reserved_30_29	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xc0	MAC_PMT_Control_Status	The PMT Control and Status Register.	28:24	RWKPTR	R	0x0	Remote Wake-up FIFO Pointer This field gives the current value (0 to 7, 15, or 31 when 4, 8, or 16 Remote Wake-up Packet Filters are selected) of the Remote Wake-up Packet Filter register pointer. When the value of this pointer is equal to maximum for the selected number of Remote Wake-up Packet Filters, the contents of the Remote Wake-up Packet Filter Register are transferred to the clk_rx_i domain when a Write occurs to that register. Value After Reset: 0x0 Testable: untestable
			23:11	Reserved_23_11	R	0x0	Reserved. Value After Reset: 0x0
			10	RWKPFE	R/W	0x0	Remote Wake-up Packet Forwarding Enable When this bit is set along with RWKPKTEN, the MAC receiver drops all received frames until it receives the expected Wake-up frame. All frames after that event including the received wake-up frame are forwarded to application. This bit is then self-cleared on receiving the wake-up packet. The application can also clear this bit before the expected wake-up frame is received. In such cases, the MAC reverts to the default behavior where packets received are forwarded to the application. This bit must only be set when RWKPKTEN is set high and PWRDWN is set low. The setting of this bit has no effect when PWRDWN is set high. Note: If Magic Packet Enable and Wake-Up Frame Enable are both set along with setting of this bit and Magic Packet is received prior to wake-up frame, this bit is self-cleared on receiving Magic Packet, the received Magic packet is dropped, and all frames after received Magic Packet are forwarded to application. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect.Values: 0x0 (DISABLE): Remote Wake-up Packet Forwarding is disabled 0x1 (ENABLE): Remote Wake-up Packet Forwarding is enabled Value After Reset: 0x0
			9	GLBLUCAST	R/W	0x0	Global Unicast When this bit set, any unicast packet filtered by the MAC (DAF) address recognition is detected as a remote wake-up packet.Values: 0x0 (DISABLE): Global unicast is disabled 0x1 (ENABLE): Global unicast is enabled Value After Reset: 0x0
			8:7	Reserved_8_7	R	0x0	Reserved. Value After Reset: 0x0
			6	RWKPRCVD	R	0x0	Remote Wake-Up Packet Received When this bit is set, it indicates that the power management event is generated because of the reception of a remote wake-up packet. This bit is cleared when this register is read. Access restriction applies. Clears on read (or write of 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Remote wake-up packet is received 0x1 (ACTIVE): Remote wake-up packet is received Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			5	MGKPRCVD	R	0x0	Magic Packet Received When this bit is set, it indicates that the power management event is generated because of the reception of a magic packet. This bit is cleared when this register is read. Access restriction applies. Clears on read (or write of 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event.Values: 0x0 (INACTIVE): No Magic packet is received 0x1 (ACTIVE): Magic packet is received Value After Reset: 0x0
			4:3	Reserved_4_3	R	0x0	Reserved. Value After Reset: 0x0
			2	RWKPKTEN	R/W	0x0	Remote Wake-Up Packet Enable When this bit is set, a power management event is generated when the MAC receives a remote wake-up packet.Values: 0x0 (DISABLE): Remote wake-up packet is disabled 0x1 (ENABLE): Remote wake-up packet is enabled Value After Reset: 0x0
			1	MGKPKTEN	R/W	0x0	Magic Packet Enable When this bit is set, a power management event is generated when the MAC receives a magic packet.Values: 0x0 (DISABLE): Magic Packet is disabled 0x1 (ENABLE): Magic Packet is enabled Value After Reset: 0x0
			0	PWRDWN	R/W	0x0	Power Down When this bit is set, the MAC receiver drops all received packets until it receives the expected magic packet or remote wake-up packet. This bit is then self-cleared and the power-down mode is disabled. The software can clear this bit before the expected magic packet or remote wake-up packet is received. The packets received by the MAC after this bit is cleared are forwarded to the application. This bit must only be set when the Magic Packet Enable, Global Unicast, or Remote Wake-Up Packet Enable bit is set high. Note: You can gate-off the CSR clock during the power-down mode. However, when the CSR clock is gated-off, you cannot perform any read or write operations on this register. Therefore, the software cannot clear this bit. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect.Values: 0x0 (DISABLE): Power down is disabled 0x1 (ENABLE): Power down is enabled Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xc4	MAC_RWK_Packet_Filter		31:0	WKUPFRMFTR	R/W	0x0	The Remote Wakeup Filter registers are implemented as 8 indirect access registers (wkuppktfilter_reg#i) and accessed by application through MAC_RWK_Packet_Filter register. When the Remote Wakeup Filters are to be programmed, the entire set of wkuppktfilter_reg registers must be written. The wkuppktfilter_reg register is programmed by sequentially writing the eight register values in MAC_RWK_Packet_Filter register for wkuppktfilter_reg0, wkuppktfilter_reg1, ..., wkuppktfilter_reg7 respectively. The wkuppktfilter_reg register is read in a similar way. The MAC updates the wkuppktfilter_reg register current pointer value in RWKPTR field of MAC_PMT_Control_Status register. The fields of Remote Wakeup Filter are described as follows: --Filter i Byte Mask: The filter i byte mask register defines the bytes of the packet that are examined by filter i (0, 1, 2, 3) to determine whether or not a packet is a wake-up packet. ----The MSB (31st bit) must be zero. ----Bit j[30:0] is the byte mask. ----If Bit j (byte number) of the byte mask is set, the CRC block processes the Filter i Offset + j of the incoming packet; otherwise Filter i Offset + j is ignored. --Filter i Command: The 4-bit filter i command controls the filter i operation. ----Bit 3 specifies the address type, defining the destination address type of the pattern. When the bit is set, the pattern applies to only multicast packets; when the bit is reset, the pattern applies only to unicast packet. ----Bit 2 (Inverse Mode), when set, reverses the logic of the CRC16 hash function signal, to reject a packet with matching CRC-16 value. Bit 2, along with Bit 1, allows a MAC to reject a subset of remote wake-up packets by creating filter logic such as "Pattern 1 AND NOT Pattern 2". ----Bit 1 (And_Previous) implements the Boolean logic. When set, the result of the current entry is logically ANDed with the result of the previous filter. This AND logic allows a filter pattern longer than 32 bytes by splitting the mask among two, three, or four filters. This depends on the number of filters that have the And_Previous bit set. The details are, The And_Previous bit setting is applicable within a set of 4 filters.Setting of And_Previous bit of filter that is not enabled has no effect, that is setting And_Previous bit of lowest number filter in the set of 4 filters has no effect. For example, setting of And_Previous bit of Filter 0 has no effect.If And_Previous bit is set for filter to form AND chained filter, the AND chain breaks at the point any filter is not enabled. For example: If Filter 2 And_Previous bit is set (bit 1 in Filter 2 command is set) but Filter 1 is not enabled (bit 0 in Filter 1 command is reset), then only Filter 2 result is

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
							considered. If Filter 2 And_Previous bit is set (bit 1 in Filter 2 command is set), Filter 3 And_Previous bit is set (bit 1 in Filter 3 command is set), but Filter 1 is not enabled (bit 0 in Filter 1 command is reset), then only Filter 2 result ANDed with Filter 3 result is considered. If Filter 2 And_Previous bit is set (bit 1 in Filter 2 command is set), Filter 3 And_Previous bit is set (bit 1 in Filter 3 command is set), but Filter 2 is not enabled (bit 0 in Filter 2 command is reset), then since setting of Filter 2 And_Previous bit has no effect only Filter 1 result ORed with Filter 3 result is considered.If filters chained by And_Previous bit setting have complementary programming, then a frame might never pass the AND chained filter. For example, if Filter 2 And_Previous bit is set (bit 1 in Filter 2 command is set), Filter 1 Address_Type bit is set (bit 3 in Filter 1 command is set) indicating multicast detection and Filter 2 Address_Type bit is reset (bit 3 in Filter 2 command is reset) indicating unicast detection or vice versa, a remote wakeup frame does not pass the AND chained filter as a remote wakeup frame cannot be of both unicast and multicast address type. ----Bit 0 is the enable for filter i. If Bit 0 is not set, filter i is disabled. --Filter i Offset: The filter i offset register defines the offset (within the packet) from which the filter i examines the packets.This 8-bit pattern-offset is the offset for the filter i first byte to be examined.The minimum allowed offset is 12, which refers to the 13th byte of the packet.The offset value 0 refers to the first byte of the packet. --Filter i CRC-16: The filter i CRC-16 register contains the CRC-16 value calculated from the pattern and the byte mask programmed in the Remote Wakeup filter register.The 16-bit CRC calculation uses the following polynomial: G(x) = x^16 + x^15 + x^2 + 1Each mask, used in the hash function calculation, is compared with a 16-bit value associated with that mask. Each filter has the following: 32-bit Mask: Each bit in this mask corresponds to one byte in the detected packet. If the bit is 1, the corresponding byte is taken into the CRC-16 calculation.8-bit Offset Pointer: Specifies the byte to start the CRC-16 computation. The pointer and the mask are used together to locate the bytes to be used in the CRC-16 calculations. RWK Packet Filter This field contains the various controls of RWK Packet filter. Value After Reset: 0x0 Testable: untestable
0xc4	RWK_Filter(#i)(#j)_CRC(fori=0;i<=1)	RWK Filter CRC-16This register contains the CRC-16 to be matched.	31:16	Filter1_CRC	R/W	0x0	Filter1 CRC-16 This filter CRC-16 contains the CRC_16 value of the pattern. The 16-bit CRC calculation uses the following polynomial: G(x) = x^16 + x^15 + x^2 + 1 Value After Reset: 0x0
			15:0	Filter0_CRC	R/W	0x0	Filter0 CRC-16 This filter CRC-16 contains the CRC_16 value of the pattern. The 16-bit CRC calculation uses the following polynomial: G(x) = x^16 + x^15 + x^2 + 1 Value After Reset: 0x0
			31:22	Reserved_31_22	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			21	LPITCSE	R/W	0x0	LPI Tx Clock Stop Enable When this bit is set, the MAC asserts sbd_tx_clk_gating_ctrl_o signal high after it enters Tx LPI mode to indicate that the Tx clock to MAC can be stopped. When this bit is reset, the MAC does not assert sbd_tx_clk_gating_ctrl_o signal high after it enters Tx LPI mode. If RGMII Interface is selected, the Tx clock is required for transmitting the LPI pattern. The Tx Clock cannot be gated and so the LPITCSE bit cannot be programmed.Values: 0x0 (DISABLE): LPI Tx Clock Stop is disabled 0x1 (ENABLE): LPI Tx Clock Stop is enabled Value After Reset: 0x0
			20	LPIATE	R/W	0x0	LPI Timer Enable This bit controls the automatic entry of the MAC Transmitter into and exit out of the LPI state. When LPIATE, LPITXA and LPIEN bits are set, the MAC Transmitter enters LPI state only when the complete MAC TX data path is IDLE for a period indicated by the MAC_LPI_Entry_Timer register. After entering LPI state, if the data path becomes non-IDLE (due to a new packet being accepted for transmission), the Transmitter exits LPI state but does not clear LPIEN bit. This enables the re-entry into LPI state when it is IDLE again. When LPIATE is 0, the LPI Auto timer is disabled and MAC Transmitter enters LPI state based on the settings of LPITXA and LPIEN bit descriptions.Values: 0x0 (DISABLE): LPI Timer is disabled 0x1 (ENABLE): LPI Timer is enabled Value After Reset: 0x0
			19	LPITXA	R/W	0x0	LPI Tx Automate This bit controls the behavior of the MAC when it is entering or coming out of the LPI mode on the Transmit side. This bit is not functional in the EQOS-CORE configurations in which the Tx clock gating is done during the LPI mode. If the LPITXA and LPIEN bits are set to 1, the MAC enters the LPI mode only after all outstanding packets (in the HW) and pending packets (in the application interface) have been transmitted. The MAC comes out of the LPI mode when the application sends any packet for transmission or the application issues a Tx FIFO Flush command. In addition, the MAC automatically clears the LPIEN bit when it exits the LPI state. If Tx FIFO Flush is set in the FTQ bit of MTL_TxQ(#i)_Operation_Mode register, when the MAC is in the LPI mode, it exits the LPI mode. When this bit is 0, the LPIEN bit directly controls behavior of the MAC when it is entering or coming out of the LPI mode.Values: 0x0 (DISABLE): LPI Tx Automate is disabled 0x1 (ENABLE): LPI Tx Automate is enabled Value After Reset: 0x0
			18	PLSEN	R/W	0x0	PHY Link Status Enable This bit enables the link status received on the RGMII, SGMII, or SMII Receive paths to be used for activating the LPI LS TIMER. When this bit is set, the MAC uses the link-status bits of the MAC_PHYIF_Control_Status register and the PLS bit for the LPI LS Timer trigger. When this bit is reset, the MAC ignores the link-status bits of the MAC_PHYIF_Control_Status register and takes only the PLS bit.Values: 0x0 (DISABLE): PHY Link Status is disabled 0x1 (ENABLE): PHY Link Status is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xd0	MAC_LPI_Control_Status	The LPI Control and Status Register controls the LPI functions and provides the LPI interrupt status. The status bits are cleared when this register is read.	17	PLS	R/W	0x0	PHY Link Status This bit indicates the link status of the PHY. The MAC Transmitter asserts the LPI pattern only when the link status is up (OKAY) at least for the time indicated by the LPI LS TIMER. When this bit is set, the link is considered to be okay (UP) and when this bit is reset, the link is considered to be down.Values: 0x0 (DISABLE): link is down 0x1 (ENABLE): link is okay (UP) Value After Reset: 0x0
			16	LPIEN	R/W	0x0	LPI Enable When this bit is set, it instructs the MAC Transmitter to enter the LPI state. When this bit is reset, it instructs the MAC to exit the LPI state and resume normal transmission. This bit is cleared when the LPITXA bit is set and the MAC exits the LPI state because of the arrival of a new packet for transmission.Values: 0x0 (DISABLE): LPI state is disabled 0x1 (ENABLE): LPI state is enabled Value After Reset: 0x0
			15:10	Reserved_15_10	R	0x0	Reserved. Value After Reset: 0x0
			9	RLPIST	R	0x0	Receive LPI State When this bit is set, it indicates that the MAC is receiving the LPI pattern on the GMII or MII interface.Values: 0x0 (INACTIVE): Receive LPI state not detected 0x1 (ACTIVE): Receive LPI state detected Value After Reset: 0x0
			8	TLPIST	R	0x0	Transmit LPI State When this bit is set, it indicates that the MAC is transmitting the LPI pattern on the GMII or MII interface.Values: 0x0 (INACTIVE): Transmit LPI state not detected 0x1 (ACTIVE): Transmit LPI state detected Value After Reset: 0x0
			7:4	Reserved_7_4	R	0x0	Reserved. Value After Reset: 0x0
			3	RLPIEX	R	0x0	Receive LPI Exit When this bit is set, it indicates that the MAC Receiver has stopped receiving the LPI pattern on the GMII or MII interface, exited the LPI state, and resumed the normal reception. This bit is cleared by a read into this register (or this bit is written to 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Note: This bit might not be set if the MAC stops receiving the LPI pattern for a very short duration, such as, less than three clock cycles of CSR clock.Values: 0x0 (INACTIVE): Receive LPI exit not detected 0x1 (ACTIVE): Receive LPI exit detected Value After Reset: 0x0
			2	RLPIEN	R	0x0	Receive LPI Entry When this bit is set, it indicates that the MAC Receiver has received an LPI pattern and entered the LPI state. This bit is cleared by a read into this register (or this bit is written to 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Note: This bit might not be set if the MAC stops receiving the LPI pattern for a very short duration, such as, less than three clock cycles of CSR clock.Values: 0x0 (INACTIVE): Receive LPI entry not detected 0x1 (ACTIVE): Receive LPI entry detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			1	TLPIEX	R	0x0	Transmit LPI Exit When this bit is set, it indicates that the MAC transmitter exited the LPI state after the application cleared the LPIEN bit and the LPI TW Timer has expired. This bit is cleared by a read into this register (or this bit is written to 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set).Values: 0x0 (INACTIVE): Transmit LPI exit not detected 0x1 (ACTIVE): Transmit LPI exit detected Value After Reset: 0x0
			0	TLPIEN	R	0x0	Transmit LPI Entry When this bit is set, it indicates that the MAC Transmitter has entered the LPI state because of the setting of the LPIEN bit. This bit is cleared by a read into this register (or this bit is written to 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set).Values: 0x0 (INACTIVE): Transmit LPI entry not detected 0x1 (ACTIVE): Transmit LPI entry detected Value After Reset: 0x0
0xd4	MAC_LPI_Timers_Control	The LPI Timers Control register controls the timeout values in the LPI states. It specifies the time for which the MAC transmits the LPI pattern and also the time for which the MAC waits before resuming the normal transmission.	31:26	Reserved_31_26	R	0x0	Reserved. Value After Reset: 0x0
			25:16	LST	R/W	0x3e8	LPI LS Timer This field specifies the minimum time (in milliseconds) for which the link status from the PHY should be up (OKAY) before the LPI pattern can be transmitted to the PHY. The MAC does not transmit the LPI pattern even when the LPIEN bit is set unless the LPI LS Timer reaches the programmed terminal count. The default value of the LPI LS Timer is 1000 (1 sec) as defined in the IEEE standard. Value After Reset: 0x3e8
			15:0	TWT	R/W	0x0	LPI TW Timer This field specifies the minimum time (in microseconds) for which the MAC waits after it stops transmitting the LPI pattern to the PHY and before it resumes the normal transmission. The TLPIEX status bit is set after the expiry of this timer. Value After Reset: 0x0
0xd8	MAC_LPI_Entry_Timer	This register controls the Tx LPI entry timer. This counter is enabled only when bit[20](LPITE) bit of MAC_LPI_Control_Status is set to 1.	31:20	Reserved_31_20	R	0x0	Reserved. Value After Reset: 0x0
			19:3	LPIET	R/W	0x0	LPI Entry Timer This field specifies the time in microseconds the MAC waits to enter LPI mode, after it has transmitted all the frames. This field is valid and used only when LPITE and LPITXA are set to 1. Bits [2:0] are read-only so that the granularity of this timer is in steps of 8 micro-seconds. Value After Reset: 0x0
			2:0	Reserved_2_0	R	0x0	Reserved. Value After Reset: 0x0
0xdc	MAC_1US_Tic_Counter	This register controls the generation of the Reference time (1 microsecond tic) for all the LPI timers. This timer has to be programmed by the software initially.	31:12	Reserved_31_12	R	0x0	Reserved. Value After Reset: 0x0
			11:0	TIC_1US_CNTR	R/W	0x63	1US TIC Counter The application must program this counter so that the number of clock cycles of CSR clock is 1us. (Subtract 1 from the value before programming). For example if the CSR clock is 100MHz then this field needs to be programmed to value 100 - 1 = 99 (which is 0x63). This is required to generate the 1US events that are used to update some of the EEE related counters. Value After Reset: 0x63
			31:22	Reserved_31_22	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xf8	MAC_PHYIF_Control_Status	The PHY Interface Control and Status register indicates the status signals received by the SGMII, RGMII, or SMII interface (selected at reset) from the PHY. This register is optional.	21:20			#VALUE!	Reserved Field: Yes
			19	LNKSTS	R	0x0	Link Status This bit indicates whether the link is up (1'b1) or down (1'b0).Values: 0x0 (INACTIVE): Link down 0x1 (ACTIVE): Link up Value After Reset: 0x0 Testable: untestable
			18:17	LNKSPEED	R	0x0	Link Speed This bit indicates the current speed of the link.Values: 0x0 (M_2500K): 2.5 MHz 0x1 (M_25M): 25 MHz 0x2 (M_125M): 125 MHz 0x3 (RSVD): Reserved Value After Reset: 0x0 Testable: untestable
			16	LNKMOD	R	0x0	Link Mode This bit indicates the current mode of operation of the link.Values: 0x0 (HDUPLX): Half-duplex mode 0x1 (FDUPLX): Full-duplex mode Value After Reset: 0x0 Testable: untestable
			15:5	Reserved_15_5	R	0x0	Reserved. Value After Reset: 0x0
			4			#VALUE!	Reserved Field: Yes
			3	Reserved_3	R	0x0	Reserved. Value After Reset: 0x0
			2			#VALUE!	Reserved Field: Yes
			1	LUD	R/W	0x0	Link Up or Down This bit indicates whether the link is up or down during transmission of configuration in the RGMII, SGMII, or SMII interface.Values: 0x0 (LINKDOWN): Link down 0x1 (LINKUP): Link up Value After Reset: 0x0
			0	TC	R/W	0x0	Transmit Configuration in RGMII, SGMII, or SMII When set, this bit enables the transmission of duplex mode, link speed, and link up or down information to the PHY in the RGMII, SMII, or SGMII port. When this bit is reset, no such information is driven to the PHY. The details of this feature are provided in the following sections: "Reduced Gigabit Media Independent Interface" "Serial Media Independent Interface" "Serial Gigabit Media Independent Interface" Values: 0x0 (DISABLE): Disable Transmit Configuration in RGMII, SGMII, or SMII 0x1 (ENABLE): Enable Transmit Configuration in RGMII, SGMII, or SMII Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x110	MAC_Version	The version register identifies the version of the DWC_ether_qos. This register contains two bytes: one that Synopsys uses to identify the IP release number, and the other that you set while configuring the IP.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:8	USERVER	R	0x10	User-defined Version (configured with coreConsultant) Value After Reset: 0x10
			7:0	SNPSVER	R	0x53	Synopsys-defined Version Value After Reset: 0x53
0x114	MAC_Debug	The Debug register provides the debug status of various MAC blocks.	31:19	Reserved_31_19	R	0x0	Reserved. Value After Reset: 0x0
			18:17	TFCSTS	R	0x0	MAC Transmit Packet Controller Status This field indicates the state of the MAC Transmit Packet Controller module.Values: 0x0 (IDLE): Idle state 0x1 (WAITING): Waiting for one of the following: Status of the previous packet OR IPG or back off period to be over 0x2 (GEN_TX_PAU): Generating and transmitting a Pause control packet (in full-duplex mode) 0x3 (TRNSFR): Transferring input packet for transmission Value After Reset: 0x0
			16	TPESTS	R	0x0	MAC GMII or MII Transmit Protocol Engine Status When this bit is set, it indicates that the MAC GMII or MII transmit protocol engine is actively transmitting data, and it is not in the Idle state.Values: 0x0 (INACTIVE): MAC GMII or MII Transmit Protocol Engine Status not detected 0x1 (ACTIVE): MAC GMII or MII Transmit Protocol Engine Status detected Value After Reset: 0x0
			15:3	Reserved_15_3	R	0x0	Reserved. Value After Reset: 0x0
			2:1	RFCFCSTS	R	0x0	MAC Receive Packet Controller FIFO Status When this bit is set, this field indicates the active state of the small FIFO Read and Write controllers of the MAC Receive Packet Controller module. Value After Reset: 0x0
			0	RPESTS	R	0x0	MAC GMII or MII Receive Protocol Engine Status When this bit is set, it indicates that the MAC GMII or MII receive protocol engine is actively receiving data, and it is not in the Idle state.Values: 0x0 (INACTIVE): MAC GMII or MII Receive Protocol Engine Status not detected 0x1 (ACTIVE): MAC GMII or MII Receive Protocol Engine Status detected Value After Reset: 0x0
			31:28	ACTPHYSEL	R	0x0	Active PHY Selected When you have multiple PHY interfaces in your configuration, this field indicates the sampled value of phy_intf_sel_i during reset de-assertion.Values: 0x0 (GMII_MII): GMII or MII 0x1 (RGMII): RGMII 0x2 (SGMII): Reserved 0x3 (TBI): Reserved 0x4 (RMII): RMII 0x5 (RTBI): Reserved 0x6 (SMII): Reserved 0x7 (REVMIII): Reserved 0x8 (B10T1S): B10T1S Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x11c	MAC_HW_Feature0	This register indicates the presence of first set of the optional features or functions of the DWC_ether_qos. The software driver can use this register to dynamically	27	SAVLANINS	R	0x1	Source Address or VLAN Insertion Enable This bit is set to 1 when the Enable SA and VLAN Insertion on Tx option is selectedValues: 0x0 (INACTIVE): Source Address or VLAN Insertion Enable option is not selected 0x1 (ACTIVE): Source Address or VLAN Insertion Enable option is selected Value After Reset: 0x1
			26:25	TSSTSSEL	R	0x1	Timestamp System Time Source This bit indicates the source of the Timestamp System Time:Values: 0x0 (INTRNL): Reserved 0x1 (EXTRNL): Internal 0x2 (BOTH): External 0x3 (RSVD): Both Value After Reset: 0x1
			24	MACADR64SEL	R	0x0	MAC Addresses 64-127 Selected This bit is set to 1 when the Enable Additional 64 MAC Address Registers (64-127) option is selectedValues: 0x0 (INACTIVE): MAC Addresses 64-127 Select option is not selected 0x1 (ACTIVE): MAC Addresses 64-127 Select option is selected Value After Reset: 0x0
			23	MACADR32SEL	R	0x0	MAC Addresses 32-63 Selected This bit is set to 1 when the Enable Additional 32 MAC Address Registers (32-63) option is selectedValues: 0x0 (INACTIVE): MAC Addresses 32-63 Select option is not selected 0x1 (ACTIVE): MAC Addresses 32-63 Select option is selected Value After Reset: 0x0
			22:18	ADDMACADRSEL	R	0x8	MAC Addresses 1-31 Selected This bit is set to 1 when the non-zero value is selected for Enable Additional 1-31 MAC Address Registers option Value After Reset: 0x8
			17	Reserved_17	R	0x0	Reserved. Value After Reset: 0x0
			16	RXCOESEL	R	0x1	Receive Checksum Offload Enabled This bit is set to 1 when the Enable Receive TCP/IP Checksum Check option is selectedValues: 0x0 (INACTIVE): Receive Checksum Offload Enable option is not selected 0x1 (ACTIVE): Receive Checksum Offload Enable option is selected Value After Reset: 0x1
			15	Reserved_15	R	0x0	Reserved. Value After Reset: 0x0
			14	TXCOESEL	R	0x1	Transmit Checksum Offload Enabled This bit is set to 1 when the Enable Transmit TCP/IP Checksum Insertion option is selectedValues: 0x0 (INACTIVE): Transmit Checksum Offload Enable option is not selected 0x1 (ACTIVE): Transmit Checksum Offload Enable option is selected Value After Reset: 0x1
			13	EEESEL	R	0x1	Energy Efficient Ethernet Enabled This bit is set to 1 when the Enable Energy Efficient Ethernet (EEE) option is selectedValues: 0x0 (INACTIVE): Energy Efficient Ethernet Enable option is not selected 0x1 (ACTIVE): Energy Efficient Ethernet Enable option is selected Value After Reset: 0x1

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
		enable or disable the programs related to the optional blocks.Note: All bits are set or reset according to the features selected while configuring the IP in coreConsultant.	12	TSSEL	R	0x1	IEEE 1588-2008 Timestamp Enabled This bit is set to 1 when the Enable IEEE 1588 Timestamp Support option is selectedValues: 0x0 (INACTIVE): IEEE 1588-2008 Timestamp Enable option is not selected 0x1 (ACTIVE): IEEE 1588-2008 Timestamp Enable option is selected Value After Reset: 0x1
			11:10	Reserved_11_10	R	0x0	Reserved. Value After Reset: 0x0
			9	ARPOFFSEL	R	0x1	ARP Offload Enabled This bit is set to 1 when the Enable IPv4 ARP Offload option is selectedValues: 0x0 (INACTIVE): ARP Offload Enable option is not selected 0x1 (ACTIVE): ARP Offload Enable option is selected Value After Reset: 0x1
			8	MMCSEL	R	0x1	RMON Module Enable This bit is set to 1 when the Enable MAC Management Counters (MMC) option is selectedValues: 0x0 (INACTIVE): RMON Module Enable option is not selected 0x1 (ACTIVE): RMON Module Enable option is selected Value After Reset: 0x1
			7	MGKSEL	R	0x1	PMT Magic Packet Enable This bit is set to 1 when the Enable Magic Packet Detection option is selectedValues: 0x0 (INACTIVE): PMT Magic Packet Enable option is not selected 0x1 (ACTIVE): PMT Magic Packet Enable option is selected Value After Reset: 0x1
			6	RWKSEL	R	0x1	PMT Remote Wake-up Packet Enable This bit is set to 1 when the Enable Remote Wake-Up Packet Detection option is selectedValues: 0x0 (INACTIVE): PMT Remote Wake-up Packet Enable option is not selected 0x1 (ACTIVE): PMT Remote Wake-up Packet Enable option is selected Value After Reset: 0x1
			5	SMASEL	R	0x1	SMA (MDIO) Interface This bit is set to 1 when the Enable Station Management (MDIO Interface) option is selectedValues: 0x0 (INACTIVE): SMA (MDIO) Interface not selected 0x1 (ACTIVE): SMA (MDIO) Interface selected Value After Reset: 0x1
			4	VLHASH	R	0x1	VLAN Hash Filter Selected This bit is set to 1 when the Enable VLAN Hash Table Based Filtering option is selectedValues: 0x0 (INACTIVE): VLAN Hash Filter not selected 0x1 (ACTIVE): VLAN Hash Filter selected Value After Reset: 0x1
			3	PCSSEL	R	0x0	PCS Registers (TBI, SGMII, or RTBI PHY interface) This bit is set to 1 when the TBI, SGMII, or RTBI PHY interface option is selectedValues: 0x0 (INACTIVE): No PCS Registers (TBI, SGMII, or RTBI PHY interface) 0x1 (ACTIVE): PCS Registers (TBI, SGMII, or RTBI PHY interface) Value After Reset: 0x0
			2	HDSEL	R	0x0	Half-duplex Support This bit is set to 1 when the half-duplex mode is selectedValues: 0x0 (INACTIVE): No Half-duplex support 0x1 (ACTIVE): Half-duplex support Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			1	GMIISEL	R	0x1	1000 Mbps Support This bit is set to 1 when 1000 Mbps is selected as the Mode of OperationValues: 0x0 (INACTIVE): No 1000 Mbps support 0x1 (ACTIVE): 1000 Mbps support Value After Reset: 0x1
			0	MIISEL	R	0x1	10 or 100 Mbps Support This bit is set to 1 when 10/100 Mbps is selected as the Mode of OperationValues: 0x0 (INACTIVE): No 10 or 100 Mbps support 0x1 (ACTIVE): 10 or 100 Mbps support Value After Reset: 0x1
			31	Reserved_31	R	0x0	Reserved. Value After Reset: 0x0
			30:27	L3L4FNUM	R	0x8	Total number of L3 or L4 Filters This field indicates the total number of L3 or L4 filters:Values: 0x0 (NOFILT): No L3 or L4 Filter 0x1 (M_1FILT): 1 L3 or L4 Filter 0x2 (M_2FILT): 2 L3 or L4 Filters 0x3 (M_3FILT): 3 L3 or L4 Filters 0x4 (M_4FILT): 4 L3 or L4 Filters 0x5 (M_5FILT): 5 L3 or L4 Filters 0x6 (M_6FILT): 6 L3 or L4 Filters 0x7 (M_7FILT): 7 L3 or L4 Filters 0x8 (M_8FILT): 8 L3 or L4 Filters Value After Reset: 0x8
			26	Reserved_26	R	0x0	Reserved. Value After Reset: 0x0
			25:24	HASHTBLSZ	R	0x1	Hash Table Size This field indicates the size of the hash table:Values: 0x0 (NO_HT): No hash table 0x1 (M_64): 64 0x2 (M_128): 128 0x3 (M_256): 256 Value After Reset: 0x1
			23	POUOST	R	0x1	One Step for PTP over UDP/IP Feature Enable This bit is set to 1 when the Enable One step timestamp for PTP over UDP/IP feature is selected.Values: 0x0 (INACTIVE): One Step for PTP over UDP/IP Feature is not selected 0x1 (ACTIVE): One Step for PTP over UDP/IP Feature is selected Value After Reset: 0x1
			22	Reserved_22	R	0x0	Reserved. Value After Reset: 0x0
			21	RAVSEL	R	0x0	Rx Side Only AV Feature Enable This bit is set to 1 when the Enable Audio Video Bridging option on Rx Side Only is selected.Values: 0x0 (INACTIVE): Rx Side Only AV Feature is not selected 0x1 (ACTIVE): Rx Side Only AV Feature is selected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x120	MAC_HW_Feature1	This register indicates the presence of second set of the optional features or functions of the DWC_ether_qos. The software driver can use this register to dynamically enable or disable the programs related to the optional blocks. Note: All bits are set or reset according to the features selected while configuring the IP in coreConsultant.	20	AVSEL	R	0x1	AV Feature Enable This bit is set to 1 when the Enable Audio Video Bridging option is selected.Values: 0x0 (INACTIVE): AV Feature is not selected 0x1 (ACTIVE): AV Feature is selected Value After Reset: 0x1
			19	DBGMEMA	R	0x1	DMA Debug Registers Enable This bit is set to 1 when the Debug Mode Enable option is selectedValues: 0x0 (INACTIVE): DMA Debug Registers option is not selected 0x1 (ACTIVE): DMA Debug Registers option is selected Value After Reset: 0x1
			18	TSOEN	R	0x1	TCP Segmentation Offload Enable This bit is set to 1 when the Enable TCP Segmentation Offloading for TCP/IP Packets option is selectedValues: 0x0 (INACTIVE): TCP Segmentation Offload Feature is not selected 0x1 (ACTIVE): TCP Segmentation Offload Feature is selected Value After Reset: 0x1
			17	SPHEN	R	0x1	Split Header Feature Enable This bit is set to 1 when the Enable Split Header Structure option is selectedValues: 0x0 (INACTIVE): Split Header Feature is not selected 0x1 (ACTIVE): Split Header Feature is selected Value After Reset: 0x1
			16	DCBEN	R	0x1	DCB Feature Enable This bit is set to 1 when the Enable Data Center Bridging option is selectedValues: 0x0 (INACTIVE): DCB Feature is not selected 0x1 (ACTIVE): DCB Feature is selected Value After Reset: 0x1
			15:14	ADDR64	R	0x1	Address Width. This field indicates the configured address width:Values: 0x0 (M_32): 32 0x1 (M_40): 40 0x2 (M_48): 48 0x3 (RSVD): Reserved Value After Reset: 0x1
			13	ADVTHWORD	R	0x1	IEEE 1588 High Word Register Enable This bit is set to 1 when the Add IEEE 1588 Higher Word Register option is selectedValues: 0x0 (INACTIVE): IEEE 1588 High Word Register option is not selected 0x1 (ACTIVE): IEEE 1588 High Word Register option is selected Value After Reset: 0x1
			12	PTOEN	R	0x1	PTP Offload Enable This bit is set to 1 when the Enable PTP Timestamp Offload Feature is selected.Values: 0x0 (INACTIVE): PTP Offload feature is not selected 0x1 (ACTIVE): PTP Offload feature is selected Value After Reset: 0x1
			11	OSTEN	R	0x1	One-Step Timestamping Enable This bit is set to 1 when the Enable One-Step Timestamp Feature is selected.Values: 0x0 (INACTIVE): One-Step Timestamping feature is not selected 0x1 (ACTIVE): One-Step Timestamping feature is selected Value After Reset: 0x1

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			10:6	TXFIFOSIZE	R	0x7	MTL Transmit FIFO Size This field contains the configured value of MTL Tx FIFO in bytes expressed as Log to base 2 minus 7, that is, Log2(TXFIFO_SIZE) -7:Values: 0x0 (M_128B): 128 bytes 0x1 (M_256B): 256 bytes 0x2 (M_512B): 512 bytes 0x3 (M_1024B): 1024 bytes 0x4 (M_2048B): 2048 bytes 0x5 (M_4096B): 4096 bytes 0x6 (M_8192B): 8192 bytes 0x7 (M_16384B): 16384 bytes 0x8 (M_32KB): 32 KB 0x9 (M_64KB): 64 KB 0xa (M_128KB): 128 KB 0xb (RSVD): Reserved Value After Reset: 0x7
			5	SPRAM	R	0x1	Single Port RAM Enable This bit is set to 1 when the Use single port RAM Feature is selected.Values: 0x0 (INACTIVE): Single Port RAM feature is not selected 0x1 (ACTIVE): Single Port RAM feature is selected Value After Reset: 0x1
			4:0	RXFIFOSIZE	R	0x7	MTL Receive FIFO Size This field contains the configured value of MTL Rx FIFO in bytes expressed as Log to base 2 minus 7, that is, Log2(RXFIFO_SIZE) -7:Values: 0x0 (M_128B): 128 bytes 0x1 (M_256B): 256 bytes 0x2 (M_512B): 512 bytes 0x3 (M_1024B): 1024 bytes 0x4 (M_2048B): 2048 bytes 0x5 (M_4096B): 4096 bytes 0x6 (M_8192B): 8192 bytes 0x7 (M_16384B): 16384 bytes 0x8 (M_32KB): 32 KB 0x9 (M_64KB): 64 KB 0xa (M_128KB): 128 KB 0xb (M_256KB): 256 KB 0xc (RSVD): Reserved Value After Reset: 0x7
			31	Reserved_31	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			30:28	AUXSNAPNUM	R	0x4	Number of Auxiliary Snapshot Inputs This field indicates the number of auxiliary snapshot inputs:Values: 0x0 (NO_AUXI): No auxiliary input 0x1 (M_1_AUXI): Reserved 0x2 (M_2_AUXI): Reserved 0x3 (M_3_AUXI): Reserved 0x4 (M_4_AUXI): 4 auxiliary input 0x5 (RSVD): Reserved Value After Reset: 0x4
			27	Reserved_27	R	0x0	Reserved. Value After Reset: 0x0
			26:24	PPSOUTNUM	R	0x1	Number of PPS Outputs This field indicates the number of PPS outputs:Values: 0x0 (NO_PPSO): No PPS output 0x1 (M_1_PPSO): 1 PPS output 0x2 (M_2_PPSO): Reserved 0x3 (M_3_PPSO): Reserved 0x4 (M_4_PPSO): Reserved 0x5 (RSVD): Reserved Value After Reset: 0x1
			23:22	TDCSZ	R	0x2	Tx DMA Descriptor Cache Size in terms of 16-bytes descriptors: 00 - Cache Not configured 01 - Four 16-bytes descriptor 10 - Eight 16-bytes descriptor 11 - Sixteen 16-bytes descriptorValues: 0x0 (NO_DCACHE): Desc Cache not configured 0x1 (M_1TDCSZ): Reserved 0x2 (M_2TDCSZ): 8 0x3 (M_3TDCSZ): Reserved Value After Reset: 0x2
			21:18	TXCHCNT	R	0x7	Number of DMA Transmit Channels This field indicates the number of DMA Transmit channels:Values: 0x0 (D_1TXCH): Reserved 0x1 (D_2TXCH): Reserved 0x2 (D_3TXCH): Reserved 0x3 (D_4TXCH): Reserved 0x4 (D_5TXCH): Reserved 0x5 (D_6TXCH): Reserved 0x6 (D_7TXCH): Reserved 0x7 (D_8TXCH): 8 DMA Tx Channels Value After Reset: 0x7

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Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x124	MAC_HW_Feature2	This register indicates the presence of third set of the optional features or functions of the DWC_ether_qos. The software driver can use this register to dynamically enable or disable the programs related to the optional blocks.	17:16	RDCSZ	R	0x2	Rx DMA Descriptor Cache Size in terms of 16-bytes descriptors: 00 - Cache Not configured 01 - Four 16-bytes descriptor 10 - Eight 16-bytes descriptor 11 - Sixteen 16-bytes descriptorValues: 0x0 (NO_DCACHE): Desc Cache not configured 0x1 (M_1RDCSZ): Reserved 0x2 (M_2RDCSZ): 8 0x3 (M_3RDCSZ): Reserved Value After Reset: 0x2
			15:12	RXCHCNT	R	0x7	Number of DMA Receive Channels This field indicates the number of DMA Receive channels:Values: 0x0 (D_1RXCH): Reserved 0x1 (D_2RXCH): Reserved 0x2 (D_3RXCH): Reserved 0x3 (D_4RXCH): Reserved 0x4 (D_5RXCH): Reserved 0x5 (D_6RXCH): Reserved 0x6 (D_7RXCH): Reserved 0x7 (D_8RXCH): 8 DMA Rx Channels Value After Reset: 0x7
			11:10	Reserved_11_10	R	0x0	Reserved. Value After Reset: 0x0
			9:6	TXQCNT	R	0x7	Number of MTL Transmit Queues This field indicates the number of MTL Transmit queues:Values: 0x0 (M_1TXQ): Reserved 0x1 (M_2TXQ): Reserved 0x2 (M_3TXQ): Reserved 0x3 (M_4TXQ): Reserved 0x4 (M_5TXQ): Reserved 0x5 (M_6TXQ): Reserved 0x6 (M_7TXQ): Reserved 0x7 (M_8TXQ): 8 MTL Tx Queues Value After Reset: 0x7
			5:4	Reserved_5_4	R	0x0	Reserved. Value After Reset: 0x0
			3:0	RXQCNT	R	0x7	Number of MTL Receive Queues This field indicates the number of MTL Receive queues:Values: 0x0 (M_1RXQ): Reserved 0x1 (M_2RXQ): Reserved 0x2 (M_3RXQ): Reserved 0x3 (M_4RXQ): Reserved 0x4 (M_5RXQ): Reserved 0x5 (M_6RXQ): Reserved 0x6 (M_7RXQ): Reserved 0x7 (M_8RXQ): 8 MTL Rx Queues Value After Reset: 0x7

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			31:30	Reserved_31_30	R	0x0	Reserved. Value After Reset: 0x0
			29:28	ASP	R	0x0	Automotive Safety Package Following are the encoding for the different Safety featuresValues: 0x0 (NONE): No Safety features selected 0x1 (ECC_ONLY): Reserved 0x2 (AS_NPPE): Reserved 0x3 (AS_PPE): Reserved Value After Reset: 0x0
			27	TBSSEL	R	0x1	Time Based Scheduling Enable This bit is set to 1 when the Time Based Scheduling feature is selected.Values: 0x0 (INACTIVE): Time Based Scheduling Enable feature is not selected 0x1 (ACTIVE): Time Based Scheduling Enable feature is selected Value After Reset: 0x1
			26	FPESEL	R	0x1	Frame Preemption Enable This bit is set to 1 when the Enable Frame preemption feature is selected.Values: 0x0 (INACTIVE): Frame Preemption Enable feature is not selected 0x1 (ACTIVE): Frame Preemption Enable feature is selected Value After Reset: 0x1
			25:22	Reserved_25_22	R	0x0	Reserved. Value After Reset: 0x0
			21:20	ESTWID	R	0x3	Width of the Time Interval field in the Gate Control List This field indicates the width of the Configured Time Interval FieldValues: 0x0 (NOWIDTH): Width not configured 0x1 (WIDTH16): Reserved 0x2 (WIDTH20): Reserved 0x3 (WIDTH24): 24 Value After Reset: 0x3
			19:17	ESTDEP	R	0x3	Depth of the Gate Control List Indicates the depth of Gate Control list, in bytes.Values: 0x0 (NODEPTH): No Depth configured 0x1 (DEPTH64): Reserved 0x2 (DEPTH128): Reserved 0x3 (DEPTH256): 256 0x4 (DEPTH512): Reserved 0x5 (DEPTH1024): Reserved 0x6 (RSVD): Reserved Value After Reset: 0x3
			16	ESTSEL	R	0x1	Enhancements to Scheduled Traffic Enable This bit is set to 1 when the Enable Enhancements to Scheduling Traffic feature is selected.Values: 0x0 (INACTIVE): Enable Enhancements to Scheduling Traffic feature is not selected 0x1 (ACTIVE): Enable Enhancements to Scheduling Traffic feature is selected Value After Reset: 0x1
			15	Reserved_15	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x128	MAC_HW_Feature3	This register indicates the presence of fourth set the optional features or functions of the DWC_ether_qos. The software driver can use this register to dynamically enable or disable the programs related to the optional blocks.	14:13	FRPES	R	0x1	Flexible Receive Parser Table Entries size This field indicates the Max Number of Parser Entries supported by Flexible Receive Parser.Values: 0x0 (M_64ENTR): Reserved 0x1 (M_128ENTR): 128 Entries 0x2 (M_256ENTR): Reserved 0x3 (RSVD): Reserved Value After Reset: 0x1
			12:11	FRPBS	R	0x2	Flexible Receive Parser Buffer size This field indicates the supported Max Number of bytes of the packet data to be Parsed by Flexible Receive Parser.Values: 0x0 (M_64BYTES): Reserved 0x1 (M_128BYTES): Reserved 0x2 (M_256BYTES): 256 Bytes 0x3 (RSVD): Reserved Value After Reset: 0x2
			10	FRPSEL	R	0x1	Flexible Receive Parser Selected This bit is set to 1 when the Enable Flexible Programmable Receive Parser option is selected.Values: 0x0 (INACTIVE): Flexible Receive Parser feature is not selected 0x1 (ACTIVE): Flexible Receive Parser feature is selected Value After Reset: 0x1
			9	PDUPSEL	R	0x1	Broadcast/Multicast Packet Duplication This bit is set to 1 when the Broadcast/Multicast Packet Duplication feature is selected.Values: 0x0 (INACTIVE): Broadcast/Multicast Packet Duplication feature is not selected 0x1 (ACTIVE): Broadcast/Multicast Packet Duplication feature is selected Value After Reset: 0x1
			8:6	Reserved_7_6	R	0x0	Reserved. Value After Reset: 0x0
			5	DVLAN	R	0x1	Double VLAN Tag Processing Selected This bit is set to 1 when the Enable Double VLAN Processing Feature is selected.Values: 0x0 (INACTIVE): Double VLAN option is not selected 0x1 (ACTIVE): Double VLAN option is selected Value After Reset: 0x1
			4	CBTISEL	R	0x1	Queue/Channel based VLAN tag insertion on Tx Enable This bit is set to 1 when the Enable Queue/Channel based VLAN tag insertion on Tx Feature is selected.Values: 0x0 (INACTIVE): Enable Queue/Channel based VLAN tag insertion on Tx feature is not selected 0x1 (ACTIVE): Enable Queue/Channel based VLAN tag insertion on Tx feature is selected Value After Reset: 0x1
			3	Reserved_3	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			2:0	NRVF	R	0x2	Number of Extended VLAN Tag Filters Enabled This field indicates the Number of Extended VLAN Tag Filters selected:Values: 0x0 (NO_ERVLAN): No Extended Rx VLAN Filters 0x1 (M_4_ERVLAN): 4 Extended Rx VLAN Filters 0x2 (M_8_ERVLAN): 8 Extended Rx VLAN Filters 0x3 (M_16_ERVLAN): Reserved 0x4 (M_24_ERVLAN): Reserved 0x5 (M_32_ERVLAN): Reserved 0x6 (RSVD): Reserved Value After Reset: 0x2
0x12c	MAC_HW_Feature4	This register indicates the presence of the fifth set of optional features or functions of the DWC_ether_qos. The software driver can use this register to dynamically enable or disable the programs related to the optional blocks.	31:2	Reserved_31_2	R	0x0	Reserved. Value After Reset: 0x0
			1:0	PCSEL	R	0x0	Policing Counters Selected This field indicates the Number of Policing Counters selected:Values: 0x0 (INACTIVE): Policing counter feature is not selected 0x1 (PCNUM_16): 16 Policing counters are selected 0x2 (PCNUM_32): 32 Policing counters are selected 0x3 (RSVD): Reserved Value After Reset: 0x0
			31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28			#VALUE!	Reserved Field: Yes
			27	PSE	R/W	0x0	Preamble Suppression Enable 1: The SMA suppresses the 32-bit preamble and transmits MDIO frames with only 1 preamble bit. 0: The SMA does not suppress the preamble. So, the MDIO frame has 32-bit preamble as defined in the IEEE specifications. Values: 0x0 (DISABLE): Preamble Suppression disabled 0x1 (ENABLE): Preamble Suppression enabled Value After Reset: 0x0
			26	BTB	R/W	0x0	Back-to-Back Transactions 1: If NTC > 0, at the end of the frame transfer, MAC acknowledges the completion of a read or write command before transferring the trailing clocks. So, application can initiate the next command and MAC can execute it immediately without waiting for the transmission of the trailing clocks of the previous frame. 0: MAC considers the read or write command to be complete, that is, MAC clears the GB field, only after generating the trailing clocks. This mode ensures that the NTC is always generated after each frame. Note: Program this field with a value 1, only when NTC > 0.Values: 0x0 (DISABLE): Back to Back transactions disabled 0x1 (ENABLE): Back to Back transactions enabled Value After Reset: 0x0
			25:21	PA	R/W	0x0	Physical Layer Address Specifies the address of the PHY device, out of the 32 possible PHY devices. Note: The value you program in the C45E field determines if the PHY device is clause 45 capable or clause 22 capable. Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			20:16	RDA	R/W	0x0	Register or Device Address For clause 22 PHY: Specifies the PHY register address of the selected PHY device. For Clause 45 PHY: Specifies the selected MDIO device. Note: The value you program in the C45E field determines if the PHY device is clause 45 capable or clause 22 capable. Values: 5'b00000: B10T1S PLCA control 5'b00001: B10T1S PLCA node control 5'b00010: B10T1S PLCA status 5'b00011: B10T1S PLCA timer 5'b00100-5'b01010: Reserved 5'b01011: B10T1S PCS control 5'b01100: B10T1S PCS status 5'b01101: B10T1S PCS diagnostic 1 5'b01110: B10T1S PCS diagnostic 2 5'b01111: B10T1S PCS Jab Timer 5'b10000-5'b10011: Reserved 5'b10100: B10T1S PMA/PMD extended ability 5'b10101: B10T1S PMA/PMD control 5'b10110: B10T1S PMA control 5'b10111: B10T1S PMA status 5'b11000: B10T1S test mode control 5'b11001: B10T1S Debug_mode 5'b11010-5'b11111: Reserved Value After Reset: 0x0
			15	Reserved_15	R	0x0	Reserved. Value After Reset: 0x0
			14:12	NTC	R/W	0x0	Number of Trailing Clocks This field controls the number of trailing clock cycles generated on the gmii_mdc_o (MDC) signal, after the end of MDIO frame transfer. The valid values are 0 to 7. When you program a value of 3'h3 in this field, MAC generates three clock cycles on the MDC line after the end of MDIO frame transfer. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x200	MAC_MDIO_Address	The MDIO Address register controls the management cycles to external PHY through a management interface.	11:8	CR	R/W	0x0	CSR Clock Range The CSR Clock Range selection determines the frequency of the MDC clock according to the CSR clock frequency used in your design: 0000: CSR clock = 60-100 MHz; MDC clock = CSR clock/42 0001: CSR clock = 100-150 MHz; MDC clock = CSR clock/62 0010: CSR clock = 20-35 MHz; MDC clock = CSR clock/16 0011: CSR clock = 35-60 MHz; MDC clock = CSR clock/26 0100: CSR clock = 150-250 MHz; MDC clock = CSR clock/102 0101: CSR clock = 250-300 MHz; MDC clock = CSR clock/124 0110: CSR clock = 300-500 MHz; MDC clock = CSR clock/204 0111: CSR clock = 500-800 MHz; MDC clock = CSR clock/324 The suggested range of CSR clock frequency applicable for each value (when Bit 11 = 0) ensures that the MDC clock is approximately between 1.0 MHz to 2.5 MHz frequency range. When Bit 11 is set, you can achieve a higher frequency of the MDC clock than the frequency limit of 2.5 MHz (specified in the IEEE 802.3) and program a clock divider of lower value. For example, when CSR clock is of 100 MHz frequency and you program these bits as 1010, the resultant MDC clock is of 12.5 MHz which is beyond the range specified in IEEE 802.3. Program the following values only if the interfacing chips support faster MDC clocks: 1000: CSR clock/4 1001: CSR clock/6 1010: CSR clock/8 1011: CSR clock/10 1100: CSR clock/12 1101: CSR clock/14 1110: CSR clock/16 1111: CSR clock/18 Value After Reset: 0x0
			7:5	Reserved_7_5	R	0x0	Reserved. Value After Reset: 0x0
			4	SKAP	R/W	0x0	Skip Address Packet 1: SMA does not send the address packets before read, write, or post-read increment address packets. This bit is valid only when C45E is set. Values: 0x0 (DISABLE): Skip Address Packet is disabled 0x1 (ENABLE): Skip Address Packet is enabled Value After Reset: 0x0
			3	GOC_1	R/W	0x0	GMII Operation Command 1 This bit is higher bit of the operation command to the PHY or RevMII, GOC_1 and GOC_O is encoded as follows: 00: Reserved 01: Write 10: Post Read Increment Address for Clause 45 PHY 11: Read When Clause 22 PHY or RevMII is enabled, only Write and Read commands are valid.Values: 0x0 (DISABLE): GMII Operation Command 1 is disabled 0x1 (ENABLE): GMII Operation Command 1 is enabled Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			2	GOC_0	R/W	0x0	GMII Operation Command 0 This is the lower bit of the operation command to the PHY or RevMII. When in SMA mode (MDIO master) this bit along with GOC_1 determines the operation to be performed to the PHY. When only RevMII is selected in configuration this bit is read-only and tied to 1.Values: 0x0 (DISABLE): GMII Operation Command 0 is disabled 0x1 (ENABLE): GMII Operation Command 0 is enabled Value After Reset: 0x0
			1	C45E	R/W	0x0	Clause 45 PHY Enable 1: Clause 45 capable PHY is connected to MDIO. 0: Clause 22 capable PHY is connected to MDIO. Values: 0x0 (DISABLE): Clause 45 PHY is disabled 0x1 (ENABLE): Clause 45 PHY is enabled Value After Reset: 0x0
			0	GB	R/W	0x0	GMII Busy The application writes 1 to this bit to instruct the SMA to initiate a read or write access to the MDIO slave. The MAC writes 0 to this bit after the MDIO frame transfer is complete. Therefore, the application must modify the fields of the MAC_MDIO_Address and MAC_MDIO_Data registers only when the value in this bit is 0. - For write operation, application must follow these steps: -- Write the 16-bit data in the GDI field of the MAC_MDIO_Data register. -- Update the RA field of the MAC_MDIO_Data register with the register address of the PHY. Note : RA field is valid only when C45E field is 1. -- Write 1 to this bit For read operation, application must follow these steps: Update the RA field of the MAC_MDIO_Data register with the register address of the PHY. Note : RA field is valid only when C45E field is 1. -- Write 1 to this bit -- Wait until MAC writes 0 to this bit. When MAC writes 0 to this bit, it implies that the data read from the PHY is available in the GD field of the MAC_MDIO_Data register and the read transfer is complete.Note: Even if the addressed PHY is not present, there is no change in the functionality of this bit. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect.Values: 0x0 (DISABLE): GMII Busy is disabled 0x1 (ENABLE): GMII Busy is enabled Value After Reset: 0x0 Testable: untestable
0x204	MAC_MDIO_Data	The MDIO Data register stores the Write data to be written to the PHY register located at the address specified in MAC_MDIO_Address. This register also stores the Read data from the PHY register located at the address specified by MDIO Address register.	31:16	RA	R/W	0x0	Register Address This field is valid only when C45E is set. It contains the Register Address in the PHY to which the MDIO frame is intended for. Value After Reset: 0x0 Testable: untestable
			15:0	GD	R/W	0x0	GMII Data This field contains the 16-bit data value read from the PHY after a Management Read operation or the 16-bit data value to be written to the PHY before a Management Write operation. Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x210	MAC_ARP_Address	The ARP Address register contains the IPv4 Destination Address of the MAC.	31:0	ARPPA	R/W	0x0	ARP Protocol Address This field contains the IPv4 Destination Address of the MAC. This address is used for perfect match with the Protocol Address of Target field in the received ARP packet. This field is available only when the Enable IPv4 ARP Offload option is selected. Value After Reset: 0x0
0x230	MAC_CSR_SW_Ctrl	This register contains software programmable controls for changing the CSR access response and status bits clearing.	31:10	Reserved_31_10	R	0x0	Reserved. Value After Reset: 0x0
			9			#VALUE!	Reserved Field: Yes
			8	SEEN	R/W	0x0	Slave Error Response Enable When this bit is set, the MAC responds with Slave Error for accesses to reserved registers in CSR space. When this bit is reset, the MAC responds with Okay response to any register accessed from CSR space.Values: 0x0 (DISABLE): Slave Error Response is disabled 0x1 (ENABLE): Slave Error Response is enabled Value After Reset: 0x0
			7:1	Reserved_7_1	R	0x0	Reserved. Value After Reset: 0x0
			0	RCWE	R/W	0x0	Register Clear on Write 1 Enable When this bit is set, the access mode of some register fields changes to Clear on Write 1, the application needs to set that respective bit to 1 to clear it. When this bit is reset, the access mode of these register fields remain as Clear on Read.Values: 0x0 (DISABLE): Register Clear on Write 1 is disabled 0x1 (ENABLE): Register Clear on Write 1 is enabled Value After Reset: 0x0
			31:20	Reserved_31_20	R	0x0	Reserved. Value After Reset: 0x0
			19	TRSP	R/W	0x0	Transmitted Respond Frame Set when a Respond mPacket is transmitted (triggered by setting SRSP field). An interrupt can be generated for this event if FPEIE bit of MAC_Interrupt_Enable is set. Access restriction applies. Clears on read (or write of 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Not transmitted Respond Frame 0x1 (ACTIVE): transmitted Respond Frame Value After Reset: 0x0
			18	TVER	R/W	0x0	Transmitted Verify Frame Set when a Verify mPacket is transmitted (triggered by setting SVER field). An interrupt can be generated for this event if FPEIE bit of MAC_Interrupt_Enable is set. Access restriction applies. Clears on read (or write of 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Not transmitted Verify Frame 0x1 (ACTIVE): transmitted Verify Frame Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x234	MAC_FPE_CTRL_STS	This register controls the operation of Frame Preemption.	17	RRSP	R/W	0x0	Received Respond Frame Set when a Respond mPacket is received. An interrupt can be generated for this event if FPEIE bit of MAC_Interrupt_Enable is set. Access restriction applies. Clears on read (or write of 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Not received Respond Frame 0x1 (ACTIVE): Received Respond Frame Value After Reset: 0x0
			16	RVER	R/W	0x0	Received Verify Frame Set when a Verify mPacket is received. An interrupt can be generated for this event if FPEIE bit of MAC_Interrupt_Enable is set. Access restriction applies. Clears on read (or write of 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Not received Verify Frame 0x1 (ACTIVE): Received Verify Frame Value After Reset: 0x0
			15:5	Reserved_15_5	R	0x0	Reserved. Value After Reset: 0x0
			4	S1_SET_0	R/W	0x0	Synopsys Reserved, Must be set to "0". This field is reserved for Synopsys Internal use, and must always be set to "0" unless instructed by Synopsys. Setting to "1" might cause unexpected behavior in the HW. Value After Reset: 0x0
			3	ARV	R/W	0x0	Autogenerate Respond mPacket on receiving Verify mPacket When set, enable the function that autogenerate a Respond mPacket on receiving Verify mPacket.Values: 0x0 (DISABLE): Autogenerate Respond mPacket on receiving Verify mPacket is disabled 0x1 (ENABLE): Autogenerate Respond mPacket on receiving Verify mPacket is enabled Value After Reset: 0x0
			2	SRSP	R/W	0x0	Send Respond mPacket When set indicates hardware to send a Respond mPacket. Reset by hardware after sending the Respond mPacket. Access restriction applies. Can be set by Software or by Hardware when ARV bit is set and a verify mPacket is received. Setting 1 sets. Self-cleared. Setting 0 has no effect.Values: 0x0 (DISABLE): Send Respond mPacket is disabled 0x1 (ENABLE): Send Respond mPacket is enabled Value After Reset: 0x0
			1	SVER	R/W	0x0	Send Verify mPacket When set indicates hardware to send a verify mPacket. Reset by hardware after sending the Verify mPacket. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect.Values: 0x0 (DISABLE): Send Verify mPacket is disabled 0x1 (ENABLE): Send Verify mPacket is enabled Value After Reset: 0x0
			0	EFPE	R/W	0x0	Enable Tx Frame Preemption When set Frame Preemption Tx functionality is enabled.Values: 0x0 (DISABLE): Tx Frame Preemption is disabled 0x1 (ENABLE): Tx Frame Preemption is enabled Value After Reset: 0x0
			31:25	Reserved_31_25	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x238	MAC_Ext_Cfg1	This register contains Split mode control field and offset field for Split Header feature.	24	SAVE	R/W	0x0	Split AV Enable When this bit is set to 1, and the received packet is an AV Type packet, the header is split at SAVO bytes from the beginning of Length/Type field of the packet, for L2 Split. When this bit is set to 0, header is split at SPLOFST bytes from the beginning of Length/Type field of the frame, for L2 Split. Value After Reset: 0x0
			23	Reserved_23	R	0x0	Reserved. Value After Reset: 0x0
			22:16	SAVO	R/W	0x2	Split AV Offset When SAVE bit is set to 1, and the received packet is an AV Type packet, these bits indicate the value of the offset from the beginning of Length/Type field at which header should be split, when appropriate SPLM is selected. The reset value of this field is 2 bytes, indicating a split at L2 header. Value is in terms of bytes. Value After Reset: 0x2
			15:10	Reserved_15_10	R	0x0	Reserved. Value After Reset: 0x0
			9:8	SPLM	R/W	0x0	Split Mode These bits indicate the mode of splitting the incoming Rx packets. They areValues: 0x0 (L3L4): Split at L3-L4 header 0x1 (L2OFST): Split at L2 header with an offset. Always Split at SPLOFST bytes from the beginning of Length-Type field of the Frame 0x2 (COMBN): Combination mode: Split similar to L3L4 for IP packets and L2OFST for non-IP packets 0x3 (RSVD): Reserved Value After Reset: 0x0
			7	Reserved_7	R	0x0	Reserved. Value After Reset: 0x0
			6:0	SPLOFST	R/W	0x2	Split Offset These bits indicate the value of offset from the beginning of Length/Type field at which header split should take place when the appropriate SPLM is selected. The reset value of this field is 2 bytes indicating a split at L2 header. Value is in terms of bytes. Value After Reset: 0x2
0x240	MAC_Presn_Time_ns	This register contains the 32-bit binary rollover equivalent time of the PTP System Time in ns.	31:0	MPTN	R	0x0	MAC 1722 Presentation Time in ns These bits indicate the value of the 32-bit binary rollover equivalent time of the PTP System Time in ns Value After Reset: 0x0
0x244	MAC_Presn_Time_Updt	This field holds the 32-bit value of MAC 1722 Presentation Time in ns, that should be added to the Current Presentation Time Counter value. Init happens when TSINIT is set, and update happens when the TSUPDT bit is set (TSINIT and TSINIT defined in MAC_Timestamp_ Control register).	31:0	MPTU	R/W	0x0	MAC 1722 Presentation Time Update This field contains the initial value or the update value for the presentation time. When you use for update, based on the value of the ADDSUB field of the MAC_System_Time_Nanoseconds_Update register, ADDSUB = 0: Add this 32-bit value in ns, to the value in the internal counter. ADDSUB = 1: Subtract this 32-bit value in ns, from the internal counter. Note: Initialization is when TSINIT = 1, in the MAC_Timestamp_Control register. Update is when TSUPDT = 1, in the MAC_Timestamp_Control register. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x300	MAC_Address0_High	The MAC Address0 High register holds the upper 16 bits of the 1st 6-byte MAC address of the station. The first DA byte that is received on the (G)MII interface corresponds to the LS byte (Bits [7:0]) of the MAC Address Low register. For example, if 0x112233445566 is received (0x11 in lane 0 of the first column) on the (G)MII as the destination address, then the MacAddress0 Register [47:0] is compared with 0x665544332211.If the MAC address registers are configured to be double-synchronized to the (G)MII clock domains, then the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the MAC Address0 Low Register are written. For proper synchronization updates, the consecutive writes to this Address Low Register should be performed after at least four clock cycles in the destination clock domain.	31	AE	R	0x1	Address Enable This bit is always set to 1.Values: 0x0 (DISABLE): INVALID : This bit must be always set to 1 0x1 (ENABLE): This bit is always set to 1 Value After Reset: 0x1
			30:24	Reserved_30_y	R	0x0	Reserved. Value After Reset: 0x0
			23:16	DCS	R/W	0x0	DMA Channel Select This field indicates the Rx DMA channel number or DMA channel numbers to which the MAC routes the received packet. When the MAC receives a packet with multicast or broadcast destination address, MAC routes the packet to multiple Rx DMA channels. The PDC field of the MAC_Ext_Configuration register determines if the MAC interprets the value of this field as a binary number or a one-hot encoded number. Value After Reset: 0x0
			15:0	ADDRHI	R/W	0xffff	MAC Address0[47:32] This field contains the upper 16 bits [47:32] of the 1st 6-byte MAC address. The MAC uses this field for filtering the received packets and inserting the MAC address in the Transmit Flow Control (Pause) Packets. Value After Reset: 0xffff
0x304	MAC_Address0_Low	The MAC Address0 Low register holds the lower 32 bits of the 1st 6-byte MAC address of the station.	31:0	ADDRLO	R/W	0xffffffff	MAC Address0[31:0] This field contains the lower 32 bits of the 1st 6-byte MAC address. The MAC uses this field for filtering the received packets and inserting the MAC address in the Transmit Flow Control (Pause) Packets. Value After Reset: 0xffffffff
0x308	MAC_Address1_High	The MAC Address1 High register holds the upper 16 bits of the 1st 6-byte MAC address of the station. The first DA byte that is received on the (G)MII interface corresponds to the LS byte (Bits [7:0]) of the MAC Address Low register. For example, if 0x112233445566 is received (0x11 in lane 0 of the first column) on the (G)MII as the destination address, then the MacAddress1 Register [47:0] is compared with 0x665544332211.If the MAC address registers are configured to be double-synchronized to the (G)MII clock domains, then the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the MAC Address1 Low Register are written. For proper synchronization updates, the consecutive writes to this Address Low Register should be performed after at least four clock cycles in the destination clock domain.	31	AE	R	0x1	Address Enable This bit is always set to 1.Values: 0x0 (DISABLE): INVALID : This bit must be always set to 1 0x1 (ENABLE): This bit is always set to 1 Value After Reset: 0x1
			30:24	Reserved_30_y	R	0x0	Reserved. Value After Reset: 0x0
			23:16	DCS	R/W	0x0	DMA Channel Select This field indicates the Rx DMA channel number or DMA channel numbers to which the MAC routes the received packet. When the MAC receives a packet with multicast or broadcast destination address, MAC routes the packet to multiple Rx DMA channels. The PDC field of the MAC_Ext_Configuration register determines if the MAC interprets the value of this field as a binary number or a one-hot encoded number. Value After Reset: 0x0
			15:0	ADDRHI	R/W	0xffff	MAC Address0[47:32] This field contains the upper 16 bits [47:32] of the 1st 6-byte MAC address. The MAC uses this field for filtering the received packets and inserting the MAC address in the Transmit Flow Control (Pause) Packets. Value After Reset: 0xffff
0x30C	MAC_Address1_Low	The MAC Address1 Low register holds the lower 32 bits of the 1st 6-byte MAC address of the station.	31:0	ADDRLO	R/W	0xffffffff	MAC Address0[31:0] This field contains the lower 32 bits of the 1st 6-byte MAC address. The MAC uses this field for filtering the received packets and inserting the MAC address in the Transmit Flow Control (Pause) Packets. Value After Reset: 0xffffffff

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x310	MAC_Address2_High	The MAC Address2 High register holds the upper 16 bits of the 1st 6-byte MAC address of the station. The first DA byte that is received on the (G)MII interface corresponds to the LS byte (Bits [7:0]) of the MAC Address Low register. For example, if 0x112233445566 is received (0x11 in lane 0 of the first column) on the (G)MII as the destination address, then the MacAddress2 Register [47:0] is compared with 0x665544332211.If the MAC address registers are configured to be double-synchronized to the (G)MII clock domains, then the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the MAC Address2 Low Register are written. For proper synchronization updates, the consecutive writes to this Address Low Register should be performed after at least four clock cycles in the destination clock domain.	31	AE	R	0x1	Address Enable This bit is always set to 1.Values: 0x0 (DISABLE): INVALID : This bit must be always set to 1 0x1 (ENABLE): This bit is always set to 1 Value After Reset: 0x1
			30:24	Reserved_30_y	R	0x0	Reserved. Value After Reset: 0x0
			23:16	DCS	R/W	0x0	DMA Channel Select This field indicates the Rx DMA channel number or DMA channel numbers to which the MAC routes the received packet. When the MAC receives a packet with multicast or broadcast destination address, MAC routes the packet to multiple Rx DMA channels. The PDC field of the MAC_Ext_Configuration register determines if the MAC interprets the value of this field as a binary number or a one-hot encoded number. Value After Reset: 0x0
			15:0	ADDRHI	R/W	0xffff	MAC Address0[47:32] This field contains the upper 16 bits [47:32] of the 1st 6-byte MAC address. The MAC uses this field for filtering the received packets and inserting the MAC address in the Transmit Flow Control (Pause) Packets. Value After Reset: 0xffff
0x314	MAC_Address2_Low	The MAC Address2 Low register holds the lower 32 bits of the 1st 6-byte MAC address of the station.	31:0	ADDRLO	R/W	0xffffffff	MAC Address0[31:0] This field contains the lower 32 bits of the 1st 6-byte MAC address. The MAC uses this field for filtering the received packets and inserting the MAC address in the Transmit Flow Control (Pause) Packets. Value After Reset: 0xffffffff
0x318	MAC_Address3_High	The MAC Address3 High register holds the upper 16 bits of the 1st 6-byte MAC address of the station. The first DA byte that is received on the (G)MII interface corresponds to the LS byte (Bits [7:0]) of the MAC Address Low register. For example, if 0x112233445566 is received (0x11 in lane 0 of the first column) on the (G)MII as the destination address, then the MacAddress3 Register [47:0] is compared with 0x665544332211.If the MAC address registers are configured to be double-synchronized to the (G)MII clock domains, then the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the MAC Address3 Low Register are written. For proper synchronization updates, the consecutive writes to this Address Low Register should be performed after at least four clock cycles in the destination clock domain.	31	AE	R	0x1	Address Enable This bit is always set to 1.Values: 0x0 (DISABLE): INVALID : This bit must be always set to 1 0x1 (ENABLE): This bit is always set to 1 Value After Reset: 0x1
			30:24	Reserved_30_y	R	0x0	Reserved. Value After Reset: 0x0
			23:16	DCS	R/W	0x0	DMA Channel Select This field indicates the Rx DMA channel number or DMA channel numbers to which the MAC routes the received packet. When the MAC receives a packet with multicast or broadcast destination address, MAC routes the packet to multiple Rx DMA channels. The PDC field of the MAC_Ext_Configuration register determines if the MAC interprets the value of this field as a binary number or a one-hot encoded number. Value After Reset: 0x0
			15:0	ADDRHI	R/W	0xffff	MAC Address0[47:32] This field contains the upper 16 bits [47:32] of the 1st 6-byte MAC address. The MAC uses this field for filtering the received packets and inserting the MAC address in the Transmit Flow Control (Pause) Packets. Value After Reset: 0xffff
0x31C	MAC_Address3_Low	The MAC Address3 Low register holds the lower 32 bits of the 1st 6-byte MAC address of the station.	31:0	ADDRLO	R/W	0xffffffff	MAC Address0[31:0] This field contains the lower 32 bits of the 1st 6-byte MAC address. The MAC uses this field for filtering the received packets and inserting the MAC address in the Transmit Flow Control (Pause) Packets. Value After Reset: 0xffffffff

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x320	MAC_Address0_High	The MAC Address4 High register holds the upper 16 bits of the 1st 6-byte MAC address of the station. The first DA byte that is received on the (G)MII interface corresponds to the LS byte (Bits [7:0]) of the MAC Address Low register. For example, if 0x112233445566 is received (0x11 in lane 0 of the first column) on the (G)MII as the destination address, then the MacAddress4 Register [47:0] is compared with 0x665544332211.If the MAC address registers are configured to be double-synchronized to the (G)MII clock domains, then the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the MAC Address4 Low Register are written. For proper synchronization updates, the consecutive writes to this Address Low Register should be performed after at least four clock cycles in the destination clock domain.	31	AE	R	0x1	Address Enable This bit is always set to 1.Values: 0x0 (DISABLE): INVALID : This bit must be always set to 1 0x1 (ENABLE): This bit is always set to 1 Value After Reset: 0x1
			30:24	Reserved_30_y	R	0x0	Reserved. Value After Reset: 0x0
			23:16	DCS	R/W	0x0	DMA Channel Select This field indicates the Rx DMA channel number or DMA channel numbers to which the MAC routes the received packet. When the MAC receives a packet with multicast or broadcast destination address, MAC routes the packet to multiple Rx DMA channels. The PDC field of the MAC_Ext_Configuration register determines if the MAC interprets the value of this field as a binary number or a one-hot encoded number. Value After Reset: 0x0
			15:0	ADDRHI	R/W	0xffff	MAC Address0[47:32] This field contains the upper 16 bits [47:32] of the 1st 6-byte MAC address. The MAC uses this field for filtering the received packets and inserting the MAC address in the Transmit Flow Control (Pause) Packets. Value After Reset: 0xffff
0x324	MAC_Address0_Low	The MAC Address4 Low register holds the lower 32 bits of the 1st 6-byte MAC address of the station.	31:0	ADDRLO	R/W	0xffffffff	MAC Address0[31:0] This field contains the lower 32 bits of the 1st 6-byte MAC address. The MAC uses this field for filtering the received packets and inserting the MAC address in the Transmit Flow Control (Pause) Packets. Value After Reset: 0xffffffff
0x328	MAC_Address1_High	The MAC Address5 High register holds the upper 16 bits of the 1st 6-byte MAC address of the station. The first DA byte that is received on the (G)MII interface corresponds to the LS byte (Bits [7:0]) of the MAC Address Low register. For example, if 0x112233445566 is received (0x11 in lane 0 of the first column) on the (G)MII as the destination address, then the MacAddress5 Register [47:0] is compared with 0x665544332211.If the MAC address registers are configured to be double-synchronized to the (G)MII clock domains, then the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the MAC Address5 Low Register are written. For proper synchronization updates, the consecutive writes to this Address Low Register should be performed after at least four clock cycles in the destination clock domain.	31	AE	R	0x1	Address Enable This bit is always set to 1.Values: 0x0 (DISABLE): INVALID : This bit must be always set to 1 0x1 (ENABLE): This bit is always set to 1 Value After Reset: 0x1
			30:24	Reserved_30_y	R	0x0	Reserved. Value After Reset: 0x0
			23:16	DCS	R/W	0x0	DMA Channel Select This field indicates the Rx DMA channel number or DMA channel numbers to which the MAC routes the received packet. When the MAC receives a packet with multicast or broadcast destination address, MAC routes the packet to multiple Rx DMA channels. The PDC field of the MAC_Ext_Configuration register determines if the MAC interprets the value of this field as a binary number or a one-hot encoded number. Value After Reset: 0x0
			15:0	ADDRHI	R/W	0xffff	MAC Address0[47:32] This field contains the upper 16 bits [47:32] of the 1st 6-byte MAC address. The MAC uses this field for filtering the received packets and inserting the MAC address in the Transmit Flow Control (Pause) Packets. Value After Reset: 0xffff
0x32C	MAC_Address1_Low	The MAC Address5 Low register holds the lower 32 bits of the 1st 6-byte MAC address of the station.	31:0	ADDRLO	R/W	0xffffffff	MAC Address0[31:0] This field contains the lower 32 bits of the 1st 6-byte MAC address. The MAC uses this field for filtering the received packets and inserting the MAC address in the Transmit Flow Control (Pause) Packets. Value After Reset: 0xffffffff

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x330	MAC_Address2_High	The MAC Address6 High register holds the upper 16 bits of the 1st 6-byte MAC address of the station. The first DA byte that is received on the (G)MII interface corresponds to the LS byte (Bits [7:0]) of the MAC Address Low register. For example, if 0x112233445566 is received (0x11 in lane 0 of the first column) on the (G)MII as the destination address, then the MacAddress6 Register [47:0] is compared with 0x665544332211.If the MAC address registers are configured to be double-synchronized to the (G)MII clock domains, then the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the MAC Address6 Low Register are written. For proper synchronization updates, the consecutive writes to this Address Low Register should be performed after at least four clock cycles in the destination clock domain.	31	AE	R	0x1	Address Enable This bit is always set to 1.Values: 0x0 (DISABLE): INVALID : This bit must be always set to 1 0x1 (ENABLE): This bit is always set to 1 Value After Reset: 0x1
			30:24	Reserved_30_y	R	0x0	Reserved. Value After Reset: 0x0
			23:16	DCS	R/W	0x0	DMA Channel Select This field indicates the Rx DMA channel number or DMA channel numbers to which the MAC routes the received packet. When the MAC receives a packet with multicast or broadcast destination address, MAC routes the packet to multiple Rx DMA channels. The PDC field of the MAC_Ext_Configuration register determines if the MAC interprets the value of this field as a binary number or a one-hot encoded number. Value After Reset: 0x0
			15:0	ADDRHI	R/W	0xffff	MAC Address0[47:32] This field contains the upper 16 bits [47:32] of the 1st 6-byte MAC address. The MAC uses this field for filtering the received packets and inserting the MAC address in the Transmit Flow Control (Pause) Packets. Value After Reset: 0xffff
0x334	MAC_Address2_Low	The MAC Address6 Low register holds the lower 32 bits of the 1st 6-byte MAC address of the station.	31:0	ADDRLO	R/W	0xffffffff	MAC Address0[31:0] This field contains the lower 32 bits of the 1st 6-byte MAC address. The MAC uses this field for filtering the received packets and inserting the MAC address in the Transmit Flow Control (Pause) Packets. Value After Reset: 0xffffffff
0x338	MAC_Address3_High	The MAC Address7 High register holds the upper 16 bits of the 1st 6-byte MAC address of the station. The first DA byte that is received on the (G)MII interface corresponds to the LS byte (Bits [7:0]) of the MAC Address Low register. For example, if 0x112233445566 is received (0x11 in lane 0 of the first column) on the (G)MII as the destination address, then the MacAddress7 Register [47:0] is compared with 0x665544332211.If the MAC address registers are configured to be double-synchronized to the (G)MII clock domains, then the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the MAC Address6 Low Register are written. For proper synchronization updates, the consecutive writes to this Address Low Register should be performed after at least four clock cycles in the destination clock domain.	31	AE	R	0x1	Address Enable This bit is always set to 1.Values: 0x0 (DISABLE): INVALID : This bit must be always set to 1 0x1 (ENABLE): This bit is always set to 1 Value After Reset: 0x1
			30:24	Reserved_30_y	R	0x0	Reserved. Value After Reset: 0x0
			23:16	DCS	R/W	0x0	DMA Channel Select This field indicates the Rx DMA channel number or DMA channel numbers to which the MAC routes the received packet. When the MAC receives a packet with multicast or broadcast destination address, MAC routes the packet to multiple Rx DMA channels. The PDC field of the MAC_Ext_Configuration register determines if the MAC interprets the value of this field as a binary number or a one-hot encoded number. Value After Reset: 0x0
			15:0	ADDRHI	R/W	0xffff	MAC Address0[47:32] This field contains the upper 16 bits [47:32] of the 1st 6-byte MAC address. The MAC uses this field for filtering the received packets and inserting the MAC address in the Transmit Flow Control (Pause) Packets. Value After Reset: 0xffff
0x33C	MAC_Address3_Low	The MAC Address7 Low register holds the lower 32 bits of the 1st 6-byte MAC address of the station.	31:0	ADDRLO	R/W	0xffffffff	MAC Address0[31:0] This field contains the lower 32 bits of the 1st 6-byte MAC address. The MAC uses this field for filtering the received packets and inserting the MAC address in the Transmit Flow Control (Pause) Packets. Value After Reset: 0xffffffff

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x340	MAC_Address3_High	The MAC Address8 High register holds the upper 16 bits of the 1st 6-byte MAC address of the station. The first DA byte that is received on the (G)MII interface corresponds to the LS byte (Bits [7:0]) of the MAC Address Low register. For example, if 0x112233445566 is received (0x11 in lane 0 of the first column) on the (G)MII as the destination address, then the MacAddress8 Register [47:0] is compared with 0x665544332211.If the MAC address registers are configured to be double-synchronized to the (G)MII clock domains, then the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the MAC Address8 Low Register are written. For proper synchronization updates, the consecutive writes to this Address Low Register should be performed after at least four clock cycles in the destination clock domain.	31	AE	R	0x1	Address Enable This bit is always set to 1.Values: 0x0 (DISABLE): INVALID : This bit must be always set to 1 0x1 (ENABLE): This bit is always set to 1 Value After Reset: 0x1
			30:24	Reserved_30_y	R	0x0	Reserved. Value After Reset: 0x0
			23:16	DCS	R/W	0x0	DMA Channel Select This field indicates the Rx DMA channel number or DMA channel numbers to which the MAC routes the received packet. When the MAC receives a packet with multicast or broadcast destination address, MAC routes the packet to multiple Rx DMA channels. The PDC field of the MAC_Ext_Configuration register determines if the MAC interprets the value of this field as a binary number or a one-hot encoded number. Value After Reset: 0x0
			15:0	ADDRHI	R/W	0xffff	MAC Address0[47:32] This field contains the upper 16 bits [47:32] of the 1st 6-byte MAC address. The MAC uses this field for filtering the received packets and inserting the MAC address in the Transmit Flow Control (Pause) Packets. Value After Reset: 0xffff
0x344	MAC_Address3_Low	The MAC Address8 Low register holds the lower 32 bits of the 1st 6-byte MAC address of the station.	31:0	ADDRLO	R/W	0xffffffff	MAC Address0[31:0] This field contains the lower 32 bits of the 1st 6-byte MAC address. The MAC uses this field for filtering the received packets and inserting the MAC address in the Transmit Flow Control (Pause) Packets. Value After Reset: 0xffffffff
			31:9	Reserved_31_9	R	0x0	Reserved. Value After Reset: 0x0
			8	UCDBC	R/W	0x0	Update MMC Counters for Dropped Broadcast Packets Note: The CNTRST bit has a higher priority than the CNTPRST bit. Therefore, when the software tries to set both bits in the same write cycle, all counters are cleared and the CNTPRST bit is not set. When set, the MAC updates all related MMC Counters for Broadcast packets that are dropped because of the setting of the DBF bit of MAC_Packet_Filter register. When reset, the MMC Counters are not updated for dropped Broadcast packets.Values: 0x0 (DISABLE): Update MMC Counters for Dropped Broadcast Packets is disabled 0x1 (ENABLE): Update MMC Counters for Dropped Broadcast Packets is enabled Value After Reset: 0x0
			7:6	Reserved_7_6	R	0x0	Reserved. Value After Reset: 0x0
			5	CNTPRSTLVL	R/W	0x0	Full-Half Preset When this bit is low and the CNTPRST bit is set, all MMC counters get preset to almost-half value. All octet counters get preset to 0x7FFF_F800 (Half 2KBytes) and all packet-counters gets preset to 0x7FFF_FFF0 (Half 16). When this bit is high and the CNTPRST bit is set, all MMC counters get preset to almost-full value. All octet counters get preset to 0xFFFF_F800 (Full 2KBytes) and all packet-counters gets preset to 0xFFFF_FFF0 (Full 16). For 16-bit counters, the almost-half preset values are 0x7800 and 0x7FF0 for the respective octet and packet counters. Similarly, the almost-full preset values for the 16-bit counters are 0xF800 and 0xFFF0.Values: 0x0 (DISABLE): Full-Half Preset is disabled 0x1 (ENABLE): Full-Half Preset is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x700	MMC_Control	This register establishes the operating mode of MMC.	4	CNTPRST	R/W	0x0	Counters Preset When this bit is set, all counters are initialized or preset to almost full or almost half according to the CNTPRSTLVL bit. This bit is cleared automatically after 1 clock cycle. This bit, along with the CNTPRSTLVL bit, is useful for debugging and testing the assertion of interrupts because of MMC counter becoming half-full or full. Access restrictions apply. initiates the counters preset when application writes 1. MAC writes 0 after the counters preset is complete. MAC ignores when application writes 0. Values: 0x0 (DISABLE): Counters Preset is disabled 0x1 (ENABLE): Counters Preset is enabled Value After Reset: 0x0 Testable: untestable
			3	CNTFREEZ	R/W	0x0	MMC Counter Freeze When this bit is set, it freezes all MMC counters to their current value. Until this bit is reset to 0, no MMC counter is updated because of any transmitted or received packet. If any MMC counter is read with the Reset on Read bit set, then that counter is also cleared in this mode.Values: 0x0 (DISABLE): MMC Counter Freeze is disabled 0x1 (ENABLE): MMC Counter Freeze is enabled Value After Reset: 0x0
			2	RSTONRD	R/W	0x0	Reset on Read When this bit is set, the MMC counters are reset to zero after Read (self-clearing after reset). The counters are cleared when the least significant byte lane (Bits[7:0]) is read.Values: 0x0 (DISABLE): Reset on Read is disabled 0x1 (ENABLE): Reset on Read is enabled Value After Reset: 0x0
			1	CNTSTOPRO	R/W	0x0	Counter Stop Rollover When this bit is set, the counter does not roll over to zero after reaching the maximum value.Values: 0x0 (DISABLE): Counter Stop Rollover is disabled 0x1 (ENABLE): Counter Stop Rollover is enabled Value After Reset: 0x0
			0	CNTRST	R/W	0x0	Counters Reset When this bit is set, all counters are reset. This bit is cleared automatically after 1 clock cycle. Access restrictions apply. initiates the counter reset when application writes 1. MAC writes 0 after the counter reset is complete. MAC ignores when application writes 0. Values: 0x0 (DISABLE): Counters are not reset 0x1 (ENABLE): All counters are reset Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			31:28	Reserved_31_28	R	0x0	Reserved. Value After Reset: 0x0
			27	RXLPITRCIS	R	0x0	MMC Receive LPI transition counter interrupt status This bit is set when the Rx_LPI_Tran_Cntr counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive LPI transition Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive LPI transition Counter Interrupt Status detected Value After Reset: 0x0
			26	RXLPIUSCIS	R	0x0	MMC Receive LPI microsecond counter interrupt status This bit is set when the Rx_LPI_USEC_Cntr counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive LPI microsecond Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive LPI microsecond Counter Interrupt Status detected Value After Reset: 0x0
			25	RXCTRLPIS	R	0x0	MMC Receive Control Packet Counter Interrupt Status This bit is set when the Rx_Control_Packets_Good counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive Control Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive Control Packet Counter Interrupt Status detected Value After Reset: 0x0
			24	RXRCVERRPIS	R	0x0	MMC Receive Error Packet Counter Interrupt Status This bit is set when the Rx_Receive_Error_Packets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive Error Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive Error Packet Counter Interrupt Status detected Value After Reset: 0x0
			23	RXWDOGPIS	R	0x0	MMC Receive Watchdog Error Packet Counter Interrupt Status This bit is set when the Rx_Watchdog_Error_Packets error counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive Watchdog Error Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive Watchdog Error Packet Counter Interrupt Status detected Value After Reset: 0x0
			22	RXVLANGBPIS	R	0x0	MMC Receive VLAN Good Bad Packet Counter Interrupt Status This bit is set when the Rx_VLAN_Packets_Good_Bad counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive VLAN Good Bad Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive VLAN Good Bad Packet Counter Interrupt Status detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			21	RXFOVPIS	R	0x0	MMC Receive FIFO Overflow Packet Counter Interrupt Status This bit is set when the Rx_FIFO_Overflow_Packets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive FIFO Overflow Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive FIFO Overflow Packet Counter Interrupt Status detected Value After Reset: 0x0
			20	RXPAUSPIS	R	0x0	MMC Receive Pause Packet Counter Interrupt Status This bit is set when the Rx_Pause_Packets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive Pause Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive Pause Packet Counter Interrupt Status detected Value After Reset: 0x0
			19	RXORANGEPIS	R	0x0	MMC Receive Out Of Range Error Packet Counter Interrupt Status. This bit is set when the Rx_Out_Of_Range_Type_Packets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive Out Of Range Error Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive Out Of Range Error Packet Counter Interrupt Status detected Value After Reset: 0x0
			18	RXLENERPIS	R	0x0	MMC Receive Length Error Packet Counter Interrupt Status This bit is set when the Rx_Length_Error_Packets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive Length Error Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive Length Error Packet Counter Interrupt Status detected Value After Reset: 0x0
			17	RXUCGPIS	R	0x0	MMC Receive Unicast Good Packet Counter Interrupt Status This bit is set when the Rx_Unicast_Packets_Good counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive Unicast Good Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive Unicast Good Packet Counter Interrupt Status detected Value After Reset: 0x0
			16	RX1024TMAXOCTGBPIS	R	0x0	MMC Receive 1024 to Maximum Octet Good Bad Packet Counter Interrupt Status This bit is set when the Rx_1024ToMaxOctets_Packets_Good_Bad counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive 1024 to Maximum Octet Good Bad Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive 1024 to Maximum Octet Good Bad Packet Counter Interrupt Status detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x704	MMC_Rx_Interrupt	This register maintains the interrupts generated from all Receive statistics counters. The MMC Receive Interrupt register maintains the interrupts that are generated when the following occur: - Receive statistic counters reach half of their maximum values (0x8000_0000 for 32 bit counter and 0x8000 for 16 bit counter). - Receive statistic counters cross their maximum values (0xFFFF_FFFF for 32 bit counter and 0xFFFF for 16 bit counter). When the Counter Stop Rollover is set, interrupts are set but the counter remains at all-ones. The MMC Receive Interrupt register is a 32 bit register. An interrupt bit is cleared when the respective MMC counter that caused the interrupt is read. The least significant byte lane (Bits[7:0]) of the respective counter must be read to clear the interrupt bit. Note: R_SS_RC means that this register bit is set internally, and it is cleared when the Counter register is read.	15	RX512T1023OCTGBPIS	R	0x0	MMC Receive 512 to 1023 Octet Good Bad Packet Counter Interrupt Status This bit is set when the Rx_512To1023Octets_Packets_Good_Bad counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive 512 to 1023 Octet Good Bad Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive 512 to 1023 Octet Good Bad Packet Counter Interrupt Status detected Value After Reset: 0x0
			14	RX256T511OCTGBPIS	R	0x0	MMC Receive 256 to 511 Octet Good Bad Packet Counter Interrupt Status This bit is set when the Rx_256To511Octets_Packets_Good_Bad counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive 256 to 511 Octet Good Bad Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive 256 to 511 Octet Good Bad Packet Counter Interrupt Status detected Value After Reset: 0x0
			13	RX128T255OCTGBPIS	R	0x0	MMC Receive 128 to 255 Octet Good Bad Packet Counter Interrupt Status This bit is set when the Rx_128To255Octets_Packets_Good_Bad counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive 128 to 255 Octet Good Bad Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive 128 to 255 Octet Good Bad Packet Counter Interrupt Status detected Value After Reset: 0x0
			12	RX65T127OCTGBPIS	R	0x0	MMC Receive 65 to 127 Octet Good Bad Packet Counter Interrupt Status This bit is set when the Rx_65To127Octets_Packets_Good_Bad counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive 65 to 127 Octet Good Bad Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive 65 to 127 Octet Good Bad Packet Counter Interrupt Status detected Value After Reset: 0x0
			11	RX64OCTGBPIS	R	0x0	MMC Receive 64 Octet Good Bad Packet Counter Interrupt Status This bit is set when the Rx_64Octets_Packets_Good_Bad counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive 64 Octet Good Bad Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive 64 Octet Good Bad Packet Counter Interrupt Status detected Value After Reset: 0x0
			10	RXOSIZEGPIS	R	0x0	MMC Receive Oversize Good Packet Counter Interrupt Status This bit is set when the Rx_Oversize_Packets_Good counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive Oversize Good Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive Oversize Good Packet Counter Interrupt Status detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			9	RXUSIZEGPIS	R	0x0	MMC Receive Undersize Good Packet Counter Interrupt Status This bit is set when the Rx_Undersize_Packets_Good counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive Undersize Good Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive Undersize Good Packet Counter Interrupt Status detected Value After Reset: 0x0
			8	RXJABERPIS	R	0x0	MMC Receive Jabber Error Packet Counter Interrupt Status This bit is set when the Rx_Jabber_Error_Packets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive Jabber Error Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive Jabber Error Packet Counter Interrupt Status detected Value After Reset: 0x0
			7	RXRUNTPIS	R	0x0	MMC Receive Runt Packet Counter Interrupt Status This bit is set when the Rx_Runt_Error_Packets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive Runt Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive Runt Packet Counter Interrupt Status detected Value After Reset: 0x0
			6	RXALGNERPIS	R	0x0	MMC Receive Alignment Error Packet Counter Interrupt Status This bit is set when the Rx_Alignment_Error_Packets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive Alignment Error Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive Alignment Error Packet Counter Interrupt Status detected Value After Reset: 0x0
			5	RXCRCERPIS	R	0x0	MMC Receive CRC Error Packet Counter Interrupt Status This bit is set when the Rx_CRC_Error_Packets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive CRC Error Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive CRC Error Packet Counter Interrupt Status detected Value After Reset: 0x0
			4	RXMCGPIS	R	0x0	MMC Receive Multicast Good Packet Counter Interrupt Status This bit is set when the Rx_Multicast_Packets_Good counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive Multicast Good Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive Multicast Good Packet Counter Interrupt Status detected Value After Reset: 0x0
			3	RXBCGPIS	R	0x0	MMC Receive Broadcast Good Packet Counter Interrupt Status This bit is set when the Rx_Broadcast_Packets_Good counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive Broadcast Good Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive Broadcast Good Packet Counter Interrupt Status detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			2	RXGOCTIS	R	0x0	MMC Receive Good Octet Counter Interrupt Status This bit is set when the Rx_Octet_Count_Good counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive Good Octet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive Good Octet Counter Interrupt Status detected Value After Reset: 0x0
			1	RXGBOCTIS	R	0x0	MMC Receive Good Bad Octet Counter Interrupt Status This bit is set when the Rx_Octet_Count_Good_Bad counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive Good Bad Octet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive Good Bad Octet Counter Interrupt Status detected Value After Reset: 0x0
			0	RXGBPKTIS	R	0x0	MMC Receive Good Bad Packet Counter Interrupt Status This bit is set when the Rx_Packets_Count_Good_Bad counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive Good Bad Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive Good Bad Packet Counter Interrupt Status detected Value After Reset: 0x0
			31:28	Reserved_31_28	R	0x0	Reserved. Value After Reset: 0x0
			27	TXLPITRCIS	R	0x0	MMC Transmit LPI transition counter interrupt status This bit is set when the Tx_LPI_Tran_Cntr counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Transmit LPI transition Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Transmit LPI transition Counter Interrupt Status detected Value After Reset: 0x0
			26	TXLPIUSCIS	R	0x0	MMC Transmit LPI microsecond counter interrupt status This bit is set when the Tx_LPI_USEC_Cntr counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Transmit LPI microsecond Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Transmit LPI microsecond Counter Interrupt Status detected Value After Reset: 0x0
			25	TXOSIZEGPIS	R	0x0	MMC Transmit Oversize Good Packet Counter Interrupt Status This bit is set when the Tx_OSize_Packets_Good counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Transmit Oversize Good Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Transmit Oversize Good Packet Counter Interrupt Status detected Value After Reset: 0x0
			24	TXVLANGPIS	R	0x0	MMC Transmit VLAN Good Packet Counter Interrupt Status This bit is set when the Tx_VLAN_Packets_Good counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Transmit VLAN Good Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Transmit VLAN Good Packet Counter Interrupt Status detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			23	TXPAUSPIS	R	0x0	MMC Transmit Pause Packet Counter Interrupt Status This bit is set when the Tx_Pause_Packets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Transmit Pause Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Transmit Pause Packet Counter Interrupt Status detected Value After Reset: 0x0
			22			#VALUE!	Reserved Field: Yes
			21	TXGPKTIS	R	0x0	MMC Transmit Good Packet Counter Interrupt Status This bit is set when the Tx_Packet_Count_Good counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Transmit Good Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Transmit Good Packet Counter Interrupt Status detected Value After Reset: 0x0
			20	TXGOCTIS	R	0x0	MMC Transmit Good Octet Counter Interrupt Status This bit is set when the Tx_Octet_Count_Good counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Transmit Good Octet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Transmit Good Octet Counter Interrupt Status detected Value After Reset: 0x0
			19:14			#VALUE!	Reserved Field: Yes
			13	TXUFLOWERPIS	R	0x0	MMC Transmit Underflow Error Packet Counter Interrupt Status This bit is set when the Tx_Underflow_Error_Packets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Transmit Underflow Error Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Transmit Underflow Error Packet Counter Interrupt Status detected Value After Reset: 0x0
			12	TXBCGBPIS	R	0x0	MMC Transmit Broadcast Good Bad Packet Counter Interrupt Status This bit is set when the Tx_Broadcast_Packets_Good_Bad counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Transmit Broadcast Good Bad Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Transmit Broadcast Good Bad Packet Counter Interrupt Status detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x708	MMC_Tx_Interrupt	This register maintains the interrupts generated from all Transmit statistics counters. The MMC Transmit Interrupt register maintains the interrupts generated when transmit statistic counters reach half their maximum values (0x8000_0000 for 32 bit counter and 0x8000 for 16 bit counter), and when they cross their maximum values (0xFFFF_FFFF for 32-bit counter and 0xFFFF for 16-bit counter). When Counter Stop Rollover is set, the interrupts are set but the counter remains at all-ones. The MMC Transmit Interrupt register is a 32 bit register. An interrupt bit is cleared when the respective MMC counter that caused the interrupt is read. The least significant byte lane (Bits[7:0]) of the respective counter must be read to clear the interrupt bit.	11	TXMCGBPIS	R	0x0	MMC Transmit Multicast Good Bad Packet Counter Interrupt Status The bit is set when the Tx_Multicast_Packets_Good_Bad counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Transmit Multicast Good Bad Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Transmit Multicast Good Bad Packet Counter Interrupt Status detected Value After Reset: 0x0
			10	TXUCGBPIS	R	0x0	MMC Transmit Unicast Good Bad Packet Counter Interrupt Status This bit is set when the Tx_Unicast_Packets_Good_Bad counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Transmit Unicast Good Bad Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Transmit Unicast Good Bad Packet Counter Interrupt Status detected Value After Reset: 0x0
			9	TX1024TMAXOCTGBPIS	R	0x0	MMC Transmit 1024 to Maximum Octet Good Bad Packet Counter Interrupt Status This bit is set when the Tx_1024ToMaxOctets_Packets_Good_Bad counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Transmit 1024 to Maximum Octet Good Bad Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Transmit 1024 to Maximum Octet Good Bad Packet Counter Interrupt Status detected Value After Reset: 0x0
			8	TX512T1023OCTGBPIS	R	0x0	MMC Transmit 512 to 1023 Octet Good Bad Packet Counter Interrupt Status This bit is set when the Tx_512To1023Octets_Packets_Good_Bad counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Transmit 512 to 1023 Octet Good Bad Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Transmit 512 to 1023 Octet Good Bad Packet Counter Interrupt Status detected Value After Reset: 0x0
			7	TX256T511OCTGBPIS	R	0x0	MMC Transmit 256 to 511 Octet Good Bad Packet Counter Interrupt Status This bit is set when the Tx_512To1023Octets_Packets_Good_Bad counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Transmit 256 to 511 Octet Good Bad Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Transmit 256 to 511 Octet Good Bad Packet Counter Interrupt Status detected Value After Reset: 0x0
			6	TX128T255OCTGBPIS	R	0x0	MMC Transmit 128 to 255 Octet Good Bad Packet Counter Interrupt Status This bit is set when the Tx_128To255Octets_Packets_Good_Bad counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Transmit 128 to 255 Octet Good Bad Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Transmit 128 to 255 Octet Good Bad Packet Counter Interrupt Status detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			5	TX65T127OCTGBPIS	R	0x0	MMC Transmit 65 to 127 Octet Good Bad Packet Counter Interrupt Status This bit is set when the Tx_65To127Octets_Packets_Good_Bad counter reaches half the maximum value, and also when it reaches the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Transmit 65 to 127 Octet Good Bad Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Transmit 65 to 127 Octet Good Bad Packet Counter Interrupt Status detected Value After Reset: 0x0
			4	TX64OCTGBPIS	R	0x0	MMC Transmit 64 Octet Good Bad Packet Counter Interrupt Status This bit is set when the Tx_64Octets_Packets_Good_Bad counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Transmit 64 Octet Good Bad Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Transmit 64 Octet Good Bad Packet Counter Interrupt Status detected Value After Reset: 0x0
			3	TXMCGPIS	R	0x0	MMC Transmit Multicast Good Packet Counter Interrupt Status This bit is set when the Tx_Multicast_Packets_Good counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Transmit Multicast Good Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Transmit Multicast Good Packet Counter Interrupt Status detected Value After Reset: 0x0
			2	TXBCGPIS	R	0x0	MMC Transmit Broadcast Good Packet Counter Interrupt Status This bit is set when the Tx_Broadcast_Packets_Good counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Transmit Broadcast Good Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Transmit Broadcast Good Packet Counter Interrupt Status detected Value After Reset: 0x0
			1	TXGBPKTIS	R	0x0	MMC Transmit Good Bad Packet Counter Interrupt Status This bit is set when the Tx_Packet_Count_Good_Bad counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Transmit Good Bad Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Transmit Good Bad Packet Counter Interrupt Status detected Value After Reset: 0x0
			0	TXGBOCTIS	R	0x0	MMC Transmit Good Bad Octet Counter Interrupt Status This bit is set when the Tx_Octet_Count_Good_Bad counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Transmit Good Bad Octet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Transmit Good Bad Octet Counter Interrupt Status detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			31:28	Reserved_31_28	R	0x0	Reserved. Value After Reset: 0x0
			27	RXLPITRCIM	R/W	0x0	MMC Receive LPI transition counter interrupt Mask Setting this bit masks the interrupt when the Rx_LPI_Tran_Cntr counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive LPI transition counter interrupt Mask is disabled 0x1 (ENABLE): MMC Receive LPI transition counter interrupt Mask is enabled Value After Reset: 0x0
			26	RXLPIUSCIM	R/W	0x0	MMC Receive LPI microsecond counter interrupt Mask Setting this bit masks the interrupt when the Rx_LPI_USEC_Cntr counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive LPI microsecond counter interrupt Mask is disabled 0x1 (ENABLE): MMC Receive LPI microsecond counter interrupt Mask is enabled Value After Reset: 0x0
			25	RXCTRLPIM	R/W	0x0	MMC Receive Control Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_Control_Packets_Good counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive Control Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive Control Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			24	RXRCVERRPIM	R/W	0x0	MMC Receive Error Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_Receive_Error_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive Error Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive Error Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			23	RXWDOGPIM	R/W	0x0	MMC Receive Watchdog Error Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_Watchdog_Error_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive Watchdog Error Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive Watchdog Error Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			22	RXVLANGBPIM	R/W	0x0	MMC Receive VLAN Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_VLAN_Packets_Good_Bad counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive VLAN Good Bad Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive VLAN Good Bad Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			21	RXFOVPIM	R/W	0x0	MMC Receive FIFO Overflow Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_FIFO_Overflow_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive FIFO Overflow Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive FIFO Overflow Packet Counter Interrupt Mask is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			20	RXPAUSPIM	R/W	0x0	MMC Receive Pause Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_Pause_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive Pause Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive Pause Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			19	RXORANGEPIM	R/W	0x0	MMC Receive Out Of Range Error Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_Out_Of_Range_Type_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive Out Of Range Error Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive Out Of Range Error Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			18	RXLENERPIM	R/W	0x0	MMC Receive Length Error Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_Length_Error_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive Length Error Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive Length Error Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			17	RXUCGPIM	R/W	0x0	MMC Receive Unicast Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_Unicast_Packets_Good counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive Unicast Good Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive Unicast Good Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			16	RX1024TMAXOCTGBPIM	R/W	0x0	MMC Receive 1024 to Maximum Octet Good Bad Packet Counter Interrupt Mask. Setting this bit masks the interrupt when the Rx_1024ToMaxOctets_Packets_Good_Bad counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive 1024 to Maximum Octet Good Bad Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive 1024 to Maximum Octet Good Bad Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			15	RX512T1023OCTGBPIM	R/W	0x0	MMC Receive 512 to 1023 Octet Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_512To1023Octets_Packets_Good_Bad counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive 512 to 1023 Octet Good Bad Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive 512 to 1023 Octet Good Bad Packet Counter Interrupt Mask is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x70c	MMC_Rx_Interrupt_Mask	This register maintains the masks for interrupts generated from all Receive statistics counters. The MMC Receive Interrupt Mask register maintains the masks for the interrupts generated when receive statistic counters reach half of their maximum value or the maximum values. This register is 32 bit wide.	14	RX256T511OCTGBPIM	R/W	0x0	MMC Receive 256 to 511 Octet Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_256To511Octets_Packets_Good_Bad counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive 256 to 511 Octet Good Bad Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive 256 to 511 Octet Good Bad Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			13	RX128T255OCTGBPIM	R/W	0x0	MMC Receive 128 to 255 Octet Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_128To255Octets_Packets_Good_Bad counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive 128 to 255 Octet Good Bad Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive 128 to 255 Octet Good Bad Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			12	RX65T127OCTGBPIM	R/W	0x0	MMC Receive 65 to 127 Octet Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_65To127Octets_Packets_Good_Bad counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive 65 to 127 Octet Good Bad Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive 65 to 127 Octet Good Bad Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			11	RX64OCTGBPIM	R/W	0x0	MMC Receive 64 Octet Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_64Octets_Packets_Good_Bad counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive 64 Octet Good Bad Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive 64 Octet Good Bad Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			10	RXOSIZEGPIM	R/W	0x0	MMC Receive Oversize Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_Oversize_Packets_Good counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive Oversize Good Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive Oversize Good Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			9	RXUSIZEGPIM	R/W	0x0	MMC Receive Undersize Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_Undersize_Packets_Good counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive Undersize Good Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive Undersize Good Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			8	RXJABERPIM	R/W	0x0	MMC Receive Jabber Error Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_Jabber_Error_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive Jabber Error Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive Jabber Error Packet Counter Interrupt Mask is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			7	RXRUNTPIM	R/W	0x0	MMC Receive Runt Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_Runt_Error_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive Runt Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive Runt Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			6	RXALGNERPIM	R/W	0x0	MMC Receive Alignment Error Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_Alignment_Error_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive Alignment Error Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive Alignment Error Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			5	RXCRCERPIM	R/W	0x0	MMC Receive CRC Error Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_CRC_Error_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive CRC Error Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive CRC Error Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			4	RXMCGPIM	R/W	0x0	MMC Receive Multicast Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_Multicast_Packets_Good counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive Multicast Good Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive Multicast Good Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			3	RXBCGPIM	R/W	0x0	MMC Receive Broadcast Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_Broadcast_Packets_Good counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive Broadcast Good Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive Broadcast Good Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			2	RXGOCTIM	R/W	0x0	MMC Receive Good Octet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_Octet_Count_Good counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive Good Octet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive Good Octet Counter Interrupt Mask is enabled Value After Reset: 0x0
			1	RXGBOCTIM	R/W	0x0	MMC Receive Good Bad Octet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_Octet_Count_Good_Bad counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive Good Bad Octet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive Good Bad Octet Counter Interrupt Mask is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	RXGBPKTIM	R/W	0x0	MMC Receive Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_Packets_Count_Good_Bad counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive Good Bad Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive Good Bad Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			31:28	Reserved_31_28	R	0x0	Reserved. Value After Reset: 0x0
			27	TXLPITRCIM	R/W	0x0	MMC Transmit LPI transition counter interrupt Mask Setting this bit masks the interrupt when the Tx_LPI_Tran_Cntr counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Transmit LPI transition counter interrupt Mask is disabled 0x1 (ENABLE): MMC Transmit LPI transition counter interrupt Mask is enabled Value After Reset: 0x0
			26	TXLPIUSCIM	R/W	0x0	MMC Transmit LPI microsecond counter interrupt Mask Setting this bit masks the interrupt when the Tx_LPI_USEC_Cntr counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Transmit LPI microsecond counter interrupt Mask is disabled 0x1 (ENABLE): MMC Transmit LPI microsecond counter interrupt Mask is enabled Value After Reset: 0x0
			25	TXOSIZEGPIM	R/W	0x0	MMC Transmit Oversize Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_OSize_Packets_Good counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Transmit Oversize Good Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Transmit Oversize Good Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			24	TXVLANGPIM	R/W	0x0	MMC Transmit VLAN Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_VLAN_Packets_Good counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Transmit VLAN Good Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Transmit VLAN Good Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			23	TXPAUSPIM	R/W	0x0	MMC Transmit Pause Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_Pause_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Transmit Pause Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Transmit Pause Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			22			#VALUE!	Reserved Field: Yes
			21	TXGPKTIM	R/W	0x0	MMC Transmit Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_Packet_Count_Good counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Transmit Good Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Transmit Good Packet Counter Interrupt Mask is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x710	MMC_Tx_Interrupt_Mask	This register maintains the masks for interrupts generated from all Transmit statistics counters. The MMC Transmit Interrupt Mask register maintains the masks for the interrupts generated when the transmit statistic counters reach half of their maximum value or the maximum values. This register is 32 bit wide.	20	TXGOCTIM	R/W	0x0	MMC Transmit Good Octet Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_Octet_Count_Good counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Transmit Good Octet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Transmit Good Octet Counter Interrupt Mask is enabled Value After Reset: 0x0
			19:14			#VALUE!	Reserved Field: Yes
			13	TXUFLOWERPIM	R/W	0x0	MMC Transmit Underflow Error Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_Underflow_Error_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Transmit Underflow Error Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Transmit Underflow Error Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			12	TXBCGBPIM	R/W	0x0	MMC Transmit Broadcast Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_Broadcast_Packets_Good_Bad counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Transmit Broadcast Good Bad Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Transmit Broadcast Good Bad Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			11	TXMCGBPIM	R/W	0x0	MMC Transmit Multicast Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_Multicast_Packets_Good_Bad counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Transmit Multicast Good Bad Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Transmit Multicast Good Bad Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			10	TXUCGBPIM	R/W	0x0	MMC Transmit Unicast Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_Unicast_Packets_Good_Bad counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Transmit Unicast Good Bad Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Transmit Unicast Good Bad Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			9	TX1024TMAXOCTGBPIM	R/W	0x0	MMC Transmit 1024 to Maximum Octet Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_1024ToMaxOctets_Packets_Good_Bad counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Transmit 1024 to Maximum Octet Good Bad Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Transmit 1024 to Maximum Octet Good Bad Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			8	TX512T1023OCTGBPIM	R/W	0x0	MMC Transmit 512 to 1023 Octet Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_512To1023Octets_Packets_Good_Bad counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Transmit 512 to 1023 Octet Good Bad Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Transmit 512 to 1023 Octet Good Bad Packet Counter Interrupt Mask is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			7	TX256T511OCTGBPIM	R/W	0x0	MMC Transmit 256 to 511 Octet Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_512To1023Octets_Packets_Good_Bad counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Transmit 256 to 511 Octet Good Bad Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Transmit 256 to 511 Octet Good Bad Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			6	TX128T255OCTGBPIM	R/W	0x0	MMC Transmit 128 to 255 Octet Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_128To255Octets_Packets_Good_Bad counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Transmit 128 to 255 Octet Good Bad Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Transmit 128 to 255 Octet Good Bad Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			5	TX65T127OCTGBPIM	R/W	0x0	MMC Transmit 65 to 127 Octet Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_65To127Octets_Packets_Good_Bad counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Transmit 65 to 127 Octet Good Bad Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Transmit 65 to 127 Octet Good Bad Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			4	TX64OCTGBPIM	R/W	0x0	MMC Transmit 64 Octet Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_64Octets_Packets_Good_Bad counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Transmit 64 Octet Good Bad Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Transmit 64 Octet Good Bad Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			3	TXMCGPIM	R/W	0x0	MMC Transmit Multicast Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_Multicast_Packets_Good counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Transmit Multicast Good Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Transmit Multicast Good Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			2	TXBCGPIM	R/W	0x0	MMC Transmit Broadcast Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_Broadcast_Packets_Good counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Transmit Broadcast Good Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Transmit Broadcast Good Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			1	TXGBPKTIM	R/W	0x0	MMC Transmit Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_Packet_Count_Good_Bad counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Transmit Good Bad Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Transmit Good Bad Packet Counter Interrupt Mask is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	TXGBOCTIM	R/W	0x0	MMC Transmit Good Bad Octet Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_Octet_Count_Good_Bad counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Transmit Good Bad Octet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Transmit Good Bad Octet Counter Interrupt Mask is enabled Value After Reset: 0x0
0x714	Tx_Octet_Count_Good_Bad	This register provides the number of bytes transmitted by the DWC_ether_qos, exclusive of preamble and retried bytes, in good and bad packets.	31:0	TXOCTGB	R	0x0	Tx Octet Count Good Bad This field indicates the number of bytes transmitted, exclusive of preamble and retried bytes, in good and bad packets. Value After Reset: 0x0
0x718	Tx_Packet_Count_Good_Bad	This register provides the number of good and bad packets transmitted by DWC_ether_qos, exclusive of retried packets.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	TXPKTGB	R	0x0	Tx Packet Count Good Bad This field indicates the number of good and bad packets transmitted, exclusive of retried packets. Value After Reset: 0x0
0x71c	Tx_Broadcast_Packets_Good	This register provides the number of good broadcast packets transmitted by DWC_ether_qos.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	TXBCASTG	R	0x0	Tx Broadcast Packets Good This field indicates the number of good broadcast packets transmitted. Value After Reset: 0x0
0x720	Tx_Multicast_Packets_Good	This register provides the number of good multicast packets transmitted by DWC_ether_qos.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	TXMCASTG	R	0x0	Tx Multicast Packets Good This field indicates the number of good multicast packets transmitted. Value After Reset: 0x0
0x724	Tx_64Octets_Packets_Good_Bad	This register provides the number of good and bad packets transmitted by DWC_ether_qos with length 64 bytes, exclusive of preamble and retried packets.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	TX64OCTGB	R	0x0	Tx 64Octets Packets Good_Bad This field indicates the number of good and bad packets transmitted with length 64 bytes, exclusive of preamble and retried packets. Value After Reset: 0x0
0x728	Tx_65To127Octets_Packets_Good_Bad	This register provides the number of good and bad packets transmitted by DWC_ether_qos with length between 65 and 127 (inclusive) bytes, exclusive of preamble and retried packets.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	TX65_127OCTGB	R	0x0	Tx 65To127Octets Packets Good BadThis field indicates the number of good and bad packets transmitted with length between 65 and 127 (inclusive) bytes, exclusive of preamble and retried packets. Value After Reset: 0x0
0x72c	Tx_128To255Octets_Packets_Good_Bad	This register provides the number of good and bad packets transmitted by DWC_ether_qos with length between 128 to 255 (inclusive) bytes, exclusive of preamble and retried packets.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	TX128_255OCTGB	R	0x0	Tx 128To255Octets Packets Good Bad This field indicates the number of good and bad packets transmitted with length between 128 and 255 (inclusive) bytes, exclusive of preamble and retried packets. Value After Reset: 0x0
		This register provides the number of good and bad	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x730	Tx_256To511Octets_Packets_Good_Bad	packets transmitted by DWC_ether_qos with length between 256 to 511 (inclusive) bytes, exclusive of preamble and retried packets.	15:0	TX256_511OCTGB	R	0x0	Tx 256To511Octets Packets Good Bad This field indicates the number of good and bad packets transmitted with length between 256 and 511 (inclusive) bytes, exclusive of preamble and retried packets. Value After Reset: 0x0
0x734	Tx_512To1023Octets_Packets_Good_Bad	This register provides the number of good and bad packets transmitted by DWC_ether_qos with length 512 to 1023 (inclusive) bytes, exclusive of preamble and retried packets.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	TX512_1023OCTGB	R	0x0	Tx 512To1023Octets Packets Good Bad This field indicates the number of good and bad packets transmitted with length between 512 and 1023 (inclusive) bytes, exclusive of preamble and retried packets. Value After Reset: 0x0
0x738	Tx_1024ToMaxOctets_Packets_Good_Bad	This register provides the number of good and bad packets transmitted by DWC_ether_qos with length 1024 to maxsize (inclusive) bytes, exclusive of preamble and retried packets.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	TX1024_MAXOCTGB	R	0x0	Tx 1024ToMaxOctets Packets Good Bad This field indicates the number of good and bad packets transmitted with length between 1024 and maxsize (inclusive) bytes, exclusive of preamble and retried packets. Value After Reset: 0x0
0x73c	Tx_Unicast_Packets_Good_Bad	This register provides the number of good and bad unicast packets transmitted by DWC_ether_qos.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	TXUCASTGB	R	0x0	Tx Unicast Packets Good Bad This field indicates the number of good and bad unicast packets transmitted. Value After Reset: 0x0
0x740	Tx_Multicast_Packets_Good_Bad	This register provides the number of good and bad multicast packets transmitted by DWC_ether_qos.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	TXMCASTGB	R	0x0	Tx Multicast Packets Good Bad This field indicates the number of good and bad multicast packets transmitted. Value After Reset: 0x0
0x744	Tx_Broadcast_Packets_Good_Bad	This register provides the number of good and bad broadcast packets transmitted by DWC_ether_qos.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	TXBCASTGB	R	0x0	Tx Broadcast Packets Good Bad This field indicates the number of good and bad broadcast packets transmitted. Value After Reset: 0x0
0x748	Tx_Underflow_Error_Packets	This register provides the number of packets aborted by DWC_ether_qos because of packets underflow error.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	TXUNDRFLW	R	0x0	Tx Underflow Error Packets This field indicates the number of packets aborted because of packets underflow error. Value After Reset: 0x0
0x764	Tx_Octet_Count_Good	This register provides the number of bytes transmitted by DWC_ether_qos, exclusive of preamble, only in good packets.	31:0	TXOCTG	R	0x0	Tx Octet Count Good This field indicates the number of bytes transmitted, exclusive of preamble, only in good packets. Value After Reset: 0x0
0x768	Tx_Packet_Count_Good	This register provides the number of good packets transmitted by DWC_ether_qos.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	TXPKTG	R	0x0	Tx Packet Count Good This field indicates the number of good packets transmitted. Value After Reset: 0x0
		This register provides the number of good Bytes	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x770	Tx_Pause_Packets	This register provides the number of good Pause packets transmitted by DWC_ether_qos.	15:0	TXPAUSE	R	0x0	Tx Pause Packets This field indicates the number of good Pause packets transmitted. Value After Reset: 0x0
0x774	Tx_VLAN_Packets_Good	This register provides the number of good VLAN packets transmitted by DWC_ether_qos.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	TXVLANG	R	0x0	Tx VLAN Packets Good This field provides the number of good VLAN packets transmitted. Value After Reset: 0x0
0x778	Tx_OSize_Packets_Good	This register provides the number of packets transmitted by DWC_ether_qos without errors and with length greater than the maxsize (1,518 or 1,522 bytes for VLAN tagged packets; 2000 bytes if enabled in S2KP bit of the MAC_Configuration register).	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	TXOSIZG	R	0x0	Tx OSize Packets Good This field indicates the number of packets transmitted without errors and with length greater than the maxsize (1,518 or 1,522 bytes for VLAN tagged packets; 2000 bytes if enabled in S2KP bit of the MAC_Configuration register). Value After Reset: 0x0
0x780	Rx_Packets_Count_Good_Bad	This register provides the number of good and bad packets received by DWC_ether_qos.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXPKTGB	R	0x0	Rx Packets Count Good Bad This field indicates the number of good and bad packets received. Value After Reset: 0x0
0x784	Rx_Octet_Count_Good_Bad	This register provides the number of bytes received by DWC_ther_qos, exclusive of preamble, in good and bad packets.	31:0	RXOCTGB	R	0x0	Rx Octet Count Good Bad This field indicates the number of bytes received, exclusive of preamble, in good and bad packets. Value After Reset: 0x0
0x788	Rx_Octet_Count_Good	This register provides the number of bytes received by DWC_ether_qos, exclusive of preamble, only in good packets.	31:0	RXOCTG	R	0x0	Rx Octet Count Good This field indicates the number of bytes received, exclusive of preamble, only in good packets. Value After Reset: 0x0
0x78c	Rx_Broadcast_Packets_Good	This register provides the number of good broadcast packets received by DWC_ether_qos.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXBCASTG	R	0x0	Rx Broadcast Packets Good This field indicates the number of good broadcast packets received. Value After Reset: 0x0
0x790	Rx_Multicast_Packets_Good	This register provides the number of good multicast packets received by DWC_ether_qos.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXMCASTG	R	0x0	Rx Multicast Packets Good This field indicates the number of good multicast packets received. Value After Reset: 0x0
0x794	Rx_CRC_Error_Packets	This register provides the number of packets received by DWC_ether_qos with CRC error.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXCRCERR	R	0x0	Rx CRC Error Packets This field indicates the number of packets received with CRC error. Value After Reset: 0x0
0x798	Rx_Alignment_Error_Packets	This register provides the number of packets received by DWC_ether_qos with alignment (dribble) error. It is valid only in 10/100 mode.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXALGNERR	R	0x0	Rx Alignment Error Packets This field indicates the number of packets received with alignment (dribble) error. It is valid only in 10/100 mode. Value After Reset: 0x0
		This register provides the number of packets received	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x79c	Rx_Runt_Error_Packets	by DWC_ether_qos with runt (length less than 64 bytes and CRC error) error.	15:0	RXRUNTERR	R	0x0	Rx Runt Error Packets This field indicates the number of packets received with runt (length less than 64 bytes and CRC error) error. Value After Reset: 0x0
0x7a0	Rx_Jabber_Error_Packets	This register provides the number of giant packets received by DWC_ether_qos with length (including CRC) greater than 1,518 bytes (1,522 bytes for VLAN tagged) and with CRC error. If Jumbo Packet mode is enabled, packets of length greater than 9,018 bytes (9,022 bytes for VLAN tagged) are considered as giant packets.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXJABERR	R	0x0	Rx Jabber Error Packets This field indicates the number of giant packets received with length (including CRC) greater than 1,518 bytes (1,522 bytes for VLAN tagged) and with CRC error. If Jumbo Packet mode is enabled, packets of length greater than 9,018 bytes (9,022 bytes for VLAN tagged) are considered as giant packets. Value After Reset: 0x0
0x7a4	Rx_Undersize_Packets_Good	This register provides the number of packets received by DWC_ether_qos with length less than 64 bytes, without any errors.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXUNDERSZG	R	0x0	Rx Undersize Packets Good This field indicates the number of packets received with length less than 64 bytes, without any errors. Value After Reset: 0x0
0x7a8	Rx_Oversize_Packets_Good	This register provides the number of packets received by DWC_ether_qos without errors, with length greater than the maxsize (1,518 bytes or 1,522 bytes for VLAN tagged packets; 2000 bytes if enabled in the S2KP bit of the MAC_Configuration register).	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXOVERSZG	R	0x0	Rx Oversize Packets Good This field indicates the number of packets received without errors, with length greater than the maxsize (1,518 bytes or 1,522 bytes for VLAN tagged packets; 2000 bytes if enabled in the S2KP bit of the MAC_Configuration register). Value After Reset: 0x0
0x7ac	Rx_64Octets_Packets_Good_Bad	This register provides the number of good and bad packets received by DWC_ether_qos with length 64 bytes, exclusive of the preamble.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RX64OCTGB	R	0x0	Rx 64 Octets Packets Good Bad This field indicates the number of good and bad packets received with length 64 bytes, exclusive of the preamble. Value After Reset: 0x0
0x7b0	Rx_65To127Octets_Packets_Good_Bad	This register provides the number of good and bad packets received by DWC_ether_qos with length between 65 and 127 (inclusive) bytes, exclusive of the preamble.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RX65_127OCTGB	R	0x0	Rx 65-127 Octets Packets Good BadThis field indicates the number of good and bad packets received with length between 65 and 127 (inclusive) bytes, exclusive of the preamble. Value After Reset: 0x0
0x7b4	Rx_128To255Octets_Packets_Good_Bad	This register provides the number of good and bad packets received by DWC_ether_qos with length between 128 and 255 (inclusive) bytes, exclusive of the preamble.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RX128_255OCTGB	R	0x0	Rx 128-255 Octets Packets Good Bad This field indicates the number of good and bad packets received with length between 128 and 255 (inclusive) bytes, exclusive of the preamble. Value After Reset: 0x0
0x7b8	Rx_256To511Octets_Packets_Good_Bad	This register provides the number of good and bad packets received by DWC_ether_qos with length between 256 and 511 (inclusive) bytes, exclusive of the preamble.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RX256_511OCTGB	R	0x0	Rx 256-511 Octets Packets Good Bad This field indicates the number of good and bad packets received with length between 256 and 511 (inclusive) bytes, exclusive of the preamble. Value After Reset: 0x0
0x7bc	Rx_512To1023Octets_Packets_Good_Bad	This register provides the number of good and bad packets received by DWC_ether_qos with length between 512 and 1023 (inclusive) bytes, exclusive of the preamble.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RX512_1023OCTGB	R	0x0	RX 512-1023 Octets Packets Good Bad This field indicates the number of good and bad packets received with length between 512 and 1023 (inclusive) bytes, exclusive of the preamble. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x7c0	Rx_1024ToMaxOctets_Packets_Good_Bad	This register provides the number of good and bad packets received by DWC_ether_qos with length between 1024 and maxsize (inclusive) bytes, exclusive of the preamble.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RX1024_MAXOCTGB	R	0x0	Rx 1024-Max Octets Good Bad This field indicates the number of good and bad packets received with length between 1024 and maxsize (inclusive) bytes, exclusive of the preamble. Value After Reset: 0x0
0x7c4	Rx_Unicast_Packets_Good	This register provides the number of good unicast packets received by DWC_ether_qos.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXUCASTG	R	0x0	Rx Unicast Packets Good This field indicates the number of good unicast packets received. Value After Reset: 0x0
0x7c8	Rx_Length_Error_Packets	This register provides the number of packets received by DWC_ether_qos with length error (Length Type field not equal to packet size), for all packets with valid length field.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXLENERR	R	0x0	Rx Length Error Packets This field indicates the number of packets received with length error (Length Type field not equal to packet size), for all packets with valid length field. Value After Reset: 0x0
0x7cc	Rx_Out_Of_Range_Type_Packets	This register provides the number of packets received by DWC_ether_qos with length field not equal to the valid packet size (greater than 1,500 but less than 1,536).	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXOUTOFRNG	R	0x0	Rx Out of Range Type Packet This field indicates the number of packets received with length field not equal to the valid packet size (greater than 1,500 but less than 1,536). Value After Reset: 0x0
0x7d0	Rx_Pause_Packets	This register provides the number of good and valid Pause packets received by DWC_ether_qos.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXPAUSEPKT	R	0x0	Rx Pause Packets This field indicates the number of good and valid Pause packets received. Value After Reset: 0x0
0x7d4	Rx_FIFO_Overflow_Packets	This register provides the number of missed received packets because of FIFO overflow in DWC_ether_qos.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXFIFOOVFL	R	0x0	Rx FIFO Overflow Packets This field indicates the number of missed received packets because of FIFO overflow. Value After Reset: 0x0
0x7d8	Rx_VLAN_Packets_Good_Bad	This register provides the number of good and bad VLAN packets received by DWC_ether_qos.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXVLANPKTGB	R	0x0	Rx VLAN Packets Good Bad This field indicates the number of good and bad VLAN packets received. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x7dc	Rx_Watchdog_Error_Packets	This register provides the number of packets received by DWC_ether_qos with error because of watchdog timeout error (packets with a data load larger than 2,048 bytes (when JE and WD bits are reset in MAC_Configuration register), 10,240 bytes (when JE bit is set and WD bit is reset in MAC_Configuration register), 16,384 bytes (when WD bit is set in MAC_Configuration register) or the value programmed in the MAC_WD_JB_Timeout register).	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXWDGERR	R	0x0	Rx Watchdog Error Packets This field indicates the number of packets received with error because of watchdog timeout error (packets with a data load larger than 2,048 bytes (when JE and WD bits are reset in MAC_Configuration register), 10,240 bytes (when JE bit is set and WD bit is reset in MAC_Configuration register), 16,384 bytes (when WD bit is set in MAC_Configuration register) or the value programmed in the MAC_WD_JB_Timeout register). Value After Reset: 0x0
0x7e0	Rx_Receive_Error_Packets	This register provides the number of packets received by DWC_ether_qos with Receive error or Packet Extension error on the GMII or MII interface.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXRCVERR	R	0x0	Rx Receive Error Packets This field indicates the number of packets received with Receive error or Packet Extension error on the GMII or MII interface. Value After Reset: 0x0
0x7e4	Rx_Control_Packets_Good	This register provides the number of good control packets received by DWC_ether_qos.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXCTRLG	R	0x0	Rx Control Packets Good This field indicates the number of good control packets received. Value After Reset: 0x0
0x7ec	Tx_LPI_USEC_Cntr	This register provides the number of microseconds Tx LPI is asserted by DWC_ether_qos.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	TXLPIUSC	R	0x0	Tx LPI Microseconds Counter This field indicates the number of microseconds Tx LPI is asserted. For every Tx LPI Entry and Exit, the Timer value can have an error of +/- 1 microsecond. Value After Reset: 0x0
0x7f0	Tx_LPI_Tran_Cntr	This register provides the number of times DWC_ether_qos has entered Tx LPI.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	TXLPITRC	R	0x0	Tx LPI Transition counter This field indicates the number of times Tx LPI Entry has occurred. Even if Tx LPI Entry occurs in Automate Mode (because of LPITXA bit set in the LPI Control and Status register), the counter increments. Value After Reset: 0x0
0x7f4	Rx_LPI_USEC_Cntr	This register provides the number of microseconds Rx LPI is sampled by DWC_ether_qos.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXLPIUSC	R	0x0	Rx LPI Microseconds Counter This field indicates the number of microseconds Rx LPI is asserted. For every Rx LPI Entry and Exit, the Timer value can have an error of +/- 1 microsecond. Value After Reset: 0x0
0x7f8	Rx_LPI_Tran_Cntr	This register provides the number of times DWC_ether_qos has entered Rx LPI.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXLPITRC	R	0x0	Rx LPI Transition counter This field indicates the number of times Rx LPI Entry has occurred. Value After Reset: 0x0
			31:30	Reserved_31_30	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			29	RXICMPEROIM	R/W	0x0	MMC Receive ICMP Error Octet Counter Interrupt Mask Setting this bit masks the interrupt when the RxICMP_Error_Octets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive ICMP Error Octet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive ICMP Error Octet Counter Interrupt Mask is enabled Value After Reset: 0x0
			28	RXICMPGOIM	R/W	0x0	MMC Receive ICMP Good Octet Counter Interrupt Mask Setting this bit masks the interrupt when the RxICMP_Good_Octets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive ICMP Good Octet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive ICMP Good Octet Counter Interrupt Mask is enabled Value After Reset: 0x0
			27	RXTCPEROIM	R/W	0x0	MMC Receive TCP Error Octet Counter Interrupt Mask Setting this bit masks the interrupt when the RxTCP_Error_Octets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive TCP Error Octet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive TCP Error Octet Counter Interrupt Mask is enabled Value After Reset: 0x0
			26	RXTCPGOIM	R/W	0x0	MMC Receive TCP Good Octet Counter Interrupt Mask Setting this bit masks the interrupt when the RxTCP_Good_Octets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive TCP Good Octet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive TCP Good Octet Counter Interrupt Mask is enabled Value After Reset: 0x0
			25	RXUDPEROIM	R/W	0x0	MMC Receive UDP Good Octet Counter Interrupt Mask Setting this bit masks the interrupt when the RxUDP_Error_Octets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive UDP Good Octet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive UDP Good Octet Counter Interrupt Mask is enabled Value After Reset: 0x0
			24	RXUDPGOIM	R/W	0x0	MMC Receive IPV6 No Payload Octet Counter Interrupt Mask Setting this bit masks the interrupt when the RxUDP_Good_Octets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive IPV6 No Payload Octet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive IPV6 No Payload Octet Counter Interrupt Mask is enabled Value After Reset: 0x0
			23	RXIPV6NOPAYOIM	R/W	0x0	MMC Receive IPV6 Header Error Octet Counter Interrupt Mask Setting this bit masks the interrupt when the RxIPv6_No_Payload_Octets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive IPV6 Header Error Octet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive IPV6 Header Error Octet Counter Interrupt Mask is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			22	RXIPV6HEROIM	R/W	0x0	MMC Receive IPV6 Good Octet Counter Interrupt Mask Setting this bit masks the interrupt when the RxIPv6_Header_Error_Octets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive IPV6 Good Octet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive IPV6 Good Octet Counter Interrupt Mask is enabled Value After Reset: 0x0
			21	RXIPV6GOIM	R/W	0x0	MMC Receive IPV6 Good Octet Counter Interrupt Mask Setting this bit masks the interrupt when the RxIPv6_Good_Octets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive IPV6 Good Octet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive IPV6 Good Octet Counter Interrupt Mask is enabled Value After Reset: 0x0
			20	RXIPV4UDSBLOIM	R/W	0x0	MMC Receive IPV4 UDP Checksum Disabled Octet Counter Interrupt Mask Setting this bit masks the interrupt when the RxIPv4_UDP_Checksum_Disable_Octets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive IPV4 UDP Checksum Disabled Octet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive IPV4 UDP Checksum Disabled Octet Counter Interrupt Mask is enabled Value After Reset: 0x0
			19	RXIPV4FRAGOIM	R/W	0x0	MMC Receive IPV4 Fragmented Octet Counter Interrupt Mask Setting this bit masks the interrupt when the RxIPv4_Fragmented_Octets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive IPV4 Fragmented Octet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive IPV4 Fragmented Octet Counter Interrupt Mask is enabled Value After Reset: 0x0
			18	RXIPV4NOPAYOIM	R/W	0x0	MMC Receive IPV4 No Payload Octet Counter Interrupt Mask Setting this bit masks the interrupt when the RxIPv4_No_Payload_Octets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive IPV4 No Payload Octet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive IPV4 No Payload Octet Counter Interrupt Mask is enabled Value After Reset: 0x0
			17	RXIPV4HEROIM	R/W	0x0	MMC Receive IPV4 Header Error Octet Counter Interrupt Mask Setting this bit masks the interrupt when the RxIPv4_Header_Error_Octets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive IPV4 Header Error Octet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive IPV4 Header Error Octet Counter Interrupt Mask is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x800	MMC_IPC_Rx_Interrupt_Mask	This register maintains the mask for the interrupt generated from the receive IPC statistic counters. The MMC Receive Checksum Off load Interrupt Mask register maintains the masks for the interrupts generated when the receive IPC (Checksum Off load) statistic counters reach half their maximum value, and when they reach their maximum values. This register is 32 bits wide.	16	RXIPV4GOIM	R/W	0x0	MMC Receive IPV4 Good Octet Counter Interrupt Mask Setting this bit masks the interrupt when the RxIPv4_Good_Octets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive IPV4 Good Octet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive IPV4 Good Octet Counter Interrupt Mask is enabled Value After Reset: 0x0
			15:14	Reserved_15_14	R	0x0	Reserved. Value After Reset: 0x0
			13	RXICMPERPIM	R/W	0x0	MMC Receive ICMP Error Packet Counter Interrupt Mask Setting this bit masks the interrupt when the RxICMP_Error_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive ICMP Error Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive ICMP Error Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			12	RXICMPGPIM	R/W	0x0	MMC Receive ICMP Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the RxICMP_Good_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive ICMP Good Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive ICMP Good Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			11	RXTCPERPIM	R/W	0x0	MMC Receive TCP Error Packet Counter Interrupt Mask Setting this bit masks the interrupt when the RxTCP_Error_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive TCP Error Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive TCP Error Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			10	RXTCPGPIM	R/W	0x0	MMC Receive TCP Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the RxTCP_Good_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive TCP Good Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive TCP Good Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			9	RXUDPERPIM	R/W	0x0	MMC Receive UDP Error Packet Counter Interrupt Mask Setting this bit masks the interrupt when the RxUDP_Error_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive UDP Error Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive UDP Error Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			8	RXUDPGPIM	R/W	0x0	MMC Receive UDP Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the RxUDP_Good_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive UDP Good Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive UDP Good Packet Counter Interrupt Mask is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			7	RXIPV6NOPAYPIM	R/W	0x0	MMC Receive IPV6 No Payload Packet Counter Interrupt Mask Setting this bit masks the interrupt when the RxIPv6_No_Payload_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive IPV6 No Payload Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive IPV6 No Payload Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			6	RXIPV6HERPIM	R/W	0x0	MMC Receive IPV6 Header Error Packet Counter Interrupt Mask Setting this bit masks the interrupt when the RxIPv6_Header_Error_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive IPV6 Header Error Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive IPV6 Header Error Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			5	RXIPV6GPIM	R/W	0x0	MMC Receive IPV6 Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the RxIPv6_Good_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive IPV6 Good Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive IPV6 Good Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			4	RXIPV4UDSBLPIM	R/W	0x0	MMC Receive IPV4 UDP Checksum Disabled Packet Counter Interrupt Mask Setting this bit masks the interrupt when the RxIPv4_UDP_Checksum_Disabled_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive IPV4 UDP Checksum Disabled Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive IPV4 UDP Checksum Disabled Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			3	RXIPV4FRAGPIM	R/W	0x0	MMC Receive IPV4 Fragmented Packet Counter Interrupt Mask Setting this bit masks the interrupt when the RxIPv4_Fragmented_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive IPV4 Fragmented Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive IPV4 Fragmented Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			2	RXIPV4NOPAYPIM	R/W	0x0	MMC Receive IPV4 No Payload Packet Counter Interrupt Mask Setting this bit masks the interrupt when the RxIPv4_No_Payload_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive IPV4 No Payload Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive IPV4 No Payload Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			1	RXIPV4HERPIM	R/W	0x0	MMC Receive IPV4 Header Error Packet Counter Interrupt Mask Setting this bit masks the interrupt when the RxIPv4_Header_Error_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive IPV4 Header Error Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive IPV4 Header Error Packet Counter Interrupt Mask is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	RXIPV4GPIM	R/W	0x0	MMC Receive IPV4 Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the RxIPv4_Good_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (DISABLE): MMC Receive IPV4 Good Packet Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Receive IPV4 Good Packet Counter Interrupt Mask is enabled Value After Reset: 0x0
			31:30	Reserved_31_30	R	0x0	Reserved. Value After Reset: 0x0
			29	RXICMPEROIS	R	0x0	MMC Receive ICMP Error Octet Counter Interrupt Status This bit is set when the RxICMP_Error_Octets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive ICMP Error Octet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive ICMP Error Octet Counter Interrupt Status detected Value After Reset: 0x0
			28	RXICMPGOIS	R	0x0	MMC Receive ICMP Good Octet Counter Interrupt Status This bit is set when the RxICMP_Good_Octets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive ICMP Good Octet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive ICMP Good Octet Counter Interrupt Status detected Value After Reset: 0x0
			27	RXTCPEROIS	R	0x0	MMC Receive TCP Error Octet Counter Interrupt Status This bit is set when the RxTCP_Error_Octets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive TCP Error Octet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive TCP Error Octet Counter Interrupt Status detected Value After Reset: 0x0
			26	RXTCPGOIS	R	0x0	MMC Receive TCP Good Octet Counter Interrupt Status This bit is set when the RxTCP_Good_Octets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive TCP Good Octet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive TCP Good Octet Counter Interrupt Status detected Value After Reset: 0x0
			25	RXUDPEROIS	R	0x0	MMC Receive UDP Error Octet Counter Interrupt Status This bit is set when the RxUDP_Error_Octets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive UDP Error Octet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive UDP Error Octet Counter Interrupt Status detected Value After Reset: 0x0
			24	RXUDPGOIS	R	0x0	MMC Receive UDP Good Octet Counter Interrupt Status This bit is set when the RxUDP_Good_Octets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive UDP Good Octet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive UDP Good Octet Counter Interrupt Status detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			23	RXIPV6NOPAYOIS	R	0x0	MMC Receive IPV6 No Payload Octet Counter Interrupt Status This bit is set when the RxIPv6_No_Payload_Octets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive IPV6 No Payload Octet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive IPV6 No Payload Octet Counter Interrupt Status detected Value After Reset: 0x0
			22	RXIPV6HEROIS	R	0x0	MMC Receive IPV6 Header Error Octet Counter Interrupt Status This bit is set when the RxIPv6_Header_Error_Octets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive IPV6 Header Error Octet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive IPV6 Header Error Octet Counter Interrupt Status detected Value After Reset: 0x0
			21	RXIPV6GOIS	R	0x0	MMC Receive IPV6 Good Octet Counter Interrupt Status This bit is set when the RxIPv6_Good_Octets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive IPV6 Good Octet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive IPV6 Good Octet Counter Interrupt Status detected Value After Reset: 0x0
			20	RXIPV4UDSBLOIS	R	0x0	MMC Receive IPV4 UDP Checksum Disabled Octet Counter Interrupt Status This bit is set when the RxIPv4_UDP_Checksum_Disable_Octets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive IPV4 UDP Checksum Disabled Octet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive IPV4 UDP Checksum Disabled Octet Counter Interrupt Status detected Value After Reset: 0x0
			19	RXIPV4FRAGOIS	R	0x0	MMC Receive IPV4 Fragmented Octet Counter Interrupt Status This bit is set when the RxIPv4_Fragmented_Octets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive IPV4 Fragmented Octet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive IPV4 Fragmented Octet Counter Interrupt Status detected Value After Reset: 0x0
			18	RXIPV4NOPAYOIS	R	0x0	MMC Receive IPV4 No Payload Octet Counter Interrupt Status This bit is set when the RxIPv4_No_Payload_Octets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive IPV4 No Payload Octet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive IPV4 No Payload Octet Counter Interrupt Status detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x808	MMC_IPC_Rx_Interrupt	This register maintains the interrupt that the receive IPC statistic counters generate.The MMC Receive Checksum Offload Interrupt register maintains the interrupts generated when receive IPC statistic counters reach half their maximum values (0x8000_0000 for 32 bit counter and 0x8000 for 16 bit counter), and when they cross their maximum values (0xFFFF_FFFF for 32 bit counter and 0xFFFF for 16 bit counter). When Counter Stop Rollover is set, the interrupts are set but the counter remains at all-ones. The MMC Receive Checksum Offload Interrupt register is 32 bit wide. When the MMC IPC counter that caused the interrupt is read, its corresponding interrupt bit is cleared. The counter's least-significant byte lane (Bits[7:0]) must be read to clear the interrupt bit.	17	RXIPV4HEROIS	R	0x0	MMC Receive IPV4 Header Error Octet Counter Interrupt Status This bit is set when the RxIPv4_Header_Error_Octets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive IPV4 Header Error Octet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive IPV4 Header Error Octet Counter Interrupt Status detected Value After Reset: 0x0
			16	RXIPV4GOIS	R	0x0	MMC Receive IPV4 Good Octet Counter Interrupt Status This bit is set when the RxIPv4_Good_Octets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive IPV4 Good Octet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive IPV4 Good Octet Counter Interrupt Status detected Value After Reset: 0x0
			15:14	Reserved_15_14	R	0x0	Reserved. Value After Reset: 0x0
			13	RXICMPERPIS	R	0x0	MMC Receive ICMP Error Packet Counter Interrupt Status This bit is set when the RxICMP_Error_Packets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive ICMP Error Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive ICMP Error Packet Counter Interrupt Status detected Value After Reset: 0x0
			12	RXICMPGPIS	R	0x0	MMC Receive ICMP Good Packet Counter Interrupt Status This bit is set when the RxICMP_Good_Packets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive ICMP Good Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive ICMP Good Packet Counter Interrupt Status detected Value After Reset: 0x0
			11	RXTCPERPIS	R	0x0	MMC Receive TCP Error Packet Counter Interrupt Status This bit is set when the RxTCP_Error_Packets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive TCP Error Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive TCP Error Packet Counter Interrupt Status detected Value After Reset: 0x0
			10	RXTCPGPIS	R	0x0	MMC Receive TCP Good Packet Counter Interrupt Status This bit is set when the RxTCP_Good_Packets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive TCP Good Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive TCP Good Packet Counter Interrupt Status detected Value After Reset: 0x0
			9	RXUDPERPIS	R	0x0	MMC Receive UDP Error Packet Counter Interrupt Status This bit is set when the RxUDP_Error_Packets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive UDP Error Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive UDP Error Packet Counter Interrupt Status detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			8	RXUDPGPIS	R	0x0	MC Receive UDP Good Packet Counter Interrupt Status This bit is set when the RxUDP_Good_Packets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive UDP Good Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive UDP Good Packet Counter Interrupt Status detected Value After Reset: 0x0
			7	RXIPV6NOPAYPIS	R	0x0	MMC Receive IPV6 No Payload Packet Counter Interrupt Status This bit is set when the RxIPv6_No_Payload_Packets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive IPV6 No Payload Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive IPV6 No Payload Packet Counter Interrupt Status detected Value After Reset: 0x0
			6	RXIPV6HERPIS	R	0x0	MMC Receive IPV6 Header Error Packet Counter Interrupt Status This bit is set when the RxIPv6_Header_Error_Packets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive IPV6 Header Error Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive IPV6 Header Error Packet Counter Interrupt Status detected Value After Reset: 0x0
			5	RXIPV6GPIS	R	0x0	MMC Receive IPV6 Good Packet Counter Interrupt Status This bit is set when the RxIPv6_Good_Packets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive IPV6 Good Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive IPV6 Good Packet Counter Interrupt Status detected Value After Reset: 0x0
			4	RXIPV4UDSBLPIS	R	0x0	MMC Receive IPV4 UDP Checksum Disabled Packet Counter Interrupt Status This bit is set when the RxIPv4_UDP_Checksum_Disabled_Packets counter reaches half of the maximum value or the maximum value.Values: 0x0 (INACTIVE): MMC Receive IPV4 UDP Checksum Disabled Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive IPV4 UDP Checksum Disabled Packet Counter Interrupt Status detected Value After Reset: 0x0
			3	RXIPV4FRAGPIS	R	0x0	MMC Receive IPV4 Fragmented Packet Counter Interrupt Status This bit is set when the RxIPv4_Fragmented_Packets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive IPV4 Fragmented Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive IPV4 Fragmented Packet Counter Interrupt Status detected Value After Reset: 0x0
			2	RXIPV4NOPAYPIS	R	0x0	MMC Receive IPV4 No Payload Packet Counter Interrupt Status This bit is set when the RxIPv4_No_Payload_Packets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive IPV4 No Payload Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive IPV4 No Payload Packet Counter Interrupt Status detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			1	RXIPV4HERPIS	R	0x0	MMC Receive IPV4 Header Error Packet Counter Interrupt Status This bit is set when the RxIPv4_Header_Error_Packets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive IPV4 Header Error Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive IPV4 Header Error Packet Counter Interrupt Status detected Value After Reset: 0x0
			0	RXIPV4GPIS	R	0x0	MMC Receive IPV4 Good Packet Counter Interrupt Status This bit is set when the RxIPv4_Good_Packets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): MMC Receive IPV4 Good Packet Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Receive IPV4 Good Packet Counter Interrupt Status detected Value After Reset: 0x0
0x810	RxIPv4_Good_Packets	This register provides the number of good IPv4 datagrams received by DWC_ether_qos with the TCP, UDP, or ICMP payload.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXIPV4GDPKT	R	0x0	RxIPv4 Good Packets This field indicates the number of good IPv4 datagrams received with the TCP, UDP, or ICMP payload. Value After Reset: 0x0
0x814	RxIPv4_Header_Error_Packets	RxIPv4 Header Error PacketsThis register provides the number of IPv4 datagrams received by DWC_ether_qos with header (checksum, length, or version mismatch) errors.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXIPV4HDRERRPKT	R	0x0	RxIPv4 Header Error Packets This field indicates the number of IPv4 datagrams received with header (checksum, length, or version mismatch) errors. Value After Reset: 0x0
0x818	RxIPv4_No_Payload_Packets	This register provides the number of IPv4 datagram packets received by DWC_ether_qos that did not have a TCP, UDP, or ICMP payload.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXIPV4NOPAYPKT	R	0x0	RxIPv4 Payload Packets This field indicates the number of IPv4 datagram packets received that did not have a TCP, UDP, or ICMP payload. Value After Reset: 0x0
0x81c	RxIPv4_Fragmented_Packets	This register provides the number of good IPv4 datagrams received by DWC_ether_qos with fragmentation.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXIPV4FRAGPKT	R	0x0	RxIPv4 Fragmented Packets This field indicates the number of good IPv4 datagrams received with fragmentation. Value After Reset: 0x0
0x820	RxIPv4_UDP_Checksum_Disabled_Packets	This register provides the number of good IPv4 datagrams received by DWC_ether_qos that had a UDP payload with checksum disabled.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXIPV4UDSBLPKT	R	0x0	RxIPv4 UDP Checksum Disabled Packets This field indicates the number of good IPv4 datagrams received that had a UDP payload with checksum disabled. Value After Reset: 0x0
0x824	RxIPv6_Good_Packets	This register provides the number of good IPv6 datagrams received by DWC_ether_qos with the TCP, UDP, or ICMP payload.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXIPV6GDPKT	R	0x0	RxIPv6 Good Packets This field indicates the number of good IPv6 datagrams received with the TCP, UDP, or ICMP payload. Value After Reset: 0x0
		This register provides the number of IPv6 datagrams	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x828	RxIPv6_Header_Error_Packets	received by DWC_ether_qos with header (length or version mismatch) errors.	15:0	RXIPV6HDRERRPKT	R	0x0	RxIPv6 Header Error Packets This field indicates the number of IPv6 datagrams received with header (length or version mismatch) errors. Value After Reset: 0x0
0x82c	RxIPv6_No_Payload_Packets	This register provides the number of IPv6 datagram packets received by DWC_ether_qos that did not have a TCP, UDP, or ICMP payload. This includes all IPv6 datagrams with fragmentation or security extension headers.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXIPV6NOPAYPKT	R	0x0	RxIPv6 Payload Packets This field indicates the number of IPv6 datagram packets received that did not have a TCP, UDP, or ICMP payload. This includes all IPv6 datagrams with fragmentation or security extension headers. Value After Reset: 0x0
0x830	RxUDP_Good_Packets	This register provides the number of good IP datagrams received by DWC_ether_qos with a good UDP payload. This counter is not updated when the RxIPv4_UDP_Checksum_Disabled_Packets counter is incremented.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXUDPGDPKT	R	0x0	RxUDP Good Packets This field indicates the number of good IP datagrams received with a good UDP payload. This counter is not updated when the RxIPv4_UDP_Checksum_Disabled_Packets counter is incremented. Value After Reset: 0x0
0x834	RxUDP_Error_Packets	This register provides the number of good IP datagrams received by DWC_ether_qos whose UDP payload has a checksum error.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXUDPERRPKT	R	0x0	RxUDP Error Packets This field indicates the number of good IP datagrams received whose UDP payload has a checksum error. Value After Reset: 0x0
0x838	RxTCP_Good_Packets	This register provides the number of good IP datagrams received by DWC_ether_qos with a good TCP payload.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXTCPGDPKT	R	0x0	RxTCP Good Packets This field indicates the number of good IP datagrams received with a good TCP payload. Value After Reset: 0x0
0x83c	RxTCP_Error_Packets	This register provides the number of good IP datagrams received by DWC_ether_qos whose TCP payload has a checksum error.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXTCPERRPKT	R	0x0	RxTCP Error Packets This field indicates the number of good IP datagrams received whose TCP payload has a checksum error. Value After Reset: 0x0
0x840	RxICMP_Good_Packets	This register provides the number of good IP datagrams received by DWC_ether_qos with a good ICMP payload.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXICMPGDPKT	R	0x0	RxICMP Good Packets This field indicates the number of good IP datagrams received with a good ICMP payload. Value After Reset: 0x0
0x844	RxICMP_Error_Packets	This register provides the number of good IP datagrams received by DWC_ether_qos whose ICMP payload has a checksum error.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	RXICMPERRPKT	R	0x0	RxICMP Error Packets This field indicates the number of good IP datagrams received whose ICMP payload has a checksum error. Value After Reset: 0x0
0x850	RxIPv4_Good_Octets	This register provides the number of bytes received by DWC_ether_qos in good IPv4 datagrams encapsulating TCP, UDP, or ICMP data. (Ethernet header, FCS, pad, or IP pad bytes are not included in this counter.)	31:0	RXIPV4GDOCT	R	0x0	RxIPv4 Good Octets This field indicates the number of bytes received in good IPv4 datagrams encapsulating TCP, UDP, or ICMP data. (Ethernet header, FCS, pad, or IP pad bytes are not included in this counter.) Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x854	RxIPv4_Header_Error_Octets	This register provides the number of bytes received by DWC_ether_qos in IPv4 datagrams with header errors (checksum, length, version mismatch). The value in the Length field of IPv4 header is used to update this counter. (Ethernet header, FCS, pad, or IP pad bytes are not included in this counter.)	31:0	RXIPV4HDRERROCT	R	0x0	RxIPv4 Header Error Octets This field indicates the number of bytes received in IPv4 datagrams with header errors (checksum, length, version mismatch). The value in the Length field of IPv4 header is used to update this counter. (Ethernet header, FCS, pad, or IP pad bytes are not included in this counter.) Value After Reset: 0x0
0x858	RxIPv4_No_Payload_Octets	This register provides the number of bytes received by DWC_ether_qos in IPv4 datagrams that did not have a TCP, UDP, or ICMP payload. The value in the Length field of IPv4 header is used to update this counter. (Ethernet header, FCS, pad, or IP pad bytes are not included in this counter.)	31:0	RXIPV4NOPAYOCT	R	0x0	RxIPv4 Payload Octets This field indicates the number of bytes received in IPv4 datagrams that did not have a TCP, UDP, or ICMP payload. The value in the Length field of IPv4 header is used to update this counter. (Ethernet header, FCS, pad, or IP pad bytes are not included in this counter.) Value After Reset: 0x0
0x85c	RxIPv4_Fragmented_Octets	This register provides the number of bytes received by DWC_ether_qos in fragmented IPv4 datagrams. The value in the Length field of IPv4 header is used to update this counter. (Ethernet header, FCS, pad, or IP pad bytes are not included in this counter.)	31:0	RXIPV4FRAGOCT	R	0x0	RxIPv4 Fragmented Octets This field indicates the number of bytes received in fragmented IPv4 datagrams. The value in the Length field of IPv4 header is used to update this counter. (Ethernet header, FCS, pad, or IP pad bytes are not included in this counter.) Value After Reset: 0x0
0x860	RxIPv4_UDP_Checksum_Disable_Octets	This register provides the number of bytes received by DWC_ether_qos in a UDP segment that had the UDP checksum disabled. This counter does not count IP Header bytes. (Ethernet header, FCS, pad, or IP pad bytes are not included in this counter.)	31:0	RXIPV4UDSBLOCT	R	0x0	RxIPv4 UDP Checksum Disable Octets This field indicates the number of bytes received in a UDP segment that had the UDP checksum disabled. This counter does not count IP Header bytes. (Ethernet header, FCS, pad, or IP pad bytes are not included in this counter.) Value After Reset: 0x0
0x864	RxIPv6_Good_Octets	This register provides the number of bytes received by DWC_ether_qos in good IPv6 datagrams encapsulating TCP, UDP, or ICMP data. (Ethernet header, FCS, pad, or IP pad bytes are not included in this counter.)	31:0	RXIPV6GDOCT	R	0x0	RxIPv6 Good Octets This field indicates the number of bytes received in good IPv6 datagrams encapsulating TCP, UDP, or ICMP data. (Ethernet header, FCS, pad, or IP pad bytes are not included in this counter.) Value After Reset: 0x0
0x868	RxIPv6_Header_Error_Octets	This register provides the number of bytes received by DWC_ether_qos in IPv6 datagrams with header errors (length, version mismatch). The value in the Length field of IPv6 header is used to update this counter. (Ethernet header, FCS, pad, or IP pad bytes are not included in this counter.)	31:0	RXIPV6HDRERROCT	R	0x0	RxIPv6 Header Error Octets This field indicates the number of bytes received in IPv6 datagrams with header errors (length, version mismatch). The value in the Length field of IPv6 header is used to update this counter. (Ethernet header, FCS, pad, or IP pad bytes are not included in this counter.) Value After Reset: 0x0
0x86c	RxIPv6_No_Payload_Octets	This register provides the number of bytes received by DWC_ether_qos in IPv6 datagrams that did not have a TCP, UDP, or ICMP payload. The value in the Length field of IPv6 header is used to update this counter. (Ethernet header, FCS, pad, or IP pad bytes are not included in this counter.)	31:0	RXIPV6NOPAYOCT	R	0x0	RxIPv6 Payload Octets This field indicates the number of bytes received in IPv6 datagrams that did not have a TCP, UDP, or ICMP payload. The value in the Length field of IPv6 header is used to update this counter. (Ethernet header, FCS, pad, or IP pad bytes are not included in this counter.) Value After Reset: 0x0
0x870	RxUDP_Good_Octets	This register provides the number of bytes received by DWC_ether_qos in a good UDP segment. This counter does not count IP header bytes.	31:0	RXUDPGDOCT	R	0x0	RxUDP Good Octets This field indicates the number of bytes received in a good UDP segment. This counter does not count IP header bytes. Value After Reset: 0x0
0x874	RxUDP_Error_Octets	This register provides the number of bytes received by DWC_ether_qos in a UDP segment that had checksum errors. This counter does not count IP header bytes.	31:0	RXUDPERROCT	R	0x0	RxUDP Error Octets This field indicates the number of bytes received in a UDP segment that had checksum errors. This counter does not count IP header bytes. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x878	RxTCP_Good_Octets	This register provides the number of bytes received by DWC_ether_qos in a good TCP segment. This counter does not count IP header bytes.	31:0	RXTCPGDOCT	R	0x0	RxTCP Good Octets This field indicates the number of bytes received in a good TCP segment. This counter does not count IP header bytes. Value After Reset: 0x0
0x87c	RxTCP_Error_Octets	This register provides the number of bytes received by DWC_ether_qos in a TCP segment that had checksum errors. This counter does not count IP header bytes.	31:0	RXTCPERROCT	R	0x0	RxTCP Error Octets This field indicates the number of bytes received in a TCP segment that had checksum errors. This counter does not count IP header bytes. Value After Reset: 0x0
0x880	RxICMP_Good_Octets	This register provides the number of bytes received by DWC_ether_qos in a good ICMP segment. This counter does not count IP header bytes.	31:0	RXICMPGDOCT	R	0x0	RxICMP Good Octets This field indicates the number of bytes received in a good ICMP segment. This counter does not count IP header bytes. Value After Reset: 0x0
0x884	RxICMP_Error_Octets	This register provides the number of bytes received by DWC_ether_qos in a ICMP segment that had checksum errors. This counter does not count IP header bytes.	31:0	RXICMPERROCT	R	0x0	RxICMP Error Octets This field indicates the number of bytes received in a ICMP segment that had checksum errors. This counter does not count IP header bytes. Value After Reset: 0x0
0x8a0	MMC_FPE_Tx_Interrupt	This register maintains the interrupts generated from all FPE related Transmit statistics counters. The MMC FPE Transmit Interrupt register maintains the interrupts generated when transmit statistic counters reach half their maximum values (0x8000_0000 for 32 bit counter and 0x8000 for 16 bit counter), and when they cross their maximum values (0xFFFF_FFFF for 32-bit counter and 0xFFFF for 16-bit counter). When Counter Stop Rollover is set, the interrupts are set but the counter remains at all-ones. The MMC FPE Transmit Interrupt register is a 32 bit register. An interrupt bit is cleared when the respective MMC counter that caused the interrupt is read. The least significant byte lane (Bits[7:0]) of the respective counter must be read to clear the interrupt bit.	31:2	Reserved_31_2	R	0x0	Reserved. Value After Reset: 0x0
			1	HRCIS	R	0x0	MMC Tx Hold Request Counter Interrupt Status This bit is set when the Tx_Hold_Req_Cntr counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. Exists when any one of the RX/TX MMC counters are enabled during FPE with AV_EST Enabled configuration.Values: 0x0 (INACTIVE): MMC Tx Hold Request Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Tx Hold Request Counter Interrupt Status detected Value After Reset: 0x0
			0	FCIS	R	0x0	MMC Tx FPE Fragment Counter Interrupt status This bit is set when the Tx_FPE_Fragment_Cntr counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. Exists when any one of the RX/TX MMC counters are enabled during FPE Enabled configuration.Values: 0x0 (INACTIVE): MMC Tx FPE Fragment Counter Interrupt status not detected 0x1 (ACTIVE): MMC Tx FPE Fragment Counter Interrupt status detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x8a4	MMC_FPE_Tx_Interrupt_Mask	This register maintains the masks for interrupts generated from all FPE related Transmit statistics counters. The MMC Receive Interrupt Mask register maintains the masks for the interrupts generated when FPE related receive statistic counters reach half of their maximum value or the maximum values. This register is 32 bit wide.	31:2	Reserved_31_2	R	0x0	Reserved. Value After Reset: 0x0
			1	HRCIM	R/W	0x0	MMC Transmit Hold Request Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_Hold_Req_Cntr counter reaches half of the maximum value or the maximum value. Exists when any one of the RX/TX MMC counters are enabled during FPE with AV_EST Enabled configuration.Values: 0x0 (DISABLE): MMC Transmit Hold Request Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Transmit Hold Request Counter Interrupt Mask is enabled Value After Reset: 0x0
			0	FCIM	R/W	0x0	MMC Transmit Fragment Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_FPE_Fragment_Cntr counter reaches half of the maximum value or the maximum value. Exists when any one of the RX/TX MMC counters are enabled during FPE Enabled configuration.Values: 0x0 (DISABLE): MMC Transmit Fragment Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Transmit Fragment Counter Interrupt Mask is enabled Value After Reset: 0x0
0x8a8	MMC_Tx_FPE_Fragment_Cntr	This register provides the number of additional mPackets transmitted due to preemption.	31:0	TXFFC	R	0x0	Tx FPE Fragment counter This field indicates the number of additional mPackets that has been transmitted due to preemption Exists when any one of the RX/TX MMC counters are enabled during FPE Enabled configuration. Value After Reset: 0x0
0x8ac	MMC_Tx_Hold_Req_Cntr	This register provides the count of number of times a hold request is given to MAC	31:0	TXHRC	R	0x0	Tx Hold Request Counter This field indicates count of number of a hold request is given to MAC. Exists when any one of the RX/TX MMC counters are enabled during FPE with AV_EST Enabled configuration. Value After Reset: 0x0
0x8c0	MMC_FPE_Rx_Interrupt	This register maintains the interrupts generated from all FPE related Receive statistics counters. The MMC FPE Receive Interrupt register maintains the interrupts generated when transmit statistic counters reach half their maximum values (0x8000_0000 for 32 bit counter and 0x8000 for 16 bit counter), and when they cross their maximum values (0xFFFF_FFFF for 32-bit counter	31:4	Reserved_31_4	R	0x0	Reserved. Value After Reset: 0x0
			3	FCIS	R	0x0	MMC Rx FPE Fragment Counter Interrupt Status This bit is set when the Rx_FPE_Fragment_Cntr counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. Exists when any one of the RX/TX MMC counters are enabled during FPE Enabled configuration.Values: 0x0 (INACTIVE): MMC Rx FPE Fragment Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Rx FPE Fragment Counter Interrupt Status detected Value After Reset: 0x0
			2	PAOCIS	R	0x0	MMC Rx Packet Assembly OK Counter Interrupt Status This bit is set when the Rx_Packet_Assemble_Ok_Cntr counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. Exists when any one of the RX/TX MMC counters are enabled during FPE Enabled configuration.Values: 0x0 (INACTIVE): MMC Rx Packet Assembly OK Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Rx Packet Assembly OK Counter Interrupt Status detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
		and 0xFFFF for 16-bit counter). When Counter Stop Rollover is set, the interrupts are set but the counter remains at all-ones. The MMC FPE Receive Interrupt register is a 32 bit register. An interrupt bit is cleared when the respective MMC counter that caused the interrupt is read. The least significant byte lane (Bits[7:0]) of the respective counter must be read to clear the interrupt bit.	1	PSECIS	R	0x0	MMC Rx Packet SMD Error Counter Interrupt Status This bit is set when the Rx_Packet_SMD_Err_Cntr counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. Exists when any one of the RX/TX MMC counters are enabled during FPE Enabled configuration.Values: 0x0 (INACTIVE): MMC Rx Packet SMD Error Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Rx Packet SMD Error Counter Interrupt Status detected Value After Reset: 0x0
			0	PAECIS	R	0x0	MMC Rx Packet Assembly Error Counter Interrupt Status This bit is set when the Rx_Packet_Assemble_Err_Cntr counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. Exists when any one of the RX/TX MMC counters are enabled during FPE Enabled configuration.Values: 0x0 (INACTIVE): MMC Rx Packet Assembly Error Counter Interrupt Status not detected 0x1 (ACTIVE): MMC Rx Packet Assembly Error Counter Interrupt Status detected Value After Reset: 0x0
0x8c4	MMC_FPE_Rx_Interrupt_Mask	This register maintains the masks for interrupts generated from all FPE related Receive statistics counters. The MMC Receive Interrupt Mask register maintains the masks for the interrupts generated when FPE related receive statistic counters reach half of their maximum value or the maximum values. This register is 32 bit wide.	31:4	Reserved_31_4	R	0x0	Reserved. Value After Reset: 0x0
			3	FCIM	R/W	0x0	MMC Rx FPE Fragment Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_FPE_Fragment_Cntr counter reaches half of the maximum value or the maximum value. Exists when any one of the RX/TX MMC counters are enabled during FPE Enabled configuration.Values: 0x0 (DISABLE): MMC Rx FPE Fragment Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Rx FPE Fragment Counter Interrupt Mask is enabled Value After Reset: 0x0
			2	PAOCIM	R/W	0x0	MMC Rx Packet Assembly OK Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_Packet_Assemble_Ok_Cntr counter reaches half of the maximum value or the maximum value. Exists when any one of the RX/TX MMC counters are enabled during FPE Enabled configuration.Values: 0x0 (DISABLE): MMC Rx Packet Assembly OK Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Rx Packet Assembly OK Counter Interrupt Mask is enabled Value After Reset: 0x0
			1	PSECIM	R/W	0x0	MMC Rx Packet SMD Error Counter Interrupt Mask Setting this bit masks the interrupt when the R Rx_Packet_SMD_Err_Cntr counter reaches half of the maximum value or the maximum value. Exists when any one of the RX/TX MMC counters are enabled during FPE Enabled configuration.Values: 0x0 (DISABLE): MMC Rx Packet SMD Error Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Rx Packet SMD Error Counter Interrupt Mask is enabled Value After Reset: 0x0
			0	PAECIM	R/W	0x0	MMC Rx Packet Assembly Error Counter Interrupt Mask Setting this bit masks the interrupt when the R Rx_Packet_Assemble_Err_Cntr counter reaches half of the maximum value or the maximum value. Exists when any one of the RX/TX MMC counters are enabled during FPE Enabled configuration.Values: 0x0 (DISABLE): MMC Rx Packet Assembly Error Counter Interrupt Mask is disabled 0x1 (ENABLE): MMC Rx Packet Assembly Error Counter Interrupt Mask is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x8c8	MMC_Rx_Packet_Assembly_Err_Cntr	This register provides the number of MAC frames with reassembly errors on the Receiver, due to mismatch in the Fragment Count value.	31:0	PAEC	R	0x0	Rx Packet Assembly Error Counter This field indicates the number of MAC frames with reassembly errors on the Receiver, due to mismatch in the Fragment Count value. Exists when any one of the RX/TX MMC counters are enabled during FPE Enabled configuration. Value After Reset: 0x0
0x8cc	MMC_Rx_Packet_SMD_Err_Cntr	This register provides the number of received MAC frames rejected due to unknown SMD value and MAC frame fragments rejected due to arriving with an SMD-C when there was no preceding preempted frame.	31:0	PSEC	R	0x0	Rx Packet SMD Error Counter This field indicates the number of MAC frames rejected due to unknown SMD value and MAC frame fragments rejected due to arriving with an SMD-C when there was no preceding preempted frame. Exists when at least one of the RX/TX MMC counters are enabled during FPE Enabled configuration. Value After Reset: 0x0
0x8d0	MMC_Rx_Packet_Assembly_OK_Cntr	This register provides the number of MAC frames that were successfully reassembled and delivered to MAC.	31:0	PAOC	R	0x0	Rx Packet Assembly OK Counter This field indicates the number of MAC frames that were successfully reassembled and delivered to MAC. Exists when at least one of the RX/TX MMC counters are enabled during FPE Enabled configuration. Value After Reset: 0x0
0x8d4	MMC_Rx_FPE_Fragment_Cntr	This register provides the number of additional mPackets received due to preemption.	31:0	FFC	R	0x0	Rx FPE Fragment Counter This field indicates the number of additional mPackets received due to preemption Exists when at least one of the RX/TX MMC counters are enabled during FPE Enabled configuration. Value After Reset: 0x0
			31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28	DMCHEN0	R/W	0x0	DMA Channel Select Enable When set, this bit enables the selection of the DMA channel number for the packet that is passed by this L3_L4 filter. The DMA channel is indicated by the DMCHN bits. When this bit is reset, the DMA channel is not decided by this filter.Values: 0x0 (DISABLE): DMA Channel Select is disabled 0x1 (ENABLE): DMA Channel Select is enabled Value After Reset: 0x0
			27	Reserved_27_y	R	0x0	Reserved. Value After Reset: 0x0
			26:24	DMCHN0	R/W	0x0	DMA Channel Number When DMCHEN is set high, this field selects the DMA Channel number to which the packet passed by this filter is routed. The width of this field depends on the number of the DMA channels present in your configuration. Value After Reset: 0x0
			23:22	Reserved_23_22	R	0x0	Reserved. Value After Reset: 0x0
			21	L4DPIM0	R/W	0x0	Layer 4 Destination Port Inverse Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Destination Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4DPM0 bit is set high.Values: 0x0 (DISABLE): Layer 4 Destination Port Inverse Match is disabled 0x1 (ENABLE): Layer 4 Destination Port Inverse Match is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			20	L4DPM0	R/W	0x0	Layer 4 Destination Port Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Destination Port number field for matching.Values: 0x0 (DISABLE): Layer 4 Destination Port Match is disabled 0x1 (ENABLE): Layer 4 Destination Port Match is enabled Value After Reset: 0x0
			19	L4SPIM0	R/W	0x0	Layer 4 Source Port Inverse Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Source Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4SPM0 bit is set high.Values: 0x0 (DISABLE): Layer 4 Source Port Inverse Match is disabled 0x1 (ENABLE): Layer 4 Source Port Inverse Match is enabled Value After Reset: 0x0
			18	L4SPM0	R/W	0x0	Layer 4 Source Port Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Source Port number field for matching.Values: 0x0 (DISABLE): Layer 4 Source Port Match is disabled 0x1 (ENABLE): Layer 4 Source Port Match is enabled Value After Reset: 0x0
			17	Reserved_17	R	0x0	Reserved. Value After Reset: 0x0
			16	L4PEN0	R/W	0x0	Layer 4 Protocol Enable When this bit is set, the Source and Destination Port number fields of UDP packets are used for matching. When this bit is reset, the Source and Destination Port number fields of TCP packets are used for matching. The Layer 4 matching is done only when the L4SPM0 or L4DPM0 bit is set.Values: 0x0 (DISABLE): Layer 4 Protocol is disabled 0x1 (ENABLE): Layer 4 Protocol is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x900	MAC_L3_L4_Control0	The Layer 3 and Layer 4 Control register controls the operations of filter 0 of Layer 3 and Layer 4.	15:11	L3HDBM0	R/W	0x0	Layer 3 IP DA Higher Bits Match IPv4 Packets: This field contains the number of higher bits of IP Destination Address that are matched in the IPv4 packets. The following list describes the values of this field: 0: No bits are masked. 1: LSb[0] is masked 2: Two LSbs [1:0] are masked .. 31: All bits except MSb are masked. IPv6 Packets: Bits[12:11] of this field correspond to Bits[6:5] of L3HSBM0 which indicate the number of lower bits of IP Source or Destination Address that are masked in the IPv6 packets. The following list describes the concatenated values of the L3HDBM0[1:0] and L3HSBM0 bits: 0: No bits are masked. 1: LSb[0] is masked. 2: Two LSbs [1:0] are masked .. 127: All bits except MSb are masked. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set. Value After Reset: 0x0
			10:6	L3HSBM0	R/W	0x0	Layer 3 IP SA Higher Bits Match IPv4 Packets: This field contains the number of lower bits of IP Source Address that are masked for matching in the IPv4 packets. The following list describes the values of this field: 0: No bits are masked. 1: LSb[0] is masked 2: Two LSbs [1:0] are masked .. 31: All bits except MSb are masked. IPv6 Packets: This field contains Bits[4:0] of L3HSBM0. These bits indicate the number of higher bits of IP Source or Destination Address matched in the IPv6 packets. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set high. Value After Reset: 0x0
			5	L3DAIM0	R/W	0x0	Layer 3 IP DA Inverse Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for inverse matching. When this bit is reset, the Layer 3 IP Destination Address field is enabled for perfect matching. This bit is valid and applicable only when the L3DAM0 bit is set high.Values: 0x0 (DISABLE): Layer 3 IP DA Inverse Match is disabled 0x1 (ENABLE): Layer 3 IP DA Inverse Match is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			4	L3DAM0	R/W	0x0	Layer 3 IP DA Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Destination Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3SAM0 bit because either IPv6 DA or SA can be checked for filtering.Values: 0x0 (DISABLE): Layer 3 IP DA Match is disabled 0x1 (ENABLE): Layer 3 IP DA Match is enabled Value After Reset: 0x0
			3	L3SAIM0	R/W	0x0	Layer 3 IP SA Inverse Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for inverse matching. When this bit reset, the Layer 3 IP Source Address field is enabled for perfect matching. This bit is valid and applicable only when the L3SAM0 bit is set.Values: 0x0 (DISABLE): Layer 3 IP SA Inverse Match is disabled 0x1 (ENABLE): Layer 3 IP SA Inverse Match is enabled Value After Reset: 0x0
			2	L3SAM0	R/W	0x0	Layer 3 IP SA Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Source Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3DAM0 bit because either IPv6 SA or DA can be checked for filtering.Values: 0x0 (DISABLE): Layer 3 IP SA Match is disabled 0x1 (ENABLE): Layer 3 IP SA Match is enabled Value After Reset: 0x0
			1	Reserved_1	R	0x0	Reserved. Value After Reset: 0x0
			0	L3PEN0	R/W	0x0	Layer 3 Protocol Enable When this bit is set, the Layer 3 IP Source or Destination Address matching is enabled for IPv6 packets. When this bit is reset, the Layer 3 IP Source or Destination Address matching is enabled for IPv4 packets. The Layer 3 matching is done only when the L3SAM0 or L3DAM0 bit is set.Values: 0x0 (DISABLE): Layer 3 Protocol is disabled 0x1 (ENABLE): Layer 3 Protocol is enabled Value After Reset: 0x0
0x904	MAC_Layer4_Address0	You can configure the Layer 3 and Layer 4 Address Registers to be double-synchronized by selecting the Synchronize Layer 3 and Layer 4 Address Registers to Rx Clock Domain option while configuring the IP. When you select this option, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the Layer 3 and Layer 4 Address Registers are written. For proper synchronization updates, you should perform consecutive writes to same Layer 3 and Layer 4 Address Registers after at least four clock cycles delay of the destination clock.	31:16	L4DP0	R/W	0x0	Layer 4 Destination Port Number Field When the L4PEN0 bit is reset and the L4DPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Destination Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4DPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Destination Port Number field in the IPv4 or IPv6 packets. Value After Reset: 0x0
			15:0	L4SP0	R/W	0x0	Layer 4 Source Port Number Field When the L4PEN0 bit is reset and the L4SPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Source Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4SPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Source Port Number field in the IPv4 or IPv6 packets. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x910	MAC_Layer3_Addr0_Reg0	For IPv4 packets, this register contains the 32-bit IP Source Address field. For IPv6 packets, it contains Bits[31:0] of the 128-bit IP Source Address or Destination Address field.	31:0	L3A00	R/W	0x0	Layer 3 Address 0 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Source Address field in the IPv4 packets. Value After Reset: 0x0
0x914	MAC_Layer3_Addr1_Reg0	For IPv4 packets, this register contains the 32-bit IP Destination Address field. For IPv6 packets, it contains Bits[63:32] of the 128-bit IP Source Address or Destination Address field.	31:0	L3A10	R/W	0x0	Layer 3 Address 1 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Destination Address field in the IPv4 packets. Value After Reset: 0x0
0x918	MAC_Layer3_Addr2_Reg0	This register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[95:64] of 128-bit IP Source Address or Destination Address field.	31:0	L3A20	R/W	0x0	Layer 3 Address 2 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used. Value After Reset: 0x0
0x91c	MAC_Layer3_Addr3_Reg0	This register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[127:96] of 128-bit IP Source Address or Destination Address field.	31:0	L3A30	R/W	0x0	Layer 3 Address 3 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used. Value After Reset: 0x0
			31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28	DMCHEN0	R/W	0x0	DMA Channel Select Enable When set, this bit enables the selection of the DMA channel number for the packet that is passed by this L3_L4 filter. The DMA channel is indicated by the DMCHN bits. When this bit is reset, the DMA channel is not decided by this filter.Values: 0x0 (DISABLE): DMA Channel Select is disabled 0x1 (ENABLE): DMA Channel Select is enabled Value After Reset: 0x0
			27	Reserved_27_y	R	0x0	Reserved. Value After Reset: 0x0
			26:24	DMCHN0	R/W	0x0	DMA Channel Number When DMCHEN is set high, this field selects the DMA Channel number to which the packet passed by this filter is routed. The width of this field depends on the number of the DMA channels present in your configuration. Value After Reset: 0x0
			23:22	Reserved_23_22	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			21	L4DPIM0	R/W	0x0	Layer 4 Destination Port Inverse Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Destination Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4DPM0 bit is set high.Values: 0x0 (DISABLE): Layer 4 Destination Port Inverse Match is disabled 0x1 (ENABLE): Layer 4 Destination Port Inverse Match is enabled Value After Reset: 0x0
			20	L4DPM0	R/W	0x0	Layer 4 Destination Port Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Destination Port number field for matching.Values: 0x0 (DISABLE): Layer 4 Destination Port Match is disabled 0x1 (ENABLE): Layer 4 Destination Port Match is enabled Value After Reset: 0x0
			19	L4SPIM0	R/W	0x0	Layer 4 Source Port Inverse Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Source Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4SPM0 bit is set high.Values: 0x0 (DISABLE): Layer 4 Source Port Inverse Match is disabled 0x1 (ENABLE): Layer 4 Source Port Inverse Match is enabled Value After Reset: 0x0
			18	L4SPM0	R/W	0x0	Layer 4 Source Port Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Source Port number field for matching.Values: 0x0 (DISABLE): Layer 4 Source Port Match is disabled 0x1 (ENABLE): Layer 4 Source Port Match is enabled Value After Reset: 0x0
			17	Reserved_17	R	0x0	Reserved. Value After Reset: 0x0
			16	L4PEN0	R/W	0x0	Layer 4 Protocol Enable When this bit is set, the Source and Destination Port number fields of UDP packets are used for matching. When this bit is reset, the Source and Destination Port number fields of TCP packets are used for matching. The Layer 4 matching is done only when the L4SPM0 or L4DPM0 bit is set.Values: 0x0 (DISABLE): Layer 4 Protocol is disabled 0x1 (ENABLE): Layer 4 Protocol is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x930	MAC_L3_L4_Control1	The Layer 3 and Layer 4 Control register controls the operations of filter 1 of Layer 3 and Layer 4.	15:11	L3HDBM0	R/W	0x0	Layer 3 IP DA Higher Bits Match IPv4 Packets: This field contains the number of higher bits of IP Destination Address that are matched in the IPv4 packets. The following list describes the values of this field: 0: No bits are masked. 1: LSb[0] is masked 2: Two LSbs [1:0] are masked .. 31: All bits except MSb are masked. IPv6 Packets: Bits[12:11] of this field correspond to Bits[6:5] of L3HSBM0 which indicate the number of lower bits of IP Source or Destination Address that are masked in the IPv6 packets. The following list describes the concatenated values of the L3HDBM0[1:0] and L3HSBM0 bits: 0: No bits are masked. 1: LSb[0] is masked. 2: Two LSbs [1:0] are masked .. 127: All bits except MSb are masked. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set. Value After Reset: 0x0
			10:6	L3HSBM0	R/W	0x0	Layer 3 IP SA Higher Bits Match IPv4 Packets: This field contains the number of lower bits of IP Source Address that are masked for matching in the IPv4 packets. The following list describes the values of this field: 0: No bits are masked. 1: LSb[0] is masked 2: Two LSbs [1:0] are masked .. 31: All bits except MSb are masked. IPv6 Packets: This field contains Bits[4:0] of L3HSBM0. These bits indicate the number of higher bits of IP Source or Destination Address matched in the IPv6 packets. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set high. Value After Reset: 0x0
			5	L3DAIM0	R/W	0x0	Layer 3 IP DA Inverse Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for inverse matching. When this bit is reset, the Layer 3 IP Destination Address field is enabled for perfect matching. This bit is valid and applicable only when the L3DAM0 bit is set high.Values: 0x0 (DISABLE): Layer 3 IP DA Inverse Match is disabled 0x1 (ENABLE): Layer 3 IP DA Inverse Match is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			4	L3DAM0	R/W	0x0	Layer 3 IP DA Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Destination Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3SAM0 bit because either IPv6 DA or SA can be checked for filtering.Values: 0x0 (DISABLE): Layer 3 IP DA Match is disabled 0x1 (ENABLE): Layer 3 IP DA Match is enabled Value After Reset: 0x0
			3	L3SAIM0	R/W	0x0	Layer 3 IP SA Inverse Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for inverse matching. When this bit reset, the Layer 3 IP Source Address field is enabled for perfect matching. This bit is valid and applicable only when the L3SAM0 bit is set.Values: 0x0 (DISABLE): Layer 3 IP SA Inverse Match is disabled 0x1 (ENABLE): Layer 3 IP SA Inverse Match is enabled Value After Reset: 0x0
			2	L3SAM0	R/W	0x0	Layer 3 IP SA Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Source Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3DAM0 bit because either IPv6 SA or DA can be checked for filtering.Values: 0x0 (DISABLE): Layer 3 IP SA Match is disabled 0x1 (ENABLE): Layer 3 IP SA Match is enabled Value After Reset: 0x0
			1	Reserved_1	R	0x0	Reserved. Value After Reset: 0x0
			0	L3PEN0	R/W	0x0	Layer 3 Protocol Enable When this bit is set, the Layer 3 IP Source or Destination Address matching is enabled for IPv6 packets. When this bit is reset, the Layer 3 IP Source or Destination Address matching is enabled for IPv4 packets. The Layer 3 matching is done only when the L3SAM0 or L3DAM0 bit is set.Values: 0x0 (DISABLE): Layer 3 Protocol is disabled 0x1 (ENABLE): Layer 3 Protocol is enabled Value After Reset: 0x0
0x934	MAC_Layer4_Address1	You can configure the Layer 3 and Layer 4 Address Registers to be double-synchronized by selecting the Synchronize Layer 3 and Layer 4 Address Registers to Rx Clock Domain option while configuring the IP. When you select this option, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:1] (in big-endian mode) of the Layer 3 and Layer 4 Address Registers are written. For proper synchronization updates, you should perform consecutive writes to same Layer 3 and Layer 4 Address Registers after at least four clock cycles delay of the destination clock.	31:16	L4DP0	R/W	0x0	Layer 4 Destination Port Number Field When the L4PEN0 bit is reset and the L4DPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Destination Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4DPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Destination Port Number field in the IPv4 or IPv6 packets. Value After Reset: 0x0
			15:0	L4SP0	R/W	0x0	Layer 4 Source Port Number Field When the L4PEN0 bit is reset and the L4SPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Source Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4SPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Source Port Number field in the IPv4 or IPv6 packets. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x940	MAC_Layer3_Addr0_Reg1	For IPv4 packets, this register contains the 32-bit IP Source Address field. For IPv6 packets, it contains Bits[31:1] of the 128-bit IP Source Address or Destination Address field.	31:0	L3A00	R/W	0x0	Layer 3 Address 0 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Source Address field in the IPv4 packets. Value After Reset: 0x0
0x944	MAC_Layer3_Addr1_Reg1	For IPv4 packets, this register contains the 32-bit IP Destination Address field. For IPv6 packets, it contains Bits[63:32] of the 128-bit IP Source Address or Destination Address field.	31:0	L3A10	R/W	0x0	Layer 3 Address 1 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Destination Address field in the IPv4 packets. Value After Reset: 0x0
0x948	MAC_Layer3_Addr2_Reg1	This register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[95:64] of 128-bit IP Source Address or Destination Address field.	31:0	L3A20	R/W	0x0	Layer 3 Address 2 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used. Value After Reset: 0x0
0x94c	MAC_Layer3_Addr3_Reg1	This register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[127:96] of 128-bit IP Source Address or Destination Address field.	31:0	L3A30	R/W	0x0	Layer 3 Address 3 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used. Value After Reset: 0x0
			31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28	DMCHEN0	R/W	0x0	DMA Channel Select Enable When set, this bit enables the selection of the DMA channel number for the packet that is passed by this L3_L4 filter. The DMA channel is indicated by the DMCHN bits. When this bit is reset, the DMA channel is not decided by this filter.Values: 0x0 (DISABLE): DMA Channel Select is disabled 0x1 (ENABLE): DMA Channel Select is enabled Value After Reset: 0x0
			27	Reserved_27_y	R	0x0	Reserved. Value After Reset: 0x0
			26:24	DMCHN0	R/W	0x0	DMA Channel Number When DMCHEN is set high, this field selects the DMA Channel number to which the packet passed by this filter is routed. The width of this field depends on the number of the DMA channels present in your configuration. Value After Reset: 0x0
			23:22	Reserved_23_22	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			21	L4DPIM0	R/W	0x0	Layer 4 Destination Port Inverse Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Destination Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4DPM0 bit is set high.Values: 0x0 (DISABLE): Layer 4 Destination Port Inverse Match is disabled 0x1 (ENABLE): Layer 4 Destination Port Inverse Match is enabled Value After Reset: 0x0
			20	L4DPM0	R/W	0x0	Layer 4 Destination Port Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Destination Port number field for matching.Values: 0x0 (DISABLE): Layer 4 Destination Port Match is disabled 0x1 (ENABLE): Layer 4 Destination Port Match is enabled Value After Reset: 0x0
			19	L4SPIM0	R/W	0x0	Layer 4 Source Port Inverse Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Source Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4SPM0 bit is set high.Values: 0x0 (DISABLE): Layer 4 Source Port Inverse Match is disabled 0x1 (ENABLE): Layer 4 Source Port Inverse Match is enabled Value After Reset: 0x0
			18	L4SPM0	R/W	0x0	Layer 4 Source Port Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Source Port number field for matching.Values: 0x0 (DISABLE): Layer 4 Source Port Match is disabled 0x1 (ENABLE): Layer 4 Source Port Match is enabled Value After Reset: 0x0
			17	Reserved_17	R	0x0	Reserved. Value After Reset: 0x0
			16	L4PEN0	R/W	0x0	Layer 4 Protocol Enable When this bit is set, the Source and Destination Port number fields of UDP packets are used for matching. When this bit is reset, the Source and Destination Port number fields of TCP packets are used for matching. The Layer 4 matching is done only when the L4SPM0 or L4DPM0 bit is set.Values: 0x0 (DISABLE): Layer 4 Protocol is disabled 0x1 (ENABLE): Layer 4 Protocol is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x960	MAC_L3_L4_Control2	The Layer 3 and Layer 4 Control register controls the operations of filter 2 of Layer 3 and Layer 4.	15:11	L3HDBM0	R/W	0x0	Layer 3 IP DA Higher Bits Match IPv4 Packets: This field contains the number of higher bits of IP Destination Address that are matched in the IPv4 packets. The following list describes the values of this field: 0: No bits are masked. 1: LSb[0] is masked 2: Two LSbs [1:0] are masked .. 31: All bits except MSb are masked. IPv6 Packets: Bits[12:11] of this field correspond to Bits[6:5] of L3HSBM0 which indicate the number of lower bits of IP Source or Destination Address that are masked in the IPv6 packets. The following list describes the concatenated values of the L3HDBM0[1:0] and L3HSBM0 bits: 0: No bits are masked. 1: LSb[0] is masked. 2: Two LSbs [1:0] are masked .. 127: All bits except MSb are masked. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set. Value After Reset: 0x0
			10:6	L3HSBM0	R/W	0x0	Layer 3 IP SA Higher Bits Match IPv4 Packets: This field contains the number of lower bits of IP Source Address that are masked for matching in the IPv4 packets. The following list describes the values of this field: 0: No bits are masked. 1: LSb[0] is masked 2: Two LSbs [1:0] are masked .. 31: All bits except MSb are masked. IPv6 Packets: This field contains Bits[4:0] of L3HSBM0. These bits indicate the number of higher bits of IP Source or Destination Address matched in the IPv6 packets. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set high. Value After Reset: 0x0
			5	L3DAIM0	R/W	0x0	Layer 3 IP DA Inverse Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for inverse matching. When this bit is reset, the Layer 3 IP Destination Address field is enabled for perfect matching. This bit is valid and applicable only when the L3DAM0 bit is set high.Values: 0x0 (DISABLE): Layer 3 IP DA Inverse Match is disabled 0x1 (ENABLE): Layer 3 IP DA Inverse Match is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			4	L3DAM0	R/W	0x0	Layer 3 IP DA Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Destination Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3SAM0 bit because either IPv6 DA or SA can be checked for filtering.Values: 0x0 (DISABLE): Layer 3 IP DA Match is disabled 0x1 (ENABLE): Layer 3 IP DA Match is enabled Value After Reset: 0x0
			3	L3SAIM0	R/W	0x0	Layer 3 IP SA Inverse Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for inverse matching. When this bit reset, the Layer 3 IP Source Address field is enabled for perfect matching. This bit is valid and applicable only when the L3SAM0 bit is set.Values: 0x0 (DISABLE): Layer 3 IP SA Inverse Match is disabled 0x1 (ENABLE): Layer 3 IP SA Inverse Match is enabled Value After Reset: 0x0
			2	L3SAM0	R/W	0x0	Layer 3 IP SA Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Source Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3DAM0 bit because either IPv6 SA or DA can be checked for filtering.Values: 0x0 (DISABLE): Layer 3 IP SA Match is disabled 0x1 (ENABLE): Layer 3 IP SA Match is enabled Value After Reset: 0x0
			1	Reserved_1	R	0x0	Reserved. Value After Reset: 0x0
			0	L3PEN0	R/W	0x0	Layer 3 Protocol Enable When this bit is set, the Layer 3 IP Source or Destination Address matching is enabled for IPv6 packets. When this bit is reset, the Layer 3 IP Source or Destination Address matching is enabled for IPv4 packets. The Layer 3 matching is done only when the L3SAM0 or L3DAM0 bit is set.Values: 0x0 (DISABLE): Layer 3 Protocol is disabled 0x1 (ENABLE): Layer 3 Protocol is enabled Value After Reset: 0x0
0x964	MAC_Layer4_Address2	You can configure the Layer 3 and Layer 4 Address Registers to be double-synchronized by selecting the Synchronize Layer 3 and Layer 4 Address Registers to Rx Clock Domain option while configuring the IP. When you select this option, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:2] (in big-endian mode) of the Layer 3 and Layer 4 Address Registers are written. For proper synchronization updates, you should perform consecutive writes to same Layer 3 and Layer 4 Address Registers after at least four clock cycles delay of the destination clock.	31:16	L4DP0	R/W	0x0	Layer 4 Destination Port Number Field When the L4PEN0 bit is reset and the L4DPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Destination Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4DPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Destination Port Number field in the IPv4 or IPv6 packets. Value After Reset: 0x0
			15:0	L4SP0	R/W	0x0	Layer 4 Source Port Number Field When the L4PEN0 bit is reset and the L4SPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Source Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4SPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Source Port Number field in the IPv4 or IPv6 packets. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x970	MAC_Layer3_Addr0_Reg2	For IPv4 packets, this register contains the 32-bit IP Source Address field. For IPv6 packets, it contains Bits[31:2] of the 128-bit IP Source Address or Destination Address field.	31:0	L3A00	R/W	0x0	Layer 3 Address 0 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Source Address field in the IPv4 packets. Value After Reset: 0x0
0x974	MAC_Layer3_Addr1_Reg2	For IPv4 packets, this register contains the 32-bit IP Destination Address field. For IPv6 packets, it contains Bits[63:32] of the 128-bit IP Source Address or Destination Address field.	31:0	L3A10	R/W	0x0	Layer 3 Address 1 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Destination Address field in the IPv4 packets. Value After Reset: 0x0
0x978	MAC_Layer3_Addr2_Reg2	This register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[95:64] of 128-bit IP Source Address or Destination Address field.	31:0	L3A20	R/W	0x0	Layer 3 Address 2 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used. Value After Reset: 0x0
0x97c	MAC_Layer3_Addr3_Reg2	This register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[127:96] of 128-bit IP Source Address or Destination Address field.	31:0	L3A30	R/W	0x0	Layer 3 Address 3 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used. Value After Reset: 0x0
			31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28	DMCHEN0	R/W	0x0	DMA Channel Select Enable When set, this bit enables the selection of the DMA channel number for the packet that is passed by this L3_L4 filter. The DMA channel is indicated by the DMCHN bits. When this bit is reset, the DMA channel is not decided by this filter.Values: 0x0 (DISABLE): DMA Channel Select is disabled 0x1 (ENABLE): DMA Channel Select is enabled Value After Reset: 0x0
			27	Reserved_27_y	R	0x0	Reserved. Value After Reset: 0x0
			26:24	DMCHN0	R/W	0x0	DMA Channel Number When DMCHEN is set high, this field selects the DMA Channel number to which the packet passed by this filter is routed. The width of this field depends on the number of the DMA channels present in your configuration. Value After Reset: 0x0
			23:22	Reserved_23_22	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			21	L4DPIM0	R/W	0x0	Layer 4 Destination Port Inverse Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Destination Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4DPM0 bit is set high.Values: 0x0 (DISABLE): Layer 4 Destination Port Inverse Match is disabled 0x1 (ENABLE): Layer 4 Destination Port Inverse Match is enabled Value After Reset: 0x0
			20	L4DPM0	R/W	0x0	Layer 4 Destination Port Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Destination Port number field for matching.Values: 0x0 (DISABLE): Layer 4 Destination Port Match is disabled 0x1 (ENABLE): Layer 4 Destination Port Match is enabled Value After Reset: 0x0
			19	L4SPIM0	R/W	0x0	Layer 4 Source Port Inverse Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Source Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4SPM0 bit is set high.Values: 0x0 (DISABLE): Layer 4 Source Port Inverse Match is disabled 0x1 (ENABLE): Layer 4 Source Port Inverse Match is enabled Value After Reset: 0x0
			18	L4SPM0	R/W	0x0	Layer 4 Source Port Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Source Port number field for matching.Values: 0x0 (DISABLE): Layer 4 Source Port Match is disabled 0x1 (ENABLE): Layer 4 Source Port Match is enabled Value After Reset: 0x0
			17	Reserved_17	R	0x0	Reserved. Value After Reset: 0x0
			16	L4PEN0	R/W	0x0	Layer 4 Protocol Enable When this bit is set, the Source and Destination Port number fields of UDP packets are used for matching. When this bit is reset, the Source and Destination Port number fields of TCP packets are used for matching. The Layer 4 matching is done only when the L4SPM0 or L4DPM0 bit is set.Values: 0x0 (DISABLE): Layer 4 Protocol is disabled 0x1 (ENABLE): Layer 4 Protocol is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x990	MAC_L3_L4_Control3	The Layer 3 and Layer 4 Control register controls the operations of filter 3 of Layer 3 and Layer 4.	15:11	L3HDBM0	R/W	0x0	Layer 3 IP DA Higher Bits Match IPv4 Packets: This field contains the number of higher bits of IP Destination Address that are matched in the IPv4 packets. The following list describes the values of this field: 0: No bits are masked. 1: LSb[0] is masked 2: Two LSbs [1:0] are masked .. 31: All bits except MSb are masked. IPv6 Packets: Bits[12:11] of this field correspond to Bits[6:5] of L3HSBM0 which indicate the number of lower bits of IP Source or Destination Address that are masked in the IPv6 packets. The following list describes the concatenated values of the L3HDBM0[1:0] and L3HSBM0 bits: 0: No bits are masked. 1: LSb[0] is masked. 2: Two LSbs [1:0] are masked .. 127: All bits except MSb are masked. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set. Value After Reset: 0x0
			10:6	L3HSBM0	R/W	0x0	Layer 3 IP SA Higher Bits Match IPv4 Packets: This field contains the number of lower bits of IP Source Address that are masked for matching in the IPv4 packets. The following list describes the values of this field: 0: No bits are masked. 1: LSb[0] is masked 2: Two LSbs [1:0] are masked .. 31: All bits except MSb are masked. IPv6 Packets: This field contains Bits[4:0] of L3HSBM0. These bits indicate the number of higher bits of IP Source or Destination Address matched in the IPv6 packets. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set high. Value After Reset: 0x0
			5	L3DAIM0	R/W	0x0	Layer 3 IP DA Inverse Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for inverse matching. When this bit is reset, the Layer 3 IP Destination Address field is enabled for perfect matching. This bit is valid and applicable only when the L3DAM0 bit is set high.Values: 0x0 (DISABLE): Layer 3 IP DA Inverse Match is disabled 0x1 (ENABLE): Layer 3 IP DA Inverse Match is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			4	L3DAM0	R/W	0x0	Layer 3 IP DA Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Destination Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3SAM0 bit because either IPv6 DA or SA can be checked for filtering.Values: 0x0 (DISABLE): Layer 3 IP DA Match is disabled 0x1 (ENABLE): Layer 3 IP DA Match is enabled Value After Reset: 0x0
			3	L3SAIM0	R/W	0x0	Layer 3 IP SA Inverse Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for inverse matching. When this bit reset, the Layer 3 IP Source Address field is enabled for perfect matching. This bit is valid and applicable only when the L3SAM0 bit is set.Values: 0x0 (DISABLE): Layer 3 IP SA Inverse Match is disabled 0x1 (ENABLE): Layer 3 IP SA Inverse Match is enabled Value After Reset: 0x0
			2	L3SAM0	R/W	0x0	Layer 3 IP SA Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Source Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3DAM0 bit because either IPv6 SA or DA can be checked for filtering.Values: 0x0 (DISABLE): Layer 3 IP SA Match is disabled 0x1 (ENABLE): Layer 3 IP SA Match is enabled Value After Reset: 0x0
			1	Reserved_1	R	0x0	Reserved. Value After Reset: 0x0
			0	L3PEN0	R/W	0x0	Layer 3 Protocol Enable When this bit is set, the Layer 3 IP Source or Destination Address matching is enabled for IPv6 packets. When this bit is reset, the Layer 3 IP Source or Destination Address matching is enabled for IPv4 packets. The Layer 3 matching is done only when the L3SAM0 or L3DAM0 bit is set.Values: 0x0 (DISABLE): Layer 3 Protocol is disabled 0x1 (ENABLE): Layer 3 Protocol is enabled Value After Reset: 0x0
0x994	MAC_Layer4_Address3	You can configure the Layer 3 and Layer 4 Address Registers to be double-synchronized by selecting the Synchronize Layer 3 and Layer 4 Address Registers to Rx Clock Domain option while configuring the IP. When you select this option, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:3] (in big-endian mode) of the Layer 3 and Layer 4 Address Registers are written. For proper synchronization updates, you should perform consecutive writes to same Layer 3 and Layer 4 Address Registers after at least four clock cycles delay of the destination clock.	31:16	L4DP0	R/W	0x0	Layer 4 Destination Port Number Field When the L4PEN0 bit is reset and the L4DPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Destination Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4DPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Destination Port Number field in the IPv4 or IPv6 packets. Value After Reset: 0x0
			15:0	L4SP0	R/W	0x0	Layer 4 Source Port Number Field When the L4PEN0 bit is reset and the L4SPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Source Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4SPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Source Port Number field in the IPv4 or IPv6 packets. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x9a0	MAC_Layer3_Addr0_Reg3	For IPv4 packets, this register contains the 32-bit IP Source Address field. For IPv6 packets, it contains Bits[31:3] of the 128-bit IP Source Address or Destination Address field.	31:0	L3A00	R/W	0x0	Layer 3 Address 0 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Source Address field in the IPv4 packets. Value After Reset: 0x0
0x9a4	MAC_Layer3_Addr1_Reg3	For IPv4 packets, this register contains the 32-bit IP Destination Address field. For IPv6 packets, it contains Bits[63:32] of the 128-bit IP Source Address or Destination Address field.	31:0	L3A10	R/W	0x0	Layer 3 Address 1 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Destination Address field in the IPv4 packets. Value After Reset: 0x0
0x9a8	MAC_Layer3_Addr2_Reg3	This register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[95:64] of 128-bit IP Source Address or Destination Address field.	31:0	L3A20	R/W	0x0	Layer 3 Address 2 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used. Value After Reset: 0x0
0x9ac	MAC_Layer3_Addr3_Reg3	This register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[127:96] of 128-bit IP Source Address or Destination Address field.	31:0	L3A30	R/W	0x0	Layer 3 Address 3 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used. Value After Reset: 0x0
			31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28	DMCHEN0	R/W	0x0	DMA Channel Select Enable When set, this bit enables the selection of the DMA channel number for the packet that is passed by this L3_L4 filter. The DMA channel is indicated by the DMCHN bits. When this bit is reset, the DMA channel is not decided by this filter.Values: 0x0 (DISABLE): DMA Channel Select is disabled 0x1 (ENABLE): DMA Channel Select is enabled Value After Reset: 0x0
			27	Reserved_27_y	R	0x0	Reserved. Value After Reset: 0x0
			26:24	DMCHN0	R/W	0x0	DMA Channel Number When DMCHEN is set high, this field selects the DMA Channel number to which the packet passed by this filter is routed. The width of this field depends on the number of the DMA channels present in your configuration. Value After Reset: 0x0
			23:22	Reserved_23_22	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			21	L4DPIM0	R/W	0x0	Layer 4 Destination Port Inverse Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Destination Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4DPM0 bit is set high.Values: 0x0 (DISABLE): Layer 4 Destination Port Inverse Match is disabled 0x1 (ENABLE): Layer 4 Destination Port Inverse Match is enabled Value After Reset: 0x0
			20	L4DPM0	R/W	0x0	Layer 4 Destination Port Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Destination Port number field for matching.Values: 0x0 (DISABLE): Layer 4 Destination Port Match is disabled 0x1 (ENABLE): Layer 4 Destination Port Match is enabled Value After Reset: 0x0
			19	L4SPIM0	R/W	0x0	Layer 4 Source Port Inverse Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Source Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4SPM0 bit is set high.Values: 0x0 (DISABLE): Layer 4 Source Port Inverse Match is disabled 0x1 (ENABLE): Layer 4 Source Port Inverse Match is enabled Value After Reset: 0x0
			18	L4SPM0	R/W	0x0	Layer 4 Source Port Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Source Port number field for matching.Values: 0x0 (DISABLE): Layer 4 Source Port Match is disabled 0x1 (ENABLE): Layer 4 Source Port Match is enabled Value After Reset: 0x0
			17	Reserved_17	R	0x0	Reserved. Value After Reset: 0x0
			16	L4PEN0	R/W	0x0	Layer 4 Protocol Enable When this bit is set, the Source and Destination Port number fields of UDP packets are used for matching. When this bit is reset, the Source and Destination Port number fields of TCP packets are used for matching. The Layer 4 matching is done only when the L4SPM0 or L4DPM0 bit is set.Values: 0x0 (DISABLE): Layer 4 Protocol is disabled 0x1 (ENABLE): Layer 4 Protocol is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x9c0	MAC_L3_L4_Control4	The Layer 3 and Layer 4 Control register controls the operations of filter 4 of Layer 3 and Layer 4.	15:11	L3HDBM0	R/W	0x0	Layer 3 IP DA Higher Bits Match IPv4 Packets: This field contains the number of higher bits of IP Destination Address that are matched in the IPv4 packets. The following list describes the values of this field: 0: No bits are masked. 1: LSb[0] is masked 2: Two LSbs [1:0] are masked .. 31: All bits except MSb are masked. IPv6 Packets: Bits[12:11] of this field correspond to Bits[6:5] of L3HSBM0 which indicate the number of lower bits of IP Source or Destination Address that are masked in the IPv6 packets. The following list describes the concatenated values of the L3HDBM0[1:0] and L3HSBM0 bits: 0: No bits are masked. 1: LSb[0] is masked. 2: Two LSbs [1:0] are masked .. 127: All bits except MSb are masked. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set. Value After Reset: 0x0
			10:6	L3HSBM0	R/W	0x0	Layer 3 IP SA Higher Bits Match IPv4 Packets: This field contains the number of lower bits of IP Source Address that are masked for matching in the IPv4 packets. The following list describes the values of this field: 0: No bits are masked. 1: LSb[0] is masked 2: Two LSbs [1:0] are masked .. 31: All bits except MSb are masked. IPv6 Packets: This field contains Bits[4:0] of L3HSBM0. These bits indicate the number of higher bits of IP Source or Destination Address matched in the IPv6 packets. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set high. Value After Reset: 0x0
			5	L3DAIM0	R/W	0x0	Layer 3 IP DA Inverse Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for inverse matching. When this bit is reset, the Layer 3 IP Destination Address field is enabled for perfect matching. This bit is valid and applicable only when the L3DAM0 bit is set high.Values: 0x0 (DISABLE): Layer 3 IP DA Inverse Match is disabled 0x1 (ENABLE): Layer 3 IP DA Inverse Match is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			4	L3DAM0	R/W	0x0	Layer 3 IP DA Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Destination Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3SAM0 bit because either IPv6 DA or SA can be checked for filtering.Values: 0x0 (DISABLE): Layer 3 IP DA Match is disabled 0x1 (ENABLE): Layer 3 IP DA Match is enabled Value After Reset: 0x0
			3	L3SAIM0	R/W	0x0	Layer 3 IP SA Inverse Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for inverse matching. When this bit reset, the Layer 3 IP Source Address field is enabled for perfect matching. This bit is valid and applicable only when the L3SAM0 bit is set.Values: 0x0 (DISABLE): Layer 3 IP SA Inverse Match is disabled 0x1 (ENABLE): Layer 3 IP SA Inverse Match is enabled Value After Reset: 0x0
			2	L3SAM0	R/W	0x0	Layer 3 IP SA Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Source Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3DAM0 bit because either IPv6 SA or DA can be checked for filtering.Values: 0x0 (DISABLE): Layer 3 IP SA Match is disabled 0x1 (ENABLE): Layer 3 IP SA Match is enabled Value After Reset: 0x0
			1	Reserved_1	R	0x0	Reserved. Value After Reset: 0x0
			0	L3PEN0	R/W	0x0	Layer 3 Protocol Enable When this bit is set, the Layer 3 IP Source or Destination Address matching is enabled for IPv6 packets. When this bit is reset, the Layer 3 IP Source or Destination Address matching is enabled for IPv4 packets. The Layer 3 matching is done only when the L3SAM0 or L3DAM0 bit is set.Values: 0x0 (DISABLE): Layer 3 Protocol is disabled 0x1 (ENABLE): Layer 3 Protocol is enabled Value After Reset: 0x0
0x9c4	MAC_Layer4_Address4	You can configure the Layer 3 and Layer 4 Address Registers to be double-synchronized by selecting the Synchronize Layer 3 and Layer 4 Address Registers to Rx Clock Domain option while configuring the IP. When you select this option, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:4] (in big-endian mode) of the Layer 3 and Layer 4 Address Registers are written. For proper synchronization updates, you should perform consecutive writes to same Layer 3 and Layer 4 Address Registers after at least four clock cycles delay of the destination clock.	31:16	L4DP0	R/W	0x0	Layer 4 Destination Port Number Field When the L4PEN0 bit is reset and the L4DPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Destination Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4DPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Destination Port Number field in the IPv4 or IPv6 packets. Value After Reset: 0x0
			15:0	L4SP0	R/W	0x0	Layer 4 Source Port Number Field When the L4PEN0 bit is reset and the L4SPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Source Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4SPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Source Port Number field in the IPv4 or IPv6 packets. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x9d0	MAC_Layer3_Addr0_Reg4	For IPv4 packets, this register contains the 32-bit IP Source Address field. For IPv6 packets, it contains Bits[31:4] of the 128-bit IP Source Address or Destination Address field.	31:0	L3A00	R/W	0x0	Layer 3 Address 0 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Source Address field in the IPv4 packets. Value After Reset: 0x0
0x9d4	MAC_Layer3_Addr1_Reg4	For IPv4 packets, this register contains the 32-bit IP Destination Address field. For IPv6 packets, it contains Bits[63:32] of the 128-bit IP Source Address or Destination Address field.	31:0	L3A10	R/W	0x0	Layer 3 Address 1 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Destination Address field in the IPv4 packets. Value After Reset: 0x0
0x9d8	MAC_Layer3_Addr2_Reg4	This register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[95:64] of 128-bit IP Source Address or Destination Address field.	31:0	L3A20	R/W	0x0	Layer 3 Address 2 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used. Value After Reset: 0x0
0x9dc	MAC_Layer3_Addr3_Reg4	This register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[127:96] of 128-bit IP Source Address or Destination Address field.	31:0	L3A30	R/W	0x0	Layer 3 Address 3 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used. Value After Reset: 0x0
			31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28	DMCHEN0	R/W	0x0	DMA Channel Select Enable When set, this bit enables the selection of the DMA channel number for the packet that is passed by this L3_L4 filter. The DMA channel is indicated by the DMCHN bits. When this bit is reset, the DMA channel is not decided by this filter.Values: 0x0 (DISABLE): DMA Channel Select is disabled 0x1 (ENABLE): DMA Channel Select is enabled Value After Reset: 0x0
			27	Reserved_27_y	R	0x0	Reserved. Value After Reset: 0x0
			26:24	DMCHN0	R/W	0x0	DMA Channel Number When DMCHEN is set high, this field selects the DMA Channel number to which the packet passed by this filter is routed. The width of this field depends on the number of the DMA channels present in your configuration. Value After Reset: 0x0
			23:22	Reserved_23_22	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			21	L4DPIM0	R/W	0x0	Layer 4 Destination Port Inverse Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Destination Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4DPM0 bit is set high.Values: 0x0 (DISABLE): Layer 4 Destination Port Inverse Match is disabled 0x1 (ENABLE): Layer 4 Destination Port Inverse Match is enabled Value After Reset: 0x0
			20	L4DPM0	R/W	0x0	Layer 4 Destination Port Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Destination Port number field for matching.Values: 0x0 (DISABLE): Layer 4 Destination Port Match is disabled 0x1 (ENABLE): Layer 4 Destination Port Match is enabled Value After Reset: 0x0
			19	L4SPIM0	R/W	0x0	Layer 4 Source Port Inverse Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Source Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4SPM0 bit is set high.Values: 0x0 (DISABLE): Layer 4 Source Port Inverse Match is disabled 0x1 (ENABLE): Layer 4 Source Port Inverse Match is enabled Value After Reset: 0x0
			18	L4SPM0	R/W	0x0	Layer 4 Source Port Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Source Port number field for matching.Values: 0x0 (DISABLE): Layer 4 Source Port Match is disabled 0x1 (ENABLE): Layer 4 Source Port Match is enabled Value After Reset: 0x0
			17	Reserved_17	R	0x0	Reserved. Value After Reset: 0x0
			16	L4PEN0	R/W	0x0	Layer 4 Protocol Enable When this bit is set, the Source and Destination Port number fields of UDP packets are used for matching. When this bit is reset, the Source and Destination Port number fields of TCP packets are used for matching. The Layer 4 matching is done only when the L4SPM0 or L4DPM0 bit is set.Values: 0x0 (DISABLE): Layer 4 Protocol is disabled 0x1 (ENABLE): Layer 4 Protocol is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x9f0	MAC_L3_L4_Control5	The Layer 3 and Layer 4 Control register controls the operations of filter 5 of Layer 3 and Layer 4.	15:11	L3HDBM0	R/W	0x0	Layer 3 IP DA Higher Bits Match IPv4 Packets: This field contains the number of higher bits of IP Destination Address that are matched in the IPv4 packets. The following list describes the values of this field: 0: No bits are masked. 1: LSb[0] is masked 2: Two LSbs [1:0] are masked .. 31: All bits except MSb are masked. IPv6 Packets: Bits[12:11] of this field correspond to Bits[6:5] of L3HSBM0 which indicate the number of lower bits of IP Source or Destination Address that are masked in the IPv6 packets. The following list describes the concatenated values of the L3HDBM0[1:0] and L3HSBM0 bits: 0: No bits are masked. 1: LSb[0] is masked. 2: Two LSbs [1:0] are masked .. 127: All bits except MSb are masked. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set. Value After Reset: 0x0
			10:6	L3HSBM0	R/W	0x0	Layer 3 IP SA Higher Bits Match IPv4 Packets: This field contains the number of lower bits of IP Source Address that are masked for matching in the IPv4 packets. The following list describes the values of this field: 0: No bits are masked. 1: LSb[0] is masked 2: Two LSbs [1:0] are masked .. 31: All bits except MSb are masked. IPv6 Packets: This field contains Bits[4:0] of L3HSBM0. These bits indicate the number of higher bits of IP Source or Destination Address matched in the IPv6 packets. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set high. Value After Reset: 0x0
			5	L3DAIM0	R/W	0x0	Layer 3 IP DA Inverse Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for inverse matching. When this bit is reset, the Layer 3 IP Destination Address field is enabled for perfect matching. This bit is valid and applicable only when the L3DAM0 bit is set high.Values: 0x0 (DISABLE): Layer 3 IP DA Inverse Match is disabled 0x1 (ENABLE): Layer 3 IP DA Inverse Match is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			4	L3DAM0	R/W	0x0	Layer 3 IP DA Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Destination Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3SAM0 bit because either IPv6 DA or SA can be checked for filtering.Values: 0x0 (DISABLE): Layer 3 IP DA Match is disabled 0x1 (ENABLE): Layer 3 IP DA Match is enabled Value After Reset: 0x0
			3	L3SAIM0	R/W	0x0	Layer 3 IP SA Inverse Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for inverse matching. When this bit reset, the Layer 3 IP Source Address field is enabled for perfect matching. This bit is valid and applicable only when the L3SAM0 bit is set.Values: 0x0 (DISABLE): Layer 3 IP SA Inverse Match is disabled 0x1 (ENABLE): Layer 3 IP SA Inverse Match is enabled Value After Reset: 0x0
			2	L3SAM0	R/W	0x0	Layer 3 IP SA Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Source Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3DAM0 bit because either IPv6 SA or DA can be checked for filtering.Values: 0x0 (DISABLE): Layer 3 IP SA Match is disabled 0x1 (ENABLE): Layer 3 IP SA Match is enabled Value After Reset: 0x0
			1	Reserved_1	R	0x0	Reserved. Value After Reset: 0x0
			0	L3PEN0	R/W	0x0	Layer 3 Protocol Enable When this bit is set, the Layer 3 IP Source or Destination Address matching is enabled for IPv6 packets. When this bit is reset, the Layer 3 IP Source or Destination Address matching is enabled for IPv4 packets. The Layer 3 matching is done only when the L3SAM0 or L3DAM0 bit is set.Values: 0x0 (DISABLE): Layer 3 Protocol is disabled 0x1 (ENABLE): Layer 3 Protocol is enabled Value After Reset: 0x0
0x9f4	MAC_Layer4_Address5	You can configure the Layer 3 and Layer 4 Address Registers to be double-synchronized by selecting the Synchronize Layer 3 and Layer 4 Address Registers to Rx Clock Domain option while configuring the IP. When you select this option, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:5] (in big-endian mode) of the Layer 3 and Layer 4 Address Registers are written. For proper synchronization updates, you should perform consecutive writes to same Layer 3 and Layer 4 Address Registers after at least four clock cycles delay of the destination clock.	31:16	L4DP0	R/W	0x0	Layer 4 Destination Port Number Field When the L4PEN0 bit is reset and the L4DPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Destination Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4DPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Destination Port Number field in the IPv4 or IPv6 packets. Value After Reset: 0x0
			15:0	L4SP0	R/W	0x0	Layer 4 Source Port Number Field When the L4PEN0 bit is reset and the L4SPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Source Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4SPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Source Port Number field in the IPv4 or IPv6 packets. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xa00	MAC_Layer3_Addr0_Reg5	For IPv4 packets, this register contains the 32-bit IP Source Address field. For IPv6 packets, it contains Bits[31:5] of the 128-bit IP Source Address or Destination Address field.	31:0	L3A00	R/W	0x0	Layer 3 Address 0 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Source Address field in the IPv4 packets. Value After Reset: 0x0
0xa04	MAC_Layer3_Addr1_Reg5	For IPv4 packets, this register contains the 32-bit IP Destination Address field. For IPv6 packets, it contains Bits[63:32] of the 128-bit IP Source Address or Destination Address field.	31:0	L3A10	R/W	0x0	Layer 3 Address 1 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Destination Address field in the IPv4 packets. Value After Reset: 0x0
0xa08	MAC_Layer3_Addr2_Reg5	This register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[95:64] of 128-bit IP Source Address or Destination Address field.	31:0	L3A20	R/W	0x0	Layer 3 Address 2 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used. Value After Reset: 0x0
0xa0c	MAC_Layer3_Addr3_Reg5	This register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[127:96] of 128-bit IP Source Address or Destination Address field.	31:0	L3A30	R/W	0x0	Layer 3 Address 3 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used. Value After Reset: 0x0
			31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28	DMCHEN0	R/W	0x0	DMA Channel Select Enable When set, this bit enables the selection of the DMA channel number for the packet that is passed by this L3_L4 filter. The DMA channel is indicated by the DMCHN bits. When this bit is reset, the DMA channel is not decided by this filter.Values: 0x0 (DISABLE): DMA Channel Select is disabled 0x1 (ENABLE): DMA Channel Select is enabled Value After Reset: 0x0
			27	Reserved_27_y	R	0x0	Reserved. Value After Reset: 0x0
			26:24	DMCHN0	R/W	0x0	DMA Channel Number When DMCHEN is set high, this field selects the DMA Channel number to which the packet passed by this filter is routed. The width of this field depends on the number of the DMA channels present in your configuration. Value After Reset: 0x0
			23:22	Reserved_23_22	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			21	L4DPIM0	R/W	0x0	Layer 4 Destination Port Inverse Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Destination Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4DPM0 bit is set high.Values: 0x0 (DISABLE): Layer 4 Destination Port Inverse Match is disabled 0x1 (ENABLE): Layer 4 Destination Port Inverse Match is enabled Value After Reset: 0x0
			20	L4DPM0	R/W	0x0	Layer 4 Destination Port Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Destination Port number field for matching.Values: 0x0 (DISABLE): Layer 4 Destination Port Match is disabled 0x1 (ENABLE): Layer 4 Destination Port Match is enabled Value After Reset: 0x0
			19	L4SPIM0	R/W	0x0	Layer 4 Source Port Inverse Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Source Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4SPM0 bit is set high.Values: 0x0 (DISABLE): Layer 4 Source Port Inverse Match is disabled 0x1 (ENABLE): Layer 4 Source Port Inverse Match is enabled Value After Reset: 0x0
			18	L4SPM0	R/W	0x0	Layer 4 Source Port Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Source Port number field for matching.Values: 0x0 (DISABLE): Layer 4 Source Port Match is disabled 0x1 (ENABLE): Layer 4 Source Port Match is enabled Value After Reset: 0x0
			17	Reserved_17	R	0x0	Reserved. Value After Reset: 0x0
			16	L4PEN0	R/W	0x0	Layer 4 Protocol Enable When this bit is set, the Source and Destination Port number fields of UDP packets are used for matching. When this bit is reset, the Source and Destination Port number fields of TCP packets are used for matching. The Layer 4 matching is done only when the L4SPM0 or L4DPM0 bit is set.Values: 0x0 (DISABLE): Layer 4 Protocol is disabled 0x1 (ENABLE): Layer 4 Protocol is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xa20	MAC_L3_L4_Control6	The Layer 3 and Layer 4 Control register controls the operations of filter 6 of Layer 3 and Layer 4.	15:11	L3HDBM0	R/W	0x0	Layer 3 IP DA Higher Bits Match IPv4 Packets: This field contains the number of higher bits of IP Destination Address that are matched in the IPv4 packets. The following list describes the values of this field: 0: No bits are masked. 1: LSb[0] is masked 2: Two LSbs [1:0] are masked .. 31: All bits except MSb are masked. IPv6 Packets: Bits[12:11] of this field correspond to Bits[6:5] of L3HSBM0 which indicate the number of lower bits of IP Source or Destination Address that are masked in the IPv6 packets. The following list describes the concatenated values of the L3HDBM0[1:0] and L3HSBM0 bits: 0: No bits are masked. 1: LSb[0] is masked. 2: Two LSbs [1:0] are masked .. 127: All bits except MSb are masked. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set. Value After Reset: 0x0
			10:6	L3HSBM0	R/W	0x0	Layer 3 IP SA Higher Bits Match IPv4 Packets: This field contains the number of lower bits of IP Source Address that are masked for matching in the IPv4 packets. The following list describes the values of this field: 0: No bits are masked. 1: LSb[0] is masked 2: Two LSbs [1:0] are masked .. 31: All bits except MSb are masked. IPv6 Packets: This field contains Bits[4:0] of L3HSBM0. These bits indicate the number of higher bits of IP Source or Destination Address matched in the IPv6 packets. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set high. Value After Reset: 0x0
			5	L3DAIM0	R/W	0x0	Layer 3 IP DA Inverse Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for inverse matching. When this bit is reset, the Layer 3 IP Destination Address field is enabled for perfect matching. This bit is valid and applicable only when the L3DAM0 bit is set high.Values: 0x0 (DISABLE): Layer 3 IP DA Inverse Match is disabled 0x1 (ENABLE): Layer 3 IP DA Inverse Match is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			4	L3DAM0	R/W	0x0	Layer 3 IP DA Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Destination Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3SAM0 bit because either IPv6 DA or SA can be checked for filtering.Values: 0x0 (DISABLE): Layer 3 IP DA Match is disabled 0x1 (ENABLE): Layer 3 IP DA Match is enabled Value After Reset: 0x0
			3	L3SAIM0	R/W	0x0	Layer 3 IP SA Inverse Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for inverse matching. When this bit reset, the Layer 3 IP Source Address field is enabled for perfect matching. This bit is valid and applicable only when the L3SAM0 bit is set.Values: 0x0 (DISABLE): Layer 3 IP SA Inverse Match is disabled 0x1 (ENABLE): Layer 3 IP SA Inverse Match is enabled Value After Reset: 0x0
			2	L3SAM0	R/W	0x0	Layer 3 IP SA Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Source Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3DAM0 bit because either IPv6 SA or DA can be checked for filtering.Values: 0x0 (DISABLE): Layer 3 IP SA Match is disabled 0x1 (ENABLE): Layer 3 IP SA Match is enabled Value After Reset: 0x0
			1	Reserved_1	R	0x0	Reserved. Value After Reset: 0x0
			0	L3PEN0	R/W	0x0	Layer 3 Protocol Enable When this bit is set, the Layer 3 IP Source or Destination Address matching is enabled for IPv6 packets. When this bit is reset, the Layer 3 IP Source or Destination Address matching is enabled for IPv4 packets. The Layer 3 matching is done only when the L3SAM0 or L3DAM0 bit is set.Values: 0x0 (DISABLE): Layer 3 Protocol is disabled 0x1 (ENABLE): Layer 3 Protocol is enabled Value After Reset: 0x0
0xa24	MAC_Layer4_Address6	You can configure the Layer 3 and Layer 4 Address Registers to be double-synchronized by selecting the Synchronize Layer 3 and Layer 4 Address Registers to Rx Clock Domain option while configuring the IP. When you select this option, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:6] (in big-endian mode) of the Layer 3 and Layer 4 Address Registers are written. For proper synchronization updates, you should perform consecutive writes to same Layer 3 and Layer 4 Address Registers after at least four clock cycles delay of the destination clock.	31:16	L4DP0	R/W	0x0	Layer 4 Destination Port Number Field When the L4PEN0 bit is reset and the L4DPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Destination Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4DPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Destination Port Number field in the IPv4 or IPv6 packets. Value After Reset: 0x0
			15:0	L4SP0	R/W	0x0	Layer 4 Source Port Number Field When the L4PEN0 bit is reset and the L4SPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Source Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4SPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Source Port Number field in the IPv4 or IPv6 packets. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xa30	MAC_Layer3_Addr0_Reg6	For IPv4 packets, this register contains the 32-bit IP Source Address field. For IPv6 packets, it contains Bits[31:6] of the 128-bit IP Source Address or Destination Address field.	31:0	L3A00	R/W	0x0	Layer 3 Address 0 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Source Address field in the IPv4 packets. Value After Reset: 0x0
0xa34	MAC_Layer3_Addr1_Reg6	For IPv4 packets, this register contains the 32-bit IP Destination Address field. For IPv6 packets, it contains Bits[63:32] of the 128-bit IP Source Address or Destination Address field.	31:0	L3A10	R/W	0x0	Layer 3 Address 1 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Destination Address field in the IPv4 packets. Value After Reset: 0x0
0xa38	MAC_Layer3_Addr2_Reg6	This register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[95:64] of 128-bit IP Source Address or Destination Address field.	31:0	L3A20	R/W	0x0	Layer 3 Address 2 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used. Value After Reset: 0x0
0xa3c	MAC_Layer3_Addr3_Reg6	This register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[127:96] of 128-bit IP Source Address or Destination Address field.	31:0	L3A30	R/W	0x0	Layer 3 Address 3 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used. Value After Reset: 0x0
			31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28	DMCHEN0	R/W	0x0	DMA Channel Select Enable When set, this bit enables the selection of the DMA channel number for the packet that is passed by this L3_L4 filter. The DMA channel is indicated by the DMCHN bits. When this bit is reset, the DMA channel is not decided by this filter.Values: 0x0 (DISABLE): DMA Channel Select is disabled 0x1 (ENABLE): DMA Channel Select is enabled Value After Reset: 0x0
			27	Reserved_27_y	R	0x0	Reserved. Value After Reset: 0x0
			26:24	DMCHN0	R/W	0x0	DMA Channel Number When DMCHEN is set high, this field selects the DMA Channel number to which the packet passed by this filter is routed. The width of this field depends on the number of the DMA channels present in your configuration. Value After Reset: 0x0
			23:22	Reserved_23_22	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			21	L4DPIM0	R/W	0x0	Layer 4 Destination Port Inverse Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Destination Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4DPM0 bit is set high.Values: 0x0 (DISABLE): Layer 4 Destination Port Inverse Match is disabled 0x1 (ENABLE): Layer 4 Destination Port Inverse Match is enabled Value After Reset: 0x0
			20	L4DPM0	R/W	0x0	Layer 4 Destination Port Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Destination Port number field for matching.Values: 0x0 (DISABLE): Layer 4 Destination Port Match is disabled 0x1 (ENABLE): Layer 4 Destination Port Match is enabled Value After Reset: 0x0
			19	L4SPIM0	R/W	0x0	Layer 4 Source Port Inverse Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Source Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4SPM0 bit is set high.Values: 0x0 (DISABLE): Layer 4 Source Port Inverse Match is disabled 0x1 (ENABLE): Layer 4 Source Port Inverse Match is enabled Value After Reset: 0x0
			18	L4SPM0	R/W	0x0	Layer 4 Source Port Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Source Port number field for matching.Values: 0x0 (DISABLE): Layer 4 Source Port Match is disabled 0x1 (ENABLE): Layer 4 Source Port Match is enabled Value After Reset: 0x0
			17	Reserved_17	R	0x0	Reserved. Value After Reset: 0x0
			16	L4PEN0	R/W	0x0	Layer 4 Protocol Enable When this bit is set, the Source and Destination Port number fields of UDP packets are used for matching. When this bit is reset, the Source and Destination Port number fields of TCP packets are used for matching. The Layer 4 matching is done only when the L4SPM0 or L4DPM0 bit is set.Values: 0x0 (DISABLE): Layer 4 Protocol is disabled 0x1 (ENABLE): Layer 4 Protocol is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xa50	MAC_L3_L4_Control7	The Layer 3 and Layer 4 Control register controls the operations of filter 7 of Layer 3 and Layer 4.	15:11	L3HDBM0	R/W	0x0	Layer 3 IP DA Higher Bits Match IPv4 Packets: This field contains the number of higher bits of IP Destination Address that are matched in the IPv4 packets. The following list describes the values of this field: 0: No bits are masked. 1: LSb[0] is masked 2: Two LSbs [1:0] are masked .. 31: All bits except MSb are masked. IPv6 Packets: Bits[12:11] of this field correspond to Bits[6:5] of L3HSBM0 which indicate the number of lower bits of IP Source or Destination Address that are masked in the IPv6 packets. The following list describes the concatenated values of the L3HDBM0[1:0] and L3HSBM0 bits: 0: No bits are masked. 1: LSb[0] is masked. 2: Two LSbs [1:0] are masked .. 127: All bits except MSb are masked. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set. Value After Reset: 0x0
			10:6	L3HSBM0	R/W	0x0	Layer 3 IP SA Higher Bits Match IPv4 Packets: This field contains the number of lower bits of IP Source Address that are masked for matching in the IPv4 packets. The following list describes the values of this field: 0: No bits are masked. 1: LSb[0] is masked 2: Two LSbs [1:0] are masked .. 31: All bits except MSb are masked. IPv6 Packets: This field contains Bits[4:0] of L3HSBM0. These bits indicate the number of higher bits of IP Source or Destination Address matched in the IPv6 packets. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set high. Value After Reset: 0x0
			5	L3DAIM0	R/W	0x0	Layer 3 IP DA Inverse Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for inverse matching. When this bit is reset, the Layer 3 IP Destination Address field is enabled for perfect matching. This bit is valid and applicable only when the L3DAM0 bit is set high.Values: 0x0 (DISABLE): Layer 3 IP DA Inverse Match is disabled 0x1 (ENABLE): Layer 3 IP DA Inverse Match is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			4	L3DAM0	R/W	0x0	Layer 3 IP DA Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Destination Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3SAM0 bit because either IPv6 DA or SA can be checked for filtering.Values: 0x0 (DISABLE): Layer 3 IP DA Match is disabled 0x1 (ENABLE): Layer 3 IP DA Match is enabled Value After Reset: 0x0
			3	L3SAIM0	R/W	0x0	Layer 3 IP SA Inverse Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for inverse matching. When this bit reset, the Layer 3 IP Source Address field is enabled for perfect matching. This bit is valid and applicable only when the L3SAM0 bit is set.Values: 0x0 (DISABLE): Layer 3 IP SA Inverse Match is disabled 0x1 (ENABLE): Layer 3 IP SA Inverse Match is enabled Value After Reset: 0x0
			2	L3SAM0	R/W	0x0	Layer 3 IP SA Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Source Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3DAM0 bit because either IPv6 SA or DA can be checked for filtering.Values: 0x0 (DISABLE): Layer 3 IP SA Match is disabled 0x1 (ENABLE): Layer 3 IP SA Match is enabled Value After Reset: 0x0
			1	Reserved_1	R	0x0	Reserved. Value After Reset: 0x0
			0	L3PEN0	R/W	0x0	Layer 3 Protocol Enable When this bit is set, the Layer 3 IP Source or Destination Address matching is enabled for IPv6 packets. When this bit is reset, the Layer 3 IP Source or Destination Address matching is enabled for IPv4 packets. The Layer 3 matching is done only when the L3SAM0 or L3DAM0 bit is set.Values: 0x0 (DISABLE): Layer 3 Protocol is disabled 0x1 (ENABLE): Layer 3 Protocol is enabled Value After Reset: 0x0
0xa54	MAC_Layer4_Address7	You can configure the Layer 3 and Layer 4 Address Registers to be double-synchronized by selecting the Synchronize Layer 3 and Layer 4 Address Registers to Rx Clock Domain option while configuring the IP. When you select this option, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:7] (in big-endian mode) of the Layer 3 and Layer 4 Address Registers are written. For proper synchronization updates, you should perform consecutive writes to same Layer 3 and Layer 4 Address Registers after at least four clock cycles delay of the destination clock.	31:16	L4DP0	R/W	0x0	Layer 4 Destination Port Number Field When the L4PEN0 bit is reset and the L4DPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Destination Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4DPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Destination Port Number field in the IPv4 or IPv6 packets. Value After Reset: 0x0
			15:0	L4SP0	R/W	0x0	Layer 4 Source Port Number Field When the L4PEN0 bit is reset and the L4SPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Source Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4SPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Source Port Number field in the IPv4 or IPv6 packets. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xa60	MAC_Layer3_Addr0_Reg7	For IPv4 packets, this register contains the 32-bit IP Source Address field. For IPv6 packets, it contains Bits[31:7] of the 128-bit IP Source Address or Destination Address field.	31:0	L3A00	R/W	0x0	Layer 3 Address 0 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Source Address field in the IPv4 packets. Value After Reset: 0x0
0xa64	MAC_Layer3_Addr1_Reg7	For IPv4 packets, this register contains the 32-bit IP Destination Address field. For IPv6 packets, it contains Bits[63:32] of the 128-bit IP Source Address or Destination Address field.	31:0	L3A10	R/W	0x0	Layer 3 Address 1 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Destination Address field in the IPv4 packets. Value After Reset: 0x0
0xa68	MAC_Layer3_Addr2_Reg7	This register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[95:64] of 128-bit IP Source Address or Destination Address field.	31:0	L3A20	R/W	0x0	Layer 3 Address 2 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used. Value After Reset: 0x0
0xa6c	MAC_Layer3_Addr3_Reg7	This register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[127:96] of 128-bit IP Source Address or Destination Address field.	31:0	L3A30	R/W	0x0	Layer 3 Address 3 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used. Value After Reset: 0x0
			31	SNPS_R	R/W	0x0	Value After Reset: 0x0 Testable: untestable
			30:20	Reserved_30_20	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xa70	MAC_Indir_Access_Ctrl	This register provides the Indirect Access control and status for MAC_<MSEl>_<AOFF> registers.	19:16	MSEL	R/W	0x0	Mode Select This field is used in indirect access of MAC_<MSEL>_<AOFF>. This field must be set along with initiation of read/write to MAC_<MSEL>_<AOFF> and should not be changed until the OB is reset. Values: 0000: MAC_TMRQ_Regs (Ethernet Type based RXQ mapping) 0001: MAC_PCCtrl_IndReg (Per Policing Counter Control Registers) 0010: MAC_PCntr_IndReg (Per Policing Counter Registers) 0011: MAC_DPCSel_IndReg (Per DA Perfect Filter Event Registers) 0100: MAC_VPCSel_IndReg (Per VLAN Filter Event Registers) 0101: MAC_LPCSel_IndReg (Per L3-L4 Filter Event Registers) 0110: MAC_FPCSel_IndReg0 (Filter Fail Event Register) 0111: MAC_PCStatus_IndReg0 (Policing Counter Status Registers) 1000-1111: Reserved for future use. Value After Reset: 0x0
			15:8	AOFF	R/W	0x0	Address Offset This field is used in indirect access of MAC_<MSEL>_<AOFF>. This field must be set along with initiation of read/write to MAC_<MSEL>_<AOFF> and should not be changed until the OB is reset. Values: When MSEL=0, 00000000: MAC_TMRQ_Regs0 00000001: MAC_TMRQ_Regs1 ... 00000111: MAC_TMRQ_Regs7 00001000-11111111: Reserved for future use. When MSEL=1, 00000000: MAC_PCCtrl_IndReg0 00000001: MAC_PCCtrl_IndReg1 ... 00011111: MAC_PCCtrl_IndReg31 00100000-11111111: Reserved for future use. When MSEL=2, 00000000: MAC_PCntr_IndReg0 00000001: MAC_PCntr_IndReg1 ... 00011111: MAC_PCntr_IndReg31 00100000-11111111: Reserved for future use. When MSEL=3, 00000000: MAC_DPCSel_IndReg0 00000001: MAC_DPCSel_IndReg1 ... 00011111: MAC_DPCSel_IndReg31 00100000-11111111: Reserved for future use. When MSEL=4, 00000000: MAC_VPCSel_IndReg0 00000001: MAC_VPCSel_IndReg1 ... 00011111: MAC_VPCSel_IndReg31 00100000-11111111: Reserved for future use.
			7:6	Reserved_7_6	R	0x0	Reserved. Value After Reset: 0x0
			5	AUTO	R/W	0x0	Auto increment 1: AOFF is incremented by 1. Software should ensure not to cause a wrap condition. Byte wise read/write is not supported when auto increment is enabled. 0: AOFF is not incremented automatically. Software should program the correct Address Offset for each access. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			4:2	Reserved_4_2	R	0x0	Reserved. Value After Reset: 0x0
			1	COM	R/W	0x0	Command type This bit indicates the register access type. 1: Indicates a read operation. 0: Indicates a write operation. Values: 0x0 (WRITE): Write operation 0x1 (READ): Read operation Value After Reset: 0x0
			0	OB	R/W	0x0	Operation Busy. This bit is set along with a read or write command for initiating the indirect access to MAC_<MSEL>_<AOFF> register. This bit is reset when the read or write command to MAC_<MSEL>_<AOFF> register is complete. The next indirect register access can be initiated only after this bit is reset. During a write operation, the bit is reset only after the data has been written into MAC_<MSEL>_<AOFF> register. During a read operation, the data should be read from MAC_Indir_Access_Data register only after this bit is reset. Value After Reset: 0x0 Testable: untestable
0xa74	MAC_Indir_Access_Data	This register holds the read/write data for Indirect Access of MAC_<MSEL_<AOFF> registers.	31:0	DATA	R/W	0x0	This field contains data to read/write for Indirect address access associated with MAC_Indir_Access_Ctrl register. Value After Reset: 0x0
0xa74	MAC_TMRQ_Regs(#i)(fori=0;i<=7)	This register contains the type, associated queue number and packet type (Pre-emption or Express) related to Type based RXQ mapping.	31:21	Reserved_31_21	R	0x0	Reserved. Value After Reset: 0x0
			20	PFEX	R/W	0x0	Preemption or Express Packet This bit indicates if it is a preemption packet or express packet. Value After Reset: 0x0
			19	Reserved_19	R	0x0	Reserved. Value After Reset: 0x0
			18:16	TMRQ	R/W	0x0	Type Match Rx Queue Number Indicates the receive queue number to which the packet needs to be forwarded on detecting a type match. Value After Reset: 0x0
			15:0	TYP	R/W	0x0	Type field Value Indicates the type value of packet that needs to be compared with received Ethernet packet. This field is valid when programmed to a value greater than or equal to 0x0600. Value After Reset: 0x0
			31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28	AV8021ASMEN	R/W	0x0	AV 802.1AS Mode Enable When this bit is set, the MAC processes only untagged PTP over Ethernet packets for providing PTP status and capturing timestamp snapshots, that is, IEEE 802.1AS mode of operation. When PTP offload feature is enabled, for the purpose of PTP offload, the transport specific field in the PTP header is generated and checked based on the value of this bit.Values: 0x0 (DISABLE): AV 802.1AS Mode is disabled 0x1 (ENABLE): AV 802.1AS Mode is enabled Value After Reset: 0x0
			27			#VALUE!	Reserved Field: Yes
			26	Reserved_26	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			25			#VALUE!	Reserved Field: Yes
			24	TXTSSTSM	R/W	0x0	Transmit Timestamp Status Mode When this bit is set, the MAC overwrites the earlier transmit timestamp status even if it is not read by the software. The MAC indicates this by setting the TXTSSMIS bit of the MAC_Tx_Timestamp_Status_Nanoseconds register. When this bit is reset, the MAC ignores the timestamp status of current packet if the timestamp status of previous packet is not read by the software. The MAC indicates this by setting the TXTSSMIS bit of the MAC_Tx_Timestamp_Status_Nanoseconds register.Values: 0x0 (DISABLE): Transmit Timestamp Status Mode is disabled 0x1 (ENABLE): Transmit Timestamp Status Mode is enabled Value After Reset: 0x0
			23:21	Reserved_23_21	R	0x0	Reserved. Value After Reset: 0x0
			20			#VALUE!	Reserved Field: Yes
			19	CSC	R/W	0x0	Enable checksum correction during OST for PTP over UDP/IPv4 packets When this bit is set, the last two bytes of PTP message sent over UDP/IPv4 is updated to keep the UDP checksum correct, for changes made to origin timestamp and/or correction field as part of one step timestamp operation. The application forms the packet with these two dummy bytes. When reset, no updates are done to keep the UDP checksum correct. The application forms the packet with UDP checksum set to 0.Values: 0x0 (DISABLE): checksum correction during OST for PTP over UDP/IPv4 packets is disabled 0x1 (ENABLE): checksum correction during OST for PTP over UDP/IPv4 packets is enabled Value After Reset: 0x0
			18	TSENMACADDR	R/W	0x0	Enable MAC Address for PTP Packet Filtering When this bit is set, the DA MAC address (that matches any MAC Address register) is used to filter the PTP packets when PTP is directly sent over Ethernet. When this bit is set, received PTP packets with DA containing a special multicast or unicast address that matches the one programmed in MAC address registers are considered for processing as indicated, when PTP is directly sent over Ethernet. For normal time stamping operation, MAC address registers 0 to 31 is considered for unicast destination address matching. For PTP offload, only MAC address register 0 is considered for unicast destination address matching.Values: 0x0 (DISABLE): MAC Address for PTP Packet Filtering is disabled 0x1 (ENABLE): MAC Address for PTP Packet Filtering is enabled Value After Reset: 0x0
			17:16	SNAPTYPSEL	R/W	0x0	Select PTP packets for Taking Snapshots These bits, along with Bits 15 and 14, decide the set of PTP packet types for which snapshot needs to be taken. The encoding is given in Timestamp Snapshot Dependency on Register Bits Table. Value After Reset: 0x0
			15	TSMSTRENA	R/W	0x0	Enable Snapshot for Messages Relevant to Master When this bit is set, the snapshot is taken only for the messages that are relevant to the master node. Otherwise, the snapshot is taken for the messages relevant to the slave node.Values: 0x0 (DISABLE): Snapshot for Messages Relevant to Master is disabled 0x1 (ENABLE): Snapshot for Messages Relevant to Master is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xb00	MAC_Timestamp_Control	This register controls the operation of the System Time generator and processing of PTP packets for timestamping in the Receiver.	14	TSEVNTENA	R/W	0x0	Enable Timestamp Snapshot for Event Messages When this bit is set, the timestamp snapshot is taken only for event messages (SYNC, Delay_Req, Pdelay_Req, or Pdelay_Resp). When this bit is reset, the snapshot is taken for all messages except Announce, Management, and Signaling. For more information about the timestamp snapshots, see Timestamp Snapshot Dependency on Register Bits Table.Values: 0x0 (DISABLE): Timestamp Snapshot for Event Messages is disabled 0x1 (ENABLE): Timestamp Snapshot for Event Messages is enabled Value After Reset: 0x0
			13	TSIPV4ENA	R/W	0x1	Enable Processing of PTP Packets Sent over IPv4-UDP When this bit is set, the MAC receiver processes the PTP packets encapsulated in IPv4-UDP packets. When this bit is reset, the MAC ignores the PTP transported over IPv4-UDP packets. This bit is set by default.Values: 0x0 (DISABLE): Processing of PTP Packets Sent over IPv4-UDP is disabled 0x1 (ENABLE): Processing of PTP Packets Sent over IPv4-UDP is enabled Value After Reset: 0x1
			12	TSIPV6ENA	R/W	0x0	Enable Processing of PTP Packets Sent over IPv6-UDP When this bit is set, the MAC receiver processes the PTP packets encapsulated in IPv6-UDP packets. When this bit is clear, the MAC ignores the PTP transported over IPv6-UDP packets.Values: 0x0 (DISABLE): Processing of PTP Packets Sent over IPv6-UDP is disabled 0x1 (ENABLE): Processing of PTP Packets Sent over IPv6-UDP is enabled Value After Reset: 0x0
			11	TSIPENA	R/W	0x0	Enable Processing of PTP over Ethernet Packets When this bit is set, the MAC receiver processes the PTP packets encapsulated directly in the Ethernet packets. When this bit is reset, the MAC ignores the PTP over Ethernet packets.Values: 0x0 (DISABLE): Processing of PTP over Ethernet Packets is disabled 0x1 (ENABLE): Processing of PTP over Ethernet Packets is enabled Value After Reset: 0x0
			10	TSVER2ENA	R/W	0x0	Enable PTP Packet Processing for Version 2 Format When this bit is set, the IEEE 1588 version 2 format is used to process the PTP packets. When this bit is reset, the IEEE 1588 version 1 format is used to process the PTP packets. The IEEE 1588 formats are described in 'PTP Processing and Control'.Values: 0x0 (DISABLE): PTP Packet Processing for Version 2 Format is disabled 0x1 (ENABLE): PTP Packet Processing for Version 2 Format is enabled Value After Reset: 0x0
			9	TSCTRLSSR	R/W	0x0	Timestamp Digital or Binary Rollover Control When this bit is set, the Timestamp Low register rolls over after 0x3B9A_C9FF value (that is, 1 nanosecond accuracy) and increments the timestamp (High) seconds. When this bit is reset, the rollover value of sub-second register is 0x7FFF_FFFF. The sub-second increment must be programmed correctly depending on the PTP reference clock frequency and the value of this bit.Values: 0x0 (DISABLE): Timestamp Digital or Binary Rollover Control is disabled 0x1 (ENABLE): Timestamp Digital or Binary Rollover Control is enabled Value After Reset: 0x0
			8	TSENALL	R/W	0x0	Enable Timestamp for All Packets When this bit is set, the timestamp snapshot is enabled for all packets received by the MAC.Values: 0x0 (DISABLE): Timestamp for All Packets disabled 0x1 (ENABLE): Timestamp for All Packets enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			7	Reserved_7	R	0x0	Reserved. Value After Reset: 0x0
			6	PTGE	R/W	0x0	Presentation Time Generation Enable When this bit is set the Presentation Time generation is enabled.Values: 0x0 (DISABLE): Presentation Time Generation is disabled 0x1 (ENABLE): Presentation Time Generation is enabled Value After Reset: 0x0
			5	TSADDREG	R/W	0x0	Update Addend Register When this bit is set, the content of the Timestamp Addend register is updated in the PTP block for fine correction. This bit is cleared when the update is complete. This bit must be zero before it is set. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect.Values: 0x0 (DISABLE): Addend Register is not updated 0x1 (ENABLE): Addend Register is updated Value After Reset: 0x0 Testable: untestable
			4			#VALUE!	Reserved Field: Yes
			3	TSUPDT	R/W	0x0	Update Timestamp When this bit is set, the system time is updated (added or subtracted) with the value specified in MAC_System_Time_Seconds_Update and MAC_System_Time_Nanoseconds_Update registers. This bit must be zero before updating it. This bit is reset when the update is complete in hardware. The Timestamp Higher Word register (if enabled while configuring the IP) is not updated. When Media Clock Generation and Recovery is enabled MAC_Presn_Time_Updt must also be updated before setting this field. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect.Values: 0x0 (DISABLE): Timestamp is not updated 0x1 (ENABLE): Timestamp is updated Value After Reset: 0x0 Testable: untestable
			2	TSINIT	R/W	0x0	Initialize Timestamp When this bit is set, the system time is initialized (overwritten) with the value specified in the MAC_System_Time_Seconds_Update and MAC_System_Time_Nanoseconds_Update registers. This bit must be zero before it is updated. This bit is reset when the initialization is complete. The Timestamp Higher Word register (if enabled while configuration the IP) can only be initialized. When Media Clock Generation and Recovery is enabled MAC_Presn_Time_Updt must also be updated before setting this field. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect.Values: 0x0 (DISABLE): Timestamp is not initialized 0x1 (ENABLE): Timestamp is initialized Value After Reset: 0x0 Testable: untestable
			1	TSCFUPDT	R/W	0x0	Fine or Coarse Timestamp Update When this bit is set, the Fine method is used to update system timestamp. When this bit is reset, Coarse method is used to update the system timestamp.Values: 0x0 (COARSE): Coarse method is used to update system timestamp 0x1 (FINE): Fine method is used to update system timestamp Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	TSENA	R/W	0x0	Enable Timestamp When this bit is set, the timestamp is added for Transmit and Receive packets. When disabled, timestamp is not added for transmit and receive packets and the Timestamp Generator is also suspended. You need to initialize the Timestamp (system time) after enabling this mode. On the Receive side, the MAC processes the 1588 packets only if this bit is set.Values: 0x0 (DISABLE): Timestamp is disabled 0x1 (ENABLE): Timestamp is enabled Value After Reset: 0x0
0xb04	MAC_Sub_Second_Increment	This register specifies the value to be added to the internal system time register every cycle of clk_ptp_ref_i clock.	31:24	Reserved_31_24	R	0x0	Reserved. Value After Reset: 0x0
			23:16	SSINC	R/W	0x0	Sub-second Increment Value The value programmed in this field is accumulated every clock cycle (of clk_ptp_i) with the contents of the sub-second register. For example, when the PTP clock is 50 MHz (period is 20 ns), you must program 20 (0x14) when the System Time Nanoseconds register has an accuracy of 1 ns [Bit 9 (TSCTRLSSR) is set in MAC_Timestamp_Control]. When TSCTRLSSR is clear, the Nanoseconds register has a resolution of 0.465 ns. In this case, you must program a value of 43 (0x2B) which is derived by 20 ns/0.465. Value After Reset: 0x0
			15:8	SNSINC	R/W	0x0	Sub-nanosecond Increment Value This field contains the sub-nanosecond increment value, represented in nanoseconds multiplied by 2^8. This value is accumulated with the sub-nanoseconds field of the subsecond register. For example, when TSCTRLSSR field in the MAC_Timestamp_Control register is 1 and if the required increment is 5.3ns, SSINC must be 0x05 and SNSINC must be 0x4C. Value After Reset: 0x0
			7:0	Reserved_7_0	R	0x0	Reserved. Value After Reset: 0x0
0xb08	MAC_System_Time_Seconds	The System Time Seconds register, along with System Time Nanoseconds register, indicates the current value of the system time maintained by the MAC. Though it is updated on a continuous basis, there is some delay from the actual time because of clock domain transfer latencies (from clk_ptp_ref_i to CSR clock).	31:0	TSS	R	0x0	Timestamp Second The value in this field indicates the current value in seconds of the System Time maintained by the MAC. Value After Reset: 0x0 Testable: untestable
			31	Reserved_31	R	0x0	Reserved. Value After Reset: 0x0
			30:0	TSSS	R	0x0	Timestamp Sub Seconds The value in this field has the sub-second representation of time, with an accuracy of 0.46 ns. When Bit 9 is set in MAC_Timestamp_Control, each bit represents 1 ns. The maximum value is 0x3B9A_C9FF after which it rolls-over to zero. Value After Reset: 0x0 Testable: untestable
0xb10	MAC_System_Time_Seconds_Update	The System Time Seconds Update register, along with the System Time Nanoseconds Update register, initializes or updates the system time maintained by the MAC. You must write both registers before setting the TSINIT or TSUPDT bits in MAC_Timestamp_Control register.	31:0	TSS	R/W	0x0	Timestamp Seconds The value in this field is the seconds part of the update. When ADDSUB is reset, this field must be programmed with the seconds part of the update value. When ADDSUB is set, this field must be programmed with the complement of the seconds part of the update value. For example, to subtract 2.000000001 seconds from the system time, the TSS field in the MAC_Timestamp_Seconds_Update register must be 0xFFFF_FFFE (that is, 2^32 - 2). Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xb14	MAC_System_Time_Nanoseconds_Update	MAC System Time Nanoseconds Update register.	31	ADDSUB	R/W	0x0	Add or Subtract Time When this bit is set, the time value is subtracted with the contents of the update register. When this bit is reset, the time value is added with the contents of the update register.Values: 0x0 (ADD): Add time 0x1 (SUB): Subtract time Value After Reset: 0x0
			30:0	TSSS	R/W	0x0	Timestamp Sub Seconds The value in this field is the sub-seconds part of the update. ADDSUB is 1: This field must be programmed with the complement of the sub-seconds part of the update value as described. Note: When ADDSUB is 1 and sub-seconds update should be 0 the TSSS field value should be programmed to following based on TSCTRLSSR: -- TSCTRLSSR=1, TSSS=10^9=0x3B9A_CA00 -- TSCTRLSSR=0, TSSS=2^31=0x8000_0000, as this cannot be accommodated in the 31 bit wide TSSS field, program (2^31)-1=0x7FFF_FFFF ADDSUB is 0: This field must be programmed with the sub-seconds part of the update value, with an accuracy based on the TSCTRLSSR bit of the MAC_Timestamp_Control register. TSCTRLSSR field in the MAC_Timestamp_Control register is 1: The programmed value must be 10^9 - <sub-second value>. Each bit represents 1 ns and the programmed value must not exceed 0x3B9A_C9FF. TSCTRLSSR field in the MAC_Timestamp_Control register is 0: The programmed value must be 2^31 - <sub-second_value>. Each bit represents an accuracy of 0.46 ns. For example, to subtract 2.000000001 seconds from the system time, then the TSSS field in the MAC_Timestamp_Nanoseconds_Update register must be 0x7FFF_FFFF (that is, 2^31 - 1), when TSCTRLSSR bit in MAC_Timestamp_Control is reset and 0x3B9A_C9FF (that is, 10^9 - 1), when TSCTRLSSR bit in MAC_Timestamp_Control is set. Value After Reset: 0x0
0xb18	MAC_Timestamp_Addend	Timestamp Addend register. This register value is used only when the system time is configured for Fine Update mode (TSCFUPDT bit in the MAC_Timestamp_Control register). The content of this register is added to a 32-bit accumulator in every clock cycle (of clk_ptp_ref_i) and the system time is updated when the accumulator overflows.	31:0	TSAR	R/W	0x0	Timestamp Addend Register This field indicates the 32-bit time value to be added to the Accumulator register to achieve time synchronization. Value After Reset: 0x0
0xb1c	MAC_System_Time_Higher_Word_Seconds	System Time - Higher Word Seconds register.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	TSHWR	R/W	0x0	Timestamp Higher Word Register This field contains the most-significant 16-bits of timestamp seconds value. This register is directly written to initialize the value and it is incremented when there is an overflow from 32-bits of the System Time Seconds register. Access restriction applies. Updated based on the event. Setting 1 sets. Setting 0 clears. Value After Reset: 0x0 Testable: untestable
			31:30	Reserved_31_30	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xb20	MAC_Timestamp_Status	Timestamp Status register. All bits except Bits[27:25] gets cleared when the application reads this register.	29:25	ATSNS	R	0x0	Number of Auxiliary Timestamp Snapshots This field indicates the number of Snapshots available in the FIFO. A value equal to the selected depth of FIFO (4, 8, or 16) indicates that the Auxiliary Snapshot FIFO is full. These bits are cleared (to 00000) when the Auxiliary snapshot FIFO clear bit is set. This bit is valid only if the Add IEEE 1588 Auxiliary Snapshot option is selected. Value After Reset: 0x0
			24	ATSSTM	R	0x0	Auxiliary Timestamp Snapshot Trigger Missed This bit is set when the Auxiliary timestamp snapshot FIFO is full and external trigger was set. This indicates that the latest snapshot is not stored in the FIFO. This bit is valid only if the Add IEEE 1588 Auxiliary Snapshot option is selected.Values: 0x0 (INACTIVE): Auxiliary Timestamp Snapshot Trigger Missed status not detected 0x1 (ACTIVE): Auxiliary Timestamp Snapshot Trigger Missed status detected Value After Reset: 0x0
			23:20	Reserved_23_20	R	0x0	Reserved. Value After Reset: 0x0
			19:16	ATSSTN	R	0x0	Auxiliary Timestamp Snapshot Trigger Identifier These bits identify the Auxiliary trigger inputs for which the timestamp available in the Auxiliary Snapshot Register is applicable. When more than one bit is set at the same time, it means that corresponding auxiliary triggers were sampled at the same clock. These bits are applicable only if the number of Auxiliary snapshots is more than one. One bit is assigned for each trigger as shown in the following list: Bit 16: Auxiliary trigger 0 Bit 17: Auxiliary trigger 1 Bit 18: Auxiliary trigger 2 Bit 19: Auxiliary trigger 3 The software can read this register to find the triggers that are set when the timestamp is taken. Value After Reset: 0x0
			15	TXTSSIS	R	0x0	Tx Timestamp Status Interrupt Status This bit is set when the captured transmit timestamp is updated in the MAC_Tx_Timestamp_Status_Nanoseconds and MAC_Tx_Timestamp_Status_Seconds registers. When PTP offload feature is enabled, this bit is set when the captured transmit timestamp is updated in the MAC_Tx_Timestamp_Status_Nanoseconds and MAC_Tx_Timestamp_Status_Seconds registers, for PTO generated Delay Request and Pdelay request packets. This bit is cleared when the MAC_Tx_Timestamp_Status_Seconds register is read (or write to MAC_Tx_Timestamp_Status_Seconds register when RCWE bit of MAC_CSR_SW_Ctrl register is set).Values: 0x0 (INACTIVE): Tx Timestamp Status Interrupt status not detected 0x1 (ACTIVE): Tx Timestamp Status Interrupt status detected Value After Reset: 0x0
			14:10	Reserved_14_10	R	0x0	Reserved. Value After Reset: 0x0
			9:4			#VALUE!	Reserved Field: Yes

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			3	TSTRGTERR0	R	0x0	Timestamp Target Time Error This bit is set when the latest target time programmed in the MAC_PPS0_Target_Time_Seconds and MAC_PPS0_Target_Time_Nanoseconds registers elapses. This bit is cleared when the application reads this bit. Access restriction applies. Clears on read (or this bit is written to 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Timestamp Target Time Error status not detected 0x1 (ACTIVE): Timestamp Target Time Error status detected Value After Reset: 0x0
			2	AUXTSTRIG	R	0x0	Auxiliary Timestamp Trigger Snapshot This bit is set high when the auxiliary snapshot is written to the FIFO. This bit is valid only if the Add IEEE 1588 Auxiliary Snapshot option is selected. Access restriction applies. Clears on read (or this bit is written to 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Auxiliary Timestamp Trigger Snapshot status not detected 0x1 (ACTIVE): Auxiliary Timestamp Trigger Snapshot status detected Value After Reset: 0x0
			1	TSTARGT0	R	0x0	Timestamp Target Time Reached When this bit is set and MCGREN0 of MAC_PPS_Control register is reset, it indicates that the value of system time is greater than or equal to the value specified in the MAC_PPS0_Target_Time_Seconds and MAC_PPS0_Target_Time_Nanoseconds registers. When this bit is set and MCGREN0 of MAC_PPS_Control register is set, it indicates that mcgr_dma_req_o[0] is asserted. Access restriction applies. Clears on read (or this bit is written to 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Timestamp Target Time Reached status not detected 0x1 (ACTIVE): Timestamp Target Time Reached status detected Value After Reset: 0x0
			0	TSSOVF	R	0x0	Timestamp Seconds Overflow When this bit is set, it indicates that the seconds value of the timestamp (when supporting version 2 format) has overflowed beyond 32'hFFFF_FFFF. Access restriction applies. Clears on read (or this bit is written to 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Timestamp Seconds Overflow status not detected 0x1 (ACTIVE): Timestamp Seconds Overflow status detected Value After Reset: 0x0
0xb24	MAC_Rx_Domain_Time_Incr	This register indicates the increment value of the PTP time maintained in Receive clock domain that is used	31:16	RXNS	R/W	0x0	Receive Domain Time Increment Value in Nanoseconds. This field indicates the Nanosecond part of the increment value of the PTP time maintained in Receive clock domain that is used for timestamping. For example, when operating in 2.5G speed the clock period is 3.2ns, so for Digital Rollover mode the RXNS and RXSNS fields must be programmed with 3 and (0.2*256)=51 respectively. In Binary Rollover mode, 3/0.466=6.44 and 0.2+0.44=0.64, so RXNS and RXSNS fields must be programmed with 6 and (0.64*256)=164 respectively. When both RXSNS and RXNS fields are all-zeros, the fixed value based on operating speed is used for PTP time increment in Receive clock domain. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
		for timestamping.	15:8	RXSNS	R/W	0x0	Receive Domain Time Increment Value in Sub-Nanoseconds. This field indicates the Sub-Nanosecond part of the increment value of the PTP time maintained in Receive clock domain that is used for timestamping. This field has granularity of 1/256 ns. When both RXSNS and RXNS fields are all-zeros, the fixed value based on operating speed is used for PTP time increment in Receive clock domain. Value After Reset: 0x0
			7:0	Reserved_7_0	R	0x0	Reserved. Value After Reset: 0x0
0xb28	MAC_Tx_Domain_Time_Incr	This register indicates the increment value of the PTP time maintained in Transmit clock domain that is used for timestamping.	31:16	TXNS	R/W	0x0	Transmit Domain Time Increment Value in Nanoseconds. This field indicates the Nanosecond part of the increment value of the PTP time maintained in Transmit clock domain that is used for timestamping. For example, when operating in 2.5G speed the clock period is 3.2ns, so for Digital Rollover mode the TXNS and RXSNS fields must be programmed with 3 and (0.2*256)=51 respectively. In Binary Rollover mode, 3/0.466=6.44 and 0.2+0.44=0.64, so TXNS and TXSNS fields must be programmed with 6 and (0.64*256)=164 respectively. When both TXSNS and TXNS fields are all-zeros, the fixed value based on operating speed is used for PTP time increment in Transmit clock domain. Value After Reset: 0x0
			15:8	TXSNS	R/W	0x0	Transmit Domain Time Increment Value in Sub-Nanoseconds. This field indicates the Sub-Nanosecond part of the increment value of the PTP time maintained in Transmit clock domain that is used for timestamping. This field has granularity of 1/256 ns. When both TXSNS and TXNS fields are all-zeros, the fixed value based on operating speed is used for PTP time increment in Transmit clock domain. Value After Reset: 0x0
			7:0	Reserved_7_0	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xb30	MAC_Tx_Timestamp_Status_Nanoseconds	This register contains the nanosecond part of timestamp captured for Transmit packets when Tx status is disabled.The MAC_Tx_Timestamp_Status_Nanoseconds register, along with MAC_Tx_Timestamp_Status_Seconds, gives the 64-bit timestamp captured for the PTP packet successfully transmitted by the MAC. This value is considered to be read by the application when the last byte of MAC_Tx_Timestamp_Status_Seconds is read. This means when bits[31:24] are read.This means when bits[7:0] are read.If the application does not read these registers and timestamp of another packet is captured, then either the current timestamp is lost (overwritten) or the new timestamp is lost (dropped), depending on the setting of the TXTSSTSM bit of the MAC_Timestamp_Control register. The status bit TXTSSIS bit [15] in MAC_Timestamp_Status register is set when the MAC transmitter captures the timestamp.	31	TXTSSMIS	R	0x0	Transmit Timestamp Status Missed When this bit is set, it indicates one of the following: The timestamp of the current packet is ignored if TXTSSTSM bit of the MAC_Timestamp_Control register is reset The timestamp of the previous packet is overwritten with timestamp of the current packet if TXTSSTSM bit of the MAC_Timestamp_Control register is set. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Transmit Timestamp Status Missed status not detected 0x1 (ACTIVE): Transmit Timestamp Status Missed status detected Value After Reset: 0x0
			30:0	TXTSSLO	R	0x0	Transmit Timestamp Status Low This field contains the 31 bits of the Nanoseconds field of the Transmit packet's captured timestamp. Value After Reset: 0x0
0xb34	MAC_Tx_Timestamp_Status_Seconds	The register contains the higher 32 bits of the timestamp (in seconds) captured when a PTP packet is transmitted.	31:0	TXTSSHI	R	0x0	Transmit Timestamp Status High This field contains the lower 32 bits of the Seconds field of Transmit packet's captured timestamp. Value After Reset: 0x0
0xb40	MAC_Auxiliary_Control	The Auxiliary Timestamp Control register controls the Auxiliary Timestamp snapshot.	31:8	Reserved_31_8	R	0x0	Reserved. Value After Reset: 0x0
			7	ATSEN3	R/W	0x0	Auxiliary Snapshot 3 Enable This bit controls the capturing of Auxiliary Snapshot Trigger 3. When this bit is set, the auxiliary snapshot of the event on ptp_aux_trig_i[3] input is enabled. When this bit is reset, the events on this input are ignored.Values: 0x0 (DISABLE): Auxiliary Snapshot $i is disabled 0x1 (ENABLE): Auxiliary Snapshot $i is enabled Value After Reset: 0x0
			6	ATSEN2	R/W	0x0	Auxiliary Snapshot 2 Enable This bit controls the capturing of Auxiliary Snapshot Trigger 2. When this bit is set, the auxiliary snapshot of the event on ptp_aux_trig_i[2] input is enabled. When this bit is reset, the events on this input are ignored.Values: 0x0 (DISABLE): Auxiliary Snapshot $i is disabled 0x1 (ENABLE): Auxiliary Snapshot $i is enabled Value After Reset: 0x0
			5	ATSEN1	R/W	0x0	Auxiliary Snapshot 1 Enable This bit controls the capturing of Auxiliary Snapshot Trigger 1. When this bit is set, the auxiliary snapshot of the event on ptp_aux_trig_i[1] input is enabled. When this bit is reset, the events on this input are ignored.Values: 0x0 (DISABLE): Auxiliary Snapshot $i is disabled 0x1 (ENABLE): Auxiliary Snapshot $i is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			4	ATSEN0	R/W	0x0	Auxiliary Snapshot 0 Enable This bit controls the capturing of Auxiliary Snapshot Trigger 0. When this bit is set, the auxiliary snapshot of the event on ptp_aux_trig_i[0] input is enabled. When this bit is reset, the events on this input are ignored.Values: 0x0 (DISABLE): Auxiliary Snapshot $i is disabled 0x1 (ENABLE): Auxiliary Snapshot $i is enabled Value After Reset: 0x0
			3:1	Reserved_3_1	R	0x0	Reserved. Value After Reset: 0x0
			0	ATSFC	R/W	0x0	Auxiliary Snapshot FIFO Clear When set, this bit resets the pointers of the Auxiliary Snapshot FIFO. This bit is cleared when the pointers are reset and the FIFO is empty. When this bit is high, the auxiliary snapshots are stored in the FIFO. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect.Values: 0x0 (DISABLE): Auxiliary Snapshot FIFO Clear is disabled 0x1 (ENABLE): Auxiliary Snapshot FIFO Clear is enabled Value After Reset: 0x0 Testable: untestable
0xb48	MAC_Auxiliary_Timestamp_Nanoseconds	The Auxiliary Timestamp Nanoseconds register, along with MAC_Auxiliary_Timestamp_Seconds, gives the 64-bit timestamp stored as auxiliary snapshot. These two registers form the read port of a 64-bit wide FIFO with a depth of 4, 8, or 16 as selected while configuring the IP. You can store multiple snapshots in this FIFO. Bits[30:25] in MAC_Timestamp_Status indicate the fill	31	Reserved_31	R	0x0	Reserved. Value After Reset: 0x0
			30:0	AUXTSLO	R	0x0	Auxiliary Timestamp Contains the lower 31 bits (nanoseconds field) of the auxiliary timestamp. Value After Reset: 0x0
0xb4c	MAC_Auxiliary_Timestamp_Seconds	The Auxiliary Timestamp - Seconds register contains the lower 32 bits of the Seconds field of the auxiliary timestamp register.	31:0	AUXTSHI	R	0x0	Auxiliary Timestamp Contains the lower 32 bits of the Seconds field of the auxiliary timestamp. Value After Reset: 0x0
0xb50	MAC_Timestamp_Ingress_Asym_Corr	The MAC Timestamp Ingress Asymmetry Correction register contains the Ingress Asymmetry Correction value to be used while updating correction field in PDelay_Resp PTP messages.	31:0	OSTIAC	R/W	0x0	One-Step Timestamp Ingress Asymmetry Correction This field contains the ingress path asymmetry value to be added to correctionField of Pdelay_Resp PTP packet. The programmed value must be in units of nanoseconds and multiplied by 2^16. For example, 2.5 ns is represented as 0x00028000. The value can also be negative, which is represented in 2's complement form with bit 31 representing the sign bit. Value After Reset: 0x0
0xb54	MAC_Timestamp_Egress_Asym_Corr	The MAC Timestamp Egress Asymmetry Correction register contains the Egress Asymmetry Correction value to be used while updating the correction field in PDelay_Req PTP messages.	31:0	OSTEAC	R/W	0x0	One-Step Timestamp Egress Asymmetry Correction This field contains the egress path asymmetry value to be subtracted from correctionField of Pdelay_Resp PTP packet. The programmed value must be the negated value in units of nanoseconds multiplied by 2^16. For example, if the required correction is +2.5 ns, the programmed value must be 0xFFFD_8000, which is the 2's complement of 0x0002_8000(2.5 * 216). Similarly, if the required correction is -3.3 ns, the programmed value is 0x0003_4CCC (3.3 * 216). Value After Reset: 0x0
0xb58	MAC_Timestamp_Ingress_Corr_Nanosecond	This register contains the correction value in nanoseconds to be used with the captured timestamp value in the ingress path.	31:0	TSIC	R/W	0x0	Timestamp Ingress Correction This field contains the ingress path correction value as defined by the Ingress Correction expression. Value After Reset: 0x0
0xb5c	MAC_Timestamp_Egress_Corr_Nanosecond	This register contains the correction value in nanoseconds to be used with the captured timestamp value in the egress path.	31:0	TSEC	R/W	0x0	Timestamp Egress Correction This field contains the nanoseconds part of the egress path correction value as defined by the Egress Correction expression. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xb60	MAC_Timestamp_Ingress_Corr_Subnanosec	This register contains the sub-nanosecond part of the correction value to be used with the captured timestamp value, for ingress direction.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:8	TSICSNS	R/W	0x0	Timestamp Ingress Correction, sub-nanoseconds This field contains the sub-nanoseconds part of the ingress path correction value as defined by the "Ingress Correction" expression. Value After Reset: 0x0
			7:0	Reserved_7_0	R	0x0	Reserved. Value After Reset: 0x0
0xb64	MAC_Timestamp_Egress_Corr_Subnanosec	This register contains the sub-nanosecond part of the correction value to be used with the captured timestamp value, for egress direction.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:8	TSECSNS	R/W	0x0	Timestamp Egress Correction, sub-nanoseconds This field contains the sub-nanoseconds part of the egress path correction value as defined by the "Egress Correction" expression. Value After Reset: 0x0
			7:0	Reserved_7_0	R	0x0	Reserved. Value After Reset: 0x0
0xb68	MAC_Timestamp_Ingress_Latency	This register holds the Ingress MAC latency.	31:28	Reserved_31_28	R	0x0	Reserved. Value After Reset: 0x0
			27:16	ITLNS	R	0x0	Ingress Timestamp Latency, in sub-nanoseconds This register holds the average latency in sub-nanoseconds between the input ports (phy_rxd_i) of MAC and the actual point (GMII/MII) where the ingress timestamp is taken. Ingress correction value is computed as described in the section, "Ingress Correction". Value After Reset: 0x0 Testable: untestable
			15:8	ITLSNS	R	0x0	Ingress Timestamp Latency, in nanoseconds This register holds the average latency in nanoseconds between the input ports (phy_rxd_i) of MAC and the actual point (GMII/MII) where the ingress timestamp is taken. Ingress correction value is computed as described in the section, "Ingress Correction". Value After Reset: 0x0 Testable: untestable
			7:0	Reserved_7_0	R	0x0	Reserved. Value After Reset: 0x0
0xb6c	MAC_Timestamp_Egress_Latency	This register holds the Egress MAC latency.	31:28	Reserved_31_28	R	0x0	Reserved. Value After Reset: 0x0
			27:16	ETLNS	R	0x0	Egress Timestamp Latency, in nanoseconds This register holds the average latency in nanoseconds between the actual point (GMII/MII) where the egress timestamp is taken and the output ports (phy_txd_o) of the MAC. Egress correction value is computed as described in the section, "Egress Correction". Value After Reset: 0x0 Testable: untestable
			15:8	ETLSNS	R	0x0	Egress Timestamp Latency, in sub-nanoseconds This register holds the average latency in sub-nanoseconds between the actual point (GMII/MII) where the egress timestamp is taken and the output ports (phy_txd_o) of the MAC. Egress correction value is computed as described in the section "Egress Correction". Value After Reset: 0x0 Testable: untestable

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Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			7:0	Reserved_7_0	R	0x0	Reserved. Value After Reset: 0x0
0xb70	MAC_PPS_Control	PPS Control register.Bits[6:4] are valid only when Flexible PPS feature is selected.	31:29			#VALUE!	Reserved Field: Yes
			28	TIMESEL	R/W	0x0	Time Select When this bit is set, 64 bit PTP time is used to capture time at MCGR trigger[0] input When this bit is reset, presentation time is used to capture time at trigger input, maintaining backward compatibility Value After Reset: 0x0
			27:21			#VALUE!	Reserved Field: Yes
			20	Reserved_20	R	0x0	Reserved. Value After Reset: 0x0
			19:13			#VALUE!	Reserved Field: Yes
			12	Reserved_12	R	0x0	Reserved. Value After Reset: 0x0
			11:8			#VALUE!	Reserved Field: Yes
			7	MCGREN0	R/W	0x0	MCGR Mode Enable for PPS0 Output This field enables the 0th PPS instance to operate in PPS or MCGR mode. When set it operates in MCGR mode and on reset it operates in PPS mode.Values: 0x0 (PPS): 0th PPS instance is enabled to operate in PPS mode 0x1 (MCGR): 0th PPS instance is enabled to operate in MCGR mode Value After Reset: 0x0
			6:5	TRGTMODSEL0	R/W	0x0	Target Time Register Mode for PPS0 Output This field indicates the Target Time registers (MAC_PPS0_Target_Time_Seconds and MAC_PPS0_Target_Time_Nanoseconds) mode for PPS0 output signal:Values: 0x0 (ONLY_INT): Target Time registers are programmed only for generating the interrupt event. The Flexible PPS function must not be enabled in this mode, otherwise spurious transitions may be observed on the corresponding ptp_pps_o output port 0x1 (MCGR): Enables MCGR Interrupt whose status bit is indicated by TSTARGT0 (MAC_Timestamp_Status[1]) 0x2 (INT_ST): Target Time registers are programmed for generating the interrupt event and starting or stopping the PPS0 output signal generation 0x3 (ONLY_ST): Target Time registers are programmed only for starting or stopping the PPS0 output signal generation. No interrupt is asserted Value After Reset: 0x0
			4	PPSEN0	R/W	0x0	Flexible PPS Output Mode Enable When this bit is 1: Bits[3:0] function as PPSCMD. 0: Bits[3:0] function as PPSCTRL (Fixed PPS mode). Values: 0x0 (DISABLE): Flexible PPS Output Mode is disabled 0x1 (ENABLE): Flexible PPS Output Mode is enabled Value After Reset: 0x0

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Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			3:0	PPSCTRL_PPSCMD	R/W	0x0	PPS Output Frequency Control This field controls the frequency of the PPS0 output (ptp_pps_o[0]) signal. The default value of PPSCTRL is 0000, and the PPS output is 1 pulse (of width clk_ptp_i) every second. For other values of PPSCTRL, the PPS output becomes a generated clock of following frequencies: 0001: The binary rollover is 2 Hz, and the digital rollover is 1 Hz. 0010: The binary rollover is 4 Hz, and the digital rollover is 2 Hz. 0011: The binary rollover is 8 Hz, and the digital rollover is 4 Hz. 0100: The binary rollover is 16 Hz, and the digital rollover is 8 Hz. .. 1111: The binary rollover is 32.768 KHz and the digital rollover is 16.384 KHz. Note: In the binary rollover mode, the PPS output (ptp_pps_o) has a duty cycle of 50 percent with these frequencies. In the digital rollover mode, the PPS output frequency is an average number. The actual clock is of different frequency that gets synchronized every second. For example: When PPSCTRL = 0001, the PPS (1 Hz) has a low period of 537 ms and a high period of 463 ms When PPSCTRL = 0010, the PPS (2 Hz) is a sequence of One clock of 50 percent duty cycle and 537 ms period Second clock of 463 ms period (268 ms low and 195 ms high) When PPSCTRL = 0011, the PPS (4 Hz) is a sequence of Three clocks of 50 percent duty cycle and 268 ms period Fourth clock of 195 ms period (134 ms low and 61 ms high) This behavior is because of the non-linear toggling of bits in the digital rollover mode in the MAC_System_Time_Nanoseconds register. or Flexible PPS Output (ptp_pps_o[0]) Control Programming these bits with a non-zero value instructs the MAC to initiate an event. When the command is transferred or synchronized to the PTP clock domain, these bits get cleared automatically. The software must ensure that these bits are programmed only when they are 'all-zero'. The following list describes the values of PPSCMD0: 0000: No Command 0001: START Single Pulse This command generates single pulse rising at the start point defined in MAC_PPS0_Target_Time_Seconds and MAC_PPS0_Target_Time_Nanoseconds register and of a duration defined in the PPS0 Width Register. 0010: START Pulse Train This command generates the train of pulses rising at the start point defined in the Target Time Registers and of a
0xb80	MAC_PPS0_Target_Time_Seconds	The PPS Target Time Seconds register, along with PPS Target Time Nanoseconds register, is used to schedule an interrupt event [Bit 1 of MAC_Timestamp_Status] when the system time exceeds the value programmed in these registers.	31:0	TSTRH0	R/W	0x0	PPS Target Time Seconds Register This field stores the time in seconds. When the timestamp value matches or exceeds both Target Timestamp registers, the MAC starts or stops the PPS signal output and generates an interrupt (if enabled) based on Target Time mode selected for the corresponding PPS output in the MAC_PPS_Control register. If PTGE field of MAC_Timestamp_Control Register is set with Presentation time control set in recovery mode, then these bits indicate the TPT being programmed by the application and in generation mode it indicates the CPT generated at the sampled trigger. Value After Reset: 0x0

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Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xb84	MAC_PPS0_Target_Time_Nanoseconds	PPS0 Target Time Nanoseconds register.	31	TRGTBUSY0	R/W	0x0	PPS Target Time Register Busy The MAC sets this bit when the PPSCMD0 field in the MAC_PPS_Control register is programmed to 010 or 011. Programming the PPSCMD0 field to 010 or 011 instructs the MAC to synchronize the Target Time Registers to the PTP clock domain. The MAC clears this bit after synchronizing the Target Time Registers to the PTP clock domain The application must not update the Target Time Registers when this bit is read as 1. Otherwise, the synchronization of the previous programmed time gets corrupted.Values: 0x0 (INACTIVE): PPS Target Time Register Busy status is not detected 0x1 (ACTIVE): PPS Target Time Register Busy is detected Value After Reset: 0x0 Testable: untestable
			30:0	TTSL0	R/W	0x0	Target Time Low for PPS Register This register stores the time in (signed) nanoseconds. When the value of the timestamp matches the value in both Target Timestamp registers, the MAC starts or stops the PPS signal output and generates an interrupt (if enabled) based on the TRGTMODSEL0 field (Bits [6:5]) in MAC_PPS_Control. When the TSCTRLSSR bit is reset in the MAC_Timestamp_Control register, this value must be (time in ns / 0.465). The actual start or stop time of the PPS signal output might have an error margin up to one unit of sub-second increment value. When the TSCTRLSSR bit is set in the MAC_Timestamp_Control register, this value must not exceed 0x3B9A_C9FF. The actual start or stop time of the PPS signal output might have an error margin up to one unit of sub-second increment value. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. Value After Reset: 0x0
0xb88	MAC_PPS0_Interval	The PPS0 Interval register contains the number of units of sub-second increment value between the rising edges of PPS0 signal output (ptp_pps_o[0]).	31:0	PPSINT0	R/W	0x0	PPS Output Signal Interval These bits store the interval between the rising edges of PPS0 signal output. The interval is stored in terms of number of units of sub-second increment value. You need to program one value less than the required interval. For example, if the PTP reference clock is 50 MHz (period of 20 ns), and desired interval between the rising edges of PPS0 signal output is 100 ns (that is, 5 units of sub-second increment value), you should program value 4 (5-1) in this register. Value After Reset: 0x0
0xb8c	MAC_PPS0_Width	The PPS0 Width register contains the number of units of sub-second increment value between the rising and corresponding falling edges of PPS0 signal output (ptp_pps_o[0]).	31:0	PPSWIDTH0	R/W	0x0	PPS Output Signal Width These bits store the width between the rising edge and corresponding falling edge of PPS0 signal output. The width is stored in terms of number of units of sub-second increment value. You need to program one value less than the required interval. For example, if PTP reference clock is 50 MHz (period of 20 ns), and width between the rising and corresponding falling edges of PPS0 signal output is 80 ns (that is, four units of sub-second increment value), you should program value 3 (4-1) in this register. Note: The value programmed in this register must be lesser than the value programmed in MAC_PPS0_Interval. Value After Reset: 0x0
			31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:8	DN	R/W	0x0	Domain Number This field indicates the domain Number in which the PTP node is operating. Value After Reset: 0x0

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Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xbc0	MAC_PTO_Control	This register controls the PTP Offload Engine operation. This register is available only when the Enable PTP Timestamp Offload feature is selected.	7	PDRDIS	R/W	0x0	Disable Peer Delay Response response generation When this bit is set, the Peer Delay Response (Pdelay_Resp) response is not be generated for received Peer Delay Request (Pdelay_Req) request packet, as required by the programmed mode. Note: Setting this bit to 1 affects the normal PTP Offload operation and the time synchronization. So, this bit must be set only if there is problem with Pdelay_Resp generation in hardware and/or Pdelay_Resp generation is handled by software.Values: 0x0 (ENABLE): Peer Delay Response response generation is enabled 0x1 (DISABLE): Peer Delay Response response generation is disabled Value After Reset: 0x0
			6	DRRDIS	R/W	0x0	Disable PTO Delay Request/Response response generation When this bit is set, the Delay Request and Delay response is not generated for received SYNC and Delay request packet respectively, as required by the programmed mode. Values: 0x0 (ENABLE): PTO Delay Request/Response response generation is enabled 0x1 (DISABLE): PTO Delay Request/Response response generation is disabled Value After Reset: 0x0
			5	APDREQTRIG	R/W	0x0	Automatic PTP Pdelay_Req message Trigger When this bit is set, one PTP Pdelay_Req message is transmitted. This bit is automatically cleared after the PTP Pdelay_Req message is transmitted. The application must set the APDREQEN bit for this operation. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect.Values: 0x0 (DISABLE): Automatic PTP Pdelay_Req message Trigger is disabled 0x1 (ENABLE): Automatic PTP Pdelay_Req message Trigger is enabled Value After Reset: 0x0 Testable: untestable
			4	ASYNCTRIG	R/W	0x0	Automatic PTP SYNC message Trigger When this bit is set, one PTP SYNC message is transmitted. This bit is automatically cleared after the PTP SYNC message is transmitted. The application must set the ASYNCEN bit for this operation. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect.Values: 0x0 (DISABLE): Automatic PTP SYNC message Trigger is disabled 0x1 (ENABLE): Automatic PTP SYNC message Trigger is enabled Value After Reset: 0x0 Testable: untestable
			3	Reserved_3	R	0x0	Reserved. Value After Reset: 0x0
			2	APDREQEN	R/W	0x0	Automatic PTP Pdelay_Req message Enable When this bit is set, PTP Pdelay_Req message is generated periodically based on interval programmed or trigger from application, when the MAC is programmed to be in Peer-to-Peer Transparent mode.Values: 0x0 (DISABLE): Automatic PTP Pdelay_Req message is disabled 0x1 (ENABLE): Automatic PTP Pdelay_Req message is enabled Value After Reset: 0x0
			1	ASYNCEN	R/W	0x0	Automatic PTP SYNC message Enable When this bit is set, PTP SYNC message is generated periodically based on interval programmed or trigger from application, when the MAC is programmed to be in Clock Master mode.Values: 0x0 (DISABLE): Automatic PTP SYNC message is disabled 0x1 (ENABLE): Automatic PTP SYNC message is enabled Value After Reset: 0x0

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Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	PTOEN	R/W	0x0	PTP Offload Enable When this bit is set, the PTP Offload feature is enabled.Values: 0x0 (DISABLE): PTP Offload feature is disabled 0x1 (ENABLE): PTP Offload feature is enabled Value After Reset: 0x0
0xbc4	MAC_Source_Port_Identity0	This register contains Bits[31:0] of the 80-bit Source Port Identity of the PTP node. This register is available only when the Enable PTP Timestamp Offload feature is selected.	31:0	SPI0	R/W	0x0	Source Port Identity 0 This field indicates bits [31:0] of sourcePortIdentity of PTP node. Value After Reset: 0x0
0xbc8	MAC_Source_Port_Identity1	This register contains Bits[63:32] of the 80-bit Source Port Identity of the PTP node. This register is available only when the Enable PTP Timestamp Offload feature is selected.	31:0	SPI1	R/W	0x0	Source Port Identity 1 This field indicates bits [63:32] of sourcePortIdentity of PTP node. Value After Reset: 0x0
0xbcc	MAC_Source_Port_Identity2	This register contains Bits[79:64] of the 80-bit Source Port Identity of the PTP node. This register is available only when the Enable PTP Timestamp Offload feature is selected.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15:0	SPI2	R/W	0x0	Source Port Identity 2 This field indicates bits [79:64] of sourcePortIdentity of PTP node. Value After Reset: 0x0
			31:24	LMPDRI	R/W	0x0	Log Min Pdelay_Req Interval This field indicates logMinPdelayReqInterval of PTP node. This is used to schedule the periodic Pdelay request packet transmission. Allowed values are -15 to 15.Negative value must be represented in 2's-complement form. For example, if the required value is -1, the value programmed must be 0xFF. Value After Reset: 0x0
			23:11	Reserved_23_11	R	0x0	Reserved. Value After Reset: 0x0

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Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xbd0	MAC_Log_Message_Interval	This register contains the periodic intervals for automatic PTP packet generation. This register is available only when the Enable PTP Timestamp Offload feature is selected.	10:8	DRSYNCR	R/W	0x0	Delay_Req to SYNC Ratio In Slave mode, it is used for controlling frequency of Delay_Req messages transmitted. 0: DelayReq generated for every received SYNC 1: DelayReq generated every alternate reception of SYNC 2: for every 4 SYNC messages 3: for every 8 SYNC messages 4: for every 16 SYNC messages 5: for every 32 SYNC messages 6-7: Reserved The master sends this information (logMinDelayReqInterval) in the DelayResp PTP messages to the slave. The DWC_ether_qos Receiver processes this value from the received DelayResp messages and updates this field accordingly. In the Slave mode, the host must not write/update this register unless it has to override the received value. In Master mode, the sum of this field and logSyncInterval (LSI) field is provided in the logMinDelayReqInterval field of the generated multicast Delay_Resp PTP message. Access restriction applies. Updated based on the event. Setting 1 sets. Setting 0 clears.Values: 0x0 (SYNC1): DelayReq generated for every received SYNC 0x1 (SYNC2): DelayReq generated every alternate reception of SYNC 0x2 (SYNC4): for every 4 SYNC messages 0x3 (SYNC8): for every 8 SYNC messages 0x4 (SYNC16): for every 16 SYNC messages 0x5 (SYNC32): for every 32 SYNC messages 0x6 (RSVD): Reserved Value After Reset: 0x0
			7:0	LSI	R/W	0x0	Log Sync Interval This field indicates the periodicity of the automatically generated SYNC message when the PTP node is Master. Allowed values are -15 to 15. Negative value must be represented in 2's-complement form. For example, if the required value is -1, the value programmed must be 0xFF. Value After Reset: 0x0
			31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15	FRPE	R/W	0x0	Flexible Rx parser Enable 1: Enables Programmable Rx Parser 0: Disables Programmable Rx Parser When the Rx parser is disabled when parsing is in progress, the Rx parser is disabled only after the current packet parsing complete. When the Rx parser is enabled, the parser gets activated for the next packet.Values: 0x0 (DISABLE): Flexible Rx parser is disabled 0x1 (ENABLE): Flexible Rx parser is enabled Value After Reset: 0x0
			14	RXPED	R/W	0x0	RxParser Software Error/Incomplete Parsing Packet Drop Enable when this bit is set to 0, packets encountering software programming errors (NPE/NVE/frame offset overflow errors) or incomplete parsing are forwarded to application with the corresponding RxParser status. When this bit is set to 1, backward compatibility is maintained where all the specified packets are dropped (when RA is not set)Values: 0x0 (DISABLE): Flexible Rx parser, packet drop in case software error is disabled 0x1 (ENABLE): Flexible Rx parser, packet drop in case software error is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xc00	MTL_Operation_Mode	The Operation Mode register establishes the Transmit and Receive operating modes and commands.	13:10	Reserved_13_10	R	0x0	Reserved. Value After Reset: 0x0
			9	CNTCLR	R/W	0x0	Counters Reset When this bit is set, all counters are reset. This bit is cleared automatically after 1 clock cycle. If this bit is set along with CNTPRST bit, CNTPRST has precedence. Access restrictions apply. MAC initiates the counters reset when application writes 1. MAC writes 0 after the counters reset is complete. MAC ignores when application writes 0. Values: 0x0 (DISABLE): Counters are not reset 0x1 (ENABLE): All counters are reset Value After Reset: 0x0 Testable: untestable
			8	CNTPRST	R/W	0x0	Counters Preset When this bit is set, MTL_TxQ[0-7]_Underflow register is initialized/preset to 12'h7F0. Missed Packet and Overflow Packet counters in MTL_RxQ[0-7]_Missed_Packet_Overflow_Cnt register is initialized/preset to 12'h7F0. Access restrictions apply. MAC initiates the counters preset when application writes 1. MAC writes 0 after the counters preset is complete. MAC ignores when application writes 0. Values: 0x0 (DISABLE): Counters Preset is disabled 0x1 (ENABLE): Counters Preset is enabled Value After Reset: 0x0 Testable: untestable
			7	Reserved_7	R	0x0	Reserved. Value After Reset: 0x0
			6:5	SCHALG	R/W	0x0	Tx Scheduling Algorithm This field indicates the algorithm for Tx scheduling:Values: 0x0 (WRR): WRR algorithm 0x1 (WFQ): WFQ algorithm when DCB feature is selected 0x2 (DWRR): DWRR algorithm when DCB feature is selected 0x3 (SP): Strict priority algorithm Value After Reset: 0x0
			4:3	Reserved_4_3	R	0x0	Reserved. Value After Reset: 0x0
			2	RAA	R/W	0x0	Receive Arbitration Algorithm This field is used to select the arbitration algorithm for the Rx side. 0: Strict priority (SP). Queue 0 has the lowest priority and the last queue has the highest priority. 1: Weighted Strict Priority (WSP) Values: 0x0 (SP): Strict priority (SP) 0x1 (WSP): Weighted Strict Priority (WSP) Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			1	DTXSTS	R/W	0x0	Drop Transmit Status 1: Tx packet status received from the MAC is dropped in the MTL 0: Tx packet status received from the MAC is forwarded to the application Values: 0x0 (DISABLE): Drop Transmit Status is disabled 0x1 (ENABLE): Drop Transmit Status is enabled Value After Reset: 0x0
			0	Reserved_0	R	0x0	Reserved. Value After Reset: 0x0
			31:19	Reserved_31_19	R	0x0	Reserved. Value After Reset: 0x0
			18:16			#VALUE!	Reserved Field: Yes
			15	STSIE	R/W	0x0	Transmit Status Available Interrupt Status Enable When this bit is set, an interrupt is generated when Transmit status is available in slave mode.Values: 0x0 (DISABLE): Transmit Packet Available Interrupt Status is disabled 0x1 (ENABLE): Transmit Packet Available Interrupt Status is enabled Value After Reset: 0x0
			14	PKTIE	R/W	0x0	Receive Packet Available Interrupt Status Enable When this bit is set, an interrupt is generated when EOP of received packet is written to the Rx FIFO.Values: 0x0 (DISABLE): Receive Packet Available Interrupt Status is disabled 0x1 (ENABLE): Receive Packet Available Interrupt Status is enabled Value After Reset: 0x0
			13:12	FIFOSEL	R/W	0x0	FIFO Selected for Access This field indicates the FIFO selected for debug access. Values: 0x0 (TXFIFO): Tx FIFO 0x1 (TXSTSFIFO): Tx Status FIFO (only read access when SLVMOD is set)/ DC Memory FIFO read-write access when DBGMOD is set 0x2 (TSOFIFO): TSO FIFO (cannot be accessed when SLVMOD is set) 0x3 (RXFIFO): Rx FIFO Value After Reset: 0x0
			11	FIFOWREN	R/W	0x0	FIFO Write Enable When this bit is set, it enables the Write operation on selected FIFO when FIFO Debug Access is enabled. This bit must not be written to 1 when FIFO Debug Access is not enabled, that is FDBGEN bit is 0. Access restriction applies. Self-cleared. Setting 0 clears. Setting 1 sets.Values: 0x0 (DISABLE): FIFO Write is disabled 0x1 (ENABLE): FIFO Write is enabled Value After Reset: 0x0 Testable: untestable
			10	FIFORDEN	R/W	0x0	FIFO Read Enable When this bit is set, it enables the Read operation on selected FIFO when FIFO Debug Access is enabled. This bit must not be written to 1 when FIFO Debug Access is not enabled, that is FDBGEN bit is 0. Access restriction applies. Self-cleared. Setting 0 clears. Setting 1 sets.Values: 0x0 (DISABLE): FIFO Read is disabled 0x1 (ENABLE): FIFO Read is enabled Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xc08	MTL_DBG_CTL	The FIFO Debug Access Control and Status register controls the operation mode of FIFO debug access.Note: Consecutive write or read to this register should be performed after at least 16 clock cycles of the slowest clock among MAC Transmitter/Receiver clock and Application/CSR clock.	9	RSTSEL	R/W	0x0	Reset Pointers of Selected FIFO When this bit is set, the pointers of the currently-selected FIFO are reset when FIFO Debug Access is enabled. This bit must not be written to 1 when FIFO Debug Access is not enabled, that is FDBGEN bit is 0. Access restriction applies. Self-cleared. Setting 0 clears. Setting 1 sets.Values: 0x0 (DISABLE): Reset Pointers of Selected FIFO is disabled 0x1 (ENABLE): Reset Pointers of Selected FIFO is enabled Value After Reset: 0x0 Testable: untestable
			8	RSTALL	R/W	0x0	Reset All Pointers When this bit is set, the pointers of all FIFOs are reset when FIFO Debug Access is enabled. This bit must not be written to 1 when FIFO Debug Access is not enabled, that is FDBGEN bit is 0. Access restriction applies. Self-cleared. Setting 0 clears. Setting 1 sets.Values: 0x0 (DISABLE): Reset All Pointers is disabled 0x1 (ENABLE): Reset All Pointers is enabled Value After Reset: 0x0 Testable: untestable
			7	Reserved_7	R	0x0	Reserved. Value After Reset: 0x0
			6:5	PKTSTATE	R/W	0x0	Encoded Packet State This field is used to write the control information to the Tx FIFO or Rx FIFO. Tx FIFO: 00: Packet Data 01: Control Word 10: SOP Data 11: EOP Data Rx FIFO: 00: Packet Data 01: Normal Status 10: Last Status 11: EOP Values: 0x0 (PKT_DATA): Packet Data 0x1 (CW_NS): Control Word/Normal Status 0x2 (SOP_LS): SOP Data/Last Status 0x3 (EOP): EOP Data/EOP Value After Reset: 0x0
			4	Reserved_4	R	0x0	Reserved. Value After Reset: 0x0
			3:2	BYTEEN	R/W	0x0	Byte Enables This field indicates the number of data bytes valid in the data register during Write operation. This is valid only when PKTSTATE is 2'b10 (EOP) and Tx FIFO or Rx FIFO is selected.Values: 0x0 (B0_VAL): Byte 0 valid 0x1 (B01_VAL): Byte 0 and Byte 1 are valid 0x2 (B012_VAL): Byte 0, Byte 1, and Byte 2 are valid 0x3 (B0123_VAL): All four bytes are valid Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			1	DBGMOD	R/W	0x0	Debug Mode Access to FIFO 1: Indicates that the current access to the FIFO is read, write, and debug access (i.e., data can be written and read to specified or all locations through FIFO Debug Interface). In this mode, the following access types are allowed: Read and Write access to Tx FIFO, TSO FIFO, DCACHE FIFO and Rx FIFO Read access is allowed to Tx Status FIFO. 0: Indicates that the current access to the FIFO is slave access bypassing the DMA (i.e., packets can be transmitted or received through FIFO Debug Interface). In this mode, the following access are allowed: Write access to the Tx FIFO Read access to the Rx FIFO and Tx Status FIFO Values: 0x0 (DISABLE): Debug Mode Access to FIFO is disabled 0x1 (ENABLE): Debug Mode Access to FIFO is enabled Value After Reset: 0x0
			0	FDBGEN	R/W	0x0	FIFO Debug Access Enable 1: Indicates that the debug mode access to the FIFO through CSR Slave Interface is enabled. 0: Indicates that the FIFO can be accessed only through a master interface. Values: 0x0 (DISABLE): FIFO Debug Access through CSR Slave Interface is disabled 0x1 (ENABLE): FIFO Debug Access through CSR Slave Interface is enabled Value After Reset: 0x0
			31:15	LOCR	R	0x0	Remaining Locations in the FIFO Slave Access Mode: This field indicates the space available in the selected FIFO. Debug Access Mode: This field contains the Write or Read pointer value of the selected FIFO during Write or Read operation, respectively. Value After Reset: 0x0
			14:10	Reserved_14_10	R	0x0	Reserved. Value After Reset: 0x0
			9	STSI	R/W	0x0	Transmit Status Available Interrupt Status When set, this bit indicates that the Slave mode Tx packet is transmitted, and the status is available in Tx Status FIFO. This bit is reset when 1 is written to this bit.Values: 0x0 (INACTIVE): Transmit Status Available Interrupt Status not detected 0x1 (ACTIVE): Transmit Status Available Interrupt Status detected Value After Reset: 0x0 Testable: untestable
			8	PKTI	R/W	0x0	Receive Packet Available Interrupt Status When set, this bit indicates that MAC layer has written the EOP of received packet to the Rx FIFO. This bit is reset when 1 is written to this bit.Values: 0x0 (INACTIVE): Receive Packet Available Interrupt Status not detected 0x1 (ACTIVE): Receive Packet Available Interrupt Status detected Value After Reset: 0x0 Testable: untestable
			7:5	Reserved_7_5	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xc0c	MTL_DBG_STS	The FIFO Debug Status register contains the status of FIFO debug access.	4:3	BYTEEN	R	0x3	Byte Enables This field indicates the number of data bytes valid in the data register during Read operation. This is valid only when PKTSTATE is 2'b10 (EOP) and Tx FIFO or Rx FIFO is selected.Values: 0x0 (B0_VAL): Byte 0 valid 0x1 (B01_VAL): Byte 0 and Byte 1 are valid 0x2 (B012_VAL): Byte 0, Byte 1, and Byte 2 are valid 0x3 (B0123_VAL): All four bytes are valid Value After Reset: 0x3
			2:1	PKTSTATE	R	0x0	Encoded Packet State This field is used to get the control or status information of the selected FIFO. Tx FIFO: 00: Packet Data 01: Control Word 10: SOP Data 11: EOP Data Rx FIFO: 00: Packet Data 01: Normal Status 10: Last Status 11: EOP This field is applicable only for Tx FIFO and Rx FIFO during Read operation.Values: 0x0 (PKT_DATA): Packet Data 0x1 (CW_NS): Control Word/Normal Status 0x2 (SOP_LS): SOP Data/Last Status 0x3 (EOP): EOP Data/EOP Value After Reset: 0x0
			0	FIFOBUSY	R	0x0	FIFO Busy When set, this bit indicates that a FIFO operation is in progress in the MAC and content of the following fields is not valid: All other fields of this register All fields of the MTL_FIFO_Debug_Data register Values: 0x0 (INACTIVE): FIFO Busy not detected 0x1 (ACTIVE): FIFO Busy detected Value After Reset: 0x0
0xc10	MTL_FIFO_Debug_Data	The FIFO Debug Data register contains the data to be written to or read from the FIFOs.	31:0	FDBGDATA	R/W	0x0	FIFO Debug Data During debug or slave access write operation, this field contains the data to be written to the Tx FIFO, Rx FIFO, or TSO FIFO. During debug or slave access read operation, this field contains the data read from the Tx FIFO, Rx FIFO, TSO FIFO, or Tx Status FIFO. Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xc20	MTL_Interrupt_Status	The software driver (application) reads this register during interrupt service routine or polling to determine the interrupt status of MTL queues and the MAC.	31:24	Reserved_31_24	R	0x0	Reserved. Value After Reset: 0x0
			23	MTLPIS	R	0x0	MTL Rx Parser Interrupt Status This bit indicates that there is an interrupt from Rx Parser Block. To reset this bit, the application must read the MTL_Rxp_Interrupt_Status register to get the exact cause of the interrupt and clear its source.Values: 0x0 (INACTIVE): MTL Rx Parser Interrupt status not detected 0x1 (ACTIVE): MTL Rx Parser Interrupt status detected Value After Reset: 0x0
			22:19	Reserved_22_19	R	0x0	Reserved. Value After Reset: 0x0
			18	ESTIS	R	0x0	EST (TAS- 802.1Qbv) Interrupt Status This bit indicates an interrupt event during the operation of 802.1Qbv. To reset this bit, the application must clear the error/event that has caused the Interrupt.Values: 0x0 (INACTIVE): EST (TAS- 802.1Qbv) Interrupt status not detected 0x1 (ACTIVE): EST (TAS- 802.1Qbv) Interrupt status detected Value After Reset: 0x0
			17	DBGIS	R	0x0	Debug Interrupt status This bit indicates an interrupt event during the slave access. To reset this bit, the application must read the FIFO Debug Access Status register to get the exact cause of the interrupt and clear its source.Values: 0x0 (INACTIVE): Debug Interrupt status not detected 0x1 (ACTIVE): Debug Interrupt status detected Value After Reset: 0x0
			16			#VALUE!	Reserved Field: Yes
			15:8	Reserved_15_8	R	0x0	Reserved. Value After Reset: 0x0
			7	Q7IS	R	0x0	Queue 7 Interrupt status This bit indicates that there is an interrupt from Queue 7. To reset this bit, the application must read the MTL_Q7_Interrupt_Control_Status Status register to get the exact cause of the interrupt and clear its source.Values: 0x0 (INACTIVE): Queue 7 Interrupt status not detected 0x1 (ACTIVE): Queue 7 Interrupt status detected Value After Reset: 0x0
			6	Q6IS	R	0x0	Queue 6 Interrupt status This bit indicates that there is an interrupt from Queue 6. To reset this bit, the application must read the MTL_Q6_Interrupt_Control_Status register to get the exact cause of the interrupt and clear its source.Values: 0x0 (INACTIVE): Queue 6 Interrupt status not detected 0x1 (ACTIVE): Queue 6 Interrupt status detected Value After Reset: 0x0
			5	Q5IS	R	0x0	Queue 5 Interrupt status This bit indicates that there is an interrupt from Queue 5. To reset this bit, the application must read the MTL_Q5_Interrupt_Control_Status register to get the exact cause of the interrupt and clear its source.Values: 0x0 (INACTIVE): Queue 5 Interrupt status not detected 0x1 (ACTIVE): Queue 5 Interrupt status detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			4	Q4IS	R	0x0	Queue 4 Interrupt status This bit indicates that there is an interrupt from Queue 4. To reset this bit, the application must read the MTL_Q4_Interrupt_Control_Status register to get the exact cause of the interrupt and clear its source.Values: 0x0 (INACTIVE): Queue 4 Interrupt status not detected 0x1 (ACTIVE): Queue 4 Interrupt status detected Value After Reset: 0x0
			3	Q3IS	R	0x0	Queue 3 Interrupt status This bit indicates that there is an interrupt from Queue 3. To reset this bit, the application must read the MTL_Q3_Interrupt_Control_Status register to get the exact cause of the interrupt and clear its source.Values: 0x0 (INACTIVE): Queue 3 Interrupt status not detected 0x1 (ACTIVE): Queue 3 Interrupt status detected Value After Reset: 0x0
			2	Q2IS	R	0x0	Queue 2 Interrupt status This bit indicates that there is an interrupt from Queue 2. To reset this bit, the application must read the MTL_Q2_Interrupt_Control_Status register to get the exact cause of the interrupt and clear its source.Values: 0x0 (INACTIVE): Queue 2 Interrupt status not detected 0x1 (ACTIVE): Queue 2 Interrupt status detected Value After Reset: 0x0
			1	Q1IS	R	0x0	Queue 1 Interrupt status This bit indicates that there is an interrupt from Queue 1. To reset this bit, the application must read the MTL_Q1_Interrupt_Control_Status register to get the exact cause of the interrupt and clear its source.Values: 0x0 (INACTIVE): Queue 1 Interrupt status not detected 0x1 (ACTIVE): Queue 1 Interrupt status detected Value After Reset: 0x0
			0	Q0IS	R	0x0	Queue 0 Interrupt status This bit indicates that there is an interrupt from Queue 0. To reset this bit, the application must read Queue 0 Interrupt Control and Status register to get the exact cause of the interrupt and clear its source.Values: 0x0 (INACTIVE): Queue 0 Interrupt status not detected 0x1 (ACTIVE): Queue 0 Interrupt status detected Value After Reset: 0x0
			31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28	Q3DDMACH	R/W	0x0	Queue 3 Enabled for Dynamic (per packet) DMA Channel Selection When set, this bit indicates that the packets received in Queue 3 are routed to a particular DMA channel as decided in the MAC Receiver based on the DMA channel number programmed in the L3-L4 filter registers, or the Ethernet DA address. When reset, this bit indicates that the packets received in Queue 3 are routed to the DMA Channel programmed in the Q3MDMACH field (Bits[26:24]).Values: 0x0 (DISABLE): Queue 3 disabled for DA-based DMA Channel Selection 0x1 (ENABLE): Queue 3 enabled for DA-based DMA Channel Selection Value After Reset: 0x0
			27	Reserved_27_y	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xc30	MTL_RxQ_DMA_Map0	This register controls the Dynamic (per packet) or Static (all packets in a Receive Queue) DMA Channel Selection for lower numbered Receive Queues.	26:24	Q3MDMACH	R/W	0x0	Queue 3 Mapped to DMA Channel This field controls the routing of the received packet in Queue 3 to the DMA channel: 000: DMA Channel 0 001: DMA Channel 1Reserved 010: DMA Channel 2Reserved 011: DMA Channel 3Reserved 100: DMA Channel 4Reserved 101: DMA Channel 5Reserved 110: DMA Channel 6Reserved 111: DMA Channel 7Reserved This field is valid when the Q3DDMACH field is reset. Value After Reset: 0x0
			23:21	Reserved_23_21	R	0x0	Reserved. Value After Reset: 0x0
			20	Q2DDMACH	R/W	0x0	Queue 2 Enabled for DA-based DMA Channel Selection When set, this bit indicates that the packets received in Queue 2 are routed to a particular DMA channel as decided in the MAC Receiver based on the DMA channel number programmed in the L3-L4 filter registers, or the Ethernet DA address. When reset, this bit indicates that the packets received in Queue 2 are routed to the DMA Channel programmed in the Q2MDMACH field (Bits[18:16]).Values: 0x0 (DISABLE): Queue 2 disabled for DA-based DMA Channel Selection 0x1 (ENABLE): Queue 2 enabled for DA-based DMA Channel Selection Value After Reset: 0x0
			19	Reserved_19_y	R	0x0	Reserved. Value After Reset: 0x0
			18:16	Q2MDMACH	R/W	0x0	Queue 2 Mapped to DMA Channel This field controls the routing of the received packet in Queue 2 to the DMA channel: 000: DMA Channel 0 001: DMA Channel 1Reserved 010: DMA Channel 2Reserved 011: DMA Channel 3Reserved 100: DMA Channel 4Reserved 101: DMA Channel 5Reserved 110: DMA Channel 6Reserved 111: DMA Channel 7Reserved This field is valid when the Q2DDMACH field is reset. Value After Reset: 0x0
			15:13	Reserved_15_13	R	0x0	Reserved. Value After Reset: 0x0

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Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			12	Q1DDMACH	R/W	0x0	Queue 1 Enabled for DA-based DMA Channel Selection When set, this bit indicates that the packets received in Queue 1 are routed to a particular DMA channel as decided in the MAC Receiver based on the DMA channel number programmed in the L3-L4 filter registers, or the Ethernet DA address. When reset, this bit indicates that the packets received in Queue 1 are routed to the DMA Channel programmed in the Q1MDMACH field (Bits[10:8]).Values: 0x0 (DISABLE): Queue 1 disabled for DA-based DMA Channel Selection 0x1 (ENABLE): Queue 1 enabled for DA-based DMA Channel Selection Value After Reset: 0x0
			11	Reserved_11_y	R	0x0	Reserved. Value After Reset: 0x0
			10:8	Q1MDMACH	R/W	0x0	Queue 1 Mapped to DMA Channel This field controls the routing of the received packet in Queue 1 to the DMA channel: 000: DMA Channel 0 001: DMA Channel 1Reserved 010: DMA Channel 2Reserved 011: DMA Channel 3Reserved 100: DMA Channel 4Reserved 101: DMA Channel 5Reserved 110: DMA Channel 6Reserved 111: DMA Channel 7Reserved This field is valid when the Q1DDMACH field is reset. Value After Reset: 0x0
			7:5	Reserved_7_5	R	0x0	Reserved. Value After Reset: 0x0
			4	Q0DDMACH	R/W	0x0	Queue 0 Enabled for DA-based DMA Channel Selection When set, this bit indicates that the packets received in Queue 0 are routed to a particular DMA channel as decided in the MAC Receiver based on the DMA channel number programmed in the L3-L4 filter registers, or the Ethernet DA address. When reset, this bit indicates that the packets received in Queue 0 are routed to the DMA Channel programmed in the Q0MDMACH field.Values: 0x0 (DISABLE): Queue 0 disabled for DA-based DMA Channel Selection 0x1 (ENABLE): Queue 0 enabled for DA-based DMA Channel Selection Value After Reset: 0x0
			3	Reserved_3_y	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			2:0	Q0MDMACH	R/W	0x0	Queue 0 Mapped to DMA Channel This field controls the routing of the packet received in Queue 0 to the DMA channel: 000: DMA Channel 0 001: DMA Channel 1Reserved 010: DMA Channel 2Reserved 011: DMA Channel 3Reserved 100: DMA Channel 4Reserved 101: DMA Channel 5Reserved 110: DMA Channel 6Reserved 111: DMA Channel 7Reserved This field is valid when the Q0DDMACH field is reset. Value After Reset: 0x0
			31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28	Q7DDMACH	R/W	0x0	Queue 7 Enabled for DA-based DMA Channel Selection When set, this bit indicates that the packets received in Queue 7 are routed to a particular DMA channel as decided in the MAC Receiver based on the DMA channel number programmed in the L3-L4 filter registers, or the Ethernet DA address. When reset, this bit indicates that the packets received in Queue 7 are routed to the DMA Channel programmed in the Q7MDMACH field.Values: 0x0 (DISABLE): Queue 5 disabled for DA-based DMA Channel Selection 0x1 (ENABLE): Queue 5 enabled for DA-based DMA Channel Selection Value After Reset: 0x0
			27	Reserved_27_y	R	0x0	Reserved. Value After Reset: 0x0
			26:24	Q7MDMACH	R/W	0x0	Queue 7 Mapped to DMA Channel This field controls the routing of the packet received in Queue 7 to the DMA channel: 000: DMA Channel 0 001: DMA Channel 1Reserved 010: DMA Channel 2Reserved 011: DMA Channel 3Reserved 100: DMA Channel 4Reserved 101: DMA Channel 5Reserved 110: DMA Channel 6Reserved 111: DMA Channel 7Reserved This field is valid when the Q7DDMACH field is reset. Value After Reset: 0x0
			23:21	Reserved_23_21	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xc34	MTL_RxQ_DMA_Map1	This register controls the Dynamic (per packet) or Static (all packets in a Receive Queue) DMA Channel Selection for higher numbered Receive Queues.	20	Q6DDMACH	R/W	0x0	Queue 6 Enabled for DA-based DMA Channel Selection When set, this bit indicates that the packets received in Queue 6 are routed to a particular DMA channel as decided in the MAC Receiver based on the DMA channel number programmed in the L3-L4 filter registers, or the Ethernet DA address. When reset, this bit indicates that the packets received in Queue 6 are routed to the DMA Channel programmed in the Q6MDMACH field.Values: 0x0 (DISABLE): Queue 6 disabled for DA-based DMA Channel Selection 0x1 (ENABLE): Queue 6 enabled for DA-based DMA Channel Selection Value After Reset: 0x0
			19	Reserved_19_y	R	0x0	Reserved. Value After Reset: 0x0
			18:16	Q6MDMACH	R/W	0x0	Queue 6 Mapped to DMA Channel This field controls the routing of the packet received in Queue 6 to the DMA channel: 000: DMA Channel 0 001: DMA Channel 1Reserved 010: DMA Channel 2Reserved 011: DMA Channel 3Reserved 100: DMA Channel 4Reserved 101: DMA Channel 5Reserved 110: DMA Channel 6Reserved 111: DMA Channel 7Reserved This field is valid when the Q6DDMACH field is reset. Value After Reset: 0x0
			15:13	Reserved_15_13	R	0x0	Reserved. Value After Reset: 0x0
			12	Q5DDMACH	R/W	0x0	Queue 5 Enabled for DA-based DMA Channel Selection When set, this bit indicates that the packets received in Queue 5 are routed to a particular DMA channel as decided in the MAC Receiver based on the DMA channel number programmed in the L3-L4 filter registers, or the Ethernet DA address. When reset, this bit indicates that the packets received in Queue 5 are routed to the DMA Channel programmed in the Q5MDMACH field.Values: 0x0 (DISABLE): Queue 5 disabled for DA-based DMA Channel Selection 0x1 (ENABLE): Queue 5 enabled for DA-based DMA Channel Selection Value After Reset: 0x0
			11	Reserved_11_y	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			10:8	Q5MDMACH	R/W	0x0	Queue 5 Mapped to DMA Channel This field controls the routing of the packets received in Queue 5 to the DMA channel: 000: DMA Channel 0 001: DMA Channel 1Reserved 010: DMA Channel 2Reserved 011: DMA Channel 3Reserved 100: DMA Channel 4Reserved 101: DMA Channel 5Reserved 110: DMA Channel 6Reserved 111: DMA Channel 7Reserved This field is valid when the Q5DDMACH field is reset. Value After Reset: 0x0
			7:5	Reserved_7_5	R	0x0	Reserved. Value After Reset: 0x0
			4	Q4DDMACH	R/W	0x0	Queue 4 Enabled for DA-based DMA Channel Selection When set, this bit indicates that the packets received in Queue 4 are routed to a particular DMA channel as decided in the MAC Receiver based on the DMA channel number programmed in the L3-L4 filter registers, or the Ethernet DA address. When reset, this bit indicates that the packets received in Queue 4 are routed to the DMA Channel programmed in the Q4MDMACH field.Values: 0x0 (DISABLE): Queue 4 disabled for DA-based DMA Channel Selection 0x1 (ENABLE): Queue 4 enabled for DA-based DMA Channel Selection Value After Reset: 0x0
			3	Reserved_3_y	R	0x0	Reserved. Value After Reset: 0x0
			2:0	Q4MDMACH	R/W	0x0	Queue 4 Mapped to DMA Channel This field controls the routing of the packet received in Queue 4 to the DMA channel: 000: DMA Channel 0 001: DMA Channel 1Reserved 010: DMA Channel 2Reserved 011: DMA Channel 3Reserved 100: DMA Channel 4Reserved 101: DMA Channel 5Reserved 110: DMA Channel 6Reserved 111: DMA Channel 7Reserved This field is valid when the Q4DDMACH field is reset. Value After Reset: 0x0
			31:8	LEOS	R/W	0x0	Launch Expiry Offset The value in units of 256 nanoseconds that has to be added to the Launch time to compute the Launch Expiry time. Value valid only when LEOV is set. Max value: 999,999,999 ns, additionally must be smaller than CTR-1 value when in ESTM mode. Because, this value is a modulo CTR value. Value After Reset: 0x0
			7	Reserved_7	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xc40	MTL_TBS_CTRL	This register controls the operation of Time Based Scheduling.	6:4	LEGOS	R/W	0x0	Launch Expiry GSN Offset The number GSN slots that has to be added to the Launch GSN to compute the Launch Expiry time. Value valid only when LEOV is set. Value After Reset: 0x0
			3:2	Reserved_3_2	R	0x0	Reserved. Value After Reset: 0x0
			1	LEOV	R/W	0x0	Launch Expiry Offset Valid When set indicates the LEOS field is valid. When not set, indicates the Launch Expiry Offset is not valid and the MTL must not check for Launch expiry time.Values: 0x0 (INVALID): LEOS field is invalid 0x1 (VALID): LEOS field is valid Value After Reset: 0x0
			0	ESTM	R/W	0x0	EST offset Mode 1: The Launch Time value used in Time Based Scheduling is interpreted as an EST offset value and is added to the Base Time Register (BTR) of the current list. 0: The Launch Time value is used as an absolute value that must be compared with the System time [39:8]. Values: 0x0 (DISABLE): EST offset Mode is disabled 0x1 (ENABLE): EST offset Mode is enabled Value After Reset: 0x0
			31:24	PTOV	R/W	0x0	PTP Time Offset Value The value of PTP Clock period multiplied by 9 in nanoseconds. This value is needed to avoid transmission overruns at the beginning of the installation of a new GCL. Value After Reset: 0x0
			23:12	CTOV	R/W	0x0	Current Time Offset Value Provides a 12 bit time offset value in nano second that is added to the current time to compensate for all the implementation pipeline delays such as the CDC sync delay, buffering delays, data path delays and so on. This offset helps to ensure that the impact of gate controls is visible on the line exactly at the pre-determined schedule (or as close to the schedule as possible). Value After Reset: 0x0
			11	Reserved_11	R	0x0	Reserved. Value After Reset: 0x0
			10:8	TILS	R/W	0x0	Time Interval Left Shift Amount This field provides the left shift amount for the programmed Time Interval values used in the Gate Control Lists. 000: No left shift needed (equal to x1ns) 001: Left shift TI by 1 bit (equal to x2ns) 010: Left shift TI by 2 bits (equal to x4ns) . . 100: Left shift TI by 7 bits (equal to x128ns) Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xc50	MTL_EST_Control	This register controls the operation of Enhancements to Scheduled Transmission (IEEE802.1Qbv).	7:6	LCSE	R/W	0x0	Loop Count to report Scheduling Error Programmable number of GCL list iterations before reporting an HLBS error defined in EST_Status register.Values: 0x0 (M_4_ITERNS): 4 iterations 0x1 (M_8_ITERNS): 8 iterations 0x2 (M_16_ITERNS): 16 iterations 0x3 (M_32_ITERNS): 32 iterations Value After Reset: 0x0
			5	DFBS	R/W	0x0	Drop Frames causing Scheduling Error When set frames reported to cause HOL Blocking due to not getting scheduled (HLBS field of EST_Status register) after 4,8,16,32 (based on LCSE field of this register) GCL iterations are dropped.Values: 0x0 (DONT_DROP): Do not Drop Frames causing Scheduling Error 0x1 (DROP): Drop Frames causing Scheduling Error Value After Reset: 0x0
			4	DDBF	R/W	0x0	Do not Drop frames during Frame Size Error When set, frames are not be dropped during Head-of-Line blocking due to Frame Size Error (HLBF field of EST_Status register).Values: 0x0 (DROP): Drop frames during Frame Size Error 0x1 (DONT_DROP): Do not Drop frames during Frame Size Error Value After Reset: 0x0
			3	QHLBF	R/W	0x0	Quick Assertion of HLBF Error When set, Time Window for Head-of-Line blocking due to Frame Size Error is 1 to 2 loop count of GCL list.On reset,Time Window for Head-of-Line blocking due to Frame Size Error is 2 to 3 loop counts of GCL list.Values: 0x0 (DISABLE): Disable Quick assertion of HLBF error 0x1 (ENABLE): Quick Assertion of HLBF Error Value After Reset: 0x0
			2	Reserved_2	R	0x0	Reserved. Value After Reset: 0x0
			1	SSWL	R/W	0x0	Switch to S/W owned list When set indicates that the software has programmed that list that it currently owns (SWOL) and the HW must switch to the new list based on the new BTR. HW clears this bit when the switch to the SWOL happens to indicate the completion of the switch or when a BTR error (BTRE in Status register) is set. When BTRE is set, this bit is cleared but SWOL is not updated as the switch was not successful. Access restrictions apply. MAC initiates switching to software owned list, when application writes 1. MAC writes 0 after switching to software owned list is complete. MAC ignores when application writes 0. Values: 0x0 (DISABLE): Switch to S/W owned list is disabled 0x1 (ENABLE): Switch to S/W owned list is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	EEST	R/W	0x0	Enable EST When reset, the gate control list processing is halted and all gates are assumed to be in Open state. Should be set for the hardware to start processing the gate control lists. During the toggle from 0 to 1, the gate control list processing starts only after the SSWL bit it set. Values: 0x0 (DISABLE): EST is disabled 0x1 (ENABLE): EST is enabled Value After Reset: 0x0
0xc54	MTL_EST_Ext_Control	This register indicates the number of Overhead bytes for EST related scheduling.	31:6	Reserved_31_6	R	0x0	Reserved. Value After Reset: 0x0
			5:0	OVHD	R/W	0x0	Overhead Bytes Value This field indicates the fixed overhead for every packet to account for EST Scheduler Delay, IPG or EIPG, and Preamble bytes. Value After Reset: 0x0
			31:20	Reserved_31_20	R	0x0	Reserved. Value After Reset: 0x0
			19:16	CGSN	R	0x0	Current GCL Slot Number Indicates the slot number of the GCL list. Slot number is a modulo 16 count of the GCL List loops executed so far. Even if a new GCL list is installed, the count is incremental. Value After Reset: 0x0
			15:8	BTRL	R	0x0	BTR Error Loop Count Provides the minimum count (N) for which the equation Current Time =< New BTR + (N * New Cycle Time) becomes true. N = "11111111" indicates the iterations exceeded the value of 128 and the hardware was not able to update New BTR to be equal to or greater than Current Time. Software intervention is needed to update the New BTR. Value cleared when BTRE field of this register is cleared. Value After Reset: 0x0
			7	SWOL	R	0x0	S/W owned list When '0' indicates Gate control list number "0" is owned by software and when "1" indicates the Gate Control list "1" is owned by the software. Any reads/writes by the software (using indirect access through GCL_Control) is directed to the list indicated by this value by default. The inverse of this value is treated as HWOL. R/W operations performed by hardware are directed to the list pointed by HWOL by default.Values: 0x0 (INACTIVE): Gate control list number "0" is owned by software 0x1 (ACTIVE): Gate control list number "1" is owned by software Value After Reset: 0x0
			6:5	Reserved_6_5	R	0x0	Reserved. Value After Reset: 0x0
			4	CGCE	R/W	0x0	Constant Gate Control Error This error occurs when the list length (LLR) is 1 and the Cycle Time (CTR) is less than or equal to the programmed Time Interval (TI) value after the optional Left Shifting. This implies, Gates are either always Closed or always Open based on the Gate Control values; the same effect can be achieved by other simpler (non TSN) programming mechanisms. In this condition an error is reported. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Constant Gate Control Error not detected 0x1 (ACTIVE): Constant Gate Control Error detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xc58	MTL_EST_Status	This register provides Status related to Enhancements to Scheduled Transmission (IEEE802.1Qbv).	3	HLBS	R	0x0	Head-Of-Line Blocking due to Scheduling Set when the frame is not able to win arbitration and get scheduled even after 4 iterations of the GCL. Indicates to software a potential programming error. The one hot encoded values of the Queue Numbers that are not able to make progress are indicated in the MTL_EST_Sch_Error register. Bit cleared when MTL_EST_Sch_Error register is all zeros. Note: A single fixed reference or tic is generated at integral multiple of loops/iterations programmed in LCSE field of MTL_EST_Control register. The HLBS is generated after (LCSE + N) loops/iterations, where N is non-integral number of loops/iterations programmed in LCSE field of MTL_EST_Control register, the packet could not be scheduled before the last 2 tics.Values: 0x0 (INACTIVE): Head-Of-Line Blocking due to Scheduling not detected 0x1 (ACTIVE): Head-Of-Line Blocking due to Scheduling detected Value After Reset: 0x0
			2	HLBF	R	0x0	Head-Of-Line Blocking due to Frame Size Set when HOL Blocking is noticed on one or more Queues as a result of none of the Time Intervals of gate open in the GCL being greater than or equal to the duration needed for frame size (or frame fragment size when preemption is enabled) transmission. The one hot encoded Queue numbers that are experiencing HLBF are indicated in the MTL_EST_Frm_Size_Error register. Additionally, the first Queue number that experienced HLBF along with the frame size is captured in MTL_EST_Frm_Size_Capture register. Bit cleared when MTL_EST_Frame_Size_ Error register is all zeros.Values: 0x0 (INACTIVE): Head-Of-Line Blocking due to Frame Size not detected 0x1 (ACTIVE): Head-Of-Line Blocking due to Frame Size detected Value After Reset: 0x0
			1	BTRE	R/W	0x0	BTR Error When "1" indicates a programming error in the BTR of SWOL where the programmed value is less than current time. If the BTRL = "11111111", SWOL is not updated and software must reprogram the BTR to a value greater than current time and then set SSWL to reinitiate the switch to SWOL. Else if the value of BTRL < "11111111", SWOL is updated and this field indicates the number of iterations (of + CycleTime) taken by hardware to update the BTR to a value greater than Current Time. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): BTR Error not detected 0x1 (ACTIVE): BTR Error detected Value After Reset: 0x0
			0	SWLC	R/W	0x0	Switch to S/W owned list Complete When "1" indicates the hardware has successfully switched to the SWOL, and the SWOL bit has been updated to that effect. Cleared when the SSWL of EST_Control register transitions from 0 to 1, or on a software write. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Switch to S/W owned list Complete not detected 0x1 (ACTIVE): Switch to S/W owned list Complete detected Value After Reset: 0x0
0xc60	MTL_EST_Sch_Error	This register provides the One Hot encoded Queue Numbers that are having the Scheduling related error (timeout).	31:8	Reserved_31_x	R	0x0	Reserved. Value After Reset: 0x0
			7:0	SEQN	R/W	0x0	Schedule Error Queue Number The One Hot Encoded Queue Numbers that have experienced error/timeout described in HLBS field of status register. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xc64	MTL_EST_Frm_Size_Error	This register provides the One Hot encoded Queue Numbers that are having the Frame Size related error.	31:8	Reserved_31_x	R	0x0	Reserved. Value After Reset: 0x0
			7:0	FEQN	R/W	0x0	Frame Size Error Queue Number The One Hot Encoded Queue Numbers that have experienced error described in HLBF field of status register. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. Value After Reset: 0x0
0xc68	MTL_EST_Frm_Size_Capture	This register captures the Frame Size and Queue Number of the first occurrence of the Frame Size related error. Up on clearing it captures the data of immediate next occurrence of a similar error.	31:19	Reserved_31_x	R	0x0	Reserved. Value After Reset: 0x0
			18:16	HBFQ	R	0x0	Queue Number of HLBF Captures the binary value of the of the first Queue (number) experiencing HLBF error (see HLBF field of status register). Value once written is not altered by any subsequent queue errors of similar nature. Once cleared the queue number of the next occurring HLBF error is captured. Width is based on the number of Tx Queues configured; remaining bits are Read-Only. Cleared when MTL_EST_Frm_Size_Error register is all zeros. Value After Reset: 0x0
			15	Reserved_15	R	0x0	Reserved. Value After Reset: 0x0
			14:0	HBFS	R	0x0	Frame Size of HLBF Captures the Frame Size of the dropped frame related to queue number indicated in HBFQ field of this register. Contents of this register must be considered invalid, if this field is zero. Cleared when MTL_EST_Frm_Size_Error register is all zeros. Value After Reset: 0x0
0xc70	MTL_EST_Intr_Enable	This register implements the Interrupt Enable bits for the various events that generate an interrupt. Bit positions have a 1 to 1 correlation with the status bit positions in MTL_ETS_Status register.	31:5	Reserved_31_5	R	0x0	Reserved. Value After Reset: 0x0
			4	CGCE	R/W	0x0	Interrupt Enable for CGCE When set, generates interrupt when the Constant Gate Control Error occurs and is indicated in the status. When reset this event does not generate an interruptValues: 0x0 (DISABLE): Interrupt for CGCE is disabled 0x1 (ENABLE): Interrupt for CGCE is enabled Value After Reset: 0x0
			3	IEHS	R/W	0x0	Interrupt Enable for HLBS When set, generates interrupt when the Head-of-Line Blocking due to Scheduling issue and is indicated in the status. When reset this event does not generate an interrupt.Values: 0x0 (DISABLE): Interrupt for HLBS is disabled 0x1 (ENABLE): Interrupt for HLBS is enabled Value After Reset: 0x0
			2	IEHF	R/W	0x0	Interrupt Enable for HLBF When set, generates interrupt when the Head-of-Line Blocking due to Frame Size error occurs and is indicated in the status. When reset this event does not generate an interrupt.Values: 0x0 (DISABLE): Interrupt for HLBF is disabled 0x1 (ENABLE): Interrupt for HLBF is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			1	IEBE	R/W	0x0	Interrupt Enable for BTR Error When set, generates interrupt when the BTR Error occurs and is indicated in the status. When reset this event does not generate an interrupt.Values: 0x0 (DISABLE): Interrupt for BTR Error is disabled 0x1 (ENABLE): Interrupt for BTR Error is enabled Value After Reset: 0x0
			0	IECC	R/W	0x0	Interrupt Enable for Switch List When set, generates interrupt when the configuration change is successful and the hardware has switched to the new list. When reset this event does not generate an interrupt.Values: 0x0 (DISABLE): Interrupt for Switch List is disabled 0x1 (ENABLE): Interrupt for Switch List is enabled Value After Reset: 0x0
			31:24	Reserved_31_24	R	0x0	Reserved. Value After Reset: 0x0
			23:21			#VALUE!	Reserved Field: Yes
			20:16	Reserved_20_y	R	0x0	Reserved. Value After Reset: 0x0
			15:8	ADDR	R/W	0x0	Gate Control List Address: (ADDR when GCRR is "0"). Provides the address (row number) of the Gate Control List at which the R/W operation has to be performed. By default the Gate Control List pointed by SWOL of MTL_EST_Status is selected for R/W, however if the DBGM bit of this register is set, a debug mode access is given to R/W from DBGB. Gate Control List Related Registers Address: (ADDR when GCRR is "1"). By default the GCL related register set pointed by SWOL of MTL_EST_Status is selected for R/W, however if the DBGM bit of this register is set, a debug mode access is given to R/W from DBGB. Lower 3 bits are only used in this mode, higher order bits are treated as dont care. 000: BTR Low (31:0) 001: BTR High (63:31) 010: CTR Low (31:0) 011: CTR High (39:32) 100: TER (22:0)(26:0)(30:0) 101: LLR (5:0)(6:0)(7:0)(8:0)(9:0) Others: Reserved Value After Reset: 0x0
			7:6	Reserved_7_6	R/W	0x0	Reserved. Value After Reset: 0x0
			5	DBGB	R/W	0x0	Debug Mode Bank Select When set to "0" indicates R/W in debug mode must be directed to Bank 0 (GCL0 and corresponding Time related registers). When set to "1" indicates R/W in debug mode must be directed to Bank 1 (GCL1 and corresponding Time related registers). This value is used when DBGM is set and overrides by value of SWOL which is normally used.Values: 0x0 (BANK0): R/W in debug mode should be directed to Bank 0 0x1 (BANK1): R/W in debug mode should be directed to Bank 1 Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xc80	MTL_EST_GCL_Control	This register provides the control information for reading/writing to the Gate Control lists.	4	DBGM	R/W	0x0	Debug Mode When set to "1" indicates R/W in debug mode where the memory bank (for GCL and Time related registers) is explicitly provided by DBGB value, when set to "0" SWOL bit is used to determine which bank to use.Values: 0x0 (DISABLE): Debug Mode is disabled 0x1 (ENABLE): Debug Mode is enabled Value After Reset: 0x0
			3	Reserved_3	R/W	0x0	Reserved. Value After Reset: 0x0
			2	GCRR	R/W	0x0	Gate Control Related Registers When set to "1" indicates the R/W access is for the GCL related registers (BTR, CTR, TER, LLR) whose address is provided by ADDR. When "0" indicates R/W must be directed to GCL from the address provided by ADDR.Values: 0x0 (DISABLE): Gate Control Related Registers are disabled 0x1 (ENABLE): Gate Control Related Registers are enabled Value After Reset: 0x0
			1	R1W0	R/W	0x0	Read '1', Write '0': 1: Read operation 0: Write operation. Values: 0x0 (WRITE): Write Operation 0x1 (READ): Read Operation Value After Reset: 0x0
			0	SRWO	R/W	0x0	Start Read or Write Operation 1: Indicates the start or progress of a read or write operation. MAC writes 0: Indicates that the read or write operation is complete or an error has occurred. Reads: Data can be read from MTL_EST_GCL_Data register when this field is 0. Writes: You must program the MTL_EST_GCL_Data register with the data before setting this bit. Access restrictions apply. MAC initiates the read or write operation when application writes 1. MAC writes 0 after the read or write operation is complete. MAC ignores when application writes 0. Values: 0x0 (DISABLE): Start Read/Write Op disabled 0x1 (ENABLE): Start Read/Write Op enabled Value After Reset: 0x0
0xc84	MTL_EST_GCL_Data	This register holds the read data or write data in case of reads and writes respectively.	31:0	GCD	R/W	0x0	Gate Control Data The data corresponding to the address selected in the GCL_Control register. Used for both Read and Write operations. Value After Reset: 0x0
			31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28	HRS	R	0x0	Hold/Release Status 1: Indicates a Set-and-Hold-MAC operation was last executed and the pMAC is in Hold State. 0: Indicates a Set-and-Release-MAC operation was last executed and the pMAC is in Release State. Values: 0x0 (SET_REL): Indicates a Set-and-Release-MAC operation was last executed and the pMAC is in Release State 0x1 (SET_HOLD): Indicates a Set-and-Hold-MAC operation was last executed and the pMAC is in Hold State Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xc90	MTL_FPE_CTRL_STS	This register controls the operation of, and provides status for Frame Preemption (IEEE802.1Qbu/802.3br).	27:16	Reserved_27_y	R	0x0	Reserved. Value After Reset: 0x0
			15:8	PEC	R/W	0x0	Preemption Classification When set indicates the corresponding Queue must be classified as preemptable, when '0' Queue is classified as express. When both EST (Qbv) and Preemption are enabled, Queue-0 is always assumed to be preemptable. When EST (Qbv) is enabled Queues categorized as preemptable here are always assumed to be in "Open" state in the Gate Control List. Value After Reset: 0x0
			7	LBHT	R/W	0x0	Level Based Hold Transition When set to 1, enables hold operation in second row of EST GCL if hold request is set in second GCL row for first loop of new GCL, even though hold request is also set in the first GCL row. This is deviation from original behavior where a transition from release request to hold request is required. When set to 0, enables hold operation only on transition from release request to hold request. Value After Reset: 0x0
			6:2	Reserved_6_2	R	0x0	Reserved. Value After Reset: 0x0
			1:0	AFSZ	R/W	0x0	Additional Fragment Size used to indicate, in units of 64 bytes, the minimum number of bytes over 64 bytes required in non-final fragments of preempted frames. The minimum non-final fragment size is (AFSZ +1) * 64 bytes Value After Reset: 0x0
0xc94	MTL_FPE_Advance	This register holds the Hold and Release Advance time.	31:16	RADV	R/W	0x0	Release Advance The maximum time in nanoseconds that can elapse between issuing a RELEASE to the MAC and the MAC being ready to resume transmission of preemptable frames, in the absence of there being any express frames available for transmission. Value After Reset: 0x0
			15:0	HADV	R/W	0x0	Hold Advance The maximum time in nanoseconds that can elapse between issuing a HOLD to the MAC and the MAC ceasing to transmit any preemptable frame that is in the process of transmission or any preemptable frames that are queued for transmission. Value After Reset: 0x0
			31	RXPI	R	0x1	RX Parser in Idle state This status bit is set to 1 when the Rx parser is in Idle State and waiting for a new packet for processing. This bit is used as a handshake with software when parser gets disables. After disabling, when bit is set then software can update the Rx parser instruction table.Values: 0x0 (INACTIVE): RX Parser not in Idle state 0x1 (ACTIVE): RX Parser in Idle state Value After Reset: 0x1
			30	ELIRS	R/W	0x0	Enable Last Instruction in RX Status When this bit is set, RDES2[31:16] indicates the index of the last instruction executed in Rx Parser, when set to 0 indicates MAC filter status. Value After Reset: 0x0
			29:23	Reserved_29_x	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xca0	MTL_RXP_Control_Status	The MTL_RXP_Control_Status register establishes the operating mode of Rx Parser and provides some status.	22:16	NPE	R/W	0x7f	Number of parsable entries in the Instruction table This control indicates the number of parsable entries in the Instruction Memory. This is used in Rx parser logic to detect programming Error. In case number of parsed entries for a packet is more than this entry then NPEOVIS bit in the MTL_RXP_Interrupt_Control_Status register is set. Value After Reset: 0x7f
			15:12	Reserved_15_12	R	0x0	Reserved. Value After Reset: 0x0
			11			#VALUE!	Reserved Field: Yes
			10:7	Reserved_10_x	R	0x0	Reserved. Value After Reset: 0x0
			6:0	NVE	R/W	0x7f	Number of valid entry address/index in the Instruction table This control indicates the number of valid entries address/index in the Instruction Memory (i.e. when NVE field in register=31, the maximum valid entry address is NVE+1 i.e. addresses/indices=0 to 32, or 33 entries). This is used in Rx parser logic to detect any programming Error. In case while parsing Table address (memory address) found to be more than this maximum valid entry address then NVEOVIS bit in the MTL_RXP_Interrupt_Control_Status register is set. Note 1: The minimum value of this must be 2. Note 2: The Flexible Receive Parser (FRP) reads two instructions in advance for better performance. Therefore, FRP can read up to NVE+2 instructions. Value After Reset: 0x7f
			31:20	Reserved_31_20	R	0x0	Reserved. Value After Reset: 0x0
			19	PDRFIE	R/W	0x0	Packet Drop due to RF Interrupt Enable When this bit is set, the PDRFIS interrupt is enabled. When this bit is reset, the PDRFIS interrupt is disabled.Values: 0x0 (DISABLE): Packet Drop due to RF Interrupt is disabled 0x1 (ENABLE): Packet Drop due to RF Interrupt is enabled Value After Reset: 0x0
			18	FOOVIE	R/W	0x0	Frame Offset Overflow Interrupt Enable When this bit is set, the FOOVIS interrupt is enabled. When this bit is reset, the FOOVIS interrupt is disabled.Values: 0x0 (DISABLE): Frame Offset Overflow Interrupt is disabled 0x1 (ENABLE): Frame Offset Overflow Interrupt is enabled Value After Reset: 0x0
			17	NPEOVIE	R/W	0x0	Number of Parsable Entries Overflow Interrupt Enable When this bit is set, the NPEOVIS interrupt is enabled. When this bit is reset, the NPEOVIS interrupt is disabled.Values: 0x0 (DISABLE): Number of Parsable Entries Overflow Interrupt is disabled 0x1 (ENABLE): Number of Parsable Entries Overflow Interrupt is enabled Value After Reset: 0x0
			16	NVEOVIE	R/W	0x0	Number of Valid Entries Overflow Interrupt Enable When this bit is set, the NVEOVIS interrupt is enabled. When this bit is reset, the NVEOVIS interrupt is disabled.Values: 0x0 (DISABLE): Number of Valid Entries Overflow Interrupt is disabled 0x1 (ENABLE): Number of Valid Entries Overflow Interrupt is enabled Value After Reset: 0x0
			15:4	Reserved_15_4	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xca4	MTL_RXP_Interrupt_Control_Status	The MTL_RXP_Interrupt_Control_Status registers provides enable control for the interrupts and provides interrupt status.	3	PDRFIS	R/W	0x0	Packet Dropped due to RF Interrupt Status If the Rx Parser result says to drop the packet by setting RF=1 in the instruction memory, then this bit is set to 1. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Packet Dropped due to RF Interrupt Status not detected 0x1 (ACTIVE): Packet Dropped due to RF Interrupt Status detected Value After Reset: 0x0 Testable: untestable
			2	FOOVIS	R/W	0x0	Frame Offset Overflow Interrupt Status While parsing if the Instruction table entry's 'Frame Offset' found to be more than EOF offset, then then this bit is set. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Frame Offset Overflow Interrupt Status not detected 0x1 (ACTIVE): Frame Offset Overflow Interrupt Status detected Value After Reset: 0x0 Testable: untestable
			1	NPEOVIS	R/W	0x0	Number of Parsable Entries Overflow Interrupt Status While parsing a packet if the number of parsed entries found to be more than NPE[] (Number of Parseable Entries in MTL_RXP_Control register),then this bit is set to 1. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Number of Parsable Entries Overflow Interrupt Status not detected 0x1 (ACTIVE): Number of Parsable Entries Overflow Interrupt Status detected Value After Reset: 0x0 Testable: untestable
			0	NVEOVIS	R/W	0x0	Number of Valid Entry Address/Index Overflow Interrupt Status While parsing if the Instruction address found to be more than NVE (Number of Valid Entry Address/index in MTL_RXP_Control register), then this bit is set to 1. For example, when NVE field in register=31, the maximum valid entry address/index is NVE+1 i.e. 32 (addresses/indices=0 to 32, or 33 entries), so NVEOVIS is set when currently processed entry indicates next address is 33 or more i.e. 34th or later entries. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Number of Valid Entries Overflow Interrupt Status not detected 0x1 (ACTIVE): Number of Valid Entries Overflow Interrupt Status detected Value After Reset: 0x0 Testable: untestable
0xca8	MTL_RXP_Drop_Cnt	The MTL_RXP_Drop_Cnt register provides the drop count of Rx Parser initiated drops.	31	RXPDCOVF	R	0x0	Rx Parser Drop Counter Overflow Bit When set, this bit indicates that the MTL_RXP_Drop_cnt (RXPDC) Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Rx Parser Drop count overflow not occurred 0x1 (ACTIVE): Rx Parser Drop count overflow occurred Value After Reset: 0x0
			30:0	RXPDC	R	0x0	Rx Parser Drop count This 31-bit counter is implemented when a Rx Parser Drops a packet due to RF =1. The counter is cleared when the register is read. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xcac	MTL_RXP_Error_Cnt	The MTL_RXP_Error_Cnt register provides the Rx Parser related error occurrence count.	31	RXPECOVF	R	0x0	Rx Parser Error Counter Overflow Bit When set, this bit indicates that the MTL_RXP_Error_cnt (RXPEC) Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Rx Parser Error count overflow not occurred 0x1 (ACTIVE): Rx Parser Error count overflow occurred Value After Reset: 0x0
			30:0	RXPEC	R	0x0	Rx Parser Error count This 31-bit counter is implemented when a Rx Parser encounters following Error scenarios Entry address >= NVE[] Number Parsed Entries >= NPE[] Entry address > EOF data entry address The counter is cleared when the register is read. Value After Reset: 0x0
0xcb0	MTL_RXP_Indirect_Acc_Control_Status	The MTL_RXP_Indirect_Acc_Control_Status register provides the Indirect Access control and status for Rx Parser memory.	31	STARTBUSY	R/W	0x0	FRP Instruction Table Access Busy Set to 1 by the software, indicates to start the Read/Write operation from/to the Rx Parser Memory. Software must read this bit as 0 before issuing read or write request to access the Parser Memory Instructions. Set to 1 by the HW, indicates that the HW is busy until it is cleared by the HW. Software must not issue read or write operation. Values: 0x0 (INACTIVE): hardware not busy 0x1 (ACTIVE): hardware is busy (Read/Write operation from/to the Rx Parser Memory) Value After Reset: 0x0 Testable: untestable
			30:23	Reserved_30_23	R	0x0	Reserved. Value After Reset: 0x0
			22:20			#VALUE!	Reserved Field: Yes
			19:17	Reserved_19_17	R	0x0	Reserved. Value After Reset: 0x0
			16	WRRDN	R/W	0x0	Read Write Control 1: Indicates the write operation to the Rx Parser Memory. 0: Indicates the read operation to the Rx Parser Memory. Values: 0x0 (READ): Read operation to the Rx Parser Memory 0x1 (WRITE): Write operation to the Rx Parser Memory Value After Reset: 0x0
			15:9	Reserved_15_x	R	0x0	Reserved. Value After Reset: 0x0
			8:0	ADDR	R/W	0x0	FRP Instruction Table Offset Address This field indicates the ADDR of the 32-bit part of a entry in Rx parser instruction table. Each entry has 128-bit (4x32-bit words). Value After Reset: 0x0
0xcb4	MTL_RXP_Indirect_Acc_Data	The MTL_RXP_Indirect_Acc_Data registers holds the data associated to Indirect Access to Rx Parser memory.	31:0	DATA	R	0x0	FRP Instruction Table Write/Read Data Software must write this register before issuing any write command. The hardware provides the read data from the Rx Parser Memory for read operation when START BUSY =0 after read command. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xcb8	MTL_RXP_Bypass_Cnt	The MTL_RXP_Bypass_Cnt register provides the bypass count of Rx Parser.	31	RXPBCOF	R	0x0	Rx Parser bypass Counter Overflow Bit When set, this bit indicates that the MTL_RXP_Bypass_cnt (RXPBC) Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Rx Parser Bypass count overflow not occurred 0x1 (ACTIVE): Rx Parser Bypass count overflow occurred Value After Reset: 0x0
			30:0	RXPBC	R	0x0	Rx Parser Bypass count This 31-bit counter is implemented when a Rx Parser bypass a packet due to AF=1 and RF =1. The counter is cleared when the register is read. Value After Reset: 0x0
0xd00	MTL_TxQ0_Operation_Mode	The Queue 0 Transmit Operation Mode register establishes the Transmit queue operating modes and commands.	31:22	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			21:16	TQS	R/W	0x0	Transmit Queue Size This field indicates the size of the allocated Transmit queues in blocks of 256 bytes, the reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on. So, program TQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (TQS+1)*256 bytes. Value After Reset: 0x0
			15:7	Reserved_15_7	R	0x0	Reserved. Value After Reset: 0x0
			6:4	TTC	R/W	0x0	Transmit Threshold Control These bits control the threshold level of the MTL Tx Queue. The transmission starts when the packet size within the MTL Tx Queue is larger than the threshold. In addition, full packets with length less than the threshold are also transmitted. These bits are used only when the TSF bit is reset.Values: 0x0 (M_32BYTES): 32 0x1 (M_64BYTES): 64 0x2 (M_96BYTES): 96 0x3 (M_128BYTES): 128 0x4 (M_192BYTES): 192 0x5 (M_256BYTES): 256 0x6 (M_384BYTES): 384 0x7 (M_512BYTES): 512 Value After Reset: 0x0
			3:2	TXQEN	R/W	0x0	Transmit Queue Enable This field is used to enable/disable the transmit queue 0. This field is Read Only in Single Queue configurations and Read Write in Multiple Queue configurations. Note: In multiple Tx queues configuration, all the queues are disabled by default. Enable the Tx queue by programming this field.Values: 0x0 (DISABLE): Not enabled 0x1 (EN_IF_AV): Enable in AV mode 0x2 (ENABLE): Enabled 0x3 (RSVD2): Reserved Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			1	TSF	R/W	0x0	Transmit Store and Forward When this bit is set, the transmission starts when a full packet resides in the MTL Tx queue. When this bit is set, the TTC values specified in Bits[6:4] of this register are ignored. This bit should be changed only when the transmission is stopped.Values: 0x0 (DISABLE): Transmit Store and Forward is disabled 0x1 (ENABLE): Transmit Store and Forward is enabled Value After Reset: 0x0
			0	FTQ	R/W	0x0	Flush Transmit Queue When this bit is set, the Tx queue controller logic is reset to its default values. Therefore, all the data in the Tx queue is lost or flushed. This bit is internally reset when the flushing operation is complete. Until this bit is reset, you must not write to the MTL_TxQ0_Operation_Mode register. The data which is already accepted by the MAC transmitter is not flushed. It is scheduled for transmission and results in underflow and runt packet transmission. Note: The flush operation is complete only when the Tx queue is empty and the application has accepted the pending Tx Status of all transmitted packets. To complete this flush operation, the PHY Tx clock (clk_tx_i) must be active. Access restrictions apply. MAC initiates the Tx queue flush when application writes 1. MAC writes 0 after the Tx queue flush is complete. MAC ignores when application writes 0. Values: 0x0 (DISABLE): Flush Transmit Queue is disabled 0x1 (ENABLE): Flush Transmit Queue is enabled Value After Reset: 0x0 Testable: untestable
0xd04	MTL_TxQ0_Underflow	The Queue 0 Underflow Counter register contains the counter for packets aborted because of Transmit queue underflow and packets missed because of Receive queue packet flush	31:12	Reserved_31_12	R	0x0	Reserved. Value After Reset: 0x0
			11	UFCNTOVF	R	0x0	Overflow Bit for Underflow Packet Counter This bit is set every time the Tx queue Underflow Packet Counter field overflows, that is, it has crossed the maximum count. In such a scenario, the overflow packet counter is reset to all-zeros and this bit indicates that the rollover happened. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Overflow not detected for Underflow Packet Counter 0x1 (ACTIVE): Overflow detected for Underflow Packet Counter Value After Reset: 0x0
			10:0	UFFRMCNT	R	0x0	Underflow Packet Counter This field indicates the number of packets aborted by the controller because of Tx Queue Underflow. This counter is incremented each time the MAC aborts outgoing packet because of underflow. The counter is cleared when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0
			31:23	Reserved_31_23	R	0x0	Reserved. Value After Reset: 0x0
			22:20	STXSTSF	R	0x0	Number of Status Words in Tx Status FIFO of Queue This field indicates the current number of status in the Tx Status FIFO of this queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of status words in Tx Status FIFO. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xd08	MTL_TxQ0_Debug	The Queue 0 Transmit Debug register gives the debug status of various blocks related to the Transmit queue.	19	Reserved_19	R	0x0	Reserved. Value After Reset: 0x0
			18:16	PTXQ	R	0x0	Number of Packets in the Transmit Queue This field indicates the current number of packets in the Tx Queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of packets in the Transmit queue. Value After Reset: 0x0
			15:6	Reserved_15_6	R	0x0	Reserved. Value After Reset: 0x0
			5	TXSTSFSTS	R	0x0	MTL Tx Status FIFO Full Status When high, this bit indicates that the MTL Tx Status FIFO is full. Therefore, the MTL cannot accept any more packets for transmission.Values: 0x0 (INACTIVE): MTL Tx Status FIFO Full status is not detected 0x1 (ACTIVE): MTL Tx Status FIFO Full status is detected Value After Reset: 0x0
			4	TXQSTS	R	0x0	MTL Tx Queue Not Empty Status When this bit is high, it indicates that the MTL Tx Queue is not empty and some data is left for transmission.Values: 0x0 (INACTIVE): MTL Tx Queue Not Empty status is not detected 0x1 (ACTIVE): MTL Tx Queue Not Empty status is detected Value After Reset: 0x0
			3	TWCSTS	R	0x0	MTL Tx Queue Write Controller Status When high, this bit indicates that the MTL Tx Queue Write Controller is active, and it is transferring the data to the Tx Queue.Values: 0x0 (INACTIVE): MTL Tx Queue Write Controller status is not detected 0x1 (ACTIVE): MTL Tx Queue Write Controller status is detected Value After Reset: 0x0
			2:1	TRCSTS	R	0x0	MTL Tx Queue Read Controller Status This field indicates the state of the Tx Queue Read Controller:Values: 0x0 (IDLE): Idle state 0x1 (READ): Read state (transferring data to the MAC transmitter) 0x2 (WAIT): Waiting for pending Tx Status from the MAC transmitter 0x3 (FLUSH): Flushing the Tx queue because of the Packet Abort request from the MAC Value After Reset: 0x0
			0	TXQPAUSED	R	0x0	Transmit Queue in Pause When this bit is high and the Rx flow control is enabled, it indicates that the Tx Queue is in the Pause condition (in the full-duplex only mode) because of the following: Reception of the PFC packet for the priorities assigned to the Tx Queue when PFC is enabled Reception of 802.3x Pause packet when PFC is disabled Values: 0x0 (INACTIVE): Transmit Queue in Pause status is not detected 0x1 (ACTIVE): Transmit Queue in Pause status is detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xd14	MTL_TxQ0_ETS_Status	The Queue 0 ETS Status register provides the average traffic transmitted in Queue 0.	31:24	Reserved_31_24	R	0x0	Reserved. Value After Reset: 0x0
			23:0	ABS	R	0x0	Average Bits per Slot This field contains the average transmitted bits per slot. When the DCB operation is enabled for Queue 0, this field is computed over every 10 million bit times slot (4 ms in 2500 Mbps; 10 ms in 1000 Mbps; 100 ms in 100 Mbps). The maximum value is 0x989680. Value After Reset: 0x0
0xd18	MTL_TxQ0_Quantum_Weight	The Queue 0 Quantum or Weights register contains the quantum value for Deficit Weighted Round Robin (DWRR), weights for the Weighted Round Robin (WRR), and Weighted Fair Queuing (WFQ) for Queue 0.	31:21	Reserved_31_21	R	0x0	Reserved. Value After Reset: 0x0
			20:0	ISCQW	R/W	0x0	Quantum or Weights When the DCB operation is enabled with DWRR algorithm for Queue 0 traffic, this field contains the quantum value in bytes to be added to credit during every queue scanning cycle. The maximum value is 0x1312D0 bytes. When DCB operation is enabled with WFQ algorithm for Queue 0 traffic, this field contains the weight for this queue. The maximum value is 0x3FFF where weight of 0 indicates 100% bandwidth. Bits[20:14] must be written to zero. The higher the programmed weights lesser the bandwidth allocated for the particular Transmit Queue. This is because the weights are used to compute the packet finish time (weights * packet_size). Lesser the finish time, higher the probability of the packet getting scheduled first and using more bandwidth. When DCB operation or generic queuing operation is enabled with WRR algorithm for Queue 0 traffic, this field contains the weight for this queue. The maximum value is 0x64. Bits [20:7] must be written to zero. Value After Reset: 0x0
			31:25	Reserved_31_25	R	0x0	Reserved. Value After Reset: 0x0
			24	RXOIE	R/W	0x0	Receive Queue Overflow Interrupt Enable When this bit is set, the Receive Queue Overflow interrupt is enabled. When this bit is reset, the Receive Queue Overflow interrupt is disabled.Values: 0x0 (DISABLE): Receive Queue Overflow Interrupt is disabled 0x1 (ENABLE): Receive Queue Overflow Interrupt is enabled Value After Reset: 0x0
			23:17	Reserved_23_17	R	0x0	Reserved. Value After Reset: 0x0
			16	RXOVFIS	R/W	0x0	Receive Queue Overflow Interrupt Status This bit indicates that the Receive Queue had an overflow while receiving the packet. If a partial packet is transferred to the application, the overflow status is set in RDES3[21]. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Queue Overflow Interrupt Status not detected 0x1 (ACTIVE): Receive Queue Overflow Interrupt Status detected Value After Reset: 0x0 Testable: untestable
			15:10	Reserved_15_10	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xd2c	MTL_Q0_Interrupt_Control_Status	This register contains the interrupt enable and status bits for the queue 0 interrupts.	9	ABPSIE	R/W	0x0	Average Bits Per Slot Interrupt Enable When this bit is set, the MAC asserts the sbd_intr_o or mci_intr_o interrupt when the average bits per slot status is updated. When this bit is cleared, the interrupt is not asserted for such an event.Values: 0x0 (DISABLE): Average Bits Per Slot Interrupt is disabled 0x1 (ENABLE): Average Bits Per Slot Interrupt is enabled Value After Reset: 0x0
			8	TXUIE	R/W	0x0	Transmit Queue Underflow Interrupt Enable When this bit is set, the Transmit Queue Underflow interrupt is enabled. When this bit is reset, the Transmit Queue Underflow interrupt is disabled.Values: 0x0 (DISABLE): Transmit Queue Underflow Interrupt Status is disabled 0x1 (ENABLE): Transmit Queue Underflow Interrupt Status is enabled Value After Reset: 0x0
			7:2	Reserved_7_2	R	0x0	Reserved. Value After Reset: 0x0
			1	ABPSIS	R/W	0x0	Average Bits Per Slot Interrupt Status When set, this bit indicates that the MAC has updated the ABS value. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Average Bits Per Slot Interrupt Status not detected 0x1 (ACTIVE): Average Bits Per Slot Interrupt Status detected Value After Reset: 0x0 Testable: untestable
			0	TXUNFIS	R/W	0x0	Transmit Queue Underflow Interrupt Status This bit indicates that the Transmit Queue had an underflow while transmitting the packet. Transmission is suspended and an Underflow Error TDES3[2] is set. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Queue Underflow Interrupt Status not detected 0x1 (ACTIVE): Transmit Queue Underflow Interrupt Status detected Value After Reset: 0x0 Testable: untestable
			31:26	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			25:20	RQS	R/W	0x0	Receive Queue Size This field indicates the size of the allocated Receive queues in blocks of 256 bytesthe reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on. So, program RQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (RQS+1)*256 bytes. Value After Reset: 0x0
			19	Reserved_19_y	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xd30	MTL_RxQ0_Operation_Mode	The Queue 0 Receive Operation Mode register establishes the Receive queue operating modes and command.	18:14	RFD	R/W	0x0	Threshold for Deactivating Flow Control (in half-duplex and full-duplex modes) These bits control the threshold (fill-level of Rx queue) at which the flow control is de-asserted after activation: 0: Full minus 1 KB, that is, FULL 1 KB 1: Full minus 1.5 KB, that is, FULL 1.5 KB 2: Full minus 2 KB, that is, FULL 2 KB 3: Full minus 2.5 KB, that is, FULL 2.5 KB ... 30: Full minus 16 KB, that is, FULL 16 KB 31: Full minus 16.5 KB, that is, FULL 16.5 KB The de-assertion is effective only after flow control is asserted. Note: The value must be programmed in such a way to make sure that the threshold is a positive number. When the EHFC is set high, these values are applicable only when the Rx queue size determined by the RQS field of this register, is equal to or greater than 4 KB. For a given queue size, the values ranges between 0 and the encoding for FULL minus (QSIZE - 0.5 KB) and all other values are illegal. Here the term FULL and QSIZE refers to the queue size determined by the RQS field of this register. Value After Reset: 0x0
			13	Reserved_13_y	R	0x0	Reserved. Value After Reset: 0x0
			12:8	RFA	R/W	0x0	Threshold for Activating Flow Control (in half-duplex and full-duplex These bits control the threshold (fill-level of Rx queue) at which the flow control is activated: For more information on encoding for this field, see RFD. Value After Reset: 0x0
			7	EHFC	R/W	0x0	Enable Hardware Flow Control When this bit is set, the flow control signal operation, based on the fill-level of Rx queue, is enabled. When reset, the flow control operation is disabled.Values: 0x0 (DISABLE): Hardware Flow Control is disabled 0x1 (ENABLE): Hardware Flow Control is enabled Value After Reset: 0x0
			6	DIS_TCP_EF	R/W	0x0	Disable Dropping of TCP/IP Checksum Error Packets When this bit is set, the MAC does not drop the packets which only have the errors detected by the Receive Checksum Offload engine. Such packets have errors only in the encapsulated payload. There are no errors (including FCS error) in the Ethernet packet received by the MAC. When this bit is reset, all error packets are dropped if the FEP bit is reset.Values: 0x0 (ENABLE): Dropping of TCP/IP Checksum Error Packets is enabled 0x1 (DISABLE): Dropping of TCP/IP Checksum Error Packets is disabled Value After Reset: 0x0
			5	RSF	R/W	0x0	Receive Queue Store and Forward When this bit is set, the DWC_ether_qos reads a packet from the Rx queue only after the complete packet has been written to it, ignoring the RTC field of this register. When this bit is reset, the Rx queue operates in the Threshold (cut-through) mode, subject to the threshold specified by the RTC field of this register.Values: 0x0 (DISABLE): Receive Queue Store and Forward is disabled 0x1 (ENABLE): Receive Queue Store and Forward is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			4	FEP	R/W	0x0	Forward Error Packets When this bit is reset, the Rx queue drops packets with error status (CRC error, GMII_ER, watchdog timeout, or overflow). However, if the start byte (write) pointer of a packet is already transferred to the read controller side (in Threshold mode), the packet is not dropped. When this bit is set, all packets except the runt error packets are forwarded to the application or DMA. If the RSF bit is set and the Rx queue overflows when a partial packet is written, the packet is dropped irrespective of the setting of this bit. However, if the RSF bit is reset and the Rx queue overflows when a partial packet is written, a partial packet might be forwarded to the application or DMA.Values: 0x0 (DISABLE): Forward Error Packets is disabled 0x1 (ENABLE): Forward Error Packets is enabled Value After Reset: 0x0
			3	FUP	R/W	0x0	Forward Undersized Good Packets When this bit is set, the Rx queue forwards the undersized good packets (packets with no error and length less than 64 bytes), including pad-bytes and CRC. When this bit is reset, the Rx queue drops all packets of less than 64 bytes, unless a packet is already transferred because of the lower value of Rx Threshold, for example, RTC = 01.Values: 0x0 (DISABLE): Forward Undersized Good Packets is disabled 0x1 (ENABLE): Forward Undersized Good Packets is enabled Value After Reset: 0x0
			2	Reserved_2	R	0x0	Reserved. Value After Reset: 0x0
			1:0	RTC	R/W	0x0	Receive Queue Threshold Control These bits control the threshold level of the MTL Rx queue (in bytes): The received packet is transferred to the application or DMA when the packet size within the MTL Rx queue is larger than the threshold. In addition, full packets with length less than the threshold are automatically transferred. This field is valid only when the RSF bit is zero. This field is ignored when the RSF bit is set to 1.Values: 0x0 (M_64BYTE): 64 0x1 (M_32BYTE): 32 0x2 (M_96BYTE): 96 0x3 (M_128BYTE): 128 Value After Reset: 0x0
0xd34	MTL_RxQ0_Missed_Packet_Overflow_Cnt	The Queue 0 Missed Packet and Overflow Counter register contains the counter for packets missed because of Receive queue packet flush and packets discarded because of Receive queue overflow.	31:28	Reserved_31_28	R	0x0	Reserved. Value After Reset: 0x0
			27	MISCNTOVF	R	0x0	Missed Packet Counter Overflow Bit When set, this bit indicates that the Rx Queue Missed Packet Counter crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Missed Packet Counter overflow not detected 0x1 (ACTIVE): Missed Packet Counter overflow detected Value After Reset: 0x0
			26:16	MISPKTCNT	R	0x0	Missed Packet Counter This field indicates the number of packets missed by the DWC_ether_qos because the application asserted ari_pkt_flush_i[] for this queue. This counter is reset when this register is read with mci_be_i[0] at 1b1. This counter is incremented by 1 when the DMA discards the packet because of buffer unavailability. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0
			15:12	Reserved_15_12	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			11	OVFCNTOVF	R	0x0	Overflow Counter Overflow Bit When set, this bit indicates that the Rx Queue Overflow Packet Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Overflow Counter overflow not detected 0x1 (ACTIVE): Overflow Counter overflow detected Value After Reset: 0x0
			10:0	OVFPKTCNT	R	0x0	Overflow Packet Counter This field indicates the number of packets discarded by the DWC_ether_qos because of Receive queue overflow. This counter is incremented each time the DWC_ether_qos discards an incoming packet because of overflow. This counter is reset when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0
0xd38	MTL_RxQ0_Debug	The Queue 0 Receive Debug register gives the debug status of various blocks related to the Receive queue.	31:30	Reserved_31_30	R	0x0	Reserved. Value After Reset: 0x0
			29:16	PRXQ	R	0x0	Number of Packets in Receive Queue This field indicates the current number of packets in the Rx Queue, that is, Max_Queue_Size/Min_Packet_Size. Value After Reset: 0x0
			15:6	Reserved_15_6	R	0x0	Reserved. Value After Reset: 0x0
			5:4	RXQSTS	R	0x0	MTL Rx Queue Fill-Level Status This field gives the status of the fill-level of the Rx Queue:Values: 0x0 (EMPTY): Rx Queue empty 0x1 (BLW_THR): Rx Queue fill-level below flow-control deactivate threshold 0x2 (ABV_THR): Rx Queue fill-level above flow-control activate threshold 0x3 (FULL): Rx Queue full Value After Reset: 0x0
			3	Reserved_3	R	0x0	Reserved. Value After Reset: 0x0
			2:1	RRCSTS	R	0x0	MTL Rx Queue Read Controller State This field gives the state of the Rx queue Read controller:Values: 0x0 (IDLE): Idle state 0x1 (READ_DATA): Reading packet data 0x2 (READ_STS): Reading packet status (or timestamp) 0x3 (FLUSH): Flushing the packet data and status Value After Reset: 0x0
			0	RWCSTS	R	0x0	MTL Rx Queue Write Controller Active Status When high, this bit indicates that the MTL Rx queue Write controller is active, and it is transferring a received packet to the Rx Queue.Values: 0x0 (INACTIVE): MTL Rx Queue Write Controller Active Status not detected 0x1 (ACTIVE): MTL Rx Queue Write Controller Active Status detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xd3c	MTL_RxQ0_Control	The Queue Receive Control register controls the receive arbitration and passing of received packets to the application.	31:4	Reserved_31_4	R	0x0	Reserved. Value After Reset: 0x0
			3	RXQ_FRM_ARBIT	R/W	0x0	Receive Queue Packet Arbitration When this bit is set, the DWC_ether_qos drives the packet data to the ARI interface such that the entire packet data of currently-selected queue is transmitted before switching to other queue. When this bit is reset, the DWC_ether_qos drives the packet data to the ARI interface such that the following amount of data of currently-selected queue is transmitted before switching to other queue: PBL amount of data (indicated by ari_qN_pbl_i[]) or Complete data of a packet The status and the timestamp are not a part of the PBL data. Therefore, the DWC_ether_qos drives the complete status (including timestamp status) during first PBL request for the packet (in store-and-forward mode) or the last PBL request for the packet (in Threshold mode).Values: 0x0 (DISABLE): Receive Queue Packet Arbitration is disabled 0x1 (ENABLE): Receive Queue Packet Arbitration is enabled Value After Reset: 0x0
			2:0	RXQ_WEGT	R/W	0x0	Receive Queue Weight This field indicates the weight assigned to the Rx Queue 0. This field needs to be programmed with one value less than the required weight, that is, reset value of 0 indicates weight of 1, value of 1 indicates weight of 2, and so on. The weight is used as the number of continuous PBL or packets requests (depending on the RXQ_FRM_ARBIT) allocated to the queue in one arbitration cycle. Note: The change in value of RXQ_WEGT takes effect only after the completion of current service round or when there is change from RAA=SP to RAA=WSP algorithm. This approach is taken so that there is smooth transition. For the RXQ_WEGT value to take effect at the start, the MTL_RxQ0_Control registers must be programmed before the MTL_Operation_Mode register. Value After Reset: 0x0
			31:22	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			21:16	TQS	R/W	0x0	Transmit Queue Size This field indicates the size of the allocated Transmit queues in blocks of 256 bytes, the reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on. So, program TQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (TQS+1)*256 bytes. Value After Reset: 0x0
			15:7	Reserved_15_7	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xd40	MTL_TxQ1_Operation_Mode	The Queue 1 Transmit Operation Mode register establishes the Transmit queue operating modes and commands.	6:4	TTC	R/W	0x0	Transmit Threshold Control These bits control the threshold level of the MTL Tx Queue. The transmission starts when the packet size within the MTL Tx Queue is larger than the threshold. In addition, full packets with length less than the threshold are also transmitted. These bits are used only when the TSF bit is reset.Values: 0x0 (M_32BYTES): 32 0x1 (M_64BYTES): 64 0x2 (M_96BYTES): 96 0x3 (M_128BYTES): 128 0x4 (M_192BYTES): 192 0x5 (M_256BYTES): 256 0x6 (M_384BYTES): 384 0x7 (M_512BYTES): 512 Value After Reset: 0x0
			3:2	TXQEN	R/W	0x0	Transmit Queue Enable This field is used to enable/disable the transmit queue 1. Note: In multiple Tx queues configuration, all the queues are disabled by default. Enable the Tx queue by programming this field.Values: 0x0 (DISABLE): Not enabled 0x1 (EN_IF_AV): Enable in AV mode 0x2 (ENABLE): Enabled 0x3 (RSVD2): Reserved Value After Reset: 0x0
			1	TSF	R/W	0x0	Transmit Store and Forward When this bit is set, the transmission starts when a full packet resides in the MTL Tx queue. When this bit is set, the TTC values specified in Bits[6:4] of this register are ignored. This bit should be changed only when the transmission is stopped.Values: 0x0 (DISABLE): Transmit Store and Forward is disabled 0x1 (ENABLE): Transmit Store and Forward is enabled Value After Reset: 0x0
			0	FTQ	R/W	0x0	Flush Transmit Queue When this bit is set, the Tx queue controller logic is reset to its default values. Therefore, all the data in the Tx queue is lost or flushed. This bit is internally reset when the flushing operation is complete. Until this bit is reset, you must not write to the MTL_TxQ1_Operation_Mode register. The data which is already accepted by the MAC transmitter is not flushed. It is scheduled for transmission and results in underflow and runt packet transmission. Note: The flush operation is complete only when the Tx queue is empty and the application has accepted the pending Tx Status of all transmitted packets. To complete this flush operation, the PHY Tx clock (clk_tx_i) must be active. Access restrictions apply. MAC initiates the Tx queue flush when application writes 1. MAC writes 0 after the Tx queue flush is complete. MAC ignores when application writes 0. Values: 0x0 (DISABLE): Flush Transmit Queue is disabled 0x1 (ENABLE): Flush Transmit Queue is enabled Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xd44	MTL_TxQ1_Underflow	The Queue 1 Underflow Counter register contains the counter for packets aborted because of Transmit queue underflow and packets missed because of Receive queue packet flush	31:12	Reserved_31_12	R	0x0	Reserved. Value After Reset: 0x0
			11	UFCNTOVF	R	0x0	Overflow Bit for Underflow Packet Counter This bit is set every time the Tx queue Underflow Packet Counter field overflows, that is, it has crossed the maximum count. In such a scenario, the overflow packet counter is reset to all-zeros and this bit indicates that the rollover happened. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Overflow not detected for Underflow Packet Counter 0x1 (ACTIVE): Overflow detected for Underflow Packet Counter Value After Reset: 0x0
			10:0	UFFRMCNT	R	0x0	Underflow Packet Counter This field indicates the number of packets aborted by the controller because of Tx Queue Underflow. This counter is incremented each time the MAC aborts outgoing packet because of underflow. The counter is cleared when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0
0xd48	MTL_TxQ1_Debug	The Queue 1 Transmit Debug register gives the debug status of various blocks related to the Transmit queue.	31:23	Reserved_31_23	R	0x0	Reserved. Value After Reset: 0x0
			22:20	STXSTSF	R	0x0	Number of Status Words in Tx Status FIFO of Queue This field indicates the current number of status in the Tx Status FIFO of this queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of status words in Tx Status FIFO. Value After Reset: 0x0
			19	Reserved_19	R	0x0	Reserved. Value After Reset: 0x0
			18:16	PTXQ	R	0x0	Number of Packets in the Transmit Queue This field indicates the current number of packets in the Tx Queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of packets in the Transmit queue. Value After Reset: 0x0
			15:6	Reserved_15_6	R	0x0	Reserved. Value After Reset: 0x0
			5	TXSTSFSTS	R	0x0	MTL Tx Status FIFO Full Status When high, this bit indicates that the MTL Tx Status FIFO is full. Therefore, the MTL cannot accept any more packets for transmission.Values: 0x0 (INACTIVE): MTL Tx Status FIFO Full status is not detected 0x1 (ACTIVE): MTL Tx Status FIFO Full status is detected Value After Reset: 0x0
			4	TXQSTS	R	0x0	MTL Tx Queue Not Empty Status When this bit is high, it indicates that the MTL Tx Queue is not empty and some data is left for transmission.Values: 0x0 (INACTIVE): MTL Tx Queue Not Empty status is not detected 0x1 (ACTIVE): MTL Tx Queue Not Empty status is detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			3	TWCSTS	R	0x0	MTL Tx Queue Write Controller Status When high, this bit indicates that the MTL Tx Queue Write Controller is active, and it is transferring the data to the Tx Queue.Values: 0x0 (INACTIVE): MTL Tx Queue Write Controller status is not detected 0x1 (ACTIVE): MTL Tx Queue Write Controller status is detected Value After Reset: 0x0
			2:1	TRCSTS	R	0x0	MTL Tx Queue Read Controller Status This field indicates the state of the Tx Queue Read Controller:Values: 0x0 (IDLE): Idle state 0x1 (READ): Read state (transferring data to the MAC transmitter) 0x2 (WAIT): Waiting for pending Tx Status from the MAC transmitter 0x3 (FLUSH): Flushing the Tx queue because of the Packet Abort request from the MAC Value After Reset: 0x0
			0	TXQPAUSED	R	0x0	Transmit Queue in Pause When this bit is high and the Rx flow control is enabled, it indicates that the Tx Queue is in the Pause condition (in the full-duplex only mode) because of the following: Reception of the PFC packet for the priorities assigned to the Tx Queue when PFC is enabled Reception of 802.3x Pause packet when PFC is disabled Values: 0x0 (INACTIVE): Transmit Queue in Pause status is not detected 0x1 (ACTIVE): Transmit Queue in Pause status is detected Value After Reset: 0x0
0xd50	MTL_TxQ1_ETS_Control	The Queue ETS Control register controls the enhanced transmission selection operation.	31:7	Reserved_31_7	R	0x0	Reserved. Value After Reset: 0x0
			6:4	SLC	R/W	0x0	Slot Count If the credit-based shaper algorithm is enabled, the software can program the number of slots (of duration programmed in DMA_CH1_Slot_Interval register) over which the average transmitted bits per slot, provided in the MTL_TxQ1_ETS_Status register, need to be computed for Queue. The encoding is as follows:Values: 0x0 (M_1_SLOT): 1 slot 0x1 (M_2_SLOT): 2 slots 0x2 (M_4_SLOT): 4 slots 0x3 (M_8_SLOT): 8 slots 0x4 (M_16_SLOT): 16 slots 0x5 (RSVD): Reserved Value After Reset: 0x0
			3	CC	R/W	0x0	Credit Control When this bit is set, the accumulated credit parameter in the credit-based shaper algorithm logic is not reset to zero when there is positive credit and no packet to transmit in Tx Queue 1. The credit accumulates even when there is no packet waiting in Tx Queue 1 and another Tx Queue is transmitting. When this bit is reset, the accumulated credit parameter in the credit-based shaper algorithm logic is set to zero when there is positive credit and no packet to transmit in Tx Queue 1. When there is no packet waiting in Tx Queue 1 and other channel is transmitting, no credit is accumulated.Values: 0x0 (DISABLE): Credit Control is disabled 0x1 (ENABLE): Credit Control is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			2	AVALG	R/W	0x0	AV Algorithm When Queue 1 is programmed for AV, this field configures the scheduling algorithm for this queue: This bit when set, indicates credit based shaper algorithm (CBS) is selected for Queue 1 traffic. When reset, strict priority is selected.Values: 0x0 (DISABLE): CBS Algorithm is disabled 0x1 (ENABLE): CBS Algorithm is enabled Value After Reset: 0x0
			1:0	Reserved_1_0	R	0x0	Reserved. Value After Reset: 0x0
0xd54	MTL_TxQ1_ETS_Status	The Queue 1 ETS Status register provides the average traffic transmitted in Queue 1.	31:24	Reserved_31_24	R	0x0	Reserved. Value After Reset: 0x0
			23:0	ABS	R	0x0	Average Bits per Slot This field contains the average transmitted bits per slot. If AV operation is enabled, this field is computed over number of slots, programmed in the SLC field of MTL_TxQ1_ETS_CONTROL register. The maximum value of this field is 0x6_4000 in 100 Mbps, 0x3E_8000 in 1000 Mbps and 9C_4000 in 2500 Mbps mode respectively. When the DCB operation is enabled for Queue, this field is computed over every 10 million bit times slot (4 ms in 2500 Mbps; 10 ms in 1000 Mbps; 100 ms in 100 Mbps). The maximum value is 0x989680. Value After Reset: 0x0
0xd58	MTL_TxQ1_Quantum_Weight	The Queue 1 idleSlopeCredit, Quantum or Weights register provides the average traffic transmitted in Queue 1.	31:21	Reserved_31_21	R	0x0	Reserved. Value After Reset: 0x0
			20:0	ISCQW	R/W	0x0	idleSlopeCredit, Quantum or Weights idleSlopeCredit When AV feature is enabled, this field contains the idleSlopeCredit value required for the credit-based shaper algorithm for Queue 1. This is the rate of change of credit in bits per cycle (40 ns for 100 Mbps; 8 ns for 1000 Mbps; 3.2 ns for 2500 Mbps) when the credit is increasing. The software should program this field with computed credit in bits per cycle scaled by 1,024. The maximum value is portTransmitRate, that is, 0x2000 in 1000/2500 Mbps mode and 0x1000 in 100 Mbps mode. Bits[20:14] must be written to zero. Quantum When the DCB operation is enabled with DWRR algorithm for Queue 1 traffic, this field contains the quantum value in bytes to be added to credit during every queue scanning cycle. The maximum value is 0x1312D0 bytes. Weights When DCB operation is enabled with WFQ algorithm for Queue 1 traffic, this field contains the weight for this queue. The maximum value is 0x3FFF where weight of 0 indicates 100% bandwidth. Bits[20:14] must be written to zero. When DCB operation or generic queuing operation is enabled with WRR algorithm for Queue 1 traffic, this field contains the weight for this queue. The maximum value is 0x64. Bits [20:7] must be written to zero. Note 1: In multiple Queue configuration this field in respective per queue register must be programmed to some non-zero value when multiple queues are enabled or single queue other than Q0 is enabled. This field need not be programmed when only Q0 is enabled. In general, when WRR algorithm is selected a non-zero value must be programmed on both Receive and Transmit. In Receive, the register is MTL_Operation_Mode register. Note 2: For WFQ algorithm, higher the programmed weights lesser the bandwidth allocated for that Transmit Queue. The finish time is not a function of particular packet alone but it is as per the formula: (previous_finish_time of particular Transmit Queue + (weights*packet_size)) Note 3: The weights programmed do not correspond to the number of packets but the fraction of bandwidth or time allocated for particular queue w.r.t. total BW or time. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xd5c	MTL_TxQ1_SendSlopeCredit	The sendSlopeCredit register contains the sendSlope credit value required for the credit-based shaper algorithm for the Queue.	31:14	Reserved_31_14	R	0x0	Reserved. Value After Reset: 0x0
			13:0	SSC	R/W	0x0	sendSlopeCredit Value When AV operation is enabled, this field contains the sendSlopeCredit value required for credit-based shaper algorithm for Tx Queue 1. This is the rate of change of credit in bits per cycle (40 ns, 8 ns and 3.2 ns for 100 Mbps, 1000 Mbps and 2500 Mbps respectively) when the credit is decreasing. The software should program this field with computed credit in bits per cycle scaled by 1,024. The maximum value is portTransmitRate, that is, 0x2000 in 1000/2500 Mbps mode and 0x1000 in 100 Mbps mode. This field should be programmed with absolute sendSlopeCredit value. The credit-based shaper logic subtracts it from the accumulated credit when Tx Queue 1 is selected for transmission. Value After Reset: 0x0
0xd60	MTL_TxQ1_HiCredit	The hiCredit register contains the hiCredit value required for the credit-based shaper algorithm for the Queue.	31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28:0	HC	R/W	0x0	hiCredit Value When the AV feature is enabled, this field contains the hiCredit value required for the credit-based shaper algorithm. This is the maximum value that can be accumulated in the credit parameter. This is specified in bits scaled by 1,024. The maximum value is maxInterferenceSize, that is, best-effort maximum packet size (16,384 bytes or 131,072 bits). The value to be specified is 131,072 * 1,024 = 134,217,728 or 0x0800_0000. Value After Reset: 0x0
0xd64	MTL_TxQ1_LoCredit	The loCredit register contains the loCredit value required for the credit-based shaper algorithm for the Queue.	31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28:0	LC	R/W	0x0	loCredit Value When AV operation is enabled, this field contains the loCredit value required for the credit-based shaper algorithm. This is the minimum value that can be accumulated in the credit parameter. This is specified in bits scaled by 1,024. The maximum value to be programmed is corresponds to twice the maxFrameSize transmitted from this queue. If the maxFrameSize is 8192 bytes, then (8192*2) * 8 * 1024 = 134,217,728 or 0x0800_0000. Because it is a negative value, the programmed value is 2's complement of the value, that is, 0x1800_0000. Value After Reset: 0x0
			31:25	Reserved_31_25	R	0x0	Reserved. Value After Reset: 0x0
			24	RXOIE	R/W	0x0	Receive Queue Overflow Interrupt Enable When this bit is set, the Receive Queue Overflow interrupt is enabled. When this bit is reset, the Receive Queue Overflow interrupt is disabled.Values: 0x0 (DISABLE): Receive Queue Overflow Interrupt is disabled 0x1 (ENABLE): Receive Queue Overflow Interrupt is enabled Value After Reset: 0x0
			23:17	Reserved_23_17	R	0x0	Reserved. Value After Reset: 0x0
			16	RXOVFIS	R/W	0x0	Receive Queue Overflow Interrupt Status This bit indicates that the Receive Queue had an overflow while receiving the packet. If a partial packet is transferred to the application, the overflow status is set in RDES3[21]. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Queue Overflow Interrupt Status not detected 0x1 (ACTIVE): Receive Queue Overflow Interrupt Status detected Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xd6c	MTL_Q1_Interrupt_Control_Status	This register contains the interrupt enable and status bits for the queue 1 interrupts.	15:10	Reserved_15_10	R	0x0	Reserved. Value After Reset: 0x0
			9	ABPSIE	R/W	0x0	Average Bits Per Slot Interrupt Enable When this bit is set, the MAC asserts the sbd_intr_o or mci_intr_o interrupt when the average bits per slot status is updated. When this bit is cleared, the interrupt is not asserted for such an event.Values: 0x0 (DISABLE): Average Bits Per Slot Interrupt is disabled 0x1 (ENABLE): Average Bits Per Slot Interrupt is enabled Value After Reset: 0x0
			8	TXUIE	R/W	0x0	Transmit Queue Underflow Interrupt Enable When this bit is set, the Transmit Queue Underflow interrupt is enabled. When this bit is reset, the Transmit Queue Underflow interrupt is disabled.Values: 0x0 (DISABLE): Transmit Queue Underflow Interrupt Status is disabled 0x1 (ENABLE): Transmit Queue Underflow Interrupt Status is enabled Value After Reset: 0x0
			7:2	Reserved_7_2	R	0x0	Reserved. Value After Reset: 0x0
			1	ABPSIS	R/W	0x0	Average Bits Per Slot Interrupt Status When set, this bit indicates that the MAC has updated the ABS value. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Average Bits Per Slot Interrupt Status not detected 0x1 (ACTIVE): Average Bits Per Slot Interrupt Status detected Value After Reset: 0x0 Testable: untestable
			0	TXUNFIS	R/W	0x0	Transmit Queue Underflow Interrupt Status This bit indicates that the Transmit Queue had an underflow while transmitting the packet. Transmission is suspended and an Underflow Error TDES3[2] is set. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Queue Underflow Interrupt Status not detected 0x1 (ACTIVE): Transmit Queue Underflow Interrupt Status detected Value After Reset: 0x0 Testable: untestable
			31:26	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			25:20	RQS	R/W	0x0	Receive Queue Size This field indicates the size of the allocated Receive queues in blocks of 256 bytesthe reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on. So, program RQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (RQS+1)*256 bytes. Value After Reset: 0x0
			19	Reserved_19_y	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xd70	MTL_RxQ1_Operation_Mode	The Queue 1 Receive Operation Mode register establishes the Receive queue operating modes and command.	18:14	RFD	R/W	0x0	Threshold for Deactivating Flow Control (in half-duplex and full-duplex modes) These bits control the threshold (fill-level of Rx queue) at which the flow control is de-asserted after activation: 0: Full minus 1 KB, that is, FULL 1 KB 1: Full minus 1.5 KB, that is, FULL 1.5 KB 2: Full minus 2 KB, that is, FULL 2 KB 3: Full minus 2.5 KB, that is, FULL 2.5 KB ... 30: Full minus 16 KB, that is, FULL 16 KB 31: Full minus 16.5 KB, that is, FULL 16.5 KB The de-assertion is effective only after flow control is asserted. Note: The value must be programmed in such a way to make sure that the threshold is a positive number. When the EHFC is set high, these values are applicable only when the Rx queue size determined by the RQS field of this register, is equal to or greater than 4 KB. For a given queue size, the values ranges between 0 and the encoding for FULL minus (QSIZE - 0.5 KB) and all other values are illegal. Here the term FULL and QSIZE refers to the queue size determined by the RQS field of this register. Value After Reset: 0x0
			13	Reserved_13_y	R	0x0	Reserved. Value After Reset: 0x0
			12:8	RFA	R/W	0x0	Threshold for Activating Flow Control (in half-duplex and full-duplex These bits control the threshold (fill-level of Rx queue) at which the flow control is activated: For more information on encoding for this field, see RFD. Value After Reset: 0x0
			7	EHFC	R/W	0x0	Enable Hardware Flow Control When this bit is set, the flow control signal operation, based on the fill-level of Rx queue, is enabled. When reset, the flow control operation is disabled.Values: 0x0 (DISABLE): Hardware Flow Control is disabled 0x1 (ENABLE): Hardware Flow Control is enabled Value After Reset: 0x0
			6	DIS_TCP_EF	R/W	0x0	Disable Dropping of TCP/IP Checksum Error Packets When this bit is set, the MAC does not drop the packets which only have the errors detected by the Receive Checksum Offload engine. Such packets have errors only in the encapsulated payload. There are no errors (including FCS error) in the Ethernet packet received by the MAC. When this bit is reset, all error packets are dropped if the FEP bit is reset.Values: 0x0 (ENABLE): Dropping of TCP/IP Checksum Error Packets is enabled 0x1 (DISABLE): Dropping of TCP/IP Checksum Error Packets is disabled Value After Reset: 0x0
			5	RSF	R/W	0x0	Receive Queue Store and Forward When this bit is set, the DWC_ether_qos reads a packet from the Rx queue only after the complete packet has been written to it, ignoring the RTC field of this register. When this bit is reset, the Rx queue operates in the Threshold (cut-through) mode, subject to the threshold specified by the RTC field of this register.Values: 0x0 (DISABLE): Receive Queue Store and Forward is disabled 0x1 (ENABLE): Receive Queue Store and Forward is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			4	FEP	R/W	0x0	Forward Error Packets When this bit is reset, the Rx queue drops packets with error status (CRC error, GMII_ER, watchdog timeout, or overflow). However, if the start byte (write) pointer of a packet is already transferred to the read controller side (in Threshold mode), the packet is not dropped. When this bit is set, all packets except the runt error packets are forwarded to the application or DMA. If the RSF bit is set and the Rx queue overflows when a partial packet is written, the packet is dropped irrespective of the setting of this bit. However, if the RSF bit is reset and the Rx queue overflows when a partial packet is written, a partial packet might be forwarded to the application or DMA.Values: 0x0 (DISABLE): Forward Error Packets is disabled 0x1 (ENABLE): Forward Error Packets is enabled Value After Reset: 0x0
			3	FUP	R/W	0x0	Forward Undersized Good Packets When this bit is set, the Rx queue forwards the undersized good packets (packets with no error and length less than 64 bytes), including pad-bytes and CRC. When this bit is reset, the Rx queue drops all packets of less than 64 bytes, unless a packet is already transferred because of the lower value of Rx Threshold, for example, RTC = 01.Values: 0x0 (DISABLE): Forward Undersized Good Packets is disabled 0x1 (ENABLE): Forward Undersized Good Packets is enabled Value After Reset: 0x0
			2	Reserved_2	R	0x0	Reserved. Value After Reset: 0x0
			1:0	RTC	R/W	0x0	Receive Queue Threshold Control These bits control the threshold level of the MTL Rx queue (in bytes): The received packet is transferred to the application or DMA when the packet size within the MTL Rx queue is larger than the threshold. In addition, full packets with length less than the threshold are automatically transferred. This field is valid only when the RSF bit is zero. This field is ignored when the RSF bit is set to 1.Values: 0x0 (M_64BYTE): 64 0x1 (M_32BYTE): 32 0x2 (M_96BYTE): 96 0x3 (M_128BYTE): 128 Value After Reset: 0x0
0xd74	MTL_RxQ1_Missed_Packet_Overflow_Cnt	The Queue 1 Missed Packet and Overflow Counter register contains the counter for packets missed because of Receive queue packet flush and packets discarded because of Receive queue overflow.	31:28	Reserved_31_28	R	0x0	Reserved. Value After Reset: 0x0
			27	MISCNTOVF	R	0x0	Missed Packet Counter Overflow Bit When set, this bit indicates that the Rx Queue Missed Packet Counter crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Missed Packet Counter overflow not detected 0x1 (ACTIVE): Missed Packet Counter overflow detected Value After Reset: 0x0
			26:16	MISPKTCNT	R	0x0	Missed Packet Counter This field indicates the number of packets missed by the DWC_ether_qos because the application asserted ari_pkt_flush_i[] for this queue. This counter is reset when this register is read with mci_be_i[0] at 1b1. This counter is incremented by 1 when the DMA discards the packet because of buffer unavailability. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0
			15:12	Reserved_15_12	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			11	OVFCNTOVF	R	0x0	Overflow Counter Overflow Bit When set, this bit indicates that the Rx Queue Overflow Packet Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Overflow Counter overflow not detected 0x1 (ACTIVE): Overflow Counter overflow detected Value After Reset: 0x0
			10:0	OVFPKTCNT	R	0x0	Overflow Packet Counter This field indicates the number of packets discarded by the DWC_ether_qos because of Receive queue overflow. This counter is incremented each time the DWC_ether_qos discards an incoming packet because of overflow. This counter is reset when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0
0xd78	MTL_RxQ1_Debug	The Queue 1 Receive Debug register gives the debug status of various blocks related to the Receive queue.	31:30	Reserved_31_30	R	0x0	Reserved. Value After Reset: 0x0
			29:16	PRXQ	R	0x0	Number of Packets in Receive Queue This field indicates the current number of packets in the Rx Queue, that is, Max_Queue_Size/Min_Packet_Size. Value After Reset: 0x0
			15:6	Reserved_15_6	R	0x0	Reserved. Value After Reset: 0x0
			5:4	RXQSTS	R	0x0	MTL Rx Queue Fill-Level Status This field gives the status of the fill-level of the Rx Queue:Values: 0x0 (EMPTY): Rx Queue empty 0x1 (BLW_THR): Rx Queue fill-level below flow-control deactivate threshold 0x2 (ABV_THR): Rx Queue fill-level above flow-control activate threshold 0x3 (FULL): Rx Queue full Value After Reset: 0x0
			3	Reserved_3	R	0x0	Reserved. Value After Reset: 0x0
			2:1	RRCSTS	R	0x0	MTL Rx Queue Read Controller State This field gives the state of the Rx queue Read controller:Values: 0x0 (IDLE): Idle state 0x1 (READ_DATA): Reading packet data 0x2 (READ_STS): Reading packet status (or timestamp) 0x3 (FLUSH): Flushing the packet data and status Value After Reset: 0x0
			0	RWCSTS	R	0x0	MTL Rx Queue Write Controller Active Status When high, this bit indicates that the MTL Rx queue Write controller is active, and it is transferring a received packet to the Rx Queue.Values: 0x0 (INACTIVE): MTL Rx Queue Write Controller Active Status not detected 0x1 (ACTIVE): MTL Rx Queue Write Controller Active Status detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xd7c	MTL_RxQ1_Control	The Queue Receive Control register controls the receive arbitration and passing of received packets to the application.	31:4	Reserved_31_4	R	0x0	Reserved. Value After Reset: 0x0
			3	RXQ_FRM_ARBIT	R/W	0x0	Receive Queue Packet Arbitration When this bit is set, the DWC_ether_qos drives the packet data to the ARI interface such that the entire packet data of currently-selected queue is transmitted before switching to other queue. When this bit is reset, the DWC_ether_qos drives the packet data to the ARI interface such that the following amount of data of currently-selected queue is transmitted before switching to other queue: PBL amount of data (indicated by ari_qN_pbl_i[]) or Complete data of a packet The status and the timestamp are not a part of the PBL data. Therefore, the DWC_ether_qos drives the complete status (including timestamp status) during first PBL request for the packet (in store-and-forward mode) or the last PBL request for the packet (in Threshold mode).Values: 0x0 (DISABLE): Receive Queue Packet Arbitration is disabled 0x1 (ENABLE): Receive Queue Packet Arbitration is enabled Value After Reset: 0x0
			2:0	RXQ_WEGT	R/W	0x0	Receive Queue Weight This field indicates the weight assigned to the Rx Queue 1. This field needs to be programmed with one value less than the required weight, that is, reset value of 0 indicates weight of 1, value of 1 indicates weight of 2, and so on. The weight is used as the number of continuous PBL or packets requests (depending on the RXQ_FRM_ARBIT) allocated to the queue in one arbitration cycle. Note: The change in value of RXQ_WEGT takes effect only after the completion of current service round or when there is change from RAA=SP to RAA=WSP algorithm. This approach is taken so that there is smooth transition. For the RXQ_WEGT value to take effect at the start, the MTL_RxQ1_Control registers must be programmed before the MTL_Operation_Mode register. Value After Reset: 0x0
			31:22	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			21:16	TQS	R/W	0x0	Transmit Queue Size This field indicates the size of the allocated Transmit queues in blocks of 256 bytes, the reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on. So, program TQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (TQS+1)*256 bytes. Value After Reset: 0x0
			15:7	Reserved_15_7	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xd80	MTL_TxQ2_Operation_Mode	The Queue 2 Transmit Operation Mode register establishes the Transmit queue operating modes and commands.	6:4	TTC	R/W	0x0	Transmit Threshold Control These bits control the threshold level of the MTL Tx Queue. The transmission starts when the packet size within the MTL Tx Queue is larger than the threshold. In addition, full packets with length less than the threshold are also transmitted. These bits are used only when the TSF bit is reset.Values: 0x0 (M_32BYTES): 32 0x1 (M_64BYTES): 64 0x2 (M_96BYTES): 96 0x3 (M_128BYTES): 128 0x4 (M_192BYTES): 192 0x5 (M_256BYTES): 256 0x6 (M_384BYTES): 384 0x7 (M_512BYTES): 512 Value After Reset: 0x0
			3:2	TXQEN	R/W	0x0	Transmit Queue Enable This field is used to enable/disable the transmit queue 1. Note: In multiple Tx queues configuration, all the queues are disabled by default. Enable the Tx queue by programming this field.Values: 0x0 (DISABLE): Not enabled 0x1 (EN_IF_AV): Enable in AV mode 0x2 (ENABLE): Enabled 0x3 (RSVD2): Reserved Value After Reset: 0x0
			1	TSF	R/W	0x0	Transmit Store and Forward When this bit is set, the transmission starts when a full packet resides in the MTL Tx queue. When this bit is set, the TTC values specified in Bits[6:4] of this register are ignored. This bit should be changed only when the transmission is stopped.Values: 0x0 (DISABLE): Transmit Store and Forward is disabled 0x1 (ENABLE): Transmit Store and Forward is enabled Value After Reset: 0x0
			0	FTQ	R/W	0x0	Flush Transmit Queue When this bit is set, the Tx queue controller logic is reset to its default values. Therefore, all the data in the Tx queue is lost or flushed. This bit is internally reset when the flushing operation is complete. Until this bit is reset, you must not write to the MTL_TxQ1_Operation_Mode register. The data which is already accepted by the MAC transmitter is not flushed. It is scheduled for transmission and results in underflow and runt packet transmission. Note: The flush operation is complete only when the Tx queue is empty and the application has accepted the pending Tx Status of all transmitted packets. To complete this flush operation, the PHY Tx clock (clk_tx_i) must be active. Access restrictions apply. MAC initiates the Tx queue flush when application writes 1. MAC writes 0 after the Tx queue flush is complete. MAC ignores when application writes 0. Values: 0x0 (DISABLE): Flush Transmit Queue is disabled 0x1 (ENABLE): Flush Transmit Queue is enabled Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xd84	MTL_TxQ2_Underflow	The Queue 2 Underflow Counter register contains the counter for packets aborted because of Transmit queue underflow and packets missed because of Receive queue packet flush	31:12	Reserved_31_12	R	0x0	Reserved. Value After Reset: 0x0
			11	UFCNTOVF	R	0x0	Overflow Bit for Underflow Packet Counter This bit is set every time the Tx queue Underflow Packet Counter field overflows, that is, it has crossed the maximum count. In such a scenario, the overflow packet counter is reset to all-zeros and this bit indicates that the rollover happened. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Overflow not detected for Underflow Packet Counter 0x1 (ACTIVE): Overflow detected for Underflow Packet Counter Value After Reset: 0x0
			10:0	UFFRMCNT	R	0x0	Underflow Packet Counter This field indicates the number of packets aborted by the controller because of Tx Queue Underflow. This counter is incremented each time the MAC aborts outgoing packet because of underflow. The counter is cleared when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0
0xd88	MTL_TxQ2_Debug	The Queue 2 Transmit Debug register gives the debug status of various blocks related to the Transmit queue.	31:23	Reserved_31_23	R	0x0	Reserved. Value After Reset: 0x0
			22:20	STXSTSF	R	0x0	Number of Status Words in Tx Status FIFO of Queue This field indicates the current number of status in the Tx Status FIFO of this queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of status words in Tx Status FIFO. Value After Reset: 0x0
			19	Reserved_19	R	0x0	Reserved. Value After Reset: 0x0
			18:16	PTXQ	R	0x0	Number of Packets in the Transmit Queue This field indicates the current number of packets in the Tx Queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of packets in the Transmit queue. Value After Reset: 0x0
			15:6	Reserved_15_6	R	0x0	Reserved. Value After Reset: 0x0
			5	TXSTSFSTS	R	0x0	MTL Tx Status FIFO Full Status When high, this bit indicates that the MTL Tx Status FIFO is full. Therefore, the MTL cannot accept any more packets for transmission.Values: 0x0 (INACTIVE): MTL Tx Status FIFO Full status is not detected 0x1 (ACTIVE): MTL Tx Status FIFO Full status is detected Value After Reset: 0x0
			4	TXQSTS	R	0x0	MTL Tx Queue Not Empty Status When this bit is high, it indicates that the MTL Tx Queue is not empty and some data is left for transmission.Values: 0x0 (INACTIVE): MTL Tx Queue Not Empty status is not detected 0x1 (ACTIVE): MTL Tx Queue Not Empty status is detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			3	TWCSTS	R	0x0	MTL Tx Queue Write Controller Status When high, this bit indicates that the MTL Tx Queue Write Controller is active, and it is transferring the data to the Tx Queue.Values: 0x0 (INACTIVE): MTL Tx Queue Write Controller status is not detected 0x1 (ACTIVE): MTL Tx Queue Write Controller status is detected Value After Reset: 0x0
			2:1	TRCSTS	R	0x0	MTL Tx Queue Read Controller Status This field indicates the state of the Tx Queue Read Controller:Values: 0x0 (IDLE): Idle state 0x1 (READ): Read state (transferring data to the MAC transmitter) 0x2 (WAIT): Waiting for pending Tx Status from the MAC transmitter 0x3 (FLUSH): Flushing the Tx queue because of the Packet Abort request from the MAC Value After Reset: 0x0
			0	TXQPAUSED	R	0x0	Transmit Queue in Pause When this bit is high and the Rx flow control is enabled, it indicates that the Tx Queue is in the Pause condition (in the full-duplex only mode) because of the following: Reception of the PFC packet for the priorities assigned to the Tx Queue when PFC is enabled Reception of 802.3x Pause packet when PFC is disabled Values: 0x0 (INACTIVE): Transmit Queue in Pause status is not detected 0x1 (ACTIVE): Transmit Queue in Pause status is detected Value After Reset: 0x0
0xd90	MTL_TxQ2_ETS_Control	The Queue ETS Control register controls the enhanced transmission selection operation.	31:7	Reserved_31_7	R	0x0	Reserved. Value After Reset: 0x0
			6:4	SLC	R/W	0x0	Slot Count If the credit-based shaper algorithm is enabled, the software can program the number of slots (of duration programmed in DMA_CH1_Slot_Interval register) over which the average transmitted bits per slot, provided in the MTL_TxQ1_ETS_Status register, need to be computed for Queue. The encoding is as follows:Values: 0x0 (M_1_SLOT): 1 slot 0x1 (M_2_SLOT): 2 slots 0x2 (M_4_SLOT): 4 slots 0x3 (M_8_SLOT): 8 slots 0x4 (M_16_SLOT): 16 slots 0x5 (RSVD): Reserved Value After Reset: 0x0
			3	CC	R/W	0x0	Credit Control When this bit is set, the accumulated credit parameter in the credit-based shaper algorithm logic is not reset to zero when there is positive credit and no packet to transmit in Tx Queue 1. The credit accumulates even when there is no packet waiting in Tx Queue 1 and another Tx Queue is transmitting. When this bit is reset, the accumulated credit parameter in the credit-based shaper algorithm logic is set to zero when there is positive credit and no packet to transmit in Tx Queue 1. When there is no packet waiting in Tx Queue 1 and other channel is transmitting, no credit is accumulated.Values: 0x0 (DISABLE): Credit Control is disabled 0x1 (ENABLE): Credit Control is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			2	AVALG	R/W	0x0	AV Algorithm When Queue 1 is programmed for AV, this field configures the scheduling algorithm for this queue: This bit when set, indicates credit based shaper algorithm (CBS) is selected for Queue 1 traffic. When reset, strict priority is selected.Values: 0x0 (DISABLE): CBS Algorithm is disabled 0x1 (ENABLE): CBS Algorithm is enabled Value After Reset: 0x0
			1:0	Reserved_1_0	R	0x0	Reserved. Value After Reset: 0x0
0xd94	MTL_TxQ2_ETS_Status	The Queue 2 ETS Status register provides the average traffic transmitted in Queue 2.	31:24	Reserved_31_24	R	0x0	Reserved. Value After Reset: 0x0
			23:0	ABS	R	0x0	Average Bits per Slot This field contains the average transmitted bits per slot. If AV operation is enabled, this field is computed over number of slots, programmed in the SLC field of MTL_TxQ1_ETS_CONTROL register. The maximum value of this field is 0x6_4000 in 100 Mbps, 0x3E_8000 in 1000 Mbps and 9C_4000 in 2500 Mbps mode respectively. When the DCB operation is enabled for Queue, this field is computed over every 10 million bit times slot (4 ms in 2500 Mbps; 10 ms in 1000 Mbps; 100 ms in 100 Mbps). The maximum value is 0x989680. Value After Reset: 0x0
0xd98	MTL_TxQ2_Quantum_Weight	The Queue 2 idleSlopeCredit, Quantum or Weights register provides the average traffic transmitted in Queue 2.	31:21	Reserved_31_21	R	0x0	Reserved. Value After Reset: 0x0
			20:0	ISCQW	R/W	0x0	idleSlopeCredit, Quantum or Weights idleSlopeCredit When AV feature is enabled, this field contains the idleSlopeCredit value required for the credit-based shaper algorithm for Queue 1. This is the rate of change of credit in bits per cycle (40 ns for 100 Mbps; 8 ns for 1000 Mbps; 3.2 ns for 2500 Mbps) when the credit is increasing. The software should program this field with computed credit in bits per cycle scaled by 1,024. The maximum value is portTransmitRate, that is, 0x2000 in 1000/2500 Mbps mode and 0x1000 in 100 Mbps mode. Bits[20:14] must be written to zero. Quantum When the DCB operation is enabled with DWRR algorithm for Queue 1 traffic, this field contains the quantum value in bytes to be added to credit during every queue scanning cycle. The maximum value is 0x1312D0 bytes. Weights When DCB operation is enabled with WFQ algorithm for Queue 1 traffic, this field contains the weight for this queue. The maximum value is 0x3FFF where weight of 0 indicates 100% bandwidth. Bits[20:14] must be written to zero. When DCB operation or generic queuing operation is enabled with WRR algorithm for Queue 1 traffic, this field contains the weight for this queue. The maximum value is 0x64. Bits [20:7] must be written to zero. Note 1: In multiple Queue configuration this field in respective per queue register must be programmed to some non-zero value when multiple queues are enabled or single queue other than Q0 is enabled. This field need not be programmed when only Q0 is enabled. In general, when WRR algorithm is selected a non-zero value must be programmed on both Receive and Transmit. In Receive, the register is MTL_Operation_Mode register. Note 2: For WFQ algorithm, higher the programmed weights lesser the bandwidth allocated for that Transmit Queue. The finish time is not a function of particular packet alone but it is as per the formula: (previous_finish_time of particular Transmit Queue + (weights*packet_size)) Note 3: The weights programmed do not correspond to the number of packets but the fraction of bandwidth or time allocated for particular queue w.r.t. total BW or time. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xd9c	MTL_TxQ2_SendSlopeCredit	The sendSlopeCredit register contains the sendSlope credit value required for the credit-based shaper algorithm for the Queue.	31:14	Reserved_31_14	R	0x0	Reserved. Value After Reset: 0x0
			13:0	SSC	R/W	0x0	sendSlopeCredit Value When AV operation is enabled, this field contains the sendSlopeCredit value required for credit-based shaper algorithm for Tx Queue 1. This is the rate of change of credit in bits per cycle (40 ns, 8 ns and 3.2 ns for 100 Mbps, 1000 Mbps and 2500 Mbps respectively) when the credit is decreasing. The software should program this field with computed credit in bits per cycle scaled by 1,024. The maximum value is portTransmitRate, that is, 0x2000 in 1000/2500 Mbps mode and 0x1000 in 100 Mbps mode. This field should be programmed with absolute sendSlopeCredit value. The credit-based shaper logic subtracts it from the accumulated credit when Tx Queue 1 is selected for transmission. Value After Reset: 0x0
0xda0	MTL_TxQ2_HiCredit	The hiCredit register contains the hiCredit value required for the credit-based shaper algorithm for the Queue.	31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28:0	HC	R/W	0x0	hiCredit Value When the AV feature is enabled, this field contains the hiCredit value required for the credit-based shaper algorithm. This is the maximum value that can be accumulated in the credit parameter. This is specified in bits scaled by 1,024. The maximum value is maxInterferenceSize, that is, best-effort maximum packet size (16,384 bytes or 131,072 bits). The value to be specified is 131,072 * 1,024 = 134,217,728 or 0x0800_0000. Value After Reset: 0x0
0xda4	MTL_TxQ2_LoCredit	The loCredit register contains the loCredit value required for the credit-based shaper algorithm for the Queue.	31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28:0	LC	R/W	0x0	loCredit Value When AV operation is enabled, this field contains the loCredit value required for the credit-based shaper algorithm. This is the minimum value that can be accumulated in the credit parameter. This is specified in bits scaled by 1,024. The maximum value to be programmed is corresponds to twice the maxFrameSize transmitted from this queue. If the maxFrameSize is 8192 bytes, then (8192*2) * 8 * 1024 = 134,217,728 or 0x0800_0000. Because it is a negative value, the programmed value is 2's complement of the value, that is, 0x1800_0000. Value After Reset: 0x0
			31:25	Reserved_31_25	R	0x0	Reserved. Value After Reset: 0x0
			24	RXOIE	R/W	0x0	Receive Queue Overflow Interrupt Enable When this bit is set, the Receive Queue Overflow interrupt is enabled. When this bit is reset, the Receive Queue Overflow interrupt is disabled.Values: 0x0 (DISABLE): Receive Queue Overflow Interrupt is disabled 0x1 (ENABLE): Receive Queue Overflow Interrupt is enabled Value After Reset: 0x0
			23:17	Reserved_23_17	R	0x0	Reserved. Value After Reset: 0x0
			16	RXOVFIS	R/W	0x0	Receive Queue Overflow Interrupt Status This bit indicates that the Receive Queue had an overflow while receiving the packet. If a partial packet is transferred to the application, the overflow status is set in RDES3[21]. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Queue Overflow Interrupt Status not detected 0x1 (ACTIVE): Receive Queue Overflow Interrupt Status detected Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xdac	MTL_Q2_Interrupt_Control_Status	This register contains the interrupt enable and status bits for the queue 2 interrupts.	15:10	Reserved_15_10	R	0x0	Reserved. Value After Reset: 0x0
			9	ABPSIE	R/W	0x0	Average Bits Per Slot Interrupt Enable When this bit is set, the MAC asserts the sbd_intr_o or mci_intr_o interrupt when the average bits per slot status is updated. When this bit is cleared, the interrupt is not asserted for such an event.Values: 0x0 (DISABLE): Average Bits Per Slot Interrupt is disabled 0x1 (ENABLE): Average Bits Per Slot Interrupt is enabled Value After Reset: 0x0
			8	TXUIE	R/W	0x0	Transmit Queue Underflow Interrupt Enable When this bit is set, the Transmit Queue Underflow interrupt is enabled. When this bit is reset, the Transmit Queue Underflow interrupt is disabled.Values: 0x0 (DISABLE): Transmit Queue Underflow Interrupt Status is disabled 0x1 (ENABLE): Transmit Queue Underflow Interrupt Status is enabled Value After Reset: 0x0
			7:2	Reserved_7_2	R	0x0	Reserved. Value After Reset: 0x0
			1	ABPSIS	R/W	0x0	Average Bits Per Slot Interrupt Status When set, this bit indicates that the MAC has updated the ABS value. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Average Bits Per Slot Interrupt Status not detected 0x1 (ACTIVE): Average Bits Per Slot Interrupt Status detected Value After Reset: 0x0 Testable: untestable
			0	TXUNFIS	R/W	0x0	Transmit Queue Underflow Interrupt Status This bit indicates that the Transmit Queue had an underflow while transmitting the packet. Transmission is suspended and an Underflow Error TDES3[2] is set. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Queue Underflow Interrupt Status not detected 0x1 (ACTIVE): Transmit Queue Underflow Interrupt Status detected Value After Reset: 0x0 Testable: untestable
			31:26	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			25:20	RQS	R/W	0x0	Receive Queue Size This field indicates the size of the allocated Receive queues in blocks of 256 bytesthe reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on. So, program RQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (RQS+1)*256 bytes. Value After Reset: 0x0
			19	Reserved_19_y	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xdb0	MTL_RxQ2_Operation_Mode	The Queue 2 Receive Operation Mode register establishes the Receive queue operating modes and command.	18:14	RFD	R/W	0x0	Threshold for Deactivating Flow Control (in half-duplex and full-duplex modes) These bits control the threshold (fill-level of Rx queue) at which the flow control is de-asserted after activation: 0: Full minus 1 KB, that is, FULL 1 KB 1: Full minus 1.5 KB, that is, FULL 1.5 KB 2: Full minus 2 KB, that is, FULL 2 KB 3: Full minus 2.5 KB, that is, FULL 2.5 KB ... 30: Full minus 16 KB, that is, FULL 16 KB 31: Full minus 16.5 KB, that is, FULL 16.5 KB The de-assertion is effective only after flow control is asserted. Note: The value must be programmed in such a way to make sure that the threshold is a positive number. When the EHFC is set high, these values are applicable only when the Rx queue size determined by the RQS field of this register, is equal to or greater than 4 KB. For a given queue size, the values ranges between 0 and the encoding for FULL minus (QSIZE - 0.5 KB) and all other values are illegal. Here the term FULL and QSIZE refers to the queue size determined by the RQS field of this register. Value After Reset: 0x0
			13	Reserved_13_y	R	0x0	Reserved. Value After Reset: 0x0
			12:8	RFA	R/W	0x0	Threshold for Activating Flow Control (in half-duplex and full-duplex These bits control the threshold (fill-level of Rx queue) at which the flow control is activated: For more information on encoding for this field, see RFD. Value After Reset: 0x0
			7	EHFC	R/W	0x0	Enable Hardware Flow Control When this bit is set, the flow control signal operation, based on the fill-level of Rx queue, is enabled. When reset, the flow control operation is disabled.Values: 0x0 (DISABLE): Hardware Flow Control is disabled 0x1 (ENABLE): Hardware Flow Control is enabled Value After Reset: 0x0
			6	DIS_TCP_EF	R/W	0x0	Disable Dropping of TCP/IP Checksum Error Packets When this bit is set, the MAC does not drop the packets which only have the errors detected by the Receive Checksum Offload engine. Such packets have errors only in the encapsulated payload. There are no errors (including FCS error) in the Ethernet packet received by the MAC. When this bit is reset, all error packets are dropped if the FEP bit is reset.Values: 0x0 (ENABLE): Dropping of TCP/IP Checksum Error Packets is enabled 0x1 (DISABLE): Dropping of TCP/IP Checksum Error Packets is disabled Value After Reset: 0x0
			5	RSF	R/W	0x0	Receive Queue Store and Forward When this bit is set, the DWC_ether_qos reads a packet from the Rx queue only after the complete packet has been written to it, ignoring the RTC field of this register. When this bit is reset, the Rx queue operates in the Threshold (cut-through) mode, subject to the threshold specified by the RTC field of this register.Values: 0x0 (DISABLE): Receive Queue Store and Forward is disabled 0x1 (ENABLE): Receive Queue Store and Forward is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			4	FEP	R/W	0x0	Forward Error Packets When this bit is reset, the Rx queue drops packets with error status (CRC error, GMII_ER, watchdog timeout, or overflow). However, if the start byte (write) pointer of a packet is already transferred to the read controller side (in Threshold mode), the packet is not dropped. When this bit is set, all packets except the runt error packets are forwarded to the application or DMA. If the RSF bit is set and the Rx queue overflows when a partial packet is written, the packet is dropped irrespective of the setting of this bit. However, if the RSF bit is reset and the Rx queue overflows when a partial packet is written, a partial packet might be forwarded to the application or DMA.Values: 0x0 (DISABLE): Forward Error Packets is disabled 0x1 (ENABLE): Forward Error Packets is enabled Value After Reset: 0x0
			3	FUP	R/W	0x0	Forward Undersized Good Packets When this bit is set, the Rx queue forwards the undersized good packets (packets with no error and length less than 64 bytes), including pad-bytes and CRC. When this bit is reset, the Rx queue drops all packets of less than 64 bytes, unless a packet is already transferred because of the lower value of Rx Threshold, for example, RTC = 01.Values: 0x0 (DISABLE): Forward Undersized Good Packets is disabled 0x1 (ENABLE): Forward Undersized Good Packets is enabled Value After Reset: 0x0
			2	Reserved_2	R	0x0	Reserved. Value After Reset: 0x0
			1:0	RTC	R/W	0x0	Receive Queue Threshold Control These bits control the threshold level of the MTL Rx queue (in bytes): The received packet is transferred to the application or DMA when the packet size within the MTL Rx queue is larger than the threshold. In addition, full packets with length less than the threshold are automatically transferred. This field is valid only when the RSF bit is zero. This field is ignored when the RSF bit is set to 1.Values: 0x0 (M_64BYTE): 64 0x1 (M_32BYTE): 32 0x2 (M_96BYTE): 96 0x3 (M_128BYTE): 128 Value After Reset: 0x0
0xdb4	MTL_RxQ2_Missed_Packet_Overflow_Cnt	The Queue 2 Missed Packet and Overflow Counter register contains the counter for packets missed because of Receive queue packet flush and packets discarded because of Receive queue overflow.	31:28	Reserved_31_28	R	0x0	Reserved. Value After Reset: 0x0
			27	MISCNTOVF	R	0x0	Missed Packet Counter Overflow Bit When set, this bit indicates that the Rx Queue Missed Packet Counter crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Missed Packet Counter overflow not detected 0x1 (ACTIVE): Missed Packet Counter overflow detected Value After Reset: 0x0
			26:16	MISPKTCNT	R	0x0	Missed Packet Counter This field indicates the number of packets missed by the DWC_ether_qos because the application asserted ari_pkt_flush_i[] for this queue. This counter is reset when this register is read with mci_be_i[0] at 1b1. This counter is incremented by 1 when the DMA discards the packet because of buffer unavailability. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0
			15:12	Reserved_15_12	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			11	OVFCNTOVF	R	0x0	Overflow Counter Overflow Bit When set, this bit indicates that the Rx Queue Overflow Packet Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Overflow Counter overflow not detected 0x1 (ACTIVE): Overflow Counter overflow detected Value After Reset: 0x0
			10:0	OVFPKTCNT	R	0x0	Overflow Packet Counter This field indicates the number of packets discarded by the DWC_ether_qos because of Receive queue overflow. This counter is incremented each time the DWC_ether_qos discards an incoming packet because of overflow. This counter is reset when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0
0xdb8	MTL_RxQ2_Debug	The Queue 2 Receive Debug register gives the debug status of various blocks related to the Receive queue.	31:30	Reserved_31_30	R	0x0	Reserved. Value After Reset: 0x0
			29:16	PRXQ	R	0x0	Number of Packets in Receive Queue This field indicates the current number of packets in the Rx Queue, that is, Max_Queue_Size/Min_Packet_Size. Value After Reset: 0x0
			15:6	Reserved_15_6	R	0x0	Reserved. Value After Reset: 0x0
			5:4	RXQSTS	R	0x0	MTL Rx Queue Fill-Level Status This field gives the status of the fill-level of the Rx Queue:Values: 0x0 (EMPTY): Rx Queue empty 0x1 (BLW_THR): Rx Queue fill-level below flow-control deactivate threshold 0x2 (ABV_THR): Rx Queue fill-level above flow-control activate threshold 0x3 (FULL): Rx Queue full Value After Reset: 0x0
			3	Reserved_3	R	0x0	Reserved. Value After Reset: 0x0
			2:1	RRCSTS	R	0x0	MTL Rx Queue Read Controller State This field gives the state of the Rx queue Read controller:Values: 0x0 (IDLE): Idle state 0x1 (READ_DATA): Reading packet data 0x2 (READ_STS): Reading packet status (or timestamp) 0x3 (FLUSH): Flushing the packet data and status Value After Reset: 0x0
			0	RWCSTS	R	0x0	MTL Rx Queue Write Controller Active Status When high, this bit indicates that the MTL Rx queue Write controller is active, and it is transferring a received packet to the Rx Queue.Values: 0x0 (INACTIVE): MTL Rx Queue Write Controller Active Status not detected 0x1 (ACTIVE): MTL Rx Queue Write Controller Active Status detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xdbc	MTL_RxQ2_Control	The Queue Receive Control register controls the receive arbitration and passing of received packets to the application.	31:4	Reserved_31_4	R	0x0	Reserved. Value After Reset: 0x0
			3	RXQ_FRM_ARBIT	R/W	0x0	Receive Queue Packet Arbitration When this bit is set, the DWC_ether_qos drives the packet data to the ARI interface such that the entire packet data of currently-selected queue is transmitted before switching to other queue. When this bit is reset, the DWC_ether_qos drives the packet data to the ARI interface such that the following amount of data of currently-selected queue is transmitted before switching to other queue: PBL amount of data (indicated by ari_qN_pbl_i[]) or Complete data of a packet The status and the timestamp are not a part of the PBL data. Therefore, the DWC_ether_qos drives the complete status (including timestamp status) during first PBL request for the packet (in store-and-forward mode) or the last PBL request for the packet (in Threshold mode).Values: 0x0 (DISABLE): Receive Queue Packet Arbitration is disabled 0x1 (ENABLE): Receive Queue Packet Arbitration is enabled Value After Reset: 0x0
			2:0	RXQ_WEGT	R/W	0x0	Receive Queue Weight This field indicates the weight assigned to the Rx Queue 1. This field needs to be programmed with one value less than the required weight, that is, reset value of 0 indicates weight of 1, value of 1 indicates weight of 2, and so on. The weight is used as the number of continuous PBL or packets requests (depending on the RXQ_FRM_ARBIT) allocated to the queue in one arbitration cycle. Note: The change in value of RXQ_WEGT takes effect only after the completion of current service round or when there is change from RAA=SP to RAA=WSP algorithm. This approach is taken so that there is smooth transition. For the RXQ_WEGT value to take effect at the start, the MTL_RxQ1_Control registers must be programmed before the MTL_Operation_Mode register. Value After Reset: 0x0
			31:22	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			21:16	TQS	R/W	0x0	Transmit Queue Size This field indicates the size of the allocated Transmit queues in blocks of 256 bytes, the reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on. So, program TQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (TQS+1)*256 bytes. Value After Reset: 0x0
			15:7	Reserved_15_7	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xdc0	MTL_TxQ3_Operation_Mode	The Queue 3 Transmit Operation Mode register establishes the Transmit queue operating modes and commands.	6:4	TTC	R/W	0x0	Transmit Threshold Control These bits control the threshold level of the MTL Tx Queue. The transmission starts when the packet size within the MTL Tx Queue is larger than the threshold. In addition, full packets with length less than the threshold are also transmitted. These bits are used only when the TSF bit is reset.Values: 0x0 (M_32BYTES): 32 0x1 (M_64BYTES): 64 0x2 (M_96BYTES): 96 0x3 (M_128BYTES): 128 0x4 (M_192BYTES): 192 0x5 (M_256BYTES): 256 0x6 (M_384BYTES): 384 0x7 (M_512BYTES): 512 Value After Reset: 0x0
			3:2	TXQEN	R/W	0x0	Transmit Queue Enable This field is used to enable/disable the transmit queue 1. Note: In multiple Tx queues configuration, all the queues are disabled by default. Enable the Tx queue by programming this field.Values: 0x0 (DISABLE): Not enabled 0x1 (EN_IF_AV): Enable in AV mode 0x2 (ENABLE): Enabled 0x3 (RSVD2): Reserved Value After Reset: 0x0
			1	TSF	R/W	0x0	Transmit Store and Forward When this bit is set, the transmission starts when a full packet resides in the MTL Tx queue. When this bit is set, the TTC values specified in Bits[6:4] of this register are ignored. This bit should be changed only when the transmission is stopped.Values: 0x0 (DISABLE): Transmit Store and Forward is disabled 0x1 (ENABLE): Transmit Store and Forward is enabled Value After Reset: 0x0
			0	FTQ	R/W	0x0	Flush Transmit Queue When this bit is set, the Tx queue controller logic is reset to its default values. Therefore, all the data in the Tx queue is lost or flushed. This bit is internally reset when the flushing operation is complete. Until this bit is reset, you must not write to the MTL_TxQ1_Operation_Mode register. The data which is already accepted by the MAC transmitter is not flushed. It is scheduled for transmission and results in underflow and runt packet transmission. Note: The flush operation is complete only when the Tx queue is empty and the application has accepted the pending Tx Status of all transmitted packets. To complete this flush operation, the PHY Tx clock (clk_tx_i) must be active. Access restrictions apply. MAC initiates the Tx queue flush when application writes 1. MAC writes 0 after the Tx queue flush is complete. MAC ignores when application writes 0. Values: 0x0 (DISABLE): Flush Transmit Queue is disabled 0x1 (ENABLE): Flush Transmit Queue is enabled Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xdc4	MTL_TxQ3_Underflow	The Queue 3 Underflow Counter register contains the counter for packets aborted because of Transmit queue underflow and packets missed because of Receive queue packet flush	31:12	Reserved_31_12	R	0x0	Reserved. Value After Reset: 0x0
			11	UFCNTOVF	R	0x0	Overflow Bit for Underflow Packet Counter This bit is set every time the Tx queue Underflow Packet Counter field overflows, that is, it has crossed the maximum count. In such a scenario, the overflow packet counter is reset to all-zeros and this bit indicates that the rollover happened. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Overflow not detected for Underflow Packet Counter 0x1 (ACTIVE): Overflow detected for Underflow Packet Counter Value After Reset: 0x0
			10:0	UFFRMCNT	R	0x0	Underflow Packet Counter This field indicates the number of packets aborted by the controller because of Tx Queue Underflow. This counter is incremented each time the MAC aborts outgoing packet because of underflow. The counter is cleared when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0
0xdc8	MTL_TxQ3_Debug	The Queue 3 Transmit Debug register gives the debug status of various blocks related to the Transmit queue.	31:23	Reserved_31_23	R	0x0	Reserved. Value After Reset: 0x0
			22:20	STXSTSF	R	0x0	Number of Status Words in Tx Status FIFO of Queue This field indicates the current number of status in the Tx Status FIFO of this queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of status words in Tx Status FIFO. Value After Reset: 0x0
			19	Reserved_19	R	0x0	Reserved. Value After Reset: 0x0
			18:16	PTXQ	R	0x0	Number of Packets in the Transmit Queue This field indicates the current number of packets in the Tx Queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of packets in the Transmit queue. Value After Reset: 0x0
			15:6	Reserved_15_6	R	0x0	Reserved. Value After Reset: 0x0
			5	TXSTSFSTS	R	0x0	MTL Tx Status FIFO Full Status When high, this bit indicates that the MTL Tx Status FIFO is full. Therefore, the MTL cannot accept any more packets for transmission.Values: 0x0 (INACTIVE): MTL Tx Status FIFO Full status is not detected 0x1 (ACTIVE): MTL Tx Status FIFO Full status is detected Value After Reset: 0x0
			4	TXQSTS	R	0x0	MTL Tx Queue Not Empty Status When this bit is high, it indicates that the MTL Tx Queue is not empty and some data is left for transmission.Values: 0x0 (INACTIVE): MTL Tx Queue Not Empty status is not detected 0x1 (ACTIVE): MTL Tx Queue Not Empty status is detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			3	TWCSTS	R	0x0	MTL Tx Queue Write Controller Status When high, this bit indicates that the MTL Tx Queue Write Controller is active, and it is transferring the data to the Tx Queue.Values: 0x0 (INACTIVE): MTL Tx Queue Write Controller status is not detected 0x1 (ACTIVE): MTL Tx Queue Write Controller status is detected Value After Reset: 0x0
			2:1	TRCSTS	R	0x0	MTL Tx Queue Read Controller Status This field indicates the state of the Tx Queue Read Controller:Values: 0x0 (IDLE): Idle state 0x1 (READ): Read state (transferring data to the MAC transmitter) 0x2 (WAIT): Waiting for pending Tx Status from the MAC transmitter 0x3 (FLUSH): Flushing the Tx queue because of the Packet Abort request from the MAC Value After Reset: 0x0
			0	TXQPAUSED	R	0x0	Transmit Queue in Pause When this bit is high and the Rx flow control is enabled, it indicates that the Tx Queue is in the Pause condition (in the full-duplex only mode) because of the following: Reception of the PFC packet for the priorities assigned to the Tx Queue when PFC is enabled Reception of 802.3x Pause packet when PFC is disabled Values: 0x0 (INACTIVE): Transmit Queue in Pause status is not detected 0x1 (ACTIVE): Transmit Queue in Pause status is detected Value After Reset: 0x0
0xdd0	MTL_TxQ3_ETS_Control	The Queue ETS Control register controls the enhanced transmission selection operation.	31:7	Reserved_31_7	R	0x0	Reserved. Value After Reset: 0x0
			6:4	SLC	R/W	0x0	Slot Count If the credit-based shaper algorithm is enabled, the software can program the number of slots (of duration programmed in DMA_CH1_Slot_Interval register) over which the average transmitted bits per slot, provided in the MTL_TxQ1_ETS_Status register, need to be computed for Queue. The encoding is as follows:Values: 0x0 (M_1_SLOT): 1 slot 0x1 (M_2_SLOT): 2 slots 0x2 (M_4_SLOT): 4 slots 0x3 (M_8_SLOT): 8 slots 0x4 (M_16_SLOT): 16 slots 0x5 (RSVD): Reserved Value After Reset: 0x0
			3	CC	R/W	0x0	Credit Control When this bit is set, the accumulated credit parameter in the credit-based shaper algorithm logic is not reset to zero when there is positive credit and no packet to transmit in Tx Queue 1. The credit accumulates even when there is no packet waiting in Tx Queue 1 and another Tx Queue is transmitting. When this bit is reset, the accumulated credit parameter in the credit-based shaper algorithm logic is set to zero when there is positive credit and no packet to transmit in Tx Queue 1. When there is no packet waiting in Tx Queue 1 and other channel is transmitting, no credit is accumulated.Values: 0x0 (DISABLE): Credit Control is disabled 0x1 (ENABLE): Credit Control is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			2	AVALG	R/W	0x0	AV Algorithm When Queue 1 is programmed for AV, this field configures the scheduling algorithm for this queue: This bit when set, indicates credit based shaper algorithm (CBS) is selected for Queue 1 traffic. When reset, strict priority is selected.Values: 0x0 (DISABLE): CBS Algorithm is disabled 0x1 (ENABLE): CBS Algorithm is enabled Value After Reset: 0x0
			1:0	Reserved_1_0	R	0x0	Reserved. Value After Reset: 0x0
0xdd4	MTL_TxQ3_ETS_Status	The Queue 3 ETS Status register provides the average traffic transmitted in Queue 3.	31:24	Reserved_31_24	R	0x0	Reserved. Value After Reset: 0x0
			23:0	ABS	R	0x0	Average Bits per Slot This field contains the average transmitted bits per slot. If AV operation is enabled, this field is computed over number of slots, programmed in the SLC field of MTL_TxQ1_ETS_CONTROL register. The maximum value of this field is 0x6_4000 in 100 Mbps, 0x3E_8000 in 1000 Mbps and 9C_4000 in 2500 Mbps mode respectively. When the DCB operation is enabled for Queue, this field is computed over every 10 million bit times slot (4 ms in 2500 Mbps; 10 ms in 1000 Mbps; 100 ms in 100 Mbps). The maximum value is 0x989680. Value After Reset: 0x0
0xdd8	MTL_TxQ3_Quantum_Weight	The Queue 3 idleSlopeCredit, Quantum or Weights register provides the average traffic transmitted in Queue 3.	31:21	Reserved_31_21	R	0x0	Reserved. Value After Reset: 0x0
			20:0	ISCQW	R/W	0x0	idleSlopeCredit, Quantum or Weights idleSlopeCredit When AV feature is enabled, this field contains the idleSlopeCredit value required for the credit-based shaper algorithm for Queue 1. This is the rate of change of credit in bits per cycle (40 ns for 100 Mbps; 8 ns for 1000 Mbps; 3.2 ns for 2500 Mbps) when the credit is increasing. The software should program this field with computed credit in bits per cycle scaled by 1,024. The maximum value is portTransmitRate, that is, 0x2000 in 1000/2500 Mbps mode and 0x1000 in 100 Mbps mode. Bits[20:14] must be written to zero. Quantum When the DCB operation is enabled with DWRR algorithm for Queue 1 traffic, this field contains the quantum value in bytes to be added to credit during every queue scanning cycle. The maximum value is 0x1312D0 bytes. Weights When DCB operation is enabled with WFQ algorithm for Queue 1 traffic, this field contains the weight for this queue. The maximum value is 0x3FFF where weight of 0 indicates 100% bandwidth. Bits[20:14] must be written to zero. When DCB operation or generic queuing operation is enabled with WRR algorithm for Queue 1 traffic, this field contains the weight for this queue. The maximum value is 0x64. Bits [20:7] must be written to zero. Note 1: In multiple Queue configuration this field in respective per queue register must be programmed to some non-zero value when multiple queues are enabled or single queue other than Q0 is enabled. This field need not be programmed when only Q0 is enabled. In general, when WRR algorithm is selected a non-zero value must be programmed on both Receive and Transmit. In Receive, the register is MTL_Operation_Mode register. Note 2: For WFQ algorithm, higher the programmed weights lesser the bandwidth allocated for that Transmit Queue. The finish time is not a function of particular packet alone but it is as per the formula: (previous_finish_time of particular Transmit Queue + (weights*packet_size)) Note 3: The weights programmed do not correspond to the number of packets but the fraction of bandwidth or time allocated for particular queue w.r.t. total BW or time. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xddc	MTL_TxQ3_SendSlopeCredit	The sendSlopeCredit register contains the sendSlope credit value required for the credit-based shaper algorithm for the Queue.	31:14	Reserved_31_14	R	0x0	Reserved. Value After Reset: 0x0
			13:0	SSC	R/W	0x0	sendSlopeCredit Value When AV operation is enabled, this field contains the sendSlopeCredit value required for credit-based shaper algorithm for Tx Queue 1. This is the rate of change of credit in bits per cycle (40 ns, 8 ns and 3.2 ns for 100 Mbps, 1000 Mbps and 2500 Mbps respectively) when the credit is decreasing. The software should program this field with computed credit in bits per cycle scaled by 1,024. The maximum value is portTransmitRate, that is, 0x2000 in 1000/2500 Mbps mode and 0x1000 in 100 Mbps mode. This field should be programmed with absolute sendSlopeCredit value. The credit-based shaper logic subtracts it from the accumulated credit when Tx Queue 1 is selected for transmission. Value After Reset: 0x0
0xde0	MTL_TxQ3_HiCredit	The hiCredit register contains the hiCredit value required for the credit-based shaper algorithm for the Queue.	31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28:0	HC	R/W	0x0	hiCredit Value When the AV feature is enabled, this field contains the hiCredit value required for the credit-based shaper algorithm. This is the maximum value that can be accumulated in the credit parameter. This is specified in bits scaled by 1,024. The maximum value is maxInterferenceSize, that is, best-effort maximum packet size (16,384 bytes or 131,072 bits). The value to be specified is 131,072 * 1,024 = 134,217,728 or 0x0800_0000. Value After Reset: 0x0
0xde4	MTL_TxQ3_LoCredit	The loCredit register contains the loCredit value required for the credit-based shaper algorithm for the Queue.	31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28:0	LC	R/W	0x0	loCredit Value When AV operation is enabled, this field contains the loCredit value required for the credit-based shaper algorithm. This is the minimum value that can be accumulated in the credit parameter. This is specified in bits scaled by 1,024. The maximum value to be programmed is corresponds to twice the maxFrameSize transmitted from this queue. If the maxFrameSize is 8192 bytes, then (8192*2) * 8 * 1024 = 134,217,728 or 0x0800_0000. Because it is a negative value, the programmed value is 2's complement of the value, that is, 0x1800_0000. Value After Reset: 0x0
			31:25	Reserved_31_25	R	0x0	Reserved. Value After Reset: 0x0
			24	RXOIE	R/W	0x0	Receive Queue Overflow Interrupt Enable When this bit is set, the Receive Queue Overflow interrupt is enabled. When this bit is reset, the Receive Queue Overflow interrupt is disabled.Values: 0x0 (DISABLE): Receive Queue Overflow Interrupt is disabled 0x1 (ENABLE): Receive Queue Overflow Interrupt is enabled Value After Reset: 0x0
			23:17	Reserved_23_17	R	0x0	Reserved. Value After Reset: 0x0
			16	RXOVFIS	R/W	0x0	Receive Queue Overflow Interrupt Status This bit indicates that the Receive Queue had an overflow while receiving the packet. If a partial packet is transferred to the application, the overflow status is set in RDES3[21]. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Queue Overflow Interrupt Status not detected 0x1 (ACTIVE): Receive Queue Overflow Interrupt Status detected Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xdec	MTL_Q3_Interrupt_Control_Status	This register contains the interrupt enable and status bits for the queue 3 interrupts.	15:10	Reserved_15_10	R	0x0	Reserved. Value After Reset: 0x0
			9	ABPSIE	R/W	0x0	Average Bits Per Slot Interrupt Enable When this bit is set, the MAC asserts the sbd_intr_o or mci_intr_o interrupt when the average bits per slot status is updated. When this bit is cleared, the interrupt is not asserted for such an event.Values: 0x0 (DISABLE): Average Bits Per Slot Interrupt is disabled 0x1 (ENABLE): Average Bits Per Slot Interrupt is enabled Value After Reset: 0x0
			8	TXUIE	R/W	0x0	Transmit Queue Underflow Interrupt Enable When this bit is set, the Transmit Queue Underflow interrupt is enabled. When this bit is reset, the Transmit Queue Underflow interrupt is disabled.Values: 0x0 (DISABLE): Transmit Queue Underflow Interrupt Status is disabled 0x1 (ENABLE): Transmit Queue Underflow Interrupt Status is enabled Value After Reset: 0x0
			7:2	Reserved_7_2	R	0x0	Reserved. Value After Reset: 0x0
			1	ABPSIS	R/W	0x0	Average Bits Per Slot Interrupt Status When set, this bit indicates that the MAC has updated the ABS value. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Average Bits Per Slot Interrupt Status not detected 0x1 (ACTIVE): Average Bits Per Slot Interrupt Status detected Value After Reset: 0x0 Testable: untestable
			0	TXUNFIS	R/W	0x0	Transmit Queue Underflow Interrupt Status This bit indicates that the Transmit Queue had an underflow while transmitting the packet. Transmission is suspended and an Underflow Error TDES3[2] is set. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Queue Underflow Interrupt Status not detected 0x1 (ACTIVE): Transmit Queue Underflow Interrupt Status detected Value After Reset: 0x0 Testable: untestable
			31:26	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			25:20	RQS	R/W	0x0	Receive Queue Size This field indicates the size of the allocated Receive queues in blocks of 256 bytesthe reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on. So, program RQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (RQS+1)*256 bytes. Value After Reset: 0x0
			19	Reserved_19_y	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xdf0	MTL_RxQ3_Operation_Mode	The Queue 3 Receive Operation Mode register establishes the Receive queue operating modes and command.	18:14	RFD	R/W	0x0	Threshold for Deactivating Flow Control (in half-duplex and full-duplex modes) These bits control the threshold (fill-level of Rx queue) at which the flow control is de-asserted after activation: 0: Full minus 1 KB, that is, FULL 1 KB 1: Full minus 1.5 KB, that is, FULL 1.5 KB 2: Full minus 2 KB, that is, FULL 2 KB 3: Full minus 2.5 KB, that is, FULL 2.5 KB ... 30: Full minus 16 KB, that is, FULL 16 KB 31: Full minus 16.5 KB, that is, FULL 16.5 KB The de-assertion is effective only after flow control is asserted. Note: The value must be programmed in such a way to make sure that the threshold is a positive number. When the EHFC is set high, these values are applicable only when the Rx queue size determined by the RQS field of this register, is equal to or greater than 4 KB. For a given queue size, the values ranges between 0 and the encoding for FULL minus (QSIZE - 0.5 KB) and all other values are illegal. Here the term FULL and QSIZE refers to the queue size determined by the RQS field of this register. Value After Reset: 0x0
			13	Reserved_13_y	R	0x0	Reserved. Value After Reset: 0x0
			12:8	RFA	R/W	0x0	Threshold for Activating Flow Control (in half-duplex and full-duplex These bits control the threshold (fill-level of Rx queue) at which the flow control is activated: For more information on encoding for this field, see RFD. Value After Reset: 0x0
			7	EHFC	R/W	0x0	Enable Hardware Flow Control When this bit is set, the flow control signal operation, based on the fill-level of Rx queue, is enabled. When reset, the flow control operation is disabled.Values: 0x0 (DISABLE): Hardware Flow Control is disabled 0x1 (ENABLE): Hardware Flow Control is enabled Value After Reset: 0x0
			6	DIS_TCP_EF	R/W	0x0	Disable Dropping of TCP/IP Checksum Error Packets When this bit is set, the MAC does not drop the packets which only have the errors detected by the Receive Checksum Offload engine. Such packets have errors only in the encapsulated payload. There are no errors (including FCS error) in the Ethernet packet received by the MAC. When this bit is reset, all error packets are dropped if the FEP bit is reset.Values: 0x0 (ENABLE): Dropping of TCP/IP Checksum Error Packets is enabled 0x1 (DISABLE): Dropping of TCP/IP Checksum Error Packets is disabled Value After Reset: 0x0
			5	RSF	R/W	0x0	Receive Queue Store and Forward When this bit is set, the DWC_ether_qos reads a packet from the Rx queue only after the complete packet has been written to it, ignoring the RTC field of this register. When this bit is reset, the Rx queue operates in the Threshold (cut-through) mode, subject to the threshold specified by the RTC field of this register.Values: 0x0 (DISABLE): Receive Queue Store and Forward is disabled 0x1 (ENABLE): Receive Queue Store and Forward is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			4	FEP	R/W	0x0	Forward Error Packets When this bit is reset, the Rx queue drops packets with error status (CRC error, GMII_ER, watchdog timeout, or overflow). However, if the start byte (write) pointer of a packet is already transferred to the read controller side (in Threshold mode), the packet is not dropped. When this bit is set, all packets except the runt error packets are forwarded to the application or DMA. If the RSF bit is set and the Rx queue overflows when a partial packet is written, the packet is dropped irrespective of the setting of this bit. However, if the RSF bit is reset and the Rx queue overflows when a partial packet is written, a partial packet might be forwarded to the application or DMA.Values: 0x0 (DISABLE): Forward Error Packets is disabled 0x1 (ENABLE): Forward Error Packets is enabled Value After Reset: 0x0
			3	FUP	R/W	0x0	Forward Undersized Good Packets When this bit is set, the Rx queue forwards the undersized good packets (packets with no error and length less than 64 bytes), including pad-bytes and CRC. When this bit is reset, the Rx queue drops all packets of less than 64 bytes, unless a packet is already transferred because of the lower value of Rx Threshold, for example, RTC = 01.Values: 0x0 (DISABLE): Forward Undersized Good Packets is disabled 0x1 (ENABLE): Forward Undersized Good Packets is enabled Value After Reset: 0x0
			2	Reserved_2	R	0x0	Reserved. Value After Reset: 0x0
			1:0	RTC	R/W	0x0	Receive Queue Threshold Control These bits control the threshold level of the MTL Rx queue (in bytes): The received packet is transferred to the application or DMA when the packet size within the MTL Rx queue is larger than the threshold. In addition, full packets with length less than the threshold are automatically transferred. This field is valid only when the RSF bit is zero. This field is ignored when the RSF bit is set to 1.Values: 0x0 (M_64BYTE): 64 0x1 (M_32BYTE): 32 0x2 (M_96BYTE): 96 0x3 (M_128BYTE): 128 Value After Reset: 0x0
0xdf4	MTL_RxQ3_Missed_Packet_Overflow_Cnt	The Queue 3 Missed Packet and Overflow Counter register contains the counter for packets missed because of Receive queue packet flush and packets discarded because of Receive queue overflow.	31:28	Reserved_31_28	R	0x0	Reserved. Value After Reset: 0x0
			27	MISCNTOVF	R	0x0	Missed Packet Counter Overflow Bit When set, this bit indicates that the Rx Queue Missed Packet Counter crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Missed Packet Counter overflow not detected 0x1 (ACTIVE): Missed Packet Counter overflow detected Value After Reset: 0x0
			26:16	MISPKTCNT	R	0x0	Missed Packet Counter This field indicates the number of packets missed by the DWC_ether_qos because the application asserted ari_pkt_flush_i[] for this queue. This counter is reset when this register is read with mci_be_i[0] at 1b1. This counter is incremented by 1 when the DMA discards the packet because of buffer unavailability. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0
			15:12	Reserved_15_12	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			11	OVFCNTOVF	R	0x0	Overflow Counter Overflow Bit When set, this bit indicates that the Rx Queue Overflow Packet Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Overflow Counter overflow not detected 0x1 (ACTIVE): Overflow Counter overflow detected Value After Reset: 0x0
			10:0	OVFPKTCNT	R	0x0	Overflow Packet Counter This field indicates the number of packets discarded by the DWC_ether_qos because of Receive queue overflow. This counter is incremented each time the DWC_ether_qos discards an incoming packet because of overflow. This counter is reset when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0
0xdf8	MTL_RxQ3_Debug	The Queue 3 Receive Debug register gives the debug status of various blocks related to the Receive queue.	31:30	Reserved_31_30	R	0x0	Reserved. Value After Reset: 0x0
			29:16	PRXQ	R	0x0	Number of Packets in Receive Queue This field indicates the current number of packets in the Rx Queue, that is, Max_Queue_Size/Min_Packet_Size. Value After Reset: 0x0
			15:6	Reserved_15_6	R	0x0	Reserved. Value After Reset: 0x0
			5:4	RXQSTS	R	0x0	MTL Rx Queue Fill-Level Status This field gives the status of the fill-level of the Rx Queue:Values: 0x0 (EMPTY): Rx Queue empty 0x1 (BLW_THR): Rx Queue fill-level below flow-control deactivate threshold 0x2 (ABV_THR): Rx Queue fill-level above flow-control activate threshold 0x3 (FULL): Rx Queue full Value After Reset: 0x0
			3	Reserved_3	R	0x0	Reserved. Value After Reset: 0x0
			2:1	RRCSTS	R	0x0	MTL Rx Queue Read Controller State This field gives the state of the Rx queue Read controller:Values: 0x0 (IDLE): Idle state 0x1 (READ_DATA): Reading packet data 0x2 (READ_STS): Reading packet status (or timestamp) 0x3 (FLUSH): Flushing the packet data and status Value After Reset: 0x0
			0	RWCSTS	R	0x0	MTL Rx Queue Write Controller Active Status When high, this bit indicates that the MTL Rx queue Write controller is active, and it is transferring a received packet to the Rx Queue.Values: 0x0 (INACTIVE): MTL Rx Queue Write Controller Active Status not detected 0x1 (ACTIVE): MTL Rx Queue Write Controller Active Status detected Value After Reset: 0x0
			31:4	Reserved_31_4	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xdfc	MTL_RxQ3_Control	The Queue Receive Control register controls the receive arbitration and passing of received packets to the application.	3	RXQ_FRM_ARBIT	R/W	0x0	Receive Queue Packet Arbitration When this bit is set, the DWC_ether_qos drives the packet data to the ARI interface such that the entire packet data of currently-selected queue is transmitted before switching to other queue. When this bit is reset, the DWC_ether_qos drives the packet data to the ARI interface such that the following amount of data of currently-selected queue is transmitted before switching to other queue: PBL amount of data (indicated by ari_qN_pbl_i[]) or Complete data of a packet The status and the timestamp are not a part of the PBL data. Therefore, the DWC_ether_qos drives the complete status (including timestamp status) during first PBL request for the packet (in store-and-forward mode) or the last PBL request for the packet (in Threshold mode).Values: 0x0 (DISABLE): Receive Queue Packet Arbitration is disabled 0x1 (ENABLE): Receive Queue Packet Arbitration is enabled Value After Reset: 0x0
			2:0	RXQ_WEGT	R/W	0x0	Receive Queue Weight This field indicates the weight assigned to the Rx Queue 1. This field needs to be programmed with one value less than the required weight, that is, reset value of 0 indicates weight of 1, value of 1 indicates weight of 2, and so on. The weight is used as the number of continuous PBL or packets requests (depending on the RXQ_FRM_ARBIT) allocated to the queue in one arbitration cycle. Note: The change in value of RXQ_WEGT takes effect only after the completion of current service round or when there is change from RAA=SP to RAA=WSP algorithm. This approach is taken so that there is smooth transition. For the RXQ_WEGT value to take effect at the start, the MTL_RxQ1_Control registers must be programmed before the MTL_Operation_Mode register. Value After Reset: 0x0
			31:22	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			21:16	TQS	R/W	0x0	Transmit Queue Size This field indicates the size of the allocated Transmit queues in blocks of 256 bytes, the reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on. So, program TQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (TQS+1)*256 bytes. Value After Reset: 0x0
			15:7	Reserved_15_7	R	0x0	Reserved. Value After Reset: 0x0
			6:4	TTC	R/W	0x0	Transmit Threshold Control These bits control the threshold level of the MTL Tx Queue. The transmission starts when the packet size within the MTL Tx Queue is larger than the threshold. In addition, full packets with length less than the threshold are also transmitted. These bits are used only when the TSF bit is reset.Values: 0x0 (M_32BYTES): 32 0x1 (M_64BYTES): 64 0x2 (M_96BYTES): 96 0x3 (M_128BYTES): 128 0x4 (M_192BYTES): 192 0x5 (M_256BYTES): 256 0x6 (M_384BYTES): 384 0x7 (M_512BYTES): 512 Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xe00	MTL_TxQ4_Operation_Mode	The Queue 4 Transmit Operation Mode register establishes the Transmit queue operating modes and commands.	3:2	TXQEN	R/W	0x0	Transmit Queue Enable This field is used to enable/disable the transmit queue 1. Note: In multiple Tx queues configuration, all the queues are disabled by default. Enable the Tx queue by programming this field.Values: 0x0 (DISABLE): Not enabled 0x1 (EN_IF_AV): Enable in AV mode 0x2 (ENABLE): Enabled 0x3 (RSVD2): Reserved Value After Reset: 0x0
			1	TSF	R/W	0x0	Transmit Store and Forward When this bit is set, the transmission starts when a full packet resides in the MTL Tx queue. When this bit is set, the TTC values specified in Bits[6:4] of this register are ignored. This bit should be changed only when the transmission is stopped.Values: 0x0 (DISABLE): Transmit Store and Forward is disabled 0x1 (ENABLE): Transmit Store and Forward is enabled Value After Reset: 0x0
			0	FTQ	R/W	0x0	Flush Transmit Queue When this bit is set, the Tx queue controller logic is reset to its default values. Therefore, all the data in the Tx queue is lost or flushed. This bit is internally reset when the flushing operation is complete. Until this bit is reset, you must not write to the MTL_TxQ1_Operation_Mode register. The data which is already accepted by the MAC transmitter is not flushed. It is scheduled for transmission and results in underflow and runt packet transmission. Note: The flush operation is complete only when the Tx queue is empty and the application has accepted the pending Tx Status of all transmitted packets. To complete this flush operation, the PHY Tx clock (clk_tx_i) must be active. Access restrictions apply. MAC initiates the Tx queue flush when application writes 1. MAC writes 0 after the Tx queue flush is complete. MAC ignores when application writes 0. Values: 0x0 (DISABLE): Flush Transmit Queue is disabled 0x1 (ENABLE): Flush Transmit Queue is enabled Value After Reset: 0x0 Testable: untestable
0xe04	MTL_TxQ4_Underflow	The Queue 4 Underflow Counter register contains the counter for packets aborted because of Transmit queue underflow and packets missed because of Receive queue packet flush	31:12	Reserved_31_12	R	0x0	Reserved. Value After Reset: 0x0
			11	UFCNTOVF	R	0x0	Overflow Bit for Underflow Packet Counter This bit is set every time the Tx queue Underflow Packet Counter field overflows, that is, it has crossed the maximum count. In such a scenario, the overflow packet counter is reset to all-zeros and this bit indicates that the rollover happened. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Overflow not detected for Underflow Packet Counter 0x1 (ACTIVE): Overflow detected for Underflow Packet Counter Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			10:0	UFFRMCNT	R	0x0	Underflow Packet Counter This field indicates the number of packets aborted by the controller because of Tx Queue Underflow. This counter is incremented each time the MAC aborts outgoing packet because of underflow. The counter is cleared when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0
0xe08	MTL_TxQ4_Debug	The Queue 4 Transmit Debug register gives the debug status of various blocks related to the Transmit queue.	31:23	Reserved_31_23	R	0x0	Reserved. Value After Reset: 0x0
			22:20	STXSTSF	R	0x0	Number of Status Words in Tx Status FIFO of Queue This field indicates the current number of status in the Tx Status FIFO of this queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of status words in Tx Status FIFO. Value After Reset: 0x0
			19	Reserved_19	R	0x0	Reserved. Value After Reset: 0x0
			18:16	PTXQ	R	0x0	Number of Packets in the Transmit Queue This field indicates the current number of packets in the Tx Queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of packets in the Transmit queue. Value After Reset: 0x0
			15:6	Reserved_15_6	R	0x0	Reserved. Value After Reset: 0x0
			5	TXSTSFSTS	R	0x0	MTL Tx Status FIFO Full Status When high, this bit indicates that the MTL Tx Status FIFO is full. Therefore, the MTL cannot accept any more packets for transmission.Values: 0x0 (INACTIVE): MTL Tx Status FIFO Full status is not detected 0x1 (ACTIVE): MTL Tx Status FIFO Full status is detected Value After Reset: 0x0
			4	TXQSTS	R	0x0	MTL Tx Queue Not Empty Status When this bit is high, it indicates that the MTL Tx Queue is not empty and some data is left for transmission.Values: 0x0 (INACTIVE): MTL Tx Queue Not Empty status is not detected 0x1 (ACTIVE): MTL Tx Queue Not Empty status is detected Value After Reset: 0x0
			3	TWCSTS	R	0x0	MTL Tx Queue Write Controller Status When high, this bit indicates that the MTL Tx Queue Write Controller is active, and it is transferring the data to the Tx Queue.Values: 0x0 (INACTIVE): MTL Tx Queue Write Controller status is not detected 0x1 (ACTIVE): MTL Tx Queue Write Controller status is detected Value After Reset: 0x0
			2:1	TRCSTS	R	0x0	MTL Tx Queue Read Controller Status This field indicates the state of the Tx Queue Read Controller:Values: 0x0 (IDLE): Idle state 0x1 (READ): Read state (transferring data to the MAC transmitter) 0x2 (WAIT): Waiting for pending Tx Status from the MAC transmitter 0x3 (FLUSH): Flushing the Tx queue because of the Packet Abort request from the MAC Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	TXQPAUSED	R	0x0	Transmit Queue in Pause When this bit is high and the Rx flow control is enabled, it indicates that the Tx Queue is in the Pause condition (in the full-duplex only mode) because of the following: Reception of the PFC packet for the priorities assigned to the Tx Queue when PFC is enabled Reception of 802.3x Pause packet when PFC is disabled Values: 0x0 (INACTIVE): Transmit Queue in Pause status is not detected 0x1 (ACTIVE): Transmit Queue in Pause status is detected Value After Reset: 0x0
0xe10	MTL_TxQ4_ETS_Control	The Queue ETS Control register controls the enhanced transmission selection operation.	31:7	Reserved_31_7	R	0x0	Reserved. Value After Reset: 0x0
			6:4	SLC	R/W	0x0	Slot Count If the credit-based shaper algorithm is enabled, the software can program the number of slots (of duration programmed in DMA_CH1_Slot_Interval register) over which the average transmitted bits per slot, provided in the MTL_TxQ1_ETS_Status register, need to be computed for Queue. The encoding is as follows:Values: 0x0 (M_1_SLOT): 1 slot 0x1 (M_2_SLOT): 2 slots 0x2 (M_4_SLOT): 4 slots 0x3 (M_8_SLOT): 8 slots 0x4 (M_16_SLOT): 16 slots 0x5 (RSVD): Reserved Value After Reset: 0x0
			3	CC	R/W	0x0	Credit Control When this bit is set, the accumulated credit parameter in the credit-based shaper algorithm logic is not reset to zero when there is positive credit and no packet to transmit in Tx Queue 1. The credit accumulates even when there is no packet waiting in Tx Queue 1 and another Tx Queue is transmitting. When this bit is reset, the accumulated credit parameter in the credit-based shaper algorithm logic is set to zero when there is positive credit and no packet to transmit in Tx Queue 1. When there is no packet waiting in Tx Queue 1 and other channel is transmitting, no credit is accumulated.Values: 0x0 (DISABLE): Credit Control is disabled 0x1 (ENABLE): Credit Control is enabled Value After Reset: 0x0
			2	AVALG	R/W	0x0	AV Algorithm When Queue 1 is programmed for AV, this field configures the scheduling algorithm for this queue: This bit when set, indicates credit based shaper algorithm (CBS) is selected for Queue 1 traffic. When reset, strict priority is selected.Values: 0x0 (DISABLE): CBS Algorithm is disabled 0x1 (ENABLE): CBS Algorithm is enabled Value After Reset: 0x0
			1:0	Reserved_1_0	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xe14	MTL_TxQ4_ETS_Status	The Queue 4 ETS Status register provides the average traffic transmitted in Queue 4.	31:24	Reserved_31_24	R	0x0	Reserved. Value After Reset: 0x0
			23:0	ABS	R	0x0	Average Bits per Slot This field contains the average transmitted bits per slot. If AV operation is enabled, this field is computed over number of slots, programmed in the SLC field of MTL_TxQ1_ETS_CONTROL register. The maximum value of this field is 0x6_4000 in 100 Mbps, 0x3E_8000 in 1000 Mbps and 9C_4000 in 2500 Mbps mode respectively. When the DCB operation is enabled for Queue, this field is computed over every 10 million bit times slot (4 ms in 2500 Mbps; 10 ms in 1000 Mbps; 100 ms in 100 Mbps). The maximum value is 0x989680. Value After Reset: 0x0
0xe18	MTL_TxQ4_Quantum_Weight	The Queue 4 idleSlopeCredit, Quantum or Weights register provides the average traffic transmitted in Queue 4.	31:21	Reserved_31_21	R	0x0	Reserved. Value After Reset: 0x0
			20:0	ISCQW	R/W	0x0	idleSlopeCredit, Quantum or Weights idleSlopeCredit When AV feature is enabled, this field contains the idleSlopeCredit value required for the credit-based shaper algorithm for Queue 1. This is the rate of change of credit in bits per cycle (40 ns for 100 Mbps; 8 ns for 1000 Mbps; 3.2 ns for 2500 Mbps) when the credit is increasing. The software should program this field with computed credit in bits per cycle scaled by 1,024. The maximum value is portTransmitRate, that is, 0x2000 in 1000/2500 Mbps mode and 0x1000 in 100 Mbps mode. Bits[20:14] must be written to zero. Quantum When the DCB operation is enabled with DWRR algorithm for Queue 1 traffic, this field contains the quantum value in bytes to be added to credit during every queue scanning cycle. The maximum value is 0x1312D0 bytes. Weights When DCB operation is enabled with WFQ algorithm for Queue 1 traffic, this field contains the weight for this queue. The maximum value is 0x3FFF where weight of 0 indicates 100% bandwidth. Bits[20:14] must be written to zero. When DCB operation or generic queuing operation is enabled with WRR algorithm for Queue 1 traffic, this field contains the weight for this queue. The maximum value is 0x64. Bits [20:7] must be written to zero. Note 1: In multiple Queue configuration this field in respective per queue register must be programmed to some non-zero value when multiple queues are enabled or single queue other than Q0 is enabled. This field need not be programmed when only Q0 is enabled. In general, when WRR algorithm is selected a non-zero value must be programmed on both Receive and Transmit. In Receive, the register is MTL_Operation_Mode register. Note 2: For WFQ algorithm, higher the programmed weights lesser the bandwidth allocated for that Transmit Queue. The finish time is not a function of particular packet alone but it is as per the formula: (previous_finish_time of particular Transmit Queue + (weights*packet_size)) Note 3: The weights programmed do not correspond to the number of packets but the fraction of bandwidth or time allocated for particular queue w.r.t. total BW or time. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xe1c	MTL_TxQ4_SendSlopeCredit	The sendSlopeCredit register contains the sendSlope credit value required for the credit-based shaper algorithm for the Queue.	31:14	Reserved_31_14	R	0x0	Reserved. Value After Reset: 0x0
			13:0	SSC	R/W	0x0	sendSlopeCredit Value When AV operation is enabled, this field contains the sendSlopeCredit value required for credit-based shaper algorithm for Tx Queue 1. This is the rate of change of credit in bits per cycle (40 ns, 8 ns and 3.2 ns for 100 Mbps, 1000 Mbps and 2500 Mbps respectively) when the credit is decreasing. The software should program this field with computed credit in bits per cycle scaled by 1,024. The maximum value is portTransmitRate, that is, 0x2000 in 1000/2500 Mbps mode and 0x1000 in 100 Mbps mode. This field should be programmed with absolute sendSlopeCredit value. The credit-based shaper logic subtracts it from the accumulated credit when Tx Queue 1 is selected for transmission. Value After Reset: 0x0
0xe20	MTL_TxQ4_HiCredit	The hiCredit register contains the hiCredit value required for the credit-based shaper algorithm for the Queue.	31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28:0	HC	R/W	0x0	hiCredit Value When the AV feature is enabled, this field contains the hiCredit value required for the credit-based shaper algorithm. This is the maximum value that can be accumulated in the credit parameter. This is specified in bits scaled by 1,024. The maximum value is maxInterferenceSize, that is, best-effort maximum packet size (16,384 bytes or 131,072 bits). The value to be specified is 131,072 * 1,024 = 134,217,728 or 0x0800_0000. Value After Reset: 0x0
0xe24	MTL_TxQ4_LoCredit	The loCredit register contains the loCredit value required for the credit-based shaper algorithm for the Queue.	31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28:0	LC	R/W	0x0	loCredit Value When AV operation is enabled, this field contains the loCredit value required for the credit-based shaper algorithm. This is the minimum value that can be accumulated in the credit parameter. This is specified in bits scaled by 1,024. The maximum value to be programmed is corresponds to twice the maxFrameSize transmitted from this queue. If the maxFrameSize is 8192 bytes, then (8192*2) * 8 * 1024 = 134,217,728 or 0x0800_0000. Because it is a negative value, the programmed value is 2's complement of the value, that is, 0x1800_0000. Value After Reset: 0x0
			31:25	Reserved_31_25	R	0x0	Reserved. Value After Reset: 0x0
			24	RXOIE	R/W	0x0	Receive Queue Overflow Interrupt Enable When this bit is set, the Receive Queue Overflow interrupt is enabled. When this bit is reset, the Receive Queue Overflow interrupt is disabled.Values: 0x0 (DISABLE): Receive Queue Overflow Interrupt is disabled 0x1 (ENABLE): Receive Queue Overflow Interrupt is enabled Value After Reset: 0x0
			23:17	Reserved_23_17	R	0x0	Reserved. Value After Reset: 0x0
			16	RXOVFIS	R/W	0x0	Receive Queue Overflow Interrupt Status This bit indicates that the Receive Queue had an overflow while receiving the packet. If a partial packet is transferred to the application, the overflow status is set in RDES3[21]. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Queue Overflow Interrupt Status not detected 0x1 (ACTIVE): Receive Queue Overflow Interrupt Status detected Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xe2c	MTL_Q4_Interrupt_Control_Status	This register contains the interrupt enable and status bits for the queue 4 interrupts.	15:10	Reserved_15_10	R	0x0	Reserved. Value After Reset: 0x0
			9	ABPSIE	R/W	0x0	Average Bits Per Slot Interrupt Enable When this bit is set, the MAC asserts the sbd_intr_o or mci_intr_o interrupt when the average bits per slot status is updated. When this bit is cleared, the interrupt is not asserted for such an event.Values: 0x0 (DISABLE): Average Bits Per Slot Interrupt is disabled 0x1 (ENABLE): Average Bits Per Slot Interrupt is enabled Value After Reset: 0x0
			8	TXUIE	R/W	0x0	Transmit Queue Underflow Interrupt Enable When this bit is set, the Transmit Queue Underflow interrupt is enabled. When this bit is reset, the Transmit Queue Underflow interrupt is disabled.Values: 0x0 (DISABLE): Transmit Queue Underflow Interrupt Status is disabled 0x1 (ENABLE): Transmit Queue Underflow Interrupt Status is enabled Value After Reset: 0x0
			7:2	Reserved_7_2	R	0x0	Reserved. Value After Reset: 0x0
			1	ABPSIS	R/W	0x0	Average Bits Per Slot Interrupt Status When set, this bit indicates that the MAC has updated the ABS value. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Average Bits Per Slot Interrupt Status not detected 0x1 (ACTIVE): Average Bits Per Slot Interrupt Status detected Value After Reset: 0x0 Testable: untestable
			0	TXUNFIS	R/W	0x0	Transmit Queue Underflow Interrupt Status This bit indicates that the Transmit Queue had an underflow while transmitting the packet. Transmission is suspended and an Underflow Error TDES3[2] is set. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Queue Underflow Interrupt Status not detected 0x1 (ACTIVE): Transmit Queue Underflow Interrupt Status detected Value After Reset: 0x0 Testable: untestable
			31:26	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			25:20	RQS	R/W	0x0	Receive Queue Size This field indicates the size of the allocated Receive queues in blocks of 256 bytesthe reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on. So, program RQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (RQS+1)*256 bytes. Value After Reset: 0x0
			19	Reserved_19_y	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xe30	MTL_RxQ4_Operation_Mode	The Queue 4 Receive Operation Mode register establishes the Receive queue operating modes and command.	18:14	RFD	R/W	0x0	Threshold for Deactivating Flow Control (in half-duplex and full-duplex modes) These bits control the threshold (fill-level of Rx queue) at which the flow control is de-asserted after activation: 0: Full minus 1 KB, that is, FULL 1 KB 1: Full minus 1.5 KB, that is, FULL 1.5 KB 2: Full minus 2 KB, that is, FULL 2 KB 3: Full minus 2.5 KB, that is, FULL 2.5 KB ... 30: Full minus 16 KB, that is, FULL 16 KB 31: Full minus 16.5 KB, that is, FULL 16.5 KB The de-assertion is effective only after flow control is asserted. Note: The value must be programmed in such a way to make sure that the threshold is a positive number. When the EHFC is set high, these values are applicable only when the Rx queue size determined by the RQS field of this register, is equal to or greater than 4 KB. For a given queue size, the values ranges between 0 and the encoding for FULL minus (QSIZE - 0.5 KB) and all other values are illegal. Here the term FULL and QSIZE refers to the queue size determined by the RQS field of this register. Value After Reset: 0x0
			13	Reserved_13_y	R	0x0	Reserved. Value After Reset: 0x0
			12:8	RFA	R/W	0x0	Threshold for Activating Flow Control (in half-duplex and full-duplex These bits control the threshold (fill-level of Rx queue) at which the flow control is activated: For more information on encoding for this field, see RFD. Value After Reset: 0x0
			7	EHFC	R/W	0x0	Enable Hardware Flow Control When this bit is set, the flow control signal operation, based on the fill-level of Rx queue, is enabled. When reset, the flow control operation is disabled.Values: 0x0 (DISABLE): Hardware Flow Control is disabled 0x1 (ENABLE): Hardware Flow Control is enabled Value After Reset: 0x0
			6	DIS_TCP_EF	R/W	0x0	Disable Dropping of TCP/IP Checksum Error Packets When this bit is set, the MAC does not drop the packets which only have the errors detected by the Receive Checksum Offload engine. Such packets have errors only in the encapsulated payload. There are no errors (including FCS error) in the Ethernet packet received by the MAC. When this bit is reset, all error packets are dropped if the FEP bit is reset.Values: 0x0 (ENABLE): Dropping of TCP/IP Checksum Error Packets is enabled 0x1 (DISABLE): Dropping of TCP/IP Checksum Error Packets is disabled Value After Reset: 0x0
			5	RSF	R/W	0x0	Receive Queue Store and Forward When this bit is set, the DWC_ether_qos reads a packet from the Rx queue only after the complete packet has been written to it, ignoring the RTC field of this register. When this bit is reset, the Rx queue operates in the Threshold (cut-through) mode, subject to the threshold specified by the RTC field of this register.Values: 0x0 (DISABLE): Receive Queue Store and Forward is disabled 0x1 (ENABLE): Receive Queue Store and Forward is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			4	FEP	R/W	0x0	Forward Error Packets When this bit is reset, the Rx queue drops packets with error status (CRC error, GMII_ER, watchdog timeout, or overflow). However, if the start byte (write) pointer of a packet is already transferred to the read controller side (in Threshold mode), the packet is not dropped. When this bit is set, all packets except the runt error packets are forwarded to the application or DMA. If the RSF bit is set and the Rx queue overflows when a partial packet is written, the packet is dropped irrespective of the setting of this bit. However, if the RSF bit is reset and the Rx queue overflows when a partial packet is written, a partial packet might be forwarded to the application or DMA.Values: 0x0 (DISABLE): Forward Error Packets is disabled 0x1 (ENABLE): Forward Error Packets is enabled Value After Reset: 0x0
			3	FUP	R/W	0x0	Forward Undersized Good Packets When this bit is set, the Rx queue forwards the undersized good packets (packets with no error and length less than 64 bytes), including pad-bytes and CRC. When this bit is reset, the Rx queue drops all packets of less than 64 bytes, unless a packet is already transferred because of the lower value of Rx Threshold, for example, RTC = 01.Values: 0x0 (DISABLE): Forward Undersized Good Packets is disabled 0x1 (ENABLE): Forward Undersized Good Packets is enabled Value After Reset: 0x0
			2	Reserved_2	R	0x0	Reserved. Value After Reset: 0x0
			1:0	RTC	R/W	0x0	Receive Queue Threshold Control These bits control the threshold level of the MTL Rx queue (in bytes): The received packet is transferred to the application or DMA when the packet size within the MTL Rx queue is larger than the threshold. In addition, full packets with length less than the threshold are automatically transferred. This field is valid only when the RSF bit is zero. This field is ignored when the RSF bit is set to 1.Values: 0x0 (M_64BYTE): 64 0x1 (M_32BYTE): 32 0x2 (M_96BYTE): 96 0x3 (M_128BYTE): 128 Value After Reset: 0x0
0xe34	MTL_RxQ4_Missed_Packet_Overflow_Cnt	The Queue 4 Missed Packet and Overflow Counter register contains the counter for packets missed because of Receive queue packet flush and packets discarded because of Receive queue overflow.	31:28	Reserved_31_28	R	0x0	Reserved. Value After Reset: 0x0
			27	MISCNTOVF	R	0x0	Missed Packet Counter Overflow Bit When set, this bit indicates that the Rx Queue Missed Packet Counter crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Missed Packet Counter overflow not detected 0x1 (ACTIVE): Missed Packet Counter overflow detected Value After Reset: 0x0
			26:16	MISPKTCNT	R	0x0	Missed Packet Counter This field indicates the number of packets missed by the DWC_ether_qos because the application asserted ari_pkt_flush_i[] for this queue. This counter is reset when this register is read with mci_be_i[0] at 1b1. This counter is incremented by 1 when the DMA discards the packet because of buffer unavailability. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0
			15:12	Reserved_15_12	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			11	OVFCNTOVF	R	0x0	Overflow Counter Overflow Bit When set, this bit indicates that the Rx Queue Overflow Packet Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Overflow Counter overflow not detected 0x1 (ACTIVE): Overflow Counter overflow detected Value After Reset: 0x0
			10:0	OVFPKTCNT	R	0x0	Overflow Packet Counter This field indicates the number of packets discarded by the DWC_ether_qos because of Receive queue overflow. This counter is incremented each time the DWC_ether_qos discards an incoming packet because of overflow. This counter is reset when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0
0xe38	MTL_RxQ4_Debug	The Queue 4 Receive Debug register gives the debug status of various blocks related to the Receive queue.	31:30	Reserved_31_30	R	0x0	Reserved. Value After Reset: 0x0
			29:16	PRXQ	R	0x0	Number of Packets in Receive Queue This field indicates the current number of packets in the Rx Queue, that is, Max_Queue_Size/Min_Packet_Size. Value After Reset: 0x0
			15:6	Reserved_15_6	R	0x0	Reserved. Value After Reset: 0x0
			5:4	RXQSTS	R	0x0	MTL Rx Queue Fill-Level Status This field gives the status of the fill-level of the Rx Queue:Values: 0x0 (EMPTY): Rx Queue empty 0x1 (BLW_THR): Rx Queue fill-level below flow-control deactivate threshold 0x2 (ABV_THR): Rx Queue fill-level above flow-control activate threshold 0x3 (FULL): Rx Queue full Value After Reset: 0x0
			3	Reserved_3	R	0x0	Reserved. Value After Reset: 0x0
			2:1	RRCSTS	R	0x0	MTL Rx Queue Read Controller State This field gives the state of the Rx queue Read controller:Values: 0x0 (IDLE): Idle state 0x1 (READ_DATA): Reading packet data 0x2 (READ_STS): Reading packet status (or timestamp) 0x3 (FLUSH): Flushing the packet data and status Value After Reset: 0x0
			0	RWCSTS	R	0x0	MTL Rx Queue Write Controller Active Status When high, this bit indicates that the MTL Rx queue Write controller is active, and it is transferring a received packet to the Rx Queue.Values: 0x0 (INACTIVE): MTL Rx Queue Write Controller Active Status not detected 0x1 (ACTIVE): MTL Rx Queue Write Controller Active Status detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xe3c	MTL_RxQ4_Control	The Queue Receive Control register controls the receive arbitration and passing of received packets to the application.	31:4	Reserved_31_4	R	0x0	Reserved. Value After Reset: 0x0
			3	RXQ_FRM_ARBIT	R/W	0x0	Receive Queue Packet Arbitration When this bit is set, the DWC_ether_qos drives the packet data to the ARI interface such that the entire packet data of currently-selected queue is transmitted before switching to other queue. When this bit is reset, the DWC_ether_qos drives the packet data to the ARI interface such that the following amount of data of currently-selected queue is transmitted before switching to other queue: PBL amount of data (indicated by ari_qN_pbl_i[]) or Complete data of a packet The status and the timestamp are not a part of the PBL data. Therefore, the DWC_ether_qos drives the complete status (including timestamp status) during first PBL request for the packet (in store-and-forward mode) or the last PBL request for the packet (in Threshold mode).Values: 0x0 (DISABLE): Receive Queue Packet Arbitration is disabled 0x1 (ENABLE): Receive Queue Packet Arbitration is enabled Value After Reset: 0x0
			2:0	RXQ_WEGT	R/W	0x0	Receive Queue Weight This field indicates the weight assigned to the Rx Queue 1. This field needs to be programmed with one value less than the required weight, that is, reset value of 0 indicates weight of 1, value of 1 indicates weight of 2, and so on. The weight is used as the number of continuous PBL or packets requests (depending on the RXQ_FRM_ARBIT) allocated to the queue in one arbitration cycle. Note: The change in value of RXQ_WEGT takes effect only after the completion of current service round or when there is change from RAA=SP to RAA=WSP algorithm. This approach is taken so that there is smooth transition. For the RXQ_WEGT value to take effect at the start, the MTL_RxQ1_Control registers must be programmed before the MTL_Operation_Mode register. Value After Reset: 0x0
			31:22	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			21:16	TQS	R/W	0x0	Transmit Queue Size This field indicates the size of the allocated Transmit queues in blocks of 256 bytes, the reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on. So, program TQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (TQS+1)*256 bytes. Value After Reset: 0x0
			15:7	Reserved_15_7	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xe40	MTL_TxQ5_Operation_Mode	The Queue 5 Transmit Operation Mode register establishes the Transmit queue operating modes and commands.	6:4	TTC	R/W	0x0	Transmit Threshold Control These bits control the threshold level of the MTL Tx Queue. The transmission starts when the packet size within the MTL Tx Queue is larger than the threshold. In addition, full packets with length less than the threshold are also transmitted. These bits are used only when the TSF bit is reset.Values: 0x0 (M_32BYTES): 32 0x1 (M_64BYTES): 64 0x2 (M_96BYTES): 96 0x3 (M_128BYTES): 128 0x4 (M_192BYTES): 192 0x5 (M_256BYTES): 256 0x6 (M_384BYTES): 384 0x7 (M_512BYTES): 512 Value After Reset: 0x0
			3:2	TXQEN	R/W	0x0	Transmit Queue Enable This field is used to enable/disable the transmit queue 1. Note: In multiple Tx queues configuration, all the queues are disabled by default. Enable the Tx queue by programming this field.Values: 0x0 (DISABLE): Not enabled 0x1 (EN_IF_AV): Enable in AV mode 0x2 (ENABLE): Enabled 0x3 (RSVD2): Reserved Value After Reset: 0x0
			1	TSF	R/W	0x0	Transmit Store and Forward When this bit is set, the transmission starts when a full packet resides in the MTL Tx queue. When this bit is set, the TTC values specified in Bits[6:4] of this register are ignored. This bit should be changed only when the transmission is stopped.Values: 0x0 (DISABLE): Transmit Store and Forward is disabled 0x1 (ENABLE): Transmit Store and Forward is enabled Value After Reset: 0x0
			0	FTQ	R/W	0x0	Flush Transmit Queue When this bit is set, the Tx queue controller logic is reset to its default values. Therefore, all the data in the Tx queue is lost or flushed. This bit is internally reset when the flushing operation is complete. Until this bit is reset, you must not write to the MTL_TxQ1_Operation_Mode register. The data which is already accepted by the MAC transmitter is not flushed. It is scheduled for transmission and results in underflow and runt packet transmission. Note: The flush operation is complete only when the Tx queue is empty and the application has accepted the pending Tx Status of all transmitted packets. To complete this flush operation, the PHY Tx clock (clk_tx_i) must be active. Access restrictions apply. MAC initiates the Tx queue flush when application writes 1. MAC writes 0 after the Tx queue flush is complete. MAC ignores when application writes 0. Values: 0x0 (DISABLE): Flush Transmit Queue is disabled 0x1 (ENABLE): Flush Transmit Queue is enabled Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xe44	MTL_TxQ5_Underflow	The Queue 5 Underflow Counter register contains the counter for packets aborted because of Transmit queue underflow and packets missed because of Receive queue packet flush	31:12	Reserved_31_12	R	0x0	Reserved. Value After Reset: 0x0
			11	UFCNTOVF	R	0x0	Overflow Bit for Underflow Packet Counter This bit is set every time the Tx queue Underflow Packet Counter field overflows, that is, it has crossed the maximum count. In such a scenario, the overflow packet counter is reset to all-zeros and this bit indicates that the rollover happened. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Overflow not detected for Underflow Packet Counter 0x1 (ACTIVE): Overflow detected for Underflow Packet Counter Value After Reset: 0x0
			10:0	UFFRMCNT	R	0x0	Underflow Packet Counter This field indicates the number of packets aborted by the controller because of Tx Queue Underflow. This counter is incremented each time the MAC aborts outgoing packet because of underflow. The counter is cleared when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0
0xe48	MTL_TxQ5_Debug	The Queue 5 Transmit Debug register gives the debug status of various blocks related to the Transmit queue.	31:23	Reserved_31_23	R	0x0	Reserved. Value After Reset: 0x0
			22:20	STXSTSF	R	0x0	Number of Status Words in Tx Status FIFO of Queue This field indicates the current number of status in the Tx Status FIFO of this queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of status words in Tx Status FIFO. Value After Reset: 0x0
			19	Reserved_19	R	0x0	Reserved. Value After Reset: 0x0
			18:16	PTXQ	R	0x0	Number of Packets in the Transmit Queue This field indicates the current number of packets in the Tx Queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of packets in the Transmit queue. Value After Reset: 0x0
			15:6	Reserved_15_6	R	0x0	Reserved. Value After Reset: 0x0
			5	TXSTSFSTS	R	0x0	MTL Tx Status FIFO Full Status When high, this bit indicates that the MTL Tx Status FIFO is full. Therefore, the MTL cannot accept any more packets for transmission.Values: 0x0 (INACTIVE): MTL Tx Status FIFO Full status is not detected 0x1 (ACTIVE): MTL Tx Status FIFO Full status is detected Value After Reset: 0x0
			4	TXQSTS	R	0x0	MTL Tx Queue Not Empty Status When this bit is high, it indicates that the MTL Tx Queue is not empty and some data is left for transmission.Values: 0x0 (INACTIVE): MTL Tx Queue Not Empty status is not detected 0x1 (ACTIVE): MTL Tx Queue Not Empty status is detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			3	TWCSTS	R	0x0	MTL Tx Queue Write Controller Status When high, this bit indicates that the MTL Tx Queue Write Controller is active, and it is transferring the data to the Tx Queue.Values: 0x0 (INACTIVE): MTL Tx Queue Write Controller status is not detected 0x1 (ACTIVE): MTL Tx Queue Write Controller status is detected Value After Reset: 0x0
			2:1	TRCSTS	R	0x0	MTL Tx Queue Read Controller Status This field indicates the state of the Tx Queue Read Controller:Values: 0x0 (IDLE): Idle state 0x1 (READ): Read state (transferring data to the MAC transmitter) 0x2 (WAIT): Waiting for pending Tx Status from the MAC transmitter 0x3 (FLUSH): Flushing the Tx queue because of the Packet Abort request from the MAC Value After Reset: 0x0
			0	TXQPAUSED	R	0x0	Transmit Queue in Pause When this bit is high and the Rx flow control is enabled, it indicates that the Tx Queue is in the Pause condition (in the full-duplex only mode) because of the following: Reception of the PFC packet for the priorities assigned to the Tx Queue when PFC is enabled Reception of 802.3x Pause packet when PFC is disabled Values: 0x0 (INACTIVE): Transmit Queue in Pause status is not detected 0x1 (ACTIVE): Transmit Queue in Pause status is detected Value After Reset: 0x0
0xe50	MTL_TxQ5_ETS_Control	The Queue ETS Control register controls the enhanced transmission selection operation.	31:7	Reserved_31_7	R	0x0	Reserved. Value After Reset: 0x0
			6:4	SLC	R/W	0x0	Slot Count If the credit-based shaper algorithm is enabled, the software can program the number of slots (of duration programmed in DMA_CH1_Slot_Interval register) over which the average transmitted bits per slot, provided in the MTL_TxQ1_ETS_Status register, need to be computed for Queue. The encoding is as follows:Values: 0x0 (M_1_SLOT): 1 slot 0x1 (M_2_SLOT): 2 slots 0x2 (M_4_SLOT): 4 slots 0x3 (M_8_SLOT): 8 slots 0x4 (M_16_SLOT): 16 slots 0x5 (RSVD): Reserved Value After Reset: 0x0
			3	CC	R/W	0x0	Credit Control When this bit is set, the accumulated credit parameter in the credit-based shaper algorithm logic is not reset to zero when there is positive credit and no packet to transmit in Tx Queue 1. The credit accumulates even when there is no packet waiting in Tx Queue 1 and another Tx Queue is transmitting. When this bit is reset, the accumulated credit parameter in the credit-based shaper algorithm logic is set to zero when there is positive credit and no packet to transmit in Tx Queue 1. When there is no packet waiting in Tx Queue 1 and other channel is transmitting, no credit is accumulated.Values: 0x0 (DISABLE): Credit Control is disabled 0x1 (ENABLE): Credit Control is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			2	AVALG	R/W	0x0	AV Algorithm When Queue 1 is programmed for AV, this field configures the scheduling algorithm for this queue: This bit when set, indicates credit based shaper algorithm (CBS) is selected for Queue 1 traffic. When reset, strict priority is selected.Values: 0x0 (DISABLE): CBS Algorithm is disabled 0x1 (ENABLE): CBS Algorithm is enabled Value After Reset: 0x0
			1:0	Reserved_1_0	R	0x0	Reserved. Value After Reset: 0x0
0xe54	MTL_TxQ5_ETS_Status	The Queue 5 ETS Status register provides the average traffic transmitted in Queue 5.	31:24	Reserved_31_24	R	0x0	Reserved. Value After Reset: 0x0
			23:0	ABS	R	0x0	Average Bits per Slot This field contains the average transmitted bits per slot. If AV operation is enabled, this field is computed over number of slots, programmed in the SLC field of MTL_TxQ1_ETS_CONTROL register. The maximum value of this field is 0x6_4000 in 100 Mbps, 0x3E_8000 in 1000 Mbps and 9C_4000 in 2500 Mbps mode respectively. When the DCB operation is enabled for Queue, this field is computed over every 10 million bit times slot (4 ms in 2500 Mbps; 10 ms in 1000 Mbps; 100 ms in 100 Mbps). The maximum value is 0x989680. Value After Reset: 0x0
0xe58	MTL_TxQ5_Quantum_Weight	The Queue 5 idleSlopeCredit, Quantum or Weights register provides the average traffic transmitted in Queue 5.	31:21	Reserved_31_21	R	0x0	Reserved. Value After Reset: 0x0
			20:0	ISCQW	R/W	0x0	idleSlopeCredit, Quantum or Weights idleSlopeCredit When AV feature is enabled, this field contains the idleSlopeCredit value required for the credit-based shaper algorithm for Queue 1. This is the rate of change of credit in bits per cycle (40 ns for 100 Mbps; 8 ns for 1000 Mbps; 3.2 ns for 2500 Mbps) when the credit is increasing. The software should program this field with computed credit in bits per cycle scaled by 1,024. The maximum value is portTransmitRate, that is, 0x2000 in 1000/2500 Mbps mode and 0x1000 in 100 Mbps mode. Bits[20:14] must be written to zero. Quantum When the DCB operation is enabled with DWRR algorithm for Queue 1 traffic, this field contains the quantum value in bytes to be added to credit during every queue scanning cycle. The maximum value is 0x1312D0 bytes. Weights When DCB operation is enabled with WFQ algorithm for Queue 1 traffic, this field contains the weight for this queue. The maximum value is 0x3FFF where weight of 0 indicates 100% bandwidth. Bits[20:14] must be written to zero. When DCB operation or generic queuing operation is enabled with WRR algorithm for Queue 1 traffic, this field contains the weight for this queue. The maximum value is 0x64. Bits [20:7] must be written to zero. Note 1: In multiple Queue configuration this field in respective per queue register must be programmed to some non-zero value when multiple queues are enabled or single queue other than Q0 is enabled. This field need not be programmed when only Q0 is enabled. In general, when WRR algorithm is selected a non-zero value must be programmed on both Receive and Transmit. In Receive, the register is MTL_Operation_Mode register. Note 2: For WFQ algorithm, higher the programmed weights lesser the bandwidth allocated for that Transmit Queue. The finish time is not a function of particular packet alone but it is as per the formula: (previous_finish_time of particular Transmit Queue + (weights*packet_size)) Note 3: The weights programmed do not correspond to the number of packets but the fraction of bandwidth or time allocated for particular queue w.r.t. total BW or time. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xe5c	MTL_TxQ5_SendSlopeCredit	The sendSlopeCredit register contains the sendSlope credit value required for the credit-based shaper algorithm for the Queue.	31:14	Reserved_31_14	R	0x0	Reserved. Value After Reset: 0x0
			13:0	SSC	R/W	0x0	sendSlopeCredit Value When AV operation is enabled, this field contains the sendSlopeCredit value required for credit-based shaper algorithm for Tx Queue 1. This is the rate of change of credit in bits per cycle (40 ns, 8 ns and 3.2 ns for 100 Mbps, 1000 Mbps and 2500 Mbps respectively) when the credit is decreasing. The software should program this field with computed credit in bits per cycle scaled by 1,024. The maximum value is portTransmitRate, that is, 0x2000 in 1000/2500 Mbps mode and 0x1000 in 100 Mbps mode. This field should be programmed with absolute sendSlopeCredit value. The credit-based shaper logic subtracts it from the accumulated credit when Tx Queue 1 is selected for transmission. Value After Reset: 0x0
0xe60	MTL_TxQ5_HiCredit	The hiCredit register contains the hiCredit value required for the credit-based shaper algorithm for the Queue.	31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28:0	HC	R/W	0x0	hiCredit Value When the AV feature is enabled, this field contains the hiCredit value required for the credit-based shaper algorithm. This is the maximum value that can be accumulated in the credit parameter. This is specified in bits scaled by 1,024. The maximum value is maxInterferenceSize, that is, best-effort maximum packet size (16,384 bytes or 131,072 bits). The value to be specified is 131,072 * 1,024 = 134,217,728 or 0x0800_0000. Value After Reset: 0x0
0xe64	MTL_TxQ5_LoCredit	The loCredit register contains the loCredit value required for the credit-based shaper algorithm for the Queue.	31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28:0	LC	R/W	0x0	loCredit Value When AV operation is enabled, this field contains the loCredit value required for the credit-based shaper algorithm. This is the minimum value that can be accumulated in the credit parameter. This is specified in bits scaled by 1,024. The maximum value to be programmed is corresponds to twice the maxFrameSize transmitted from this queue. If the maxFrameSize is 8192 bytes, then (8192*2) * 8 * 1024 = 134,217,728 or 0x0800_0000. Because it is a negative value, the programmed value is 2's complement of the value, that is, 0x1800_0000. Value After Reset: 0x0
			31:25	Reserved_31_25	R	0x0	Reserved. Value After Reset: 0x0
			24	RXOIE	R/W	0x0	Receive Queue Overflow Interrupt Enable When this bit is set, the Receive Queue Overflow interrupt is enabled. When this bit is reset, the Receive Queue Overflow interrupt is disabled.Values: 0x0 (DISABLE): Receive Queue Overflow Interrupt is disabled 0x1 (ENABLE): Receive Queue Overflow Interrupt is enabled Value After Reset: 0x0
			23:17	Reserved_23_17	R	0x0	Reserved. Value After Reset: 0x0
			16	RXOVFIS	R/W	0x0	Receive Queue Overflow Interrupt Status This bit indicates that the Receive Queue had an overflow while receiving the packet. If a partial packet is transferred to the application, the overflow status is set in RDES3[21]. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Queue Overflow Interrupt Status not detected 0x1 (ACTIVE): Receive Queue Overflow Interrupt Status detected Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xe6c	MTL_Q5_Interrupt_Control_Status	This register contains the interrupt enable and status bits for the queue 5 interrupts.	15:10	Reserved_15_10	R	0x0	Reserved. Value After Reset: 0x0
			9	ABPSIE	R/W	0x0	Average Bits Per Slot Interrupt Enable When this bit is set, the MAC asserts the sbd_intr_o or mci_intr_o interrupt when the average bits per slot status is updated. When this bit is cleared, the interrupt is not asserted for such an event.Values: 0x0 (DISABLE): Average Bits Per Slot Interrupt is disabled 0x1 (ENABLE): Average Bits Per Slot Interrupt is enabled Value After Reset: 0x0
			8	TXUIE	R/W	0x0	Transmit Queue Underflow Interrupt Enable When this bit is set, the Transmit Queue Underflow interrupt is enabled. When this bit is reset, the Transmit Queue Underflow interrupt is disabled.Values: 0x0 (DISABLE): Transmit Queue Underflow Interrupt Status is disabled 0x1 (ENABLE): Transmit Queue Underflow Interrupt Status is enabled Value After Reset: 0x0
			7:2	Reserved_7_2	R	0x0	Reserved. Value After Reset: 0x0
			1	ABPSIS	R/W	0x0	Average Bits Per Slot Interrupt Status When set, this bit indicates that the MAC has updated the ABS value. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Average Bits Per Slot Interrupt Status not detected 0x1 (ACTIVE): Average Bits Per Slot Interrupt Status detected Value After Reset: 0x0 Testable: untestable
			0	TXUNFIS	R/W	0x0	Transmit Queue Underflow Interrupt Status This bit indicates that the Transmit Queue had an underflow while transmitting the packet. Transmission is suspended and an Underflow Error TDES3[2] is set. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Queue Underflow Interrupt Status not detected 0x1 (ACTIVE): Transmit Queue Underflow Interrupt Status detected Value After Reset: 0x0 Testable: untestable
			31:26	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			25:20	RQS	R/W	0x0	Receive Queue Size This field indicates the size of the allocated Receive queues in blocks of 256 bytesthe reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on. So, program RQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (RQS+1)*256 bytes. Value After Reset: 0x0
			19	Reserved_19_y	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xe70	MTL_RxQ5_Operation_Mode	The Queue 5 Receive Operation Mode register establishes the Receive queue operating modes and command.	18:14	RFD	R/W	0x0	Threshold for Deactivating Flow Control (in half-duplex and full-duplex modes) These bits control the threshold (fill-level of Rx queue) at which the flow control is de-asserted after activation: 0: Full minus 1 KB, that is, FULL 1 KB 1: Full minus 1.5 KB, that is, FULL 1.5 KB 2: Full minus 2 KB, that is, FULL 2 KB 3: Full minus 2.5 KB, that is, FULL 2.5 KB ... 30: Full minus 16 KB, that is, FULL 16 KB 31: Full minus 16.5 KB, that is, FULL 16.5 KB The de-assertion is effective only after flow control is asserted. Note: The value must be programmed in such a way to make sure that the threshold is a positive number. When the EHFC is set high, these values are applicable only when the Rx queue size determined by the RQS field of this register, is equal to or greater than 4 KB. For a given queue size, the values ranges between 0 and the encoding for FULL minus (QSIZE - 0.5 KB) and all other values are illegal. Here the term FULL and QSIZE refers to the queue size determined by the RQS field of this register. Value After Reset: 0x0
			13	Reserved_13_y	R	0x0	Reserved. Value After Reset: 0x0
			12:8	RFA	R/W	0x0	Threshold for Activating Flow Control (in half-duplex and full-duplex These bits control the threshold (fill-level of Rx queue) at which the flow control is activated: For more information on encoding for this field, see RFD. Value After Reset: 0x0
			7	EHFC	R/W	0x0	Enable Hardware Flow Control When this bit is set, the flow control signal operation, based on the fill-level of Rx queue, is enabled. When reset, the flow control operation is disabled.Values: 0x0 (DISABLE): Hardware Flow Control is disabled 0x1 (ENABLE): Hardware Flow Control is enabled Value After Reset: 0x0
			6	DIS_TCP_EF	R/W	0x0	Disable Dropping of TCP/IP Checksum Error Packets When this bit is set, the MAC does not drop the packets which only have the errors detected by the Receive Checksum Offload engine. Such packets have errors only in the encapsulated payload. There are no errors (including FCS error) in the Ethernet packet received by the MAC. When this bit is reset, all error packets are dropped if the FEP bit is reset.Values: 0x0 (ENABLE): Dropping of TCP/IP Checksum Error Packets is enabled 0x1 (DISABLE): Dropping of TCP/IP Checksum Error Packets is disabled Value After Reset: 0x0
			5	RSF	R/W	0x0	Receive Queue Store and Forward When this bit is set, the DWC_ether_qos reads a packet from the Rx queue only after the complete packet has been written to it, ignoring the RTC field of this register. When this bit is reset, the Rx queue operates in the Threshold (cut-through) mode, subject to the threshold specified by the RTC field of this register.Values: 0x0 (DISABLE): Receive Queue Store and Forward is disabled 0x1 (ENABLE): Receive Queue Store and Forward is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			4	FEP	R/W	0x0	Forward Error Packets When this bit is reset, the Rx queue drops packets with error status (CRC error, GMII_ER, watchdog timeout, or overflow). However, if the start byte (write) pointer of a packet is already transferred to the read controller side (in Threshold mode), the packet is not dropped. When this bit is set, all packets except the runt error packets are forwarded to the application or DMA. If the RSF bit is set and the Rx queue overflows when a partial packet is written, the packet is dropped irrespective of the setting of this bit. However, if the RSF bit is reset and the Rx queue overflows when a partial packet is written, a partial packet might be forwarded to the application or DMA.Values: 0x0 (DISABLE): Forward Error Packets is disabled 0x1 (ENABLE): Forward Error Packets is enabled Value After Reset: 0x0
			3	FUP	R/W	0x0	Forward Undersized Good Packets When this bit is set, the Rx queue forwards the undersized good packets (packets with no error and length less than 64 bytes), including pad-bytes and CRC. When this bit is reset, the Rx queue drops all packets of less than 64 bytes, unless a packet is already transferred because of the lower value of Rx Threshold, for example, RTC = 01.Values: 0x0 (DISABLE): Forward Undersized Good Packets is disabled 0x1 (ENABLE): Forward Undersized Good Packets is enabled Value After Reset: 0x0
			2	Reserved_2	R	0x0	Reserved. Value After Reset: 0x0
			1:0	RTC	R/W	0x0	Receive Queue Threshold Control These bits control the threshold level of the MTL Rx queue (in bytes): The received packet is transferred to the application or DMA when the packet size within the MTL Rx queue is larger than the threshold. In addition, full packets with length less than the threshold are automatically transferred. This field is valid only when the RSF bit is zero. This field is ignored when the RSF bit is set to 1.Values: 0x0 (M_64BYTE): 64 0x1 (M_32BYTE): 32 0x2 (M_96BYTE): 96 0x3 (M_128BYTE): 128 Value After Reset: 0x0
0xe74	MTL_RxQ5_Missed_Packet_Overflow_Cnt	The Queue 5 Missed Packet and Overflow Counter register contains the counter for packets missed because of Receive queue packet flush and packets discarded because of Receive queue overflow.	31:28	Reserved_31_28	R	0x0	Reserved. Value After Reset: 0x0
			27	MISCNTOVF	R	0x0	Missed Packet Counter Overflow Bit When set, this bit indicates that the Rx Queue Missed Packet Counter crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Missed Packet Counter overflow not detected 0x1 (ACTIVE): Missed Packet Counter overflow detected Value After Reset: 0x0
			26:16	MISPKTCNT	R	0x0	Missed Packet Counter This field indicates the number of packets missed by the DWC_ether_qos because the application asserted ari_pkt_flush_i[] for this queue. This counter is reset when this register is read with mci_be_i[0] at 1b1. This counter is incremented by 1 when the DMA discards the packet because of buffer unavailability. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0
			15:12	Reserved_15_12	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			11	OVFCNTOVF	R	0x0	Overflow Counter Overflow Bit When set, this bit indicates that the Rx Queue Overflow Packet Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Overflow Counter overflow not detected 0x1 (ACTIVE): Overflow Counter overflow detected Value After Reset: 0x0
			10:0	OVFPKTCNT	R	0x0	Overflow Packet Counter This field indicates the number of packets discarded by the DWC_ether_qos because of Receive queue overflow. This counter is incremented each time the DWC_ether_qos discards an incoming packet because of overflow. This counter is reset when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0
0xe78	MTL_RxQ5_Debug	The Queue 5 Receive Debug register gives the debug status of various blocks related to the Receive queue.	31:30	Reserved_31_30	R	0x0	Reserved. Value After Reset: 0x0
			29:16	PRXQ	R	0x0	Number of Packets in Receive Queue This field indicates the current number of packets in the Rx Queue, that is, Max_Queue_Size/Min_Packet_Size. Value After Reset: 0x0
			15:6	Reserved_15_6	R	0x0	Reserved. Value After Reset: 0x0
			5:4	RXQSTS	R	0x0	MTL Rx Queue Fill-Level Status This field gives the status of the fill-level of the Rx Queue:Values: 0x0 (EMPTY): Rx Queue empty 0x1 (BLW_THR): Rx Queue fill-level below flow-control deactivate threshold 0x2 (ABV_THR): Rx Queue fill-level above flow-control activate threshold 0x3 (FULL): Rx Queue full Value After Reset: 0x0
			3	Reserved_3	R	0x0	Reserved. Value After Reset: 0x0
			2:1	RRCSTS	R	0x0	MTL Rx Queue Read Controller State This field gives the state of the Rx queue Read controller:Values: 0x0 (IDLE): Idle state 0x1 (READ_DATA): Reading packet data 0x2 (READ_STS): Reading packet status (or timestamp) 0x3 (FLUSH): Flushing the packet data and status Value After Reset: 0x0
			0	RWCSTS	R	0x0	MTL Rx Queue Write Controller Active Status When high, this bit indicates that the MTL Rx queue Write controller is active, and it is transferring a received packet to the Rx Queue.Values: 0x0 (INACTIVE): MTL Rx Queue Write Controller Active Status not detected 0x1 (ACTIVE): MTL Rx Queue Write Controller Active Status detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xe7c	MTL_RxQ5_Control	The Queue Receive Control register controls the receive arbitration and passing of received packets to the application.	31:4	Reserved_31_4	R	0x0	Reserved. Value After Reset: 0x0
			3	RXQ_FRM_ARBIT	R/W	0x0	Receive Queue Packet Arbitration When this bit is set, the DWC_ether_qos drives the packet data to the ARI interface such that the entire packet data of currently-selected queue is transmitted before switching to other queue. When this bit is reset, the DWC_ether_qos drives the packet data to the ARI interface such that the following amount of data of currently-selected queue is transmitted before switching to other queue: PBL amount of data (indicated by ari_qN_pbl_i[]) or Complete data of a packet The status and the timestamp are not a part of the PBL data. Therefore, the DWC_ether_qos drives the complete status (including timestamp status) during first PBL request for the packet (in store-and-forward mode) or the last PBL request for the packet (in Threshold mode).Values: 0x0 (DISABLE): Receive Queue Packet Arbitration is disabled 0x1 (ENABLE): Receive Queue Packet Arbitration is enabled Value After Reset: 0x0
			2:0	RXQ_WEGT	R/W	0x0	Receive Queue Weight This field indicates the weight assigned to the Rx Queue 1. This field needs to be programmed with one value less than the required weight, that is, reset value of 0 indicates weight of 1, value of 1 indicates weight of 2, and so on. The weight is used as the number of continuous PBL or packets requests (depending on the RXQ_FRM_ARBIT) allocated to the queue in one arbitration cycle. Note: The change in value of RXQ_WEGT takes effect only after the completion of current service round or when there is change from RAA=SP to RAA=WSP algorithm. This approach is taken so that there is smooth transition. For the RXQ_WEGT value to take effect at the start, the MTL_RxQ1_Control registers must be programmed before the MTL_Operation_Mode register. Value After Reset: 0x0
			31:22	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			21:16	TQS	R/W	0x0	Transmit Queue Size This field indicates the size of the allocated Transmit queues in blocks of 256 bytes, the reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on. So, program TQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (TQS+1)*256 bytes. Value After Reset: 0x0
			15:7	Reserved_15_7	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xe80	MTL_TxQ6_Operation_Mode	The Queue 6 Transmit Operation Mode register establishes the Transmit queue operating modes and commands.	6:4	TTC	R/W	0x0	Transmit Threshold Control These bits control the threshold level of the MTL Tx Queue. The transmission starts when the packet size within the MTL Tx Queue is larger than the threshold. In addition, full packets with length less than the threshold are also transmitted. These bits are used only when the TSF bit is reset.Values: 0x0 (M_32BYTES): 32 0x1 (M_64BYTES): 64 0x2 (M_96BYTES): 96 0x3 (M_128BYTES): 128 0x4 (M_192BYTES): 192 0x5 (M_256BYTES): 256 0x6 (M_384BYTES): 384 0x7 (M_512BYTES): 512 Value After Reset: 0x0
			3:2	TXQEN	R/W	0x0	Transmit Queue Enable This field is used to enable/disable the transmit queue 1. Note: In multiple Tx queues configuration, all the queues are disabled by default. Enable the Tx queue by programming this field.Values: 0x0 (DISABLE): Not enabled 0x1 (EN_IF_AV): Enable in AV mode 0x2 (ENABLE): Enabled 0x3 (RSVD2): Reserved Value After Reset: 0x0
			1	TSF	R/W	0x0	Transmit Store and Forward When this bit is set, the transmission starts when a full packet resides in the MTL Tx queue. When this bit is set, the TTC values specified in Bits[6:4] of this register are ignored. This bit should be changed only when the transmission is stopped.Values: 0x0 (DISABLE): Transmit Store and Forward is disabled 0x1 (ENABLE): Transmit Store and Forward is enabled Value After Reset: 0x0
			0	FTQ	R/W	0x0	Flush Transmit Queue When this bit is set, the Tx queue controller logic is reset to its default values. Therefore, all the data in the Tx queue is lost or flushed. This bit is internally reset when the flushing operation is complete. Until this bit is reset, you must not write to the MTL_TxQ1_Operation_Mode register. The data which is already accepted by the MAC transmitter is not flushed. It is scheduled for transmission and results in underflow and runt packet transmission. Note: The flush operation is complete only when the Tx queue is empty and the application has accepted the pending Tx Status of all transmitted packets. To complete this flush operation, the PHY Tx clock (clk_tx_i) must be active. Access restrictions apply. MAC initiates the Tx queue flush when application writes 1. MAC writes 0 after the Tx queue flush is complete. MAC ignores when application writes 0. Values: 0x0 (DISABLE): Flush Transmit Queue is disabled 0x1 (ENABLE): Flush Transmit Queue is enabled Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xe84	MTL_TxQ6_Underflow	The Queue 6 Underflow Counter register contains the counter for packets aborted because of Transmit queue underflow and packets missed because of Receive queue packet flush	31:12	Reserved_31_12	R	0x0	Reserved. Value After Reset: 0x0
			11	UFCNTOVF	R	0x0	Overflow Bit for Underflow Packet Counter This bit is set every time the Tx queue Underflow Packet Counter field overflows, that is, it has crossed the maximum count. In such a scenario, the overflow packet counter is reset to all-zeros and this bit indicates that the rollover happened. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Overflow not detected for Underflow Packet Counter 0x1 (ACTIVE): Overflow detected for Underflow Packet Counter Value After Reset: 0x0
			10:0	UFFRMCNT	R	0x0	Underflow Packet Counter This field indicates the number of packets aborted by the controller because of Tx Queue Underflow. This counter is incremented each time the MAC aborts outgoing packet because of underflow. The counter is cleared when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0
0xe88	MTL_TxQ6_Debug	The Queue 6 Transmit Debug register gives the debug status of various blocks related to the Transmit queue.	31:23	Reserved_31_23	R	0x0	Reserved. Value After Reset: 0x0
			22:20	STXSTSF	R	0x0	Number of Status Words in Tx Status FIFO of Queue This field indicates the current number of status in the Tx Status FIFO of this queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of status words in Tx Status FIFO. Value After Reset: 0x0
			19	Reserved_19	R	0x0	Reserved. Value After Reset: 0x0
			18:16	PTXQ	R	0x0	Number of Packets in the Transmit Queue This field indicates the current number of packets in the Tx Queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of packets in the Transmit queue. Value After Reset: 0x0
			15:6	Reserved_15_6	R	0x0	Reserved. Value After Reset: 0x0
			5	TXSTSFSTS	R	0x0	MTL Tx Status FIFO Full Status When high, this bit indicates that the MTL Tx Status FIFO is full. Therefore, the MTL cannot accept any more packets for transmission.Values: 0x0 (INACTIVE): MTL Tx Status FIFO Full status is not detected 0x1 (ACTIVE): MTL Tx Status FIFO Full status is detected Value After Reset: 0x0
			4	TXQSTS	R	0x0	MTL Tx Queue Not Empty Status When this bit is high, it indicates that the MTL Tx Queue is not empty and some data is left for transmission.Values: 0x0 (INACTIVE): MTL Tx Queue Not Empty status is not detected 0x1 (ACTIVE): MTL Tx Queue Not Empty status is detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			3	TWCSTS	R	0x0	MTL Tx Queue Write Controller Status When high, this bit indicates that the MTL Tx Queue Write Controller is active, and it is transferring the data to the Tx Queue.Values: 0x0 (INACTIVE): MTL Tx Queue Write Controller status is not detected 0x1 (ACTIVE): MTL Tx Queue Write Controller status is detected Value After Reset: 0x0
			2:1	TRCSTS	R	0x0	MTL Tx Queue Read Controller Status This field indicates the state of the Tx Queue Read Controller:Values: 0x0 (IDLE): Idle state 0x1 (READ): Read state (transferring data to the MAC transmitter) 0x2 (WAIT): Waiting for pending Tx Status from the MAC transmitter 0x3 (FLUSH): Flushing the Tx queue because of the Packet Abort request from the MAC Value After Reset: 0x0
			0	TXQPAUSED	R	0x0	Transmit Queue in Pause When this bit is high and the Rx flow control is enabled, it indicates that the Tx Queue is in the Pause condition (in the full-duplex only mode) because of the following: Reception of the PFC packet for the priorities assigned to the Tx Queue when PFC is enabled Reception of 802.3x Pause packet when PFC is disabled Values: 0x0 (INACTIVE): Transmit Queue in Pause status is not detected 0x1 (ACTIVE): Transmit Queue in Pause status is detected Value After Reset: 0x0
0xe90	MTL_TxQ6_ETS_Control	The Queue ETS Control register controls the enhanced transmission selection operation.	31:7	Reserved_31_7	R	0x0	Reserved. Value After Reset: 0x0
			6:4	SLC	R/W	0x0	Slot Count If the credit-based shaper algorithm is enabled, the software can program the number of slots (of duration programmed in DMA_CH1_Slot_Interval register) over which the average transmitted bits per slot, provided in the MTL_TxQ1_ETS_Status register, need to be computed for Queue. The encoding is as follows:Values: 0x0 (M_1_SLOT): 1 slot 0x1 (M_2_SLOT): 2 slots 0x2 (M_4_SLOT): 4 slots 0x3 (M_8_SLOT): 8 slots 0x4 (M_16_SLOT): 16 slots 0x5 (RSVD): Reserved Value After Reset: 0x0
			3	CC	R/W	0x0	Credit Control When this bit is set, the accumulated credit parameter in the credit-based shaper algorithm logic is not reset to zero when there is positive credit and no packet to transmit in Tx Queue 1. The credit accumulates even when there is no packet waiting in Tx Queue 1 and another Tx Queue is transmitting. When this bit is reset, the accumulated credit parameter in the credit-based shaper algorithm logic is set to zero when there is positive credit and no packet to transmit in Tx Queue 1. When there is no packet waiting in Tx Queue 1 and other channel is transmitting, no credit is accumulated.Values: 0x0 (DISABLE): Credit Control is disabled 0x1 (ENABLE): Credit Control is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			2	AVALG	R/W	0x0	AV Algorithm When Queue 1 is programmed for AV, this field configures the scheduling algorithm for this queue: This bit when set, indicates credit based shaper algorithm (CBS) is selected for Queue 1 traffic. When reset, strict priority is selected.Values: 0x0 (DISABLE): CBS Algorithm is disabled 0x1 (ENABLE): CBS Algorithm is enabled Value After Reset: 0x0
			1:0	Reserved_1_0	R	0x0	Reserved. Value After Reset: 0x0
0xe94	MTL_TxQ6_ETS_Status	The Queue 6 ETS Status register provides the average traffic transmitted in Queue 6.	31:24	Reserved_31_24	R	0x0	Reserved. Value After Reset: 0x0
			23:0	ABS	R	0x0	Average Bits per Slot This field contains the average transmitted bits per slot. If AV operation is enabled, this field is computed over number of slots, programmed in the SLC field of MTL_TxQ1_ETS_CONTROL register. The maximum value of this field is 0x6_4000 in 100 Mbps, 0x3E_8000 in 1000 Mbps and 9C_4000 in 2500 Mbps mode respectively. When the DCB operation is enabled for Queue, this field is computed over every 10 million bit times slot (4 ms in 2500 Mbps; 10 ms in 1000 Mbps; 100 ms in 100 Mbps). The maximum value is 0x989680. Value After Reset: 0x0
0xe98	MTL_TxQ6_Quantum_Weight	The Queue 6 idleSlopeCredit, Quantum or Weights register provides the average traffic transmitted in Queue 6.	31:21	Reserved_31_21	R	0x0	Reserved. Value After Reset: 0x0
			20:0	ISCQW	R/W	0x0	idleSlopeCredit, Quantum or Weights idleSlopeCredit When AV feature is enabled, this field contains the idleSlopeCredit value required for the credit-based shaper algorithm for Queue 1. This is the rate of change of credit in bits per cycle (40 ns for 100 Mbps; 8 ns for 1000 Mbps; 3.2 ns for 2500 Mbps) when the credit is increasing. The software should program this field with computed credit in bits per cycle scaled by 1,024. The maximum value is portTransmitRate, that is, 0x2000 in 1000/2500 Mbps mode and 0x1000 in 100 Mbps mode. Bits[20:14] must be written to zero. Quantum When the DCB operation is enabled with DWRR algorithm for Queue 1 traffic, this field contains the quantum value in bytes to be added to credit during every queue scanning cycle. The maximum value is 0x1312D0 bytes. Weights When DCB operation is enabled with WFQ algorithm for Queue 1 traffic, this field contains the weight for this queue. The maximum value is 0x3FFF where weight of 0 indicates 100% bandwidth. Bits[20:14] must be written to zero. When DCB operation or generic queuing operation is enabled with WRR algorithm for Queue 1 traffic, this field contains the weight for this queue. The maximum value is 0x64. Bits [20:7] must be written to zero. Note 1: In multiple Queue configuration this field in respective per queue register must be programmed to some non-zero value when multiple queues are enabled or single queue other than Q0 is enabled. This field need not be programmed when only Q0 is enabled. In general, when WRR algorithm is selected a non-zero value must be programmed on both Receive and Transmit. In Receive, the register is MTL_Operation_Mode register. Note 2: For WFQ algorithm, higher the programmed weights lesser the bandwidth allocated for that Transmit Queue. The finish time is not a function of particular packet alone but it is as per the formula: (previous_finish_time of particular Transmit Queue + (weights*packet_size)) Note 3: The weights programmed do not correspond to the number of packets but the fraction of bandwidth or time allocated for particular queue w.r.t. total BW or time. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xe9c	MTL_TxQ6_SendSlopeCredit	The sendSlopeCredit register contains the sendSlope credit value required for the credit-based shaper algorithm for the Queue.	31:14	Reserved_31_14	R	0x0	Reserved. Value After Reset: 0x0
			13:0	SSC	R/W	0x0	sendSlopeCredit Value When AV operation is enabled, this field contains the sendSlopeCredit value required for credit-based shaper algorithm for Tx Queue 1. This is the rate of change of credit in bits per cycle (40 ns, 8 ns and 3.2 ns for 100 Mbps, 1000 Mbps and 2500 Mbps respectively) when the credit is decreasing. The software should program this field with computed credit in bits per cycle scaled by 1,024. The maximum value is portTransmitRate, that is, 0x2000 in 1000/2500 Mbps mode and 0x1000 in 100 Mbps mode. This field should be programmed with absolute sendSlopeCredit value. The credit-based shaper logic subtracts it from the accumulated credit when Tx Queue 1 is selected for transmission. Value After Reset: 0x0
0xea0	MTL_TxQ6_HiCredit	The hiCredit register contains the hiCredit value required for the credit-based shaper algorithm for the Queue.	31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28:0	HC	R/W	0x0	hiCredit Value When the AV feature is enabled, this field contains the hiCredit value required for the credit-based shaper algorithm. This is the maximum value that can be accumulated in the credit parameter. This is specified in bits scaled by 1,024. The maximum value is maxInterferenceSize, that is, best-effort maximum packet size (16,384 bytes or 131,072 bits). The value to be specified is 131,072 * 1,024 = 134,217,728 or 0x0800_0000. Value After Reset: 0x0
0xea4	MTL_TxQ6_LoCredit	The loCredit register contains the loCredit value required for the credit-based shaper algorithm for the Queue.	31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28:0	LC	R/W	0x0	loCredit Value When AV operation is enabled, this field contains the loCredit value required for the credit-based shaper algorithm. This is the minimum value that can be accumulated in the credit parameter. This is specified in bits scaled by 1,024. The maximum value to be programmed is corresponds to twice the maxFrameSize transmitted from this queue. If the maxFrameSize is 8192 bytes, then (8192*2) * 8 * 1024 = 134,217,728 or 0x0800_0000. Because it is a negative value, the programmed value is 2's complement of the value, that is, 0x1800_0000. Value After Reset: 0x0
			31:25	Reserved_31_25	R	0x0	Reserved. Value After Reset: 0x0
			24	RXOIE	R/W	0x0	Receive Queue Overflow Interrupt Enable When this bit is set, the Receive Queue Overflow interrupt is enabled. When this bit is reset, the Receive Queue Overflow interrupt is disabled.Values: 0x0 (DISABLE): Receive Queue Overflow Interrupt is disabled 0x1 (ENABLE): Receive Queue Overflow Interrupt is enabled Value After Reset: 0x0
			23:17	Reserved_23_17	R	0x0	Reserved. Value After Reset: 0x0
			16	RXOVFIS	R/W	0x0	Receive Queue Overflow Interrupt Status This bit indicates that the Receive Queue had an overflow while receiving the packet. If a partial packet is transferred to the application, the overflow status is set in RDES3[21]. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Queue Overflow Interrupt Status not detected 0x1 (ACTIVE): Receive Queue Overflow Interrupt Status detected Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xeac	MTL_Q6_Interrupt_Control_Status	This register contains the interrupt enable and status bits for the queue 6 interrupts.	15:10	Reserved_15_10	R	0x0	Reserved. Value After Reset: 0x0
			9	ABPSIE	R/W	0x0	Average Bits Per Slot Interrupt Enable When this bit is set, the MAC asserts the sbd_intr_o or mci_intr_o interrupt when the average bits per slot status is updated. When this bit is cleared, the interrupt is not asserted for such an event.Values: 0x0 (DISABLE): Average Bits Per Slot Interrupt is disabled 0x1 (ENABLE): Average Bits Per Slot Interrupt is enabled Value After Reset: 0x0
			8	TXUIE	R/W	0x0	Transmit Queue Underflow Interrupt Enable When this bit is set, the Transmit Queue Underflow interrupt is enabled. When this bit is reset, the Transmit Queue Underflow interrupt is disabled.Values: 0x0 (DISABLE): Transmit Queue Underflow Interrupt Status is disabled 0x1 (ENABLE): Transmit Queue Underflow Interrupt Status is enabled Value After Reset: 0x0
			7:2	Reserved_7_2	R	0x0	Reserved. Value After Reset: 0x0
			1	ABPSIS	R/W	0x0	Average Bits Per Slot Interrupt Status When set, this bit indicates that the MAC has updated the ABS value. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Average Bits Per Slot Interrupt Status not detected 0x1 (ACTIVE): Average Bits Per Slot Interrupt Status detected Value After Reset: 0x0 Testable: untestable
			0	TXUNFIS	R/W	0x0	Transmit Queue Underflow Interrupt Status This bit indicates that the Transmit Queue had an underflow while transmitting the packet. Transmission is suspended and an Underflow Error TDES3[2] is set. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Queue Underflow Interrupt Status not detected 0x1 (ACTIVE): Transmit Queue Underflow Interrupt Status detected Value After Reset: 0x0 Testable: untestable
			31:26	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			25:20	RQS	R/W	0x0	Receive Queue Size This field indicates the size of the allocated Receive queues in blocks of 256 bytesthe reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on. So, program RQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (RQS+1)*256 bytes. Value After Reset: 0x0
			19	Reserved_19_y	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xeb0	MTL_RxQ6_Operation_Mode	The Queue 6 Receive Operation Mode register establishes the Receive queue operating modes and command.	18:14	RFD	R/W	0x0	Threshold for Deactivating Flow Control (in half-duplex and full-duplex modes) These bits control the threshold (fill-level of Rx queue) at which the flow control is de-asserted after activation: 0: Full minus 1 KB, that is, FULL 1 KB 1: Full minus 1.5 KB, that is, FULL 1.5 KB 2: Full minus 2 KB, that is, FULL 2 KB 3: Full minus 2.5 KB, that is, FULL 2.5 KB ... 30: Full minus 16 KB, that is, FULL 16 KB 31: Full minus 16.5 KB, that is, FULL 16.5 KB The de-assertion is effective only after flow control is asserted. Note: The value must be programmed in such a way to make sure that the threshold is a positive number. When the EHFC is set high, these values are applicable only when the Rx queue size determined by the RQS field of this register, is equal to or greater than 4 KB. For a given queue size, the values ranges between 0 and the encoding for FULL minus (QSIZE - 0.5 KB) and all other values are illegal. Here the term FULL and QSIZE refers to the queue size determined by the RQS field of this register. Value After Reset: 0x0
			13	Reserved_13_y	R	0x0	Reserved. Value After Reset: 0x0
			12:8	RFA	R/W	0x0	Threshold for Activating Flow Control (in half-duplex and full-duplex These bits control the threshold (fill-level of Rx queue) at which the flow control is activated: For more information on encoding for this field, see RFD. Value After Reset: 0x0
			7	EHFC	R/W	0x0	Enable Hardware Flow Control When this bit is set, the flow control signal operation, based on the fill-level of Rx queue, is enabled. When reset, the flow control operation is disabled.Values: 0x0 (DISABLE): Hardware Flow Control is disabled 0x1 (ENABLE): Hardware Flow Control is enabled Value After Reset: 0x0
			6	DIS_TCP_EF	R/W	0x0	Disable Dropping of TCP/IP Checksum Error Packets When this bit is set, the MAC does not drop the packets which only have the errors detected by the Receive Checksum Offload engine. Such packets have errors only in the encapsulated payload. There are no errors (including FCS error) in the Ethernet packet received by the MAC. When this bit is reset, all error packets are dropped if the FEP bit is reset.Values: 0x0 (ENABLE): Dropping of TCP/IP Checksum Error Packets is enabled 0x1 (DISABLE): Dropping of TCP/IP Checksum Error Packets is disabled Value After Reset: 0x0
			5	RSF	R/W	0x0	Receive Queue Store and Forward When this bit is set, the DWC_ether_qos reads a packet from the Rx queue only after the complete packet has been written to it, ignoring the RTC field of this register. When this bit is reset, the Rx queue operates in the Threshold (cut-through) mode, subject to the threshold specified by the RTC field of this register.Values: 0x0 (DISABLE): Receive Queue Store and Forward is disabled 0x1 (ENABLE): Receive Queue Store and Forward is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			4	FEP	R/W	0x0	Forward Error Packets When this bit is reset, the Rx queue drops packets with error status (CRC error, GMII_ER, watchdog timeout, or overflow). However, if the start byte (write) pointer of a packet is already transferred to the read controller side (in Threshold mode), the packet is not dropped. When this bit is set, all packets except the runt error packets are forwarded to the application or DMA. If the RSF bit is set and the Rx queue overflows when a partial packet is written, the packet is dropped irrespective of the setting of this bit. However, if the RSF bit is reset and the Rx queue overflows when a partial packet is written, a partial packet might be forwarded to the application or DMA.Values: 0x0 (DISABLE): Forward Error Packets is disabled 0x1 (ENABLE): Forward Error Packets is enabled Value After Reset: 0x0
			3	FUP	R/W	0x0	Forward Undersized Good Packets When this bit is set, the Rx queue forwards the undersized good packets (packets with no error and length less than 64 bytes), including pad-bytes and CRC. When this bit is reset, the Rx queue drops all packets of less than 64 bytes, unless a packet is already transferred because of the lower value of Rx Threshold, for example, RTC = 01.Values: 0x0 (DISABLE): Forward Undersized Good Packets is disabled 0x1 (ENABLE): Forward Undersized Good Packets is enabled Value After Reset: 0x0
			2	Reserved_2	R	0x0	Reserved. Value After Reset: 0x0
			1:0	RTC	R/W	0x0	Receive Queue Threshold Control These bits control the threshold level of the MTL Rx queue (in bytes): The received packet is transferred to the application or DMA when the packet size within the MTL Rx queue is larger than the threshold. In addition, full packets with length less than the threshold are automatically transferred. This field is valid only when the RSF bit is zero. This field is ignored when the RSF bit is set to 1.Values: 0x0 (M_64BYTE): 64 0x1 (M_32BYTE): 32 0x2 (M_96BYTE): 96 0x3 (M_128BYTE): 128 Value After Reset: 0x0
0xeb4	MTL_RxQ6_Missed_Packet_Overflow_Cnt	The Queue 6 Missed Packet and Overflow Counter register contains the counter for packets missed because of Receive queue packet flush and packets discarded because of Receive queue overflow.	31:28	Reserved_31_28	R	0x0	Reserved. Value After Reset: 0x0
			27	MISCNTOVF	R	0x0	Missed Packet Counter Overflow Bit When set, this bit indicates that the Rx Queue Missed Packet Counter crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Missed Packet Counter overflow not detected 0x1 (ACTIVE): Missed Packet Counter overflow detected Value After Reset: 0x0
			26:16	MISPKTCNT	R	0x0	Missed Packet Counter This field indicates the number of packets missed by the DWC_ether_qos because the application asserted ari_pkt_flush_i[] for this queue. This counter is reset when this register is read with mci_be_i[0] at 1b1. This counter is incremented by 1 when the DMA discards the packet because of buffer unavailability. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0
			15:12	Reserved_15_12	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			11	OVFCNTOVF	R	0x0	Overflow Counter Overflow Bit When set, this bit indicates that the Rx Queue Overflow Packet Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Overflow Counter overflow not detected 0x1 (ACTIVE): Overflow Counter overflow detected Value After Reset: 0x0
			10:0	OVFPKTCNT	R	0x0	Overflow Packet Counter This field indicates the number of packets discarded by the DWC_ether_qos because of Receive queue overflow. This counter is incremented each time the DWC_ether_qos discards an incoming packet because of overflow. This counter is reset when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0
0xeb8	MTL_RxQ6_Debug	The Queue 6 Receive Debug register gives the debug status of various blocks related to the Receive queue.	31:30	Reserved_31_30	R	0x0	Reserved. Value After Reset: 0x0
			29:16	PRXQ	R	0x0	Number of Packets in Receive Queue This field indicates the current number of packets in the Rx Queue, that is, Max_Queue_Size/Min_Packet_Size. Value After Reset: 0x0
			15:6	Reserved_15_6	R	0x0	Reserved. Value After Reset: 0x0
			5:4	RXQSTS	R	0x0	MTL Rx Queue Fill-Level Status This field gives the status of the fill-level of the Rx Queue:Values: 0x0 (EMPTY): Rx Queue empty 0x1 (BLW_THR): Rx Queue fill-level below flow-control deactivate threshold 0x2 (ABV_THR): Rx Queue fill-level above flow-control activate threshold 0x3 (FULL): Rx Queue full Value After Reset: 0x0
			3	Reserved_3	R	0x0	Reserved. Value After Reset: 0x0
			2:1	RRCSTS	R	0x0	MTL Rx Queue Read Controller State This field gives the state of the Rx queue Read controller:Values: 0x0 (IDLE): Idle state 0x1 (READ_DATA): Reading packet data 0x2 (READ_STS): Reading packet status (or timestamp) 0x3 (FLUSH): Flushing the packet data and status Value After Reset: 0x0
			0	RWCSTS	R	0x0	MTL Rx Queue Write Controller Active Status When high, this bit indicates that the MTL Rx queue Write controller is active, and it is transferring a received packet to the Rx Queue.Values: 0x0 (INACTIVE): MTL Rx Queue Write Controller Active Status not detected 0x1 (ACTIVE): MTL Rx Queue Write Controller Active Status detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xebc	MTL_RxQ6_Control	The Queue Receive Control register controls the receive arbitration and passing of received packets to the application.	31:4	Reserved_31_4	R	0x0	Reserved. Value After Reset: 0x0
			3	RXQ_FRM_ARBIT	R/W	0x0	Receive Queue Packet Arbitration When this bit is set, the DWC_ether_qos drives the packet data to the ARI interface such that the entire packet data of currently-selected queue is transmitted before switching to other queue. When this bit is reset, the DWC_ether_qos drives the packet data to the ARI interface such that the following amount of data of currently-selected queue is transmitted before switching to other queue: PBL amount of data (indicated by ari_qN_pbl_i[]) or Complete data of a packet The status and the timestamp are not a part of the PBL data. Therefore, the DWC_ether_qos drives the complete status (including timestamp status) during first PBL request for the packet (in store-and-forward mode) or the last PBL request for the packet (in Threshold mode).Values: 0x0 (DISABLE): Receive Queue Packet Arbitration is disabled 0x1 (ENABLE): Receive Queue Packet Arbitration is enabled Value After Reset: 0x0
			2:0	RXQ_WEGT	R/W	0x0	Receive Queue Weight This field indicates the weight assigned to the Rx Queue 1. This field needs to be programmed with one value less than the required weight, that is, reset value of 0 indicates weight of 1, value of 1 indicates weight of 2, and so on. The weight is used as the number of continuous PBL or packets requests (depending on the RXQ_FRM_ARBIT) allocated to the queue in one arbitration cycle. Note: The change in value of RXQ_WEGT takes effect only after the completion of current service round or when there is change from RAA=SP to RAA=WSP algorithm. This approach is taken so that there is smooth transition. For the RXQ_WEGT value to take effect at the start, the MTL_RxQ1_Control registers must be programmed before the MTL_Operation_Mode register. Value After Reset: 0x0
			31:22	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			21:16	TQS	R/W	0x0	Transmit Queue Size This field indicates the size of the allocated Transmit queues in blocks of 256 bytes, the reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on. So, program TQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (TQS+1)*256 bytes. Value After Reset: 0x0
			15:7	Reserved_15_7	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xec0	MTL_TxQ7_Operation_Mode	The Queue 7 Transmit Operation Mode register establishes the Transmit queue operating modes and commands.	6:4	TTC	R/W	0x0	Transmit Threshold Control These bits control the threshold level of the MTL Tx Queue. The transmission starts when the packet size within the MTL Tx Queue is larger than the threshold. In addition, full packets with length less than the threshold are also transmitted. These bits are used only when the TSF bit is reset.Values: 0x0 (M_32BYTES): 32 0x1 (M_64BYTES): 64 0x2 (M_96BYTES): 96 0x3 (M_128BYTES): 128 0x4 (M_192BYTES): 192 0x5 (M_256BYTES): 256 0x6 (M_384BYTES): 384 0x7 (M_512BYTES): 512 Value After Reset: 0x0
			3:2	TXQEN	R/W	0x0	Transmit Queue Enable This field is used to enable/disable the transmit queue 1. Note: In multiple Tx queues configuration, all the queues are disabled by default. Enable the Tx queue by programming this field.Values: 0x0 (DISABLE): Not enabled 0x1 (EN_IF_AV): Enable in AV mode 0x2 (ENABLE): Enabled 0x3 (RSVD2): Reserved Value After Reset: 0x0
			1	TSF	R/W	0x0	Transmit Store and Forward When this bit is set, the transmission starts when a full packet resides in the MTL Tx queue. When this bit is set, the TTC values specified in Bits[6:4] of this register are ignored. This bit should be changed only when the transmission is stopped.Values: 0x0 (DISABLE): Transmit Store and Forward is disabled 0x1 (ENABLE): Transmit Store and Forward is enabled Value After Reset: 0x0
			0	FTQ	R/W	0x0	Flush Transmit Queue When this bit is set, the Tx queue controller logic is reset to its default values. Therefore, all the data in the Tx queue is lost or flushed. This bit is internally reset when the flushing operation is complete. Until this bit is reset, you must not write to the MTL_TxQ1_Operation_Mode register. The data which is already accepted by the MAC transmitter is not flushed. It is scheduled for transmission and results in underflow and runt packet transmission. Note: The flush operation is complete only when the Tx queue is empty and the application has accepted the pending Tx Status of all transmitted packets. To complete this flush operation, the PHY Tx clock (clk_tx_i) must be active. Access restrictions apply. MAC initiates the Tx queue flush when application writes 1. MAC writes 0 after the Tx queue flush is complete. MAC ignores when application writes 0. Values: 0x0 (DISABLE): Flush Transmit Queue is disabled 0x1 (ENABLE): Flush Transmit Queue is enabled Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xec4	MTL_TxQ7_Underflow	The Queue 7 Underflow Counter register contains the counter for packets aborted because of Transmit queue underflow and packets missed because of Receive queue packet flush	31:12	Reserved_31_12	R	0x0	Reserved. Value After Reset: 0x0
			11	UFCNTOVF	R	0x0	Overflow Bit for Underflow Packet Counter This bit is set every time the Tx queue Underflow Packet Counter field overflows, that is, it has crossed the maximum count. In such a scenario, the overflow packet counter is reset to all-zeros and this bit indicates that the rollover happened. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Overflow not detected for Underflow Packet Counter 0x1 (ACTIVE): Overflow detected for Underflow Packet Counter Value After Reset: 0x0
			10:0	UFFRMCNT	R	0x0	Underflow Packet Counter This field indicates the number of packets aborted by the controller because of Tx Queue Underflow. This counter is incremented each time the MAC aborts outgoing packet because of underflow. The counter is cleared when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0
0xec8	MTL_TxQ7_Debug	The Queue 7 Transmit Debug register gives the debug status of various blocks related to the Transmit queue.	31:23	Reserved_31_23	R	0x0	Reserved. Value After Reset: 0x0
			22:20	STXSTSF	R	0x0	Number of Status Words in Tx Status FIFO of Queue This field indicates the current number of status in the Tx Status FIFO of this queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of status words in Tx Status FIFO. Value After Reset: 0x0
			19	Reserved_19	R	0x0	Reserved. Value After Reset: 0x0
			18:16	PTXQ	R	0x0	Number of Packets in the Transmit Queue This field indicates the current number of packets in the Tx Queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of packets in the Transmit queue. Value After Reset: 0x0
			15:6	Reserved_15_6	R	0x0	Reserved. Value After Reset: 0x0
			5	TXSTSFSTS	R	0x0	MTL Tx Status FIFO Full Status When high, this bit indicates that the MTL Tx Status FIFO is full. Therefore, the MTL cannot accept any more packets for transmission.Values: 0x0 (INACTIVE): MTL Tx Status FIFO Full status is not detected 0x1 (ACTIVE): MTL Tx Status FIFO Full status is detected Value After Reset: 0x0
			4	TXQSTS	R	0x0	MTL Tx Queue Not Empty Status When this bit is high, it indicates that the MTL Tx Queue is not empty and some data is left for transmission.Values: 0x0 (INACTIVE): MTL Tx Queue Not Empty status is not detected 0x1 (ACTIVE): MTL Tx Queue Not Empty status is detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			3	TWCSTS	R	0x0	MTL Tx Queue Write Controller Status When high, this bit indicates that the MTL Tx Queue Write Controller is active, and it is transferring the data to the Tx Queue.Values: 0x0 (INACTIVE): MTL Tx Queue Write Controller status is not detected 0x1 (ACTIVE): MTL Tx Queue Write Controller status is detected Value After Reset: 0x0
			2:1	TRCSTS	R	0x0	MTL Tx Queue Read Controller Status This field indicates the state of the Tx Queue Read Controller:Values: 0x0 (IDLE): Idle state 0x1 (READ): Read state (transferring data to the MAC transmitter) 0x2 (WAIT): Waiting for pending Tx Status from the MAC transmitter 0x3 (FLUSH): Flushing the Tx queue because of the Packet Abort request from the MAC Value After Reset: 0x0
			0	TXQPAUSED	R	0x0	Transmit Queue in Pause When this bit is high and the Rx flow control is enabled, it indicates that the Tx Queue is in the Pause condition (in the full-duplex only mode) because of the following: Reception of the PFC packet for the priorities assigned to the Tx Queue when PFC is enabled Reception of 802.3x Pause packet when PFC is disabled Values: 0x0 (INACTIVE): Transmit Queue in Pause status is not detected 0x1 (ACTIVE): Transmit Queue in Pause status is detected Value After Reset: 0x0
0xed0	MTL_TxQ7_ETS_Control	The Queue ETS Control register controls the enhanced transmission selection operation.	31:7	Reserved_31_7	R	0x0	Reserved. Value After Reset: 0x0
			6:4	SLC	R/W	0x0	Slot Count If the credit-based shaper algorithm is enabled, the software can program the number of slots (of duration programmed in DMA_CH1_Slot_Interval register) over which the average transmitted bits per slot, provided in the MTL_TxQ1_ETS_Status register, need to be computed for Queue. The encoding is as follows:Values: 0x0 (M_1_SLOT): 1 slot 0x1 (M_2_SLOT): 2 slots 0x2 (M_4_SLOT): 4 slots 0x3 (M_8_SLOT): 8 slots 0x4 (M_16_SLOT): 16 slots 0x5 (RSVD): Reserved Value After Reset: 0x0
			3	CC	R/W	0x0	Credit Control When this bit is set, the accumulated credit parameter in the credit-based shaper algorithm logic is not reset to zero when there is positive credit and no packet to transmit in Tx Queue 1. The credit accumulates even when there is no packet waiting in Tx Queue 1 and another Tx Queue is transmitting. When this bit is reset, the accumulated credit parameter in the credit-based shaper algorithm logic is set to zero when there is positive credit and no packet to transmit in Tx Queue 1. When there is no packet waiting in Tx Queue 1 and other channel is transmitting, no credit is accumulated.Values: 0x0 (DISABLE): Credit Control is disabled 0x1 (ENABLE): Credit Control is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			2	AVALG	R/W	0x0	AV Algorithm When Queue 1 is programmed for AV, this field configures the scheduling algorithm for this queue: This bit when set, indicates credit based shaper algorithm (CBS) is selected for Queue 1 traffic. When reset, strict priority is selected.Values: 0x0 (DISABLE): CBS Algorithm is disabled 0x1 (ENABLE): CBS Algorithm is enabled Value After Reset: 0x0
			1:0	Reserved_1_0	R	0x0	Reserved. Value After Reset: 0x0
0xed4	MTL_TxQ7_ETS_Status	The Queue 7 ETS Status register provides the average traffic transmitted in Queue 7.	31:24	Reserved_31_24	R	0x0	Reserved. Value After Reset: 0x0
			23:0	ABS	R	0x0	Average Bits per Slot This field contains the average transmitted bits per slot. If AV operation is enabled, this field is computed over number of slots, programmed in the SLC field of MTL_TxQ1_ETS_CONTROL register. The maximum value of this field is 0x6_4000 in 100 Mbps, 0x3E_8000 in 1000 Mbps and 9C_4000 in 2500 Mbps mode respectively. When the DCB operation is enabled for Queue, this field is computed over every 10 million bit times slot (4 ms in 2500 Mbps; 10 ms in 1000 Mbps; 100 ms in 100 Mbps). The maximum value is 0x989680. Value After Reset: 0x0
0xed8	MTL_TxQ7_Quantum_Weight	The Queue 7 idleSlopeCredit, Quantum or Weights register provides the average traffic transmitted in Queue 7.	31:21	Reserved_31_21	R	0x0	Reserved. Value After Reset: 0x0
			20:0	ISCQW	R/W	0x0	idleSlopeCredit, Quantum or Weights idleSlopeCredit When AV feature is enabled, this field contains the idleSlopeCredit value required for the credit-based shaper algorithm for Queue 1. This is the rate of change of credit in bits per cycle (40 ns for 100 Mbps; 8 ns for 1000 Mbps; 3.2 ns for 2500 Mbps) when the credit is increasing. The software should program this field with computed credit in bits per cycle scaled by 1,024. The maximum value is portTransmitRate, that is, 0x2000 in 1000/2500 Mbps mode and 0x1000 in 100 Mbps mode. Bits[20:14] must be written to zero. Quantum When the DCB operation is enabled with DWRR algorithm for Queue 1 traffic, this field contains the quantum value in bytes to be added to credit during every queue scanning cycle. The maximum value is 0x1312D0 bytes. Weights When DCB operation is enabled with WFQ algorithm for Queue 1 traffic, this field contains the weight for this queue. The maximum value is 0x3FFF where weight of 0 indicates 100% bandwidth. Bits[20:14] must be written to zero. When DCB operation or generic queuing operation is enabled with WRR algorithm for Queue 1 traffic, this field contains the weight for this queue. The maximum value is 0x64. Bits [20:7] must be written to zero. Note 1: In multiple Queue configuration this field in respective per queue register must be programmed to some non-zero value when multiple queues are enabled or single queue other than Q0 is enabled. This field need not be programmed when only Q0 is enabled. In general, when WRR algorithm is selected a non-zero value must be programmed on both Receive and Transmit. In Receive, the register is MTL_Operation_Mode register. Note 2: For WFQ algorithm, higher the programmed weights lesser the bandwidth allocated for that Transmit Queue. The finish time is not a function of particular packet alone but it is as per the formula: (previous_finish_time of particular Transmit Queue + (weights*packet_size)) Note 3: The weights programmed do not correspond to the number of packets but the fraction of bandwidth or time allocated for particular queue w.r.t. total BW or time. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xedc	MTL_TxQ7_SendSlopeCredit	The sendSlopeCredit register contains the sendSlope credit value required for the credit-based shaper algorithm for the Queue.	31:14	Reserved_31_14	R	0x0	Reserved. Value After Reset: 0x0
			13:0	SSC	R/W	0x0	sendSlopeCredit Value When AV operation is enabled, this field contains the sendSlopeCredit value required for credit-based shaper algorithm for Tx Queue 1. This is the rate of change of credit in bits per cycle (40 ns, 8 ns and 3.2 ns for 100 Mbps, 1000 Mbps and 2500 Mbps respectively) when the credit is decreasing. The software should program this field with computed credit in bits per cycle scaled by 1,024. The maximum value is portTransmitRate, that is, 0x2000 in 1000/2500 Mbps mode and 0x1000 in 100 Mbps mode. This field should be programmed with absolute sendSlopeCredit value. The credit-based shaper logic subtracts it from the accumulated credit when Tx Queue 1 is selected for transmission. Value After Reset: 0x0
0xee0	MTL_TxQ7_HiCredit	The hiCredit register contains the hiCredit value required for the credit-based shaper algorithm for the Queue.	31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28:0	HC	R/W	0x0	hiCredit Value When the AV feature is enabled, this field contains the hiCredit value required for the credit-based shaper algorithm. This is the maximum value that can be accumulated in the credit parameter. This is specified in bits scaled by 1,024. The maximum value is maxInterferenceSize, that is, best-effort maximum packet size (16,384 bytes or 131,072 bits). The value to be specified is 131,072 * 1,024 = 134,217,728 or 0x0800_0000. Value After Reset: 0x0
0xee4	MTL_TxQ7_LoCredit	The loCredit register contains the loCredit value required for the credit-based shaper algorithm for the Queue.	31:29	Reserved_31_29	R	0x0	Reserved. Value After Reset: 0x0
			28:0	LC	R/W	0x0	loCredit Value When AV operation is enabled, this field contains the loCredit value required for the credit-based shaper algorithm. This is the minimum value that can be accumulated in the credit parameter. This is specified in bits scaled by 1,024. The maximum value to be programmed is corresponds to twice the maxFrameSize transmitted from this queue. If the maxFrameSize is 8192 bytes, then (8192*2) * 8 * 1024 = 134,217,728 or 0x0800_0000. Because it is a negative value, the programmed value is 2's complement of the value, that is, 0x1800_0000. Value After Reset: 0x0
			31:25	Reserved_31_25	R	0x0	Reserved. Value After Reset: 0x0
			24	RXOIE	R/W	0x0	Receive Queue Overflow Interrupt Enable When this bit is set, the Receive Queue Overflow interrupt is enabled. When this bit is reset, the Receive Queue Overflow interrupt is disabled.Values: 0x0 (DISABLE): Receive Queue Overflow Interrupt is disabled 0x1 (ENABLE): Receive Queue Overflow Interrupt is enabled Value After Reset: 0x0
			23:17	Reserved_23_17	R	0x0	Reserved. Value After Reset: 0x0
			16	RXOVFIS	R/W	0x0	Receive Queue Overflow Interrupt Status This bit indicates that the Receive Queue had an overflow while receiving the packet. If a partial packet is transferred to the application, the overflow status is set in RDES3[21]. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Queue Overflow Interrupt Status not detected 0x1 (ACTIVE): Receive Queue Overflow Interrupt Status detected Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xeec	MTL_Q7_Interrupt_Control_Status	This register contains the interrupt enable and status bits for the queue 7 interrupts.	15:10	Reserved_15_10	R	0x0	Reserved. Value After Reset: 0x0
			9	ABPSIE	R/W	0x0	Average Bits Per Slot Interrupt Enable When this bit is set, the MAC asserts the sbd_intr_o or mci_intr_o interrupt when the average bits per slot status is updated. When this bit is cleared, the interrupt is not asserted for such an event.Values: 0x0 (DISABLE): Average Bits Per Slot Interrupt is disabled 0x1 (ENABLE): Average Bits Per Slot Interrupt is enabled Value After Reset: 0x0
			8	TXUIE	R/W	0x0	Transmit Queue Underflow Interrupt Enable When this bit is set, the Transmit Queue Underflow interrupt is enabled. When this bit is reset, the Transmit Queue Underflow interrupt is disabled.Values: 0x0 (DISABLE): Transmit Queue Underflow Interrupt Status is disabled 0x1 (ENABLE): Transmit Queue Underflow Interrupt Status is enabled Value After Reset: 0x0
			7:2	Reserved_7_2	R	0x0	Reserved. Value After Reset: 0x0
			1	ABPSIS	R/W	0x0	Average Bits Per Slot Interrupt Status When set, this bit indicates that the MAC has updated the ABS value. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Average Bits Per Slot Interrupt Status not detected 0x1 (ACTIVE): Average Bits Per Slot Interrupt Status detected Value After Reset: 0x0 Testable: untestable
			0	TXUNFIS	R/W	0x0	Transmit Queue Underflow Interrupt Status This bit indicates that the Transmit Queue had an underflow while transmitting the packet. Transmission is suspended and an Underflow Error TDES3[2] is set. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Queue Underflow Interrupt Status not detected 0x1 (ACTIVE): Transmit Queue Underflow Interrupt Status detected Value After Reset: 0x0 Testable: untestable
			31:26	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			25:20	RQS	R/W	0x0	Receive Queue Size This field indicates the size of the allocated Receive queues in blocks of 256 bytesthe reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on. So, program RQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (RQS+1)*256 bytes. Value After Reset: 0x0
			19	Reserved_19_y	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xef0	MTL_RxQ7_Operation_Mode	The Queue 7 Receive Operation Mode register establishes the Receive queue operating modes and command.	18:14	RFD	R/W	0x0	Threshold for Deactivating Flow Control (in half-duplex and full-duplex modes) These bits control the threshold (fill-level of Rx queue) at which the flow control is de-asserted after activation: 0: Full minus 1 KB, that is, FULL 1 KB 1: Full minus 1.5 KB, that is, FULL 1.5 KB 2: Full minus 2 KB, that is, FULL 2 KB 3: Full minus 2.5 KB, that is, FULL 2.5 KB ... 30: Full minus 16 KB, that is, FULL 16 KB 31: Full minus 16.5 KB, that is, FULL 16.5 KB The de-assertion is effective only after flow control is asserted. Note: The value must be programmed in such a way to make sure that the threshold is a positive number. When the EHFC is set high, these values are applicable only when the Rx queue size determined by the RQS field of this register, is equal to or greater than 4 KB. For a given queue size, the values ranges between 0 and the encoding for FULL minus (QSIZE - 0.5 KB) and all other values are illegal. Here the term FULL and QSIZE refers to the queue size determined by the RQS field of this register. Value After Reset: 0x0
			13	Reserved_13_y	R	0x0	Reserved. Value After Reset: 0x0
			12:8	RFA	R/W	0x0	Threshold for Activating Flow Control (in half-duplex and full-duplex These bits control the threshold (fill-level of Rx queue) at which the flow control is activated: For more information on encoding for this field, see RFD. Value After Reset: 0x0
			7	EHFC	R/W	0x0	Enable Hardware Flow Control When this bit is set, the flow control signal operation, based on the fill-level of Rx queue, is enabled. When reset, the flow control operation is disabled.Values: 0x0 (DISABLE): Hardware Flow Control is disabled 0x1 (ENABLE): Hardware Flow Control is enabled Value After Reset: 0x0
			6	DIS_TCP_EF	R/W	0x0	Disable Dropping of TCP/IP Checksum Error Packets When this bit is set, the MAC does not drop the packets which only have the errors detected by the Receive Checksum Offload engine. Such packets have errors only in the encapsulated payload. There are no errors (including FCS error) in the Ethernet packet received by the MAC. When this bit is reset, all error packets are dropped if the FEP bit is reset.Values: 0x0 (ENABLE): Dropping of TCP/IP Checksum Error Packets is enabled 0x1 (DISABLE): Dropping of TCP/IP Checksum Error Packets is disabled Value After Reset: 0x0
			5	RSF	R/W	0x0	Receive Queue Store and Forward When this bit is set, the DWC_ether_qos reads a packet from the Rx queue only after the complete packet has been written to it, ignoring the RTC field of this register. When this bit is reset, the Rx queue operates in the Threshold (cut-through) mode, subject to the threshold specified by the RTC field of this register.Values: 0x0 (DISABLE): Receive Queue Store and Forward is disabled 0x1 (ENABLE): Receive Queue Store and Forward is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			4	FEP	R/W	0x0	Forward Error Packets When this bit is reset, the Rx queue drops packets with error status (CRC error, GMII_ER, watchdog timeout, or overflow). However, if the start byte (write) pointer of a packet is already transferred to the read controller side (in Threshold mode), the packet is not dropped. When this bit is set, all packets except the runt error packets are forwarded to the application or DMA. If the RSF bit is set and the Rx queue overflows when a partial packet is written, the packet is dropped irrespective of the setting of this bit. However, if the RSF bit is reset and the Rx queue overflows when a partial packet is written, a partial packet might be forwarded to the application or DMA.Values: 0x0 (DISABLE): Forward Error Packets is disabled 0x1 (ENABLE): Forward Error Packets is enabled Value After Reset: 0x0
			3	FUP	R/W	0x0	Forward Undersized Good Packets When this bit is set, the Rx queue forwards the undersized good packets (packets with no error and length less than 64 bytes), including pad-bytes and CRC. When this bit is reset, the Rx queue drops all packets of less than 64 bytes, unless a packet is already transferred because of the lower value of Rx Threshold, for example, RTC = 01.Values: 0x0 (DISABLE): Forward Undersized Good Packets is disabled 0x1 (ENABLE): Forward Undersized Good Packets is enabled Value After Reset: 0x0
			2	Reserved_2	R	0x0	Reserved. Value After Reset: 0x0
			1:0	RTC	R/W	0x0	Receive Queue Threshold Control These bits control the threshold level of the MTL Rx queue (in bytes): The received packet is transferred to the application or DMA when the packet size within the MTL Rx queue is larger than the threshold. In addition, full packets with length less than the threshold are automatically transferred. This field is valid only when the RSF bit is zero. This field is ignored when the RSF bit is set to 1.Values: 0x0 (M_64BYTE): 64 0x1 (M_32BYTE): 32 0x2 (M_96BYTE): 96 0x3 (M_128BYTE): 128 Value After Reset: 0x0
0xef4	MTL_RxQ7_Missed_Packet_Overflow_Cnt	The Queue 7 Missed Packet and Overflow Counter register contains the counter for packets missed because of Receive queue packet flush and packets discarded because of Receive queue overflow.	31:28	Reserved_31_28	R	0x0	Reserved. Value After Reset: 0x0
			27	MISCNTOVF	R	0x0	Missed Packet Counter Overflow Bit When set, this bit indicates that the Rx Queue Missed Packet Counter crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Missed Packet Counter overflow not detected 0x1 (ACTIVE): Missed Packet Counter overflow detected Value After Reset: 0x0
			26:16	MISPKTCNT	R	0x0	Missed Packet Counter This field indicates the number of packets missed by the DWC_ether_qos because the application asserted ari_pkt_flush_i[] for this queue. This counter is reset when this register is read with mci_be_i[0] at 1b1. This counter is incremented by 1 when the DMA discards the packet because of buffer unavailability. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0
			15:12	Reserved_15_12	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			11	OVFCNTOVF	R	0x0	Overflow Counter Overflow Bit When set, this bit indicates that the Rx Queue Overflow Packet Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Overflow Counter overflow not detected 0x1 (ACTIVE): Overflow Counter overflow detected Value After Reset: 0x0
			10:0	OVFPKTCNT	R	0x0	Overflow Packet Counter This field indicates the number of packets discarded by the DWC_ether_qos because of Receive queue overflow. This counter is incremented each time the DWC_ether_qos discards an incoming packet because of overflow. This counter is reset when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0
0xef8	MTL_RxQ7_Debug	The Queue 7 Receive Debug register gives the debug status of various blocks related to the Receive queue.	31:30	Reserved_31_30	R	0x0	Reserved. Value After Reset: 0x0
			29:16	PRXQ	R	0x0	Number of Packets in Receive Queue This field indicates the current number of packets in the Rx Queue, that is, Max_Queue_Size/Min_Packet_Size. Value After Reset: 0x0
			15:6	Reserved_15_6	R	0x0	Reserved. Value After Reset: 0x0
			5:4	RXQSTS	R	0x0	MTL Rx Queue Fill-Level Status This field gives the status of the fill-level of the Rx Queue:Values: 0x0 (EMPTY): Rx Queue empty 0x1 (BLW_THR): Rx Queue fill-level below flow-control deactivate threshold 0x2 (ABV_THR): Rx Queue fill-level above flow-control activate threshold 0x3 (FULL): Rx Queue full Value After Reset: 0x0
			3	Reserved_3	R	0x0	Reserved. Value After Reset: 0x0
			2:1	RRCSTS	R	0x0	MTL Rx Queue Read Controller State This field gives the state of the Rx queue Read controller:Values: 0x0 (IDLE): Idle state 0x1 (READ_DATA): Reading packet data 0x2 (READ_STS): Reading packet status (or timestamp) 0x3 (FLUSH): Flushing the packet data and status Value After Reset: 0x0
			0	RWCSTS	R	0x0	MTL Rx Queue Write Controller Active Status When high, this bit indicates that the MTL Rx queue Write controller is active, and it is transferring a received packet to the Rx Queue.Values: 0x0 (INACTIVE): MTL Rx Queue Write Controller Active Status not detected 0x1 (ACTIVE): MTL Rx Queue Write Controller Active Status detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0xefc	MTL_RxQ7_Control	The Queue Receive Control register controls the receive arbitration and passing of received packets to the application.	31:4	Reserved_31_4	R	0x0	Reserved. Value After Reset: 0x0
			3	RXQ_FRM_ARBIT	R/W	0x0	Receive Queue Packet Arbitration When this bit is set, the DWC_ether_qos drives the packet data to the ARI interface such that the entire packet data of currently-selected queue is transmitted before switching to other queue. When this bit is reset, the DWC_ether_qos drives the packet data to the ARI interface such that the following amount of data of currently-selected queue is transmitted before switching to other queue: PBL amount of data (indicated by ari_qN_pbl_i[]) or Complete data of a packet The status and the timestamp are not a part of the PBL data. Therefore, the DWC_ether_qos drives the complete status (including timestamp status) during first PBL request for the packet (in store-and-forward mode) or the last PBL request for the packet (in Threshold mode).Values: 0x0 (DISABLE): Receive Queue Packet Arbitration is disabled 0x1 (ENABLE): Receive Queue Packet Arbitration is enabled Value After Reset: 0x0
			2:0	RXQ_WEGT	R/W	0x0	Receive Queue Weight This field indicates the weight assigned to the Rx Queue 1. This field needs to be programmed with one value less than the required weight, that is, reset value of 0 indicates weight of 1, value of 1 indicates weight of 2, and so on. The weight is used as the number of continuous PBL or packets requests (depending on the RXQ_FRM_ARBIT) allocated to the queue in one arbitration cycle. Note: The change in value of RXQ_WEGT takes effect only after the completion of current service round or when there is change from RAA=SP to RAA=WSP algorithm. This approach is taken so that there is smooth transition. For the RXQ_WEGT value to take effect at the start, the MTL_RxQ1_Control registers must be programmed before the MTL_Operation_Mode register. Value After Reset: 0x0
			31:24	Reserved_31_24	R	0x0	Reserved. Value After Reset: 0x0
			23:22	RNDF	R/W	0x0	Rx DMA's Maximum Number of Descriptors to be fetched in a burst 00: DWC_EQOS_DCRX_DEP 01: DWC_EQOS_DCRX_DEP/2 10: DWC_EQOS_DCRX_DEP/4 11: DWC_EQOS_DCRX_DEP/8 (Reserved when DWC_EQOS_DCRX_DEP = 4) Value After Reset: 0x0
			21:20	TNDF	R/W	0x0	Tx DMA's Maximum Number of Descriptors to be fetched in a burst 00: DWC_EQOS_DCTX_DEP 01: DWC_EQOS_DCTX_DEP/2 10: DWC_EQOS_DCTX_DEP/4 11: DWC_EQOS_DCTX_DEP/8 (Reserved when DWC_EQOS_DCTX_DEP = 4) Value After Reset: 0x0
			19	DCHE	R/W	0x0	Descriptor Cache Enable 1: Enables prefetching of the descriptors to the descriptor cache. 0: Disables the descriptor cache feature. Values: 0x0 (DISABLE): Descriptor Cache Support is disabled 0x1 (ENABLE): Descriptor Cache Support is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1000	DMA_Mode	The Bus Mode register establishes the bus operating modes for the DMA.	18	Reserved_18	R	0x0	Reserved. Value After Reset: 0x0
			17:16	INTM	R/W	0x0	Interrupt Mode This field defines the interrupt mode of DWC_ether_qos. The behavior of the following outputs change depending on the following settings: sbd_perch_tx_intr_o[] (Transmit Per Channel Interrupt) sbd_perch_rx_intr_o[] (Receive Per Channel Interrupt) sbd_intr_o (Common Interrupt) It also changes the behavior of the RI/TI bits in the DMA_CH(#i)_Status. 00: sbd_perch_* are pulse signals for each TX/RX packet transfer completion events (irrespective of whether corresponding interrupts are enabled) for which IOC bits are enabled in descriptor. sbd_intr_o is also asserted when corresponding interrupts are enabled and cleared only when software clears the corresponding RI/TI status bits. 01: sbd_perch_* are level signals asserted on TX/RX packet transfer completion event when corresponding interrupts are enabled and de-asserted when the software clears the corresponding RI/TI status bits. The sbd_intr_o is not asserted for these TX/RX packet transfer completion events. 10: sbd_perch_* are level signals asserted on TX/RX packet transfer completion event when corresponding interrupts are enabled and de-asserted when the software clears the corresponding RI/TI status bits. However, the signal is asserted again if the same event occurred again before it was cleared. The sbd_intr_o is not asserted for these TX/RX packet transfer completion events. 11: Reserved For more details, see Table "DWC_ether_qos Transfer Complete Interrupt Behavior".Values: 0x0 (MODE0): See above description 0x1 (MODE1): See above description 0x2 (MODE2): See above description 0x3 (RSVD): Reserved Value After Reset: 0x0
			15	Reserved_15	R	0x0	Reserved. Value After Reset: 0x0
			14:12			#VALUE!	Reserved Field: Yes
			11	TXPR	R/W	0x0	Transmit Priority 1: Indicates that the Tx DMA has priority over the Rx DMA during arbitration for the system-side bus. When you enable descriptor cache feature, descriptor reads from DCACHE memory. Values: 0x0 (DISABLE): Transmit Priority is disabled 0x1 (ENABLE): Transmit Priority is enabled Value After Reset: 0x0
			10	SCSW	R/W	0x0	SCSW is Synopsys Reserved, This field must be set to "0". This field is reserved for Synopsys Internal use, and must always be set to "0" unless instructed by Synopsys. Setting this field to "1" might cause unexpected behavior in the HW.Values: 0x0 (DISABLE): Synopsys reserved field disabled 0x1 (ENABLE): Synopsys reserved field enabled up on Synopsys request Value After Reset: 0x0
			9			#VALUE!	Reserved Field: Yes

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			8	DSPW	R/W	0x0	Descriptor Posted Write 0: Indicates that the descriptor writes are non-posted. 1: Indicates that descriptor writes are non-posted only when IOC = 1 (Interrupt on completion) in the last descriptor. Otherwise, the descriptor writes are always posted. Values: 0x0 (DISABLE): Descriptor Posted Write is disabled 0x1 (ENABLE): Descriptor Posted Write is enabled Value After Reset: 0x0
			7:5	Reserved_7_5	R	0x0	Reserved. Value After Reset: 0x0
			4:2	TAA	R/W	0x0	Transmit Arbitration Algorithm This field is used to select the arbitration policy for the Transmit side when you select multiple Tx DMAs.Values: 0x0 (FP): Fixed priority (Channel 0 has the lowest priority and the last channel has the highest priority) 0x1 (WSP): Weighted Strict Priority (WSP) 0x2 (WRR): Weighted Round-Robin (WRR) 0x3 (RSVD): Reserved (for 3'b011 to 3'b111) Value After Reset: 0x0
			1			#VALUE!	Reserved Field: Yes
			0	SWR	R/W	0x0	Software Reset 1: Indicates that MAC and the DMA controller reset the logic and all internal registers of the DMA, MTL, and MAC. This bit is automatically cleared after the reset operation is complete in all DWC_ether_qos clock domains. Before reprogramming any DWC_ether_qos register, read a value of zero in this bit. After writing 1 to this bit, wait for at the least 4 CSR clock cycles, before reading this bit. Note: The reset operation is complete only when DWC_ether_qos has de-asserted all resets in all active clock domains. Therefore, it is essential that all PHY input clocks of the selected PHY interface are present for the completion of the software reset. The time required to complete the software reset operation depends on the frequency of the slowest active clock. Access restrictions apply. Initiates the software reset when application writes 1. MAC writes 0 after the software reset is complete. MAC ignores when application writes 0. Values: 0x0 (DISABLE): Software Reset is disabled 0x1 (ENABLE): Software Reset is enabled Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			31	EN_LPI	R/W	0x0	Enable Low Power Interface (LPI) 1: Enables the LPI mode supported by the EQOS-AXI configuration and accepts the LPI request from the AXI system clock controller. 0: Disables the LPI mode. Denies the LPI request from the AXI system clock controller. Values: 0x0 (DISABLE): Low Power Interface (LPI) is disabled 0x1 (ENABLE): Low Power Interface (LPI) is enabled Value After Reset: 0x0
			30	LPI_XIT_PKT	R/W	0x0	Unlock on Magic Packet or Remote Wake-Up Packet 1: Enables the AXI master to come out of the LPI mode only when the magic packet or remote wake-up packet is received. 0: Enables the AXI master to come out of the LPI mode when any packet is received. Values: 0x0 (DISABLE): Unlock on Magic Packet or Remote Wake-Up Packet is disabled 0x1 (ENABLE): Unlock on Magic Packet or Remote Wake-Up Packet is enabled Value After Reset: 0x0
			29	Reserved_29_y	R	0x0	Reserved. Value After Reset: 0x0
			28:24	WR_OSR_LMT	R/W	0x1	AXI Maximum Write Outstanding Request Limit This value limits the maximum outstanding request on the AXI write interface. Maximum outstanding requests = WR_OSR_LMT + 1 Value After Reset: 0x1
			23:21	Reserved_23_y	R	0x0	Reserved. Value After Reset: 0x0
			20:16	RD_OSR_LMT	R/W	0x1	AXI Maximum Read Outstanding Request Limit This value limits the maximum outstanding request on the AXI read interface. Maximum outstanding requests = RD_OSR_LMT + 1 Value After Reset: 0x1
			15:14			#VALUE!	Reserved Field: Yes
			13	ONEKBBE	R/W	0x0	1 KB Boundary Crossing Enable for the EQOS-AXI Master 1: The burst transfers performed by the EQOS-AXI master do not cross 1 KB boundary. 0: The burst transfers performed by the EQOS-AXI master do not cross 4 KB boundary. Values: 0x0 (DISABLE): 1 KB Boundary Crossing for the EQOS-AXI Master Beats is disabled 0x1 (ENABLE): 1 KB Boundary Crossing for the EQOS-AXI Master Beats is enabled Value After Reset: 0x0
			12	AAL	R/W	0x0	Address-Aligned Beats 1: The EQOS-AXI master performs address-aligned burst transfers on Read and Write channels. 0: The EQOS-AXI master performs burst transfers on Read and Write channels without aligning to address boundaries. Values: 0x0 (DISABLE): Address-Aligned Beats is disabled 0x1 (ENABLE): Address-Aligned Beats is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1004	DMA_SysBus_Mode	System Bus Mode This register controls the behavior of the AXI master. It mainly controls burst splitting and the number of outstanding requests.	11	EAME	R/W	0x0	Enhanced Address Mode Enable. When set to 1: the DMA master enables the enhanced address mode (40-bit or 48-bit addressing mode). In this mode, the DMA engine uses either the 40- or 48-bit address, depending on the configuration. 0: the DMA master enables the normal address mode (32-bit). Values: 0x0 (DISABLE): Enhanced Address Mode is disabled 0x1 (ENABLE): Enhanced Address Mode is enabled Value After Reset: 0x0
			10	AALE	R/W	0x0	Automatic AXI LPI enable 1: Enables the AXI master to enter into LPI state when there is no activity in the DWC_ether_qos for certain number of system clock cycles. You can program the number of clock cycles in the LPIEI field of AXI_LPI_Entry_Interval register. Values: 0x0 (DISABLE): Automatic AXI LPI is disabled 0x1 (ENABLE): Automatic AXI LPI is enabled Value After Reset: 0x0
			9:8	Reserved_9_8	R	0x0	Reserved. Value After Reset: 0x0
			7	BLEN256	R/W	0x0	AXI Burst Length 256 When set to 1: the EQOS-AXI master can select a burst length of 256 on the AXI interface. 0: the EQOS-AXI master cannot select a burst length of greater than 128 on the AXI interface. Values: 0x0 (DISABLE): No effect 0x1 (ENABLE): AXI Burst Length 256 Value After Reset: 0x0
			6	BLEN128	R/W	0x0	AXI Burst Length 128 When set to 1: the EQOS-AXI master can select a burst length of 128 on the AXI interface. 0: the EQOS-AXI master cannot select a burst length of greater than 64 on the AXI interface. Values: 0x0 (DISABLE): No effect 0x1 (ENABLE): AXI Burst Length 128 Value After Reset: 0x0
			5	BLEN64	R/W	0x0	AXI Burst Length 64 When set to 1: the EQOS-AXI master can select a burst length of 64 on the AXI interface. 0: the EQOS-AXI master cannot select a burst length of greater than 32 on the AXI interface. Values: 0x0 (DISABLE): No effect 0x1 (ENABLE): AXI Burst Length 64 Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			4	BLEN32	R/W	0x0	AXI Burst Length 32 When set to 1: the EQOS-AXI master can select a burst length of 32 on the AXI interface. 0: the EQOS-AXI master cannot select a burst length of greater than 16 on the AXI interface. Values: 0x0 (DISABLE): No effect 0x1 (ENABLE): AXI Burst Length 32 Value After Reset: 0x0
			3	BLEN16	R/W	0x0	AXI Burst Length 16 When this bit is 1 or the FB bit is 1, the EQOS-AXI master can select a burst length of 16 on the AXI interface. When the FB bit is 0, setting this bit has no effect. Values: 0x0 (DISABLE): No effect 0x1 (ENABLE): AXI Burst Length 16 Value After Reset: 0x0
			2	BLEN8	R/W	0x0	AXI Burst Length 8 When this bit is 1 or the FB bit is 1, the EQOS-AXI master can select a burst length of 8 on the AXI interface. When the FB bit is 0, setting this bit has no effect. Values: 0x0 (DISABLE): No effect 0x1 (ENABLE): AXI Burst Length 8 Value After Reset: 0x0
			1	BLEN4	R/W	0x0	AXI Burst Length 4 When this bit is 1 or the FB bit is 1, the EQOS-AXI master can select a burst length of 4 on the AXI interface. When the FB bit is 0, setting this bit has no effect. Values: 0x0 (DISABLE): No effect 0x1 (ENABLE): AXI Burst Length 4 Value After Reset: 0x0
			0	FB	R/W	0x0	Fixed Burst Length 1: the EQOS-AXI master initiates burst transfers of specified lengths: Burst transfers of fixed burst lengths as indicated by the BLEN256, BLEN128, BLEN64, BLEN32, BLEN16, BLEN8, or BLEN4 field Burst transfers of length 1 0: EQOS-AXI master initiates burst transfers that are equal to or less than the maximum allowed burst length programmed in Bits[7:4]. Values: 0x0 (DISABLE): Fixed Burst Length is disabled 0x1 (ENABLE): Fixed Burst Length is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1008	DMA_Interrupt_Status	The application reads this Interrupt Status register during interrupt service routine or polling to determine the interrupt status of DMA channels, MTL queues, and the MAC.	31:18	Reserved_31_18	R	0x0	Reserved. Value After Reset: 0x0
			17	MACIS	R	0x0	MAC Interrupt Status Indicates an interrupt event in the MAC. To reset this bit to 1'b0, application must read the corresponding register in the MAC, get the exact cause of the interrupt and clear the source of the interrupt.Values: 0x0 (INACTIVE): MAC Interrupt Status not detected 0x1 (ACTIVE): MAC Interrupt Status detected Value After Reset: 0x0
			16	MTLIS	R	0x0	MTL Interrupt Status Indicates an interrupt event in the MTL. To reset this bit to 1'b0, application must read the corresponding register in the MTL, get the exact cause of the interrupt and clear the source of the interrupt.Values: 0x0 (INACTIVE): MTL Interrupt Status not detected 0x1 (ACTIVE): MTL Interrupt Status detected Value After Reset: 0x0
			15:8	Reserved_15_8	R	0x0	Reserved. Value After Reset: 0x0
			7	DC7IS	R	0x0	DMA Channel 7 Interrupt Status Indicates an interrupt event in DMA Channel 7. To reset this bit to 1'b0, application must read the corresponding register in DMA Channel 7, get the exact cause of the interrupt and clear the source of the interrupt.Values: 0x0 (INACTIVE): DMA Channel 7 Interrupt Status not detected 0x1 (ACTIVE): DMA Channel 7 Interrupt Status detected Value After Reset: 0x0
			6	DC6IS	R	0x0	DMA Channel 6 Interrupt Status Indicates an interrupt event in DMA Channel 6. To reset this bit to 1'b0, the application must read the corresponding register in DMA Channel 6, get the exact cause of the interrupt and clear the source of the interrupt.Values: 0x0 (INACTIVE): DMA Channel 6 Interrupt Status not detected 0x1 (ACTIVE): DMA Channel 6 Interrupt Status detected Value After Reset: 0x0
			5	DC5IS	R	0x0	DMA Channel 5 Interrupt Status This bit indicates an interrupt event in DMA Channel 5. To reset this bit to 1'b0, application must read the corresponding register in DMA Channel 5, get the exact cause of the interrupt and clear the source of the interrupt.Values: 0x0 (INACTIVE): DMA Channel 5 Interrupt Status not detected 0x1 (ACTIVE): DMA Channel 5 Interrupt Status detected Value After Reset: 0x0
			4	DC4IS	R	0x0	DMA Channel 4 Interrupt Status This bit indicates an interrupt event in DMA Channel 4. To reset this bit to 1'b0, the software must read the corresponding register in DMA Channel 4 to get the exact cause of the interrupt and clear its source.Values: 0x0 (INACTIVE): DMA Channel 4 Interrupt Status not detected 0x1 (ACTIVE): DMA Channel 4 Interrupt Status detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			3	DC3IS	R	0x0	DMA Channel 3 Interrupt Status This bit indicates an interrupt event in DMA Channel 3. To reset this bit to 1'b0, the software must read the corresponding register in DMA Channel 3 to get the exact cause of the interrupt and clear its source.Values: 0x0 (INACTIVE): DMA Channel 3 Interrupt Status not detected 0x1 (ACTIVE): DMA Channel 3 Interrupt Status detected Value After Reset: 0x0
			2	DC2IS	R	0x0	DMA Channel 2 Interrupt Status This bit indicates an interrupt event in DMA Channel 2. To reset this bit to 1'b0, the software must read the corresponding register in DMA Channel 2 to get the exact cause of the interrupt and clear its source.Values: 0x0 (INACTIVE): DMA Channel 2 Interrupt Status not detected 0x1 (ACTIVE): DMA Channel 2 Interrupt Status detected Value After Reset: 0x0
			1	DC1IS	R	0x0	DMA Channel 1 Interrupt Status This bit indicates an interrupt event in DMA Channel 1. To reset this bit to 1'b0, the software must read the corresponding register in DMA Channel 1 to get the exact cause of the interrupt and clear its source.Values: 0x0 (INACTIVE): DMA Channel 1 Interrupt Status not detected 0x1 (ACTIVE): DMA Channel 1 Interrupt Status detected Value After Reset: 0x0
			0	DC0IS	R	0x0	DMA Channel 0 Interrupt Status This bit indicates an interrupt event in DMA Channel 0. To reset this bit to 1'b0, the software must read the corresponding register in DMA Channel 0 to get the exact cause of the interrupt and clear its source.Values: 0x0 (INACTIVE): DMA Channel 0 Interrupt Status not detected 0x1 (ACTIVE): DMA Channel 0 Interrupt Status detected Value After Reset: 0x0
			31:28	TPS2	R	0x0	DMA Channel 2 Transmit Process State This field indicates the Tx DMA FSM state for Channel 2. The MSB of this field always returns 0. This field does not generate an interrupt.Values: 0x0 (STOP): Stopped (Reset or Stop Transmit Command issued) 0x1 (RUN_FTTD): Running (Fetching Tx Transfer Descriptor) 0x2 (RUN_WS): Running (Waiting for status) 0x3 (RUN_RDS): Running (Reading Data from system memory buffer and queuing it to the Tx buffer (Tx FIFO)) 0x4 (TSTMP_WS): Timestamp write state 0x5 (RSVD): Reserved for future use 0x6 (SUSPND): Suspended (Tx Descriptor Unavailable or Tx Buffer Underflow) 0x7 (RUN_CTD): Running (Closing Tx Descriptor) Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x100c	DMA_Debug_Status0	The Debug Status 0 register gives the Receive and Transmit process status for DMA Channel 0-Channel 2 for debugging purpose.	27:24	RPS2	R	0x0	DMA Channel 2 Receive Process State This field indicates the Rx DMA FSM state for Channel 2. The MSB of this field always returns 0. This field does not generate an interrupt.Values: 0x0 (STOP): Stopped (Reset or Stop Receive Command issued) 0x1 (RUN_FRTD): Running (Fetching Rx Transfer Descriptor) 0x2 (RSVD): Reserved for future use 0x3 (RUN_WRP): Running (Waiting for Rx packet) 0x4 (SUSPND): Suspended (Rx Descriptor Unavailable) 0x5 (RUN_CRD): Running (Closing the Rx Descriptor) 0x6 (TSTMP): Timestamp write state 0x7 (RUN_TRP): Running (Transferring the received packet data from the Rx buffer to the system memory) Value After Reset: 0x0
			23:20	TPS1	R	0x0	DMA Channel 1 Transmit Process State This field indicates the Tx DMA FSM state for Channel 1. The MSB of this field always returns 0. This field does not generate an interrupt.Values: 0x0 (STOP): Stopped (Reset or Stop Transmit Command issued) 0x1 (RUN_FTTD): Running (Fetching Tx Transfer Descriptor) 0x2 (RUN_WS): Running (Waiting for status) 0x3 (RUN_RDS): Running (Reading Data from system memory buffer and queuing it to the Tx buffer (Tx FIFO)) 0x4 (TSTMP_WS): Timestamp write state 0x5 (RSVD): Reserved for future use 0x6 (SUSPND): Suspended (Tx Descriptor Unavailable or Tx Buffer Underflow) 0x7 (RUN_CTD): Running (Closing Tx Descriptor) Value After Reset: 0x0
			19:16	RPS1	R	0x0	DMA Channel 1 Receive Process State This field indicates the Rx DMA FSM state for Channel 1. The MSB of this field always returns 0. This field does not generate an interrupt.Values: 0x0 (STOP): Stopped (Reset or Stop Receive Command issued) 0x1 (RUN_FRTD): Running (Fetching Rx Transfer Descriptor) 0x2 (RSVD): Reserved for future use 0x3 (RUN_WRP): Running (Waiting for Rx packet) 0x4 (SUSPND): Suspended (Rx Descriptor Unavailable) 0x5 (RUN_CRD): Running (Closing the Rx Descriptor) 0x6 (TSTMP): Timestamp write state 0x7 (RUN_TRP): Running (Transferring the received packet data from the Rx buffer to the system memory) Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			15:12	TPS0	R	0x0	DMA Channel 0 Transmit Process State This field indicates the Tx DMA FSM state for Channel 0. The MSB of this field always returns 0. This field does not generate an interrupt.Values: 0x0 (STOP): Stopped (Reset or Stop Transmit Command issued) 0x1 (RUN_FTTD): Running (Fetching Tx Transfer Descriptor) 0x2 (RUN_WS): Running (Waiting for status) 0x3 (RUN_RDS): Running (Reading Data from system memory buffer and queuing it to the Tx buffer (Tx FIFO)) 0x4 (TSTMP_WS): Timestamp write state 0x5 (RSVD): Reserved for future use 0x6 (SUSPND): Suspended (Tx Descriptor Unavailable or Tx Buffer Underflow) 0x7 (RUN_CTD): Running (Closing Tx Descriptor) Value After Reset: 0x0
			11:8	RPS0	R	0x0	DMA Channel 0 Receive Process State This field indicates the Rx DMA FSM state for Channel 0. The MSB of this field always returns 0. This field does not generate an interrupt.Values: 0x0 (STOP): Stopped (Reset or Stop Receive Command issued) 0x1 (RUN_FRTD): Running (Fetching Rx Transfer Descriptor) 0x2 (RSVD): Reserved for future use 0x3 (RUN_WRP): Running (Waiting for Rx packet) 0x4 (SUSPND): Suspended (Rx Descriptor Unavailable) 0x5 (RUN_CRD): Running (Closing the Rx Descriptor) 0x6 (TSTMP): Timestamp write state 0x7 (RUN_TRP): Running (Transferring the received packet data from the Rx buffer to the system memory) Value After Reset: 0x0
			7:2	Reserved_7_2	R	0x0	Reserved. Value After Reset: 0x0
			1	AXRHSTS	R	0x0	AXI Master Read Channel Status 1: Indicates that the read channel of the AXI master is active, and it is transferring the data. Values: 0x0 (INACTIVE): AXI Master Read Channel Status not detected 0x1 (ACTIVE): AXI Master Read Channel Status detected Value After Reset: 0x0
			0	AXWHSTS	R	0x0	AXI Master Write Channel When high, this bit indicates that the write channel of the AXI master is active, and it is transferring data.Values: 0x0 (INACTIVE): AXI Master Write Channel or AHB Master Status not detected 0x1 (ACTIVE): AXI Master Write Channel or AHB Master Status detected Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			31:28	TPS6	R	0x0	DMA Channel 6 Transmit Process State This field indicates the Tx DMA FSM state for Channel 6. The MSB of this field always returns 0. This field does not generate an interrupt.Values: 0x0 (STOP): Stopped (Reset or Stop Transmit Command issued) 0x1 (RUN_FTTD): Running (Fetching Tx Transfer Descriptor) 0x2 (RUN_WS): Running (Waiting for status) 0x3 (RUN_RDS): Running (Reading Data from system memory buffer and queuing it to the Tx buffer (Tx FIFO)) 0x4 (TSTMP_WS): Timestamp write state 0x5 (RSVD): Reserved for future use 0x6 (SUSPND): Suspended (Tx Descriptor Unavailable or Tx Buffer Underflow) 0x7 (RUN_CTD): Running (Closing Tx Descriptor) Value After Reset: 0x0
			27:24	RPS6	R	0x0	DMA Channel 6 Receive Process State This field indicates the Rx DMA FSM state for Channel 6. The MSB of this field always returns 0. This field does not generate an interrupt.Values: 0x0 (STOP): Stopped (Reset or Stop Receive Command issued) 0x1 (RUN_FRTD): Running (Fetching Rx Transfer Descriptor) 0x2 (RSVD): Reserved for future use 0x3 (RUN_WRP): Running (Waiting for Rx packet) 0x4 (SUSPND): Suspended (Rx Descriptor Unavailable) 0x5 (RUN_CRD): Running (Closing the Rx Descriptor) 0x6 (TSTMP): Timestamp write state 0x7 (RUN_TRP): Running (Transferring the received packet data from the Rx buffer to the system memory) Value After Reset: 0x0
			23:20	TPS5	R	0x0	DMA Channel 5 Transmit Process State This field indicates the Tx DMA FSM state for Channel 5. The MSB of this field always returns 0. This field does not generate an interrupt.Values: 0x0 (STOP): Stopped (Reset or Stop Transmit Command issued) 0x1 (RUN_FTTD): Running (Fetching Tx Transfer Descriptor) 0x2 (RUN_WS): Running (Waiting for status) 0x3 (RUN_RDS): Running (Reading Data from system memory buffer and queuing it to the Tx buffer (Tx FIFO)) 0x4 (TSTMP_WS): Timestamp write state 0x5 (RSVD): Reserved for future use 0x6 (SUSPND): Suspended (Tx Descriptor Unavailable or Tx Buffer Underflow) 0x7 (RUN_CTD): Running (Closing Tx Descriptor) Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1010	DMA_Debug_Status1	The Debug Status1 register gives the Receive and Transmit process status for DMA Channel 3-Channel 6.	19:16	RPS5	R	0x0	DMA Channel 5 Receive Process State This field indicates the Rx DMA FSM state for Channel 5. The MSB of this field always returns 0. This field does not generate an interrupt.Values: 0x0 (STOP): Stopped (Reset or Stop Receive Command issued) 0x1 (RUN_FRTD): Running (Fetching Rx Transfer Descriptor) 0x2 (RSVD): Reserved for future use 0x3 (RUN_WRP): Running (Waiting for Rx packet) 0x4 (SUSPND): Suspended (Rx Descriptor Unavailable) 0x5 (RUN_CRD): Running (Closing the Rx Descriptor) 0x6 (TSTMP): Timestamp write state 0x7 (RUN_TRP): Running (Transferring the received packet data from the Rx buffer to the system memory) Value After Reset: 0x0
			15:12	TPS4	R	0x0	DMA Channel 4 Transmit Process State This field indicates the Tx DMA FSM state for Channel 4. The MSB of this field always returns 0. This field does not generate an interrupt.Values: 0x0 (STOP): Stopped (Reset or Stop Transmit Command issued) 0x1 (RUN_FTTD): Running (Fetching Tx Transfer Descriptor) 0x2 (RUN_WS): Running (Waiting for status) 0x3 (RUN_RDS): Running (Reading Data from system memory buffer and queuing it to the Tx buffer (Tx FIFO)) 0x4 (TSTMP_WS): Timestamp write state 0x5 (RSVD): Reserved for future use 0x6 (SUSPND): Suspended (Tx Descriptor Unavailable or Tx Buffer Underflow) 0x7 (RUN_CTD): Running (Closing Tx Descriptor) Value After Reset: 0x0
			11:8	RPS4	R	0x0	DMA Channel 4 Receive Process State This field indicates the Rx DMA FSM state for Channel 4. The MSB of this field always returns 0. This field does not generate an interrupt.Values: 0x0 (STOP): Stopped (Reset or Stop Receive Command issued) 0x1 (RUN_FRTD): Running (Fetching Rx Transfer Descriptor) 0x2 (RSVD): Reserved for future use 0x3 (RUN_WRP): Running (Waiting for Rx packet) 0x4 (SUSPND): Suspended (Rx Descriptor Unavailable) 0x5 (RUN_CRD): Running (Closing the Rx Descriptor) 0x6 (TSTMP): Timestamp write state 0x7 (RUN_TRP): Running (Transferring the received packet data from the Rx buffer to the system memory) Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			7:4	TPS3	R	0x0	DMA Channel 3 Transmit Process State This field indicates the Tx DMA FSM state for Channel 3. The MSB of this field always returns 0. This field does not generate an interrupt.Values: 0x0 (STOP): Stopped (Reset or Stop Transmit Command issued) 0x1 (RUN_FTTD): Running (Fetching Tx Transfer Descriptor) 0x2 (RUN_WS): Running (Waiting for status) 0x3 (RUN_RDS): Running (Reading Data from system memory buffer and queuing it to the Tx buffer (Tx FIFO)) 0x4 (TSTMP_WS): Timestamp write state 0x5 (RSVD): Reserved for future use 0x6 (SUSPND): Suspended (Tx Descriptor Unavailable or Tx Buffer Underflow) 0x7 (RUN_CTD): Running (Closing Tx Descriptor) Value After Reset: 0x0
			3:0	RPS3	R	0x0	DMA Channel 3 Receive Process State This field indicates the Rx DMA FSM state for Channel 3. The MSB of this field always returns 0. This field does not generate an interrupt.Values: 0x0 (STOP): Stopped (Reset or Stop Receive Command issued) 0x1 (RUN_FRTD): Running (Fetching Rx Transfer Descriptor) 0x2 (RSVD): Reserved for future use 0x3 (RUN_WRP): Running (Waiting for Rx packet) 0x4 (SUSPND): Suspended (Rx Descriptor Unavailable) 0x5 (RUN_CRD): Running (Closing the Rx Descriptor) 0x6 (TSTMP): Timestamp write state 0x7 (RUN_TRP): Running (Transferring the received packet data from the Rx buffer to the system memory) Value After Reset: 0x0
0x1014	DMA_Debug_Status2	The Debug Status Register 2 gives the Receive and Transmit process status for DMA Channel 7.	31:8	Reserved_31_8	R	0x0	Reserved. Value After Reset: 0x0
			7:4	TPS7	R	0x0	DMA Channel 7 Transmit Process State This field indicates the Tx DMA FSM state for Channel 7. The MSB of this field always returns 0. This field does not generate an interrupt.Values: 0x0 (STOP): Stopped (Reset or Stop Transmit Command issued) 0x1 (RUN_FTTD): Running (Fetching Tx Transfer Descriptor) 0x2 (RUN_WS): Running (Waiting for status) 0x3 (RUN_RDS): Running (Reading Data from system memory buffer and queuing it to the Tx buffer (Tx FIFO)) 0x4 (TSTMP_WS): Timestamp write state 0x5 (RSVD): Reserved for future use 0x6 (SUSPND): Suspended (Tx Descriptor Unavailable or Tx Buffer Underflow) 0x7 (RUN_CTD): Running (Closing Tx Descriptor) Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			3:0	RPS7	R	0x0	DMA Channel 7 Receive Process State This field indicates the Rx DMA FSM state for Channel 7. The MSB of this field always returns 0. This field does not generate an interrupt.Values: 0x0 (STOP): Stopped (Reset or Stop Receive Command issued) 0x1 (RUN_FRTD): Running (Fetching Rx Transfer Descriptor) 0x2 (RSVD): Reserved for future use 0x3 (RUN_WRP): Running (Waiting for Rx packet) 0x4 (SUSPND): Suspended (Rx Descriptor Unavailable) 0x5 (RUN_CRD): Running (Closing the Rx Descriptor) 0x6 (TSTMP): Timestamp write state 0x7 (RUN_TRP): Running (Transferring the received packet data from the Rx buffer to the system memory) Value After Reset: 0x0
0x1020	AXI4_Tx_AR_ACE_Control	This register is used to control the AXI4 Cache Coherency Signals for read transactions by all the Transmit DMA channels. The following signals of the AXI4 interface are driven with different values as programmed for corresponding type (descriptor, buffer1, buffer2) of access.arcache_m_o[3:0]ardomain_m_o[1:0]	31:22	Reserved_31_22	R	0x0	Reserved. Value After Reset: 0x0
			21:20	THD	R/W	0x0	Transmit DMA First Packet Buffer or TSO Header Domain Control When TSO is NOT enabled, This field is used to drive ardomain_m_o[1:0] signal when Transmit DMA is accessing First Buffer of the Packet (First valid buffer with FD being set in the TDES3 of the Descriptor). When TSO is enabled, This field is used to drive ardomain_m_o[1:0] signal when the Transmit DMA is accessing the TSO Header data. Value After Reset: 0x0
			19:16	THC	R/W	0x0	Transmit DMA First Packet Buffer or TSO Header Cache Control When TSO is NOT enabled, This field is used to drive arcache_m_o[3:0] signal when Transmit DMA is accessing First Buffer of the Packet (First valid buffer with FD being set in the TDES3 of the Descriptor). When TSO is enabled, This field is used to drive arcache_m_o[3:0] signal when the Transmit DMA is accessing the TSO Header data. Value After Reset: 0x0
			15:14	Reserved_15_14	R	0x0	Reserved. Value After Reset: 0x0
			13:12	TED	R/W	0x0	Transmit DMA Extended Packet Buffer or TSO Payload Domain Control When TSO is NOT enabled, This field is used to drive ardomain_m_o[1:0] signal when Transmit DMA is accessing the extended buffers (when packet is distributed across multiple buffers). When TSO is enabled, This field is used to drive ardomain_m_o[1:0] signal when the Transmit DMA is accessing the TSO payload data. Value After Reset: 0x0
			11:8	TEC	R/W	0x0	Transmit DMA Extended Packet Buffer or TSO Payload Cache Control When TSO is NOT enabled, This field is used to drive arcache_m_o[3:0] signal when Transmit DMA is accessing the extended buffers (when packet is distributed across multiple buffers). When TSO is enabled, This field is used to drive arcache_m_o[3:0] signal when the Transmit DMA is accessing the TSO payload data. Value After Reset: 0x0
			7:6	Reserved_7_6	R	0x0	Reserved. Value After Reset: 0x0
			5:4	TDRD	R/W	0x0	Transmit DMA Read Descriptor Domain Control This field is used to drive ardomain_m_o[1:0] signal when Transmit DMA engines access the Descriptor. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			3:0	TDRC	R/W	0x0	Transmit DMA Read Descriptor Cache Control This field is used to drive arcache_m_o[3:0] signal when Transmit DMA engines access the Descriptor. Value After Reset: 0x0
0x1024	AXI4_Rx_AW_ACE_Control	This register is used to control the AXI4 Cache Coherency Signals for write transactions by all the Receive DMA channels. The following signals of the AXI4 interface are driven with different values as programmedfor corresponding type (descriptor, buffer1, buffer2) of access.awcache_m_o[3:0]awdomain_m_o[1:0]	31:30	Reserved_31_30	R	0x0	Reserved. Value After Reset: 0x0
			29:28	RDD	R/W	0x0	Receive DMA Buffer Domain Control This field is used to drive the awdomain_m_o[1:0] signal when Receive DMA is accessing the Buffer when Header and payload are NOT separated. Value After Reset: 0x0
			27:24	RDC	R/W	0x0	Receive DMA Buffer Cache Control This field is used to drive awcache_m_o[3:0] signal when Receive DMA is accessing the Buffer when Header and payload are NOT separated. Value After Reset: 0x0
			23:22	Reserved_23_22	R	0x0	Reserved. Value After Reset: 0x0
			21:20	RHD	R/W	0x0	Receive DMA Header Domain Control This field is used to drive awdomain_m_o[1:0] and signal when Receive DMA is accessing the header Buffer when Header and payload are separated. Value After Reset: 0x0
			19:16	RHC	R/W	0x0	Receive DMA Header Cache Control This field is used to drive awcache_m_o[3:0] and signal when Receive DMA is accessing the header Buffer when Header and payload are separated. Value After Reset: 0x0
			15:14	Reserved_15_14	R	0x0	Reserved. Value After Reset: 0x0
			13:12	RPD	R/W	0x0	Receive DMA Payload Domain Control This field is used to drive awdomain_m_o[1:0] signal when Receive DMA is accessing the Payload Buffer when Header and payload are separated. Value After Reset: 0x0
			11:8	RPC	R/W	0x0	Receive DMA Payload Cache Control This field is used to drive awcache_m_o[3:0] signal when Receive DMA is accessing the Payload Buffer when Header and payload are separated. Value After Reset: 0x0
			7:6	Reserved_7_6	R	0x0	Reserved. Value After Reset: 0x0
			5:4	RDWD	R/W	0x0	Receive DMA Write Descriptor Domain Control This field is used to drive awdomain_m_o[1:0] signal when Receive DMA accesses the Descriptor. Value After Reset: 0x0
			3:0	RDWC	R/W	0x0	Receive DMA Write Descriptor Cache Control This field is used to drive awcache_m_o[3:0] signal when Receive DMA accesses the Descriptor. Value After Reset: 0x0
			31:23	Reserved_31_23	R	0x0	Reserved. Value After Reset: 0x0
			22:20	WRP	R/W	0x0	DMA Write Protection control This field is used to drive awprot_m_o[2:0] signal on the AXI Write Channel. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1028	AXI4_TxRx_AWAR_ACE_Control	This register is used to control the AXI4 Cache Coherency Signals for Descriptor write transactions by all the TxDMA channels and Descriptor read transactions by all the RxDMA channels. It also controls the values to be driven on awprot_m_o and arprot_m_o.	19	Reserved_19	R	0x0	Reserved. Value After Reset: 0x0
			18:16	RDP	R/W	0x0	DMA Read Protection control This field is used to drive arprot_m_o[2:0] signal during all read requests. Value After Reset: 0x0
			15:14	Reserved_15_14	R	0x0	Reserved. Value After Reset: 0x0
			13:12	RDRD	R/W	0x0	Receive DMA Read Descriptor Domain control This field is used to drive ardomain_m_o[1:0] signal when Receive DMA engines read the Descriptor. Value After Reset: 0x0
			11:8	RDRC	R/W	0x0	Receive DMA Read Descriptor Cache control This field is used to drive arcache_m_o[3:0] signal when Receive DMA engines read the Descriptor. Value After Reset: 0x0
			7:6	Reserved_7_6	R	0x0	Reserved. Value After Reset: 0x0
			5:4	TDWD	R/W	0x0	Transmit DMA Write Descriptor Domain control This field is used to drive awdomain_m_o[1:0] signal when Transmit DMA write to the Descriptor. Value After Reset: 0x0
			3:0	TDWC	R/W	0x0	Transmit DMA Write Descriptor Cache control This field is used to drive awcache_m_o[3:0] signal when Transmit DMA writes to the Descriptor. Value After Reset: 0x0
0x1040	AXI_LPI_Entry_Interval	This register is used to control the AXI LPI entry interval.	31:4	Reserved_31_4	R	0x0	Reserved. Value After Reset: 0x0
			3:0	LPIEI	R/W	0x0	LPI Entry Interval Contains the number of system clock cycles, multiplied by 64, to wait for an activity in the DWC_ether_qos to enter into the AXI low power state 0 indicates 64 clock cycles Value After Reset: 0x0
0x1050	DMA_TBS_CTRL0	This register is used to control the TBS attributes.	31:8	FTOS	R/W	0x0	Fetch Time Offset The value in units of 256 ns, that has to be deducted from the Launch time to compute the Fetch Time. Max value: 999,999,999 ns, additionally must be smaller than CTR-1 value when in ESTM mode. Because, this value is a modulo CTR value. Value After Reset: 0x0
			7	Reserved_7	R	0x0	Reserved. Value After Reset: 0x0
			6:4	FGOS	R/W	0x0	Fetch GSN Offset The number GSN slots that must be deducted from the Launch GSN to compute the Fetch GSN. Value valid only when FTOV is set. Value After Reset: 0x0
			3:1	Reserved_3_1	R	0x0	Reserved. Value After Reset: 0x0
			0	FTOV	R/W	0x0	Fetch Time Offset Valid When set indicates the FTOS field is valid. When not set, indicates the Fetch Offset is not valid and the DMA engine can fetch the frames from host memory without any time restrictions.Values: 0x0 (INVALID): Fetch Time Offset is invalid 0x1 (VALID): Fetch Time Offset is valid Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1054	DMA_TBS_CTRL1	This register is used to control the TBS attributes.	31:8	FTOS	R/W	0x0	Fetch Time Offset The value in units of 256 nanoseconds, that has to be deducted from the Launch time to compute the Fetch Time. Max value: 999,999,999 ns, additionally must be smaller than CTR-1 value when in ESTM mode. Because, this value is a modulo CTR value. Value After Reset: 0x0
			7	Reserved_7	R	0x0	Reserved. Value After Reset: 0x0
			6:4	FGOS	R/W	0x0	Fetch GSN Offset The number GSN slots that must be deducted from the Launch GSN to compute the Fetch GSN. Value valid only when FTOV is set. Value After Reset: 0x0
			3:1	Reserved_3_1	R	0x0	Reserved. Value After Reset: 0x0
			0	FTOV	R/W	0x0	Fetch Time Offset Valid When set indicates the FTOS field is valid. When not set, indicates the Fetch Offset is not valid and the DMA engine can fetch the frames from host memory without any time restrictions.Values: 0x0 (INVALID): Fetch Time Offset is invalid 0x1 (VALID): Fetch Time Offset is valid Value After Reset: 0x0
0x1058	DMA_TBS_CTRL2	This register is used to control the TBS attributes.	31:8	FTOS	R/W	0x0	Fetch Time Offset The value in units of 256 nanoseconds, that has to be deducted from the Launch time to compute the Fetch Time. Max value: 999,999,999 ns, additionally must be smaller than CTR-1 value when in ESTM mode. Because, this value is a modulo CTR value. Value After Reset: 0x0
			7	Reserved_7	R	0x0	Reserved. Value After Reset: 0x0
			6:4	FGOS	R/W	0x0	Fetch GSN Offset The number GSN slots that must be deducted from the Launch GSN to compute the Fetch GSN. Value valid only when FTOV is set. Value After Reset: 0x0
			3:1	Reserved_3_1	R	0x0	Reserved. Value After Reset: 0x0
			0	FTOV	R/W	0x0	Fetch Time Offset Valid When set indicates the FTOS field is valid. When not set, indicates the Fetch Offset is not valid and the DMA engine can fetch the frames from host memory without any time restrictions.Values: 0x0 (INVALID): Fetch Time Offset is invalid 0x1 (VALID): Fetch Time Offset is valid Value After Reset: 0x0
			31:8	FTOS	R/W	0x0	Fetch Time Offset The value in units of 256 nanoseconds, that has to be deducted from the Launch time to compute the Fetch Time. Max value: 999,999,999 ns, additionally must be smaller than CTR-1 value when in ESTM mode. Because, this value is a modulo CTR value. Value After Reset: 0x0
			7	Reserved_7	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x105c	DMA_TBS_CTRL3	This register is used to control the TBS attributes.	6:4	FGOS	R/W	0x0	Fetch GSN Offset The number GSN slots that must be deducted from the Launch GSN to compute the Fetch GSN. Value valid only when FTOV is set. Value After Reset: 0x0
			3:1	Reserved_3_1	R	0x0	Reserved. Value After Reset: 0x0
			0	FTOV	R/W	0x0	Fetch Time Offset Valid When set indicates the FTOS field is valid. When not set, indicates the Fetch Offset is not valid and the DMA engine can fetch the frames from host memory without any time restrictions.Values: 0x0 (INVALID): Fetch Time Offset is invalid 0x1 (VALID): Fetch Time Offset is valid Value After Reset: 0x0
0x1100	DMA_CH0_Control	The DMA Channel0 Control register specifies the MSS value for segmentation, length to skip between two descriptors, and also the features such as header splitting and 8xPBL mode.	31:25	Reserved_31_25	R	0x0	Reserved. Value After Reset: 0x0
			24	SPH	R/W	0x0	Split Headers 1: DMA splits the header and payload in the receive path. The DMA writes the header to the Buffer Address1 of RDES0. The DMA writes the payload to the buffer to which the Buffer Address2 is pointing. The software must ensure that the header fits into the Receive buffers. If the header length exceeds the receive buffer size, the DMA does not split the header and payload. This bit is available only when you select Enable Split Header Structure feature.Values: 0x0 (DISABLE): Split Headers feature is disabled 0x1 (ENABLE): Split Headers feature is enabled Value After Reset: 0x0
			23:21	Reserved_23_21	R	0x0	Reserved. Value After Reset: 0x0
			20:18	DSL	R/W	0x0	Descriptor Skip Length This bit specifies the Lword (128-bit) number of locations to skip between two descriptors. The address skipping starts from the end of the current descriptor to the start of the next descriptor. When the DSL value is equal to zero, the DMA takes the descriptor table as contiguous. Value After Reset: 0x0
			17	Reserved_17	R	0x0	Reserved. Value After Reset: 0x0
			16	PBLx8	R/W	0x0	8xPBL mode 1: Indicates that the PBL value programmed in Bits[21:16] in DMA_CH0_Tx_Control and Bits[21:16] in DMA_CH0_Rx_Control is multiplied by eight times. Therefore, the DMA transfers the data in 8, 16, 32, 64, 128, and 256 beats depending on the PBL value. Values: 0x0 (DISABLE): 8xPBL mode is disabled 0x1 (ENABLE): 8xPBL mode is enabled Value After Reset: 0x0
			15:14	Reserved_15_14	R	0x0	Reserved. Value After Reset: 0x0
			13:0	MSS	R/W	0x0	Maximum Segment Size This field specifies the maximum segment size that must be used while segmenting the packet. This field is valid only if the TSE bit of DMA_CH0_Tx_Control register is set. Program a value in this field that is more than the configured datawidth in bytes. Use MSS value of 64 bytes or more. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			31	Reserved_31	R	0x0	Reserved. Value After Reset: 0x0
			30:29	TFSEL	R/W	0x0	TBS Fetch Select Select bits for one of the four DMA_TBS_CTRL register fields (FTOS,FGSN,FTOV) for the channel Values: 2'b00: DMA_TBS_CTRL0 2'b01: DMA_TBS_CTRL1 2'b10: DMA_TBS_CTRL2 2'b11: DMA_TBS_CTRL3 (Reserved if DWC_EQOS_NUM_DMA_TX_CH=3) Value After Reset: 0x0
			28	EDSE	R/W	0x0	Enhanced Descriptor Enable 1: The corresponding channel uses Enhanced Descriptors that are 32 Bytes for both Normal and Context Descriptors. 0: The corresponding channel uses the descriptors that are 16 Bytes. Values: 0x0 (DISABLE): Enhanced Descriptor is disabled 0x1 (ENABLE): Enhanced Descriptor is enabled Value After Reset: 0x0
			27:24	TQOS	R/W	0x0	Transmit QOS. This field is used to drive arqos_m_o[3:0] or awqos_m_o[3:0] output signals for all transactions of DMA Tx Channel0. Value After Reset: 0x0
			23	Reserved_23	R	0x0	Reserved. Value After Reset: 0x0
			22			#VALUE!	Reserved Field: Yes
			21:16	TxPBL	R/W	0x0	Transmit Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in DMA_CH0_Control register. 2. Set the TxPBL. Note: The maximum value of TxPBL must be less than or equal to half the Tx Queue size (TQS field of MTL_TxQ0_Operation_Mode register) in terms of beats. This is required so that the Tx Queue has space to store at least another Tx PBL worth of data while the MTL Tx Queue Controller is transferring data to MAC. For example, in 32-bit data width configurations the total locations in Tx Queue of size 512 bytes is 128, TxPBL and 8xPBL needs to be programmed to less than or equal to 64. Value After Reset: 0x0
			15	IPBL	R/W	0x0	Ignore PBL Requirement 1: DMA does not check for PBL number of locations in the MTL before initiating a transfer. If space is not available, the MTL might use handshaking to slow the DMA. Note: Do not use this mode when you enable multiple Transmit DMA channels. Because, it might block other Transmit and Receive DMA Channels from accessing the Read Data Channel of the AXI bus until space is available in Transmit Queue for current transfer.Values: 0x0 (DISABLE): Ignore PBL Requirement is disabled 0x1 (ENABLE): Ignore PBL Requirement is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1104	DMA_CH0_Tx_Control	The DMA Channel0 Transmit Control register controls the Tx features such as PBL, TCP segmentation, and Tx Channel weights.	14:13	TSE_MODE	R/W	0x0	TSE Mode 00: TSO/USO (segmentation functionality is enabled). In this mode, the setting of TSE bit enables the TSO/USO segmentation. 01: UFO with Checksum (UDP Fragmentation over IPv4 with Checksum). In this mode, the setting of TSE bit enables the UDP fragmentation functionality with Checksum for all the UDP packets. 10: UFO without Checksum (UDP Fragmentation over IPv4 with Checksum). In this mode, the setting of TSE bit enables the UDP fragmentation functionality without Checksum for all the UDP packets. 11: Reserved Values: 0x0 (TSO_USO): TSO/USO 0x1 (UFOWC): UFO with Checksum 0x2 (UFOWOC): UFO without Checksum 0x3 (RSVD): Reserved Value After Reset: 0x0
			12	TSE	R/W	0x0	TCP Segmentation Enabled 1: DMA performs the TCP segmentation or UDP Segmentation/Fragmentation for packets in this channel. The TCP segmentation or UDP packet's segmentation/Fragmentation is performed only for those packets for which the TSE bit (TDES0[19]) = 1 in the Tx Normal descriptor. When this bit is 1, the TxPBL value must be greater than 4. Values: 0x0 (DISABLE): TCP Segmentation is disabled 0x1 (ENABLE): TCP Segmentation is enabled Value After Reset: 0x0
			11:5	Reserved_11_5	R	0x0	Reserved. Value After Reset: 0x0
			4	OSF	R/W	0x0	Operate on Second Packet 1: Instructs the DMA to process the second packet of the Tx data even before the status for the first packet is obtained. Values: 0x0 (DISABLE): Operate on Second Packet disabled 0x1 (ENABLE): Operate on Second Packet enabled Value After Reset: 0x0
			3:1	TCW	R/W	0x0	Transmit Channel Weight This field indicates the weight assigned to the corresponding Tx channel. When reset is complete, this field is set to 0 for all channels by default, resulting in equal weights to all channels. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	ST	R/W	0x0	Start or Stop Transmission Command 1: Transmission is placed in the running state. The DMA checks the Tx list at the current position for a packet to be transmitted. The DMA tries to acquire descriptor from either of the following positions: The current position in the list This is the base address of the Transmit list set by the DMA_CH0_TxDesc_List_Address register. The position at which the transmission was previously stopped If the DMA does not own the current descriptor, the transmission enters the Suspended state and the TBU bit of the DMA_CH0_Status register is set. The Start Transmission command is effective only when the transmission is stopped. If the command is issued before setting the DMA_CH0_TxDesc_List_Address register, the DMA behavior is unpredictable. When this bit is reset, the transmission process is placed in the Stopped state after completing the transmission of the current packet. The Next Descriptor position in the Transmit list is saved, and it becomes the current position when the transmission is restarted. To change the list address, you need to program DMA_CH0_TxDesc_List_Address register with a new value when this bit is reset. The new value is considered when this bit is set again. The stop transmission command is effective only when the transmission of the current packet is complete or the transmission is in the Suspended state.Values: 0x0 (STOP): Stop Transmission Command 0x1 (START): Start Transmission Command Value After Reset: 0x0 Testable: untestable
			31	RPF	R/W	0x0	Rx Packet Flush. 1: DWC_ether_qos automatically flushes the packet from the Rx Queues destined to this DMA Rx Channel, when it is stopped. When this bit remains set and the DMA is re-started by the software driver, the packets residing in the Rx Queues that were received when this RxDMA was stopped, still get flushed out. The packets received by the MAC only after the DMA is re-started is routed to the Rx DMA.The flushing is done on the Read side of the Rx Queue. When the packet is duplicated to multiple Rx DMA, it is flushed by the Rx DMA to which it is routed only when it is stopped to prevent head-of-line blocking. The packet duplicated to multiple Rx DMA is not flushed when the Rx DMA is re-started, irrespective of whether the packet was received during one or more Rx DMAs were stopped. 0: DWC_ether_qos does not flush the packet in the Rx Queue destined to this RxDMA Channel when it is in STOP state. This might cause head-of-line blocking in the corresponding RxQueue. Note: The stopping of packet flow from a Rx DMA Channel to the application by setting RPF works only when there is one-to-one mapping of Rx Queue to Rx DMA channels. In Dynamic mapping mode, setting RPF bit in any DMA_CH0_Rx_Control register might flush packets from unintended Rx Queues which are destined to the stopped Rx DMA Channel.Values: 0x0 (DISABLE): Rx Packet Flush is disabled 0x1 (ENABLE): Rx Packet Flush is enabled Value After Reset: 0x0
			30:28	Reserved_30_28	R	0x0	Reserved. Value After Reset: 0x0
			27:24	RQOS	R/W	0x0	Rx AXI4 QOS. This field is used to drive arqos_m_o[3:0] or awqos_m_o[3:0] output signals for all transactions of DMA Rx Channel0. Value After Reset: 0x0
			23	Reserved_23	R	0x0	Reserved. Value After Reset: 0x0
			22			#VALUE!	Reserved Field: Yes

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1108	DMA_CH0_Rx_Control	The DMA Channel0 Receive Control register controls the Rx features such as PBL, buffer size, and extended status.	21:16	RxPBL	R/W	0x0	Receive Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in the DMA_CH0_Control register. 2. Set the RxPBL. Note: The maximum value of RxPBL must be less than or equal to half the Rx Queue size (RQS field of MTL_RxQ0_Operation_Mode register) in terms of beats. This is required so that the Rx Queue has space to store at least another Rx PBL worth of data while the Rx DMA is transferring a block of data. For example, in 32-bit data width configurations the total locations in Rx Queue of size 512 bytes is 128, so RxPBL and 8xPBL needs to be programmed to less than or equal to 64. Value After Reset: 0x0
			15	Reserved_15	R	0x0	Reserved. Value After Reset: 0x0
			14:5	RBSZ_13_y	R/W	0x0	Receive Buffer size High RBSZ[13:0] is split into two fields higher RBSZ_13_y and lower RBSZ_x_0. The RBSZ[13:0] field indicates the size of the Rx buffers specified in bytes. The maximum buffer size is limited to 16K bytes. The buffer size is applicable to payload buffers when split headers are enabled. Note: The buffer size must be a multiple of 16. This is required even if the value of buffer address pointer is not aligned to data bus width. Hence the lower RBSZ_x_0 bits are read-only and the value is considered as all-zero. Thus the RBSZ_13_y indicates the buffer size in terms of locations. Value After Reset: 0x0
			4:1	RBSZ_x_0	R	0x0	Receive Buffer size Low RBSZ[13:0] is split into two fields RBSZ_13_y and RBSZ_x_0. The RBSZ_x_0 is the lower field. This field is of width 4 bits. This field is read-only (RO). Value After Reset: 0x0
			0	SR	R/W	0x0	Start or Stop Receive 1: DMA attempts to acquire the descriptor from the Receive list and processes the incoming packets. The DMA tries to acquire descriptor from either of the following positions: -- The current position in the list This is the address set by the DMA_CH0_RxDesc_List_Address register. -- The position at which the Rx process was previously stopped If the DMA does not own the current descriptor, the reception is suspended and the RBU bit of the DMA_CH0_Status register is set. The Start Receive command is effective only when the reception is stopped. If the command is issued before setting the DMA_CH0_RxDesc_List_Address register, the DMA behavior is unpredictable. 0: Rx DMA operation stops after the transfer of the current packet. The next descriptor position in the Receive list is saved, and it becomes the current position after the Rx process is restarted. The Stop Receive command is effective only when the Rx process is in the Running (waiting for Rx packet) or Suspended state. Values: 0x0 (STOP): Stop Receive 0x1 (START): Start Receive Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1110	DMA_CH0_TxDesc_List_HAddress	The Channel0 Tx Descriptor List HAddress register has the higher 8 or 16 bits of the start address of the Transmit descriptor list.You can write to this register only when the Tx DMA has stopped, that is, the ST bit is set to zero in DMA_CH0_Tx_Control register. When stopped, this register can be written with a new descriptor list address. When you set the ST bit to 1, the DMA takes the newly-programmed descriptor base address. If this register is not changed when the ST bit is set to 0, the DMA takes the descriptor address where it was stopped earlier. This register must be programmed with desired the higher address before programming the DMA_CH0_TxDesc_List_Address register.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	TDESHA	R/W	0x0	Start of Transmit List This field contains the most-significant 8 or 16 bits of the 40- or 48-bit base address of the first descriptor in the Transmit descriptor list. Value After Reset: 0x0
0x1114	DMA_CH0_TxDesc_List_Address	The Channel0 Tx Descriptor List Address register points the DMA to the start of Transmit descriptor list. The descriptor lists reside in the physical memory space of the application and must be Lword (128-bit). The DMA internally converts it to bus width aligned address by making the corresponding LSB to low.You can write to this register only when the Tx DMA has stopped, that is, the ST bit is set to zero in DMA_CH0_Tx_Control register. When stopped, this register can be written with a new descriptor list address. When you set the ST bit to 1, the DMA takes the newly-programmed descriptor base address. If this register is not changed when the ST bit is set to 0, the DMA takes the descriptor address where it was stopped earlier.	31:4	TDESLA	R/W	0x0	Start of Transmit List This field contains the base address of the first descriptor in the Transmit descriptor list. The DMA ignores the LSB bits (3:0) and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0 Testable: untestable
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x1118	DMA_CH0_RxDesc_List_HAddress	The Channel0 Rx Descriptor List HAddress register has the higher 8 or 16 bits of the start address of the Receive descriptor list.Writing to this register is permitted only when reception is stopped. When stopped, this register must be written to before the receive Start command is given. You can write to this register only when Rx DMA has stopped, that is, SR bit is set to zero in the DMA_CH0_RxDesc_List_Address registerWhen you set the SR bit to 1, the DMA takes the newly programmed descriptor base address.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	RDESHA	R/W	0x0	Start of Receive List This field contains the most-significant 8 or 16 bits of the 40-bit or 48-bit base address of the first descriptor in the Rx Descriptor list. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x111c	DMA_CH0_RxDesc_List_Address	The Channel0 Rx Descriptor List Address register points the DMA to the start of Receive descriptor list.This register points to the start of the Receive Descriptor List. The descriptor lists reside in the physical memory space of the application and must be Lword (128-bit). The DMA internally converts it to bus width aligned address by making the corresponding LS bits low. Writing to this register is permitted only when reception is stopped. When stopped, this register must be written to before the receive Start command is given. You can write to this register only when Rx DMA has stopped, that is, SR bit is set to zero in DMA_CH0_Rx_Control register. When stopped, this register can be written with a new descriptor list address.When you set the SR bit to 1, the DMA takes the newly programmed descriptor base address.	31:4	RDESLA	R/W	0x0	Start of Receive List This field contains the base address of the first descriptor in the Rx Descriptor list. The DMA ignores the LSB bits (3:0) and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0 Testable: untestable
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x1120	DMA_CH0_TxDesc_Tail_Pointer	The Channel0 Tx Descriptor Tail Pointer register points to an offset from the base and indicates the location of the last valid descriptor.	31:4	TDTP	R/W	0x0	Transmit Descriptor Tail Pointer This field contains the tail pointer for the Tx descriptor ring. The software writes the tail pointer to add more descriptors to the Tx channel. The hardware tries to transmit all packets referenced by the descriptors between the head and the tail pointer registers. The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x1128	DMA_CH0_RxDesc_Tail_Pointer	The Channel0 Rx Descriptor Tail Pointer Points to an offset from the base and indicates the location of the last valid descriptor.	31:4	RDTP	R/W	0x0	Receive Descriptor Tail Pointer This field contains the tail pointer for the Rx descriptor ring. The software writes the tail pointer to add more descriptors to the Rx channel. The hardware tries to write all received packets to the descriptors referenced between the head and the tail pointer registers. The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x112c	DMA_CH0_TxDesc_Ring_Length	The Tx Descriptor Ring Length register contains the length of the Transmit descriptor ring.	31:10	Reserved_31_10	R	0x0	Reserved. Value After Reset: 0x0
			9:0	TDRL	R/W	0x0	Transmit Descriptor Ring Length This field sets the maximum number of Tx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. It is recommended that a minimum ring descriptor length of 4. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1130	DMA_CH0_Rx_Control2	The Channel0 Receive Control register controls the Rx features such as Rx Descriptor Ring Length and Alternate Rx Buffer Size.	31:24	Reserved_31_24	R	0x0	Reserved. Value After Reset: 0x0
			23:18	ARBS	R/W	0x0	Alternate Receive Buffer Size Indicates size in bytes for Buffer 1 when ARBS is programmed to a non-zero value (when split header feature is not enabled). When split header feature is enabled, ARBS indicates the buffer size for header data. The maximum alternate buffer is limited to 1008 bytes. When ARBS=0, Rx Buffer1 and Rx Buffer2 sizes are based on RBSZ field of DMA_CH0_Rx_Control. Width of ARBS field is 6 bits. Value After Reset: 0x0
			17:10	Reserved_x_10	R	0x0	Reserved. Value After Reset: 0x0
			9:0	RDRL	R/W	0x0	Receive Descriptor Ring Length This register sets the maximum number of Rx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9. Value After Reset: 0x0
			31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15	NIE	R/W	0x0	Normal Interrupt Summary Enable When this bit is set, the normal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: Bit 0: Transmit Interrupt Bit 2: Transmit Buffer Unavailable Bit 6: Receive Interrupt Bit 11: Early Receive Interrupt When this bit is reset, the normal interrupt summary is disabled.Values: 0x0 (DISABLE): Normal Interrupt Summary is disabled 0x1 (ENABLE): Normal Interrupt Summary is enabled Value After Reset: 0x0
			14	AIE	R/W	0x0	Abnormal Interrupt Summary Enable When this bit is set, the abnormal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: Bit 1: Transmit Process Stopped Bit 7: Rx Buffer Unavailable Bit 8: Receive Process Stopped Bit 9: Receive Watchdog Timeout Bit 10: Early Transmit Interrupt Bit 12: Fatal Bus Error Bit 13: Context Descriptor Error When this bit is reset, the abnormal interrupt summary is disabled.Values: 0x0 (DISABLE): Abnormal Interrupt Summary is disabled 0x1 (ENABLE): Abnormal Interrupt Summary is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1134	DMA_CH0_Interrupt_Enable	The Channel0 Interrupt Enable register enables the interrupts reported by the Status register.	13	CDEE	R/W	0x0	Context Descriptor Error Enable When this bit is set along with the AIE bit, the Descriptor error interrupt is enabled. When this bit is reset, the Descriptor error interrupt is disabled.Values: 0x0 (DISABLE): Context Descriptor Error is disabled 0x1 (ENABLE): Context Descriptor Error is enabled Value After Reset: 0x0
			12	FBEE	R/W	0x0	Fatal Bus Error Enable When this bit is set along with the AIE bit, the Fatal Bus error interrupt is enabled. When this bit is reset, the Fatal Bus Error error interrupt is disabled.Values: 0x0 (DISABLE): Fatal Bus Error is disabled 0x1 (ENABLE): Fatal Bus Error is enabled Value After Reset: 0x0
			11	ERIE	R/W	0x0	Early Receive Interrupt Enable When this bit is set along with the NIE bit, the Early Receive interrupt is enabled. When this bit is reset, the Early Receive interrupt is disabled.Values: 0x0 (DISABLE): Early Receive Interrupt is disabled 0x1 (ENABLE): Early Receive Interrupt is enabled Value After Reset: 0x0
			10	ETIE	R/W	0x0	Early Transmit Interrupt Enable When this bit is set along with the AIE bit, the Early Transmit interrupt is enabled. When this bit is reset, the Early Transmit interrupt is disabled.Values: 0x0 (DISABLE): Early Transmit Interrupt is disabled 0x1 (ENABLE): Early Transmit Interrupt is enabled Value After Reset: 0x0
			9	RWTE	R/W	0x0	Receive Watchdog Timeout Enable When this bit is set along with the AIE bit, the Receive Watchdog Timeout interrupt is enabled. When this bit is reset, the Receive Watchdog Timeout interrupt is disabled.Values: 0x0 (DISABLE): Receive Watchdog Timeout is disabled 0x1 (ENABLE): Receive Watchdog Timeout is enabled Value After Reset: 0x0
			8	RSE	R/W	0x0	Receive Stopped Enable When this bit is set along with the AIE bit, the Receive Stopped Interrupt is enabled. When this bit is reset, the Receive Stopped interrupt is disabled.Values: 0x0 (DISABLE): Receive Stopped is disabled 0x1 (ENABLE): Receive Stopped is enabled Value After Reset: 0x0
			7	RBUE	R/W	0x0	Receive Buffer Unavailable Enable When this bit is set along with the AIE bit, the Receive Buffer Unavailable interrupt is enabled. When this bit is reset, the Receive Buffer Unavailable interrupt is disabled.Values: 0x0 (DISABLE): Receive Buffer Unavailable is disabled 0x1 (ENABLE): Receive Buffer Unavailable is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			6	RIE	R/W	0x0	Receive Interrupt Enable When this bit is set along with the NIE bit, the Receive Interrupt is enabled. When this bit is reset, the Receive Interrupt is disabled.Values: 0x0 (DISABLE): Receive Interrupt is disabled 0x1 (ENABLE): Receive Interrupt is enabled Value After Reset: 0x0
			5:3	Reserved_5_3	R	0x0	Reserved. Value After Reset: 0x0
			2	TBUE	R/W	0x0	Transmit Buffer Unavailable Enable When this bit is set along with the NIE bit, the Transmit Buffer Unavailable interrupt is enabled. When this bit is reset, the Transmit Buffer Unavailable interrupt is disabled.Values: 0x0 (DISABLE): Transmit Buffer Unavailable is disabled 0x1 (ENABLE): Transmit Buffer Unavailable is enabled Value After Reset: 0x0
			1	TXSE	R/W	0x0	Transmit Stopped Enable When this bit is set along with the AIE bit, the Transmission Stopped interrupt is enabled. When this bit is reset, the Transmission Stopped interrupt is disabled.Values: 0x0 (DISABLE): Transmit Stopped is disabled 0x1 (ENABLE): Transmit Stopped is enabled Value After Reset: 0x0
			0	TIE	R/W	0x0	Transmit Interrupt Enable When this bit is set along with the NIE bit, the Transmit Interrupt is enabled. When this bit is reset, the Transmit Interrupt is disabled.Values: 0x0 (DISABLE): Transmit Interrupt is disabled 0x1 (ENABLE): Transmit Interrupt is enabled Value After Reset: 0x0
0x1138	DMA_CH0_Rx_Interrupt_Watchdog_Timer	The Receive Interrupt Watchdog Timer register indicates the watchdog timeout for Receive Interrupt (RI) from the DMA. When this register is written with a non-zero value, it enables the watchdog timer for the RI bit of the DMA_CH0_Status register.	31:18	Reserved_31_18	R	0x0	Reserved. Value After Reset: 0x0
			17:16	RWTU	R/W	0x0	Receive Interrupt Watchdog Timer Count Units This fields indicates the number of system clock cycles corresponding to one unit in RWT field. 2'b00: 256 2'b01: 512 2'b10: 1024 2'b11: 2048 For example, when RWT=2 and RWTU=1, the watchdog timer is set for 2*512=1024 system clock cycles. Value After Reset: 0x0 Testable: untestable
			15:8	Reserved_15_8	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			7:0	RWT	R/W	0x0	Receive Interrupt Watchdog Timer Count This field indicates the number of system clock cycles, multiplied by factor indicated in RWTU field, for which the watchdog timer is set. The watchdog timer is triggered with the programmed value after the Rx DMA completes the transfer of a packet for which the RI bit is not set in the DMA_CH0_Status register, because of the setting of Interrupt Enable bit in the corresponding descriptor RDES3[30]. When the watchdog timer runs out, the RI bit is set and the timer is stopped. The watchdog timer is reset when the RI bit is set high because of automatic setting of RI as per the Interrupt Enable bit RDES3[30] of any received packet. Value After Reset: 0x0 Testable: untestable
0x113c	DMA_CH0_Slot_Function_Control_Status	The Slot Function Control and Status register contains the control bits for slot function and the status for Transmit path.	31:20	Reserved_31_20	R	0x0	Reserved. Value After Reset: 0x0
			19:16	RSN	R	0x0	Reference Slot Number This field gives the current value of the reference slot number in the DMA. It is used for slot comparison. Value After Reset: 0x0 Testable: untestable
			15:4	SIV	R/W	0x7c	Slot Interval Value This field controls the period of the slot interval in which the TxDMA fetches the scheduled packets. A value of 0 specifies the slot interval of 1 us while the maximum value 4095 specifies the slot interval of 4096us. The default/reset value is 0x07C which corresponds to slot interval of 125us Value After Reset: 0x7c
			3:2	Reserved_3_2	R	0x0	Reserved. Value After Reset: 0x0
			1	ASC	R/W	0x0	Advance Slot Check When set, this bit enables the DMA to fetch the data from the buffer when the slot number (SLOTNUM) programmed in the Tx descriptor is equal to the reference slot number given in the RSN field or ahead of the reference slot number by up to two slots This bit is applicable only when the ESC bit is set.Values: 0x0 (DISABLE): Advance Slot Check is disabled 0x1 (ENABLE): Advance Slot Check is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	ESC	R/W	0x0	Enable Slot Comparison When set, this bit enables the checking of the slot numbers programmed in the Tx descriptor with the current reference given in the RSN field. The DMA fetches the data from the corresponding buffer only when the slot number is equal to the reference slot number or ahead of the reference slot number by one slot When reset, this bit disables the checking of the slot numbers. The DMA fetches the data immediately after the descriptor is processed. Note: The UFO (UDP Fragmentation over IPv4)/TSO/USO must not be enabled along with TBS/AVB Slot number check. The UFO/TSO/USO involves multiple packets/segments/fragments transmission for single packet received from application and the slot number check are applicable for fetching of only first segment/fragment. As a result it might be difficult for software to specify slot number for subsequent packets.Values: 0x0 (DISABLE): Slot Comparison is disabled 0x1 (ENABLE): Slot Comparison is enabled Value After Reset: 0x0
0x1144	DMA_CH0_Current_App_Tx Desc	The Channel0 Current Application Transmit Descriptor register points to the current Transmit descriptor read by the DMA.	31:0	CURTDESAPTR	R	0x0	Application Transmit Descriptor Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x114c	DMA_CH0_Current_App_Rx Desc	The Channel0 Current Application Receive Descriptor register points to the current Receive descriptor read by the DMA.	31:0	CURRDESAPTR	R	0x0	Application Receive Descriptor Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x1150	DMA_CH0_Current_App_Tx Buffer_H	The Channel0 Current Application Transmit Buffer Address High register has the higher 8 or 16 bits of the current address of the Transmit buffer address read by the DMA.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	CURTBUFAPTRH	R	0x0	Application Transmit Buffer Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x1154	DMA_CH0_Current_App_Tx Buffer	The Channel0 Current Application Transmit Buffer Address register points to the current Tx buffer address read by the DMA.	31:0	CURTBUFAPTR	R	0x0	Application Transmit Buffer Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x1158	DMA_CH0_Current_App_Rx Buffer_H	The Channel0 Current Application Receive Buffer Address High register has the higher 8 or 16 bits of the current address of the Receive buffer address read by the DMA.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	CURRBUFAPTRH	R	0x0	Application Receive Buffer Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x115c	DMA_CH0_Current_App_Rx Buffer	The Channel 0 Current Application Receive Buffer Address register points to the current Rx buffer address read by the DMA.	31:0	CURRBUFAPTR	R	0x0	Application Receive Buffer Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset. Value After Reset: 0x0
			31:22	Reserved_31_22	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			21:19	REB	R	0x0	Rx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AXI interface. Bit 21 1'b1: Error during data transfer by Rx DMA 1'b0: No Error during data transfer by Rx DMA Bit 20 1'b1: Error during descriptor access 1'b0: Error during data buffer access Bit 19 1'b1: Error during read transfer 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt. Value After Reset: 0x0 Testable: untestable
			18:16	TEB	R	0x0	Tx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AXI interface. Bit 18 1'b1: Error during data transfer by Tx DMA 1'b0: No Error during data transfer by Tx DMA Bit 17 1'b1: Error during descriptor access 1'b0: Error during data buffer access Bit 16 1'b1: Error during read transfer 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt. Value After Reset: 0x0 Testable: untestable
			15	NIS	R/W	0x0	Normal Interrupt Summary Normal Interrupt Summary bit value is the logical OR of the following bits when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: Bit 0: Transmit Interrupt Bit 2: Transmit Buffer Unavailable Bit 6: Receive Interrupt Bit 11: Early Receive Interrupt Only unmasked bits (interrupts for which interrupt enable is set in DMA_CH0_Interrupt_Enable register) affect the Normal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit which causes NIS to be set is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Normal Interrupt Summary status not detected 0x1 (ACTIVE): Normal Interrupt Summary status detected Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1160	DMA_CH0_Status	The software driver (application) reads the Status register during interrupt service routine or polling to determine the status of the DMA.Note: The number of DMA_CH0_Status register in the configuration is the higher of number of Rx DMA Channels and Tx DMA Channels.	14	AIS	R/W	0x0	Abnormal Interrupt Summary Abnormal Interrupt Summary bit value is the logical OR of the following when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: Bit 1: Transmit Process Stopped Bit 7: Receive Buffer Unavailable Bit 8: Receive Process Stopped Bit 10: Early Transmit Interrupt Bit 12: Fatal Bus Error Bit 13: Context Descriptor Error Only unmasked bits affect the Abnormal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit, which causes AIS to be set, is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Abnormal Interrupt Summary status not detected 0x1 (ACTIVE): Abnormal Interrupt Summary status detected Value After Reset: 0x0 Testable: untestable
			13	CDE	R/W	0x0	Context Descriptor Error This bit indicates that the DMA Tx/Rx engine received a descriptor error, which indicates invalid context in the middle of packet flow (intermediate descriptor) or all one's descriptor in Tx case and on Rx side it indicates DMA has read a descriptor with either of the buffer address as ones which is considered to be invalid. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Context Descriptor Error status not detected 0x1 (ACTIVE): Context Descriptor Error status detected Value After Reset: 0x0 Testable: untestable
			12	FBE	R/W	0x0	Fatal Bus Error This bit indicates that a bus error occurred (as described in the EB field). When this bit is set, the corresponding DMA channel engine disables all bus accesses. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Fatal Bus Error status not detected 0x1 (ACTIVE): Fatal Bus Error status detected Value After Reset: 0x0 Testable: untestable
			11	ERI	R/W	0x0	Early Receive Interrupt This bit when set indicates that the RxDMA has completed the transfer of packet data to the memory. In configurations supporting ERIC, ERIC = 0: This bit is set only after the Rx DMA has completely filled up a receive buffer with packet data. ERIC = 1: This bit is set after every burst transfer of data from the Rx DMA to the buffer. The setting of RI bit automatically clears this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Early Receive Interrupt status not detected 0x1 (ACTIVE): Early Receive Interrupt status detected Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			10	ETI	R/W	0x0	Early Transmit Interrupt This bit when set indicates that the TxDMA has completed the transfer of packet data to the MTL TXFIFO memory. In configurations supporting ERIC: ETIC = 0: This bit is set only after the Tx DMA has transferred a complete packet to MTL. ETIC = 1: This bit is set after completion of (partial) packet data transfer from buffers in the Transmit descriptor in which IOC = 1. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Early Transmit Interrupt status not detected 0x1 (ACTIVE): Early Transmit Interrupt status detected Value After Reset: 0x0 Testable: untestable
			9	RWT	R/W	0x0	Receive Watchdog Timeout This bit is asserted when a packet with length greater than 2,048 bytes (10,240 bytes when Jumbo Packet mode is enabled) is received.Values: 0x0 (INACTIVE): Receive Watchdog Timeout status not detected 0x1 (ACTIVE): Receive Watchdog Timeout status detected Value After Reset: 0x0 Testable: untestable
			8	RPS	R/W	0x0	Receive Process Stopped This bit is asserted when the Rx process enters the Stopped state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Process Stopped status not detected 0x1 (ACTIVE): Receive Process Stopped status detected Value After Reset: 0x0 Testable: untestable
			7	RBU	R/W	0x0	Receive Buffer Unavailable This bit indicates that the application owns the next descriptor in the Receive list, and the DMA cannot acquire it. The Rx process is suspended. To resume processing Rx descriptors, the application must change the ownership of the descriptor and issue a Receive Poll Demand command. If this command is not issued, the Rx process resumes when the next recognized incoming packet is received. In ring mode, the application must advance the Receive Descriptor Tail Pointer register of a channel. This bit is set only when the DMA owns the previous Rx descriptor. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Buffer Unavailable status not detected 0x1 (ACTIVE): Receive Buffer Unavailable status detected Value After Reset: 0x0 Testable: untestable
			6	RI	R/W	0x0	Receive Interrupt This bit indicates that the packet reception is complete. When packet reception is complete, Bit 31 of RDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor. The reception remains in the Running state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Interrupt status not detected 0x1 (ACTIVE): Receive Interrupt status detected Value After Reset: 0x0 Testable: untestable
			5:3	Reserved_5_3	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			2	TBU	R/W	0x0	Transmit Buffer Unavailable This bit indicates that the application owns the next descriptor in the Transmit list, and the DMA cannot acquire it. Transmission is suspended. The TPS0 field of the DMA_Debug_Status0-DMA_Debug_Status2 register explains the Transmit Process state transitions. To resume processing the Transmit descriptors, the application must do the following: Change the ownership of the descriptor by setting Bit 31 of TDES3. Issue a Transmit Poll Demand command. For ring mode, the application must advance the Transmit Descriptor Tail Pointer register of a channel. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Buffer Unavailable status not detected 0x1 (ACTIVE): Transmit Buffer Unavailable status detected Value After Reset: 0x0 Testable: untestable
			1	TPS	R/W	0x0	Transmit Process Stopped This bit is set when the transmission is stopped. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Process Stopped status not detected 0x1 (ACTIVE): Transmit Process Stopped status detected Value After Reset: 0x0 Testable: untestable
			0	TI	R/W	0x0	Transmit Interrupt This bit indicates that the packet transmission is complete. When transmission is complete, Bit 31 of TDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Interrupt status not detected 0x1 (ACTIVE): Transmit Interrupt status detected Value After Reset: 0x0 Testable: untestable
0x1164	DMA_CH0_Miss_Frame_Cnt	This register has the number of packet counter that got dropped by the DMA either due to Bus Error or due to programming RPF field in DMA_CH0_Rx_Control register.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15	MFCO	R	0x0	Overflow status of the MFC Counter When this bit is set then the MFC counter does not get incremented further. The bit gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Miss Frame Counter overflow not occurred 0x1 (ACTIVE): Miss Frame Counter overflow occurred Value After Reset: 0x0
			14:11	Reserved_14_11	R	0x0	Reserved. Value After Reset: 0x0
			10:0	MFC	R	0x0	Dropped Packet Counters This counter indicates the number of packet counters that are dropped by the DMA either because of bus error or because of programing RPF field in DMA_CH0_Rx_Control register. The counter gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1168	DMA_CH0_RXP_Accept_Cnt	The DMA_CH0_RXP_Accept_Cnt registers provides the count of the number of frames accepted by Rx Parser.	31	RXPACOF	R	0x0	Rx Parser Accept Counter Overflow Bit When set, this bit indicates that the RXPAC Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Rx Parser Accept Counter overflow not occurred 0x1 (ACTIVE): Rx Parser Accept Counter overflow occurred Value After Reset: 0x0
			30:0	RXPAC	R	0x0	Rx Parser Accept Counter This 31-bit counter is implemented when a Rx Parser Accept a packet due to AF =1. The counter is cleared when the register is read. Value After Reset: 0x0
0x1180	DMA_CH1_Control	The DMA Channel1 Control register specifies the MSS value for segmentation, length to skip between two descriptors, and also the features such as header splitting and 8xPBL mode.	31:25	Reserved_31_25	R	0x0	Reserved. Value After Reset: 0x0
			24	SPH	R/W	0x0	Split Headers 1: DMA splits the header and payload in the receive path. The DMA writes the header to the Buffer Address1 of RDES0. The DMA writes the payload to the buffer to which the Buffer Address2 is pointing. The software must ensure that the header fits into the Receive buffers. If the header length exceeds the receive buffer size, the DMA does not split the header and payload. This bit is available only when you select Enable Split Header Structure feature.Values: 0x0 (DISABLE): Split Headers feature is disabled 0x1 (ENABLE): Split Headers feature is enabled Value After Reset: 0x0
			23:21	Reserved_23_21	R	0x0	Reserved. Value After Reset: 0x0
			20:18	DSL	R/W	0x0	Descriptor Skip Length This bit specifies the Lword (128-bit) number of locations to skip between two descriptors. The address skipping starts from the end of the current descriptor to the start of the next descriptor. When the DSL value is equal to zero, the DMA takes the descriptor table as contiguous. Value After Reset: 0x0
			17	Reserved_17	R	0x0	Reserved. Value After Reset: 0x0
			16	PBLx8	R/W	0x0	8xPBL mode 1: Indicates that the PBL value programmed in Bits[21:16] in DMA_CH0_Tx_Control and Bits[21:16] in DMA_CH0_Rx_Control is multiplied by eight times. Therefore, the DMA transfers the data in 8, 16, 32, 64, 128, and 256 beats depending on the PBL value. Values: 0x0 (DISABLE): 8xPBL mode is disabled 0x1 (ENABLE): 8xPBL mode is enabled Value After Reset: 0x0
			15:14	Reserved_15_14	R	0x0	Reserved. Value After Reset: 0x0
			13:0	MSS	R/W	0x0	Maximum Segment Size This field specifies the maximum segment size that must be used while segmenting the packet. This field is valid only if the TSE bit of DMA_CH0_Tx_Control register is set. Program a value in this field that is more than the configured datawidth in bytes. Use MSS value of 64 bytes or more. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			31	Reserved_31	R	0x0	Reserved. Value After Reset: 0x0
			30:29	TFSEL	R/W	0x0	TBS Fetch Select Select bits for one of the four DMA_TBS_CTRL register fields (FTOS,FGSN,FTOV) for the channel Values: 2'b00: DMA_TBS_CTRL0 2'b01: DMA_TBS_CTRL1 2'b10: DMA_TBS_CTRL2 2'b11: DMA_TBS_CTRL3 (Reserved if DWC_EQOS_NUM_DMA_TX_CH=3) Value After Reset: 0x0
			28	EDSE	R/W	0x0	Enhanced Descriptor Enable 1: The corresponding channel uses Enhanced Descriptors that are 32 Bytes for both Normal and Context Descriptors. 0: The corresponding channel uses the descriptors that are 16 Bytes. Values: 0x0 (DISABLE): Enhanced Descriptor is disabled 0x1 (ENABLE): Enhanced Descriptor is enabled Value After Reset: 0x0
			27:24	TQOS	R/W	0x0	Transmit QOS. This field is used to drive arqos_m_o[3:0] or awqos_m_o[3:0] output signals for all transactions of DMA Tx Channel0. Value After Reset: 0x0
			23	Reserved_23	R	0x0	Reserved. Value After Reset: 0x0
			22			#VALUE!	Reserved Field: Yes
			21:16	TxPBL	R/W	0x0	Transmit Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in DMA_CH0_Control register. 2. Set the TxPBL. Note: The maximum value of TxPBL must be less than or equal to half the Tx Queue size (TQS field of MTL_TxQ0_Operation_Mode register) in terms of beats. This is required so that the Tx Queue has space to store at least another Tx PBL worth of data while the MTL Tx Queue Controller is transferring data to MAC. For example, in 32-bit data width configurations the total locations in Tx Queue of size 512 bytes is 128, TxPBL and 8xPBL needs to be programmed to less than or equal to 64. Value After Reset: 0x0
			15	IPBL	R/W	0x0	Ignore PBL Requirement 1: DMA does not check for PBL number of locations in the MTL before initiating a transfer. If space is not available, the MTL might use handshaking to slow the DMA. Note: Do not use this mode when you enable multiple Transmit DMA channels. Because, it might block other Transmit and Receive DMA Channels from accessing the Read Data Channel of the AXI bus until space is available in Transmit Queue for current transfer.Values: 0x0 (DISABLE): Ignore PBL Requirement is disabled 0x1 (ENABLE): Ignore PBL Requirement is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1184	DMA_CH1_Tx_Control	The DMA Channel1 Transmit Control register controls the Tx features such as PBL, TCP segmentation, and Tx Channel weights.	14:13	TSE_MODE	R/W	0x0	TSE Mode 00: TSO/USO (segmentation functionality is enabled). In this mode, the setting of TSE bit enables the TSO/USO segmentation. 01: UFO with Checksum (UDP Fragmentation over IPv4 with Checksum). In this mode, the setting of TSE bit enables the UDP fragmentation functionality with Checksum for all the UDP packets. 10: UFO without Checksum (UDP Fragmentation over IPv4 with Checksum). In this mode, the setting of TSE bit enables the UDP fragmentation functionality without Checksum for all the UDP packets. 11: Reserved Values: 0x0 (TSO_USO): TSO/USO 0x1 (UFOWC): UFO with Checksum 0x2 (UFOWOC): UFO without Checksum 0x3 (RSVD): Reserved Value After Reset: 0x0
			12	TSE	R/W	0x0	TCP Segmentation Enabled 1: DMA performs the TCP segmentation or UDP Segmentation/Fragmentation for packets in this channel. The TCP segmentation or UDP packet's segmentation/Fragmentation is performed only for those packets for which the TSE bit (TDES0[19]) = 1 in the Tx Normal descriptor. When this bit is 1, the TxPBL value must be greater than 4. Values: 0x0 (DISABLE): TCP Segmentation is disabled 0x1 (ENABLE): TCP Segmentation is enabled Value After Reset: 0x0
			11:5	Reserved_11_5	R	0x0	Reserved. Value After Reset: 0x0
			4	OSF	R/W	0x0	Operate on Second Packet 1: Instructs the DMA to process the second packet of the Tx data even before the status for the first packet is obtained. Values: 0x0 (DISABLE): Operate on Second Packet disabled 0x1 (ENABLE): Operate on Second Packet enabled Value After Reset: 0x0
			3:1	TCW	R/W	0x0	Transmit Channel Weight This field indicates the weight assigned to the corresponding Tx channel. When reset is complete, this field is set to 0 for all channels by default, resulting in equal weights to all channels. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	ST	R/W	0x0	Start or Stop Transmission Command 1: Transmission is placed in the running state. The DMA checks the Tx list at the current position for a packet to be transmitted. The DMA tries to acquire descriptor from either of the following positions: The current position in the list This is the base address of the Transmit list set by the DMA_CH0_TxDesc_List_Address register. The position at which the transmission was previously stopped If the DMA does not own the current descriptor, the transmission enters the Suspended state and the TBU bit of the DMA_CH0_Status register is set. The Start Transmission command is effective only when the transmission is stopped. If the command is issued before setting the DMA_CH0_TxDesc_List_Address register, the DMA behavior is unpredictable. When this bit is reset, the transmission process is placed in the Stopped state after completing the transmission of the current packet. The Next Descriptor position in the Transmit list is saved, and it becomes the current position when the transmission is restarted. To change the list address, you need to program DMA_CH0_TxDesc_List_Address register with a new value when this bit is reset. The new value is considered when this bit is set again. The stop transmission command is effective only when the transmission of the current packet is complete or the transmission is in the Suspended state.Values: 0x0 (STOP): Stop Transmission Command 0x1 (START): Start Transmission Command Value After Reset: 0x0 Testable: untestable
			31	RPF	R/W	0x0	Rx Packet Flush. 1: DWC_ether_qos automatically flushes the packet from the Rx Queues destined to this DMA Rx Channel, when it is stopped. When this bit remains set and the DMA is re-started by the software driver, the packets residing in the Rx Queues that were received when this RxDMA was stopped, still get flushed out. The packets received by the MAC only after the DMA is re-started is routed to the Rx DMA.The flushing is done on the Read side of the Rx Queue. When the packet is duplicated to multiple Rx DMA, it is flushed by the Rx DMA to which it is routed only when it is stopped to prevent head-of-line blocking. The packet duplicated to multiple Rx DMA is not flushed when the Rx DMA is re-started, irrespective of whether the packet was received during one or more Rx DMAs were stopped. 0: DWC_ether_qos does not flush the packet in the Rx Queue destined to this RxDMA Channel when it is in STOP state. This might cause head-of-line blocking in the corresponding RxQueue. Note: The stopping of packet flow from a Rx DMA Channel to the application by setting RPF works only when there is one-to-one mapping of Rx Queue to Rx DMA channels. In Dynamic mapping mode, setting RPF bit in any DMA_CH0_Rx_Control register might flush packets from unintended Rx Queues which are destined to the stopped Rx DMA Channel.Values: 0x0 (DISABLE): Rx Packet Flush is disabled 0x1 (ENABLE): Rx Packet Flush is enabled Value After Reset: 0x0
			30:28	Reserved_30_28	R	0x0	Reserved. Value After Reset: 0x0
			27:24	RQOS	R/W	0x0	Rx AXI4 QOS. This field is used to drive arqos_m_o[3:0] or awqos_m_o[3:0] output signals for all transactions of DMA Rx Channel0. Value After Reset: 0x0
			23	Reserved_23	R	0x0	Reserved. Value After Reset: 0x0
			22			#VALUE!	Reserved Field: Yes

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1188	DMA_CH1_Rx_Control	The DMA Channel1 Receive Control register controls the Rx features such as PBL, buffer size, and extended status.	21:16	RxPBL	R/W	0x0	Receive Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in the DMA_CH0_Control register. 2. Set the RxPBL. Note: The maximum value of RxPBL must be less than or equal to half the Rx Queue size (RQS field of MTL_RxQ0_Operation_Mode register) in terms of beats. This is required so that the Rx Queue has space to store at least another Rx PBL worth of data while the Rx DMA is transferring a block of data. For example, in 32-bit data width configurations the total locations in Rx Queue of size 512 bytes is 128, so RxPBL and 8xPBL needs to be programmed to less than or equal to 64. Value After Reset: 0x0
			15	Reserved_15	R	0x0	Reserved. Value After Reset: 0x0
			14:5	RBSZ_13_y	R/W	0x0	Receive Buffer size High RBSZ[13:0] is split into two fields higher RBSZ_13_y and lower RBSZ_x_0. The RBSZ[13:0] field indicates the size of the Rx buffers specified in bytes. The maximum buffer size is limited to 16K bytes. The buffer size is applicable to payload buffers when split headers are enabled. Note: The buffer size must be a multiple of 16. This is required even if the value of buffer address pointer is not aligned to data bus width. Hence the lower RBSZ_x_0 bits are read-only and the value is considered as all-zero. Thus the RBSZ_13_y indicates the buffer size in terms of locations. Value After Reset: 0x0
			4:1	RBSZ_x_0	R	0x0	Receive Buffer size Low RBSZ[13:0] is split into two fields RBSZ_13_y and RBSZ_x_0. The RBSZ_x_0 is the lower field. This field is of width 4 bits. This field is read-only (RO). Value After Reset: 0x0
			0	SR	R/W	0x0	Start or Stop Receive 1: DMA attempts to acquire the descriptor from the Receive list and processes the incoming packets. The DMA tries to acquire descriptor from either of the following positions: -- The current position in the list This is the address set by the DMA_CH0_RxDesc_List_Address register. -- The position at which the Rx process was previously stopped If the DMA does not own the current descriptor, the reception is suspended and the RBU bit of the DMA_CH0_Status register is set. The Start Receive command is effective only when the reception is stopped. If the command is issued before setting the DMA_CH0_RxDesc_List_Address register, the DMA behavior is unpredictable. 0: Rx DMA operation stops after the transfer of the current packet. The next descriptor position in the Receive list is saved, and it becomes the current position after the Rx process is restarted. The Stop Receive command is effective only when the Rx process is in the Running (waiting for Rx packet) or Suspended state. Values: 0x0 (STOP): Stop Receive 0x1 (START): Start Receive Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1190	DMA_CH1_TxDesc_List_HAddress	The Channel1 Tx Descriptor List HAddress register has the higher 8 or 16 bits of the start address of the Transmit descriptor list.You can write to this register only when the Tx DMA has stopped, that is, the ST bit is set to zero in DMA_CH1_Tx_Control register. When stopped, this register can be written with a new descriptor list address. When you set the ST bit to 1, the DMA takes the newly-programmed descriptor base address. If this register is not changed when the ST bit is set to 1, the DMA takes the descriptor address where it was stopped earlier. This register must be programmed with desired the higher address before programming the DMA_CH1_TxDesc_List_Address register.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	TDESHA	R/W	0x0	Start of Transmit List This field contains the most-significant 8 or 16 bits of the 40- or 48-bit base address of the first descriptor in the Transmit descriptor list. Value After Reset: 0x0
0x1194	DMA_CH1_TxDesc_List_Address	The Channel1 Tx Descriptor List Address register points the DMA to the start of Transmit descriptor list. The descriptor lists reside in the physical memory space of the application and must be Lword (128-bit). The DMA internally converts it to bus width aligned address by making the corresponding LSB to low.You can write to this register only when the Tx DMA has stopped, that is, the ST bit is set to zero in DMA_CH1_Tx_Control register. When stopped, this register can be written with a new descriptor list address. When you set the ST bit to 1, the DMA takes the newly-programmed descriptor base address. If this register is not changed when the ST bit is set to 1, the DMA takes the descriptor address where it was stopped earlier.	31:4	TDESLA	R/W	0x0	Start of Transmit List This field contains the base address of the first descriptor in the Transmit descriptor list. The DMA ignores the LSB bits (3:0) and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0 Testable: untestable
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x1198	DMA_CH1_RxDesc_List_HAddress	The Channel1 Rx Descriptor List HAddress register has the higher 8 or 16 bits of the start address of the Receive descriptor list.Writing to this register is permitted only when reception is stopped. When stopped, this register must be written to before the receive Start command is given. You can write to this register only when Rx DMA has stopped, that is, SR bit is set to zero in the DMA_CH1_RxDesc_List_Address registerWhen you set the SR bit to 1, the DMA takes the newly programmed descriptor base address.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	RDESHA	R/W	0x0	Start of Receive List This field contains the most-significant 8 or 16 bits of the 40-bit or 48-bit base address of the first descriptor in the Rx Descriptor list. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x119c	DMA_CH1_RxDesc_List_Address	The Channel1 Rx Descriptor List Address register points the DMA to the start of Receive descriptor list.This register points to the start of the Receive Descriptor List. The descriptor lists reside in the physical memory space of the application and must be Lword (128-bit). The DMA internally converts it to bus width aligned address by making the corresponding LS bits low. Writing to this register is permitted only when reception is stopped. When stopped, this register must be written to before the receive Start command is given. You can write to this register only when Rx DMA has stopped, that is, SR bit is set to zero in DMA_CH1_Rx_Control register. When stopped, this register can be written with a new descriptor list address.When you set the SR bit to 1, the DMA takes the newly programmed descriptor base address.	31:4	RDESLA	R/W	0x0	Start of Receive List This field contains the base address of the first descriptor in the Rx Descriptor list. The DMA ignores the LSB bits (3:0) and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0 Testable: untestable
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x11a0	DMA_CH1_TxDesc_Tail_Pointer	The Channel1 Tx Descriptor Tail Pointer register points to an offset from the base and indicates the location of the last valid descriptor.	31:4	TDTP	R/W	0x0	Transmit Descriptor Tail Pointer This field contains the tail pointer for the Tx descriptor ring. The software writes the tail pointer to add more descriptors to the Tx channel. The hardware tries to transmit all packets referenced by the descriptors between the head and the tail pointer registers. The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x11a8	DMA_CH1_RxDesc_Tail_Pointer	The Channel1 Rx Descriptor Tail Pointer Points to an offset from the base and indicates the location of the last valid descriptor.	31:4	RDTP	R/W	0x0	Receive Descriptor Tail Pointer This field contains the tail pointer for the Rx descriptor ring. The software writes the tail pointer to add more descriptors to the Rx channel. The hardware tries to write all received packets to the descriptors referenced between the head and the tail pointer registers. The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x11ac	DMA_CH1_TxDesc_Ring_Length	The Tx Descriptor Ring Length register contains the length of the Transmit descriptor ring.	31:10	Reserved_31_10	R	0x0	Reserved. Value After Reset: 0x0
			9:0	TDRL	R/W	0x0	Transmit Descriptor Ring Length This field sets the maximum number of Tx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. It is recommended that a minimum ring descriptor length of 4. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x11b0	DMA_CH1_Rx_Control2	The Channel1 Receive Control register controls the Rx features such as Rx Descriptor Ring Length and Alternate Rx Buffer Size.	31:24	Reserved_31_24	R	0x0	Reserved. Value After Reset: 0x0
			23:18	ARBS	R/W	0x0	Alternate Receive Buffer Size Indicates size in bytes for Buffer 1 when ARBS is programmed to a non-zero value (when split header feature is not enabled). When split header feature is enabled, ARBS indicates the buffer size for header data. The maximum alternate buffer is limited to 1008 bytes. When ARBS=0, Rx Buffer1 and Rx Buffer2 sizes are based on RBSZ field of DMA_CH0_Rx_Control. Width of ARBS field is 6 bits. Value After Reset: 0x0
			17:10	Reserved_x_10	R	0x0	Reserved. Value After Reset: 0x0
			9:0	RDRL	R/W	0x0	Receive Descriptor Ring Length This register sets the maximum number of Rx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9. Value After Reset: 0x0
			31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15	NIE	R/W	0x0	Normal Interrupt Summary Enable When this bit is set, the normal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: Bit 0: Transmit Interrupt Bit 2: Transmit Buffer Unavailable Bit 6: Receive Interrupt Bit 11: Early Receive Interrupt When this bit is reset, the normal interrupt summary is disabled.Values: 0x0 (DISABLE): Normal Interrupt Summary is disabled 0x1 (ENABLE): Normal Interrupt Summary is enabled Value After Reset: 0x0
			14	AIE	R/W	0x0	Abnormal Interrupt Summary Enable When this bit is set, the abnormal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: Bit 1: Transmit Process Stopped Bit 7: Rx Buffer Unavailable Bit 8: Receive Process Stopped Bit 9: Receive Watchdog Timeout Bit 10: Early Transmit Interrupt Bit 12: Fatal Bus Error Bit 13: Context Descriptor Error When this bit is reset, the abnormal interrupt summary is disabled.Values: 0x0 (DISABLE): Abnormal Interrupt Summary is disabled 0x1 (ENABLE): Abnormal Interrupt Summary is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x11b4	DMA_CH1_Interrupt_Enable	The Channel1 Interrupt Enable register enables the interrupts reported by the Status register.	13	CDEE	R/W	0x0	Context Descriptor Error Enable When this bit is set along with the AIE bit, the Descriptor error interrupt is enabled. When this bit is reset, the Descriptor error interrupt is disabled.Values: 0x0 (DISABLE): Context Descriptor Error is disabled 0x1 (ENABLE): Context Descriptor Error is enabled Value After Reset: 0x0
			12	FBEE	R/W	0x0	Fatal Bus Error Enable When this bit is set along with the AIE bit, the Fatal Bus error interrupt is enabled. When this bit is reset, the Fatal Bus Error error interrupt is disabled.Values: 0x0 (DISABLE): Fatal Bus Error is disabled 0x1 (ENABLE): Fatal Bus Error is enabled Value After Reset: 0x0
			11	ERIE	R/W	0x0	Early Receive Interrupt Enable When this bit is set along with the NIE bit, the Early Receive interrupt is enabled. When this bit is reset, the Early Receive interrupt is disabled.Values: 0x0 (DISABLE): Early Receive Interrupt is disabled 0x1 (ENABLE): Early Receive Interrupt is enabled Value After Reset: 0x0
			10	ETIE	R/W	0x0	Early Transmit Interrupt Enable When this bit is set along with the AIE bit, the Early Transmit interrupt is enabled. When this bit is reset, the Early Transmit interrupt is disabled.Values: 0x0 (DISABLE): Early Transmit Interrupt is disabled 0x1 (ENABLE): Early Transmit Interrupt is enabled Value After Reset: 0x0
			9	RWTE	R/W	0x0	Receive Watchdog Timeout Enable When this bit is set along with the AIE bit, the Receive Watchdog Timeout interrupt is enabled. When this bit is reset, the Receive Watchdog Timeout interrupt is disabled.Values: 0x0 (DISABLE): Receive Watchdog Timeout is disabled 0x1 (ENABLE): Receive Watchdog Timeout is enabled Value After Reset: 0x0
			8	RSE	R/W	0x0	Receive Stopped Enable When this bit is set along with the AIE bit, the Receive Stopped Interrupt is enabled. When this bit is reset, the Receive Stopped interrupt is disabled.Values: 0x0 (DISABLE): Receive Stopped is disabled 0x1 (ENABLE): Receive Stopped is enabled Value After Reset: 0x0
			7	RBUE	R/W	0x0	Receive Buffer Unavailable Enable When this bit is set along with the AIE bit, the Receive Buffer Unavailable interrupt is enabled. When this bit is reset, the Receive Buffer Unavailable interrupt is disabled.Values: 0x0 (DISABLE): Receive Buffer Unavailable is disabled 0x1 (ENABLE): Receive Buffer Unavailable is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			6	RIE	R/W	0x0	Receive Interrupt Enable When this bit is set along with the NIE bit, the Receive Interrupt is enabled. When this bit is reset, the Receive Interrupt is disabled.Values: 0x0 (DISABLE): Receive Interrupt is disabled 0x1 (ENABLE): Receive Interrupt is enabled Value After Reset: 0x0
			5:3	Reserved_5_3	R	0x0	Reserved. Value After Reset: 0x0
			2	TBUE	R/W	0x0	Transmit Buffer Unavailable Enable When this bit is set along with the NIE bit, the Transmit Buffer Unavailable interrupt is enabled. When this bit is reset, the Transmit Buffer Unavailable interrupt is disabled.Values: 0x0 (DISABLE): Transmit Buffer Unavailable is disabled 0x1 (ENABLE): Transmit Buffer Unavailable is enabled Value After Reset: 0x0
			1	TXSE	R/W	0x0	Transmit Stopped Enable When this bit is set along with the AIE bit, the Transmission Stopped interrupt is enabled. When this bit is reset, the Transmission Stopped interrupt is disabled.Values: 0x0 (DISABLE): Transmit Stopped is disabled 0x1 (ENABLE): Transmit Stopped is enabled Value After Reset: 0x0
			0	TIE	R/W	0x0	Transmit Interrupt Enable When this bit is set along with the NIE bit, the Transmit Interrupt is enabled. When this bit is reset, the Transmit Interrupt is disabled.Values: 0x0 (DISABLE): Transmit Interrupt is disabled 0x1 (ENABLE): Transmit Interrupt is enabled Value After Reset: 0x0
0x11b8	DMA_CH1_Rx_Interrupt_Watchdog_Timer	The Receive Interrupt Watchdog Timer register indicates the watchdog timeout for Receive Interrupt (RI) from the DMA. When this register is written with a non-zero value, it enables the watchdog timer for the RI bit of the DMA_CH1_Status register.	31:18	Reserved_31_18	R	0x0	Reserved. Value After Reset: 0x0
			17:16	RWTU	R/W	0x0	Receive Interrupt Watchdog Timer Count Units This fields indicates the number of system clock cycles corresponding to one unit in RWT field. 2'b00: 256 2'b01: 512 2'b10: 1024 2'b11: 2048 For example, when RWT=2 and RWTU=1, the watchdog timer is set for 2*512=1024 system clock cycles. Value After Reset: 0x0 Testable: untestable
			15:8	Reserved_15_8	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			7:0	RWT	R/W	0x0	Receive Interrupt Watchdog Timer Count This field indicates the number of system clock cycles, multiplied by factor indicated in RWTU field, for which the watchdog timer is set. The watchdog timer is triggered with the programmed value after the Rx DMA completes the transfer of a packet for which the RI bit is not set in the DMA_CH0_Status register, because of the setting of Interrupt Enable bit in the corresponding descriptor RDES3[30]. When the watchdog timer runs out, the RI bit is set and the timer is stopped. The watchdog timer is reset when the RI bit is set high because of automatic setting of RI as per the Interrupt Enable bit RDES3[30] of any received packet. Value After Reset: 0x0 Testable: untestable
0x11bc	DMA_CH1_Slot_Function_Control_Status	The Slot Function Control and Status register contains the control bits for slot function and the status for Transmit path.	31:20	Reserved_31_20	R	0x0	Reserved. Value After Reset: 0x0
			19:16	RSN	R	0x0	Reference Slot Number This field gives the current value of the reference slot number in the DMA. It is used for slot comparison. Value After Reset: 0x0 Testable: untestable
			15:4	SIV	R/W	0x7c	Slot Interval Value This field controls the period of the slot interval in which the TxDMA fetches the scheduled packets. A value of 0 specifies the slot interval of 1 us while the maximum value 4095 specifies the slot interval of 4096us. The default/reset value is 0x07C which corresponds to slot interval of 125us Value After Reset: 0x7c
			3:2	Reserved_3_2	R	0x0	Reserved. Value After Reset: 0x0
			1	ASC	R/W	0x0	Advance Slot Check When set, this bit enables the DMA to fetch the data from the buffer when the slot number (SLOTNUM) programmed in the Tx descriptor is equal to the reference slot number given in the RSN field or ahead of the reference slot number by up to two slots This bit is applicable only when the ESC bit is set.Values: 0x0 (DISABLE): Advance Slot Check is disabled 0x1 (ENABLE): Advance Slot Check is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	ESC	R/W	0x0	Enable Slot Comparison When set, this bit enables the checking of the slot numbers programmed in the Tx descriptor with the current reference given in the RSN field. The DMA fetches the data from the corresponding buffer only when the slot number is equal to the reference slot number or ahead of the reference slot number by one slot When reset, this bit disables the checking of the slot numbers. The DMA fetches the data immediately after the descriptor is processed. Note: The UFO (UDP Fragmentation over IPv4)/TSO/USO must not be enabled along with TBS/AVB Slot number check. The UFO/TSO/USO involves multiple packets/segments/fragments transmission for single packet received from application and the slot number check are applicable for fetching of only first segment/fragment. As a result it might be difficult for software to specify slot number for subsequent packets.Values: 0x0 (DISABLE): Slot Comparison is disabled 0x1 (ENABLE): Slot Comparison is enabled Value After Reset: 0x0
0x11c4	DMA_CH1_Current_App_Tx Desc	The Channel1 Current Application Transmit Descriptor register points to the current Transmit descriptor read by the DMA.	31:0	CURTDESAPTR	R	0x0	Application Transmit Descriptor Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x11cc	DMA_CH1_Current_App_Rx Desc	The Channel1 Current Application Receive Descriptor register points to the current Receive descriptor read by the DMA.	31:0	CURRDESAPTR	R	0x0	Application Receive Descriptor Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x11d0	DMA_CH1_Current_App_Tx Buffer_H	The Channel1 Current Application Transmit Buffer Address High register has the higher 8 or 16 bits of the current address of the Transmit buffer address read by the DMA.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	CURTBUFAPTRH	R	0x0	Application Transmit Buffer Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x11d4	DMA_CH1_Current_App_Tx Buffer	The Channel1 Current Application Transmit Buffer Address register points to the current Tx buffer address read by the DMA.	31:0	CURTBUFAPTR	R	0x0	Application Transmit Buffer Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x11d8	DMA_CH1_Current_App_Rx Buffer_H	The Channel1 Current Application Receive Buffer Address High register has the higher 8 or 16 bits of the current address of the Receive buffer address read by the DMA.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	CURRBUFAPTRH	R	0x0	Application Receive Buffer Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x11dc	DMA_CH1_Current_App_Rx Buffer	The Channel 1 Current Application Receive Buffer Address register points to the current Rx buffer address read by the DMA.	31:0	CURRBUFAPTR	R	0x0	Application Receive Buffer Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			31:22	Reserved_31_22	R	0x0	Reserved. Value After Reset: 0x0
			21:19	REB	R	0x0	Rx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AXI interface. Bit 21 1'b1: Error during data transfer by Rx DMA 1'b0: No Error during data transfer by Rx DMA Bit 20 1'b1: Error during descriptor access 1'b0: Error during data buffer access Bit 19 1'b1: Error during read transfer 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt. Value After Reset: 0x0 Testable: untestable
			18:16	TEB	R	0x0	Tx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AXI interface. Bit 18 1'b1: Error during data transfer by Tx DMA 1'b0: No Error during data transfer by Tx DMA Bit 17 1'b1: Error during descriptor access 1'b0: Error during data buffer access Bit 16 1'b1: Error during read transfer 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt. Value After Reset: 0x0 Testable: untestable
			15	NIS	R/W	0x0	Normal Interrupt Summary Normal Interrupt Summary bit value is the logical OR of the following bits when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: Bit 0: Transmit Interrupt Bit 2: Transmit Buffer Unavailable Bit 6: Receive Interrupt Bit 11: Early Receive Interrupt Only unmasked bits (interrupts for which interrupt enable is set in DMA_CH0_Interrupt_Enable register) affect the Normal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit which causes NIS to be set is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Normal Interrupt Summary status not detected 0x1 (ACTIVE): Normal Interrupt Summary status detected Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x11e0	DMA_CH1_Status	The software driver (application) reads the Status register during interrupt service routine or polling to determine the status of the DMA.Note: The number of DMA_CH1_Status register in the configuration is the higher of number of Rx DMA Channels and Tx DMA Channels.	14	AIS	R/W	0x0	Abnormal Interrupt Summary Abnormal Interrupt Summary bit value is the logical OR of the following when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: Bit 1: Transmit Process Stopped Bit 7: Receive Buffer Unavailable Bit 8: Receive Process Stopped Bit 10: Early Transmit Interrupt Bit 12: Fatal Bus Error Bit 13: Context Descriptor Error Only unmasked bits affect the Abnormal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit, which causes AIS to be set, is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Abnormal Interrupt Summary status not detected 0x1 (ACTIVE): Abnormal Interrupt Summary status detected Value After Reset: 0x0 Testable: untestable
			13	CDE	R/W	0x0	Context Descriptor Error This bit indicates that the DMA Tx/Rx engine received a descriptor error, which indicates invalid context in the middle of packet flow (intermediate descriptor) or all one's descriptor in Tx case and on Rx side it indicates DMA has read a descriptor with either of the buffer address as ones which is considered to be invalid. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Context Descriptor Error status not detected 0x1 (ACTIVE): Context Descriptor Error status detected Value After Reset: 0x0 Testable: untestable
			12	FBE	R/W	0x0	Fatal Bus Error This bit indicates that a bus error occurred (as described in the EB field). When this bit is set, the corresponding DMA channel engine disables all bus accesses. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Fatal Bus Error status not detected 0x1 (ACTIVE): Fatal Bus Error status detected Value After Reset: 0x0 Testable: untestable
			11	ERI	R/W	0x0	Early Receive Interrupt This bit when set indicates that the RxDMA has completed the transfer of packet data to the memory. In configurations supporting ERIC, ERIC = 0: This bit is set only after the Rx DMA has completely filled up a receive buffer with packet data. ERIC = 1: This bit is set after every burst transfer of data from the Rx DMA to the buffer. The setting of RI bit automatically clears this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Early Receive Interrupt status not detected 0x1 (ACTIVE): Early Receive Interrupt status detected Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			10	ETI	R/W	0x0	Early Transmit Interrupt This bit when set indicates that the TxDMA has completed the transfer of packet data to the MTL TXFIFO memory. In configurations supporting ERIC: ETIC = 0: This bit is set only after the Tx DMA has transferred a complete packet to MTL. ETIC = 1: This bit is set after completion of (partial) packet data transfer from buffers in the Transmit descriptor in which IOC = 1. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Early Transmit Interrupt status not detected 0x1 (ACTIVE): Early Transmit Interrupt status detected Value After Reset: 0x0 Testable: untestable
			9	RWT	R/W	0x0	Receive Watchdog Timeout This bit is asserted when a packet with length greater than 2,048 bytes (10,240 bytes when Jumbo Packet mode is enabled) is received.Values: 0x0 (INACTIVE): Receive Watchdog Timeout status not detected 0x1 (ACTIVE): Receive Watchdog Timeout status detected Value After Reset: 0x0 Testable: untestable
			8	RPS	R/W	0x0	Receive Process Stopped This bit is asserted when the Rx process enters the Stopped state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Process Stopped status not detected 0x1 (ACTIVE): Receive Process Stopped status detected Value After Reset: 0x0 Testable: untestable
			7	RBU	R/W	0x0	Receive Buffer Unavailable This bit indicates that the application owns the next descriptor in the Receive list, and the DMA cannot acquire it. The Rx process is suspended. To resume processing Rx descriptors, the application must change the ownership of the descriptor and issue a Receive Poll Demand command. If this command is not issued, the Rx process resumes when the next recognized incoming packet is received. In ring mode, the application must advance the Receive Descriptor Tail Pointer register of a channel. This bit is set only when the DMA owns the previous Rx descriptor. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Buffer Unavailable status not detected 0x1 (ACTIVE): Receive Buffer Unavailable status detected Value After Reset: 0x0 Testable: untestable
			6	RI	R/W	0x0	Receive Interrupt This bit indicates that the packet reception is complete. When packet reception is complete, Bit 31 of RDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor. The reception remains in the Running state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Interrupt status not detected 0x1 (ACTIVE): Receive Interrupt status detected Value After Reset: 0x0 Testable: untestable
			5:3	Reserved_5_3	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			2	TBU	R/W	0x0	Transmit Buffer Unavailable This bit indicates that the application owns the next descriptor in the Transmit list, and the DMA cannot acquire it. Transmission is suspended. The TPS0 field of the DMA_Debug_Status0-DMA_Debug_Status2 register explains the Transmit Process state transitions. To resume processing the Transmit descriptors, the application must do the following: Change the ownership of the descriptor by setting Bit 31 of TDES3. Issue a Transmit Poll Demand command. For ring mode, the application must advance the Transmit Descriptor Tail Pointer register of a channel. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Buffer Unavailable status not detected 0x1 (ACTIVE): Transmit Buffer Unavailable status detected Value After Reset: 0x0 Testable: untestable
			1	TPS	R/W	0x0	Transmit Process Stopped This bit is set when the transmission is stopped. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Process Stopped status not detected 0x1 (ACTIVE): Transmit Process Stopped status detected Value After Reset: 0x0 Testable: untestable
			0	TI	R/W	0x0	Transmit Interrupt This bit indicates that the packet transmission is complete. When transmission is complete, Bit 31 of TDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Interrupt status not detected 0x1 (ACTIVE): Transmit Interrupt status detected Value After Reset: 0x0 Testable: untestable
0x11e4	DMA_CH1_Miss_Frame_Cnt	This register has the number of packet counter that got dropped by the DMA either due to Bus Error or due to programming RPF field in DMA_CH1_Rx_Control register.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15	MFCO	R	0x0	Overflow status of the MFC Counter When this bit is set then the MFC counter does not get incremented further. The bit gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Miss Frame Counter overflow not occurred 0x1 (ACTIVE): Miss Frame Counter overflow occurred Value After Reset: 0x0
			14:11	Reserved_14_11	R	0x0	Reserved. Value After Reset: 0x0
			10:0	MFC	R	0x0	Dropped Packet Counters This counter indicates the number of packet counters that are dropped by the DMA either because of bus error or because of programing RPF field in DMA_CH0_Rx_Control register. The counter gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x11e8	DMA_CH1_RXP_Accept_Cnt	The DMA_CH1_RXP_Accept_Cnt registers provides the count of the number of frames accepted by Rx Parser.	31	RXPACOF	R	0x0	Rx Parser Accept Counter Overflow Bit When set, this bit indicates that the RXPAC Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Rx Parser Accept Counter overflow not occurred 0x1 (ACTIVE): Rx Parser Accept Counter overflow occurred Value After Reset: 0x0
			30:0	RXPAC	R	0x0	Rx Parser Accept Counter This 31-bit counter is implemented when a Rx Parser Accept a packet due to AF =1. The counter is cleared when the register is read. Value After Reset: 0x0
0x1200	DMA_CH2_Control	The DMA Channel2 Control register specifies the MSS value for segmentation, length to skip between two descriptors, and also the features such as header splitting and 8xPBL mode.	31:25	Reserved_31_25	R	0x0	Reserved. Value After Reset: 0x0
			24	SPH	R/W	0x0	Split Headers 1: DMA splits the header and payload in the receive path. The DMA writes the header to the Buffer Address1 of RDES0. The DMA writes the payload to the buffer to which the Buffer Address2 is pointing. The software must ensure that the header fits into the Receive buffers. If the header length exceeds the receive buffer size, the DMA does not split the header and payload. This bit is available only when you select Enable Split Header Structure feature.Values: 0x0 (DISABLE): Split Headers feature is disabled 0x1 (ENABLE): Split Headers feature is enabled Value After Reset: 0x0
			23:21	Reserved_23_21	R	0x0	Reserved. Value After Reset: 0x0
			20:18	DSL	R/W	0x0	Descriptor Skip Length This bit specifies the Lword (128-bit) number of locations to skip between two descriptors. The address skipping starts from the end of the current descriptor to the start of the next descriptor. When the DSL value is equal to zero, the DMA takes the descriptor table as contiguous. Value After Reset: 0x0
			17	Reserved_17	R	0x0	Reserved. Value After Reset: 0x0
			16	PBLx8	R/W	0x0	8xPBL mode 1: Indicates that the PBL value programmed in Bits[21:16] in DMA_CH0_Tx_Control and Bits[21:16] in DMA_CH0_Rx_Control is multiplied by eight times. Therefore, the DMA transfers the data in 8, 16, 32, 64, 128, and 256 beats depending on the PBL value. Values: 0x0 (DISABLE): 8xPBL mode is disabled 0x1 (ENABLE): 8xPBL mode is enabled Value After Reset: 0x0
			15:14	Reserved_15_14	R	0x0	Reserved. Value After Reset: 0x0
			13:0	MSS	R/W	0x0	Maximum Segment Size This field specifies the maximum segment size that must be used while segmenting the packet. This field is valid only if the TSE bit of DMA_CH0_Tx_Control register is set. Program a value in this field that is more than the configured datawidth in bytes. Use MSS value of 64 bytes or more. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			31	Reserved_31	R	0x0	Reserved. Value After Reset: 0x0
			30:29	TFSEL	R/W	0x0	TBS Fetch Select Select bits for one of the four DMA_TBS_CTRL register fields (FTOS,FGSN,FTOV) for the channel Values: 2'b00: DMA_TBS_CTRL0 2'b01: DMA_TBS_CTRL1 2'b10: DMA_TBS_CTRL2 2'b11: DMA_TBS_CTRL3 (Reserved if DWC_EQOS_NUM_DMA_TX_CH=3) Value After Reset: 0x0
			28	EDSE	R/W	0x0	Enhanced Descriptor Enable 1: The corresponding channel uses Enhanced Descriptors that are 32 Bytes for both Normal and Context Descriptors. 0: The corresponding channel uses the descriptors that are 16 Bytes. Values: 0x0 (DISABLE): Enhanced Descriptor is disabled 0x1 (ENABLE): Enhanced Descriptor is enabled Value After Reset: 0x0
			27:24	TQOS	R/W	0x0	Transmit QOS. This field is used to drive arqos_m_o[3:0] or awqos_m_o[3:0] output signals for all transactions of DMA Tx Channel0. Value After Reset: 0x0
			23	Reserved_23	R	0x0	Reserved. Value After Reset: 0x0
			22			#VALUE!	Reserved Field: Yes
			21:16	TxPBL	R/W	0x0	Transmit Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in DMA_CH0_Control register. 2. Set the TxPBL. Note: The maximum value of TxPBL must be less than or equal to half the Tx Queue size (TQS field of MTL_TxQ0_Operation_Mode register) in terms of beats. This is required so that the Tx Queue has space to store at least another Tx PBL worth of data while the MTL Tx Queue Controller is transferring data to MAC. For example, in 32-bit data width configurations the total locations in Tx Queue of size 512 bytes is 128, TxPBL and 8xPBL needs to be programmed to less than or equal to 64. Value After Reset: 0x0
			15	IPBL	R/W	0x0	Ignore PBL Requirement 1: DMA does not check for PBL number of locations in the MTL before initiating a transfer. If space is not available, the MTL might use handshaking to slow the DMA. Note: Do not use this mode when you enable multiple Transmit DMA channels. Because, it might block other Transmit and Receive DMA Channels from accessing the Read Data Channel of the AXI bus until space is available in Transmit Queue for current transfer.Values: 0x0 (DISABLE): Ignore PBL Requirement is disabled 0x1 (ENABLE): Ignore PBL Requirement is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1204	DMA_CH2_Tx_Control	The DMA Channel2 Transmit Control register controls the Tx features such as PBL, TCP segmentation, and Tx Channel weights.	14:13	TSE_MODE	R/W	0x0	TSE Mode 00: TSO/USO (segmentation functionality is enabled). In this mode, the setting of TSE bit enables the TSO/USO segmentation. 01: UFO with Checksum (UDP Fragmentation over IPv4 with Checksum). In this mode, the setting of TSE bit enables the UDP fragmentation functionality with Checksum for all the UDP packets. 10: UFO without Checksum (UDP Fragmentation over IPv4 with Checksum). In this mode, the setting of TSE bit enables the UDP fragmentation functionality without Checksum for all the UDP packets. 11: Reserved Values: 0x0 (TSO_USO): TSO/USO 0x1 (UFOWC): UFO with Checksum 0x2 (UFOWOC): UFO without Checksum 0x3 (RSVD): Reserved Value After Reset: 0x0
			12	TSE	R/W	0x0	TCP Segmentation Enabled 1: DMA performs the TCP segmentation or UDP Segmentation/Fragmentation for packets in this channel. The TCP segmentation or UDP packet's segmentation/Fragmentation is performed only for those packets for which the TSE bit (TDES0[19]) = 1 in the Tx Normal descriptor. When this bit is 1, the TxPBL value must be greater than 4. Values: 0x0 (DISABLE): TCP Segmentation is disabled 0x1 (ENABLE): TCP Segmentation is enabled Value After Reset: 0x0
			11:5	Reserved_11_5	R	0x0	Reserved. Value After Reset: 0x0
			4	OSF	R/W	0x0	Operate on Second Packet 1: Instructs the DMA to process the second packet of the Tx data even before the status for the first packet is obtained. Values: 0x0 (DISABLE): Operate on Second Packet disabled 0x1 (ENABLE): Operate on Second Packet enabled Value After Reset: 0x0
			3:1	TCW	R/W	0x0	Transmit Channel Weight This field indicates the weight assigned to the corresponding Tx channel. When reset is complete, this field is set to 0 for all channels by default, resulting in equal weights to all channels. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	ST	R/W	0x0	Start or Stop Transmission Command 1: Transmission is placed in the running state. The DMA checks the Tx list at the current position for a packet to be transmitted. The DMA tries to acquire descriptor from either of the following positions: The current position in the list This is the base address of the Transmit list set by the DMA_CH0_TxDesc_List_Address register. The position at which the transmission was previously stopped If the DMA does not own the current descriptor, the transmission enters the Suspended state and the TBU bit of the DMA_CH0_Status register is set. The Start Transmission command is effective only when the transmission is stopped. If the command is issued before setting the DMA_CH0_TxDesc_List_Address register, the DMA behavior is unpredictable. When this bit is reset, the transmission process is placed in the Stopped state after completing the transmission of the current packet. The Next Descriptor position in the Transmit list is saved, and it becomes the current position when the transmission is restarted. To change the list address, you need to program DMA_CH0_TxDesc_List_Address register with a new value when this bit is reset. The new value is considered when this bit is set again. The stop transmission command is effective only when the transmission of the current packet is complete or the transmission is in the Suspended state.Values: 0x0 (STOP): Stop Transmission Command 0x1 (START): Start Transmission Command Value After Reset: 0x0 Testable: untestable
			31	RPF	R/W	0x0	Rx Packet Flush. 1: DWC_ether_qos automatically flushes the packet from the Rx Queues destined to this DMA Rx Channel, when it is stopped. When this bit remains set and the DMA is re-started by the software driver, the packets residing in the Rx Queues that were received when this RxDMA was stopped, still get flushed out. The packets received by the MAC only after the DMA is re-started is routed to the Rx DMA.The flushing is done on the Read side of the Rx Queue. When the packet is duplicated to multiple Rx DMA, it is flushed by the Rx DMA to which it is routed only when it is stopped to prevent head-of-line blocking. The packet duplicated to multiple Rx DMA is not flushed when the Rx DMA is re-started, irrespective of whether the packet was received during one or more Rx DMAs were stopped. 0: DWC_ether_qos does not flush the packet in the Rx Queue destined to this RxDMA Channel when it is in STOP state. This might cause head-of-line blocking in the corresponding RxQueue. Note: The stopping of packet flow from a Rx DMA Channel to the application by setting RPF works only when there is one-to-one mapping of Rx Queue to Rx DMA channels. In Dynamic mapping mode, setting RPF bit in any DMA_CH0_Rx_Control register might flush packets from unintended Rx Queues which are destined to the stopped Rx DMA Channel.Values: 0x0 (DISABLE): Rx Packet Flush is disabled 0x1 (ENABLE): Rx Packet Flush is enabled Value After Reset: 0x0
			30:28	Reserved_30_28	R	0x0	Reserved. Value After Reset: 0x0
			27:24	RQOS	R/W	0x0	Rx AXI4 QOS. This field is used to drive arqos_m_o[3:0] or awqos_m_o[3:0] output signals for all transactions of DMA Rx Channel0. Value After Reset: 0x0
			23	Reserved_23	R	0x0	Reserved. Value After Reset: 0x0
			22			#VALUE!	Reserved Field: Yes

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1208	DMA_CH2_Rx_Control	The DMA Channel2 Receive Control register controls the Rx features such as PBL, buffer size, and extended status.	21:16	RxPBL	R/W	0x0	Receive Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in the DMA_CH0_Control register. 2. Set the RxPBL. Note: The maximum value of RxPBL must be less than or equal to half the Rx Queue size (RQS field of MTL_RxQ0_Operation_Mode register) in terms of beats. This is required so that the Rx Queue has space to store at least another Rx PBL worth of data while the Rx DMA is transferring a block of data. For example, in 32-bit data width configurations the total locations in Rx Queue of size 512 bytes is 128, so RxPBL and 8xPBL needs to be programmed to less than or equal to 64. Value After Reset: 0x0
			15	Reserved_15	R	0x0	Reserved. Value After Reset: 0x0
			14:5	RBSZ_13_y	R/W	0x0	Receive Buffer size High RBSZ[13:0] is split into two fields higher RBSZ_13_y and lower RBSZ_x_0. The RBSZ[13:0] field indicates the size of the Rx buffers specified in bytes. The maximum buffer size is limited to 16K bytes. The buffer size is applicable to payload buffers when split headers are enabled. Note: The buffer size must be a multiple of 16. This is required even if the value of buffer address pointer is not aligned to data bus width. Hence the lower RBSZ_x_0 bits are read-only and the value is considered as all-zero. Thus the RBSZ_13_y indicates the buffer size in terms of locations. Value After Reset: 0x0
			4:1	RBSZ_x_0	R	0x0	Receive Buffer size Low RBSZ[13:0] is split into two fields RBSZ_13_y and RBSZ_x_0. The RBSZ_x_0 is the lower field. This field is of width 4 bits. This field is read-only (RO). Value After Reset: 0x0
			0	SR	R/W	0x0	Start or Stop Receive 1: DMA attempts to acquire the descriptor from the Receive list and processes the incoming packets. The DMA tries to acquire descriptor from either of the following positions: -- The current position in the list This is the address set by the DMA_CH0_RxDesc_List_Address register. -- The position at which the Rx process was previously stopped If the DMA does not own the current descriptor, the reception is suspended and the RBU bit of the DMA_CH0_Status register is set. The Start Receive command is effective only when the reception is stopped. If the command is issued before setting the DMA_CH0_RxDesc_List_Address register, the DMA behavior is unpredictable. 0: Rx DMA operation stops after the transfer of the current packet. The next descriptor position in the Receive list is saved, and it becomes the current position after the Rx process is restarted. The Stop Receive command is effective only when the Rx process is in the Running (waiting for Rx packet) or Suspended state. Values: 0x0 (STOP): Stop Receive 0x1 (START): Start Receive Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1210	DMA_CH2_TxDesc_List_HAddress	The Channel2 Tx Descriptor List HAddress register has the higher 8 or 16 bits of the start address of the Transmit descriptor list.You can write to this register only when the Tx DMA has stopped, that is, the ST bit is set to zero in DMA_CH2_Tx_Control register. When stopped, this register can be written with a new descriptor list address. When you set the ST bit to 1, the DMA takes the newly-programmed descriptor base address. If this register is not changed when the ST bit is set to 2, the DMA takes the descriptor address where it was stopped earlier. This register must be programmed with desired the higher address before programming the DMA_CH2_TxDesc_List_Address register.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	TDESHA	R/W	0x0	Start of Transmit List This field contains the most-significant 8 or 16 bits of the 40- or 48-bit base address of the first descriptor in the Transmit descriptor list. Value After Reset: 0x0
0x1214	DMA_CH2_TxDesc_List_Address	The Channel2 Tx Descriptor List Address register points the DMA to the start of Transmit descriptor list. The descriptor lists reside in the physical memory space of the application and must be Lword (128-bit). The DMA internally converts it to bus width aligned address by making the corresponding LSB to low.You can write to this register only when the Tx DMA has stopped, that is, the ST bit is set to zero in DMA_CH2_Tx_Control register. When stopped, this register can be written with a new descriptor list address. When you set the ST bit to 1, the DMA takes the newly-programmed descriptor base address. If this register is not changed when the ST bit is set to 2, the DMA takes the descriptor address where it was stopped earlier.	31:4	TDESLA	R/W	0x0	Start of Transmit List This field contains the base address of the first descriptor in the Transmit descriptor list. The DMA ignores the LSB bits (3:0) and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0 Testable: untestable
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x1218	DMA_CH2_RxDesc_List_HAddress	The Channel2 Rx Descriptor List HAddress register has the higher 8 or 16 bits of the start address of the Receive descriptor list.Writing to this register is permitted only when reception is stopped. When stopped, this register must be written to before the receive Start command is given. You can write to this register only when Rx DMA has stopped, that is, SR bit is set to zero in the DMA_CH2_RxDesc_List_Address registerWhen you set the SR bit to 1, the DMA takes the newly programmed descriptor base address.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	RDESHA	R/W	0x0	Start of Receive List This field contains the most-significant 8 or 16 bits of the 40-bit or 48-bit base address of the first descriptor in the Rx Descriptor list. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x121c	DMA_CH2_RxDesc_List_Address	The Channel2 Rx Descriptor List Address register points the DMA to the start of Receive descriptor list.This register points to the start of the Receive Descriptor List. The descriptor lists reside in the physical memory space of the application and must be Lword (128-bit). The DMA internally converts it to bus width aligned address by making the corresponding LS bits low. Writing to this register is permitted only when reception is stopped. When stopped, this register must be written to before the receive Start command is given. You can write to this register only when Rx DMA has stopped, that is, SR bit is set to zero in DMA_CH2_Rx_Control register. When stopped, this register can be written with a new descriptor list address.When you set the SR bit to 1, the DMA takes the newly programmed descriptor base address.	31:4	RDESLA	R/W	0x0	Start of Receive List This field contains the base address of the first descriptor in the Rx Descriptor list. The DMA ignores the LSB bits (3:0) and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0 Testable: untestable
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x1220	DMA_CH2_TxDesc_Tail_Pointer	The Channel2 Tx Descriptor Tail Pointer register points to an offset from the base and indicates the location of the last valid descriptor.	31:4	TDTP	R/W	0x0	Transmit Descriptor Tail Pointer This field contains the tail pointer for the Tx descriptor ring. The software writes the tail pointer to add more descriptors to the Tx channel. The hardware tries to transmit all packets referenced by the descriptors between the head and the tail pointer registers. The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x1228	DMA_CH2_RxDesc_Tail_Pointer	The Channel2 Rx Descriptor Tail Pointer Points to an offset from the base and indicates the location of the last valid descriptor.	31:4	RDTP	R/W	0x0	Receive Descriptor Tail Pointer This field contains the tail pointer for the Rx descriptor ring. The software writes the tail pointer to add more descriptors to the Rx channel. The hardware tries to write all received packets to the descriptors referenced between the head and the tail pointer registers. The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x122c	DMA_CH2_TxDesc_Ring_Length	The Tx Descriptor Ring Length register contains the length of the Transmit descriptor ring.	31:10	Reserved_31_10	R	0x0	Reserved. Value After Reset: 0x0
			9:0	TDRL	R/W	0x0	Transmit Descriptor Ring Length This field sets the maximum number of Tx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. It is recommended that a minimum ring descriptor length of 4. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1230	DMA_CH2_Rx_Control2	The Channel2 Receive Control register controls the Rx features such as Rx Descriptor Ring Length and Alternate Rx Buffer Size.	31:24	Reserved_31_24	R	0x0	Reserved. Value After Reset: 0x0
			23:18	ARBS	R/W	0x0	Alternate Receive Buffer Size Indicates size in bytes for Buffer 1 when ARBS is programmed to a non-zero value (when split header feature is not enabled). When split header feature is enabled, ARBS indicates the buffer size for header data. The maximum alternate buffer is limited to 1008 bytes. When ARBS=0, Rx Buffer1 and Rx Buffer2 sizes are based on RBSZ field of DMA_CH0_Rx_Control. Width of ARBS field is 6 bits. Value After Reset: 0x0
			17:10	Reserved_x_10	R	0x0	Reserved. Value After Reset: 0x0
			9:0	RDRL	R/W	0x0	Receive Descriptor Ring Length This register sets the maximum number of Rx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9. Value After Reset: 0x0
			31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15	NIE	R/W	0x0	Normal Interrupt Summary Enable When this bit is set, the normal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: Bit 0: Transmit Interrupt Bit 2: Transmit Buffer Unavailable Bit 6: Receive Interrupt Bit 11: Early Receive Interrupt When this bit is reset, the normal interrupt summary is disabled.Values: 0x0 (DISABLE): Normal Interrupt Summary is disabled 0x1 (ENABLE): Normal Interrupt Summary is enabled Value After Reset: 0x0
			14	AIE	R/W	0x0	Abnormal Interrupt Summary Enable When this bit is set, the abnormal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: Bit 1: Transmit Process Stopped Bit 7: Rx Buffer Unavailable Bit 8: Receive Process Stopped Bit 9: Receive Watchdog Timeout Bit 10: Early Transmit Interrupt Bit 12: Fatal Bus Error Bit 13: Context Descriptor Error When this bit is reset, the abnormal interrupt summary is disabled.Values: 0x0 (DISABLE): Abnormal Interrupt Summary is disabled 0x1 (ENABLE): Abnormal Interrupt Summary is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1234	DMA_CH2_Interrupt_Enable	The Channel2 Interrupt Enable register enables the interrupts reported by the Status register.	13	CDEE	R/W	0x0	Context Descriptor Error Enable When this bit is set along with the AIE bit, the Descriptor error interrupt is enabled. When this bit is reset, the Descriptor error interrupt is disabled.Values: 0x0 (DISABLE): Context Descriptor Error is disabled 0x1 (ENABLE): Context Descriptor Error is enabled Value After Reset: 0x0
			12	FBEE	R/W	0x0	Fatal Bus Error Enable When this bit is set along with the AIE bit, the Fatal Bus error interrupt is enabled. When this bit is reset, the Fatal Bus Error error interrupt is disabled.Values: 0x0 (DISABLE): Fatal Bus Error is disabled 0x1 (ENABLE): Fatal Bus Error is enabled Value After Reset: 0x0
			11	ERIE	R/W	0x0	Early Receive Interrupt Enable When this bit is set along with the NIE bit, the Early Receive interrupt is enabled. When this bit is reset, the Early Receive interrupt is disabled.Values: 0x0 (DISABLE): Early Receive Interrupt is disabled 0x1 (ENABLE): Early Receive Interrupt is enabled Value After Reset: 0x0
			10	ETIE	R/W	0x0	Early Transmit Interrupt Enable When this bit is set along with the AIE bit, the Early Transmit interrupt is enabled. When this bit is reset, the Early Transmit interrupt is disabled.Values: 0x0 (DISABLE): Early Transmit Interrupt is disabled 0x1 (ENABLE): Early Transmit Interrupt is enabled Value After Reset: 0x0
			9	RWTE	R/W	0x0	Receive Watchdog Timeout Enable When this bit is set along with the AIE bit, the Receive Watchdog Timeout interrupt is enabled. When this bit is reset, the Receive Watchdog Timeout interrupt is disabled.Values: 0x0 (DISABLE): Receive Watchdog Timeout is disabled 0x1 (ENABLE): Receive Watchdog Timeout is enabled Value After Reset: 0x0
			8	RSE	R/W	0x0	Receive Stopped Enable When this bit is set along with the AIE bit, the Receive Stopped Interrupt is enabled. When this bit is reset, the Receive Stopped interrupt is disabled.Values: 0x0 (DISABLE): Receive Stopped is disabled 0x1 (ENABLE): Receive Stopped is enabled Value After Reset: 0x0
			7	RBUE	R/W	0x0	Receive Buffer Unavailable Enable When this bit is set along with the AIE bit, the Receive Buffer Unavailable interrupt is enabled. When this bit is reset, the Receive Buffer Unavailable interrupt is disabled.Values: 0x0 (DISABLE): Receive Buffer Unavailable is disabled 0x1 (ENABLE): Receive Buffer Unavailable is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			6	RIE	R/W	0x0	Receive Interrupt Enable When this bit is set along with the NIE bit, the Receive Interrupt is enabled. When this bit is reset, the Receive Interrupt is disabled.Values: 0x0 (DISABLE): Receive Interrupt is disabled 0x1 (ENABLE): Receive Interrupt is enabled Value After Reset: 0x0
			5:3	Reserved_5_3	R	0x0	Reserved. Value After Reset: 0x0
			2	TBUE	R/W	0x0	Transmit Buffer Unavailable Enable When this bit is set along with the NIE bit, the Transmit Buffer Unavailable interrupt is enabled. When this bit is reset, the Transmit Buffer Unavailable interrupt is disabled.Values: 0x0 (DISABLE): Transmit Buffer Unavailable is disabled 0x1 (ENABLE): Transmit Buffer Unavailable is enabled Value After Reset: 0x0
			1	TXSE	R/W	0x0	Transmit Stopped Enable When this bit is set along with the AIE bit, the Transmission Stopped interrupt is enabled. When this bit is reset, the Transmission Stopped interrupt is disabled.Values: 0x0 (DISABLE): Transmit Stopped is disabled 0x1 (ENABLE): Transmit Stopped is enabled Value After Reset: 0x0
			0	TIE	R/W	0x0	Transmit Interrupt Enable When this bit is set along with the NIE bit, the Transmit Interrupt is enabled. When this bit is reset, the Transmit Interrupt is disabled.Values: 0x0 (DISABLE): Transmit Interrupt is disabled 0x1 (ENABLE): Transmit Interrupt is enabled Value After Reset: 0x0
0x1238	DMA_CH2_Rx_Interrupt_Watchdog_Timer	The Receive Interrupt Watchdog Timer register indicates the watchdog timeout for Receive Interrupt (RI) from the DMA. When this register is written with a non-zero value, it enables the watchdog timer for the RI bit of the DMA_CH2_Status register.	31:18	Reserved_31_18	R	0x0	Reserved. Value After Reset: 0x0
			17:16	RWTU	R/W	0x0	Receive Interrupt Watchdog Timer Count Units This fields indicates the number of system clock cycles corresponding to one unit in RWT field. 2'b00: 256 2'b01: 512 2'b10: 1024 2'b11: 2048 For example, when RWT=2 and RWTU=1, the watchdog timer is set for 2*512=1024 system clock cycles. Value After Reset: 0x0 Testable: untestable
			15:8	Reserved_15_8	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			7:0	RWT	R/W	0x0	Receive Interrupt Watchdog Timer Count This field indicates the number of system clock cycles, multiplied by factor indicated in RWTU field, for which the watchdog timer is set. The watchdog timer is triggered with the programmed value after the Rx DMA completes the transfer of a packet for which the RI bit is not set in the DMA_CH0_Status register, because of the setting of Interrupt Enable bit in the corresponding descriptor RDES3[30]. When the watchdog timer runs out, the RI bit is set and the timer is stopped. The watchdog timer is reset when the RI bit is set high because of automatic setting of RI as per the Interrupt Enable bit RDES3[30] of any received packet. Value After Reset: 0x0 Testable: untestable
0x123c	DMA_CH2_Slot_Function_Control_Status	The Slot Function Control and Status register contains the control bits for slot function and the status for Transmit path.	31:20	Reserved_31_20	R	0x0	Reserved. Value After Reset: 0x0
			19:16	RSN	R	0x0	Reference Slot Number This field gives the current value of the reference slot number in the DMA. It is used for slot comparison. Value After Reset: 0x0 Testable: untestable
			15:4	SIV	R/W	0x7c	Slot Interval Value This field controls the period of the slot interval in which the TxDMA fetches the scheduled packets. A value of 0 specifies the slot interval of 1 us while the maximum value 4095 specifies the slot interval of 4096us. The default/reset value is 0x07C which corresponds to slot interval of 125us Value After Reset: 0x7c
			3:2	Reserved_3_2	R	0x0	Reserved. Value After Reset: 0x0
			1	ASC	R/W	0x0	Advance Slot Check When set, this bit enables the DMA to fetch the data from the buffer when the slot number (SLOTNUM) programmed in the Tx descriptor is equal to the reference slot number given in the RSN field or ahead of the reference slot number by up to two slots This bit is applicable only when the ESC bit is set.Values: 0x0 (DISABLE): Advance Slot Check is disabled 0x1 (ENABLE): Advance Slot Check is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	ESC	R/W	0x0	Enable Slot Comparison When set, this bit enables the checking of the slot numbers programmed in the Tx descriptor with the current reference given in the RSN field. The DMA fetches the data from the corresponding buffer only when the slot number is equal to the reference slot number or ahead of the reference slot number by one slot When reset, this bit disables the checking of the slot numbers. The DMA fetches the data immediately after the descriptor is processed. Note: The UFO (UDP Fragmentation over IPv4)/TSO/USO must not be enabled along with TBS/AVB Slot number check. The UFO/TSO/USO involves multiple packets/segments/fragments transmission for single packet received from application and the slot number check are applicable for fetching of only first segment/fragment. As a result it might be difficult for software to specify slot number for subsequent packets.Values: 0x0 (DISABLE): Slot Comparison is disabled 0x1 (ENABLE): Slot Comparison is enabled Value After Reset: 0x0
0x1244	DMA_CH2_Current_App_Tx Desc	The Channel2 Current Application Transmit Descriptor register points to the current Transmit descriptor read by the DMA.	31:0	CURTDESAPTR	R	0x0	Application Transmit Descriptor Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x124c	DMA_CH2_Current_App_Rx Desc	The Channel2 Current Application Receive Descriptor register points to the current Receive descriptor read by the DMA.	31:0	CURRDESAPTR	R	0x0	Application Receive Descriptor Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x1250	DMA_CH2_Current_App_Tx Buffer_H	The Channel2 Current Application Transmit Buffer Address High register has the higher 8 or 16 bits of the current address of the Transmit buffer address read by the DMA.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	CURTBUFAPTRH	R	0x0	Application Transmit Buffer Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x1254	DMA_CH2_Current_App_Tx Buffer	The Channel2 Current Application Transmit Buffer Address register points to the current Tx buffer address read by the DMA.	31:0	CURTBUFAPTR	R	0x0	Application Transmit Buffer Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x1258	DMA_CH2_Current_App_Rx Buffer_H	The Channel2 Current Application Receive Buffer Address High register has the higher 8 or 16 bits of the current address of the Receive buffer address read by the DMA.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	CURRBUFAPTRH	R	0x0	Application Receive Buffer Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x125c	DMA_CH2_Current_App_Rx Buffer	The Channel 2 Current Application Receive Buffer Address register points to the current Rx buffer address read by the DMA.	31:0	CURRBUFAPTR	R	0x0	Application Receive Buffer Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			31:22	Reserved_31_22	R	0x0	Reserved. Value After Reset: 0x0
			21:19	REB	R	0x0	Rx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AXI interface. Bit 21 1'b1: Error during data transfer by Rx DMA 1'b0: No Error during data transfer by Rx DMA Bit 20 1'b1: Error during descriptor access 1'b0: Error during data buffer access Bit 19 1'b1: Error during read transfer 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt. Value After Reset: 0x0 Testable: untestable
			18:16	TEB	R	0x0	Tx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AXI interface. Bit 18 1'b1: Error during data transfer by Tx DMA 1'b0: No Error during data transfer by Tx DMA Bit 17 1'b1: Error during descriptor access 1'b0: Error during data buffer access Bit 16 1'b1: Error during read transfer 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt. Value After Reset: 0x0 Testable: untestable
			15	NIS	R/W	0x0	Normal Interrupt Summary Normal Interrupt Summary bit value is the logical OR of the following bits when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: Bit 0: Transmit Interrupt Bit 2: Transmit Buffer Unavailable Bit 6: Receive Interrupt Bit 11: Early Receive Interrupt Only unmasked bits (interrupts for which interrupt enable is set in DMA_CH0_Interrupt_Enable register) affect the Normal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit which causes NIS to be set is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Normal Interrupt Summary status not detected 0x1 (ACTIVE): Normal Interrupt Summary status detected Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1260	DMA_CH2_Status	The software driver (application) reads the Status register during interrupt service routine or polling to determine the status of the DMA.Note: The number of DMA_CH2_Status register in the configuration is the higher of number of Rx DMA Channels and Tx DMA Channels.	14	AIS	R/W	0x0	Abnormal Interrupt Summary Abnormal Interrupt Summary bit value is the logical OR of the following when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: Bit 1: Transmit Process Stopped Bit 7: Receive Buffer Unavailable Bit 8: Receive Process Stopped Bit 10: Early Transmit Interrupt Bit 12: Fatal Bus Error Bit 13: Context Descriptor Error Only unmasked bits affect the Abnormal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit, which causes AIS to be set, is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Abnormal Interrupt Summary status not detected 0x1 (ACTIVE): Abnormal Interrupt Summary status detected Value After Reset: 0x0 Testable: untestable
			13	CDE	R/W	0x0	Context Descriptor Error This bit indicates that the DMA Tx/Rx engine received a descriptor error, which indicates invalid context in the middle of packet flow (intermediate descriptor) or all one's descriptor in Tx case and on Rx side it indicates DMA has read a descriptor with either of the buffer address as ones which is considered to be invalid. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Context Descriptor Error status not detected 0x1 (ACTIVE): Context Descriptor Error status detected Value After Reset: 0x0 Testable: untestable
			12	FBE	R/W	0x0	Fatal Bus Error This bit indicates that a bus error occurred (as described in the EB field). When this bit is set, the corresponding DMA channel engine disables all bus accesses. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Fatal Bus Error status not detected 0x1 (ACTIVE): Fatal Bus Error status detected Value After Reset: 0x0 Testable: untestable
			11	ERI	R/W	0x0	Early Receive Interrupt This bit when set indicates that the RxDMA has completed the transfer of packet data to the memory. In configurations supporting ERIC, ERIC = 0: This bit is set only after the Rx DMA has completely filled up a receive buffer with packet data. ERIC = 1: This bit is set after every burst transfer of data from the Rx DMA to the buffer. The setting of RI bit automatically clears this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Early Receive Interrupt status not detected 0x1 (ACTIVE): Early Receive Interrupt status detected Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			10	ETI	R/W	0x0	Early Transmit Interrupt This bit when set indicates that the TxDMA has completed the transfer of packet data to the MTL TXFIFO memory. In configurations supporting ERIC: ETIC = 0: This bit is set only after the Tx DMA has transferred a complete packet to MTL. ETIC = 1: This bit is set after completion of (partial) packet data transfer from buffers in the Transmit descriptor in which IOC = 1. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Early Transmit Interrupt status not detected 0x1 (ACTIVE): Early Transmit Interrupt status detected Value After Reset: 0x0 Testable: untestable
			9	RWT	R/W	0x0	Receive Watchdog Timeout This bit is asserted when a packet with length greater than 2,048 bytes (10,240 bytes when Jumbo Packet mode is enabled) is received.Values: 0x0 (INACTIVE): Receive Watchdog Timeout status not detected 0x1 (ACTIVE): Receive Watchdog Timeout status detected Value After Reset: 0x0 Testable: untestable
			8	RPS	R/W	0x0	Receive Process Stopped This bit is asserted when the Rx process enters the Stopped state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Process Stopped status not detected 0x1 (ACTIVE): Receive Process Stopped status detected Value After Reset: 0x0 Testable: untestable
			7	RBU	R/W	0x0	Receive Buffer Unavailable This bit indicates that the application owns the next descriptor in the Receive list, and the DMA cannot acquire it. The Rx process is suspended. To resume processing Rx descriptors, the application must change the ownership of the descriptor and issue a Receive Poll Demand command. If this command is not issued, the Rx process resumes when the next recognized incoming packet is received. In ring mode, the application must advance the Receive Descriptor Tail Pointer register of a channel. This bit is set only when the DMA owns the previous Rx descriptor. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Buffer Unavailable status not detected 0x1 (ACTIVE): Receive Buffer Unavailable status detected Value After Reset: 0x0 Testable: untestable
			6	RI	R/W	0x0	Receive Interrupt This bit indicates that the packet reception is complete. When packet reception is complete, Bit 31 of RDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor. The reception remains in the Running state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Interrupt status not detected 0x1 (ACTIVE): Receive Interrupt status detected Value After Reset: 0x0 Testable: untestable
			5:3	Reserved_5_3	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			2	TBU	R/W	0x0	Transmit Buffer Unavailable This bit indicates that the application owns the next descriptor in the Transmit list, and the DMA cannot acquire it. Transmission is suspended. The TPS0 field of the DMA_Debug_Status0-DMA_Debug_Status2 register explains the Transmit Process state transitions. To resume processing the Transmit descriptors, the application must do the following: Change the ownership of the descriptor by setting Bit 31 of TDES3. Issue a Transmit Poll Demand command. For ring mode, the application must advance the Transmit Descriptor Tail Pointer register of a channel. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Buffer Unavailable status not detected 0x1 (ACTIVE): Transmit Buffer Unavailable status detected Value After Reset: 0x0 Testable: untestable
			1	TPS	R/W	0x0	Transmit Process Stopped This bit is set when the transmission is stopped. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Process Stopped status not detected 0x1 (ACTIVE): Transmit Process Stopped status detected Value After Reset: 0x0 Testable: untestable
			0	TI	R/W	0x0	Transmit Interrupt This bit indicates that the packet transmission is complete. When transmission is complete, Bit 31 of TDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Interrupt status not detected 0x1 (ACTIVE): Transmit Interrupt status detected Value After Reset: 0x0 Testable: untestable
0x1264	DMA_CH2_Miss_Frame_Cnt	This register has the number of packet counter that got dropped by the DMA either due to Bus Error or due to programming RPF field in DMA_CH2_Rx_Control register.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15	MFCO	R	0x0	Overflow status of the MFC Counter When this bit is set then the MFC counter does not get incremented further. The bit gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Miss Frame Counter overflow not occurred 0x1 (ACTIVE): Miss Frame Counter overflow occurred Value After Reset: 0x0
			14:11	Reserved_14_11	R	0x0	Reserved. Value After Reset: 0x0
			10:0	MFC	R	0x0	Dropped Packet Counters This counter indicates the number of packet counters that are dropped by the DMA either because of bus error or because of programing RPF field in DMA_CH0_Rx_Control register. The counter gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1268	DMA_CH2_RXP_Accept_Cnt	The DMA_CH2_RXP_Accept_Cnt registers provides the count of the number of frames accepted by Rx Parser.	31	RXPACOF	R	0x0	Rx Parser Accept Counter Overflow Bit When set, this bit indicates that the RXPAC Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Rx Parser Accept Counter overflow not occurred 0x1 (ACTIVE): Rx Parser Accept Counter overflow occurred Value After Reset: 0x0
			30:0	RXPAC	R	0x0	Rx Parser Accept Counter This 31-bit counter is implemented when a Rx Parser Accept a packet due to AF =1. The counter is cleared when the register is read. Value After Reset: 0x0
0x1280	DMA_CH3_Control	The DMA Channel3 Control register specifies the MSS value for segmentation, length to skip between two descriptors, and also the features such as header splitting and 8xPBL mode.	31:25	Reserved_31_25	R	0x0	Reserved. Value After Reset: 0x0
			24	SPH	R/W	0x0	Split Headers 1: DMA splits the header and payload in the receive path. The DMA writes the header to the Buffer Address1 of RDES0. The DMA writes the payload to the buffer to which the Buffer Address2 is pointing. The software must ensure that the header fits into the Receive buffers. If the header length exceeds the receive buffer size, the DMA does not split the header and payload. This bit is available only when you select Enable Split Header Structure feature.Values: 0x0 (DISABLE): Split Headers feature is disabled 0x1 (ENABLE): Split Headers feature is enabled Value After Reset: 0x0
			23:21	Reserved_23_21	R	0x0	Reserved. Value After Reset: 0x0
			20:18	DSL	R/W	0x0	Descriptor Skip Length This bit specifies the Lword (128-bit) number of locations to skip between two descriptors. The address skipping starts from the end of the current descriptor to the start of the next descriptor. When the DSL value is equal to zero, the DMA takes the descriptor table as contiguous. Value After Reset: 0x0
			17	Reserved_17	R	0x0	Reserved. Value After Reset: 0x0
			16	PBLx8	R/W	0x0	8xPBL mode 1: Indicates that the PBL value programmed in Bits[21:16] in DMA_CH0_Tx_Control and Bits[21:16] in DMA_CH0_Rx_Control is multiplied by eight times. Therefore, the DMA transfers the data in 8, 16, 32, 64, 128, and 256 beats depending on the PBL value. Values: 0x0 (DISABLE): 8xPBL mode is disabled 0x1 (ENABLE): 8xPBL mode is enabled Value After Reset: 0x0
			15:14	Reserved_15_14	R	0x0	Reserved. Value After Reset: 0x0
			13:0	MSS	R/W	0x0	Maximum Segment Size This field specifies the maximum segment size that must be used while segmenting the packet. This field is valid only if the TSE bit of DMA_CH0_Tx_Control register is set. Program a value in this field that is more than the configured datawidth in bytes. Use MSS value of 64 bytes or more. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			31	Reserved_31	R	0x0	Reserved. Value After Reset: 0x0
			30:29	TFSEL	R/W	0x0	TBS Fetch Select Select bits for one of the four DMA_TBS_CTRL register fields (FTOS,FGSN,FTOV) for the channel Values: 2'b00: DMA_TBS_CTRL0 2'b01: DMA_TBS_CTRL1 2'b10: DMA_TBS_CTRL2 2'b11: DMA_TBS_CTRL3 (Reserved if DWC_EQOS_NUM_DMA_TX_CH=3) Value After Reset: 0x0
			28	EDSE	R/W	0x0	Enhanced Descriptor Enable 1: The corresponding channel uses Enhanced Descriptors that are 32 Bytes for both Normal and Context Descriptors. 0: The corresponding channel uses the descriptors that are 16 Bytes. Values: 0x0 (DISABLE): Enhanced Descriptor is disabled 0x1 (ENABLE): Enhanced Descriptor is enabled Value After Reset: 0x0
			27:24	TQOS	R/W	0x0	Transmit QOS. This field is used to drive arqos_m_o[3:0] or awqos_m_o[3:0] output signals for all transactions of DMA Tx Channel0. Value After Reset: 0x0
			23	Reserved_23	R	0x0	Reserved. Value After Reset: 0x0
			22			#VALUE!	Reserved Field: Yes
			21:16	TxPBL	R/W	0x0	Transmit Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in DMA_CH0_Control register. 2. Set the TxPBL. Note: The maximum value of TxPBL must be less than or equal to half the Tx Queue size (TQS field of MTL_TxQ0_Operation_Mode register) in terms of beats. This is required so that the Tx Queue has space to store at least another Tx PBL worth of data while the MTL Tx Queue Controller is transferring data to MAC. For example, in 32-bit data width configurations the total locations in Tx Queue of size 512 bytes is 128, TxPBL and 8xPBL needs to be programmed to less than or equal to 64. Value After Reset: 0x0
			15	IPBL	R/W	0x0	Ignore PBL Requirement 1: DMA does not check for PBL number of locations in the MTL before initiating a transfer. If space is not available, the MTL might use handshaking to slow the DMA. Note: Do not use this mode when you enable multiple Transmit DMA channels. Because, it might block other Transmit and Receive DMA Channels from accessing the Read Data Channel of the AXI bus until space is available in Transmit Queue for current transfer.Values: 0x0 (DISABLE): Ignore PBL Requirement is disabled 0x1 (ENABLE): Ignore PBL Requirement is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1284	DMA_CH3_Tx_Control	The DMA Channel3 Transmit Control register controls the Tx features such as PBL, TCP segmentation, and Tx Channel weights.	14:13	TSE_MODE	R/W	0x0	TSE Mode 00: TSO/USO (segmentation functionality is enabled). In this mode, the setting of TSE bit enables the TSO/USO segmentation. 01: UFO with Checksum (UDP Fragmentation over IPv4 with Checksum). In this mode, the setting of TSE bit enables the UDP fragmentation functionality with Checksum for all the UDP packets. 10: UFO without Checksum (UDP Fragmentation over IPv4 with Checksum). In this mode, the setting of TSE bit enables the UDP fragmentation functionality without Checksum for all the UDP packets. 11: Reserved Values: 0x0 (TSO_USO): TSO/USO 0x1 (UFOWC): UFO with Checksum 0x2 (UFOWOC): UFO without Checksum 0x3 (RSVD): Reserved Value After Reset: 0x0
			12	TSE	R/W	0x0	TCP Segmentation Enabled 1: DMA performs the TCP segmentation or UDP Segmentation/Fragmentation for packets in this channel. The TCP segmentation or UDP packet's segmentation/Fragmentation is performed only for those packets for which the TSE bit (TDES0[19]) = 1 in the Tx Normal descriptor. When this bit is 1, the TxPBL value must be greater than 4. Values: 0x0 (DISABLE): TCP Segmentation is disabled 0x1 (ENABLE): TCP Segmentation is enabled Value After Reset: 0x0
			11:5	Reserved_11_5	R	0x0	Reserved. Value After Reset: 0x0
			4	OSF	R/W	0x0	Operate on Second Packet 1: Instructs the DMA to process the second packet of the Tx data even before the status for the first packet is obtained. Values: 0x0 (DISABLE): Operate on Second Packet disabled 0x1 (ENABLE): Operate on Second Packet enabled Value After Reset: 0x0
			3:1	TCW	R/W	0x0	Transmit Channel Weight This field indicates the weight assigned to the corresponding Tx channel. When reset is complete, this field is set to 0 for all channels by default, resulting in equal weights to all channels. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	ST	R/W	0x0	Start or Stop Transmission Command 1: Transmission is placed in the running state. The DMA checks the Tx list at the current position for a packet to be transmitted. The DMA tries to acquire descriptor from either of the following positions: The current position in the list This is the base address of the Transmit list set by the DMA_CH0_TxDesc_List_Address register. The position at which the transmission was previously stopped If the DMA does not own the current descriptor, the transmission enters the Suspended state and the TBU bit of the DMA_CH0_Status register is set. The Start Transmission command is effective only when the transmission is stopped. If the command is issued before setting the DMA_CH0_TxDesc_List_Address register, the DMA behavior is unpredictable. When this bit is reset, the transmission process is placed in the Stopped state after completing the transmission of the current packet. The Next Descriptor position in the Transmit list is saved, and it becomes the current position when the transmission is restarted. To change the list address, you need to program DMA_CH0_TxDesc_List_Address register with a new value when this bit is reset. The new value is considered when this bit is set again. The stop transmission command is effective only when the transmission of the current packet is complete or the transmission is in the Suspended state.Values: 0x0 (STOP): Stop Transmission Command 0x1 (START): Start Transmission Command Value After Reset: 0x0 Testable: untestable
			31	RPF	R/W	0x0	Rx Packet Flush. 1: DWC_ether_qos automatically flushes the packet from the Rx Queues destined to this DMA Rx Channel, when it is stopped. When this bit remains set and the DMA is re-started by the software driver, the packets residing in the Rx Queues that were received when this RxDMA was stopped, still get flushed out. The packets received by the MAC only after the DMA is re-started is routed to the Rx DMA.The flushing is done on the Read side of the Rx Queue. When the packet is duplicated to multiple Rx DMA, it is flushed by the Rx DMA to which it is routed only when it is stopped to prevent head-of-line blocking. The packet duplicated to multiple Rx DMA is not flushed when the Rx DMA is re-started, irrespective of whether the packet was received during one or more Rx DMAs were stopped. 0: DWC_ether_qos does not flush the packet in the Rx Queue destined to this RxDMA Channel when it is in STOP state. This might cause head-of-line blocking in the corresponding RxQueue. Note: The stopping of packet flow from a Rx DMA Channel to the application by setting RPF works only when there is one-to-one mapping of Rx Queue to Rx DMA channels. In Dynamic mapping mode, setting RPF bit in any DMA_CH0_Rx_Control register might flush packets from unintended Rx Queues which are destined to the stopped Rx DMA Channel.Values: 0x0 (DISABLE): Rx Packet Flush is disabled 0x1 (ENABLE): Rx Packet Flush is enabled Value After Reset: 0x0
			30:28	Reserved_30_28	R	0x0	Reserved. Value After Reset: 0x0
			27:24	RQOS	R/W	0x0	Rx AXI4 QOS. This field is used to drive arqos_m_o[3:0] or awqos_m_o[3:0] output signals for all transactions of DMA Rx Channel0. Value After Reset: 0x0
			23	Reserved_23	R	0x0	Reserved. Value After Reset: 0x0
			22			#VALUE!	Reserved Field: Yes

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1288	DMA_CH3_Rx_Control	The DMA Channel3 Receive Control register controls the Rx features such as PBL, buffer size, and extended status.	21:16	RxPBL	R/W	0x0	Receive Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in the DMA_CH0_Control register. 2. Set the RxPBL. Note: The maximum value of RxPBL must be less than or equal to half the Rx Queue size (RQS field of MTL_RxQ0_Operation_Mode register) in terms of beats. This is required so that the Rx Queue has space to store at least another Rx PBL worth of data while the Rx DMA is transferring a block of data. For example, in 32-bit data width configurations the total locations in Rx Queue of size 512 bytes is 128, so RxPBL and 8xPBL needs to be programmed to less than or equal to 64. Value After Reset: 0x0
			15	Reserved_15	R	0x0	Reserved. Value After Reset: 0x0
			14:5	RBSZ_13_y	R/W	0x0	Receive Buffer size High RBSZ[13:0] is split into two fields higher RBSZ_13_y and lower RBSZ_x_0. The RBSZ[13:0] field indicates the size of the Rx buffers specified in bytes. The maximum buffer size is limited to 16K bytes. The buffer size is applicable to payload buffers when split headers are enabled. Note: The buffer size must be a multiple of 16. This is required even if the value of buffer address pointer is not aligned to data bus width. Hence the lower RBSZ_x_0 bits are read-only and the value is considered as all-zero. Thus the RBSZ_13_y indicates the buffer size in terms of locations. Value After Reset: 0x0
			4:1	RBSZ_x_0	R	0x0	Receive Buffer size Low RBSZ[13:0] is split into two fields RBSZ_13_y and RBSZ_x_0. The RBSZ_x_0 is the lower field. This field is of width 4 bits. This field is read-only (RO). Value After Reset: 0x0
			0	SR	R/W	0x0	Start or Stop Receive 1: DMA attempts to acquire the descriptor from the Receive list and processes the incoming packets. The DMA tries to acquire descriptor from either of the following positions: -- The current position in the list This is the address set by the DMA_CH0_RxDesc_List_Address register. -- The position at which the Rx process was previously stopped If the DMA does not own the current descriptor, the reception is suspended and the RBU bit of the DMA_CH0_Status register is set. The Start Receive command is effective only when the reception is stopped. If the command is issued before setting the DMA_CH0_RxDesc_List_Address register, the DMA behavior is unpredictable. 0: Rx DMA operation stops after the transfer of the current packet. The next descriptor position in the Receive list is saved, and it becomes the current position after the Rx process is restarted. The Stop Receive command is effective only when the Rx process is in the Running (waiting for Rx packet) or Suspended state. Values: 0x0 (STOP): Stop Receive 0x1 (START): Start Receive Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1290	DMA_CH3_TxDesc_List_HAddress	The Channel3 Tx Descriptor List HAddress register has the higher 8 or 16 bits of the start address of the Transmit descriptor list.You can write to this register only when the Tx DMA has stopped, that is, the ST bit is set to zero in DMA_CH3_Tx_Control register. When stopped, this register can be written with a new descriptor list address. When you set the ST bit to 1, the DMA takes the newly-programmed descriptor base address. If this register is not changed when the ST bit is set to 3, the DMA takes the descriptor address where it was stopped earlier. This register must be programmed with desired the higher address before programming the DMA_CH3_TxDesc_List_Address register.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	TDESHA	R/W	0x0	Start of Transmit List This field contains the most-significant 8 or 16 bits of the 40- or 48-bit base address of the first descriptor in the Transmit descriptor list. Value After Reset: 0x0
0x1294	DMA_CH3_TxDesc_List_Address	The Channel3 Tx Descriptor List Address register points the DMA to the start of Transmit descriptor list. The descriptor lists reside in the physical memory space of the application and must be Lword (128-bit). The DMA internally converts it to bus width aligned address by making the corresponding LSB to low.You can write to this register only when the Tx DMA has stopped, that is, the ST bit is set to zero in DMA_CH3_Tx_Control register. When stopped, this register can be written with a new descriptor list address. When you set the ST bit to 1, the DMA takes the newly-programmed descriptor base address. If this register is not changed when the ST bit is set to 3, the DMA takes the descriptor address where it was stopped earlier.	31:4	TDESLA	R/W	0x0	Start of Transmit List This field contains the base address of the first descriptor in the Transmit descriptor list. The DMA ignores the LSB bits (3:0) and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0 Testable: untestable
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x1298	DMA_CH3_RxDesc_List_HAddress	The Channel3 Rx Descriptor List HAddress register has the higher 8 or 16 bits of the start address of the Receive descriptor list.Writing to this register is permitted only when reception is stopped. When stopped, this register must be written to before the receive Start command is given. You can write to this register only when Rx DMA has stopped, that is, SR bit is set to zero in the DMA_CH3_RxDesc_List_Address registerWhen you set the SR bit to 1, the DMA takes the newly programmed descriptor base address.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	RDESHA	R/W	0x0	Start of Receive List This field contains the most-significant 8 or 16 bits of the 40-bit or 48-bit base address of the first descriptor in the Rx Descriptor list. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x129c	DMA_CH3_RxDesc_List_Address	The Channel3 Rx Descriptor List Address register points the DMA to the start of Receive descriptor list.This register points to the start of the Receive Descriptor List. The descriptor lists reside in the physical memory space of the application and must be Lword (128-bit). The DMA internally converts it to bus width aligned address by making the corresponding LS bits low. Writing to this register is permitted only when reception is stopped. When stopped, this register must be written to before the receive Start command is given. You can write to this register only when Rx DMA has stopped, that is, SR bit is set to zero in DMA_CH3_Rx_Control register. When stopped, this register can be written with a new descriptor list address.When you set the SR bit to 1, the DMA takes the newly programmed descriptor base address.	31:4	RDESLA	R/W	0x0	Start of Receive List This field contains the base address of the first descriptor in the Rx Descriptor list. The DMA ignores the LSB bits (3:0) and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0 Testable: untestable
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x12a0	DMA_CH3_TxDesc_Tail_Pointer	The Channel3 Tx Descriptor Tail Pointer register points to an offset from the base and indicates the location of the last valid descriptor.	31:4	TDTP	R/W	0x0	Transmit Descriptor Tail Pointer This field contains the tail pointer for the Tx descriptor ring. The software writes the tail pointer to add more descriptors to the Tx channel. The hardware tries to transmit all packets referenced by the descriptors between the head and the tail pointer registers. The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x12a8	DMA_CH3_RxDesc_Tail_Pointer	The Channel3 Rx Descriptor Tail Pointer Points to an offset from the base and indicates the location of the last valid descriptor.	31:4	RDTP	R/W	0x0	Receive Descriptor Tail Pointer This field contains the tail pointer for the Rx descriptor ring. The software writes the tail pointer to add more descriptors to the Rx channel. The hardware tries to write all received packets to the descriptors referenced between the head and the tail pointer registers. The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x12ac	DMA_CH3_TxDesc_Ring_Length	The Tx Descriptor Ring Length register contains the length of the Transmit descriptor ring.	31:10	Reserved_31_10	R	0x0	Reserved. Value After Reset: 0x0
			9:0	TDRL	R/W	0x0	Transmit Descriptor Ring Length This field sets the maximum number of Tx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. It is recommended that a minimum ring descriptor length of 4. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x12b0	DMA_CH3_Rx_Control2	The Channel3 Receive Control register controls the Rx features such as Rx Descriptor Ring Length and Alternate Rx Buffer Size.	31:24	Reserved_31_24	R	0x0	Reserved. Value After Reset: 0x0
			23:18	ARBS	R/W	0x0	Alternate Receive Buffer Size Indicates size in bytes for Buffer 1 when ARBS is programmed to a non-zero value (when split header feature is not enabled). When split header feature is enabled, ARBS indicates the buffer size for header data. The maximum alternate buffer is limited to 1008 bytes. When ARBS=0, Rx Buffer1 and Rx Buffer2 sizes are based on RBSZ field of DMA_CH0_Rx_Control. Width of ARBS field is 6 bits. Value After Reset: 0x0
			17:10	Reserved_x_10	R	0x0	Reserved. Value After Reset: 0x0
			9:0	RDRL	R/W	0x0	Receive Descriptor Ring Length This register sets the maximum number of Rx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9. Value After Reset: 0x0
			31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15	NIE	R/W	0x0	Normal Interrupt Summary Enable When this bit is set, the normal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: Bit 0: Transmit Interrupt Bit 2: Transmit Buffer Unavailable Bit 6: Receive Interrupt Bit 11: Early Receive Interrupt When this bit is reset, the normal interrupt summary is disabled.Values: 0x0 (DISABLE): Normal Interrupt Summary is disabled 0x1 (ENABLE): Normal Interrupt Summary is enabled Value After Reset: 0x0
			14	AIE	R/W	0x0	Abnormal Interrupt Summary Enable When this bit is set, the abnormal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: Bit 1: Transmit Process Stopped Bit 7: Rx Buffer Unavailable Bit 8: Receive Process Stopped Bit 9: Receive Watchdog Timeout Bit 10: Early Transmit Interrupt Bit 12: Fatal Bus Error Bit 13: Context Descriptor Error When this bit is reset, the abnormal interrupt summary is disabled.Values: 0x0 (DISABLE): Abnormal Interrupt Summary is disabled 0x1 (ENABLE): Abnormal Interrupt Summary is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x12b4	DMA_CH3_Interrupt_Enable	The Channel3 Interrupt Enable register enables the interrupts reported by the Status register.	13	CDEE	R/W	0x0	Context Descriptor Error Enable When this bit is set along with the AIE bit, the Descriptor error interrupt is enabled. When this bit is reset, the Descriptor error interrupt is disabled.Values: 0x0 (DISABLE): Context Descriptor Error is disabled 0x1 (ENABLE): Context Descriptor Error is enabled Value After Reset: 0x0
			12	FBEE	R/W	0x0	Fatal Bus Error Enable When this bit is set along with the AIE bit, the Fatal Bus error interrupt is enabled. When this bit is reset, the Fatal Bus Error error interrupt is disabled.Values: 0x0 (DISABLE): Fatal Bus Error is disabled 0x1 (ENABLE): Fatal Bus Error is enabled Value After Reset: 0x0
			11	ERIE	R/W	0x0	Early Receive Interrupt Enable When this bit is set along with the NIE bit, the Early Receive interrupt is enabled. When this bit is reset, the Early Receive interrupt is disabled.Values: 0x0 (DISABLE): Early Receive Interrupt is disabled 0x1 (ENABLE): Early Receive Interrupt is enabled Value After Reset: 0x0
			10	ETIE	R/W	0x0	Early Transmit Interrupt Enable When this bit is set along with the AIE bit, the Early Transmit interrupt is enabled. When this bit is reset, the Early Transmit interrupt is disabled.Values: 0x0 (DISABLE): Early Transmit Interrupt is disabled 0x1 (ENABLE): Early Transmit Interrupt is enabled Value After Reset: 0x0
			9	RWTE	R/W	0x0	Receive Watchdog Timeout Enable When this bit is set along with the AIE bit, the Receive Watchdog Timeout interrupt is enabled. When this bit is reset, the Receive Watchdog Timeout interrupt is disabled.Values: 0x0 (DISABLE): Receive Watchdog Timeout is disabled 0x1 (ENABLE): Receive Watchdog Timeout is enabled Value After Reset: 0x0
			8	RSE	R/W	0x0	Receive Stopped Enable When this bit is set along with the AIE bit, the Receive Stopped Interrupt is enabled. When this bit is reset, the Receive Stopped interrupt is disabled.Values: 0x0 (DISABLE): Receive Stopped is disabled 0x1 (ENABLE): Receive Stopped is enabled Value After Reset: 0x0
			7	RBUE	R/W	0x0	Receive Buffer Unavailable Enable When this bit is set along with the AIE bit, the Receive Buffer Unavailable interrupt is enabled. When this bit is reset, the Receive Buffer Unavailable interrupt is disabled.Values: 0x0 (DISABLE): Receive Buffer Unavailable is disabled 0x1 (ENABLE): Receive Buffer Unavailable is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			6	RIE	R/W	0x0	Receive Interrupt Enable When this bit is set along with the NIE bit, the Receive Interrupt is enabled. When this bit is reset, the Receive Interrupt is disabled.Values: 0x0 (DISABLE): Receive Interrupt is disabled 0x1 (ENABLE): Receive Interrupt is enabled Value After Reset: 0x0
			5:3	Reserved_5_3	R	0x0	Reserved. Value After Reset: 0x0
			2	TBUE	R/W	0x0	Transmit Buffer Unavailable Enable When this bit is set along with the NIE bit, the Transmit Buffer Unavailable interrupt is enabled. When this bit is reset, the Transmit Buffer Unavailable interrupt is disabled.Values: 0x0 (DISABLE): Transmit Buffer Unavailable is disabled 0x1 (ENABLE): Transmit Buffer Unavailable is enabled Value After Reset: 0x0
			1	TXSE	R/W	0x0	Transmit Stopped Enable When this bit is set along with the AIE bit, the Transmission Stopped interrupt is enabled. When this bit is reset, the Transmission Stopped interrupt is disabled.Values: 0x0 (DISABLE): Transmit Stopped is disabled 0x1 (ENABLE): Transmit Stopped is enabled Value After Reset: 0x0
			0	TIE	R/W	0x0	Transmit Interrupt Enable When this bit is set along with the NIE bit, the Transmit Interrupt is enabled. When this bit is reset, the Transmit Interrupt is disabled.Values: 0x0 (DISABLE): Transmit Interrupt is disabled 0x1 (ENABLE): Transmit Interrupt is enabled Value After Reset: 0x0
0x12b8	DMA_CH3_Rx_Interrupt_Watchdog_Timer	The Receive Interrupt Watchdog Timer register indicates the watchdog timeout for Receive Interrupt (RI) from the DMA. When this register is written with a non-zero value, it enables the watchdog timer for the RI bit of the DMA_CH3_Status register.	31:18	Reserved_31_18	R	0x0	Reserved. Value After Reset: 0x0
			17:16	RWTU	R/W	0x0	Receive Interrupt Watchdog Timer Count Units This fields indicates the number of system clock cycles corresponding to one unit in RWT field. 2'b00: 256 2'b01: 512 2'b10: 1024 2'b11: 2048 For example, when RWT=2 and RWTU=1, the watchdog timer is set for 2*512=1024 system clock cycles. Value After Reset: 0x0 Testable: untestable
			15:8	Reserved_15_8	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			7:0	RWT	R/W	0x0	Receive Interrupt Watchdog Timer Count This field indicates the number of system clock cycles, multiplied by factor indicated in RWTU field, for which the watchdog timer is set. The watchdog timer is triggered with the programmed value after the Rx DMA completes the transfer of a packet for which the RI bit is not set in the DMA_CH0_Status register, because of the setting of Interrupt Enable bit in the corresponding descriptor RDES3[30]. When the watchdog timer runs out, the RI bit is set and the timer is stopped. The watchdog timer is reset when the RI bit is set high because of automatic setting of RI as per the Interrupt Enable bit RDES3[30] of any received packet. Value After Reset: 0x0 Testable: untestable
0x12bc	DMA_CH3_Slot_Function_Control_Status	The Slot Function Control and Status register contains the control bits for slot function and the status for Transmit path.	31:20	Reserved_31_20	R	0x0	Reserved. Value After Reset: 0x0
			19:16	RSN	R	0x0	Reference Slot Number This field gives the current value of the reference slot number in the DMA. It is used for slot comparison. Value After Reset: 0x0 Testable: untestable
			15:4	SIV	R/W	0x7c	Slot Interval Value This field controls the period of the slot interval in which the TxDMA fetches the scheduled packets. A value of 0 specifies the slot interval of 1 us while the maximum value 4095 specifies the slot interval of 4096us. The default/reset value is 0x07C which corresponds to slot interval of 125us Value After Reset: 0x7c
			3:2	Reserved_3_2	R	0x0	Reserved. Value After Reset: 0x0
			1	ASC	R/W	0x0	Advance Slot Check When set, this bit enables the DMA to fetch the data from the buffer when the slot number (SLOTNUM) programmed in the Tx descriptor is equal to the reference slot number given in the RSN field or ahead of the reference slot number by up to two slots This bit is applicable only when the ESC bit is set.Values: 0x0 (DISABLE): Advance Slot Check is disabled 0x1 (ENABLE): Advance Slot Check is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	ESC	R/W	0x0	Enable Slot Comparison When set, this bit enables the checking of the slot numbers programmed in the Tx descriptor with the current reference given in the RSN field. The DMA fetches the data from the corresponding buffer only when the slot number is equal to the reference slot number or ahead of the reference slot number by one slot When reset, this bit disables the checking of the slot numbers. The DMA fetches the data immediately after the descriptor is processed. Note: The UFO (UDP Fragmentation over IPv4)/TSO/USO must not be enabled along with TBS/AVB Slot number check. The UFO/TSO/USO involves multiple packets/segments/fragments transmission for single packet received from application and the slot number check are applicable for fetching of only first segment/fragment. As a result it might be difficult for software to specify slot number for subsequent packets.Values: 0x0 (DISABLE): Slot Comparison is disabled 0x1 (ENABLE): Slot Comparison is enabled Value After Reset: 0x0
0x12c4	DMA_CH3_Current_App_Tx Desc	The Channel3 Current Application Transmit Descriptor register points to the current Transmit descriptor read by the DMA.	31:0	CURTDESAPTR	R	0x0	Application Transmit Descriptor Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x12cc	DMA_CH3_Current_App_Rx Desc	The Channel3 Current Application Receive Descriptor register points to the current Receive descriptor read by the DMA.	31:0	CURRDESAPTR	R	0x0	Application Receive Descriptor Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x12d0	DMA_CH3_Current_App_Tx Buffer_H	The Channel3 Current Application Transmit Buffer Address High register has the higher 8 or 16 bits of the current address of the Transmit buffer address read by the DMA.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	CURTBUFAPTRH	R	0x0	Application Transmit Buffer Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x12d4	DMA_CH3_Current_App_Tx Buffer	The Channel3 Current Application Transmit Buffer Address register points to the current Tx buffer address read by the DMA.	31:0	CURTBUFAPTR	R	0x0	Application Transmit Buffer Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x12d8	DMA_CH3_Current_App_Rx Buffer_H	The Channel3 Current Application Receive Buffer Address High register has the higher 8 or 16 bits of the current address of the Receive buffer address read by the DMA.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	CURRBUFAPTRH	R	0x0	Application Receive Buffer Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x12dc	DMA_CH3_Current_App_Rx Buffer	The Channel 3 Current Application Receive Buffer Address register points to the current Rx buffer address read by the DMA.	31:0	CURRBUFAPTR	R	0x0	Application Receive Buffer Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			31:22	Reserved_31_22	R	0x0	Reserved. Value After Reset: 0x0
			21:19	REB	R	0x0	Rx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AXI interface. Bit 21 1'b1: Error during data transfer by Rx DMA 1'b0: No Error during data transfer by Rx DMA Bit 20 1'b1: Error during descriptor access 1'b0: Error during data buffer access Bit 19 1'b1: Error during read transfer 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt. Value After Reset: 0x0 Testable: untestable
			18:16	TEB	R	0x0	Tx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AXI interface. Bit 18 1'b1: Error during data transfer by Tx DMA 1'b0: No Error during data transfer by Tx DMA Bit 17 1'b1: Error during descriptor access 1'b0: Error during data buffer access Bit 16 1'b1: Error during read transfer 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt. Value After Reset: 0x0 Testable: untestable
			15	NIS	R/W	0x0	Normal Interrupt Summary Normal Interrupt Summary bit value is the logical OR of the following bits when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: Bit 0: Transmit Interrupt Bit 2: Transmit Buffer Unavailable Bit 6: Receive Interrupt Bit 11: Early Receive Interrupt Only unmasked bits (interrupts for which interrupt enable is set in DMA_CH0_Interrupt_Enable register) affect the Normal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit which causes NIS to be set is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Normal Interrupt Summary status not detected 0x1 (ACTIVE): Normal Interrupt Summary status detected Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x12e0	DMA_CH3_Status	The software driver (application) reads the Status register during interrupt service routine or polling to determine the status of the DMA.Note: The number of DMA_CH3_Status register in the configuration is the higher of number of Rx DMA Channels and Tx DMA Channels.	14	AIS	R/W	0x0	Abnormal Interrupt Summary Abnormal Interrupt Summary bit value is the logical OR of the following when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: Bit 1: Transmit Process Stopped Bit 7: Receive Buffer Unavailable Bit 8: Receive Process Stopped Bit 10: Early Transmit Interrupt Bit 12: Fatal Bus Error Bit 13: Context Descriptor Error Only unmasked bits affect the Abnormal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit, which causes AIS to be set, is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Abnormal Interrupt Summary status not detected 0x1 (ACTIVE): Abnormal Interrupt Summary status detected Value After Reset: 0x0 Testable: untestable
			13	CDE	R/W	0x0	Context Descriptor Error This bit indicates that the DMA Tx/Rx engine received a descriptor error, which indicates invalid context in the middle of packet flow (intermediate descriptor) or all one's descriptor in Tx case and on Rx side it indicates DMA has read a descriptor with either of the buffer address as ones which is considered to be invalid. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Context Descriptor Error status not detected 0x1 (ACTIVE): Context Descriptor Error status detected Value After Reset: 0x0 Testable: untestable
			12	FBE	R/W	0x0	Fatal Bus Error This bit indicates that a bus error occurred (as described in the EB field). When this bit is set, the corresponding DMA channel engine disables all bus accesses. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Fatal Bus Error status not detected 0x1 (ACTIVE): Fatal Bus Error status detected Value After Reset: 0x0 Testable: untestable
			11	ERI	R/W	0x0	Early Receive Interrupt This bit when set indicates that the RxDMA has completed the transfer of packet data to the memory. In configurations supporting ERIC, ERIC = 0: This bit is set only after the Rx DMA has completely filled up a receive buffer with packet data. ERIC = 1: This bit is set after every burst transfer of data from the Rx DMA to the buffer. The setting of RI bit automatically clears this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Early Receive Interrupt status not detected 0x1 (ACTIVE): Early Receive Interrupt status detected Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			10	ETI	R/W	0x0	Early Transmit Interrupt This bit when set indicates that the TxDMA has completed the transfer of packet data to the MTL TXFIFO memory. In configurations supporting ERIC: ETIC = 0: This bit is set only after the Tx DMA has transferred a complete packet to MTL. ETIC = 1: This bit is set after completion of (partial) packet data transfer from buffers in the Transmit descriptor in which IOC = 1. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Early Transmit Interrupt status not detected 0x1 (ACTIVE): Early Transmit Interrupt status detected Value After Reset: 0x0 Testable: untestable
			9	RWT	R/W	0x0	Receive Watchdog Timeout This bit is asserted when a packet with length greater than 2,048 bytes (10,240 bytes when Jumbo Packet mode is enabled) is received.Values: 0x0 (INACTIVE): Receive Watchdog Timeout status not detected 0x1 (ACTIVE): Receive Watchdog Timeout status detected Value After Reset: 0x0 Testable: untestable
			8	RPS	R/W	0x0	Receive Process Stopped This bit is asserted when the Rx process enters the Stopped state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Process Stopped status not detected 0x1 (ACTIVE): Receive Process Stopped status detected Value After Reset: 0x0 Testable: untestable
			7	RBU	R/W	0x0	Receive Buffer Unavailable This bit indicates that the application owns the next descriptor in the Receive list, and the DMA cannot acquire it. The Rx process is suspended. To resume processing Rx descriptors, the application must change the ownership of the descriptor and issue a Receive Poll Demand command. If this command is not issued, the Rx process resumes when the next recognized incoming packet is received. In ring mode, the application must advance the Receive Descriptor Tail Pointer register of a channel. This bit is set only when the DMA owns the previous Rx descriptor. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Buffer Unavailable status not detected 0x1 (ACTIVE): Receive Buffer Unavailable status detected Value After Reset: 0x0 Testable: untestable
			6	RI	R/W	0x0	Receive Interrupt This bit indicates that the packet reception is complete. When packet reception is complete, Bit 31 of RDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor. The reception remains in the Running state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Interrupt status not detected 0x1 (ACTIVE): Receive Interrupt status detected Value After Reset: 0x0 Testable: untestable
			5:3	Reserved_5_3	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			2	TBU	R/W	0x0	Transmit Buffer Unavailable This bit indicates that the application owns the next descriptor in the Transmit list, and the DMA cannot acquire it. Transmission is suspended. The TPS0 field of the DMA_Debug_Status0-DMA_Debug_Status2 register explains the Transmit Process state transitions. To resume processing the Transmit descriptors, the application must do the following: Change the ownership of the descriptor by setting Bit 31 of TDES3. Issue a Transmit Poll Demand command. For ring mode, the application must advance the Transmit Descriptor Tail Pointer register of a channel. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Buffer Unavailable status not detected 0x1 (ACTIVE): Transmit Buffer Unavailable status detected Value After Reset: 0x0 Testable: untestable
			1	TPS	R/W	0x0	Transmit Process Stopped This bit is set when the transmission is stopped. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Process Stopped status not detected 0x1 (ACTIVE): Transmit Process Stopped status detected Value After Reset: 0x0 Testable: untestable
			0	TI	R/W	0x0	Transmit Interrupt This bit indicates that the packet transmission is complete. When transmission is complete, Bit 31 of TDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Interrupt status not detected 0x1 (ACTIVE): Transmit Interrupt status detected Value After Reset: 0x0 Testable: untestable
0x12e4	DMA_CH3_Miss_Frame_Cnt	This register has the number of packet counter that got dropped by the DMA either due to Bus Error or due to programming RPF field in DMA_CH3_Rx_Control register.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15	MFCO	R	0x0	Overflow status of the MFC Counter When this bit is set then the MFC counter does not get incremented further. The bit gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Miss Frame Counter overflow not occurred 0x1 (ACTIVE): Miss Frame Counter overflow occurred Value After Reset: 0x0
			14:11	Reserved_14_11	R	0x0	Reserved. Value After Reset: 0x0
			10:0	MFC	R	0x0	Dropped Packet Counters This counter indicates the number of packet counters that are dropped by the DMA either because of bus error or because of programing RPF field in DMA_CH0_Rx_Control register. The counter gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x12e8	DMA_CH3_RXP_Accept_Cnt	The DMA_CH3_RXP_Accept_Cnt registers provides the count of the number of frames accepted by Rx Parser.	31	RXPACOF	R	0x0	Rx Parser Accept Counter Overflow Bit When set, this bit indicates that the RXPAC Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Rx Parser Accept Counter overflow not occurred 0x1 (ACTIVE): Rx Parser Accept Counter overflow occurred Value After Reset: 0x0
			30:0	RXPAC	R	0x0	Rx Parser Accept Counter This 31-bit counter is implemented when a Rx Parser Accept a packet due to AF =1. The counter is cleared when the register is read. Value After Reset: 0x0
0x1300	DMA_CH4_Control	The DMA Channel4 Control register specifies the MSS value for segmentation, length to skip between two descriptors, and also the features such as header splitting and 8xPBL mode.	31:25	Reserved_31_25	R	0x0	Reserved. Value After Reset: 0x0
			24	SPH	R/W	0x0	Split Headers 1: DMA splits the header and payload in the receive path. The DMA writes the header to the Buffer Address1 of RDES0. The DMA writes the payload to the buffer to which the Buffer Address2 is pointing. The software must ensure that the header fits into the Receive buffers. If the header length exceeds the receive buffer size, the DMA does not split the header and payload. This bit is available only when you select Enable Split Header Structure feature.Values: 0x0 (DISABLE): Split Headers feature is disabled 0x1 (ENABLE): Split Headers feature is enabled Value After Reset: 0x0
			23:21	Reserved_23_21	R	0x0	Reserved. Value After Reset: 0x0
			20:18	DSL	R/W	0x0	Descriptor Skip Length This bit specifies the Lword (128-bit) number of locations to skip between two descriptors. The address skipping starts from the end of the current descriptor to the start of the next descriptor. When the DSL value is equal to zero, the DMA takes the descriptor table as contiguous. Value After Reset: 0x0
			17	Reserved_17	R	0x0	Reserved. Value After Reset: 0x0
			16	PBLx8	R/W	0x0	8xPBL mode 1: Indicates that the PBL value programmed in Bits[21:16] in DMA_CH0_Tx_Control and Bits[21:16] in DMA_CH0_Rx_Control is multiplied by eight times. Therefore, the DMA transfers the data in 8, 16, 32, 64, 128, and 256 beats depending on the PBL value. Values: 0x0 (DISABLE): 8xPBL mode is disabled 0x1 (ENABLE): 8xPBL mode is enabled Value After Reset: 0x0
			15:14	Reserved_15_14	R	0x0	Reserved. Value After Reset: 0x0
			13:0	MSS	R/W	0x0	Maximum Segment Size This field specifies the maximum segment size that must be used while segmenting the packet. This field is valid only if the TSE bit of DMA_CH0_Tx_Control register is set. Program a value in this field that is more than the configured datawidth in bytes. Use MSS value of 64 bytes or more. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			31	Reserved_31	R	0x0	Reserved. Value After Reset: 0x0
			30:29	TFSEL	R/W	0x0	TBS Fetch Select Select bits for one of the four DMA_TBS_CTRL register fields (FTOS,FGSN,FTOV) for the channel Values: 2'b00: DMA_TBS_CTRL0 2'b01: DMA_TBS_CTRL1 2'b10: DMA_TBS_CTRL2 2'b11: DMA_TBS_CTRL3 (Reserved if DWC_EQOS_NUM_DMA_TX_CH=3) Value After Reset: 0x0
			28	EDSE	R/W	0x0	Enhanced Descriptor Enable 1: The corresponding channel uses Enhanced Descriptors that are 32 Bytes for both Normal and Context Descriptors. 0: The corresponding channel uses the descriptors that are 16 Bytes. Values: 0x0 (DISABLE): Enhanced Descriptor is disabled 0x1 (ENABLE): Enhanced Descriptor is enabled Value After Reset: 0x0
			27:24	TQOS	R/W	0x0	Transmit QOS. This field is used to drive arqos_m_o[3:0] or awqos_m_o[3:0] output signals for all transactions of DMA Tx Channel0. Value After Reset: 0x0
			23	Reserved_23	R	0x0	Reserved. Value After Reset: 0x0
			22			#VALUE!	Reserved Field: Yes
			21:16	TxPBL	R/W	0x0	Transmit Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in DMA_CH0_Control register. 2. Set the TxPBL. Note: The maximum value of TxPBL must be less than or equal to half the Tx Queue size (TQS field of MTL_TxQ0_Operation_Mode register) in terms of beats. This is required so that the Tx Queue has space to store at least another Tx PBL worth of data while the MTL Tx Queue Controller is transferring data to MAC. For example, in 32-bit data width configurations the total locations in Tx Queue of size 512 bytes is 128, TxPBL and 8xPBL needs to be programmed to less than or equal to 64. Value After Reset: 0x0
			15	IPBL	R/W	0x0	Ignore PBL Requirement 1: DMA does not check for PBL number of locations in the MTL before initiating a transfer. If space is not available, the MTL might use handshaking to slow the DMA. Note: Do not use this mode when you enable multiple Transmit DMA channels. Because, it might block other Transmit and Receive DMA Channels from accessing the Read Data Channel of the AXI bus until space is available in Transmit Queue for current transfer.Values: 0x0 (DISABLE): Ignore PBL Requirement is disabled 0x1 (ENABLE): Ignore PBL Requirement is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1304	DMA_CH4_Tx_Control	The DMA Channel4 Transmit Control register controls the Tx features such as PBL, TCP segmentation, and Tx Channel weights.	14:13	TSE_MODE	R/W	0x0	TSE Mode 00: TSO/USO (segmentation functionality is enabled). In this mode, the setting of TSE bit enables the TSO/USO segmentation. 01: UFO with Checksum (UDP Fragmentation over IPv4 with Checksum). In this mode, the setting of TSE bit enables the UDP fragmentation functionality with Checksum for all the UDP packets. 10: UFO without Checksum (UDP Fragmentation over IPv4 with Checksum). In this mode, the setting of TSE bit enables the UDP fragmentation functionality without Checksum for all the UDP packets. 11: Reserved Values: 0x0 (TSO_USO): TSO/USO 0x1 (UFOWC): UFO with Checksum 0x2 (UFOWOC): UFO without Checksum 0x3 (RSVD): Reserved Value After Reset: 0x0
			12	TSE	R/W	0x0	TCP Segmentation Enabled 1: DMA performs the TCP segmentation or UDP Segmentation/Fragmentation for packets in this channel. The TCP segmentation or UDP packet's segmentation/Fragmentation is performed only for those packets for which the TSE bit (TDES0[19]) = 1 in the Tx Normal descriptor. When this bit is 1, the TxPBL value must be greater than 4. Values: 0x0 (DISABLE): TCP Segmentation is disabled 0x1 (ENABLE): TCP Segmentation is enabled Value After Reset: 0x0
			11:5	Reserved_11_5	R	0x0	Reserved. Value After Reset: 0x0
			4	OSF	R/W	0x0	Operate on Second Packet 1: Instructs the DMA to process the second packet of the Tx data even before the status for the first packet is obtained. Values: 0x0 (DISABLE): Operate on Second Packet disabled 0x1 (ENABLE): Operate on Second Packet enabled Value After Reset: 0x0
			3:1	TCW	R/W	0x0	Transmit Channel Weight This field indicates the weight assigned to the corresponding Tx channel. When reset is complete, this field is set to 0 for all channels by default, resulting in equal weights to all channels. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	ST	R/W	0x0	Start or Stop Transmission Command 1: Transmission is placed in the running state. The DMA checks the Tx list at the current position for a packet to be transmitted. The DMA tries to acquire descriptor from either of the following positions: The current position in the list This is the base address of the Transmit list set by the DMA_CH0_TxDesc_List_Address register. The position at which the transmission was previously stopped If the DMA does not own the current descriptor, the transmission enters the Suspended state and the TBU bit of the DMA_CH0_Status register is set. The Start Transmission command is effective only when the transmission is stopped. If the command is issued before setting the DMA_CH0_TxDesc_List_Address register, the DMA behavior is unpredictable. When this bit is reset, the transmission process is placed in the Stopped state after completing the transmission of the current packet. The Next Descriptor position in the Transmit list is saved, and it becomes the current position when the transmission is restarted. To change the list address, you need to program DMA_CH0_TxDesc_List_Address register with a new value when this bit is reset. The new value is considered when this bit is set again. The stop transmission command is effective only when the transmission of the current packet is complete or the transmission is in the Suspended state.Values: 0x0 (STOP): Stop Transmission Command 0x1 (START): Start Transmission Command Value After Reset: 0x0 Testable: untestable
			31	RPF	R/W	0x0	Rx Packet Flush. 1: DWC_ether_qos automatically flushes the packet from the Rx Queues destined to this DMA Rx Channel, when it is stopped. When this bit remains set and the DMA is re-started by the software driver, the packets residing in the Rx Queues that were received when this RxDMA was stopped, still get flushed out. The packets received by the MAC only after the DMA is re-started is routed to the Rx DMA.The flushing is done on the Read side of the Rx Queue. When the packet is duplicated to multiple Rx DMA, it is flushed by the Rx DMA to which it is routed only when it is stopped to prevent head-of-line blocking. The packet duplicated to multiple Rx DMA is not flushed when the Rx DMA is re-started, irrespective of whether the packet was received during one or more Rx DMAs were stopped. 0: DWC_ether_qos does not flush the packet in the Rx Queue destined to this RxDMA Channel when it is in STOP state. This might cause head-of-line blocking in the corresponding RxQueue. Note: The stopping of packet flow from a Rx DMA Channel to the application by setting RPF works only when there is one-to-one mapping of Rx Queue to Rx DMA channels. In Dynamic mapping mode, setting RPF bit in any DMA_CH0_Rx_Control register might flush packets from unintended Rx Queues which are destined to the stopped Rx DMA Channel.Values: 0x0 (DISABLE): Rx Packet Flush is disabled 0x1 (ENABLE): Rx Packet Flush is enabled Value After Reset: 0x0
			30:28	Reserved_30_28	R	0x0	Reserved. Value After Reset: 0x0
			27:24	RQOS	R/W	0x0	Rx AXI4 QOS. This field is used to drive arqos_m_o[3:0] or awqos_m_o[3:0] output signals for all transactions of DMA Rx Channel0. Value After Reset: 0x0
			23	Reserved_23	R	0x0	Reserved. Value After Reset: 0x0
			22			#VALUE!	Reserved Field: Yes

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1308	DMA_CH4_Rx_Control	The DMA Channel4 Receive Control register controls the Rx features such as PBL, buffer size, and extended status.	21:16	RxPBL	R/W	0x0	Receive Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in the DMA_CH0_Control register. 2. Set the RxPBL. Note: The maximum value of RxPBL must be less than or equal to half the Rx Queue size (RQS field of MTL_RxQ0_Operation_Mode register) in terms of beats. This is required so that the Rx Queue has space to store at least another Rx PBL worth of data while the Rx DMA is transferring a block of data. For example, in 32-bit data width configurations the total locations in Rx Queue of size 512 bytes is 128, so RxPBL and 8xPBL needs to be programmed to less than or equal to 64. Value After Reset: 0x0
			15	Reserved_15	R	0x0	Reserved. Value After Reset: 0x0
			14:5	RBSZ_13_y	R/W	0x0	Receive Buffer size High RBSZ[13:0] is split into two fields higher RBSZ_13_y and lower RBSZ_x_0. The RBSZ[13:0] field indicates the size of the Rx buffers specified in bytes. The maximum buffer size is limited to 16K bytes. The buffer size is applicable to payload buffers when split headers are enabled. Note: The buffer size must be a multiple of 16. This is required even if the value of buffer address pointer is not aligned to data bus width. Hence the lower RBSZ_x_0 bits are read-only and the value is considered as all-zero. Thus the RBSZ_13_y indicates the buffer size in terms of locations. Value After Reset: 0x0
			4:1	RBSZ_x_0	R	0x0	Receive Buffer size Low RBSZ[13:0] is split into two fields RBSZ_13_y and RBSZ_x_0. The RBSZ_x_0 is the lower field. This field is of width 4 bits. This field is read-only (RO). Value After Reset: 0x0
			0	SR	R/W	0x0	Start or Stop Receive 1: DMA attempts to acquire the descriptor from the Receive list and processes the incoming packets. The DMA tries to acquire descriptor from either of the following positions: -- The current position in the list This is the address set by the DMA_CH0_RxDesc_List_Address register. -- The position at which the Rx process was previously stopped If the DMA does not own the current descriptor, the reception is suspended and the RBU bit of the DMA_CH0_Status register is set. The Start Receive command is effective only when the reception is stopped. If the command is issued before setting the DMA_CH0_RxDesc_List_Address register, the DMA behavior is unpredictable. 0: Rx DMA operation stops after the transfer of the current packet. The next descriptor position in the Receive list is saved, and it becomes the current position after the Rx process is restarted. The Stop Receive command is effective only when the Rx process is in the Running (waiting for Rx packet) or Suspended state. Values: 0x0 (STOP): Stop Receive 0x1 (START): Start Receive Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1310	DMA_CH4_TxDesc_List_HAddress	The Channel4 Tx Descriptor List HAddress register has the higher 8 or 16 bits of the start address of the Transmit descriptor list.You can write to this register only when the Tx DMA has stopped, that is, the ST bit is set to zero in DMA_CH4_Tx_Control register. When stopped, this register can be written with a new descriptor list address. When you set the ST bit to 1, the DMA takes the newly-programmed descriptor base address. If this register is not changed when the ST bit is set to 4, the DMA takes the descriptor address where it was stopped earlier. This register must be programmed with desired the higher address before programming the DMA_CH4_TxDesc_List_Address register.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	TDESHA	R/W	0x0	Start of Transmit List This field contains the most-significant 8 or 16 bits of the 40- or 48-bit base address of the first descriptor in the Transmit descriptor list. Value After Reset: 0x0
0x1314	DMA_CH4_TxDesc_List_Address	The Channel4 Tx Descriptor List Address register points the DMA to the start of Transmit descriptor list. The descriptor lists reside in the physical memory space of the application and must be Lword (128-bit). The DMA internally converts it to bus width aligned address by making the corresponding LSB to low.You can write to this register only when the Tx DMA has stopped, that is, the ST bit is set to zero in DMA_CH4_Tx_Control register. When stopped, this register can be written with a new descriptor list address. When you set the ST bit to 1, the DMA takes the newly-programmed descriptor base address. If this register is not changed when the ST bit is set to 4, the DMA takes the descriptor address where it was stopped earlier.	31:4	TDESLA	R/W	0x0	Start of Transmit List This field contains the base address of the first descriptor in the Transmit descriptor list. The DMA ignores the LSB bits (3:0) and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0 Testable: untestable
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x1318	DMA_CH4_RxDesc_List_HAddress	The Channel4 Rx Descriptor List HAddress register has the higher 8 or 16 bits of the start address of the Receive descriptor list.Writing to this register is permitted only when reception is stopped. When stopped, this register must be written to before the receive Start command is given. You can write to this register only when Rx DMA has stopped, that is, SR bit is set to zero in the DMA_CH4_RxDesc_List_Address registerWhen you set the SR bit to 1, the DMA takes the newly programmed descriptor base address.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	RDESHA	R/W	0x0	Start of Receive List This field contains the most-significant 8 or 16 bits of the 40-bit or 48-bit base address of the first descriptor in the Rx Descriptor list. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x131c	DMA_CH4_RxDesc_List_Address	The Channel4 Rx Descriptor List Address register points the DMA to the start of Receive descriptor list.This register points to the start of the Receive Descriptor List. The descriptor lists reside in the physical memory space of the application and must be Lword (128-bit). The DMA internally converts it to bus width aligned address by making the corresponding LS bits low. Writing to this register is permitted only when reception is stopped. When stopped, this register must be written to before the receive Start command is given. You can write to this register only when Rx DMA has stopped, that is, SR bit is set to zero in DMA_CH4_Rx_Control register. When stopped, this register can be written with a new descriptor list address.When you set the SR bit to 1, the DMA takes the newly programmed descriptor base address.	31:4	RDESLA	R/W	0x0	Start of Receive List This field contains the base address of the first descriptor in the Rx Descriptor list. The DMA ignores the LSB bits (3:0) and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0 Testable: untestable
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x1320	DMA_CH4_TxDesc_Tail_Pointer	The Channel4 Tx Descriptor Tail Pointer register points to an offset from the base and indicates the location of the last valid descriptor.	31:4	TDTP	R/W	0x0	Transmit Descriptor Tail Pointer This field contains the tail pointer for the Tx descriptor ring. The software writes the tail pointer to add more descriptors to the Tx channel. The hardware tries to transmit all packets referenced by the descriptors between the head and the tail pointer registers. The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x1328	DMA_CH4_RxDesc_Tail_Pointer	The Channel4 Rx Descriptor Tail Pointer Points to an offset from the base and indicates the location of the last valid descriptor.	31:4	RDTP	R/W	0x0	Receive Descriptor Tail Pointer This field contains the tail pointer for the Rx descriptor ring. The software writes the tail pointer to add more descriptors to the Rx channel. The hardware tries to write all received packets to the descriptors referenced between the head and the tail pointer registers. The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x132c	DMA_CH4_TxDesc_Ring_Length	The Tx Descriptor Ring Length register contains the length of the Transmit descriptor ring.	31:10	Reserved_31_10	R	0x0	Reserved. Value After Reset: 0x0
			9:0	TDRL	R/W	0x0	Transmit Descriptor Ring Length This field sets the maximum number of Tx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. It is recommended that a minimum ring descriptor length of 4. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1330	DMA_CH4_Rx_Control2	The Channel4 Receive Control register controls the Rx features such as Rx Descriptor Ring Length and Alternate Rx Buffer Size.	31:24	Reserved_31_24	R	0x0	Reserved. Value After Reset: 0x0
			23:18	ARBS	R/W	0x0	Alternate Receive Buffer Size Indicates size in bytes for Buffer 1 when ARBS is programmed to a non-zero value (when split header feature is not enabled). When split header feature is enabled, ARBS indicates the buffer size for header data. The maximum alternate buffer is limited to 1008 bytes. When ARBS=0, Rx Buffer1 and Rx Buffer2 sizes are based on RBSZ field of DMA_CH0_Rx_Control. Width of ARBS field is 6 bits. Value After Reset: 0x0
			17:10	Reserved_x_10	R	0x0	Reserved. Value After Reset: 0x0
			9:0	RDRL	R/W	0x0	Receive Descriptor Ring Length This register sets the maximum number of Rx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9. Value After Reset: 0x0
			31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15	NIE	R/W	0x0	Normal Interrupt Summary Enable When this bit is set, the normal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: Bit 0: Transmit Interrupt Bit 2: Transmit Buffer Unavailable Bit 6: Receive Interrupt Bit 11: Early Receive Interrupt When this bit is reset, the normal interrupt summary is disabled.Values: 0x0 (DISABLE): Normal Interrupt Summary is disabled 0x1 (ENABLE): Normal Interrupt Summary is enabled Value After Reset: 0x0
			14	AIE	R/W	0x0	Abnormal Interrupt Summary Enable When this bit is set, the abnormal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: Bit 1: Transmit Process Stopped Bit 7: Rx Buffer Unavailable Bit 8: Receive Process Stopped Bit 9: Receive Watchdog Timeout Bit 10: Early Transmit Interrupt Bit 12: Fatal Bus Error Bit 13: Context Descriptor Error When this bit is reset, the abnormal interrupt summary is disabled.Values: 0x0 (DISABLE): Abnormal Interrupt Summary is disabled 0x1 (ENABLE): Abnormal Interrupt Summary is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1334	DMA_CH4_Interrupt_Enable	The Channel4 Interrupt Enable register enables the interrupts reported by the Status register.	13	CDEE	R/W	0x0	Context Descriptor Error Enable When this bit is set along with the AIE bit, the Descriptor error interrupt is enabled. When this bit is reset, the Descriptor error interrupt is disabled.Values: 0x0 (DISABLE): Context Descriptor Error is disabled 0x1 (ENABLE): Context Descriptor Error is enabled Value After Reset: 0x0
			12	FBEE	R/W	0x0	Fatal Bus Error Enable When this bit is set along with the AIE bit, the Fatal Bus error interrupt is enabled. When this bit is reset, the Fatal Bus Error error interrupt is disabled.Values: 0x0 (DISABLE): Fatal Bus Error is disabled 0x1 (ENABLE): Fatal Bus Error is enabled Value After Reset: 0x0
			11	ERIE	R/W	0x0	Early Receive Interrupt Enable When this bit is set along with the NIE bit, the Early Receive interrupt is enabled. When this bit is reset, the Early Receive interrupt is disabled.Values: 0x0 (DISABLE): Early Receive Interrupt is disabled 0x1 (ENABLE): Early Receive Interrupt is enabled Value After Reset: 0x0
			10	ETIE	R/W	0x0	Early Transmit Interrupt Enable When this bit is set along with the AIE bit, the Early Transmit interrupt is enabled. When this bit is reset, the Early Transmit interrupt is disabled.Values: 0x0 (DISABLE): Early Transmit Interrupt is disabled 0x1 (ENABLE): Early Transmit Interrupt is enabled Value After Reset: 0x0
			9	RWTE	R/W	0x0	Receive Watchdog Timeout Enable When this bit is set along with the AIE bit, the Receive Watchdog Timeout interrupt is enabled. When this bit is reset, the Receive Watchdog Timeout interrupt is disabled.Values: 0x0 (DISABLE): Receive Watchdog Timeout is disabled 0x1 (ENABLE): Receive Watchdog Timeout is enabled Value After Reset: 0x0
			8	RSE	R/W	0x0	Receive Stopped Enable When this bit is set along with the AIE bit, the Receive Stopped Interrupt is enabled. When this bit is reset, the Receive Stopped interrupt is disabled.Values: 0x0 (DISABLE): Receive Stopped is disabled 0x1 (ENABLE): Receive Stopped is enabled Value After Reset: 0x0
			7	RBUE	R/W	0x0	Receive Buffer Unavailable Enable When this bit is set along with the AIE bit, the Receive Buffer Unavailable interrupt is enabled. When this bit is reset, the Receive Buffer Unavailable interrupt is disabled.Values: 0x0 (DISABLE): Receive Buffer Unavailable is disabled 0x1 (ENABLE): Receive Buffer Unavailable is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			6	RIE	R/W	0x0	Receive Interrupt Enable When this bit is set along with the NIE bit, the Receive Interrupt is enabled. When this bit is reset, the Receive Interrupt is disabled.Values: 0x0 (DISABLE): Receive Interrupt is disabled 0x1 (ENABLE): Receive Interrupt is enabled Value After Reset: 0x0
			5:3	Reserved_5_3	R	0x0	Reserved. Value After Reset: 0x0
			2	TBUE	R/W	0x0	Transmit Buffer Unavailable Enable When this bit is set along with the NIE bit, the Transmit Buffer Unavailable interrupt is enabled. When this bit is reset, the Transmit Buffer Unavailable interrupt is disabled.Values: 0x0 (DISABLE): Transmit Buffer Unavailable is disabled 0x1 (ENABLE): Transmit Buffer Unavailable is enabled Value After Reset: 0x0
			1	TXSE	R/W	0x0	Transmit Stopped Enable When this bit is set along with the AIE bit, the Transmission Stopped interrupt is enabled. When this bit is reset, the Transmission Stopped interrupt is disabled.Values: 0x0 (DISABLE): Transmit Stopped is disabled 0x1 (ENABLE): Transmit Stopped is enabled Value After Reset: 0x0
			0	TIE	R/W	0x0	Transmit Interrupt Enable When this bit is set along with the NIE bit, the Transmit Interrupt is enabled. When this bit is reset, the Transmit Interrupt is disabled.Values: 0x0 (DISABLE): Transmit Interrupt is disabled 0x1 (ENABLE): Transmit Interrupt is enabled Value After Reset: 0x0
0x1338	DMA_CH4_Rx_Interrupt_Watchdog_Timer	The Receive Interrupt Watchdog Timer register indicates the watchdog timeout for Receive Interrupt (RI) from the DMA. When this register is written with a non-zero value, it enables the watchdog timer for the RI bit of the DMA_CH4_Status register.	31:18	Reserved_31_18	R	0x0	Reserved. Value After Reset: 0x0
			17:16	RWTU	R/W	0x0	Receive Interrupt Watchdog Timer Count Units This fields indicates the number of system clock cycles corresponding to one unit in RWT field. 2'b00: 256 2'b01: 512 2'b10: 1024 2'b11: 2048 For example, when RWT=2 and RWTU=1, the watchdog timer is set for 2*512=1024 system clock cycles. Value After Reset: 0x0 Testable: untestable
			15:8	Reserved_15_8	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			7:0	RWT	R/W	0x0	Receive Interrupt Watchdog Timer Count This field indicates the number of system clock cycles, multiplied by factor indicated in RWTU field, for which the watchdog timer is set. The watchdog timer is triggered with the programmed value after the Rx DMA completes the transfer of a packet for which the RI bit is not set in the DMA_CH0_Status register, because of the setting of Interrupt Enable bit in the corresponding descriptor RDES3[30]. When the watchdog timer runs out, the RI bit is set and the timer is stopped. The watchdog timer is reset when the RI bit is set high because of automatic setting of RI as per the Interrupt Enable bit RDES3[30] of any received packet. Value After Reset: 0x0 Testable: untestable
0x133c	DMA_CH4_Slot_Function_Control_Status	The Slot Function Control and Status register contains the control bits for slot function and the status for Transmit path.	31:20	Reserved_31_20	R	0x0	Reserved. Value After Reset: 0x0
			19:16	RSN	R	0x0	Reference Slot Number This field gives the current value of the reference slot number in the DMA. It is used for slot comparison. Value After Reset: 0x0 Testable: untestable
			15:4	SIV	R/W	0x7c	Slot Interval Value This field controls the period of the slot interval in which the TxDMA fetches the scheduled packets. A value of 0 specifies the slot interval of 1 us while the maximum value 4095 specifies the slot interval of 4096us. The default/reset value is 0x07C which corresponds to slot interval of 125us Value After Reset: 0x7c
			3:2	Reserved_3_2	R	0x0	Reserved. Value After Reset: 0x0
			1	ASC	R/W	0x0	Advance Slot Check When set, this bit enables the DMA to fetch the data from the buffer when the slot number (SLOTNUM) programmed in the Tx descriptor is equal to the reference slot number given in the RSN field or ahead of the reference slot number by up to two slots This bit is applicable only when the ESC bit is set.Values: 0x0 (DISABLE): Advance Slot Check is disabled 0x1 (ENABLE): Advance Slot Check is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	ESC	R/W	0x0	Enable Slot Comparison When set, this bit enables the checking of the slot numbers programmed in the Tx descriptor with the current reference given in the RSN field. The DMA fetches the data from the corresponding buffer only when the slot number is equal to the reference slot number or ahead of the reference slot number by one slot When reset, this bit disables the checking of the slot numbers. The DMA fetches the data immediately after the descriptor is processed. Note: The UFO (UDP Fragmentation over IPv4)/TSO/USO must not be enabled along with TBS/AVB Slot number check. The UFO/TSO/USO involves multiple packets/segments/fragments transmission for single packet received from application and the slot number check are applicable for fetching of only first segment/fragment. As a result it might be difficult for software to specify slot number for subsequent packets.Values: 0x0 (DISABLE): Slot Comparison is disabled 0x1 (ENABLE): Slot Comparison is enabled Value After Reset: 0x0
0x1344	DMA_CH4_Current_App_Tx Desc	The Channel4 Current Application Transmit Descriptor register points to the current Transmit descriptor read by the DMA.	31:0	CURTDESAPTR	R	0x0	Application Transmit Descriptor Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x134c	DMA_CH4_Current_App_Rx Desc	The Channel4 Current Application Receive Descriptor register points to the current Receive descriptor read by the DMA.	31:0	CURRDESAPTR	R	0x0	Application Receive Descriptor Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x1350	DMA_CH4_Current_App_Tx Buffer_H	The Channel4 Current Application Transmit Buffer Address High register has the higher 8 or 16 bits of the current address of the Transmit buffer address read by the DMA.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	CURTBUFAPTRH	R	0x0	Application Transmit Buffer Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x1354	DMA_CH4_Current_App_Tx Buffer	The Channel4 Current Application Transmit Buffer Address register points to the current Tx buffer address read by the DMA.	31:0	CURTBUFAPTR	R	0x0	Application Transmit Buffer Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x1358	DMA_CH4_Current_App_Rx Buffer_H	The Channel4 Current Application Receive Buffer Address High register has the higher 8 or 16 bits of the current address of the Receive buffer address read by the DMA.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	CURRBUFAPTRH	R	0x0	Application Receive Buffer Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x135c	DMA_CH4_Current_App_Rx Buffer	The Channel 4 Current Application Receive Buffer Address register points to the current Rx buffer address read by the DMA.	31:0	CURRBUFAPTR	R	0x0	Application Receive Buffer Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset. Value After Reset: 0x0
			31:22	Reserved_31_22	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			21:19	REB	R	0x0	Rx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AXI interface. Bit 21 1'b1: Error during data transfer by Rx DMA 1'b0: No Error during data transfer by Rx DMA Bit 20 1'b1: Error during descriptor access 1'b0: Error during data buffer access Bit 19 1'b1: Error during read transfer 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt. Value After Reset: 0x0 Testable: untestable
			18:16	TEB	R	0x0	Tx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AXI interface. Bit 18 1'b1: Error during data transfer by Tx DMA 1'b0: No Error during data transfer by Tx DMA Bit 17 1'b1: Error during descriptor access 1'b0: Error during data buffer access Bit 16 1'b1: Error during read transfer 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt. Value After Reset: 0x0 Testable: untestable
			15	NIS	R/W	0x0	Normal Interrupt Summary Normal Interrupt Summary bit value is the logical OR of the following bits when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: Bit 0: Transmit Interrupt Bit 2: Transmit Buffer Unavailable Bit 6: Receive Interrupt Bit 11: Early Receive Interrupt Only unmasked bits (interrupts for which interrupt enable is set in DMA_CH0_Interrupt_Enable register) affect the Normal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit which causes NIS to be set is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Normal Interrupt Summary status not detected 0x1 (ACTIVE): Normal Interrupt Summary status detected Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1360	DMA_CH4_Status	The software driver (application) reads the Status register during interrupt service routine or polling to determine the status of the DMA.Note: The number of DMA_CH4_Status register in the configuration is the higher of number of Rx DMA Channels and Tx DMA Channels.	14	AIS	R/W	0x0	Abnormal Interrupt Summary Abnormal Interrupt Summary bit value is the logical OR of the following when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: Bit 1: Transmit Process Stopped Bit 7: Receive Buffer Unavailable Bit 8: Receive Process Stopped Bit 10: Early Transmit Interrupt Bit 12: Fatal Bus Error Bit 13: Context Descriptor Error Only unmasked bits affect the Abnormal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit, which causes AIS to be set, is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Abnormal Interrupt Summary status not detected 0x1 (ACTIVE): Abnormal Interrupt Summary status detected Value After Reset: 0x0 Testable: untestable
			13	CDE	R/W	0x0	Context Descriptor Error This bit indicates that the DMA Tx/Rx engine received a descriptor error, which indicates invalid context in the middle of packet flow (intermediate descriptor) or all one's descriptor in Tx case and on Rx side it indicates DMA has read a descriptor with either of the buffer address as ones which is considered to be invalid. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Context Descriptor Error status not detected 0x1 (ACTIVE): Context Descriptor Error status detected Value After Reset: 0x0 Testable: untestable
			12	FBE	R/W	0x0	Fatal Bus Error This bit indicates that a bus error occurred (as described in the EB field). When this bit is set, the corresponding DMA channel engine disables all bus accesses. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Fatal Bus Error status not detected 0x1 (ACTIVE): Fatal Bus Error status detected Value After Reset: 0x0 Testable: untestable
			11	ERI	R/W	0x0	Early Receive Interrupt This bit when set indicates that the RxDMA has completed the transfer of packet data to the memory. In configurations supporting ERIC, ERIC = 0: This bit is set only after the Rx DMA has completely filled up a receive buffer with packet data. ERIC = 1: This bit is set after every burst transfer of data from the Rx DMA to the buffer. The setting of RI bit automatically clears this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Early Receive Interrupt status not detected 0x1 (ACTIVE): Early Receive Interrupt status detected Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			10	ETI	R/W	0x0	Early Transmit Interrupt This bit when set indicates that the TxDMA has completed the transfer of packet data to the MTL TXFIFO memory. In configurations supporting ERIC: ETIC = 0: This bit is set only after the Tx DMA has transferred a complete packet to MTL. ETIC = 1: This bit is set after completion of (partial) packet data transfer from buffers in the Transmit descriptor in which IOC = 1. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Early Transmit Interrupt status not detected 0x1 (ACTIVE): Early Transmit Interrupt status detected Value After Reset: 0x0 Testable: untestable
			9	RWT	R/W	0x0	Receive Watchdog Timeout This bit is asserted when a packet with length greater than 2,048 bytes (10,240 bytes when Jumbo Packet mode is enabled) is received.Values: 0x0 (INACTIVE): Receive Watchdog Timeout status not detected 0x1 (ACTIVE): Receive Watchdog Timeout status detected Value After Reset: 0x0 Testable: untestable
			8	RPS	R/W	0x0	Receive Process Stopped This bit is asserted when the Rx process enters the Stopped state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Process Stopped status not detected 0x1 (ACTIVE): Receive Process Stopped status detected Value After Reset: 0x0 Testable: untestable
			7	RBU	R/W	0x0	Receive Buffer Unavailable This bit indicates that the application owns the next descriptor in the Receive list, and the DMA cannot acquire it. The Rx process is suspended. To resume processing Rx descriptors, the application must change the ownership of the descriptor and issue a Receive Poll Demand command. If this command is not issued, the Rx process resumes when the next recognized incoming packet is received. In ring mode, the application must advance the Receive Descriptor Tail Pointer register of a channel. This bit is set only when the DMA owns the previous Rx descriptor. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Buffer Unavailable status not detected 0x1 (ACTIVE): Receive Buffer Unavailable status detected Value After Reset: 0x0 Testable: untestable
			6	RI	R/W	0x0	Receive Interrupt This bit indicates that the packet reception is complete. When packet reception is complete, Bit 31 of RDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor. The reception remains in the Running state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Interrupt status not detected 0x1 (ACTIVE): Receive Interrupt status detected Value After Reset: 0x0 Testable: untestable
			5:3	Reserved_5_3	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			2	TBU	R/W	0x0	Transmit Buffer Unavailable This bit indicates that the application owns the next descriptor in the Transmit list, and the DMA cannot acquire it. Transmission is suspended. The TPS0 field of the DMA_Debug_Status0-DMA_Debug_Status2 register explains the Transmit Process state transitions. To resume processing the Transmit descriptors, the application must do the following: Change the ownership of the descriptor by setting Bit 31 of TDES3. Issue a Transmit Poll Demand command. For ring mode, the application must advance the Transmit Descriptor Tail Pointer register of a channel. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Buffer Unavailable status not detected 0x1 (ACTIVE): Transmit Buffer Unavailable status detected Value After Reset: 0x0 Testable: untestable
			1	TPS	R/W	0x0	Transmit Process Stopped This bit is set when the transmission is stopped. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Process Stopped status not detected 0x1 (ACTIVE): Transmit Process Stopped status detected Value After Reset: 0x0 Testable: untestable
			0	TI	R/W	0x0	Transmit Interrupt This bit indicates that the packet transmission is complete. When transmission is complete, Bit 31 of TDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Interrupt status not detected 0x1 (ACTIVE): Transmit Interrupt status detected Value After Reset: 0x0 Testable: untestable
0x1364	DMA_CH4_Miss_Frame_Cnt	This register has the number of packet counter that got dropped by the DMA either due to Bus Error or due to programming RPF field in DMA_CH4_Rx_Control register.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15	MFCO	R	0x0	Overflow status of the MFC Counter When this bit is set then the MFC counter does not get incremented further. The bit gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Miss Frame Counter overflow not occurred 0x1 (ACTIVE): Miss Frame Counter overflow occurred Value After Reset: 0x0
			14:11	Reserved_14_11	R	0x0	Reserved. Value After Reset: 0x0
			10:0	MFC	R	0x0	Dropped Packet Counters This counter indicates the number of packet counters that are dropped by the DMA either because of bus error or because of programing RPF field in DMA_CH0_Rx_Control register. The counter gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1368	DMA_CH4_RXP_Accept_Cnt	The DMA_CH4_RXP_Accept_Cnt registers provides the count of the number of frames accepted by Rx Parser.	31	RXPACOF	R	0x0	Rx Parser Accept Counter Overflow Bit When set, this bit indicates that the RXPAC Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Rx Parser Accept Counter overflow not occurred 0x1 (ACTIVE): Rx Parser Accept Counter overflow occurred Value After Reset: 0x0
			30:0	RXPAC	R	0x0	Rx Parser Accept Counter This 31-bit counter is implemented when a Rx Parser Accept a packet due to AF =1. The counter is cleared when the register is read. Value After Reset: 0x0
0x1380	DMA_CH5_Control	The DMA Channel5 Control register specifies the MSS value for segmentation, length to skip between two descriptors, and also the features such as header splitting and 8xPBL mode.	31:25	Reserved_31_25	R	0x0	Reserved. Value After Reset: 0x0
			24	SPH	R/W	0x0	Split Headers 1: DMA splits the header and payload in the receive path. The DMA writes the header to the Buffer Address1 of RDES0. The DMA writes the payload to the buffer to which the Buffer Address2 is pointing. The software must ensure that the header fits into the Receive buffers. If the header length exceeds the receive buffer size, the DMA does not split the header and payload. This bit is available only when you select Enable Split Header Structure feature.Values: 0x0 (DISABLE): Split Headers feature is disabled 0x1 (ENABLE): Split Headers feature is enabled Value After Reset: 0x0
			23:21	Reserved_23_21	R	0x0	Reserved. Value After Reset: 0x0
			20:18	DSL	R/W	0x0	Descriptor Skip Length This bit specifies the Lword (128-bit) number of locations to skip between two descriptors. The address skipping starts from the end of the current descriptor to the start of the next descriptor. When the DSL value is equal to zero, the DMA takes the descriptor table as contiguous. Value After Reset: 0x0
			17	Reserved_17	R	0x0	Reserved. Value After Reset: 0x0
			16	PBLx8	R/W	0x0	8xPBL mode 1: Indicates that the PBL value programmed in Bits[21:16] in DMA_CH0_Tx_Control and Bits[21:16] in DMA_CH0_Rx_Control is multiplied by eight times. Therefore, the DMA transfers the data in 8, 16, 32, 64, 128, and 256 beats depending on the PBL value. Values: 0x0 (DISABLE): 8xPBL mode is disabled 0x1 (ENABLE): 8xPBL mode is enabled Value After Reset: 0x0
			15:14	Reserved_15_14	R	0x0	Reserved. Value After Reset: 0x0
			13:0	MSS	R/W	0x0	Maximum Segment Size This field specifies the maximum segment size that must be used while segmenting the packet. This field is valid only if the TSE bit of DMA_CH0_Tx_Control register is set. Program a value in this field that is more than the configured datawidth in bytes. Use MSS value of 64 bytes or more. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			31	Reserved_31	R	0x0	Reserved. Value After Reset: 0x0
			30:29	TFSEL	R/W	0x0	TBS Fetch Select Select bits for one of the four DMA_TBS_CTRL register fields (FTOS,FGSN,FTOV) for the channel Values: 2'b00: DMA_TBS_CTRL0 2'b01: DMA_TBS_CTRL1 2'b10: DMA_TBS_CTRL2 2'b11: DMA_TBS_CTRL3 (Reserved if DWC_EQOS_NUM_DMA_TX_CH=3) Value After Reset: 0x0
			28	EDSE	R/W	0x0	Enhanced Descriptor Enable 1: The corresponding channel uses Enhanced Descriptors that are 32 Bytes for both Normal and Context Descriptors. 0: The corresponding channel uses the descriptors that are 16 Bytes. Values: 0x0 (DISABLE): Enhanced Descriptor is disabled 0x1 (ENABLE): Enhanced Descriptor is enabled Value After Reset: 0x0
			27:24	TQOS	R/W	0x0	Transmit QOS. This field is used to drive arqos_m_o[3:0] or awqos_m_o[3:0] output signals for all transactions of DMA Tx Channel0. Value After Reset: 0x0
			23	Reserved_23	R	0x0	Reserved. Value After Reset: 0x0
			22			#VALUE!	Reserved Field: Yes
			21:16	TxPBL	R/W	0x0	Transmit Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in DMA_CH0_Control register. 2. Set the TxPBL. Note: The maximum value of TxPBL must be less than or equal to half the Tx Queue size (TQS field of MTL_TxQ0_Operation_Mode register) in terms of beats. This is required so that the Tx Queue has space to store at least another Tx PBL worth of data while the MTL Tx Queue Controller is transferring data to MAC. For example, in 32-bit data width configurations the total locations in Tx Queue of size 512 bytes is 128, TxPBL and 8xPBL needs to be programmed to less than or equal to 64. Value After Reset: 0x0
			15	IPBL	R/W	0x0	Ignore PBL Requirement 1: DMA does not check for PBL number of locations in the MTL before initiating a transfer. If space is not available, the MTL might use handshaking to slow the DMA. Note: Do not use this mode when you enable multiple Transmit DMA channels. Because, it might block other Transmit and Receive DMA Channels from accessing the Read Data Channel of the AXI bus until space is available in Transmit Queue for current transfer.Values: 0x0 (DISABLE): Ignore PBL Requirement is disabled 0x1 (ENABLE): Ignore PBL Requirement is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1384	DMA_CH5_Tx_Control	The DMA Channel5 Transmit Control register controls the Tx features such as PBL, TCP segmentation, and Tx Channel weights.	14:13	TSE_MODE	R/W	0x0	TSE Mode 00: TSO/USO (segmentation functionality is enabled). In this mode, the setting of TSE bit enables the TSO/USO segmentation. 01: UFO with Checksum (UDP Fragmentation over IPv4 with Checksum). In this mode, the setting of TSE bit enables the UDP fragmentation functionality with Checksum for all the UDP packets. 10: UFO without Checksum (UDP Fragmentation over IPv4 with Checksum). In this mode, the setting of TSE bit enables the UDP fragmentation functionality without Checksum for all the UDP packets. 11: Reserved Values: 0x0 (TSO_USO): TSO/USO 0x1 (UFOWC): UFO with Checksum 0x2 (UFOWOC): UFO without Checksum 0x3 (RSVD): Reserved Value After Reset: 0x0
			12	TSE	R/W	0x0	TCP Segmentation Enabled 1: DMA performs the TCP segmentation or UDP Segmentation/Fragmentation for packets in this channel. The TCP segmentation or UDP packet's segmentation/Fragmentation is performed only for those packets for which the TSE bit (TDES0[19]) = 1 in the Tx Normal descriptor. When this bit is 1, the TxPBL value must be greater than 4. Values: 0x0 (DISABLE): TCP Segmentation is disabled 0x1 (ENABLE): TCP Segmentation is enabled Value After Reset: 0x0
			11:5	Reserved_11_5	R	0x0	Reserved. Value After Reset: 0x0
			4	OSF	R/W	0x0	Operate on Second Packet 1: Instructs the DMA to process the second packet of the Tx data even before the status for the first packet is obtained. Values: 0x0 (DISABLE): Operate on Second Packet disabled 0x1 (ENABLE): Operate on Second Packet enabled Value After Reset: 0x0
			3:1	TCW	R/W	0x0	Transmit Channel Weight This field indicates the weight assigned to the corresponding Tx channel. When reset is complete, this field is set to 0 for all channels by default, resulting in equal weights to all channels. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	ST	R/W	0x0	Start or Stop Transmission Command 1: Transmission is placed in the running state. The DMA checks the Tx list at the current position for a packet to be transmitted. The DMA tries to acquire descriptor from either of the following positions: The current position in the list This is the base address of the Transmit list set by the DMA_CH0_TxDesc_List_Address register. The position at which the transmission was previously stopped If the DMA does not own the current descriptor, the transmission enters the Suspended state and the TBU bit of the DMA_CH0_Status register is set. The Start Transmission command is effective only when the transmission is stopped. If the command is issued before setting the DMA_CH0_TxDesc_List_Address register, the DMA behavior is unpredictable. When this bit is reset, the transmission process is placed in the Stopped state after completing the transmission of the current packet. The Next Descriptor position in the Transmit list is saved, and it becomes the current position when the transmission is restarted. To change the list address, you need to program DMA_CH0_TxDesc_List_Address register with a new value when this bit is reset. The new value is considered when this bit is set again. The stop transmission command is effective only when the transmission of the current packet is complete or the transmission is in the Suspended state.Values: 0x0 (STOP): Stop Transmission Command 0x1 (START): Start Transmission Command Value After Reset: 0x0 Testable: untestable
			31	RPF	R/W	0x0	Rx Packet Flush. 1: DWC_ether_qos automatically flushes the packet from the Rx Queues destined to this DMA Rx Channel, when it is stopped. When this bit remains set and the DMA is re-started by the software driver, the packets residing in the Rx Queues that were received when this RxDMA was stopped, still get flushed out. The packets received by the MAC only after the DMA is re-started is routed to the Rx DMA.The flushing is done on the Read side of the Rx Queue. When the packet is duplicated to multiple Rx DMA, it is flushed by the Rx DMA to which it is routed only when it is stopped to prevent head-of-line blocking. The packet duplicated to multiple Rx DMA is not flushed when the Rx DMA is re-started, irrespective of whether the packet was received during one or more Rx DMAs were stopped. 0: DWC_ether_qos does not flush the packet in the Rx Queue destined to this RxDMA Channel when it is in STOP state. This might cause head-of-line blocking in the corresponding RxQueue. Note: The stopping of packet flow from a Rx DMA Channel to the application by setting RPF works only when there is one-to-one mapping of Rx Queue to Rx DMA channels. In Dynamic mapping mode, setting RPF bit in any DMA_CH0_Rx_Control register might flush packets from unintended Rx Queues which are destined to the stopped Rx DMA Channel.Values: 0x0 (DISABLE): Rx Packet Flush is disabled 0x1 (ENABLE): Rx Packet Flush is enabled Value After Reset: 0x0
			30:28	Reserved_30_28	R	0x0	Reserved. Value After Reset: 0x0
			27:24	RQOS	R/W	0x0	Rx AXI4 QOS. This field is used to drive arqos_m_o[3:0] or awqos_m_o[3:0] output signals for all transactions of DMA Rx Channel0. Value After Reset: 0x0
			23	Reserved_23	R	0x0	Reserved. Value After Reset: 0x0
			22			#VALUE!	Reserved Field: Yes

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1388	DMA_CH5_Rx_Control	The DMA Channel5 Receive Control register controls the Rx features such as PBL, buffer size, and extended status.	21:16	RxPBL	R/W	0x0	Receive Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in the DMA_CH0_Control register. 2. Set the RxPBL. Note: The maximum value of RxPBL must be less than or equal to half the Rx Queue size (RQS field of MTL_RxQ0_Operation_Mode register) in terms of beats. This is required so that the Rx Queue has space to store at least another Rx PBL worth of data while the Rx DMA is transferring a block of data. For example, in 32-bit data width configurations the total locations in Rx Queue of size 512 bytes is 128, so RxPBL and 8xPBL needs to be programmed to less than or equal to 64. Value After Reset: 0x0
			15	Reserved_15	R	0x0	Reserved. Value After Reset: 0x0
			14:5	RBSZ_13_y	R/W	0x0	Receive Buffer size High RBSZ[13:0] is split into two fields higher RBSZ_13_y and lower RBSZ_x_0. The RBSZ[13:0] field indicates the size of the Rx buffers specified in bytes. The maximum buffer size is limited to 16K bytes. The buffer size is applicable to payload buffers when split headers are enabled. Note: The buffer size must be a multiple of 16. This is required even if the value of buffer address pointer is not aligned to data bus width. Hence the lower RBSZ_x_0 bits are read-only and the value is considered as all-zero. Thus the RBSZ_13_y indicates the buffer size in terms of locations. Value After Reset: 0x0
			4:1	RBSZ_x_0	R	0x0	Receive Buffer size Low RBSZ[13:0] is split into two fields RBSZ_13_y and RBSZ_x_0. The RBSZ_x_0 is the lower field. This field is of width 4 bits. This field is read-only (RO). Value After Reset: 0x0
			0	SR	R/W	0x0	Start or Stop Receive 1: DMA attempts to acquire the descriptor from the Receive list and processes the incoming packets. The DMA tries to acquire descriptor from either of the following positions: -- The current position in the list This is the address set by the DMA_CH0_RxDesc_List_Address register. -- The position at which the Rx process was previously stopped If the DMA does not own the current descriptor, the reception is suspended and the RBU bit of the DMA_CH0_Status register is set. The Start Receive command is effective only when the reception is stopped. If the command is issued before setting the DMA_CH0_RxDesc_List_Address register, the DMA behavior is unpredictable. 0: Rx DMA operation stops after the transfer of the current packet. The next descriptor position in the Receive list is saved, and it becomes the current position after the Rx process is restarted. The Stop Receive command is effective only when the Rx process is in the Running (waiting for Rx packet) or Suspended state. Values: 0x0 (STOP): Stop Receive 0x1 (START): Start Receive Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1390	DMA_CH5_TxDesc_List_HAddress	The Channel5 Tx Descriptor List HAddress register has the higher 8 or 16 bits of the start address of the Transmit descriptor list.You can write to this register only when the Tx DMA has stopped, that is, the ST bit is set to zero in DMA_CH5_Tx_Control register. When stopped, this register can be written with a new descriptor list address. When you set the ST bit to 1, the DMA takes the newly-programmed descriptor base address. If this register is not changed when the ST bit is set to 5, the DMA takes the descriptor address where it was stopped earlier. This register must be programmed with desired the higher address before programming the DMA_CH5_TxDesc_List_Address register.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	TDESHA	R/W	0x0	Start of Transmit List This field contains the most-significant 8 or 16 bits of the 40- or 48-bit base address of the first descriptor in the Transmit descriptor list. Value After Reset: 0x0
0x1394	DMA_CH5_TxDesc_List_Address	The Channel5 Tx Descriptor List Address register points the DMA to the start of Transmit descriptor list. The descriptor lists reside in the physical memory space of the application and must be Lword (128-bit). The DMA internally converts it to bus width aligned address by making the corresponding LSB to low.You can write to this register only when the Tx DMA has stopped, that is, the ST bit is set to zero in DMA_CH5_Tx_Control register. When stopped, this register can be written with a new descriptor list address. When you set the ST bit to 1, the DMA takes the newly-programmed descriptor base address. If this register is not changed when the ST bit is set to 5, the DMA takes the descriptor address where it was stopped earlier.	31:4	TDESLA	R/W	0x0	Start of Transmit List This field contains the base address of the first descriptor in the Transmit descriptor list. The DMA ignores the LSB bits (3:0) and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0 Testable: untestable
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x1398	DMA_CH5_RxDesc_List_HAddress	The Channel5 Rx Descriptor List HAddress register has the higher 8 or 16 bits of the start address of the Receive descriptor list.Writing to this register is permitted only when reception is stopped. When stopped, this register must be written to before the receive Start command is given. You can write to this register only when Rx DMA has stopped, that is, SR bit is set to zero in the DMA_CH5_RxDesc_List_Address registerWhen you set the SR bit to 1, the DMA takes the newly programmed descriptor base address.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	RDESHA	R/W	0x0	Start of Receive List This field contains the most-significant 8 or 16 bits of the 40-bit or 48-bit base address of the first descriptor in the Rx Descriptor list. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x139c	DMA_CH5_RxDesc_List_Address	The Channel5 Rx Descriptor List Address register points the DMA to the start of Receive descriptor list.This register points to the start of the Receive Descriptor List. The descriptor lists reside in the physical memory space of the application and must be Lword (128-bit). The DMA internally converts it to bus width aligned address by making the corresponding LS bits low. Writing to this register is permitted only when reception is stopped. When stopped, this register must be written to before the receive Start command is given. You can write to this register only when Rx DMA has stopped, that is, SR bit is set to zero in DMA_CH5_Rx_Control register. When stopped, this register can be written with a new descriptor list address.When you set the SR bit to 1, the DMA takes the newly programmed descriptor base address.	31:4	RDESLA	R/W	0x0	Start of Receive List This field contains the base address of the first descriptor in the Rx Descriptor list. The DMA ignores the LSB bits (3:0) and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0 Testable: untestable
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x13a0	DMA_CH5_TxDesc_Tail_Pointer	The Channel5 Tx Descriptor Tail Pointer register points to an offset from the base and indicates the location of the last valid descriptor.	31:4	TDTP	R/W	0x0	Transmit Descriptor Tail Pointer This field contains the tail pointer for the Tx descriptor ring. The software writes the tail pointer to add more descriptors to the Tx channel. The hardware tries to transmit all packets referenced by the descriptors between the head and the tail pointer registers. The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x13a8	DMA_CH5_RxDesc_Tail_Pointer	The Channel5 Rx Descriptor Tail Pointer Points to an offset from the base and indicates the location of the last valid descriptor.	31:4	RDTP	R/W	0x0	Receive Descriptor Tail Pointer This field contains the tail pointer for the Rx descriptor ring. The software writes the tail pointer to add more descriptors to the Rx channel. The hardware tries to write all received packets to the descriptors referenced between the head and the tail pointer registers. The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x13ac	DMA_CH5_TxDesc_Ring_Length	The Tx Descriptor Ring Length register contains the length of the Transmit descriptor ring.	31:10	Reserved_31_10	R	0x0	Reserved. Value After Reset: 0x0
			9:0	TDRL	R/W	0x0	Transmit Descriptor Ring Length This field sets the maximum number of Tx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. It is recommended that a minimum ring descriptor length of 4. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x13b0	DMA_CH5_Rx_Control2	The Channel5 Receive Control register controls the Rx features such as Rx Descriptor Ring Length and Alternate Rx Buffer Size.	31:24	Reserved_31_24	R	0x0	Reserved. Value After Reset: 0x0
			23:18	ARBS	R/W	0x0	Alternate Receive Buffer Size Indicates size in bytes for Buffer 1 when ARBS is programmed to a non-zero value (when split header feature is not enabled). When split header feature is enabled, ARBS indicates the buffer size for header data. The maximum alternate buffer is limited to 1008 bytes. When ARBS=0, Rx Buffer1 and Rx Buffer2 sizes are based on RBSZ field of DMA_CH0_Rx_Control. Width of ARBS field is 6 bits. Value After Reset: 0x0
			17:10	Reserved_x_10	R	0x0	Reserved. Value After Reset: 0x0
			9:0	RDRL	R/W	0x0	Receive Descriptor Ring Length This register sets the maximum number of Rx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9. Value After Reset: 0x0
			31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15	NIE	R/W	0x0	Normal Interrupt Summary Enable When this bit is set, the normal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: Bit 0: Transmit Interrupt Bit 2: Transmit Buffer Unavailable Bit 6: Receive Interrupt Bit 11: Early Receive Interrupt When this bit is reset, the normal interrupt summary is disabled.Values: 0x0 (DISABLE): Normal Interrupt Summary is disabled 0x1 (ENABLE): Normal Interrupt Summary is enabled Value After Reset: 0x0
			14	AIE	R/W	0x0	Abnormal Interrupt Summary Enable When this bit is set, the abnormal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: Bit 1: Transmit Process Stopped Bit 7: Rx Buffer Unavailable Bit 8: Receive Process Stopped Bit 9: Receive Watchdog Timeout Bit 10: Early Transmit Interrupt Bit 12: Fatal Bus Error Bit 13: Context Descriptor Error When this bit is reset, the abnormal interrupt summary is disabled.Values: 0x0 (DISABLE): Abnormal Interrupt Summary is disabled 0x1 (ENABLE): Abnormal Interrupt Summary is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x13b4	DMA_CH5_Interrupt_Enable	The Channel5 Interrupt Enable register enables the interrupts reported by the Status register.	13	CDEE	R/W	0x0	Context Descriptor Error Enable When this bit is set along with the AIE bit, the Descriptor error interrupt is enabled. When this bit is reset, the Descriptor error interrupt is disabled.Values: 0x0 (DISABLE): Context Descriptor Error is disabled 0x1 (ENABLE): Context Descriptor Error is enabled Value After Reset: 0x0
			12	FBEE	R/W	0x0	Fatal Bus Error Enable When this bit is set along with the AIE bit, the Fatal Bus error interrupt is enabled. When this bit is reset, the Fatal Bus Error error interrupt is disabled.Values: 0x0 (DISABLE): Fatal Bus Error is disabled 0x1 (ENABLE): Fatal Bus Error is enabled Value After Reset: 0x0
			11	ERIE	R/W	0x0	Early Receive Interrupt Enable When this bit is set along with the NIE bit, the Early Receive interrupt is enabled. When this bit is reset, the Early Receive interrupt is disabled.Values: 0x0 (DISABLE): Early Receive Interrupt is disabled 0x1 (ENABLE): Early Receive Interrupt is enabled Value After Reset: 0x0
			10	ETIE	R/W	0x0	Early Transmit Interrupt Enable When this bit is set along with the AIE bit, the Early Transmit interrupt is enabled. When this bit is reset, the Early Transmit interrupt is disabled.Values: 0x0 (DISABLE): Early Transmit Interrupt is disabled 0x1 (ENABLE): Early Transmit Interrupt is enabled Value After Reset: 0x0
			9	RWTE	R/W	0x0	Receive Watchdog Timeout Enable When this bit is set along with the AIE bit, the Receive Watchdog Timeout interrupt is enabled. When this bit is reset, the Receive Watchdog Timeout interrupt is disabled.Values: 0x0 (DISABLE): Receive Watchdog Timeout is disabled 0x1 (ENABLE): Receive Watchdog Timeout is enabled Value After Reset: 0x0
			8	RSE	R/W	0x0	Receive Stopped Enable When this bit is set along with the AIE bit, the Receive Stopped Interrupt is enabled. When this bit is reset, the Receive Stopped interrupt is disabled.Values: 0x0 (DISABLE): Receive Stopped is disabled 0x1 (ENABLE): Receive Stopped is enabled Value After Reset: 0x0
			7	RBUE	R/W	0x0	Receive Buffer Unavailable Enable When this bit is set along with the AIE bit, the Receive Buffer Unavailable interrupt is enabled. When this bit is reset, the Receive Buffer Unavailable interrupt is disabled.Values: 0x0 (DISABLE): Receive Buffer Unavailable is disabled 0x1 (ENABLE): Receive Buffer Unavailable is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			6	RIE	R/W	0x0	Receive Interrupt Enable When this bit is set along with the NIE bit, the Receive Interrupt is enabled. When this bit is reset, the Receive Interrupt is disabled.Values: 0x0 (DISABLE): Receive Interrupt is disabled 0x1 (ENABLE): Receive Interrupt is enabled Value After Reset: 0x0
			5:3	Reserved_5_3	R	0x0	Reserved. Value After Reset: 0x0
			2	TBUE	R/W	0x0	Transmit Buffer Unavailable Enable When this bit is set along with the NIE bit, the Transmit Buffer Unavailable interrupt is enabled. When this bit is reset, the Transmit Buffer Unavailable interrupt is disabled.Values: 0x0 (DISABLE): Transmit Buffer Unavailable is disabled 0x1 (ENABLE): Transmit Buffer Unavailable is enabled Value After Reset: 0x0
			1	TXSE	R/W	0x0	Transmit Stopped Enable When this bit is set along with the AIE bit, the Transmission Stopped interrupt is enabled. When this bit is reset, the Transmission Stopped interrupt is disabled.Values: 0x0 (DISABLE): Transmit Stopped is disabled 0x1 (ENABLE): Transmit Stopped is enabled Value After Reset: 0x0
			0	TIE	R/W	0x0	Transmit Interrupt Enable When this bit is set along with the NIE bit, the Transmit Interrupt is enabled. When this bit is reset, the Transmit Interrupt is disabled.Values: 0x0 (DISABLE): Transmit Interrupt is disabled 0x1 (ENABLE): Transmit Interrupt is enabled Value After Reset: 0x0
0x13b8	DMA_CH5_Rx_Interrupt_Watchdog_Timer	The Receive Interrupt Watchdog Timer register indicates the watchdog timeout for Receive Interrupt (RI) from the DMA. When this register is written with a non-zero value, it enables the watchdog timer for the RI bit of the DMA_CH5_Status register.	31:18	Reserved_31_18	R	0x0	Reserved. Value After Reset: 0x0
			17:16	RWTU	R/W	0x0	Receive Interrupt Watchdog Timer Count Units This fields indicates the number of system clock cycles corresponding to one unit in RWT field. 2'b00: 256 2'b01: 512 2'b10: 1024 2'b11: 2048 For example, when RWT=2 and RWTU=1, the watchdog timer is set for 2*512=1024 system clock cycles. Value After Reset: 0x0 Testable: untestable
			15:8	Reserved_15_8	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			7:0	RWT	R/W	0x0	Receive Interrupt Watchdog Timer Count This field indicates the number of system clock cycles, multiplied by factor indicated in RWTU field, for which the watchdog timer is set. The watchdog timer is triggered with the programmed value after the Rx DMA completes the transfer of a packet for which the RI bit is not set in the DMA_CH0_Status register, because of the setting of Interrupt Enable bit in the corresponding descriptor RDES3[30]. When the watchdog timer runs out, the RI bit is set and the timer is stopped. The watchdog timer is reset when the RI bit is set high because of automatic setting of RI as per the Interrupt Enable bit RDES3[30] of any received packet. Value After Reset: 0x0 Testable: untestable
0x13bc	DMA_CH5_Slot_Function_Control_Status	The Slot Function Control and Status register contains the control bits for slot function and the status for Transmit path.	31:20	Reserved_31_20	R	0x0	Reserved. Value After Reset: 0x0
			19:16	RSN	R	0x0	Reference Slot Number This field gives the current value of the reference slot number in the DMA. It is used for slot comparison. Value After Reset: 0x0 Testable: untestable
			15:4	SIV	R/W	0x7c	Slot Interval Value This field controls the period of the slot interval in which the TxDMA fetches the scheduled packets. A value of 0 specifies the slot interval of 1 us while the maximum value 4095 specifies the slot interval of 4096us. The default/reset value is 0x07C which corresponds to slot interval of 125us Value After Reset: 0x7c
			3:2	Reserved_3_2	R	0x0	Reserved. Value After Reset: 0x0
			1	ASC	R/W	0x0	Advance Slot Check When set, this bit enables the DMA to fetch the data from the buffer when the slot number (SLOTNUM) programmed in the Tx descriptor is equal to the reference slot number given in the RSN field or ahead of the reference slot number by up to two slots This bit is applicable only when the ESC bit is set.Values: 0x0 (DISABLE): Advance Slot Check is disabled 0x1 (ENABLE): Advance Slot Check is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	ESC	R/W	0x0	Enable Slot Comparison When set, this bit enables the checking of the slot numbers programmed in the Tx descriptor with the current reference given in the RSN field. The DMA fetches the data from the corresponding buffer only when the slot number is equal to the reference slot number or ahead of the reference slot number by one slot When reset, this bit disables the checking of the slot numbers. The DMA fetches the data immediately after the descriptor is processed. Note: The UFO (UDP Fragmentation over IPv4)/TSO/USO must not be enabled along with TBS/AVB Slot number check. The UFO/TSO/USO involves multiple packets/segments/fragments transmission for single packet received from application and the slot number check are applicable for fetching of only first segment/fragment. As a result it might be difficult for software to specify slot number for subsequent packets.Values: 0x0 (DISABLE): Slot Comparison is disabled 0x1 (ENABLE): Slot Comparison is enabled Value After Reset: 0x0
0x13c4	DMA_CH5_Current_App_Tx Desc	The Channel5 Current Application Transmit Descriptor register points to the current Transmit descriptor read by the DMA.	31:0	CURTDESAPTR	R	0x0	Application Transmit Descriptor Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x13cc	DMA_CH5_Current_App_Rx Desc	The Channel5 Current Application Receive Descriptor register points to the current Receive descriptor read by the DMA.	31:0	CURRDESAPTR	R	0x0	Application Receive Descriptor Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x13d0	DMA_CH5_Current_App_Tx Buffer_H	The Channel5 Current Application Transmit Buffer Address High register has the higher 8 or 16 bits of the current address of the Transmit buffer address read by the DMA.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	CURTBUFAPTRH	R	0x0	Application Transmit Buffer Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x13d4	DMA_CH5_Current_App_Tx Buffer	The Channel5 Current Application Transmit Buffer Address register points to the current Tx buffer address read by the DMA.	31:0	CURTBUFAPTR	R	0x0	Application Transmit Buffer Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x13d8	DMA_CH5_Current_App_Rx Buffer_H	The Channel5 Current Application Receive Buffer Address High register has the higher 8 or 16 bits of the current address of the Receive buffer address read by the DMA.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	CURRBUFAPTRH	R	0x0	Application Receive Buffer Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x13dc	DMA_CH5_Current_App_Rx Buffer	The Channel 5 Current Application Receive Buffer Address register points to the current Rx buffer address read by the DMA.	31:0	CURRBUFAPTR	R	0x0	Application Receive Buffer Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			31:22	Reserved_31_22	R	0x0	Reserved. Value After Reset: 0x0
			21:19	REB	R	0x0	Rx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AXI interface. Bit 21 1'b1: Error during data transfer by Rx DMA 1'b0: No Error during data transfer by Rx DMA Bit 20 1'b1: Error during descriptor access 1'b0: Error during data buffer access Bit 19 1'b1: Error during read transfer 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt. Value After Reset: 0x0 Testable: untestable
			18:16	TEB	R	0x0	Tx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AXI interface. Bit 18 1'b1: Error during data transfer by Tx DMA 1'b0: No Error during data transfer by Tx DMA Bit 17 1'b1: Error during descriptor access 1'b0: Error during data buffer access Bit 16 1'b1: Error during read transfer 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt. Value After Reset: 0x0 Testable: untestable
			15	NIS	R/W	0x0	Normal Interrupt Summary Normal Interrupt Summary bit value is the logical OR of the following bits when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: Bit 0: Transmit Interrupt Bit 2: Transmit Buffer Unavailable Bit 6: Receive Interrupt Bit 11: Early Receive Interrupt Only unmasked bits (interrupts for which interrupt enable is set in DMA_CH0_Interrupt_Enable register) affect the Normal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit which causes NIS to be set is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Normal Interrupt Summary status not detected 0x1 (ACTIVE): Normal Interrupt Summary status detected Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x13e0	DMA_CH5_Status	The software driver (application) reads the Status register during interrupt service routine or polling to determine the status of the DMA.Note: The number of DMA_CH5_Status register in the configuration is the higher of number of Rx DMA Channels and Tx DMA Channels.	14	AIS	R/W	0x0	Abnormal Interrupt Summary Abnormal Interrupt Summary bit value is the logical OR of the following when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: Bit 1: Transmit Process Stopped Bit 7: Receive Buffer Unavailable Bit 8: Receive Process Stopped Bit 10: Early Transmit Interrupt Bit 12: Fatal Bus Error Bit 13: Context Descriptor Error Only unmasked bits affect the Abnormal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit, which causes AIS to be set, is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Abnormal Interrupt Summary status not detected 0x1 (ACTIVE): Abnormal Interrupt Summary status detected Value After Reset: 0x0 Testable: untestable
			13	CDE	R/W	0x0	Context Descriptor Error This bit indicates that the DMA Tx/Rx engine received a descriptor error, which indicates invalid context in the middle of packet flow (intermediate descriptor) or all one's descriptor in Tx case and on Rx side it indicates DMA has read a descriptor with either of the buffer address as ones which is considered to be invalid. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Context Descriptor Error status not detected 0x1 (ACTIVE): Context Descriptor Error status detected Value After Reset: 0x0 Testable: untestable
			12	FBE	R/W	0x0	Fatal Bus Error This bit indicates that a bus error occurred (as described in the EB field). When this bit is set, the corresponding DMA channel engine disables all bus accesses. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Fatal Bus Error status not detected 0x1 (ACTIVE): Fatal Bus Error status detected Value After Reset: 0x0 Testable: untestable
			11	ERI	R/W	0x0	Early Receive Interrupt This bit when set indicates that the RxDMA has completed the transfer of packet data to the memory. In configurations supporting ERIC, ERIC = 0: This bit is set only after the Rx DMA has completely filled up a receive buffer with packet data. ERIC = 1: This bit is set after every burst transfer of data from the Rx DMA to the buffer. The setting of RI bit automatically clears this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Early Receive Interrupt status not detected 0x1 (ACTIVE): Early Receive Interrupt status detected Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			10	ETI	R/W	0x0	Early Transmit Interrupt This bit when set indicates that the TxDMA has completed the transfer of packet data to the MTL TXFIFO memory. In configurations supporting ERIC: ETIC = 0: This bit is set only after the Tx DMA has transferred a complete packet to MTL. ETIC = 1: This bit is set after completion of (partial) packet data transfer from buffers in the Transmit descriptor in which IOC = 1. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Early Transmit Interrupt status not detected 0x1 (ACTIVE): Early Transmit Interrupt status detected Value After Reset: 0x0 Testable: untestable
			9	RWT	R/W	0x0	Receive Watchdog Timeout This bit is asserted when a packet with length greater than 2,048 bytes (10,240 bytes when Jumbo Packet mode is enabled) is received.Values: 0x0 (INACTIVE): Receive Watchdog Timeout status not detected 0x1 (ACTIVE): Receive Watchdog Timeout status detected Value After Reset: 0x0 Testable: untestable
			8	RPS	R/W	0x0	Receive Process Stopped This bit is asserted when the Rx process enters the Stopped state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Process Stopped status not detected 0x1 (ACTIVE): Receive Process Stopped status detected Value After Reset: 0x0 Testable: untestable
			7	RBU	R/W	0x0	Receive Buffer Unavailable This bit indicates that the application owns the next descriptor in the Receive list, and the DMA cannot acquire it. The Rx process is suspended. To resume processing Rx descriptors, the application must change the ownership of the descriptor and issue a Receive Poll Demand command. If this command is not issued, the Rx process resumes when the next recognized incoming packet is received. In ring mode, the application must advance the Receive Descriptor Tail Pointer register of a channel. This bit is set only when the DMA owns the previous Rx descriptor. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Buffer Unavailable status not detected 0x1 (ACTIVE): Receive Buffer Unavailable status detected Value After Reset: 0x0 Testable: untestable
			6	RI	R/W	0x0	Receive Interrupt This bit indicates that the packet reception is complete. When packet reception is complete, Bit 31 of RDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor. The reception remains in the Running state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Interrupt status not detected 0x1 (ACTIVE): Receive Interrupt status detected Value After Reset: 0x0 Testable: untestable
			5:3	Reserved_5_3	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			2	TBU	R/W	0x0	Transmit Buffer Unavailable This bit indicates that the application owns the next descriptor in the Transmit list, and the DMA cannot acquire it. Transmission is suspended. The TPS0 field of the DMA_Debug_Status0-DMA_Debug_Status2 register explains the Transmit Process state transitions. To resume processing the Transmit descriptors, the application must do the following: Change the ownership of the descriptor by setting Bit 31 of TDES3. Issue a Transmit Poll Demand command. For ring mode, the application must advance the Transmit Descriptor Tail Pointer register of a channel. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Buffer Unavailable status not detected 0x1 (ACTIVE): Transmit Buffer Unavailable status detected Value After Reset: 0x0 Testable: untestable
			1	TPS	R/W	0x0	Transmit Process Stopped This bit is set when the transmission is stopped. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Process Stopped status not detected 0x1 (ACTIVE): Transmit Process Stopped status detected Value After Reset: 0x0 Testable: untestable
			0	TI	R/W	0x0	Transmit Interrupt This bit indicates that the packet transmission is complete. When transmission is complete, Bit 31 of TDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Interrupt status not detected 0x1 (ACTIVE): Transmit Interrupt status detected Value After Reset: 0x0 Testable: untestable
0x13e4	DMA_CH5_Miss_Frame_Cnt	This register has the number of packet counter that got dropped by the DMA either due to Bus Error or due to programming RPF field in DMA_CH5_Rx_Control register.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15	MFCO	R	0x0	Overflow status of the MFC Counter When this bit is set then the MFC counter does not get incremented further. The bit gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Miss Frame Counter overflow not occurred 0x1 (ACTIVE): Miss Frame Counter overflow occurred Value After Reset: 0x0
			14:11	Reserved_14_11	R	0x0	Reserved. Value After Reset: 0x0
			10:0	MFC	R	0x0	Dropped Packet Counters This counter indicates the number of packet counters that are dropped by the DMA either because of bus error or because of programing RPF field in DMA_CH0_Rx_Control register. The counter gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x13e8	DMA_CH5_RXP_Accept_Cnt	The DMA_CH5_RXP_Accept_Cnt registers provides the count of the number of frames accepted by Rx Parser.	31	RXPACOF	R	0x0	Rx Parser Accept Counter Overflow Bit When set, this bit indicates that the RXPAC Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Rx Parser Accept Counter overflow not occurred 0x1 (ACTIVE): Rx Parser Accept Counter overflow occurred Value After Reset: 0x0
			30:0	RXPAC	R	0x0	Rx Parser Accept Counter This 31-bit counter is implemented when a Rx Parser Accept a packet due to AF =1. The counter is cleared when the register is read. Value After Reset: 0x0
0x1400	DMA_CH6_Control	The DMA Channel6 Control register specifies the MSS value for segmentation, length to skip between two descriptors, and also the features such as header splitting and 8xPBL mode.	31:25	Reserved_31_25	R	0x0	Reserved. Value After Reset: 0x0
			24	SPH	R/W	0x0	Split Headers 1: DMA splits the header and payload in the receive path. The DMA writes the header to the Buffer Address1 of RDES0. The DMA writes the payload to the buffer to which the Buffer Address2 is pointing. The software must ensure that the header fits into the Receive buffers. If the header length exceeds the receive buffer size, the DMA does not split the header and payload. This bit is available only when you select Enable Split Header Structure feature.Values: 0x0 (DISABLE): Split Headers feature is disabled 0x1 (ENABLE): Split Headers feature is enabled Value After Reset: 0x0
			23:21	Reserved_23_21	R	0x0	Reserved. Value After Reset: 0x0
			20:18	DSL	R/W	0x0	Descriptor Skip Length This bit specifies the Lword (128-bit) number of locations to skip between two descriptors. The address skipping starts from the end of the current descriptor to the start of the next descriptor. When the DSL value is equal to zero, the DMA takes the descriptor table as contiguous. Value After Reset: 0x0
			17	Reserved_17	R	0x0	Reserved. Value After Reset: 0x0
			16	PBLx8	R/W	0x0	8xPBL mode 1: Indicates that the PBL value programmed in Bits[21:16] in DMA_CH0_Tx_Control and Bits[21:16] in DMA_CH0_Rx_Control is multiplied by eight times. Therefore, the DMA transfers the data in 8, 16, 32, 64, 128, and 256 beats depending on the PBL value. Values: 0x0 (DISABLE): 8xPBL mode is disabled 0x1 (ENABLE): 8xPBL mode is enabled Value After Reset: 0x0
			15:14	Reserved_15_14	R	0x0	Reserved. Value After Reset: 0x0
			13:0	MSS	R/W	0x0	Maximum Segment Size This field specifies the maximum segment size that must be used while segmenting the packet. This field is valid only if the TSE bit of DMA_CH0_Tx_Control register is set. Program a value in this field that is more than the configured datawidth in bytes. Use MSS value of 64 bytes or more. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			31	Reserved_31	R	0x0	Reserved. Value After Reset: 0x0
			30:29	TFSEL	R/W	0x0	TBS Fetch Select Select bits for one of the four DMA_TBS_CTRL register fields (FTOS,FGSN,FTOV) for the channel Values: 2'b00: DMA_TBS_CTRL0 2'b01: DMA_TBS_CTRL1 2'b10: DMA_TBS_CTRL2 2'b11: DMA_TBS_CTRL3 (Reserved if DWC_EQOS_NUM_DMA_TX_CH=3) Value After Reset: 0x0
			28	EDSE	R/W	0x0	Enhanced Descriptor Enable 1: The corresponding channel uses Enhanced Descriptors that are 32 Bytes for both Normal and Context Descriptors. 0: The corresponding channel uses the descriptors that are 16 Bytes. Values: 0x0 (DISABLE): Enhanced Descriptor is disabled 0x1 (ENABLE): Enhanced Descriptor is enabled Value After Reset: 0x0
			27:24	TQOS	R/W	0x0	Transmit QOS. This field is used to drive arqos_m_o[3:0] or awqos_m_o[3:0] output signals for all transactions of DMA Tx Channel0. Value After Reset: 0x0
			23	Reserved_23	R	0x0	Reserved. Value After Reset: 0x0
			22			#VALUE!	Reserved Field: Yes
			21:16	TxPBL	R/W	0x0	Transmit Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in DMA_CH0_Control register. 2. Set the TxPBL. Note: The maximum value of TxPBL must be less than or equal to half the Tx Queue size (TQS field of MTL_TxQ0_Operation_Mode register) in terms of beats. This is required so that the Tx Queue has space to store at least another Tx PBL worth of data while the MTL Tx Queue Controller is transferring data to MAC. For example, in 32-bit data width configurations the total locations in Tx Queue of size 512 bytes is 128, TxPBL and 8xPBL needs to be programmed to less than or equal to 64. Value After Reset: 0x0
			15	IPBL	R/W	0x0	Ignore PBL Requirement 1: DMA does not check for PBL number of locations in the MTL before initiating a transfer. If space is not available, the MTL might use handshaking to slow the DMA. Note: Do not use this mode when you enable multiple Transmit DMA channels. Because, it might block other Transmit and Receive DMA Channels from accessing the Read Data Channel of the AXI bus until space is available in Transmit Queue for current transfer.Values: 0x0 (DISABLE): Ignore PBL Requirement is disabled 0x1 (ENABLE): Ignore PBL Requirement is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1404	DMA_CH6_Tx_Control	The DMA Channel6 Transmit Control register controls the Tx features such as PBL, TCP segmentation, and Tx Channel weights.	14:13	TSE_MODE	R/W	0x0	TSE Mode 00: TSO/USO (segmentation functionality is enabled). In this mode, the setting of TSE bit enables the TSO/USO segmentation. 01: UFO with Checksum (UDP Fragmentation over IPv4 with Checksum). In this mode, the setting of TSE bit enables the UDP fragmentation functionality with Checksum for all the UDP packets. 10: UFO without Checksum (UDP Fragmentation over IPv4 with Checksum). In this mode, the setting of TSE bit enables the UDP fragmentation functionality without Checksum for all the UDP packets. 11: Reserved Values: 0x0 (TSO_USO): TSO/USO 0x1 (UFOWC): UFO with Checksum 0x2 (UFOWOC): UFO without Checksum 0x3 (RSVD): Reserved Value After Reset: 0x0
			12	TSE	R/W	0x0	TCP Segmentation Enabled 1: DMA performs the TCP segmentation or UDP Segmentation/Fragmentation for packets in this channel. The TCP segmentation or UDP packet's segmentation/Fragmentation is performed only for those packets for which the TSE bit (TDES0[19]) = 1 in the Tx Normal descriptor. When this bit is 1, the TxPBL value must be greater than 4. Values: 0x0 (DISABLE): TCP Segmentation is disabled 0x1 (ENABLE): TCP Segmentation is enabled Value After Reset: 0x0
			11:5	Reserved_11_5	R	0x0	Reserved. Value After Reset: 0x0
			4	OSF	R/W	0x0	Operate on Second Packet 1: Instructs the DMA to process the second packet of the Tx data even before the status for the first packet is obtained. Values: 0x0 (DISABLE): Operate on Second Packet disabled 0x1 (ENABLE): Operate on Second Packet enabled Value After Reset: 0x0
			3:1	TCW	R/W	0x0	Transmit Channel Weight This field indicates the weight assigned to the corresponding Tx channel. When reset is complete, this field is set to 0 for all channels by default, resulting in equal weights to all channels. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	ST	R/W	0x0	Start or Stop Transmission Command 1: Transmission is placed in the running state. The DMA checks the Tx list at the current position for a packet to be transmitted. The DMA tries to acquire descriptor from either of the following positions: The current position in the list This is the base address of the Transmit list set by the DMA_CH0_TxDesc_List_Address register. The position at which the transmission was previously stopped If the DMA does not own the current descriptor, the transmission enters the Suspended state and the TBU bit of the DMA_CH0_Status register is set. The Start Transmission command is effective only when the transmission is stopped. If the command is issued before setting the DMA_CH0_TxDesc_List_Address register, the DMA behavior is unpredictable. When this bit is reset, the transmission process is placed in the Stopped state after completing the transmission of the current packet. The Next Descriptor position in the Transmit list is saved, and it becomes the current position when the transmission is restarted. To change the list address, you need to program DMA_CH0_TxDesc_List_Address register with a new value when this bit is reset. The new value is considered when this bit is set again. The stop transmission command is effective only when the transmission of the current packet is complete or the transmission is in the Suspended state.Values: 0x0 (STOP): Stop Transmission Command 0x1 (START): Start Transmission Command Value After Reset: 0x0 Testable: untestable
			31	RPF	R/W	0x0	Rx Packet Flush. 1: DWC_ether_qos automatically flushes the packet from the Rx Queues destined to this DMA Rx Channel, when it is stopped. When this bit remains set and the DMA is re-started by the software driver, the packets residing in the Rx Queues that were received when this RxDMA was stopped, still get flushed out. The packets received by the MAC only after the DMA is re-started is routed to the Rx DMA.The flushing is done on the Read side of the Rx Queue. When the packet is duplicated to multiple Rx DMA, it is flushed by the Rx DMA to which it is routed only when it is stopped to prevent head-of-line blocking. The packet duplicated to multiple Rx DMA is not flushed when the Rx DMA is re-started, irrespective of whether the packet was received during one or more Rx DMAs were stopped. 0: DWC_ether_qos does not flush the packet in the Rx Queue destined to this RxDMA Channel when it is in STOP state. This might cause head-of-line blocking in the corresponding RxQueue. Note: The stopping of packet flow from a Rx DMA Channel to the application by setting RPF works only when there is one-to-one mapping of Rx Queue to Rx DMA channels. In Dynamic mapping mode, setting RPF bit in any DMA_CH0_Rx_Control register might flush packets from unintended Rx Queues which are destined to the stopped Rx DMA Channel.Values: 0x0 (DISABLE): Rx Packet Flush is disabled 0x1 (ENABLE): Rx Packet Flush is enabled Value After Reset: 0x0
			30:28	Reserved_30_28	R	0x0	Reserved. Value After Reset: 0x0
			27:24	RQOS	R/W	0x0	Rx AXI4 QOS. This field is used to drive arqos_m_o[3:0] or awqos_m_o[3:0] output signals for all transactions of DMA Rx Channel0. Value After Reset: 0x0
			23	Reserved_23	R	0x0	Reserved. Value After Reset: 0x0
			22			#VALUE!	Reserved Field: Yes

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1408	DMA_CH6_Rx_Control	The DMA Channel6 Receive Control register controls the Rx features such as PBL, buffer size, and extended status.	21:16	RxPBL	R/W	0x0	Receive Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in the DMA_CH0_Control register. 2. Set the RxPBL. Note: The maximum value of RxPBL must be less than or equal to half the Rx Queue size (RQS field of MTL_RxQ0_Operation_Mode register) in terms of beats. This is required so that the Rx Queue has space to store at least another Rx PBL worth of data while the Rx DMA is transferring a block of data. For example, in 32-bit data width configurations the total locations in Rx Queue of size 512 bytes is 128, so RxPBL and 8xPBL needs to be programmed to less than or equal to 64. Value After Reset: 0x0
			15	Reserved_15	R	0x0	Reserved. Value After Reset: 0x0
			14:5	RBSZ_13_y	R/W	0x0	Receive Buffer size High RBSZ[13:0] is split into two fields higher RBSZ_13_y and lower RBSZ_x_0. The RBSZ[13:0] field indicates the size of the Rx buffers specified in bytes. The maximum buffer size is limited to 16K bytes. The buffer size is applicable to payload buffers when split headers are enabled. Note: The buffer size must be a multiple of 16. This is required even if the value of buffer address pointer is not aligned to data bus width. Hence the lower RBSZ_x_0 bits are read-only and the value is considered as all-zero. Thus the RBSZ_13_y indicates the buffer size in terms of locations. Value After Reset: 0x0
			4:1	RBSZ_x_0	R	0x0	Receive Buffer size Low RBSZ[13:0] is split into two fields RBSZ_13_y and RBSZ_x_0. The RBSZ_x_0 is the lower field. This field is of width 4 bits. This field is read-only (RO). Value After Reset: 0x0
			0	SR	R/W	0x0	Start or Stop Receive 1: DMA attempts to acquire the descriptor from the Receive list and processes the incoming packets. The DMA tries to acquire descriptor from either of the following positions: -- The current position in the list This is the address set by the DMA_CH0_RxDesc_List_Address register. -- The position at which the Rx process was previously stopped If the DMA does not own the current descriptor, the reception is suspended and the RBU bit of the DMA_CH0_Status register is set. The Start Receive command is effective only when the reception is stopped. If the command is issued before setting the DMA_CH0_RxDesc_List_Address register, the DMA behavior is unpredictable. 0: Rx DMA operation stops after the transfer of the current packet. The next descriptor position in the Receive list is saved, and it becomes the current position after the Rx process is restarted. The Stop Receive command is effective only when the Rx process is in the Running (waiting for Rx packet) or Suspended state. Values: 0x0 (STOP): Stop Receive 0x1 (START): Start Receive Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1410	DMA_CH6_TxDesc_List_HAddress	The Channel6 Tx Descriptor List HAddress register has the higher 8 or 16 bits of the start address of the Transmit descriptor list.You can write to this register only when the Tx DMA has stopped, that is, the ST bit is set to zero in DMA_CH6_Tx_Control register. When stopped, this register can be written with a new descriptor list address. When you set the ST bit to 1, the DMA takes the newly-programmed descriptor base address. If this register is not changed when the ST bit is set to 6, the DMA takes the descriptor address where it was stopped earlier. This register must be programmed with desired the higher address before programming the DMA_CH6_TxDesc_List_Address register.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	TDESHA	R/W	0x0	Start of Transmit List This field contains the most-significant 8 or 16 bits of the 40- or 48-bit base address of the first descriptor in the Transmit descriptor list. Value After Reset: 0x0
0x1414	DMA_CH6_TxDesc_List_Address	The Channel6 Tx Descriptor List Address register points the DMA to the start of Transmit descriptor list. The descriptor lists reside in the physical memory space of the application and must be Lword (128-bit). The DMA internally converts it to bus width aligned address by making the corresponding LSB to low.You can write to this register only when the Tx DMA has stopped, that is, the ST bit is set to zero in DMA_CH6_Tx_Control register. When stopped, this register can be written with a new descriptor list address. When you set the ST bit to 1, the DMA takes the newly-programmed descriptor base address. If this register is not changed when the ST bit is set to 6, the DMA takes the descriptor address where it was stopped earlier.	31:4	TDESLA	R/W	0x0	Start of Transmit List This field contains the base address of the first descriptor in the Transmit descriptor list. The DMA ignores the LSB bits (3:0) and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0 Testable: untestable
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x1418	DMA_CH6_RxDesc_List_HAddress	The Channel6 Rx Descriptor List HAddress register has the higher 8 or 16 bits of the start address of the Receive descriptor list.Writing to this register is permitted only when reception is stopped. When stopped, this register must be written to before the receive Start command is given. You can write to this register only when Rx DMA has stopped, that is, SR bit is set to zero in the DMA_CH6_RxDesc_List_Address registerWhen you set the SR bit to 1, the DMA takes the newly programmed descriptor base address.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	RDESHA	R/W	0x0	Start of Receive List This field contains the most-significant 8 or 16 bits of the 40-bit or 48-bit base address of the first descriptor in the Rx Descriptor list. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x141c	DMA_CH6_RxDesc_List_Address	The Channel6 Rx Descriptor List Address register points the DMA to the start of Receive descriptor list.This register points to the start of the Receive Descriptor List. The descriptor lists reside in the physical memory space of the application and must be Lword (128-bit). The DMA internally converts it to bus width aligned address by making the corresponding LS bits low. Writing to this register is permitted only when reception is stopped. When stopped, this register must be written to before the receive Start command is given. You can write to this register only when Rx DMA has stopped, that is, SR bit is set to zero in DMA_CH6_Rx_Control register. When stopped, this register can be written with a new descriptor list address.When you set the SR bit to 1, the DMA takes the newly programmed descriptor base address.	31:4	RDESLA	R/W	0x0	Start of Receive List This field contains the base address of the first descriptor in the Rx Descriptor list. The DMA ignores the LSB bits (3:0) and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0 Testable: untestable
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x1420	DMA_CH6_TxDesc_Tail_Pointer	The Channel6 Tx Descriptor Tail Pointer register points to an offset from the base and indicates the location of the last valid descriptor.	31:4	TDTP	R/W	0x0	Transmit Descriptor Tail Pointer This field contains the tail pointer for the Tx descriptor ring. The software writes the tail pointer to add more descriptors to the Tx channel. The hardware tries to transmit all packets referenced by the descriptors between the head and the tail pointer registers. The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x1428	DMA_CH6_RxDesc_Tail_Pointer	The Channel6 Rx Descriptor Tail Pointer Points to an offset from the base and indicates the location of the last valid descriptor.	31:4	RDTP	R/W	0x0	Receive Descriptor Tail Pointer This field contains the tail pointer for the Rx descriptor ring. The software writes the tail pointer to add more descriptors to the Rx channel. The hardware tries to write all received packets to the descriptors referenced between the head and the tail pointer registers. The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x142c	DMA_CH6_TxDesc_Ring_Length	The Tx Descriptor Ring Length register contains the length of the Transmit descriptor ring.	31:10	Reserved_31_10	R	0x0	Reserved. Value After Reset: 0x0
			9:0	TDRL	R/W	0x0	Transmit Descriptor Ring Length This field sets the maximum number of Tx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. It is recommended that a minimum ring descriptor length of 4. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1430	DMA_CH6_Rx_Control2	The Channel6 Receive Control register controls the Rx features such as Rx Descriptor Ring Length and Alternate Rx Buffer Size.	31:24	Reserved_31_24	R	0x0	Reserved. Value After Reset: 0x0
			23:18	ARBS	R/W	0x0	Alternate Receive Buffer Size Indicates size in bytes for Buffer 1 when ARBS is programmed to a non-zero value (when split header feature is not enabled). When split header feature is enabled, ARBS indicates the buffer size for header data. The maximum alternate buffer is limited to 1008 bytes. When ARBS=0, Rx Buffer1 and Rx Buffer2 sizes are based on RBSZ field of DMA_CH0_Rx_Control. Width of ARBS field is 6 bits. Value After Reset: 0x0
			17:10	Reserved_x_10	R	0x0	Reserved. Value After Reset: 0x0
			9:0	RDRL	R/W	0x0	Receive Descriptor Ring Length This register sets the maximum number of Rx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9. Value After Reset: 0x0
			31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15	NIE	R/W	0x0	Normal Interrupt Summary Enable When this bit is set, the normal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: Bit 0: Transmit Interrupt Bit 2: Transmit Buffer Unavailable Bit 6: Receive Interrupt Bit 11: Early Receive Interrupt When this bit is reset, the normal interrupt summary is disabled.Values: 0x0 (DISABLE): Normal Interrupt Summary is disabled 0x1 (ENABLE): Normal Interrupt Summary is enabled Value After Reset: 0x0
			14	AIE	R/W	0x0	Abnormal Interrupt Summary Enable When this bit is set, the abnormal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: Bit 1: Transmit Process Stopped Bit 7: Rx Buffer Unavailable Bit 8: Receive Process Stopped Bit 9: Receive Watchdog Timeout Bit 10: Early Transmit Interrupt Bit 12: Fatal Bus Error Bit 13: Context Descriptor Error When this bit is reset, the abnormal interrupt summary is disabled.Values: 0x0 (DISABLE): Abnormal Interrupt Summary is disabled 0x1 (ENABLE): Abnormal Interrupt Summary is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1434	DMA_CH6_Interrupt_Enable	The Channel6 Interrupt Enable register enables the interrupts reported by the Status register.	13	CDEE	R/W	0x0	Context Descriptor Error Enable When this bit is set along with the AIE bit, the Descriptor error interrupt is enabled. When this bit is reset, the Descriptor error interrupt is disabled.Values: 0x0 (DISABLE): Context Descriptor Error is disabled 0x1 (ENABLE): Context Descriptor Error is enabled Value After Reset: 0x0
			12	FBEE	R/W	0x0	Fatal Bus Error Enable When this bit is set along with the AIE bit, the Fatal Bus error interrupt is enabled. When this bit is reset, the Fatal Bus Error error interrupt is disabled.Values: 0x0 (DISABLE): Fatal Bus Error is disabled 0x1 (ENABLE): Fatal Bus Error is enabled Value After Reset: 0x0
			11	ERIE	R/W	0x0	Early Receive Interrupt Enable When this bit is set along with the NIE bit, the Early Receive interrupt is enabled. When this bit is reset, the Early Receive interrupt is disabled.Values: 0x0 (DISABLE): Early Receive Interrupt is disabled 0x1 (ENABLE): Early Receive Interrupt is enabled Value After Reset: 0x0
			10	ETIE	R/W	0x0	Early Transmit Interrupt Enable When this bit is set along with the AIE bit, the Early Transmit interrupt is enabled. When this bit is reset, the Early Transmit interrupt is disabled.Values: 0x0 (DISABLE): Early Transmit Interrupt is disabled 0x1 (ENABLE): Early Transmit Interrupt is enabled Value After Reset: 0x0
			9	RWTE	R/W	0x0	Receive Watchdog Timeout Enable When this bit is set along with the AIE bit, the Receive Watchdog Timeout interrupt is enabled. When this bit is reset, the Receive Watchdog Timeout interrupt is disabled.Values: 0x0 (DISABLE): Receive Watchdog Timeout is disabled 0x1 (ENABLE): Receive Watchdog Timeout is enabled Value After Reset: 0x0
			8	RSE	R/W	0x0	Receive Stopped Enable When this bit is set along with the AIE bit, the Receive Stopped Interrupt is enabled. When this bit is reset, the Receive Stopped interrupt is disabled.Values: 0x0 (DISABLE): Receive Stopped is disabled 0x1 (ENABLE): Receive Stopped is enabled Value After Reset: 0x0
			7	RBUE	R/W	0x0	Receive Buffer Unavailable Enable When this bit is set along with the AIE bit, the Receive Buffer Unavailable interrupt is enabled. When this bit is reset, the Receive Buffer Unavailable interrupt is disabled.Values: 0x0 (DISABLE): Receive Buffer Unavailable is disabled 0x1 (ENABLE): Receive Buffer Unavailable is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			6	RIE	R/W	0x0	Receive Interrupt Enable When this bit is set along with the NIE bit, the Receive Interrupt is enabled. When this bit is reset, the Receive Interrupt is disabled.Values: 0x0 (DISABLE): Receive Interrupt is disabled 0x1 (ENABLE): Receive Interrupt is enabled Value After Reset: 0x0
			5:3	Reserved_5_3	R	0x0	Reserved. Value After Reset: 0x0
			2	TBUE	R/W	0x0	Transmit Buffer Unavailable Enable When this bit is set along with the NIE bit, the Transmit Buffer Unavailable interrupt is enabled. When this bit is reset, the Transmit Buffer Unavailable interrupt is disabled.Values: 0x0 (DISABLE): Transmit Buffer Unavailable is disabled 0x1 (ENABLE): Transmit Buffer Unavailable is enabled Value After Reset: 0x0
			1	TXSE	R/W	0x0	Transmit Stopped Enable When this bit is set along with the AIE bit, the Transmission Stopped interrupt is enabled. When this bit is reset, the Transmission Stopped interrupt is disabled.Values: 0x0 (DISABLE): Transmit Stopped is disabled 0x1 (ENABLE): Transmit Stopped is enabled Value After Reset: 0x0
			0	TIE	R/W	0x0	Transmit Interrupt Enable When this bit is set along with the NIE bit, the Transmit Interrupt is enabled. When this bit is reset, the Transmit Interrupt is disabled.Values: 0x0 (DISABLE): Transmit Interrupt is disabled 0x1 (ENABLE): Transmit Interrupt is enabled Value After Reset: 0x0
0x1438	DMA_CH6_Rx_Interrupt_Watchdog_Timer	The Receive Interrupt Watchdog Timer register indicates the watchdog timeout for Receive Interrupt (RI) from the DMA. When this register is written with a non-zero value, it enables the watchdog timer for the RI bit of the DMA_CH6_Status register.	31:18	Reserved_31_18	R	0x0	Reserved. Value After Reset: 0x0
			17:16	RWTU	R/W	0x0	Receive Interrupt Watchdog Timer Count Units This fields indicates the number of system clock cycles corresponding to one unit in RWT field. 2'b00: 256 2'b01: 512 2'b10: 1024 2'b11: 2048 For example, when RWT=2 and RWTU=1, the watchdog timer is set for 2*512=1024 system clock cycles. Value After Reset: 0x0 Testable: untestable
			15:8	Reserved_15_8	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			7:0	RWT	R/W	0x0	Receive Interrupt Watchdog Timer Count This field indicates the number of system clock cycles, multiplied by factor indicated in RWTU field, for which the watchdog timer is set. The watchdog timer is triggered with the programmed value after the Rx DMA completes the transfer of a packet for which the RI bit is not set in the DMA_CH0_Status register, because of the setting of Interrupt Enable bit in the corresponding descriptor RDES3[30]. When the watchdog timer runs out, the RI bit is set and the timer is stopped. The watchdog timer is reset when the RI bit is set high because of automatic setting of RI as per the Interrupt Enable bit RDES3[30] of any received packet. Value After Reset: 0x0 Testable: untestable
0x143c	DMA_CH6_Slot_Function_Control_Status	The Slot Function Control and Status register contains the control bits for slot function and the status for Transmit path.	31:20	Reserved_31_20	R	0x0	Reserved. Value After Reset: 0x0
			19:16	RSN	R	0x0	Reference Slot Number This field gives the current value of the reference slot number in the DMA. It is used for slot comparison. Value After Reset: 0x0 Testable: untestable
			15:4	SIV	R/W	0x7c	Slot Interval Value This field controls the period of the slot interval in which the TxDMA fetches the scheduled packets. A value of 0 specifies the slot interval of 1 us while the maximum value 4095 specifies the slot interval of 4096us. The default/reset value is 0x07C which corresponds to slot interval of 125us Value After Reset: 0x7c
			3:2	Reserved_3_2	R	0x0	Reserved. Value After Reset: 0x0
			1	ASC	R/W	0x0	Advance Slot Check When set, this bit enables the DMA to fetch the data from the buffer when the slot number (SLOTNUM) programmed in the Tx descriptor is equal to the reference slot number given in the RSN field or ahead of the reference slot number by up to two slots This bit is applicable only when the ESC bit is set.Values: 0x0 (DISABLE): Advance Slot Check is disabled 0x1 (ENABLE): Advance Slot Check is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	ESC	R/W	0x0	Enable Slot Comparison When set, this bit enables the checking of the slot numbers programmed in the Tx descriptor with the current reference given in the RSN field. The DMA fetches the data from the corresponding buffer only when the slot number is equal to the reference slot number or ahead of the reference slot number by one slot When reset, this bit disables the checking of the slot numbers. The DMA fetches the data immediately after the descriptor is processed. Note: The UFO (UDP Fragmentation over IPv4)/TSO/USO must not be enabled along with TBS/AVB Slot number check. The UFO/TSO/USO involves multiple packets/segments/fragments transmission for single packet received from application and the slot number check are applicable for fetching of only first segment/fragment. As a result it might be difficult for software to specify slot number for subsequent packets.Values: 0x0 (DISABLE): Slot Comparison is disabled 0x1 (ENABLE): Slot Comparison is enabled Value After Reset: 0x0
0x1444	DMA_CH6_Current_App_Tx Desc	The Channel6 Current Application Transmit Descriptor register points to the current Transmit descriptor read by the DMA.	31:0	CURTDESAPTR	R	0x0	Application Transmit Descriptor Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x144c	DMA_CH6_Current_App_Rx Desc	The Channel6 Current Application Receive Descriptor register points to the current Receive descriptor read by the DMA.	31:0	CURRDESAPTR	R	0x0	Application Receive Descriptor Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x1450	DMA_CH6_Current_App_Tx Buffer_H	The Channel6 Current Application Transmit Buffer Address High register has the higher 8 or 16 bits of the current address of the Transmit buffer address read by the DMA.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	CURTBUFAPTRH	R	0x0	Application Transmit Buffer Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x1454	DMA_CH6_Current_App_Tx Buffer	The Channel6 Current Application Transmit Buffer Address register points to the current Tx buffer address read by the DMA.	31:0	CURTBUFAPTR	R	0x0	Application Transmit Buffer Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x1458	DMA_CH6_Current_App_Rx Buffer_H	The Channel6 Current Application Receive Buffer Address High register has the higher 8 or 16 bits of the current address of the Receive buffer address read by the DMA.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	CURRBUFAPTRH	R	0x0	Application Receive Buffer Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x145c	DMA_CH6_Current_App_Rx Buffer	The Channel 6 Current Application Receive Buffer Address register points to the current Rx buffer address read by the DMA.	31:0	CURRBUFAPTR	R	0x0	Application Receive Buffer Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			31:22	Reserved_31_22	R	0x0	Reserved. Value After Reset: 0x0
			21:19	REB	R	0x0	Rx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AXI interface. Bit 21 1'b1: Error during data transfer by Rx DMA 1'b0: No Error during data transfer by Rx DMA Bit 20 1'b1: Error during descriptor access 1'b0: Error during data buffer access Bit 19 1'b1: Error during read transfer 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt. Value After Reset: 0x0 Testable: untestable
			18:16	TEB	R	0x0	Tx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AXI interface. Bit 18 1'b1: Error during data transfer by Tx DMA 1'b0: No Error during data transfer by Tx DMA Bit 17 1'b1: Error during descriptor access 1'b0: Error during data buffer access Bit 16 1'b1: Error during read transfer 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt. Value After Reset: 0x0 Testable: untestable
			15	NIS	R/W	0x0	Normal Interrupt Summary Normal Interrupt Summary bit value is the logical OR of the following bits when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: Bit 0: Transmit Interrupt Bit 2: Transmit Buffer Unavailable Bit 6: Receive Interrupt Bit 11: Early Receive Interrupt Only unmasked bits (interrupts for which interrupt enable is set in DMA_CH0_Interrupt_Enable register) affect the Normal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit which causes NIS to be set is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Normal Interrupt Summary status not detected 0x1 (ACTIVE): Normal Interrupt Summary status detected Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1460	DMA_CH6_Status	The software driver (application) reads the Status register during interrupt service routine or polling to determine the status of the DMA.Note: The number of DMA_CH6_Status register in the configuration is the higher of number of Rx DMA Channels and Tx DMA Channels.	14	AIS	R/W	0x0	Abnormal Interrupt Summary Abnormal Interrupt Summary bit value is the logical OR of the following when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: Bit 1: Transmit Process Stopped Bit 7: Receive Buffer Unavailable Bit 8: Receive Process Stopped Bit 10: Early Transmit Interrupt Bit 12: Fatal Bus Error Bit 13: Context Descriptor Error Only unmasked bits affect the Abnormal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit, which causes AIS to be set, is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Abnormal Interrupt Summary status not detected 0x1 (ACTIVE): Abnormal Interrupt Summary status detected Value After Reset: 0x0 Testable: untestable
			13	CDE	R/W	0x0	Context Descriptor Error This bit indicates that the DMA Tx/Rx engine received a descriptor error, which indicates invalid context in the middle of packet flow (intermediate descriptor) or all one's descriptor in Tx case and on Rx side it indicates DMA has read a descriptor with either of the buffer address as ones which is considered to be invalid. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Context Descriptor Error status not detected 0x1 (ACTIVE): Context Descriptor Error status detected Value After Reset: 0x0 Testable: untestable
			12	FBE	R/W	0x0	Fatal Bus Error This bit indicates that a bus error occurred (as described in the EB field). When this bit is set, the corresponding DMA channel engine disables all bus accesses. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Fatal Bus Error status not detected 0x1 (ACTIVE): Fatal Bus Error status detected Value After Reset: 0x0 Testable: untestable
			11	ERI	R/W	0x0	Early Receive Interrupt This bit when set indicates that the RxDMA has completed the transfer of packet data to the memory. In configurations supporting ERIC, ERIC = 0: This bit is set only after the Rx DMA has completely filled up a receive buffer with packet data. ERIC = 1: This bit is set after every burst transfer of data from the Rx DMA to the buffer. The setting of RI bit automatically clears this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Early Receive Interrupt status not detected 0x1 (ACTIVE): Early Receive Interrupt status detected Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			10	ETI	R/W	0x0	Early Transmit Interrupt This bit when set indicates that the TxDMA has completed the transfer of packet data to the MTL TXFIFO memory. In configurations supporting ERIC: ETIC = 0: This bit is set only after the Tx DMA has transferred a complete packet to MTL. ETIC = 1: This bit is set after completion of (partial) packet data transfer from buffers in the Transmit descriptor in which IOC = 1. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Early Transmit Interrupt status not detected 0x1 (ACTIVE): Early Transmit Interrupt status detected Value After Reset: 0x0 Testable: untestable
			9	RWT	R/W	0x0	Receive Watchdog Timeout This bit is asserted when a packet with length greater than 2,048 bytes (10,240 bytes when Jumbo Packet mode is enabled) is received.Values: 0x0 (INACTIVE): Receive Watchdog Timeout status not detected 0x1 (ACTIVE): Receive Watchdog Timeout status detected Value After Reset: 0x0 Testable: untestable
			8	RPS	R/W	0x0	Receive Process Stopped This bit is asserted when the Rx process enters the Stopped state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Process Stopped status not detected 0x1 (ACTIVE): Receive Process Stopped status detected Value After Reset: 0x0 Testable: untestable
			7	RBU	R/W	0x0	Receive Buffer Unavailable This bit indicates that the application owns the next descriptor in the Receive list, and the DMA cannot acquire it. The Rx process is suspended. To resume processing Rx descriptors, the application must change the ownership of the descriptor and issue a Receive Poll Demand command. If this command is not issued, the Rx process resumes when the next recognized incoming packet is received. In ring mode, the application must advance the Receive Descriptor Tail Pointer register of a channel. This bit is set only when the DMA owns the previous Rx descriptor. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Buffer Unavailable status not detected 0x1 (ACTIVE): Receive Buffer Unavailable status detected Value After Reset: 0x0 Testable: untestable
			6	RI	R/W	0x0	Receive Interrupt This bit indicates that the packet reception is complete. When packet reception is complete, Bit 31 of RDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor. The reception remains in the Running state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Interrupt status not detected 0x1 (ACTIVE): Receive Interrupt status detected Value After Reset: 0x0 Testable: untestable
			5:3	Reserved_5_3	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			2	TBU	R/W	0x0	Transmit Buffer Unavailable This bit indicates that the application owns the next descriptor in the Transmit list, and the DMA cannot acquire it. Transmission is suspended. The TPS0 field of the DMA_Debug_Status0-DMA_Debug_Status2 register explains the Transmit Process state transitions. To resume processing the Transmit descriptors, the application must do the following: Change the ownership of the descriptor by setting Bit 31 of TDES3. Issue a Transmit Poll Demand command. For ring mode, the application must advance the Transmit Descriptor Tail Pointer register of a channel. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Buffer Unavailable status not detected 0x1 (ACTIVE): Transmit Buffer Unavailable status detected Value After Reset: 0x0 Testable: untestable
			1	TPS	R/W	0x0	Transmit Process Stopped This bit is set when the transmission is stopped. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Process Stopped status not detected 0x1 (ACTIVE): Transmit Process Stopped status detected Value After Reset: 0x0 Testable: untestable
			0	TI	R/W	0x0	Transmit Interrupt This bit indicates that the packet transmission is complete. When transmission is complete, Bit 31 of TDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Interrupt status not detected 0x1 (ACTIVE): Transmit Interrupt status detected Value After Reset: 0x0 Testable: untestable
0x1464	DMA_CH6_Miss_Frame_Cnt	This register has the number of packet counter that got dropped by the DMA either due to Bus Error or due to programming RPF field in DMA_CH6_Rx_Control register.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15	MFCO	R	0x0	Overflow status of the MFC Counter When this bit is set then the MFC counter does not get incremented further. The bit gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Miss Frame Counter overflow not occurred 0x1 (ACTIVE): Miss Frame Counter overflow occurred Value After Reset: 0x0
			14:11	Reserved_14_11	R	0x0	Reserved. Value After Reset: 0x0
			10:0	MFC	R	0x0	Dropped Packet Counters This counter indicates the number of packet counters that are dropped by the DMA either because of bus error or because of programing RPF field in DMA_CH0_Rx_Control register. The counter gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1468	DMA_CH6_RXP_Accept_Cnt	The DMA_CH6_RXP_Accept_Cnt registers provides the count of the number of frames accepted by Rx Parser.	31	RXPACOF	R	0x0	Rx Parser Accept Counter Overflow Bit When set, this bit indicates that the RXPAC Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Rx Parser Accept Counter overflow not occurred 0x1 (ACTIVE): Rx Parser Accept Counter overflow occurred Value After Reset: 0x0
			30:0	RXPAC	R	0x0	Rx Parser Accept Counter This 31-bit counter is implemented when a Rx Parser Accept a packet due to AF =1. The counter is cleared when the register is read. Value After Reset: 0x0
0x1480	DMA_CH7_Control	The DMA Channel7 Control register specifies the MSS value for segmentation, length to skip between two descriptors, and also the features such as header splitting and 8xPBL mode.	31:25	Reserved_31_25	R	0x0	Reserved. Value After Reset: 0x0
			24	SPH	R/W	0x0	Split Headers 1: DMA splits the header and payload in the receive path. The DMA writes the header to the Buffer Address1 of RDES0. The DMA writes the payload to the buffer to which the Buffer Address2 is pointing. The software must ensure that the header fits into the Receive buffers. If the header length exceeds the receive buffer size, the DMA does not split the header and payload. This bit is available only when you select Enable Split Header Structure feature.Values: 0x0 (DISABLE): Split Headers feature is disabled 0x1 (ENABLE): Split Headers feature is enabled Value After Reset: 0x0
			23:21	Reserved_23_21	R	0x0	Reserved. Value After Reset: 0x0
			20:18	DSL	R/W	0x0	Descriptor Skip Length This bit specifies the Lword (128-bit) number of locations to skip between two descriptors. The address skipping starts from the end of the current descriptor to the start of the next descriptor. When the DSL value is equal to zero, the DMA takes the descriptor table as contiguous. Value After Reset: 0x0
			17	Reserved_17	R	0x0	Reserved. Value After Reset: 0x0
			16	PBLx8	R/W	0x0	8xPBL mode 1: Indicates that the PBL value programmed in Bits[21:16] in DMA_CH0_Tx_Control and Bits[21:16] in DMA_CH0_Rx_Control is multiplied by eight times. Therefore, the DMA transfers the data in 8, 16, 32, 64, 128, and 256 beats depending on the PBL value. Values: 0x0 (DISABLE): 8xPBL mode is disabled 0x1 (ENABLE): 8xPBL mode is enabled Value After Reset: 0x0
			15:14	Reserved_15_14	R	0x0	Reserved. Value After Reset: 0x0
			13:0	MSS	R/W	0x0	Maximum Segment Size This field specifies the maximum segment size that must be used while segmenting the packet. This field is valid only if the TSE bit of DMA_CH0_Tx_Control register is set. Program a value in this field that is more than the configured datawidth in bytes. Use MSS value of 64 bytes or more. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			31	Reserved_31	R	0x0	Reserved. Value After Reset: 0x0
			30:29	TFSEL	R/W	0x0	TBS Fetch Select Select bits for one of the four DMA_TBS_CTRL register fields (FTOS,FGSN,FTOV) for the channel Values: 2'b00: DMA_TBS_CTRL0 2'b01: DMA_TBS_CTRL1 2'b10: DMA_TBS_CTRL2 2'b11: DMA_TBS_CTRL3 (Reserved if DWC_EQOS_NUM_DMA_TX_CH=3) Value After Reset: 0x0
			28	EDSE	R/W	0x0	Enhanced Descriptor Enable 1: The corresponding channel uses Enhanced Descriptors that are 32 Bytes for both Normal and Context Descriptors. 0: The corresponding channel uses the descriptors that are 16 Bytes. Values: 0x0 (DISABLE): Enhanced Descriptor is disabled 0x1 (ENABLE): Enhanced Descriptor is enabled Value After Reset: 0x0
			27:24	TQOS	R/W	0x0	Transmit QOS. This field is used to drive arqos_m_o[3:0] or awqos_m_o[3:0] output signals for all transactions of DMA Tx Channel0. Value After Reset: 0x0
			23	Reserved_23	R	0x0	Reserved. Value After Reset: 0x0
			22			#VALUE!	Reserved Field: Yes
			21:16	TxPBL	R/W	0x0	Transmit Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in DMA_CH0_Control register. 2. Set the TxPBL. Note: The maximum value of TxPBL must be less than or equal to half the Tx Queue size (TQS field of MTL_TxQ0_Operation_Mode register) in terms of beats. This is required so that the Tx Queue has space to store at least another Tx PBL worth of data while the MTL Tx Queue Controller is transferring data to MAC. For example, in 32-bit data width configurations the total locations in Tx Queue of size 512 bytes is 128, TxPBL and 8xPBL needs to be programmed to less than or equal to 64. Value After Reset: 0x0
			15	IPBL	R/W	0x0	Ignore PBL Requirement 1: DMA does not check for PBL number of locations in the MTL before initiating a transfer. If space is not available, the MTL might use handshaking to slow the DMA. Note: Do not use this mode when you enable multiple Transmit DMA channels. Because, it might block other Transmit and Receive DMA Channels from accessing the Read Data Channel of the AXI bus until space is available in Transmit Queue for current transfer.Values: 0x0 (DISABLE): Ignore PBL Requirement is disabled 0x1 (ENABLE): Ignore PBL Requirement is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1484	DMA_CH7_Tx_Control	The DMA Channel7 Transmit Control register controls the Tx features such as PBL, TCP segmentation, and Tx Channel weights.	14:13	TSE_MODE	R/W	0x0	TSE Mode 00: TSO/USO (segmentation functionality is enabled). In this mode, the setting of TSE bit enables the TSO/USO segmentation. 01: UFO with Checksum (UDP Fragmentation over IPv4 with Checksum). In this mode, the setting of TSE bit enables the UDP fragmentation functionality with Checksum for all the UDP packets. 10: UFO without Checksum (UDP Fragmentation over IPv4 with Checksum). In this mode, the setting of TSE bit enables the UDP fragmentation functionality without Checksum for all the UDP packets. 11: Reserved Values: 0x0 (TSO_USO): TSO/USO 0x1 (UFOWC): UFO with Checksum 0x2 (UFOWOC): UFO without Checksum 0x3 (RSVD): Reserved Value After Reset: 0x0
			12	TSE	R/W	0x0	TCP Segmentation Enabled 1: DMA performs the TCP segmentation or UDP Segmentation/Fragmentation for packets in this channel. The TCP segmentation or UDP packet's segmentation/Fragmentation is performed only for those packets for which the TSE bit (TDES0[19]) = 1 in the Tx Normal descriptor. When this bit is 1, the TxPBL value must be greater than 4. Values: 0x0 (DISABLE): TCP Segmentation is disabled 0x1 (ENABLE): TCP Segmentation is enabled Value After Reset: 0x0
			11:5	Reserved_11_5	R	0x0	Reserved. Value After Reset: 0x0
			4	OSF	R/W	0x0	Operate on Second Packet 1: Instructs the DMA to process the second packet of the Tx data even before the status for the first packet is obtained. Values: 0x0 (DISABLE): Operate on Second Packet disabled 0x1 (ENABLE): Operate on Second Packet enabled Value After Reset: 0x0
			3:1	TCW	R/W	0x0	Transmit Channel Weight This field indicates the weight assigned to the corresponding Tx channel. When reset is complete, this field is set to 0 for all channels by default, resulting in equal weights to all channels. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	ST	R/W	0x0	Start or Stop Transmission Command 1: Transmission is placed in the running state. The DMA checks the Tx list at the current position for a packet to be transmitted. The DMA tries to acquire descriptor from either of the following positions: The current position in the list This is the base address of the Transmit list set by the DMA_CH0_TxDesc_List_Address register. The position at which the transmission was previously stopped If the DMA does not own the current descriptor, the transmission enters the Suspended state and the TBU bit of the DMA_CH0_Status register is set. The Start Transmission command is effective only when the transmission is stopped. If the command is issued before setting the DMA_CH0_TxDesc_List_Address register, the DMA behavior is unpredictable. When this bit is reset, the transmission process is placed in the Stopped state after completing the transmission of the current packet. The Next Descriptor position in the Transmit list is saved, and it becomes the current position when the transmission is restarted. To change the list address, you need to program DMA_CH0_TxDesc_List_Address register with a new value when this bit is reset. The new value is considered when this bit is set again. The stop transmission command is effective only when the transmission of the current packet is complete or the transmission is in the Suspended state.Values: 0x0 (STOP): Stop Transmission Command 0x1 (START): Start Transmission Command Value After Reset: 0x0 Testable: untestable
			31	RPF	R/W	0x0	Rx Packet Flush. 1: DWC_ether_qos automatically flushes the packet from the Rx Queues destined to this DMA Rx Channel, when it is stopped. When this bit remains set and the DMA is re-started by the software driver, the packets residing in the Rx Queues that were received when this RxDMA was stopped, still get flushed out. The packets received by the MAC only after the DMA is re-started is routed to the Rx DMA.The flushing is done on the Read side of the Rx Queue. When the packet is duplicated to multiple Rx DMA, it is flushed by the Rx DMA to which it is routed only when it is stopped to prevent head-of-line blocking. The packet duplicated to multiple Rx DMA is not flushed when the Rx DMA is re-started, irrespective of whether the packet was received during one or more Rx DMAs were stopped. 0: DWC_ether_qos does not flush the packet in the Rx Queue destined to this RxDMA Channel when it is in STOP state. This might cause head-of-line blocking in the corresponding RxQueue. Note: The stopping of packet flow from a Rx DMA Channel to the application by setting RPF works only when there is one-to-one mapping of Rx Queue to Rx DMA channels. In Dynamic mapping mode, setting RPF bit in any DMA_CH0_Rx_Control register might flush packets from unintended Rx Queues which are destined to the stopped Rx DMA Channel.Values: 0x0 (DISABLE): Rx Packet Flush is disabled 0x1 (ENABLE): Rx Packet Flush is enabled Value After Reset: 0x0
			30:28	Reserved_30_28	R	0x0	Reserved. Value After Reset: 0x0
			27:24	RQOS	R/W	0x0	Rx AXI4 QOS. This field is used to drive arqos_m_o[3:0] or awqos_m_o[3:0] output signals for all transactions of DMA Rx Channel0. Value After Reset: 0x0
			23	Reserved_23	R	0x0	Reserved. Value After Reset: 0x0
			22			#VALUE!	Reserved Field: Yes

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1488	DMA_CH7_Rx_Control	The DMA Channel7 Receive Control register controls the Rx features such as PBL, buffer size, and extended status.	21:16	RxPBL	R/W	0x0	Receive Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in the DMA_CH0_Control register. 2. Set the RxPBL. Note: The maximum value of RxPBL must be less than or equal to half the Rx Queue size (RQS field of MTL_RxQ0_Operation_Mode register) in terms of beats. This is required so that the Rx Queue has space to store at least another Rx PBL worth of data while the Rx DMA is transferring a block of data. For example, in 32-bit data width configurations the total locations in Rx Queue of size 512 bytes is 128, so RxPBL and 8xPBL needs to be programmed to less than or equal to 64. Value After Reset: 0x0
			15	Reserved_15	R	0x0	Reserved. Value After Reset: 0x0
			14:5	RBSZ_13_y	R/W	0x0	Receive Buffer size High RBSZ[13:0] is split into two fields higher RBSZ_13_y and lower RBSZ_x_0. The RBSZ[13:0] field indicates the size of the Rx buffers specified in bytes. The maximum buffer size is limited to 16K bytes. The buffer size is applicable to payload buffers when split headers are enabled. Note: The buffer size must be a multiple of 16. This is required even if the value of buffer address pointer is not aligned to data bus width. Hence the lower RBSZ_x_0 bits are read-only and the value is considered as all-zero. Thus the RBSZ_13_y indicates the buffer size in terms of locations. Value After Reset: 0x0
			4:1	RBSZ_x_0	R	0x0	Receive Buffer size Low RBSZ[13:0] is split into two fields RBSZ_13_y and RBSZ_x_0. The RBSZ_x_0 is the lower field. This field is of width 4 bits. This field is read-only (RO). Value After Reset: 0x0
			0	SR	R/W	0x0	Start or Stop Receive 1: DMA attempts to acquire the descriptor from the Receive list and processes the incoming packets. The DMA tries to acquire descriptor from either of the following positions: -- The current position in the list This is the address set by the DMA_CH0_RxDesc_List_Address register. -- The position at which the Rx process was previously stopped If the DMA does not own the current descriptor, the reception is suspended and the RBU bit of the DMA_CH0_Status register is set. The Start Receive command is effective only when the reception is stopped. If the command is issued before setting the DMA_CH0_RxDesc_List_Address register, the DMA behavior is unpredictable. 0: Rx DMA operation stops after the transfer of the current packet. The next descriptor position in the Receive list is saved, and it becomes the current position after the Rx process is restarted. The Stop Receive command is effective only when the Rx process is in the Running (waiting for Rx packet) or Suspended state. Values: 0x0 (STOP): Stop Receive 0x1 (START): Start Receive Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x1490	DMA_CH7_TxDesc_List_HAddress	The Channel7 Tx Descriptor List HAddress register has the higher 8 or 16 bits of the start address of the Transmit descriptor list.You can write to this register only when the Tx DMA has stopped, that is, the ST bit is set to zero in DMA_CH7_Tx_Control register. When stopped, this register can be written with a new descriptor list address. When you set the ST bit to 1, the DMA takes the newly-programmed descriptor base address. If this register is not changed when the ST bit is set to 7, the DMA takes the descriptor address where it was stopped earlier. This register must be programmed with desired the higher address before programming the DMA_CH7_TxDesc_List_Address register.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	TDESHA	R/W	0x0	Start of Transmit List This field contains the most-significant 8 or 16 bits of the 40- or 48-bit base address of the first descriptor in the Transmit descriptor list. Value After Reset: 0x0
0x1494	DMA_CH7_TxDesc_List_Address	The Channel7 Tx Descriptor List Address register points the DMA to the start of Transmit descriptor list. The descriptor lists reside in the physical memory space of the application and must be Lword (128-bit). The DMA internally converts it to bus width aligned address by making the corresponding LSB to low.You can write to this register only when the Tx DMA has stopped, that is, the ST bit is set to zero in DMA_CH7_Tx_Control register. When stopped, this register can be written with a new descriptor list address. When you set the ST bit to 1, the DMA takes the newly-programmed descriptor base address. If this register is not changed when the ST bit is set to 7, the DMA takes the descriptor address where it was stopped earlier.	31:4	TDESLA	R/W	0x0	Start of Transmit List This field contains the base address of the first descriptor in the Transmit descriptor list. The DMA ignores the LSB bits (3:0) and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0 Testable: untestable
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x1498	DMA_CH7_RxDesc_List_HAddress	The Channel7 Rx Descriptor List HAddress register has the higher 8 or 16 bits of the start address of the Receive descriptor list.Writing to this register is permitted only when reception is stopped. When stopped, this register must be written to before the receive Start command is given. You can write to this register only when Rx DMA has stopped, that is, SR bit is set to zero in the DMA_CH7_RxDesc_List_Address registerWhen you set the SR bit to 1, the DMA takes the newly programmed descriptor base address.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	RDESHA	R/W	0x0	Start of Receive List This field contains the most-significant 8 or 16 bits of the 40-bit or 48-bit base address of the first descriptor in the Rx Descriptor list. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x149c	DMA_CH7_RxDesc_List_Address	The Channel7 Rx Descriptor List Address register points the DMA to the start of Receive descriptor list.This register points to the start of the Receive Descriptor List. The descriptor lists reside in the physical memory space of the application and must be Lword (128-bit). The DMA internally converts it to bus width aligned address by making the corresponding LS bits low. Writing to this register is permitted only when reception is stopped. When stopped, this register must be written to before the receive Start command is given. You can write to this register only when Rx DMA has stopped, that is, SR bit is set to zero in DMA_CH7_Rx_Control register. When stopped, this register can be written with a new descriptor list address.When you set the SR bit to 1, the DMA takes the newly programmed descriptor base address.	31:4	RDESLA	R/W	0x0	Start of Receive List This field contains the base address of the first descriptor in the Rx Descriptor list. The DMA ignores the LSB bits (3:0) and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0 Testable: untestable
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x14a0	DMA_CH7_TxDesc_Tail_Pointer	The Channel7 Tx Descriptor Tail Pointer register points to an offset from the base and indicates the location of the last valid descriptor.	31:4	TDTP	R/W	0x0	Transmit Descriptor Tail Pointer This field contains the tail pointer for the Tx descriptor ring. The software writes the tail pointer to add more descriptors to the Tx channel. The hardware tries to transmit all packets referenced by the descriptors between the head and the tail pointer registers. The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x14a8	DMA_CH7_RxDesc_Tail_Pointer	The Channel7 Rx Descriptor Tail Pointer Points to an offset from the base and indicates the location of the last valid descriptor.	31:4	RDTP	R/W	0x0	Receive Descriptor Tail Pointer This field contains the tail pointer for the Rx descriptor ring. The software writes the tail pointer to add more descriptors to the Rx channel. The hardware tries to write all received packets to the descriptors referenced between the head and the tail pointer registers. The width of this field depends on the configuration: 31:4 for 128-bit configuration Value After Reset: 0x0
			3:0	Reserved_LSb	R	0x0	Reserved. Value After Reset: 0x0
0x14ac	DMA_CH7_TxDesc_Ring_Length	The Tx Descriptor Ring Length register contains the length of the Transmit descriptor ring.	31:10	Reserved_31_10	R	0x0	Reserved. Value After Reset: 0x0
			9:0	TDRL	R/W	0x0	Transmit Descriptor Ring Length This field sets the maximum number of Tx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. It is recommended that a minimum ring descriptor length of 4. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x14b0	DMA_CH7_Rx_Control2	The Channel7 Receive Control register controls the Rx features such as Rx Descriptor Ring Length and Alternate Rx Buffer Size.	31:24	Reserved_31_24	R	0x0	Reserved. Value After Reset: 0x0
			23:18	ARBS	R/W	0x0	Alternate Receive Buffer Size Indicates size in bytes for Buffer 1 when ARBS is programmed to a non-zero value (when split header feature is not enabled). When split header feature is enabled, ARBS indicates the buffer size for header data. The maximum alternate buffer is limited to 1008 bytes. When ARBS=0, Rx Buffer1 and Rx Buffer2 sizes are based on RBSZ field of DMA_CH0_Rx_Control. Width of ARBS field is 6 bits. Value After Reset: 0x0
			17:10	Reserved_x_10	R	0x0	Reserved. Value After Reset: 0x0
			9:0	RDRL	R/W	0x0	Receive Descriptor Ring Length This register sets the maximum number of Rx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9. Value After Reset: 0x0
			31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15	NIE	R/W	0x0	Normal Interrupt Summary Enable When this bit is set, the normal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: Bit 0: Transmit Interrupt Bit 2: Transmit Buffer Unavailable Bit 6: Receive Interrupt Bit 11: Early Receive Interrupt When this bit is reset, the normal interrupt summary is disabled.Values: 0x0 (DISABLE): Normal Interrupt Summary is disabled 0x1 (ENABLE): Normal Interrupt Summary is enabled Value After Reset: 0x0
			14	AIE	R/W	0x0	Abnormal Interrupt Summary Enable When this bit is set, the abnormal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: Bit 1: Transmit Process Stopped Bit 7: Rx Buffer Unavailable Bit 8: Receive Process Stopped Bit 9: Receive Watchdog Timeout Bit 10: Early Transmit Interrupt Bit 12: Fatal Bus Error Bit 13: Context Descriptor Error When this bit is reset, the abnormal interrupt summary is disabled.Values: 0x0 (DISABLE): Abnormal Interrupt Summary is disabled 0x1 (ENABLE): Abnormal Interrupt Summary is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x14b4	DMA_CH7_Interrupt_Enable	The Channel7 Interrupt Enable register enables the interrupts reported by the Status register.	13	CDEE	R/W	0x0	Context Descriptor Error Enable When this bit is set along with the AIE bit, the Descriptor error interrupt is enabled. When this bit is reset, the Descriptor error interrupt is disabled.Values: 0x0 (DISABLE): Context Descriptor Error is disabled 0x1 (ENABLE): Context Descriptor Error is enabled Value After Reset: 0x0
			12	FBEE	R/W	0x0	Fatal Bus Error Enable When this bit is set along with the AIE bit, the Fatal Bus error interrupt is enabled. When this bit is reset, the Fatal Bus Error error interrupt is disabled.Values: 0x0 (DISABLE): Fatal Bus Error is disabled 0x1 (ENABLE): Fatal Bus Error is enabled Value After Reset: 0x0
			11	ERIE	R/W	0x0	Early Receive Interrupt Enable When this bit is set along with the NIE bit, the Early Receive interrupt is enabled. When this bit is reset, the Early Receive interrupt is disabled.Values: 0x0 (DISABLE): Early Receive Interrupt is disabled 0x1 (ENABLE): Early Receive Interrupt is enabled Value After Reset: 0x0
			10	ETIE	R/W	0x0	Early Transmit Interrupt Enable When this bit is set along with the AIE bit, the Early Transmit interrupt is enabled. When this bit is reset, the Early Transmit interrupt is disabled.Values: 0x0 (DISABLE): Early Transmit Interrupt is disabled 0x1 (ENABLE): Early Transmit Interrupt is enabled Value After Reset: 0x0
			9	RWTE	R/W	0x0	Receive Watchdog Timeout Enable When this bit is set along with the AIE bit, the Receive Watchdog Timeout interrupt is enabled. When this bit is reset, the Receive Watchdog Timeout interrupt is disabled.Values: 0x0 (DISABLE): Receive Watchdog Timeout is disabled 0x1 (ENABLE): Receive Watchdog Timeout is enabled Value After Reset: 0x0
			8	RSE	R/W	0x0	Receive Stopped Enable When this bit is set along with the AIE bit, the Receive Stopped Interrupt is enabled. When this bit is reset, the Receive Stopped interrupt is disabled.Values: 0x0 (DISABLE): Receive Stopped is disabled 0x1 (ENABLE): Receive Stopped is enabled Value After Reset: 0x0
			7	RBUE	R/W	0x0	Receive Buffer Unavailable Enable When this bit is set along with the AIE bit, the Receive Buffer Unavailable interrupt is enabled. When this bit is reset, the Receive Buffer Unavailable interrupt is disabled.Values: 0x0 (DISABLE): Receive Buffer Unavailable is disabled 0x1 (ENABLE): Receive Buffer Unavailable is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			6	RIE	R/W	0x0	Receive Interrupt Enable When this bit is set along with the NIE bit, the Receive Interrupt is enabled. When this bit is reset, the Receive Interrupt is disabled.Values: 0x0 (DISABLE): Receive Interrupt is disabled 0x1 (ENABLE): Receive Interrupt is enabled Value After Reset: 0x0
			5:3	Reserved_5_3	R	0x0	Reserved. Value After Reset: 0x0
			2	TBUE	R/W	0x0	Transmit Buffer Unavailable Enable When this bit is set along with the NIE bit, the Transmit Buffer Unavailable interrupt is enabled. When this bit is reset, the Transmit Buffer Unavailable interrupt is disabled.Values: 0x0 (DISABLE): Transmit Buffer Unavailable is disabled 0x1 (ENABLE): Transmit Buffer Unavailable is enabled Value After Reset: 0x0
			1	TXSE	R/W	0x0	Transmit Stopped Enable When this bit is set along with the AIE bit, the Transmission Stopped interrupt is enabled. When this bit is reset, the Transmission Stopped interrupt is disabled.Values: 0x0 (DISABLE): Transmit Stopped is disabled 0x1 (ENABLE): Transmit Stopped is enabled Value After Reset: 0x0
			0	TIE	R/W	0x0	Transmit Interrupt Enable When this bit is set along with the NIE bit, the Transmit Interrupt is enabled. When this bit is reset, the Transmit Interrupt is disabled.Values: 0x0 (DISABLE): Transmit Interrupt is disabled 0x1 (ENABLE): Transmit Interrupt is enabled Value After Reset: 0x0
0x14b8	DMA_CH7_Rx_Interrupt_Watchdog_Timer	The Receive Interrupt Watchdog Timer register indicates the watchdog timeout for Receive Interrupt (RI) from the DMA. When this register is written with a non-zero value, it enables the watchdog timer for the RI bit of the DMA_CH7_Status register.	31:18	Reserved_31_18	R	0x0	Reserved. Value After Reset: 0x0
			17:16	RWTU	R/W	0x0	Receive Interrupt Watchdog Timer Count Units This fields indicates the number of system clock cycles corresponding to one unit in RWT field. 2'b00: 256 2'b01: 512 2'b10: 1024 2'b11: 2048 For example, when RWT=2 and RWTU=1, the watchdog timer is set for 2*512=1024 system clock cycles. Value After Reset: 0x0 Testable: untestable
			15:8	Reserved_15_8	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			7:0	RWT	R/W	0x0	Receive Interrupt Watchdog Timer Count This field indicates the number of system clock cycles, multiplied by factor indicated in RWTU field, for which the watchdog timer is set. The watchdog timer is triggered with the programmed value after the Rx DMA completes the transfer of a packet for which the RI bit is not set in the DMA_CH0_Status register, because of the setting of Interrupt Enable bit in the corresponding descriptor RDES3[30]. When the watchdog timer runs out, the RI bit is set and the timer is stopped. The watchdog timer is reset when the RI bit is set high because of automatic setting of RI as per the Interrupt Enable bit RDES3[30] of any received packet. Value After Reset: 0x0 Testable: untestable
0x14bc	DMA_CH7_Slot_Function_Control_Status	The Slot Function Control and Status register contains the control bits for slot function and the status for Transmit path.	31:20	Reserved_31_20	R	0x0	Reserved. Value After Reset: 0x0
			19:16	RSN	R	0x0	Reference Slot Number This field gives the current value of the reference slot number in the DMA. It is used for slot comparison. Value After Reset: 0x0 Testable: untestable
			15:4	SIV	R/W	0x7c	Slot Interval Value This field controls the period of the slot interval in which the TxDMA fetches the scheduled packets. A value of 0 specifies the slot interval of 1 us while the maximum value 4095 specifies the slot interval of 4096us. The default/reset value is 0x07C which corresponds to slot interval of 125us Value After Reset: 0x7c
			3:2	Reserved_3_2	R	0x0	Reserved. Value After Reset: 0x0
			1	ASC	R/W	0x0	Advance Slot Check When set, this bit enables the DMA to fetch the data from the buffer when the slot number (SLOTNUM) programmed in the Tx descriptor is equal to the reference slot number given in the RSN field or ahead of the reference slot number by up to two slots This bit is applicable only when the ESC bit is set.Values: 0x0 (DISABLE): Advance Slot Check is disabled 0x1 (ENABLE): Advance Slot Check is enabled Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			0	ESC	R/W	0x0	Enable Slot Comparison When set, this bit enables the checking of the slot numbers programmed in the Tx descriptor with the current reference given in the RSN field. The DMA fetches the data from the corresponding buffer only when the slot number is equal to the reference slot number or ahead of the reference slot number by one slot When reset, this bit disables the checking of the slot numbers. The DMA fetches the data immediately after the descriptor is processed. Note: The UFO (UDP Fragmentation over IPv4)/TSO/USO must not be enabled along with TBS/AVB Slot number check. The UFO/TSO/USO involves multiple packets/segments/fragments transmission for single packet received from application and the slot number check are applicable for fetching of only first segment/fragment. As a result it might be difficult for software to specify slot number for subsequent packets.Values: 0x0 (DISABLE): Slot Comparison is disabled 0x1 (ENABLE): Slot Comparison is enabled Value After Reset: 0x0
0x14c4	DMA_CH7_Current_App_Tx Desc	The Channel7 Current Application Transmit Descriptor register points to the current Transmit descriptor read by the DMA.	31:0	CURTDESAPTR	R	0x0	Application Transmit Descriptor Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x14cc	DMA_CH7_Current_App_Rx Desc	The Channel7 Current Application Receive Descriptor register points to the current Receive descriptor read by the DMA.	31:0	CURRDESAPTR	R	0x0	Application Receive Descriptor Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x14d0	DMA_CH7_Current_App_Tx Buffer_H	The Channel7 Current Application Transmit Buffer Address High register has the higher 8 or 16 bits of the current address of the Transmit buffer address read by the DMA.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	CURTBUFAPTRH	R	0x0	Application Transmit Buffer Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x14d4	DMA_CH7_Current_App_Tx Buffer	The Channel7 Current Application Transmit Buffer Address register points to the current Tx buffer address read by the DMA.	31:0	CURTBUFAPTR	R	0x0	Application Transmit Buffer Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x14d8	DMA_CH7_Current_App_Rx Buffer_H	The Channel7 Current Application Receive Buffer Address High register has the higher 8 or 16 bits of the current address of the Receive buffer address read by the DMA.	31:8	Reserved_31_y	R	0x0	Reserved. Value After Reset: 0x0
			7:0	CURRBUFAPTRH	R	0x0	Application Receive Buffer Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset. Value After Reset: 0x0
0x14dc	DMA_CH7_Current_App_Rx Buffer	The Channel 7 Current Application Receive Buffer Address register points to the current Rx buffer address read by the DMA.	31:0	CURRBUFAPTR	R	0x0	Application Receive Buffer Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			31:22	Reserved_31_22	R	0x0	Reserved. Value After Reset: 0x0
			21:19	REB	R	0x0	Rx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AXI interface. Bit 21 1'b1: Error during data transfer by Rx DMA 1'b0: No Error during data transfer by Rx DMA Bit 20 1'b1: Error during descriptor access 1'b0: Error during data buffer access Bit 19 1'b1: Error during read transfer 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt. Value After Reset: 0x0 Testable: untestable
			18:16	TEB	R	0x0	Tx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AXI interface. Bit 18 1'b1: Error during data transfer by Tx DMA 1'b0: No Error during data transfer by Tx DMA Bit 17 1'b1: Error during descriptor access 1'b0: Error during data buffer access Bit 16 1'b1: Error during read transfer 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt. Value After Reset: 0x0 Testable: untestable
			15	NIS	R/W	0x0	Normal Interrupt Summary Normal Interrupt Summary bit value is the logical OR of the following bits when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: Bit 0: Transmit Interrupt Bit 2: Transmit Buffer Unavailable Bit 6: Receive Interrupt Bit 11: Early Receive Interrupt Only unmasked bits (interrupts for which interrupt enable is set in DMA_CH0_Interrupt_Enable register) affect the Normal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit which causes NIS to be set is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Normal Interrupt Summary status not detected 0x1 (ACTIVE): Normal Interrupt Summary status detected Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x14e0	DMA_CH7_Status	The software driver (application) reads the Status register during interrupt service routine or polling to determine the status of the DMA.Note: The number of DMA_CH7_Status register in the configuration is the higher of number of Rx DMA Channels and Tx DMA Channels.	14	AIS	R/W	0x0	Abnormal Interrupt Summary Abnormal Interrupt Summary bit value is the logical OR of the following when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: Bit 1: Transmit Process Stopped Bit 7: Receive Buffer Unavailable Bit 8: Receive Process Stopped Bit 10: Early Transmit Interrupt Bit 12: Fatal Bus Error Bit 13: Context Descriptor Error Only unmasked bits affect the Abnormal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit, which causes AIS to be set, is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Abnormal Interrupt Summary status not detected 0x1 (ACTIVE): Abnormal Interrupt Summary status detected Value After Reset: 0x0 Testable: untestable
			13	CDE	R/W	0x0	Context Descriptor Error This bit indicates that the DMA Tx/Rx engine received a descriptor error, which indicates invalid context in the middle of packet flow (intermediate descriptor) or all one's descriptor in Tx case and on Rx side it indicates DMA has read a descriptor with either of the buffer address as ones which is considered to be invalid. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Context Descriptor Error status not detected 0x1 (ACTIVE): Context Descriptor Error status detected Value After Reset: 0x0 Testable: untestable
			12	FBE	R/W	0x0	Fatal Bus Error This bit indicates that a bus error occurred (as described in the EB field). When this bit is set, the corresponding DMA channel engine disables all bus accesses. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Fatal Bus Error status not detected 0x1 (ACTIVE): Fatal Bus Error status detected Value After Reset: 0x0 Testable: untestable
			11	ERI	R/W	0x0	Early Receive Interrupt This bit when set indicates that the RxDMA has completed the transfer of packet data to the memory. In configurations supporting ERIC, ERIC = 0: This bit is set only after the Rx DMA has completely filled up a receive buffer with packet data. ERIC = 1: This bit is set after every burst transfer of data from the Rx DMA to the buffer. The setting of RI bit automatically clears this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Early Receive Interrupt status not detected 0x1 (ACTIVE): Early Receive Interrupt status detected Value After Reset: 0x0 Testable: untestable

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			10	ETI	R/W	0x0	Early Transmit Interrupt This bit when set indicates that the TxDMA has completed the transfer of packet data to the MTL TXFIFO memory. In configurations supporting ERIC: ETIC = 0: This bit is set only after the Tx DMA has transferred a complete packet to MTL. ETIC = 1: This bit is set after completion of (partial) packet data transfer from buffers in the Transmit descriptor in which IOC = 1. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Early Transmit Interrupt status not detected 0x1 (ACTIVE): Early Transmit Interrupt status detected Value After Reset: 0x0 Testable: untestable
			9	RWT	R/W	0x0	Receive Watchdog Timeout This bit is asserted when a packet with length greater than 2,048 bytes (10,240 bytes when Jumbo Packet mode is enabled) is received.Values: 0x0 (INACTIVE): Receive Watchdog Timeout status not detected 0x1 (ACTIVE): Receive Watchdog Timeout status detected Value After Reset: 0x0 Testable: untestable
			8	RPS	R/W	0x0	Receive Process Stopped This bit is asserted when the Rx process enters the Stopped state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Process Stopped status not detected 0x1 (ACTIVE): Receive Process Stopped status detected Value After Reset: 0x0 Testable: untestable
			7	RBU	R/W	0x0	Receive Buffer Unavailable This bit indicates that the application owns the next descriptor in the Receive list, and the DMA cannot acquire it. The Rx process is suspended. To resume processing Rx descriptors, the application must change the ownership of the descriptor and issue a Receive Poll Demand command. If this command is not issued, the Rx process resumes when the next recognized incoming packet is received. In ring mode, the application must advance the Receive Descriptor Tail Pointer register of a channel. This bit is set only when the DMA owns the previous Rx descriptor. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Buffer Unavailable status not detected 0x1 (ACTIVE): Receive Buffer Unavailable status detected Value After Reset: 0x0 Testable: untestable
			6	RI	R/W	0x0	Receive Interrupt This bit indicates that the packet reception is complete. When packet reception is complete, Bit 31 of RDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor. The reception remains in the Running state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Receive Interrupt status not detected 0x1 (ACTIVE): Receive Interrupt status detected Value After Reset: 0x0 Testable: untestable
			5:3	Reserved_5_3	R	0x0	Reserved. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
			2	TBU	R/W	0x0	Transmit Buffer Unavailable This bit indicates that the application owns the next descriptor in the Transmit list, and the DMA cannot acquire it. Transmission is suspended. The TPS0 field of the DMA_Debug_Status0-DMA_Debug_Status2 register explains the Transmit Process state transitions. To resume processing the Transmit descriptors, the application must do the following: Change the ownership of the descriptor by setting Bit 31 of TDES3. Issue a Transmit Poll Demand command. For ring mode, the application must advance the Transmit Descriptor Tail Pointer register of a channel. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Buffer Unavailable status not detected 0x1 (ACTIVE): Transmit Buffer Unavailable status detected Value After Reset: 0x0 Testable: untestable
			1	TPS	R/W	0x0	Transmit Process Stopped This bit is set when the transmission is stopped. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Process Stopped status not detected 0x1 (ACTIVE): Transmit Process Stopped status detected Value After Reset: 0x0 Testable: untestable
			0	TI	R/W	0x0	Transmit Interrupt This bit indicates that the packet transmission is complete. When transmission is complete, Bit 31 of TDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.Values: 0x0 (INACTIVE): Transmit Interrupt status not detected 0x1 (ACTIVE): Transmit Interrupt status detected Value After Reset: 0x0 Testable: untestable
0x14e4	DMA_CH7_Miss_Frame_Cnt	This register has the number of packet counter that got dropped by the DMA either due to Bus Error or due to programming RPF field in DMA_CH7_Rx_Control register.	31:16	Reserved_31_16	R	0x0	Reserved. Value After Reset: 0x0
			15	MFCO	R	0x0	Overflow status of the MFC Counter When this bit is set then the MFC counter does not get incremented further. The bit gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Miss Frame Counter overflow not occurred 0x1 (ACTIVE): Miss Frame Counter overflow occurred Value After Reset: 0x0
			14:11	Reserved_14_11	R	0x0	Reserved. Value After Reset: 0x0
			10:0	MFC	R	0x0	Dropped Packet Counters This counter indicates the number of packet counters that are dropped by the DMA either because of bus error or because of programing RPF field in DMA_CH0_Rx_Control register. The counter gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event. Value After Reset: 0x0

Base Address: 0x3501_0000							
Offset	Register Name	Register Description	Bits	Field Name	Memory Access	Default Value	Field Description
0x14e8	DMA_CH7_RXP_Accept_Cnt	The DMA_CH7_RXP_Accept_Cnt registers provides the count of the number of frames accepted by Rx Parser.	31	RXPACOF	R	0x0	Rx Parser Accept Counter Overflow Bit When set, this bit indicates that the RXPAC Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event.Values: 0x0 (INACTIVE): Rx Parser Accept Counter overflow not occurred 0x1 (ACTIVE): Rx Parser Accept Counter overflow occurred Value After Reset: 0x0
			30:0	RXPAC	R	0x0	Rx Parser Accept Counter This 31-bit counter is implemented when a Rx Parser Accept a packet due to AF =1. The counter is cleared when the register is read. Value After Reset: 0x0

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